YOGI VEMANA UNIVERSITY
Vemanapuram, Kadapa-516 005, A.P., INDIA
(A State University, Accredited with "A" Grade by NAAC)
M.Sc. Psychology Syllabus

Course structure
(For the DDE students admitted during the academic year 2022 – 23 onwards)

Course Code	Title of the Course	No. of credits	No. of hours per week	Practical/ Project	Internal Assessment	Semester End Exams	Total
SEMESTER-I							
PSY15141	General Psychology	04	04		25	75	100
PSY 15142	Social Psychology	04	04		25	75	100
PSY 15143	Abnormal Psychology	04	04		25	75	100
PSY 15144	Psychological Measurement and Testing	04	04		25	75	100
PSY 15145	Environmental Psychology	04	04		25	75	100
PSY 15146	Research methodology	04	16		25	75	100
	Sub-total	24	36		150	450	600
SEMESTER-II							
PSY 25141	Cognitive Psychology	04	04		25	75	100
PSY 25142	Applied Social Psychology	04	04		25	75	100
PSY 25143	Clinical Psychology	04	04		25	75	100
PSY 25144	Positive Psychology	04	04		25	75	100
PSY 25145	Psychology of Stress	04	04		25	75	100
PSY 25146	Statistics in Psychology	04	16		25	75	100
	Sub-total	24	36		150	450	600
SEMESTER-III							
PSY 35141	Personality-I	04	04		25	75	100
PSY 35142	Developmental Psychology-I	04	04		25	75	100
PSY 35143	Counselling Psychology-I	04	04		25	75	100
PSY 35144	Health Psychology	04	04		25	75	100
PSY 35145	Educational Psychology	04	04		25	75	100
PSY 35146 (Skill oriented course)	Internship with Registered Hospitals/NGO/ Organizations /In-situ training at Psychiatry wards (240 hrs.) Or Self - Learning Course (MOOCS)	04	16		50	50	100
	Sub-total	24	36		150	450	600
SEMESTER-IV							
PSY 45141	Theories of Personality-II	24	36		25	75	100
PSY 45142	Developmental Psychology-II	04	04		25	75	100
PSY 45143	Counselling Psychology- II	04	04		25	75	100
PSY 45144	Rehabilitation Psychology	04	04		25	75	100
PSY 45145	Industrial /Organizational Psychology	04	04		25	75	100
PSY 45146 Multi-Disciplinary/ Project	Project Work- Community Service project (240 hrs.) (Observational report/ Case study of an organization/ Field survey/ Survey /In-situ report)	04	16		50	50	100
	Sub-total	24	36		150	450	600
	Grand Total	**96**	**144**		**600**	**1800**	**2400**

PSY15141 GENERAL PSYCHOLOGY

Objectives:

General Psychology will helps the Students to demonstrate acquisition of both factual knowledge and the ability to conceptualize and apply this knowledge to their own behavior, ways of interacting with others, and to their roles in culture and society

Unit – I: Psychology as science

Lesson-1 Introduction to Psychology- Definition, nature, scope and aims of Psychology- The rise of experimental psychology.

Lesson-2 Branches and Fields of Psychology- Early Divisions, Basic Fields, Applied Field, Both basic and Applied Fields.

Lesson-3 Schools of psychology- Structuralism, Functionalism, Behaviorism, Psychoanalytic.

Lesson-4 Methods of Psychology – Introspection method, observation method, Survey method, case study method, experimental method and Correlational method.

Unit-II: Physiological basis of behavior

Lesson-5 Neuron- The structure of neuron-Types of neurons-Varieties of neuron transmitters.

Lesson-6 Nervous system -Central and Peripheral nervous system- Autonomic Nervous system.

Lesson-7 Brain - Structure and functions of brain- Limbic System.

Lesson-8 Endocrine glands – Effects of over and under secretion of glands.

Unit - III: Sensation, Attention & Perception

Lesson-9 Sensation: Sensory process-Senses and sensory organs, general characteristics of Sensation and Sensory thresholds, Subliminal Perception, Signal Detection Theory.

Lesson-10 Attention: Selective attention – Kinds of Attention- Barriers of Attention- Concepts and Mechanisms- Selective Attention.

Lesson-11 Determinants and Theories -Broadbent, Treisman and Deutsch and Deutsch, Norman Models, Sustained Attention, Theories Expectancy, Signal Detection, Arousal, Habituation; Intensive Properties of Attention, Fluctuations of Attention, Attention Span.

Lesson-12 Perception: Meaning of Sensation- Types of Sensation - Definition of Perception – Factors influencing Perception (Physiological and Psychological) - Perception and Illusion

Unit – IV: Motivation and Emotion

Lesson-13 Motivation-definition and Nature - Instinct, drive, needs, incentive, motive, Classification of motives.

Lesson-14 Theories of Motivation- Instinct theories – Freud & Mc Dougal; Need hierarchy theory – Maslow

Lesson-15 Emotion: Nature, kinds of emotions- physiological basis of emotion.

Lesson-16 Theories of emotion: The James – Lange theory – The Cannon – Bard theory.

References:

- Feldman, Robert. S (2004). Understanding Psychology 6/E W. (2004). (n.p.): McGraw-Hill Education (India) Pvt Limited,New Delhi.
- Hilgard, E.R, Atkinson, R.C. &Atkison, R.R., (1976). Introduction to Psychology(6thedition), New Delhi: Oxford and IBM Publishing Company Private Limited.
- Robert S. Feldman (2011). Understanding Psychology. (n.d.). United States: McGraw-HillEducation (India) Pvt Limited.

- Robert S. Feldman (2021). Understanding Psychology (15th Edition). New Delhi: Tata – McGraw Hill Education Pvt.Ltd

Outcomes: To understand the nature and origin of psychology
To know the nature of Sensation, Perception, Learning and Memory
To grasp the meaning and processes of cognition, intelligence and creativity
To understand the basic aspects and theories of Motivation and Emotion
To know the nature of Personality and its Assessments

Unit – I: Psychology as Science

PSY15141

Lesson-1: Introduction to Psychology- Definition, nature, scope and aims of Psychology- The rise of experimental psychology.

INTRODUCTION

Do you know that the word psychology is derived from two Greek words “psyche” and “logos”? Psyche means soul (life) and logos means knowledge (explanation). or the study of the soul. The founders of psychology were both philosophers and psychologists and both tried to find answers to the philosophical questions in a scientific manner, like distinction between nature and nurture, existence of free will, etc. The word ‘psychology’ (Psychologia in Latin) dates back to 16th century, but was popularized in German by the early 18th century philospher Christian Worff. One of the most influential early users of the term in English was Herbert Spencer, who published the first volume of his Principles of Psychology in 1870.

Psychologists preferred using the word mind rather than soul. Psychology was one of the last special sciences that separated from philosophy in the nineteenth century. Today, psychology is very popular among students and has become a part of everyday lives. So, in this unit, we will talk about interesting facts and information about the origin of psychology. This unit will explain what psychology is all about and what are the different approaches applied by psychologists to understand human behavior. The unit will further elucidate various subfields of psychology and the methods used to conduct research in psychology. It will also discuss the diverse areas where psychologists work. Finally, development of psychology in India will be discussed.

DEFINITION OF PSYCHOLOGY

It is difficult to define psychology as it has a very vast scope. As the discipline was emerging, there were long debates as what should be studied in psychology and how it should be studied. The most popular definition is that it is the science of human and animal behavior and it includes the application of this science to different problems of the human beings.The most accepted definition today is psychology is the science of behavior and mental processes. Thus, there are three parts to the definition. Firstly, psychology is a science because it systematically tries to gather data by carefully observing and recording events or behaviors of humans and animals

under controlled experimental conditions.Thus, measurement of behavior is as objective as possible in psychology.The knowledge that is gained collecting data with the help of experiments and other methods of observation, is then applied to solve real-life problems.

Secondly, the term behavior in the definition implies the overt actions and reactions. Whatever the animals or humans do, that is of interest,can be recorded, such as actions, thoughts, emotions, attitudes, etc. Lastly, the term mental processes refer to all the covert or internal processes, such as thinking, feeling and remembering. Succinctly said, psychology is a science that seeks to understand and predict human and animal behavior (Morgan, King, & Robinson, 1984).

PSYCHOLOGY; MEANING

- The term psychology is derived from Greek words Psyche and Logos.
- Psyche means, **Soul/Mind** and Logos means **Science/Study.** Thus the word psychology means **Science of the Soul or Study of the soul.**
- Psychology is the branch of Philosophy.
- First stage 'study of the soul' *philosophy dominated stage. ? Raised; what is soul? How can it be studied?
- Second stage 'study of the mind' *philosopher come psychologist period. Mind mysterious then soul. ? Raised; what is mindl? How can it be studied?
- Psychology deals with mental processes (thoughts, feelings sensations, perceptions and wishes). So psychology is defined as the science of the mind.
- Mental processes include reasoning, learning, thinking, problem solving, perceptions etc..
- Third stage (1890-1894) **'Psychology as the study of consciousness'** *said by William James, Wilhelm Wundt & Edward Bradford Titchener.
- J.B.Watson(1913) also defines psychology as the science of human behaviour.
- Psychology as the science of soul: The word psychology comes from two Greek words: psyche meaning soul and logos meaning science. Thus, psychology means the science of soul. Philosophers like Democritus (about 460 BC), Plato (427–347 BC) and Aristotle

(384–322 BC) interpreted psychology as the science of soul. Aristotle defined soul as 'an entity which realizes an idea— not separable from body—and its abode in the human heart.' According to McDougall (1871–1938), soul is 'a vital principle inhabiting and animating each human body and somehow the ground of each individual's experience.' The nature of soul could not be defined. Therefore, the meaning of psychology in terms of soul was rejected.

- Psychology as the science of mind: A German philosopher, Kant (1724– 1804) criticized the idea of psychology as a science of soul, and instead laid stress on the mind aspect of science. Other philosophers in the Middle Ages also considered psychology as the science of mind. However, they failed to give the exact nature and form of the mind of an individual. According to Descartes (1596– 1650), mind is a kind of matter located in the brain. In modern times, E B Titchener (1867–1927) divided mind into three elements: sensations, images and affections. Some thinkers consider mind as nothing but an assemblage of ideas in the human brain, gathered by experience. Thus, opinions differ as to the location of mind in the human body. Charak (First century AD), the court physician of King Kanishka, considered the human heart to be the seat of the human mind. Thus, the concept of mind was also not clear and psychology as the science of mind could not earn much acceptance.

- Psychology as the science of consciousness: Psychologists like William Wundt (1832–1920), William James (1842–1910) and others considered psychology as the science of consciousness or immediate experiences. By consciousness, these scholars meant awareness of wakefulness. But there were several interpretations of consciousness and this concept was rejected. Freud (1856–1939) criticized this approach and laid stress on the unconscious.

- Psychology as the science of behaviour: The latest and most modem concept of psychology is in terms of behaviour. The term behaviour was popularized by J B Watson (1878–1958). According to him, 'psychology is the positive science of behaviour.

- Behaviour is the out word organic of mental processes or behaviour as an act which can be Observed and measured Objectively- **J.B. Watson.**

- Behavior is any Overt (observable) response or activity by an organism. From the word behavior one can extract two meaning, because there are two kind of behavior Overt and Covert behavior.

- **Overt behavior** are those which is visible to other in other word seen behaviors, Example: The way we Talk, Walk, Dress, Eat and so on… in short these are the action we see and observe in individual and group around us..

- **Covert behavior** are those which is not obvious to the people around us. Simply they are unseen behavior. For example, Feeling such as anger, jealousy, kindness, pity, happiness and sad.

- Other Modern Definitions of Psychology

  1. 'Psychology is the study of human nature.'—Boring and Langfield

  2. 'Psychology is the study of human behaviour and human relationships.'— Crow and Crow

  3. 'Psychology is the science of the facts or phenomena of self.' —John Dewey

  4. 'Psychology is the scientific study of behaviour of living creatures in their contact with the outer world.' —Kurt Koffka

  5. 'Psychology is the science which aims to give us better understanding; and control of the behaviour of the organism as a whole.' —William McDougall

  6. 'Psychology today concerns itself with the scientific investigation of behaviour.' —N L Munn 7. 'Psychology is the science of behaviour and experience.' —Frederic B Skinner

  8. 'Psychology undertakes a scientific study of the individual considered as a unit as he really is in his dealings with other individuals and with the world.' —R S Woodworth

- In 1879, at University of Leipzig, Wundt founded the first formal or psychological laboratory for psychological research.

Conclusion of the Meaning …...

Psychology is considered a science of behavior or a scientific study of the behavioral activities & experiences".

Nature of Psychology

- McDougall, pillsbury, watson and other modern psychologists considered psychology as the science of human behaviour.
- Meaning of Science: A science is a systematic body of knowledge relating to a subject.
- A science is a particular branch of science such as Physics, Chemistry, Biology.
- A Science is the study of some aspects of human behaviour. For example Sociology or Anthropology.
- Anthropology study of human beings, especially of their origin,development,customs and beliefs. Sociology study of societies and social behaviour.

Characteristics of Science

- A Science seeks truth.
- Scientific results are systematic and self consistent.
- Scientific laws or principles helpful future predictions.
- Scientific results are applicable to future researches.

What kind of science is Psychology

- Physical science such as Physics, Chemistry, Mathematics deal with phenomena of matter.
- Botany, Zoology, Physiology deal with phenomena of Life. They are called Biological sciences.
- Psychology closely related to Physical and Biological Sciences.
- Physical science study the nature of Physical Phenomena. They help psychology to explain behavior of an organism which are reactions to Physical stimuli.
- Biological science study there a close relation between nervous system and glandular system with the behavior of an individual.

Scope of Psychology

- Psychology primarily deals with human mental processes.
- Human mental processes has three fundamental characteristics. These are

(1). Cognition
(2). Affection
(3). Conation

- Cognition include sensation, perception, imagination, memory, thinking, reasoning.
- Affection include feelings, sentiments and emotions.
- Conation include reflexes, instincts, and general tendencies.
- Formerly psychology was concerned completely with the conscious phase of the human mind. But today it studies the conscious as well as un conscious process of our mental health.

Lesson-2: Branches and Fields of Psychology- Early Divisions, Basic Fields, Applied Field, both basic and Applied Fields.

SUBFIELDS OF PSYCHOLOGY

There are different sub-fields of psychology. Psychology involves a wide range of studies of mind, behaviour, perception, learning, to name a few. Before explaining the sub-fields, let us first differentiate between a psychologist, psychiatrist and a psychiatric social worker. A psychologist has a doctorate degree in psychology and not a medical degree. After academic training and specialization in a specific area, the psychologist works in a particular vocational setting or a career. On the other hand, a psychiatrist has a medical degree in psychiatry, and makes diagnosis and treats patients of psychological disorders by prescribing medicines. A closely related field is of psychiatric social worker who has earned a Master's degree in this area and is mainly concerned working on the environmental conditions that may have an impact on psychological disorders.

Psychologists examine different patterns of behaviours and study how external and internal factors influence the mind and behavior. There are varied interests of different psychologists. Some may have interest in social psychology, while others may be interested in cognitive psychology, while some may be interested in community and developmental psychology both

General Psychology:- General psychology studies the various mental processes of a normal individuals. It uses introspection, observation and experimental method of investigation.

Cognitive Psychology:- Cognitive psychology is the study of mental processes such as "attention, language use, memory, perception, problem solving, creativity, and thinking.“

Biological Psychology:- Behavioral neuroscience, also known as biological psychology, biopsychology, or psychobiology is the application of the principles of biology (in particular neurobiology), to the study of physiological, genetic, and developmental mechanisms of behavior in humans and nonhuman animals.

Abnormal Psychology:- It deals with mental diseases, their classification.

Health Psychology:- Health psychology is the study of psychological and behavioral processes in health, illness, and healthcare.

Geo Psychology:- This branch of Psychology describes and explain the relation of physical environment, particularly weather, climate, soil and landscape with behavior.

Para Psychology:- This new branch of Psychology deals with extra-sensory perception, precognition, cases of claimed rebirth, telepathy and allied phenomena.

Legal Psychology:- Legal psychology studies the behavior of clients, criminals, witnesses(a person who sees an event a crime) etc

Clinical Psychology:-This branch of psychology deals with causes and treatment of mental diseases of patients coming to clinic or hospital.

Biopsychology: Studies the biological bases of behaviour. The intimate relationship between psychology and the biological sciences is quite obvious. All behaviour occurs through bodily processes. The brain plays a very important role in coordinating and organising the functions of the different organs of the body. In fact, it is the seat of all forms of complex behaviour. It is impossible to understand and explain behaviour without an understanding of the structure and functioning of the brain. Along with the brain, the entire nervous system plays a crucial role in behaviour. Hence, there is an intimate relationship between psychology and neurology, neurophysiology, neurochemistry and other branches of knowledge which are directly involved with the study of the nervous system, particularly the brain. Genetics, the branch of the biology which deals with the nature of inheritance of different qualities is also an important discipline from the point of view of psychology. Over the years, geneticists have carried out important

researches, bringing out the role of heredity in determining behaviour. This has been particularly so in the case of abnormal behaviour like neurosis, mental retardation, psychosis etc. Studies on the role of heredity have also indicated the importance of the genes in determining the intelligence level. In recent years, the role of chemical factors especially the hormones, secreted by endocrine glands have been shown to play an important role in behaviour. Emotional behaviour, temperament etc. are to a considerable extent, influenced by the hormones of the endocrine glands.

Comparative Psychology: Studies and compares the behaviour of different species, especially animals. That is why some authors used to call this field as animal psychology. By studying animal behaviour, these psychologists gather important information which can be compared with and applied to human behaviour. For example, investigating how the queen bee directs, control, and gets things done by the worker bees, may provide meaningful information about leadership.

Cultural Psychology: Studies the ways in which culture, subculture, and ethnic group membership affect behaviour. These psychologists do cross cultural research and compare behaviour of people in different nations.

Experimental Psychology: Investigates all aspects of psychological processes like perception, learning, and motivation. The major research method used by these psychologists include controlled experiments. But, as Morgan et al. (1986) put it, experimental method is also used by psychologists other than experimental psychologists. For instance, social psychologists may do experiments to determine the effects of various group pressures and influences on a person's behaviour. So, in spite of its name, it is not the method that distinguishes experimental psychology from other sub-fields. Instead, experimental psychology is distinguished by what it studies—the fundamental processes of learning, and memory, thinking, sensation and perception, motivation, emotion, and the physiological or biological bases of behaviour.

Gender Psychology: Does research on differences between males and females, the acquisition of gender identity, and the role of gender throughout life.

Learning Psychology: Studies how and why learning occurs. These psychologists develop theories of learning and apply the laws and principles of learning to solve a variety of human problems.

Personality Psychology: Studies personality traits and dynamics. These psychologists develop theories of personality and tests for assessing personality traits. They also identify the causes of problems related to personality development.

Physiological Psychology: Physiological psychologists investigate the role of biochemical changes within our nervous systems and bodies in everything we do, sense, feel, or think. Mostly, they use experimental method and do basic research on the brain, nervous system, and other physical origins of behaviour.

Physiological Psychology is not only a part of psychology, but also is considered to be part of the broader field called neurobiology which studies the nervous system and its functions. As we know, Physiological Psychology is categorised under 'experimental' psychology. That is why some authors called this branch as 'experimental and Physiological Psychology'. On the other hand, some authors have categorized 'experimental Psychology' as a separate branch of psychology.

Sensation and Perception Psychology: Studies the sense organs and the process of perception. Psychologists working in this field, investigate the mechanisms of sensation and develop theories about how perception or misperception (illusion) occurs. They also study how do we perceive depth, movement, and individual differences in perception. Researches in this field has given rise to many laws and principles that help us understanding the ways we adjust to the visual world in a meaningful way.

Applied Fields

Social Psychology: Investigates human social behaviour, including attitudes, conformity, persuasion, prejudice, friendship, aggression, helping and so forth. Emphasises on all aspects of social behaviour such as how we think about and interact with others, how we influence and are influenced by others. For example, social psychologists study how we perceive others and how those perceptions affect our attitude and behaviour towards them.

This field has developed by the joint contribution of sociologies and social psychologists and their research interest overlaps. However, their focus differs in the sense that while the former are concerned primarily with social institutions, the later focus typically up on the individual.

The social psychologists who are working on the applied side of this field, have developed and standardised techniques to measure attitudes and opinions. Their survey research on political opinion, consumer attitudes and attitudes related to important social issues provide important

information to politicians, business executives, and community leaders who benefit from these, while making decisions.

Clinical Psychology: Does psychotherapy; investigates clinical problems; develops methods of treatment. This field emphasises on the diagnosis, causes, and treatment of severe psychological disorders and emotional troubles.

Confusion between the fields of clinical psychology and psychiatry occurs because both clinical psychologists and psychiatrists provide psychotherapy. And both usually work together in many hospitals/clinics. That is why many people get confused regarding the difference between the two. Well, they belong to two different groups of professionals and differ in their educational background as well as ways of diagnosis and treatment.

Psychiatrists are physicians. After completing medical studies, they do Doctor of Medicine (M.D.) in psychiatry and specialise in the treatment of mental disorders, whereas, clinical psychologists hold a master's degree M.A/M.Sc and/ or a doctorate degree (Doctor of Philosophy [Ph.D] or Doctor of Psychology [Psy.D]) in clinical psychology.

Because of this difference in training, clinical psychologists who do not have medical training, cannot prescribe drugs to treat behaviour disorders. Also, whenever there is a possibility of medical disorder, a patient should be examined by a psychiatrist or other physician. Moreover, mostly, only a psychiatrist can refer a patient to a hospital for treatment and care. Clinical psychologists carry out research to find out better ways of diagnosing, treating, and preventing psychological disorders. They also rely heavily on standardised tests for identifying the causes of these disorders. They use psychotherapy, for which they are trained, for the treatment of mental disorders. But clinical psychologists are not authorised to prescribe drugs to treat behaviour disorders, as they do not have medical training. Also, they cannot refer a patient to a hospital, for care and treatment. Whenever there is a possibility of a medical disorder, a patient should be examined by a psychiatrist or other physician.

Community Psychology: Promotes community-wide mental health through research, prevention, education, and consultation. Community psychologists apply psychological principles, ideas, and points of view to help solve social problems and to help individuals in adapting to their work and living groups.

Consumer Psychology: Researches packaging, advertising, marketing methods, and characteristics of consumers. This field is an offshoot of social psychology. Counseling

Psychology: Does psychotherapy and personal counseling; researches emotional disturbances and counseling methods. This branch deals with helping people/individuals with personal problems including interpersonal relations, career choice, mild emotional troubles or behavioural problems such as over eating, slow learning or lack of concentration. Counseling psychologists assist individuals having specific problem like how to plan career, how to develop more effective interpersonal skills(e.g. communication skills). Now a days, there are many specialised fields within the counseling psychology and experts are working as marriage counselors, family counselors, school counselors etc. The work of the counseling psychologist is quite similar to that of the clinical psychologist. The difference between them is that counseling psychologists generally work with people who have milder emotional and personal problems. They may use psychotherapy in an attempt to help with these problems. Counseling psychologists are often consulted by people with specific questions, such as a choice of career or educational program.

Educational Psychology: Investigates classroom dynamics, teaching styles, and learning; develops educational tests, evaluates educational programs. Investigates all aspects of educational process ranging from curriculum design to techniques of instruction to learning disabilities. This branch deals with broader problem of increasing the efficiency of learning in school by applying psychological knowledge about/of learning and motivation to the curriculum. Another specialised sub-field called School Psychology may be included in educational psychology.

Engineering Psychology: Does applied research on the design of machinery, computers, airplanes, automobiles, and so on, for business, industry, and the military. Psychologists working in this field also write instruction manual in such a manner that can be understood by laypersons so that they can operate complex machinery and home appliances.

Forensic Psychology: Investigates problems of crime and crime prevention, rehabilitation programs, prisons, courtroom dynamics; selects candidates for police work. Forensic psychologists mostly work within the judicial system in such areas as assessing emotional and psychological state of under trials and victims, evaluation of rehabilitation programmes; eyewitness testimony and evidence; jury selection; and police training etc.

Industrial/Organisational Psychology: Investigates all aspects of behaviour in work setting ranging from selection and recruitment of employees, performance appraisal, work motivation

to leadership. The first application of psychology to the problems of industries and organisations was selection and recruitment of employees by using intelligence, aptitude tests.

Medical Psychology: Applies psychology to manage medical problems, such as the emotional impact of illness, self-screening for cancer, compliance in taking medicines. Job of these psychologists overlaps with part of health psychology.

School Psychology: These psychologists do psychological testing, referrals, emotional and vocational counseling of students; detect and treat learning disabilities, and help improve classroom learning. The job of school psychologists include diagnosing learning difficulties and trying to remedy them.

There are some fields which are categorised under both basic and applied fields, as per Feldman (Coon & Mitterer, 2007). These are described below.

Developmental Psychology: Conducts research on infant, child, adolescent, and adult development; does clinical work with disturbed children; acts as consultant to parents and schools. Emphasises on how people change physically, cognitively and socially over the entire life span. Developmental psychologists try to understand complex behaviours by studying their beginnings and the orderly ways in which they change with time. If we can trace the origin and developmental sequence of a certain behaviour, we will have a better understanding of it. Child psychology, the study of children's behaviour, consists of a large part of developmental psychology, because changes in behaviour occurs in an accelerated manner. But developmental changes also occur in adolescence, adulthood, and old age; and so the study of these changes is also a part of developmental psychology.

Developmental psychology has both research and applied aspects. For instance, a great deal of research has been done on the development of thinking in children. Progressive and systematic changes take place in their thinking during the first few years of life. On the applied side, developmental psychologists are often concerned with children who have behaviour problems or psychological disorders. The kinds of behaviours found in disturbed children are frequently quite different from the behaviours found in disturbed adults, and different methods are used to treat them.

Environmental Psychology: Studies the effects of urban noise, crowding, attitudes toward the environment, and human use of space. These psychologists act as consultants on environmental issues. Health Psychology: Studies the relationship between behaviour and health; uses psychological principles to promote health and prevent illness.

Positive psychology: This is an emerging field of 21st century. This field is defined as the science of happiness and human strengths. Psychologists working in this field are concerned with the positive aspects of human nature such as hope, optimism, passion, love, gratitude, forgiveness, humility etc. They try to find out what makes a good life. The work of these psychologists overlaps with that of humanistic psychologists who have similar viewpoints.

Lesson-3 : Schools of psychology- Structuralism, Functionalism, Behaviorism, Psychoanalytic.

SCHOOLS OF PSYCHOLOGY

- **Wilhelm Wundt:-** Wilhelm Wundt was German physiologist, Philosopher and Professor, know today as one of the founders of modern psychology.
- **Born:-** 16 August 1832.
- **Died:-** 31 August 1920.
- **Education:-** University of Heidelberg.
- **Edward B. Tichener:-** Edward B. Tichener Professor of Psychology at Cornell University.
- **Born:-** 1867.
- **Died:-** 1927.
- **Wilhelm Wundt:-** Wilhelm Wundt was German physiologist, Philosopher and Professor, know today as one of the founders of modern psychology.
- In 1879 at university of Leipzig ,Wundt founded the first psychological laboratory for psychological research and also called as father of experimental psychology.

- Wundt focused his experiences on conscious experience, involving ones thoughts, feelings, sensations , perceptions and ideas.
- They believed that the mind is made up of building blocks in the various types of sensation and perception and that these building blocks could be discovered through introspection or looking into one's own mind. Introspection, which required subjects to look inward and observe and report on the working of their mind.
- Aimed to study identify the building blocks of conscious experience.
- Sensations, feelings and images as building blocks.
- Method was used Introspection a process used structuralists that involved participants detailed verbal reports of his/her mental processes (Sensations, feelings and images) in carefully controlled Psychological experiments.
- Tichener's Method.
- Edward B. Tichener (1867-1927)Professor of Psychology at Cornell University.
- According to him psychology may be regarded as the science of consciousness or study of experience.
- Tichener's analyzed consciousness there basic elements.

(1). Sensations.

(2). Images.

(3). Affections.

Sensations:- A sensation is simplest form of mental activity and originate from the sense organs. Basic elements of perception sight, sound, taste, smell, and touch.

❖ **Sense organs**	❖ **Sensations**	❖ **Sense organs**	❖ **Sensations**
❖ Eye	❖ Visual ❖ Light ❖ Color ❖ Shape ❖ Distance	❖ Tongue	❖ Acid ❖ Sweet ❖ Bitter ❖ Salt
❖ Ear	❖ Auditory ❖ Pitch ❖ Intensity ❖ Quality	❖ Skin	❖ Tactual ❖ Pressure ❖ Pain ❖ Temperature
❖ Nose	❖ Olfactory ❖ Various kinds Odours.	❖	❖

- ❖ **Images:-** Elements of Ideas- reflects of experiences not actually present at the time. Example- Memory of a sight.
- ❖ **Affections:-** Feelings- the elements of emotions- Love, hate, anger, sadness etc….

Dis Advantages:-

- ❖ The method Introspection was criticized for being less scientific.
- ❖ Introspection method could not be verified by outside observers. Hence criticized by other Psychologists.

- Structuralism was too concerned with internal behavior, which is not directly observable and can not be accurately measured.

Functionalism :-

- Functionalism was rooted in Darwin's theory of evolution. Evolution is based on individual differences and the survival of adaptive features.
- Spencer saw evolution as the progressive development of the physical world, biological organisms, the human mind, and human culture and societies.
- "Adaptation" becomes a popular approach to measuring intelligence, and "individual differences" become a valued part of mental research.
- William James's focus on how mental activities help a person adapt to his or her environment is known as functionalism.
- As James saw it, psychology's purpose was to study the function of behavior in the world, and as such, his perspective was known as **functionalism**.
- Functionalism focused on how mental activities helped an organism fit into its environment.
- Functionalism school of psychology concerned with How the mind helps us adopt to our environment.
- What difference between Structuralism and Functionalism
- Structuralism focus on introspection and breaking down mental events to the smallest elements.
- Functionalism focused on the wholeness of an mental events, taking in to the impact of the environment on behavior.
- PSYCHOANALYTIC THEORY

FOUNDER SIGMUND FREUD

- **MAY 6, 1856** – Sigmund Freud was Born in Freyberg Town, Crèche Republic.
- **1881** – He graduated from medical faculty, university of Vienna.

- **1900** – He released ‘interpretation of dreams’
- **SEPTEMBER 23, 1939** –Freud passed away in Hampstead house.
- STRUCTURE OF PERSONALITY
- Human mind can be divided in to three parts, Id, Ego, Superego.

Id

- •Operates on PLEASURE PRINCIPE – to gain pleasure, avoid pain
- •Driven by sexual and aggressive urge

Ego:

- Operates on REALITY PRINCIPLES – does realistic and logical thinking.
- Source of decision making and rational thought.
- The balance between Id and Superego.

Superego

- Partially unconscious.
- Operates on MORAL PRINCIPLES.
- Able to differentiate between good and bad, right and wrong.
- If people follow their superego, they will feel proud but if they don’t follow, they will feel guilty and anxious.
- LEVEL OF MENTAL LIFE

THE CONSCIOUS MIND

- It includes everything that we are aware of. This is the aspect of our mental processing that we can think and talk about rationally.
- Events & experiences that are easily remembered or retrieved are considered to be within one’s conscious awareness. Example:-Telephone numbers, birthday of self & dates of special holidays.
- The part of the mind we are aware of everyday thoughts and feelings.

THE PRECONSCIOUS MIND

- It includes all memories that may have been forgotten or are not in present awareness but with attention can be readily recalled into consciousness.
- It is thought to be partially under the control of the super-ego, which helps to suppress unacceptable thoughts and behaviors.
- Thoughts and memories not accessible at all the times but easily Recalled.
- Facts stored in a part of the brain, which are not conscious but are available for possible use in the future (E.g. A person will never think of her home address at that moment but when her friend ask for it, she can easily recall it).

THE UNCONSCIOUS MIND

- It consists of unpleasant & non essential memories that have been repressed & can be retrieved through therapy, hypnosis & with other substances that alter awareness.
- Also contain painful memories, experiences, emotions,guilt feelings and unresolved conflicts which we deliberately suppress.
- The deep dark shameful part. Repressed thoughts, memories and feelings.

BEHAVIORISM

- John B.Watson (1878-1958) founder of the Behaviorism.
- J.B. Watson stated the psychology should only study observable behavior.
- Psychology= Scientific study of behavior.
- Behavioral psychology is the study of external behavior.
- Behavior is the response of an organism to stimuli.

- Behavior is the out word organic of mental processes or behavior as an act which can be Observed and measured Objectively- **J.B. Watson.**
- Behavior is any Overt (observable) response or activity by an organism. From the word behavior one can extract two meaning, because there are two kind of behavior Overt and Covert behavior.
- **Overt behavior** are those which is visible to other in other word seen behaviors, Example: The way we Talk, Walk, Dress, Eat and so on… in short these are the action we see and observe in individual and group around us..
- **Covert behavior** are those which is not obvious to the people around us. Simply they are unseen behavior. For example, Feeling such as anger, jealousy, kindness, pity, happiness and sad.
- History of Behaviorism
- Pavlov (1927), a Russian physiologist discovered classical conditioning in dogs.
- "Give me a dozen healthy infants, well formed, and my own special world to bring them up in and I'll guarantee to take any one at random and train him to be any type of specialist I might select – a doctor, a lawyer, artist…" Bachelors of science in Clinical Psychology -Watson 1924.

Lesson-4 : Methods of Psychology – Introspection method, observation method, Survey method, case study method, experimental method and Correlational method.

METHODS OF PSYCHOLOGY

Observation Method

- Observation is a very powerful tool of psychological enquiry. It is an effective method of describing behavior.
- In our daily life, we remain busy with observing numerous things throughout the day. Many times, we do not take notice of what we are seeing or what we have seen. We see but we do not observe.

- A scientific observation differs from day-to-day observation in many respects. These are

(1). Selection

(2). Recording

(3). Analysis of Data.

(1).**Selection:-** Psychologists do not observe all the behavior of the subjects. Rather, they select a particular behavior for observation.

(2).**Recording :-** While observing, a researcher records the selected behavior using different means, such as, taking notes describing each activity in greater detail using short hand or symbols, photographs, video recording, etc.

(3). **Analysis of Data :-** After the observations have been made, psychologists analyze whatever they have recorded with a view to derive some meaning out of it.

Types of Observation

- **Naturalistic Observation :-** Systematic study of their observation in natural setting. Naturalistic Observation is often used in the study of animal behavior, it is some times applied to human beings as well as especially to behavior in public places such as airport, shopping etc ..

- **Controlled Observation**:- Observer look at the behavior under controlled lab conditions.

- **Participant Observation :-** In the participant observation , observer is in direct contact with the group he/she is observing. This is called as participant observation.

  In the school students studying certain behavior like play activity

- **Non-Participant observation :-** In the Non- participant observation , observer is not in direct contact with the group he/she is observing. This is called as Non-Participant observation.

- In clinical setting One group is out side the glass and observer is behind the glass. The group can not see the observer or psychologist but observer can see the group.

Introspection Method:

- The term introspection method is derived from two words “intro” and “section.” “Intro” means “within” and “section” means “Looking” or “Observing.”

- Thus introspection means “Looking within” “or Observing within” one self to experience one’s own mental state.

- It is a kind of self observation or self examination.
- It consists in trying to understand the nature of mental processes by looking within one's own mind. For instance, I want to the nature of anger. When I am angry I look within myself and try to analyse the peculiarity of the emotion.
- Wood worth calls introspection as self observation and Angel defined it as "looking inward".

Drawbacks:

- Mental processes are vague. So it is difficult to introspect them. Only a few trained individuals can use introspection.
- Mental processes are fleeting by nature. They tend to vanish when we try to introspect them.
- Introspection is purely subjective. For instance, two persons cannot have the same emotional experience during anger or fear.
- **Not universally applicable:** Introspection method cannot be used to study animals, children's and insane people.

Merits of introspection

- Introspection is simplest and most economical method as it requires no laboratory or finance.
- Self evaluation and self suggestions help the teachers for the improvement of their teaching.
- There is no substitute to this method for studying conscious experiences like feelings, thought processes, mental imaginary etc…

Experimental method

- The Experimental method is considered the most scientific and objective method of studying behaviour.
- Wilhelm Wundt is considered as the father of experimental psychology.
- The word "Experiment" comes from a Latin word experimentum meaning "a trial or test."

- An experimental method deal with various kinds of psychological testing's under the controlled laboratory situations. In the laboratory used scientific instruments for investigation in to the behavior of animals and human beings.
- In the science while conducting such experiments in the laboratory or outside in a natural environment, we may want to learn the effect of friction on motion, the effect of sun light on growth of plants etc…
- Psychological experiments performed in this method essentially require two persons, the experimenter and the subject or the person whose behaviour is to be observed.
- In scientific experiment they are two kinds of variables. There are (1). Independent variable (2). Dependent variable. (3). Intervening or Controlled variable.
- Variables: A variable is something that can be changed….

- **Manipulation:** It refers to introduction, removal or change of a variable.
- **Independent Variable:** The variable the change/cause because of the change in the independent variable.
- **Dependent Variables:** The dependent variable is the changes because of the change in the independent variable. The experimenter introduces, removes or changes the independent variable.
- If I change the amount of water a plant receives, the height of the plant will change.
- **Independent Variable-** Water because we are changing it.
- **Dependent variable-** Dependent variable is height of the plant, because it is affected by the water.
- **Extraneous Variables:-** (external to the environment), must be controlled to isolate the effect of the independent variable on the Dependent variables.

Merits of experimental method

- It is the most systematic method and condition can be completely controlled.
- The data collected can be analyzed in an objective manner and so it is reliable.
- A cause effect relationship of individual's behaviour can be established.

- The findings of experimental method are open to critical examination and verification.
- It provides innovative ideas for further experimentation.

Demerits of experimental method

- As human behaviour is changeable, identical behaviour of the same individual not occur even under identical conditions at different times.
- This experimental method is costly and time consuming.
- Experimental method in educational psychology stems from the fact that experiments cannot handle too well the patterns of covert behaviour of children in laboratory.

Correlational Method

- Correlational Research In psychological research, we often wish to determine the relationship between two variables for prediction purposes.
- For example, you may be interested in knowing whether "the amount of study time" is related to the "student's academic achievement".
- Rather, you simply find out the relationship between the two variables to determine whether they are associated or not.
- The strength and direction of the relationship between the two variables is represented by a number, known as correlation coefficient.
- Its value can range from +1.0 through 0.0 to –1.0.
- Coefficient of correlation is of three types:

(1).Positive Correlation.
(2).Negative Correlation.
(3). Zero Correlation.

(1).Positive correlation: - A positive correlation indicates that as the value of one variable (X) increases, the value of the other variable (Y) will also increase. Similarly when variable X decreases, a decrease in Y too takes place. Suppose, it is found that **more time the students spend on studying**, the **higher was their achievement score**. Also the **less they studied**, the **lower was their achievement** score.

Negative correlation:- Negative correlation tells us that as the value of one variable (X) increases, the value of the other (Y) decreases.

- For example, you may hypothesise that as the **hours of study time increase**, the **number of hours spent in other activities will decrease.** Here, you are expecting a negative correlation, ranging between 0 and –1.0.

Zero Correlation:

- It is also possible that sometimes no correlation may exist between the two variables. This is called zero correlation. Generally, it is difficult to find zero correlation but the correlations found may be close to zero, e.g., -.02 or +.03.
- This indicates that no significant relationship exists between two variables or the two variables are unrelated.

Survey Method

- **Survey Method:-** The method of study which is used to gather information about opinions from a group of participants selected for study.
- You may have read in the newspapers or seen on the television that during elections surveys are conducted to find out if people would vote for a particular political party, or favour a particular candidate.
- The survey research uses different techniques for collecting information. Included among these techniques are: personal Interviews, questionnaires, telephonic surveys. These techniques are discussed here in some detail.
- **Interview technique:-** Two or more persons sit face-to-face with each other, in which one person (generally called interviewer) asks the questions and the other person (called interviewee or respondent) answers the questions related to a problem.
- **Interviewer:-** the one who generally asks questions to gather information from the respondent in interview.
- **Interviewee:-** the one who answer the questions asked by the interviewer.
- It is generally conducted face-to-face but sometimes it can also take place over the phone. There can be two broad types of interviews: structured or standardised, and unstructured or non-standardised.

Survey method: Interview technique

- **Structured interview**

Interview schedule: List of questions prepared beforehand.

Questions have a specific sequence which is followed.

In some cases responses are also specified in advance [Closed ended].

- **Unstructured interview**

Questions are not written or decided in advance.

Interviewer has the liberty to ask questions as per choice.

Interviewer has the liberty to phrase the questions as per choice.

Respondent too is free to answer questions as per choice.

- An interview may have the following combinations of participants in an interview situation:
- **(a) Individual to Individual :** It is a situation where one interviewer interviews another person.
- **(b) Individual to Group** : In this situation, one interviewer interviews a group of persons. One variant of it is called a Focus Group Discussion (FGD).
- **(c) Group to Individuals:** It is a situation where one group of interviewers interview one person. You may experience this type of situation when you appear for a job interview.
- **(d) Group to Group:** It is a situation where one group of interviewers interview another group of interviewees.

Survey method: Questionnaire Technique

- **Questionnaire:-** A predetermined set of questions to which the respondent marks the answers on paper.
- **Open ended:-** Respondent is free to write whatever answer she or he considers appropriate subjective answers.
- **Closed ended:-** Respondent has to select the answers from the options available in the questionnaire.
- Ex:- Yes/No, True/false, multiple choices, rating scales[3/5/7/9 point rating scales]or rate the multiple options according to preference.

Telephone Survey

- Telephone Surveys are also conducted through telephone, and now-a-days you must have seen programmes asking you to send your views through mobile phones' SMS. The telephone survey helps in reducing time.

CASE STUDY METHOD

- Case studies are in-depth investigation of a single person, group, event (or) community.
- Typically data are gathered from a variety of sources and by using several different methods as observation and interview.
- The case study research method originated in clinical medicine
- The case study is a research method but also a method of data collection and analysis
- In psychology case studies are often confined to the study of a particular individual behaviour

OBJECTIVES OF CASE STUDY

- Diagnosis on a treatment behavioural problems.
- The complete and detailed study of case may involve the use of observation, interview medical examination and use of various tests of intelligence personality, aptitudes and inter.
- The case study method is primarily use to collect detailed information on the behaviour problems of mal-adjusted and deviant cases.
- The mal-adjustment may be in the form of anti-social behaviour, emotional disturbances (or) in the area of learning and backwardness in school subjects.

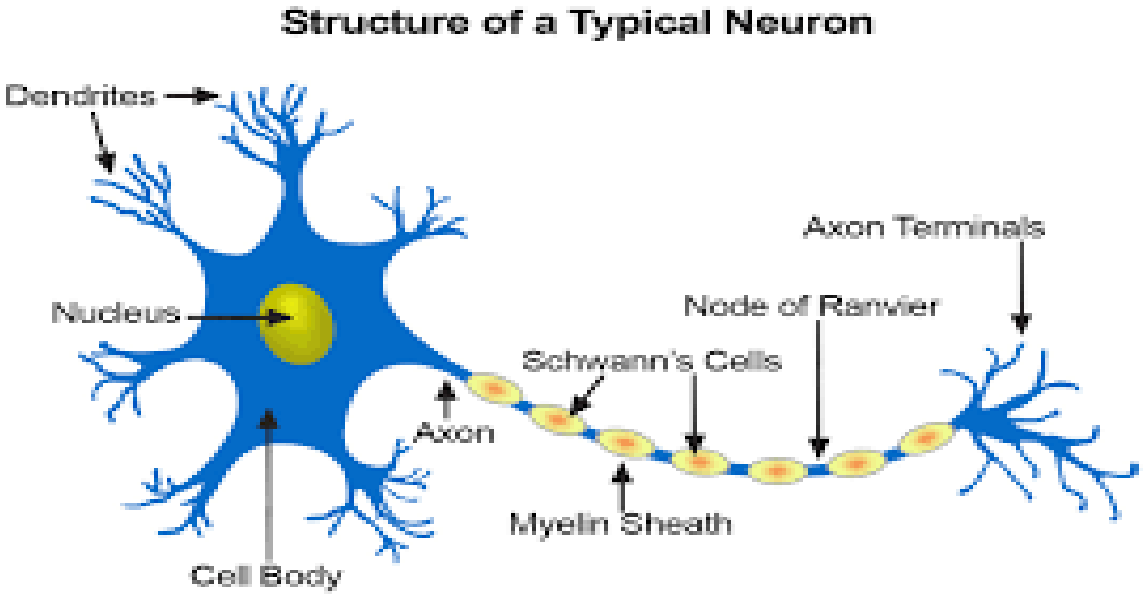
Unit-II: Physiological basis of behavior

Lesson 5 : Neuron- The structure of neuron-Types of neurons-Varieties of neuron transmitters.

STRUCTURE OF NERONS

NEUROTRANSMETERS NERVOUS SYSTEM

- The nervous system consists of two kinds of cells: neurons and glia. Neurons receive information and transmit it to other cells. According to one estimate, the adult human brain contains approximately 100 billion neurons (R. W. Williams & Herrup, 1988).
- Glial cells provide support and protection for neurons. They are thus known as the "supporting cells" of the nervous system.
- Although glia cells DO NOT carry nerve impulses (action potentials) they do have many important functions. In fact, without glia, the neurons would not work properly.



The main functions of glial cells are:

- To surround neurons and hold them in place.
- To supply nutrients and oxygen to neurons.
- To destroy and remove the carcasses of dead neurons (clean up).

Six types of neuroglia, four(astrocytes, oligodendrocytes, microglia, and ependymal) cells are found only in the CNS. The remaining two types Schwann cells and capsular (satellite) cells are present in the PNS.

- Each neuron consists of three basic parts (1).Cell body(soma), (2).Axon (3). Dendrites.
- According to one estimate, the adult human brain contains approximately 100 billion neurons (R. W. Williams & Herrup, 1988).

Cell body (soma).

- The cell body, or soma, contains the nucleus, ribosomes, mitochondria, and other structures found in most cells. Much of the metabolic work of the neuron occurs here. Like the dendrites, the cell body is covered with synapses on its surface in many neurons.

Axon

- The term axon comes from a Greek word meaning "axis."In many neurons the axon is covered by a sheath of fatty material known as myelin. The myelin sheath is interrupted by small gaps(nodes of Ranvier). The myelin sheath and gaps in it play an important role in the neurons ability to transmit information.
- Damage the myelin sheath surrounding axons can seriously affect synaptic transmission. The end of axon divides into several small branches . These in turn, end in round structures known as axon terminals. Axon terminals of neuron closely approach other cells is known as the synapse.
- Axons end in axonal terminals
- Axonal terminals contain vesicles with neurotransmitters
- Axonal terminals are separated from the next neuron by a gap

Synapse – junction between nerves.

This is the point from which the axon releases chemicals that cross through the junction between one neuron and the next.

Dendrites

- Dendrites are branching fibers that get narrower near their ends. (The term dendrite comes from a Greek root word meaning "tree"; a dendrite is shaped like a tree.) The dendrite's, at which the dendrite receives information from other neurons.

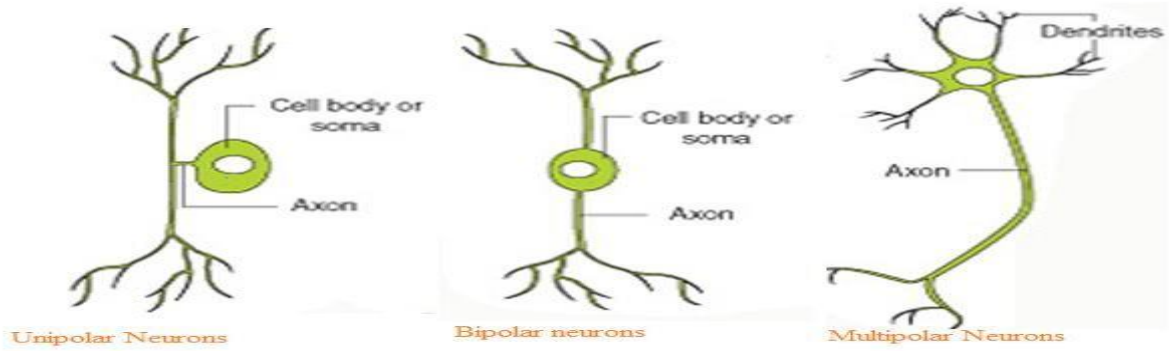
STRUCTURAL TYPES OF NEURON

Neurons are divided on the basis of structures and size.

Three General Types of Neurons

- Unipolar: A neuron from which only a single axon leaves the cell body Most sensory neurons have this shape.
- Bipolar: A neuron from which two processes leave the cell body. It's found in the retina of our eyes.
- Multipolar: A neuron from which multiple branches leave the cell body. It got many neuronal processes. Multi means many. This is a very important nerve because all motor neurons have this shape and many interneurons also have.

Structural Classification of Neurons.



Types of Neurotransmitters

- At a synapse, one neuron releases chemicals that affect a second neuron. Those chemicals are known as neurotransmitters.
- Neurotransmitters are the chemicals that allow neurons to send signal across synapse to another neuron.
- Neurotransmitters play a crucial role in communication between neurons.
- Neurotransmitters have a profound effects on our physical and psychological functioning.
- Our ability to perceive feel, think, move, act& react depends on the delicate balance of neurotransmitters in the nervous system.
- Neurotransmitters are present in only minuscule amount in the human body. Many behaviors are the complex interaction of different neurotransmitters.

Acetylcholine:-

Neurotransmitters that cause muscle contraction and is involved in memory function.

- Location: found in all motor neurons and many neurons on the brain.
- Function: stimulate muscle to contract, including the heart and stomach muscles. It is important in memory, learning & general intellectual functioning.
- Associated disorders: Alzheimer's disease.

Dopamine:

- Definition : Neurotransmitters involved in the regulation of bodily movement thought processes and rewarding sensations.
- Function: involved in movement, attention, learning and pleasurable and rewarding sensations.
- Associated disorders: poor balance and difficulty in initiating movements, schizophrenia, depression and mania. Mental disorders.

Serotonin:

- Definition: Neurotransmitters involved in sleep and emotions.
- Location: found in different areas.
- Function: involved in sleep, mood and emotional states.
- Associated disorders: schizophrenia, depression and mania.

Nor epinephrine:

- Definition : Neurotransmitters involved in learning and memory also a hormone manufactured by adrenal glands.
- Location: found in different brain areas.
- Function: key player in learning and memory retrieval. Communicates with the sympathetic nervous system.
- Associated disorders: anxiety disorder, depression and other stress related conditions.

GABA (Gamma- amino butyric acid):

- Definition : Neurotransmitters that usually communicates an inhibitory message.
- Location : primarily in the brain.
- Function : communicates an inhibitory message to other neurons.
- Associated disorders: anxiety disorders with too little GABA.

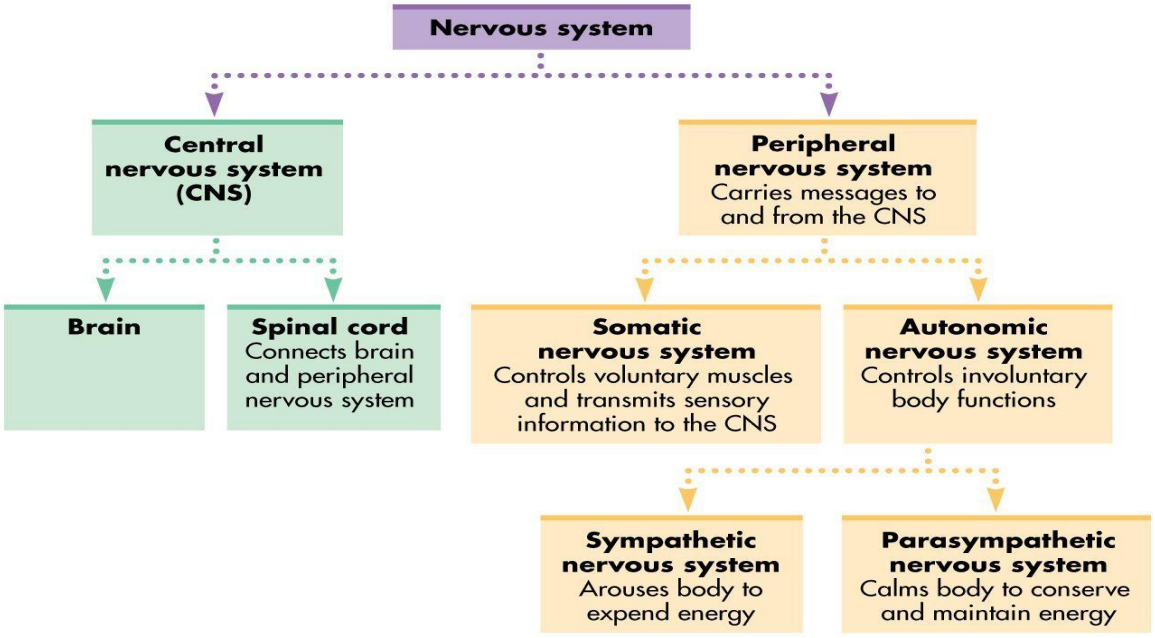
Endorphins:

- Definition: Neurotransmitters that regulate pain perceptions. A term derived from the phrase endogenous morphine's.(the term endogenous means produced internally in the body.)
- Function : reduce the perception of pain.
- Implementing in the pain reducing effects of acupuncture.
- Sometimes play role in the placebo effect.
- Also associated with positive mood.

Lesson 6 : Nervous system -Central and Peripheral nervous system- Autonomic Nervous system.

NERVOUS SYSTEM

- The nervous system coordinates voluntary and involuntary actions in the body.
- The nervous system can be divided into two parts: Central Nervous System (CNS) and Peripheral Nervous System (PNS).
- PNS can be further classified into Somatic and Autonomic nervous system. Somatic nervous system is concerned with voluntary actions, while the autonomic nervous system performs functions on which we have no voluntary control.



Central nervous system

The central nervous system consists of the brain the spinal cord and thousands of nerves and nerves fibres.

- This part of the nervous system performs a wide variety of functions:

  [A] It receives information about the outside world through neurons in the sense organs.

  [B] It coordinates and interprets the information from different parts of the body

  [C] It gives rise to activity.

The central nervous system works as an organizer and an integrator of information

- The central nervous system has two main parts;

  [a] The brain and

  [b] The spinal cord.

Brain:-

- Brain is the most important part of the human body.
- Brain situated in a cranial box (cranium)which is made of bones.
- Brain is covered by three membranes of connecting tissue, termed as meninges.
- Brain is located in the bony skull. Within the skull, the brain is protected by three layers of tissues called meninges. Outer most layer is called dura matter and inner most layer is called pia matter. Between these two layers is a soft membrane called arechnoid.

Arechnoid is filled with CSF (cerebro spinal fluid). Brain is composed of white and grey matter.

- Weight of brain- in adult male 1400 gm, female 1250gm.
- Meningitis:- any inflammation (painful especially as a reaction to injury or infection)of meninges is called meningitis
- The brain is divided into three main parts viz .the forebrain ,the midbrain and the hindbrain.
- Forebrain is the highest and most evolved part of the brain .It plays an important role in behavior patterns including those of emotion, motivation ,language ,thinking and learning . Some of the most important areas of forebrain are Cerebrum, Thalamus and Hypothalamus.
- Forebrain is the highest and most evolved part of the brain .It plays an important role in behavior patterns including those of emotion, motivation ,language ,thinking and learning . Some of the most important areas of forebrain are Cerebrum, Thalamus and Hypothalamus.
- Cerebral hemispheres(Cerebrum):- The largest division of the brain. It is divided into two hemispheres, each of which is divided into four lobes.
- The Left Hemisphere

  -Processes logical tasks (*reading, writing, speaking, mathematics, and comprehension skills*)

  -Controls the right side of our body
- The Right Hemisphere

  -Processes non-verbal tasks/perceptual (*spatial relationships, musical/artistic ability and mental imagery*)

  -Controls the left side of our body

  -May also be related to some negative emotions

(On an IQ test, non-verbal tasks include building with blocks to make a printed design, discerning patterns, and determining how to balance a scale. These tasks are measures of non-verbal reasoning skills. Nonverbal intelligence describes thinking skills and problem-

solving abilities that do not fundamentally require verbal language production and comprehension.)

- The surface is made of ridges (gyri) and grooves (sulci).
- Ridges are known as gyri, where as grooves are known as sulci. These gyri and sulci increase the surface area of brain.

❖ Functions of the cerebrum

Frontal Lobe- associated with reasoning, planning, parts of speech, movement, emotions, and problem solving.

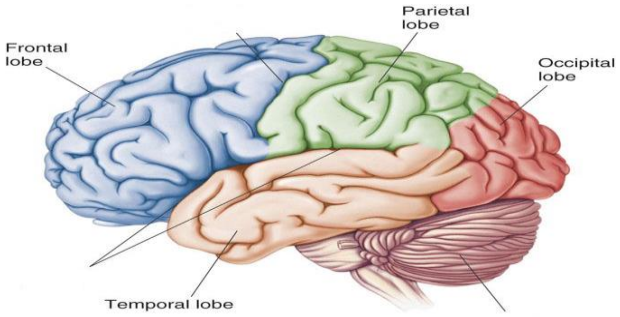
Parietal Lobe-it is primarily responsible for bodily sensations and knowledge of direction. associated with movement, orientation, recognition, perception of stimuli.

Occipital Lobe- associated with visual processing .

Temporal Lobe- associated with perception and recognition of auditory stimuli, memory, and speech.

Specialized Areas of the Cerebrum

- Primary motor area – sends impulses to skeletal muscles.
- Broca's area – involved in our ability to speak.
- Somatic sensory area – receives impulses from the body's sensory receptors.



Thalamus- filters sensory information, controls mood states and body movement associated with emotive states. The thalamus is the relay center(receive and pass on information or message) for all sensory impulses except olfactory (sense of smell) and motor areas of the cortex.

❖ It receives information for all of the senses EXCEPT for smell.

❖ The Hypothalamus lies below (*hypo*) the thalamus. The hypothalamus regulates behavior and emotional expression, body temperature, and many metabolic activities. Attached to the pituitary gland, it also controls hormonal secretions of this gland. It helps control the endocrine system by giving directions to the pituitary gland.

- ❖ An important part of the limbic system (emotions)
- ❖ The pituitary gland is attached to the hypothalamus

Midbrain

- ❖ The midbrain lies between the forebrain and the hindbrain. It contains areas involved mainly in sensory and motor processes. The principal functions of the midbrain are audition, vision and muscle control.

Hindbrain

- ❖ The hindbrain connects the spinal cord with the rest of the brain. The three most important areas of the hindbrain are Cerebellum , Pons and Medulla oblongata.
- ❖ Cerebellum – The cerebellum is situated below the cerebrum. It is ovoid in shape and has two hemispheres, separated by a narrow median strip called the vermis.

Functions of cerebellum

- ❖ The cerebellum is concerned with the coordination of voluntary muscular movement. It coordinates activities associated with the maintenance of the balance and equilibrium of the body.
- Balance and posture: Your cerebellum works with sensory input from your eyes and ears to keep you upright and steady.
- Motor learning: This involves the learning and fine-tuning of various movements. Examples include the specific, precise(accurate) movements used for writing or for riding a bicycle.
- Speech: The cerebellum is also involved in the movements associated with speaking.
- ❖ It is also involved in certain cognitive functions, such as language.
- language
- processing of emotions
- attention
- pleasure or reward response
- fear response

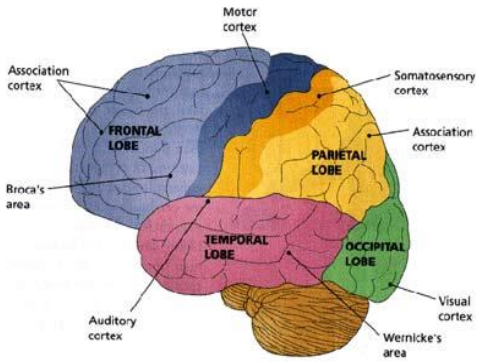
- Damage to the cerebellum results in clumsy(awkward in movements) uncoordinated muscular movement, staggering gait and inability to carry out smooth, steady, precise movements.
- a head injury
- stroke
- a brain tumor
- autoimmune conditions, such as multiple sclerosis(abnormal hardening of body tissue).
- neurodegenerative conditions, such as Parkinson's disease or Huntington's disease Infections.

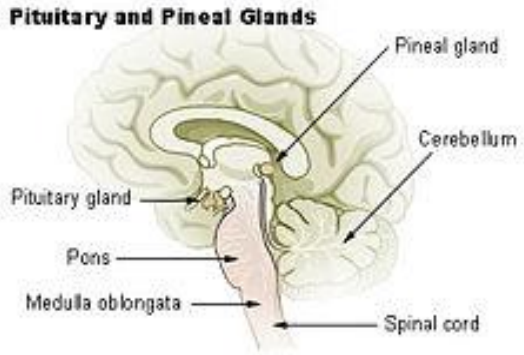
Parkinson's disease is a progressive nervous system disorder that affects movement. Symptoms start gradually, sometimes starting with a barely noticeable tremor (a disorder that causes tremors and muscle rigidity) in just one hand. Tremors are common, but the disorder also commonly causes stiffness or slowing of movement.

What can you do to protect your cerebellum?

- Keeping your cerebellum and the rest of your brain healthy and free of injury is key to your physical and emotional health and overall sense of well-being. Here are some tips to keep in mind for good brain health:
- Exercise regularly: Not only is exercise good for your overall health, but it also helps to stimulate blood flow to your brain.
- Eat a healthy diet: All parts of your body can benefit from a healthy diet. Focus on fresh fruits and vegetables, whole grains, nuts, seeds, fish, and lean meat.
- Limit alcohol consumption: Drinking too much alcohol can damage your cerebellum. It can also increase your risk of stroke.
- Avoid smoking: Smoking is associated with many health conditions, including high blood pressure and stroke.
- **Pons**- Relay station between cerebrum and cerebellum. It controls mastication,(namaladamu), movements of the eye and also influences salivary secreation.It act as a sensory relay station of hearing.

- **<u>Medulla oblongata-</u>** Conscious control of skeletal muscles, balance, co-ordination regulating sound impulses in the inner ear, regulation of automatic responses such as heart rate, swallowing, vomiting, coughing and sneezing.
- The medulla oblongata connects the rest of the brain to the spinal cord. It regulates breathing, swallowing, coughing, sneezing, vomiting, heartbeat, and blood pressure.

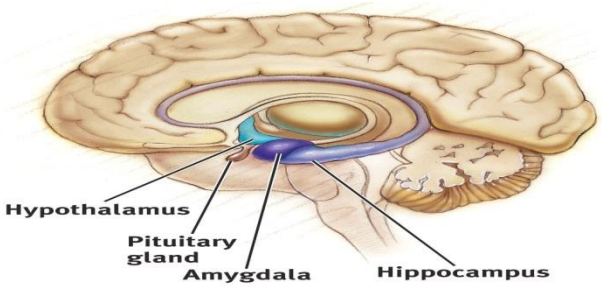




Limbic System

- The Limbic System is a doughnut-shaped system of neural structures at the border of the brainstem and cerebrum, associated with emotions such as fear, aggression and drives for food and sex.
- It includes the hippocampus, amygdala, and hypothalamus.
- The Hypothalamus lies below (*hypo*) the thalamus. It directs several maintenance activities like eating, drinking, body temperature, and control of emotions. Attached to the pituitary gland, it also controls hormonal secretions of this gland. It helps control the endocrine system by giving directions to the pituitary gland.
- *Amygdala*- mediates and controls major affective mood states such as friendship , love, affection, fear, rage and aggression.
- Hippocampus- It is an important part of the limbic system, a cortical region that regulates motivation, emotion, learning, and memory.

Damaged by disease or injury, it can influence a person's memories as well as their ability to form new memories.



Spinal cord

- The spinal cord is an extension of the brain and continuous above with the medulla oblongata. It is a long, thin bundle of nervous tissue. It is a rough cylindrical structure lying within the case of the backbone. It is connected to the peripheral nervous system through 31 pairs of spinal nerves .The principal function of the spinal cord is to distribute motor fibres to the effectors of the body[glands and muscles] and to collect somatosensory information to be passed to the brain. The spinal cord receives and transmits electric signals throughout the enter body. It is also responsible for certain reflex actions.

- The spinal card centre of reflex actions. It converts sensory impulses in to motor impulses. The spinal card is the channel of communication from the brain and to the body. It not only connects to the brain with several parts of the body but it also controls reflex actions.

- Reflex actions: reflex action is an involuntary activity produced in a tissue in response to an afferent impulse. A sense organ is stimulated by a stimulus. A sensory neuron carries the nerve impulse from the sense organ to a sensory centre or nerve cell in the spinal card or the brain stem. There is a synaptic connection between a sensory neuron and a motor neuron. The motor neuron carries the nerve current from the central nervous system to a muscle or gland. This combination of a sensory neuron with a motor neuron through a synaptic junction is called the reflex action.

- A reflex action consists of 1. receptor 2. afferent or sensory nerve. 3. the synapse where the sensory stimulus is converted in to a motor impulse 4. efferent neuron or motor neuron.5. muscle.

The Peripheral Nervous System

- The PNS is composed of all the neurons and nerve fibers that connect the CNS to the rest of the body.

- The PNS provides information to the CNS from sensory receptors (eyes, ears, skin, etc.) and relays back motor commands from the brain to the muscles and glands.
- The peripheral nervous system consists of 1. the afferent or sensory nerves and 2. the efferent or motor nerves: the afferent (sensory) system, which carries messages from other parts of the body to the central nervous system, and the efferent (motor) system, which carries messages from the central nervous system to other parts of the body.
- The neurons of the peripheral nervous system do not make complex decisions about the information they carry. ... For example, if a mosquito lands on a person's arm, sensory neurons in the skin send a message to the spinal cord and then the brain, where the message is understood, and a reaction formulated.
- Peripheral Neuropathy:- a result of damage to peripheral nerves -unable to transmit information to the Central Nervous system (CNS). Often causes weakness, numbness and pain, usually in your hands and feet. It can also affect other areas of your body.
- The PNS is divided into Somatic Nervous System and Autonomic Nervous System.
- The autonomic nervous system is further divided into Sympathetic and Parasympathetic systems.

The Somatic Nervous System

- This system consists of two types of nerves, called cranial nerves and spinal nerves. There are twelve sets of cranial nerves which either emanate from or reach different locations of the brain.
- There are three types of cranial nerves - sensory, motor, and mixed. Sensory nerves collect sensory information from receptors of the head region (vision, audition, smell, taste, touch, etc.) and carry them to the brain.
- The motor nerves carry motor impulses originating from the brain to muscles of the head region. For example, movements of the eyeballs are controlled by motor cranial nerves.
- Mixed nerves have both sensory and motor fibers, which conduct sensory and motor information to and from the brain.
- There are thirty one sets of spinal nerves coming out of or reaching to the spinal cord. Each set has sensory and motor nerves.

- Spinal nerves have two functions. The sensory fibers of the spinal nerves collect sensory information from all over the body (except the head region) and send them to the spinal cord from where they are then carried out to the brain. In addition, motor impulses coming down from the brain are sent to the muscles by the motor fibers of the spinal nerves.

The Autonomic Nervous System

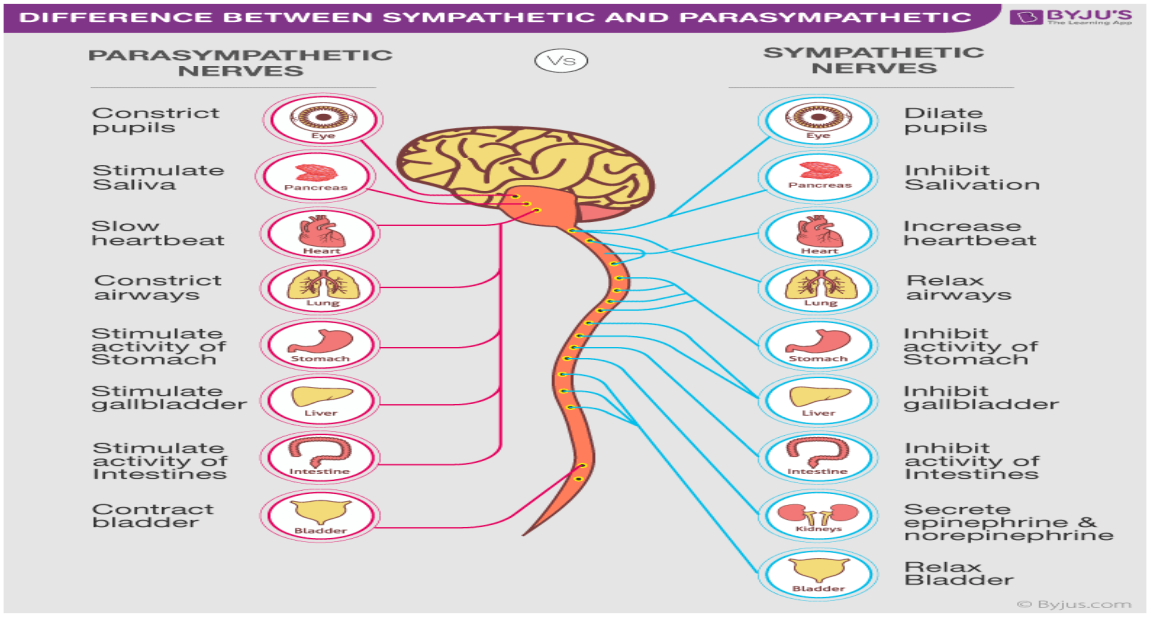
- This system governs activities which are normally not under direct control of individuals. It controls such internal functions as breathing, blood circulation, salivation, stomach contraction, and emotional reactions. These activities of the autonomic system are under the control of different structures of the brain.

- The autonomic nervous system is a part of the peripheral nervous system. it controls the involuntary bodily functions such as sweating, gland secretions, blood pressure, and the heart. It is divided into the 'sympathetic' and 'parasympathetic' divisions.

- The Autonomic Nervous System has two divisions: Sympathetic division and Parasympathetic division. Although the effect of one division is opposite to the effect of the other, both work together to maintain a state of equilibrium.

- The sympathetic division deals with emergencies when the action must be quick and powerful, such as in situations of fight or flight. During this period, the digestion stops, blood flows from internal organs to the muscles, and breathing rate, oxygen supply, heart rate, and blood sugar level increases.

- The Parasympathetic division is mainly concerned with conservation of energy. It monitors the routine functions of the internal system of the body. When the emergency is over, the parasympathetic division takes over; it decelerates the sympathetic activation and calms down the individual to a normal condition. As a result all body functions like heart beat, breathing, and blood flow return to their normal levels.



Lesson 7 : Brain - Structure and functions of brain- Limbic System.

Brain:-

- Brain is the most important part of the human body.
- Brain situated in a cranial box (cranium)which is made of bones.
- Brain is covered by three membranes of connecting tissue, termed as meninges.
- Brain is located in the bony skull. Within the skull, the brain is protected by three layers of tissues called meninges. Outer most layer is called dura matter and inner most layer is called pia matter. Between these two layers is a soft membrane called arechnoid. Arechnoid is filled with CSF (cerebro spinal fluid). Brain is composed of white and grey matter.
- Weight of brain- in adult male 1400 gm, female 1250gm.
- Meningitis:- any inflammation (painful especially as a reaction to injury or infection)of meninges is called meningitis
- The brain is divided into three main parts viz .the forebrain ,the midbrain and the hindbrain.

- Forebrain is the highest and most evolved part of the brain .It plays an important role in behavior patterns including those of emotion, motivation ,language ,thinking and learning . Some of the most important areas of forebrain are Cerebrum, Thalamus and Hypothalamus.

- Forebrain is the highest and most evolved part of the brain .It plays an important role in behavior patterns including those of emotion, motivation ,language ,thinking and learning . Some of the most important areas of forebrain are Cerebrum, Thalamus and Hypothalamus.

- Cerebral hemispheres(Cerebrum):- The largest division of the brain. It is divided into two hemispheres, each of which is divided into four lobes.

- The Left Hemisphere

  -Processes logical tasks (*reading, writing, speaking, mathematics, and comprehension skills*)

  -Controls the right side of our body

- The Right Hemisphere

  -Processes non-verbal tasks/perceptual (*spatial relationships, musical/artistic ability and mental imagery*)

  -Controls the left side of our body

  -May also be related to some negative emotions

(On an IQ test, non-verbal tasks include building with blocks to make a printed design, discerning patterns, and determining how to balance a scale. These tasks are measures of non-verbal reasoning skills. Nonverbal intelligence describes thinking skills and problem-solving abilities that do not fundamentally require verbal language production and comprehension.)

- The surface is made of ridges (gyri) and grooves (sulci).
- Ridges are known as gyri, where as grooves are known as sulci. These gyri and sulci increase the surface area of brain.

- Functions of the cerebrum

  Frontal Lobe- associated with reasoning, planning, parts of speech, movement, emotions, and problem solving.

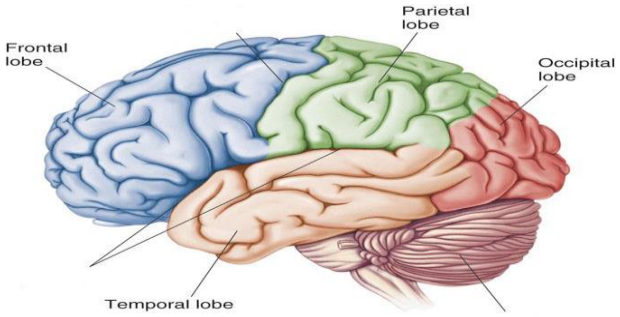
Parietal Lobe-it is primarily responsible for bodily sensations and knowledge of direction. associated with movement, orientation, recognition, perception of stimuli.

Occipital Lobe- associated with visual processing .

Temporal Lobe- associated with perception and recognition of auditory stimuli, memory, and speech.

Specialized Areas of the Cerebrum

- Primary motor area – sends impulses to skeletal muscles.
- Broca's area – involved in our ability to speak.
- Somatic sensory area – receives impulses from the body's sensory receptors.



Thalamus- filters sensory information, controls mood states and body movement associated with emotive states. The thalamus is the relay center(receive and pass on information or message) for all sensory impulses except olfactory (sense of smell) and motor areas of the cortex.

- It receives information for all of the senses EXCEPT for smell.
- The Hypothalamus lies below (*hypo*) the thalamus. The hypothalamus regulates behavior and emotional expression, body temperature, and many metabolic activities. Attached to the pituitary gland, it also controls hormonal secretions of this gland. It helps control the endocrine system by giving directions to the pituitary gland.
- An important part of the limbic system (emotions)
- The pituitary gland is attached to the hypothalamus.

Midbrain

- ❖ The midbrain lies between the forebrain and the hindbrain. It contains areas involved mainly in sensory and motor processes. The principal functions of the midbrain are audition, vision and muscle control.

Hindbrain

- ❖ The hindbrain connects the spinal cord with the rest of the brain. The three most important areas of the hindbrain are Cerebellum , Pons and Medulla oblongata.
- ❖ Cerebellum – The cerebellum is situated below the cerebrum. It is ovoid in shape and has two hemispheres, separated by a narrow median strip called the vermis.

Functions of cerebellum

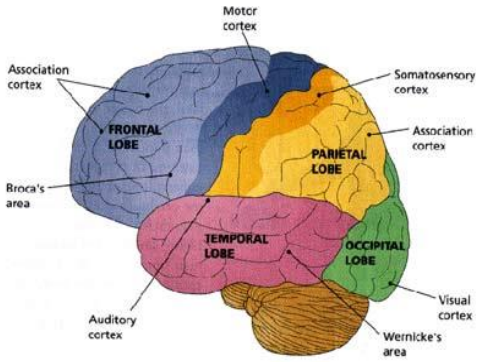
- ❖ The cerebellum is concerned with the coordination of voluntary muscular movement. It coordinates activities associated with the maintenance of the balance and equilibrium of the body.
- Balance and posture: Your cerebellum works with sensory input from your eyes and ears to keep you upright and steady.
- Motor learning: This involves the learning and fine-tuning of various movements. Examples include the specific, precise(accurate) movements used for writing or for riding a bicycle.
- Speech: The cerebellum is also involved in the movements associated with speaking.
- ❖ It is also involved in certain cognitive functions, such as language.
- language
- processing of emotions
- attention
- pleasure or reward response
- fear response
- Damage to the cerebellum results in clumsy(awkward in movements) uncoordinated muscular movement, staggering gait and inability to carry out smooth, steady, precise movements.
- a head injury

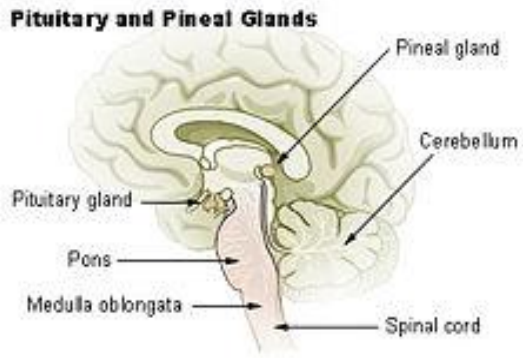
- stroke
- a brain tumor
- autoimmune conditions, such as multiple sclerosis(abnormal hardening of body tissue).
- neurodegenerative conditions, such as Parkinson's disease or Huntington's disease Infections.

Parkinson's disease is a progressive nervous system disorder that affects movement. Symptoms start gradually, sometimes starting with a barely noticeable tremor (a disorder that causes tremors and muscle rigidity) in just one hand. Tremors are common, but the disorder also commonly causes stiffness or slowing of movement.

What can you do to protect your cerebellum?

- Keeping your cerebellum and the rest of your brain healthy and free of injury is key to your physical and emotional health and overall sense of well-being. Here are some tips to keep in mind for good brain health:
- Exercise regularly: Not only is exercise good for your overall health, but it also helps to stimulate blood flow to your brain.
- Eat a healthy diet: All parts of your body can benefit from a healthy diet. Focus on fresh fruits and vegetables, whole grains, nuts, seeds, fish, and lean meat.
- Limit alcohol consumption: Drinking too much alcohol can damage your cerebellum. It can also increase your risk of stroke.
- Avoid smoking: Smoking is associated with many health conditions, including high blood pressure and stroke.
- **Pons**- Relay station between cerebrum and cerebellum. It controls mastication,(namaladamu), movements of the eye and also influences salivary secreation.It act as a sensory relay station of hearing.
- **Medulla oblongata-** Conscious control of skeletal muscles, balance, co-ordination regulating sound impulses in the inner ear, regulation of automatic responses such as heart rate, swallowing, vomiting, coughing and sneezing.
- The medulla oblongata connects the rest of the brain to the spinal cord. It regulates breathing, swallowing, coughing, sneezing, vomiting, heartbeat, and blood pressure.





Limbic System

- The Limbic System is a doughnut-shaped system of neural structures at the border of the brainstem and cerebrum, associated with emotions such as fear, aggression and drives for food and sex.
- It includes the hippocampus, amygdala, and hypothalamus.
- The Hypothalamus lies below (*hypo*) the thalamus. It directs several maintenance activities like eating, drinking, body temperature, and control of emotions. Attached to the pituitary gland, it also controls hormonal secretions of this gland. It helps control the endocrine system by giving directions to the pituitary gland.
- *Amygdala*- mediates and controls major affective mood states such as friendship , love, affection, fear, rage and aggression.
- Hippocampus- It is an important part of the limbic system, a cortical region that regulates motivation, emotion, learning, and memory.

Lesson-8 : Endocrine glands – Effects of over and under secretion of glands.

Endocrine glands

Endocrine System:

- The **endocrine system** includes the endocrine glands and their hormones.
- The function of the endocrine system is to secrete **hormones** into the bloodstream.
- **Hormone**: A Chemical messenger which targets a specific group of cells, in order to cause that group of cells do some activity or stop doing an activity.
- **Hormone**: A Chemical messenger ,affect certain target tissues or organs and regulate their activities.

- **Endocrine** Glands are called ductless glands, Release hormones directly into the bloodstream, Blood transports hormones throughout the body, Each hormone acts on only a certain kind of tissue called its **target tissue.**
- Glands are situated at different centers of the body. They may be divided in to:

Exocrine glands:

Exocrine glands release their secretions into ducts, or tubes

- Liver→ Bile released into the gallbladder, then through a duct into the small intestine.
- Your **gallbladder** is a small, pear-shaped organ on the right side of your abdomen, just beneath your liver. The **gallbladder** holds a digestive fluid called bile that's released into your small intestine.
- Function of galbladder: The **gallbladder** is a small pouch that sits just under the liver. The **gallbladder** stores bile produced by the liver. After meals, the **gallbladder** is empty and flat, like a deflated balloon. Before a meal, the **gallbladder** may be full of bile and about the size of a small pear.
- Pancreas→ releases pancreatic juice into the small intestine via a duct

Endocrine Glands are called ductless glands

- Release hormones directly into the bloodstream
- Blood transports hormones throughout the body
- Each hormone acts on only a certain kind of tissue called its **target tissue**(Group of cells that respond to specific hormones).

The Important ductless glands are

- Pituitary
- Thyroid
- Parathyroid
- Adrenal
- Pancreas (Islets of Langerhans)
- Testes and Ovaries

Pituitary gland:

- The pituitary, a pea-sized gland at the base of the brain, produces a number of **hormones**. Each of these **hormones** affects a specific part of the **body** (a target organ or tissue). Because the pituitary controls the function of most other endocrine glands, it is often called the master gland. The biological **tissue**(s) most adversely affected by exposure to a chemical substance. **Target organs** include the bone marrow, skin, and nervous system after either acute or chronic excessive exposure.

- The pituitary body has two parts:

(1). Anterior lobe

(2). Posterior lobe

Secretions from the anterior lobe of the pituitary gland…

Growth Hormone (GH): essential for the growth and development of bones, muscles, and other organs. It also enhances (or stimulate) protein synthesis, and increases fat breakdown to provide the energy necessary for tissue growth.

Oversecretion of growth hormone:

- Giantism in childhood and acromegaly in adults (bones of face, hands and feet enlarge). Gigantism occurs in patients who had excessive growth hormone in childhood. The pituitary tumor cells secrete too much growth hormone (GH), leading to many changes in the body. Symptoms:

- **Acromegaly is a rare hormonal condition that results from an excess amount of growth hormone (GH) in the body.** The initial symptom is typically enlargement of the hands and feet. There may also be an enlargement of the forehead, jaw, and nose. Symptoms of acromegaly include abnormal enlargement in **bones** of the hands, arms, feet, legs, and head. People with **acromegaly** may suffer with headaches, aches and pains in their bones and joints, and increased sweating.

Common symptoms of acromegaly are:

- enlarged bones in the face, feet, and hands
- excessive hair growth in women
- an enlarged jaw or tongue
- weight gain
- swollen and painful joints that limit movement

- spaces between the teeth
- splayed fingers and toes
- a hoarse, deep voice
- fatigue
- headaches
- an inability to sleep
- muscle weakness
- profuse sweating
- body odor
- enlarged sebaceous glands, which are glands that produce oils in the skin
- thickened skin

Undersecretion of growth hormone:

- Dwarfism in childhood. Dwarfism is a growth disorder characterized by shorter than average body height. **Dwarfism** is generally defined as an adult height of 4 feet 10 inches (147 centimeters) or less. The average adult height among people with **dwarfism** is 4 feet (122 cm). Adults typically are about 4 feet tall.
- **Adrenocorticotropin (ACTH):** Adrenocorticotropic hormone (ACTH) is produced by the pituitary gland. Its key function is to stimulate the production and release of cortisol from the cortex (outer part) of the adrenal gland. **Cortisol:** regulates carbohydrate, protein, and fat metabolism; helps the body cope during times of stress.
- **Thyroid-Stimulating Hormone (TSH):** essential for the growth and development of the thyroid gland. Thyroid-stimulating hormone (also known as thyrotropin, thyrotropic hormone, or abbreviated TSH) is a **pituitary** hormone that stimulates the **thyroid gland** to produce thyroxine (T_4), and then triiodothyronine (T_3) which stimulates the metabolism of almost every tissue in the **body**.

Symptoms of hyperthyroidism, also known as overactive thyroid, include:

- Anxiety
- Weight loss

- Tremors in the hands
- Increased heart rate
- Puffiness
- Bulging of the eyes
- Difficulty sleeping

Symptoms of hypothyroidism, also known as underactive thyroid, include:

- Weight gain
- Tiredness
- Hair loss
- Low tolerance for cold temperatures
- Irregular menstrual periods
- Constipation

- **Follicle-Stimulating Hormone (FSH):** is a gonadotropic hormone. It stimulates the growth ovarian follicles in the female and the production of sperm in the male.

FSH plays an important role in sexual development and functioning.

- In women, FSH helps control the menstrual cycle and stimulates the growth of eggs in the ovaries.
- In men, FSH helps control the production of sperm.
- Too much or too little FSH can cause a variety of problems, including infertility (the inability to get pregnant), menstrual difficulties in women, low sex drive in men, and early or delayed puberty in children.

- **Luteinizing Hormone (LH):** is a gonadotropic hormone stimulating the development of corpus luteum in the female ovarian follicles and the production of testosterone in the male. The yellow corpus luteum remains after ovulation; it produces estrogen and progesterone.

- **Corpus luteum:** a hormone-secreting structure that develops in an ovary after an ovum has been discharged but degenerates after a few days unless pregnancy has begun.
- LH helps your reproductive system: specifically, a woman's ovaries and a man's testes.
- **Prolactin (PRL):** stimulates the development and growth of the mammary glands and milk production during pregnancy. The sucking motion of the baby stimulates prolactin secretion.

Symptoms of excess prolactin in women who have not gone through menopause include:

- Irregular periods
- Periods that have stopped completely before the age of 40. This is known as premature menopause.
- Infertility
- Breast tenderness

Symptoms of excess prolactin in men include:

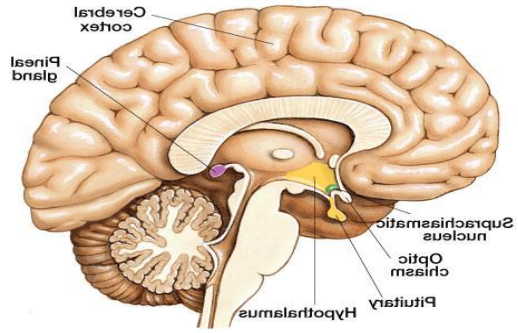
- Nipple discharge
- Breast enlargement
- Low sex drive
- Erectile dysfunction
- Decrease in body hair
- **Melanocyte-stimulating hormone (MSH):** regulates skin pigmentation and promotes the deposit of melanine in the skin after exposure to sunlight. Melanocyte-stimulating hormone describes a group of hormones produced by the pituitary gland, hypothalamus and skin cells. It is important for protecting the skin from UV rays, development of pigmentation and control of appetite.
- Ultraviolet (UV) rays, increase **MSH** production by the skin and pituitary gland. It has a vital role in making pigmentation (coloring) of the skin, hair, and eyes. Specialized skin cells known as melanocytes produce melanin, which helps protect cells from skin cancer.

Secretions from the posterior lobe of the pituitary gland…

- **Antidiuretic Hormone (ADH):** stimulates the reabsorption of water by the renal tubules. Hyposecretion of this hormone can result in diabetes insipidus. Antidiuretic hormone (ADH) is a hormone that helps your kidneys manage the amount of water in your body. The ADH test measures how much ADH is in your blood
- **Oxytocin:** Stimulates the uterus to contract during labor, delivery, and parturition. A synthetic version of this hormone, used to induce labor, is called Pitocin. It also stimulates the mammary glands to release milk.

Secretions from the pineal gland…

- The pineal gland is pine-cone-shaped and only about 1 cm in diameter.
- Pienal gland produces a hormone known as melatonin
- **Melatonin:** communicates information about environmental lighting to various parts of the body. Has some effect on sleep/awake cycles and other biological events connected to them, such as a lower production of gastric secretions at night.
- Serotonin: a neurotransmitter that regulates intestinal movements and affects appetite, mood, sleep, anger, and metabolism.



Secretions of the thyroid gland…

- The thyroid gland plays a vital role in metabolism and regulates the body's metabolic processes. **Located in the neck, Releases hormone thyroxin, Function is to regulate rates of metabolism in the body.**

Oversecretion of thyroxin:

Results in nervousness and weight loss

Undersecretion of thyroxin:

Results in cretinism (mental retardation, small size) in children.

Iodine deficiency in your diet results in **goiter** (enlargement of thyroid gland).

cretinism : Congenital **(of a disease or physical abnormality)** iodine deficiency syndrome (formerly known as **cretinism**) is a medical condition present at birth marked by impaired physical and mental. most common cause of congenital hypothyroidism is dietary iodine deficiency or the hyposecretion of Thyroxine hormone. It has affected many people worldwide and continues to be a major public health problem in many countries

- **Causes of cretinisn: Cretinism** is a condition of severe physical and mental retardation due to iodine deficiency, and specifically due to deficiency of thyroid hormones during early pregnancy.
- What is difference between cretinism and dwarfism?
- In several hormonal disorders and hereditary conditions **dwarfism** is associated with subnormal intelligence. Inadequate production of thyroid hormone during gestation and early infancy results **in a** condition known as **cretinism**, which is characterized by growth retardation and severe mental retardation.
- **Congenital hypothyroidism vs. myxedema**
- Myxedema (swelling of the skin) is a term used to describe a severely underactive thyroid gland in an adult. Congenital hypothyroidism refers to a thyroid deficiency in an infant.
- Myxedema can also be used to describe skin changes caused by low thyroid hormone levels.

Secretions of the parathyroid gland…

- The two pairs of parathyroid glands are located on the dorsal or back side of the thyroid gland. They secrete arathormone,which plays a role in the metabolism of phosphorus.
- Too little results in cramping; too much results in osteoporosis or kidney stones.

- Undersecretion of parathormone results in nerve disorders, brittle bones and clotting problems.
- Overuse of a muscle, dehydration, muscle strain or simply holding a position for a prolonged period can cause a muscle **cramp**. In many cases, however, the cause isn't known. Although most muscle **cramps** are harmless, some may be related to an underlying medical condition, such as: Inadequate blood supply.

Cramping:

- 1 : a painful involuntary spasmodic(irregular bursts) contraction of a muscle.
- 2: a temporary paralysis of muscles from overuse — compare writer's **cramp**.
- 3a : sharp abdominal pain —usually used in plural.
- b : persistent and often intense though dull lower abdominal pain associated with dysmenorrhea —usually used in plural.
- **Symptoms** of menstrual **cramps** include: **cramping** pain in your lower abdomen that can be intense. Pain that starts 1 to 3 days before your period, peaks 24 hours after the onset of your period and subsides in 2 to 3 days. Dull, continuous ache(a continuous or prolonged dull pain in a part of one's body.

Some women also have:

- Nausea.(a feeling of sickness with an inclination to vomit.)
- Headache.
- Dizziness.(a sensation of spinning around and losing one's balance.)
- Diseases related to para thyroid gland:
- **Rickets** : It is a disease characterized mainly by bone deformities in young children's.
- Characteristic features:
- A]Deformed bones b]Thick wrist and ankles c] Retarded growth.
- **Osteomalacia :** This is due to inadequate absorption of calcium due to deficiency of vitamin D and calcium in the diet.
- **Hypoparathyroidism :** It is an abnormally low level of calcium in the blood.
- **Osteoporosis** causes bones to become weak and brittle so brittle that a fall or even mild stresses such as bending over.
- **Osteoporosis**-related fractures most commonly occur in the hip, wrist or spine.

- The **main cause of osteoporosis** is **bone loss** due to a drop in your body's estrogen levels. Estrogen is a hormone that helps build and maintain your bones. The **most common cause** of estrogen loss in women is menopause.

Pancreas:

The islets of Langerhans are small clusters of cells located in the pancreas.

Secretions from the islets of Langerhans…

Beta cells secrete the hormone insulin, which is essential for the maintenance of normal blood sugar levels. Inadequate levels result in diabetes mellitus.

Alpha cells facilitate the breakdown of glycogen to glucose. This elevates the blood sugar.

Delta cells suppress the release of glucagon and insulin.

- Oversecretion of insulin results in **low** blood sugar
- Undersecretion of insulin results in **high** blood sugar
- This can lead to a condition known as **diabetes**

The adrenal glands…

The triangular-shaped adrenal glands are located on the top of each kidney. The inside is called the medulla and the outside layer is called the cortex.

- Undersecretion of adrenaline results in the **inability** to deal with stress

Secretions from the adrenal cortex…

- Cortisol: regulates carbohydrate, protein, and fat metabolism; has an anti-inflammatory effect; helps the body cope during times of stress.

Anti-inflammatory effect;

- Non-steroidal **anti-inflammatory** drugs (NSAIDs) are medicines that are widely used to relieve pain, reduce **inflammation**, and bring down a high temperature. They're often used to relieve symptoms of headaches, painful periods, sprains and strains, colds and flu, arthritis, and other causes of long-term pain.
- Hyposecretion results in Addison's disease; hypersecretion results in Cushing's disease.
- Corticosterone:like cortisol, it is a steroid; influences potassium and sodium metabolism.

- Aldosterone: essential in regulating electrolyte and water balance by promoting sodium and chloride retention and potassium excretion.
- Androgens: several hormones including testosterone; they promote the development of secondary sex characteristics in the male.

Secretions from the adrenal medulla…

- Dopamine is used to treat shock. It dilates the arteries, elevates systolic blood pressure, increases cardiac output, and increases urinary output.
- Epinephrine is also called adrenalin. It elevates systolic blood pressure, increases heart rate and cardiac output, speeds up the release of glucose from the liver… giving a spurt of energy, dilates the bronchial tubes and relaxes airways, and dilates the pupils to see more clearly. It is often used to counteract an allergic reaction.
- Norepinephrine, like epinephrine, is released when the body is under stress. It creates the underlying influence in the *fight or flight response*. As a drug, however, it actually triggers a drop in heart rate.

Secretions of the ovaries…

- The ovaries produce several estrogen hormones and progesterone. These hormones prepare the uterus for pregnancy, promote the development of mammary glands, play a role in sex drive, and develop secondary sex characteristics in the female.
- Estrogen is essential for the growth, development, and maintenance of female sex organs.

Secretions of the testes…

- The testes produce the male sex hormone called testosterone. It is essential for normal growth and development of the male sex organs. Testosterone is responsible for the erection of the penis.

Thymus gland:

The thymus gland has two lobes, and is part of the lymphatic system. It is a ductless gland, and secretes thymosin. This is necessary for the Thymus' normal production of T cells for the immune system.

Unit - III: Sensation, Attention &Perception

Lesson-9 Sensation: Sensory process-Senses and sensory organs, general characteristics of Sensation and Sensory thresholds, Subliminal Perception, Signal Detection Theory.

Sensation

Meaning

- Any experience that takes place through a sense organ is called Sensation.
- Sensation is also called as the gate ways of knowledge, because all our knowledge is based upon the functioning of sense organs.
- It is the elementary raw material of our knowledge. It is the simplest form of cognition

Sl. No.	Type of Sensation	Sense Organ	Senses	Knowledge gaining
1.	Visual sensation	Eye	Sight	83%
2.	Auditory Sensation	Ear	Hear	11%
3	Olfactory Sensation	Nose	Smell	3.5%
4	Taste Sensation	Tongue	Taste	1.0%
5	Tactual Sensation	Skin	Touch	1.5%

Definitions:

- A sensation is an elementary cognitive experience – **Dr. Jalota.**
- Sensations are first things in the way of consciousness – **James.**
- Sensation is defined as the simplest of all conscious experience – **Duglas and Halland.**

- Sensation is the process of receiving, translating and transmitting information to the brain from the external and internal environment – **Huffman.**

NATURE OF SENSATION

- A Sensation is an inner experience aroused by the respective stimulus through sense organs.
- It is the elementary raw material of our knowledge. It is the simplest form of cognition.
- Everything we experience comes to us through our sense organs. For instance, a ray of light acts upon the eye and produces the sensation of colour.
- A Sensation is the simplest form of mental activity.
- Sensations originate from the sense organs. Sensations may be classified according to sense organs.

Sense organs	**Sensations**	**Sense organs**	**Sensations**
Eye	Visual Light Color Shape Distance	Tongue	Acid Sweet Bitter Salt
Ear	Auditory Pitch Intensity Quality	Skin	Tactual Pressure Pain Temperature
Nose	Olfactory Various kinds Odours.		

KINDS OF SENSATIONS

There are three types of sensations

- **ORGANIC SENSATIONS:** Organic sensations are produced by internal organs. Hunger, thirst, etc. are examples of organic sensations.
- Organic sensations have the following characteristics. They have no special sense-organs. They are not produced by external stimuli. They are produced by changes in the internal organs of the organism.
- In the case of hunger, these are caused by contractions of the internal muscles. Thirst is caused by a feeling of dryness and stickiness in the mouth.
- which are the soft internal **organs** of the body, including the lungs, the heart, and the **organs** of the digestive, excretory, reproductive, and circulatory systems.
- The **organs** located within the three central cavities of the body (chest, abdomen and pelvis). The main cardiovascular, pulmonary, gastrointestinal, and genitourinary **organs.**
- **Synonyms:** Internal organs; Viscera
- **Sensory Neuron Motor Neuron**Neurons that carry sensory impulse from sensory organs to the central nervous system are known as sensory neurons
- A neuron that carries motor impulses from the central nervous system to specific effectors is known as motor neurons.
- They are located in the dorsal root ganglion of the spinal nerve
- They are located in the ventral root ganglion of the spinal cord.
- It is unipolarIt is multipolarComprises of a short axon Comprises of a long axon
- An adult has an average of 10 million sensory nerves in the body. Half million of motor neurons are found in the body
- Found in eyes, skin, ears, tongue and noseFound in muscles and glands.
- **Motor sensations [kinaesthetic sensations]:** These are produced by changes in the organs of movement viz, mscles, tendons and joints. Motor sensations are produced by the strain in the muscles, tendons and joints.
- Neurons that carry sensory impulse from sensory organs to the central nervous system are known as sensory neurons

- A neuron that carries motor impulses from the central nervous system to specific effectors is known as motor neurons.

- **Special sensations:** sensations of colours, sounds, tastes, smells, pressures, heat cold etc. are special sensations. They originate from eye [visual],ear [auditory],skin [tactual], nose [olfactory] and tongue [gustatory]. Special sensations are produced by the stimulation of the special sense-organs, viz., the eye, the ear, the tongue, the nose and the skin by special kinds of stimuli.

Attributes of sensations:

- Sensations have following attributes: a) Quality. B) Intensity. C) Extensity. D). Duration

Quality:

- Sensations differ in quality. Sensation of colours, sounds, tastes, smells heat and cold; differ from one another in quality. They are produced by different kinds of stimuli and they have different sense-organs. For example Sensations of colours are produced by the action of light waves on the retina of the eye-ball. Sensations of sounds are produced by the action of air waves on the auditory organ.

Intensity:

- All sensations differ in intensity; lights, sounds, tastes, smells, temperatures, pains, pressures, hunger, thirst, fatigue, etc., all vary from very weak to very intensive nature.

- Sensations of the same quality may differ in intensity. A light may be dim or bright, a sound faint or loud, a smell mild or strong, a taste weak or strong, a pressure light or heavy. A dim light produces a faint sensation of light.

A bright light produces an intense sensation of light. The other conditions remaining the same, greater intensities of stimuli produce greater intensities of sensations.

Extensity:

- Visual and tactual sensation have extensity. This is admitted by all psychologists. Look at the moon and then at a star. You have two visual sensations. The first visual sensation has greater extensity than the second. Touch a book with a finger-tip, and then with your palm.

- You have two tactual sensations. The first tactual sensation has greater extensity than the second. Extensity is the attribute of a sensation which is due to the area of the sensitive

surface which is stimulated by a stimulus. Extensity is different from intensity. Put a coin on your skin.

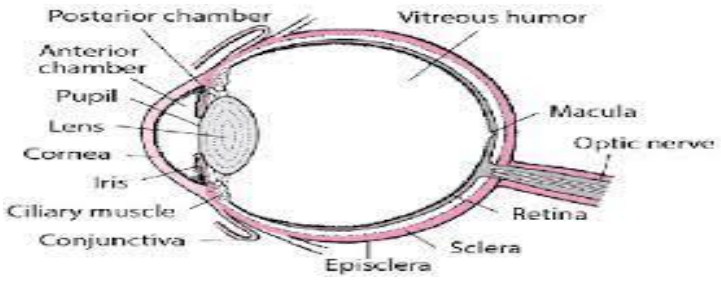
Duration of Protensity:

- Every sensation has a sensible duration. A sound sensation may last for a shorter or longer duration. The sound which lasts five seconds feels differently when it ceases from a sound which lasts twenty seconds. This is a difference in duration.

The Human Eye

- As you can see, our eye is made up of three layers. In the outer layer, there is a transparent cornea and a tough sclera that surrounds the rest of the eye. It protects the eye and maintains its shape. The middle layer is called choroid, which is richly supplied with blood vessels. The inner layer is known as retina. It contains the photoreceptors (rods and cones) and an elaborate network of interconnecting neurons.

- The eye is generally compared with a camera. For example, the eye and camera have a lens. The lens divides the eye into two unequal chambers, namely aqueous chamber and vitreous chamber. The aqueous chamber is located between the cornea and the lens. It is smaller in size and is filled with a water-like substance, called aqueous humor.



- The vitreous chamber is located between the lens and the retina. It is filled with a jelly like protein, called vitreous humor. These fluids help in holding the lens at its appropriate place and in proper shape.

- They also allow enough flexibility for the occurrence of accommodation a process through which the lens changes its shape in order to focus the objects at varying distances. This process is regulated by ciliary muscles, which are attached to the lens. These muscles flatten the lens to focus the distant objects and thicken it to focus the near objects.

- Like a camera, the eye also has a mechanism to control the amount of light entering into it. The iris serves this purpose. It is a disc-like coloured membrane lying between the cornea and the lens.
- It controls the amount of light entering the eye by regulating pupil dilation. In dim light the pupil dilates; in bright light it contracts.
- **Retina is the inner most layer of an eye. It is made up of five types of photosensitive cells among which rods and cones are most important.** Rods are the receptors for scotopic vision (night vision). They operate at low intensities of light, and lead to achromatic (colourless) vision.
- Cones are the receptors for photopic (day light) vision. They operate at high levels of illumination, and lead to chromatic (colour) vision.
- The rods are responsible for vision in night as work in low light condition (cannot distinguish different wavelength of light).
- The cones are responsible for color discrimination. There are three types of cones (differ on the basis of photoreceptor protein they make).
- 1. L-cones sense long wavelenght (red light).
- 2. M-cones sense medium wavelength (green light).
- 3. S-cones sense shorter wavelenght (blue light).
- When you see an object light enters your eye and stimulates the cone cells Your brain interpret the signals′ from the cone cell and enable you to see the colour of object They work together enabling us′ to see whole spectrum of colours. Faulty cones thus are′ responsible for color blindness.
- Color blindness means that you have trouble seeing red, green, or blue or a mix of these colors. It's rare that a person sees no color at all. Color blindness is also called a color vision′ problem.

CAUSES OF COLOUR BLINDNESS:

- 1. Usually genetic condition. Red/Green or Blue colour blindness is passed down′ from parents.

- The gene responsible is on X chromosome.′ Mutations capable of causing color blindness originate′ from at least 19 different chromosomes and many different genes. More males are affected the females are basically′ carrier. There is 50% chance of mother passing this condition to′ her son.
- Each eye contains about 100 million rods and about 7 million cones. The cones are highly concentrated in the central region of the retina surrounding the fovea, which is a small circular region of the size of a pea. It is also known as the yellow spot. It is the region of maximum visual acuity.
- Besides photoreceptors, retina also contains a bundle of axons of a cell (called ganglion cell) that forms the optic nerve, which leads to the brain.
- The **blind spot** is where the optic nerve and blood vessels leave the **eyeball**. The optic nerve is connected to the brain. It carries images to the brain, where they're processed.

Vision problems

- **Hyperopia:** Difficulty focusing nearby objects (farsightedness)
- **Myopia:** Difficulty focusing distant objects (nearsightedness)
- **Astigmatism:** Corneal, lens, or eye defect that causes some areas of vision to be out of focus; relatively common
- **Presbyopia:** Farsightedness caused by aging.

The Sense of Hearing:

- The ear is the sense-organ for sounds. The structure of an ear is divided into three segments, called the external ear, the middle ear, and the inner ear.

The ear consists of three parts:

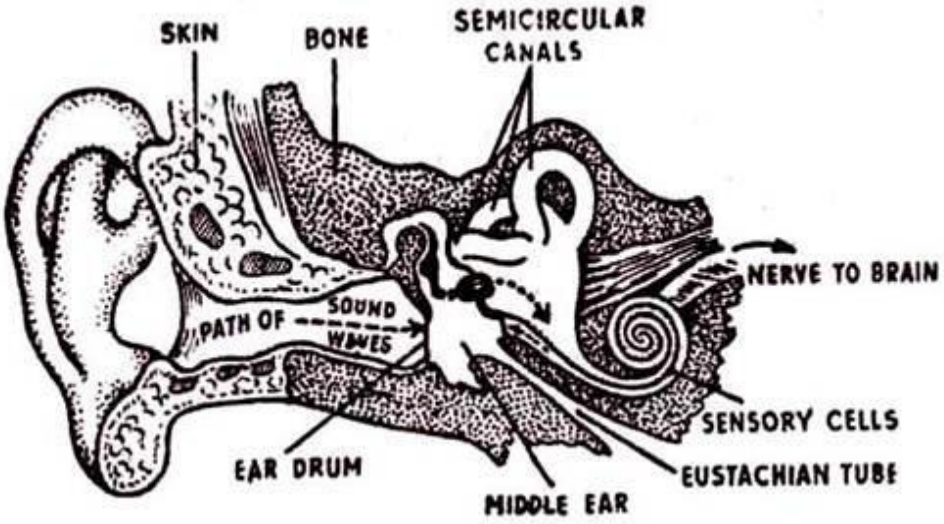
(1) The external ear of auricle,
(2) The middle ear or tympanum,
(3) The internal ear or labyrinth.

(1)The external ear:

- The external ear gathers the sound waves and transmits them to the middle ear. The sound waves strike the ear-drum called the tympanic membrane, and set it in vibration. This

vibration is conducted by means of three tiny bones, called the hammer, the anvil and the stirrup, to the membrane at the close of the middle ear.

- The labyrinth (internal ear)consists of three parts, viz., (i) the vestibule, (ii) the semicircular canals, and (iii) the cochlea. The cochlea contains a membrane called the basilar membrane, on which is seated the organ of corti, consisting of rods and cells.



- This is the real organ of hearing. The nerve-fibres issuing out of the nerve-cells compose the auditory nerve. The vibration of the membrane at the end of the middle ear sets the membrane in the cochlea into sympathetic vibration.
- The nerve-cells which are located along the harp of the cochlea translate these vibrations into the nerve currents which are conducted by the auditory nerve to the brain. Then sensations of sound are produced in the mind.

Smell

- **Smell :** The stimulus for smell sensation consists of molecules of various substances contained in the air. They enter the nasal passage where they dissolve in moist nasal tissues. This brings them in contact with receptor cells contained in olfactory epithelium.
- The sense of smell, you get, from stimulation of receptor cells present in the nose. Smell provides information about chemicals suspended in air which excite receptors located at the top of our nasal cavity. Humans have only about 50 million olfactory receptors whereas, dogs possess more than 200 million such receptors. Dogs are more sensitive to smell therefore, they are put in dogsquad to detect crime and criminals in police department. It is indicated that human beings can recognise and distinguish among

approximately 10,000 different odours. The sense of smell also shows sensory adaptation like other senses.

Taste

- The sensory receptors for taste are located inside small bumps on the tongue, known as papillae. In each papilla there is a cluster of taste buds. Human beings possess almost 10,000 taste buds. While people claim to distinguish a large number of flavours in food, there are only four basic tastes, namely sweet, sour, bitter and salty. The four primary tastes, sweet, sour, salt, and bitter cannot be reduced to any other simpler tastes. The tip of the tongue is more sensitive to sweet, its back to bitter, and its sides to sour, but it is equally sensitive to salt over its whole surface.

- Taste is a chemical sense. Tiny nubs, papilla, or prominences on the surface of the tongue contain taste buds or taste bulbs in which hair cells connect with the ends of a sensory nerve. Taste buds are branches of sense cells. They are not located on the surface of the tongue. They art placed in little pits which extend down the surface.

The metallic and alkaline tastes are not recognized as primary tastes now.

- A chemical substance in liquid forms or soluble in the saliva of the mouth is the stimulus. It must be liquid or soluble in order to affect the ends of the hair cells and be tasted. There is a chemical action between the sense-organ and the stimulus. It is conducted to the brain by the gustatory nerve.

Primary Tastes:

- Sweet, sour, salt and bitter are the four primary tastes. Wundt added metallic and alkaline tastes also. But the metallic taste is a combination of taste with smell and muscular sensations. Strong alkalies may make the tongue slippery and may also produce puckering of the surface of the tongue.

- **Touch and other skin senses :** Skin is a sensory organ that produces sensations of touch (pressure), warmth, cold, and pain. In our skin there are specialised receptors for each one of these sensations. The receptors of touch are not evenly distributed in our skin. That is why some areas of our body (e.g., face, fingertips) are more sensitive than others (e.g., legs). Pain sensation has no specific stimulus. Hence, determining its mechanism has been fairly difficult.

Sensory threshold

- In psychophysics, sensory threshold is the weakest stimulus that an organism can sense. Unless otherwise indicated, it is usually defined as the weakest stimulus that can be detected half the time, for example, as indicated by a point on a probability curve.
- Absolute threshold is the smallest level of energy required by an external stimulus to be detectable by the human senses, including vision, hearing, taste, smell and touch.
- SIGNAL DETECTION THEORY

  Recent development in psychophysics takes suitably the problem of response bias attitude in sensory experience. Green and Swets (1966) described detection experiment on the basis of which a theory was developed known as 'signal detection theory'. This theory suggests that these are no absolute thresholds for sensations. Rather, detection of stimuli depends on their physical energy and on external factors such as the relative costs and benefits associated with detecting their presence..

  Signal detection theory has a somewhat different testing technique in which catch trials are a regular part of the procedure rather than just an occasional check to keep the subjects on their toes. To know whether the subject can detect the presence of a stimulus, a fairly weak stimulus is selected and presented on half of its trials. On the other half of the trials 'no stimulus' is presented.

  Both the stimuli that is the 'weak stimulus' or 'no stimulus' are presented in random order. Two kinds of errors may be committed by the subject. One is a 'miss', not reporting a stimulus when present; another is a 'false alarm', reporting a stimulus when, in fact, not present.

  Two kinds of correct responses may also be given. Reporting a stimulus when it is actually present 'hit' and not reporting it when none is present 'correct negative'.

  The detection experiment gives a basis for getting at the non-sensory factors underlying response bias. One is differential payoff based on 'payoff matrix'. That is , in this you pay a subject for every 'hit' and 'correct negative' and penalise for every 'miss' and 'false alarm' as per prescribed schedule of gains and losses. This is called 'payoff matrix'. The four possible outcomes of the detection experiment are

	Responds Yes	**Responds No**

Stimulus present	Hit	Miss
Stimulus absence	False alarm Correct negative	False alarm Correct negative
	Subject says 'yes' Subject says 'no'	**Subject says 'yes' Subject says 'no'**
Stimulus present	+10 Rs. (Hit) – 10 Rs. (Miss)	+10 Rs. (Hit) – 10 Rs. (Miss)
Stimulus absence	– 1 Re. (False alarm) + 5 (Correct negative)	– 1 Re. (False alarm) + 5 (Correct negative)

The 'pay off matrix' where the subject gains Rs. 10/- for every 'hit' and Rs. 5/- for every correct negative; loses Rs. 10 /- . for every 'miss' and Re. 1/- for every 'false alarm', This will ; will lead to a 'bias' toward 'yes' judgements. With this payoff matrix, the subject will do well to adopt a liberal criterion and give a 'yes' judgement whenever the subject is in doubt.

Take for instance an example, Suppose in such an experiment, on 50 trials the subject has no sensory information on the basis of which she can decide whether the stimulus is present or absent. If she consistently says 'yes' she will be on the average be correct on 25 trials (thus collecting Rs.250/-) and wrong on the other 25 (thus losing Rs.25/-) and her net gain will be Rs.225. In contrast, consistent 'no' judgements will lead to a net loss of Rs. 125.00 (+ Rs. 125 for the correct negative and – Rs. 250 for the false alarm). If the stimulus is presented half of the trials, the payoff bias can be calculated easily by comparing the sum of the values under the 'yes' column with the sum under the 'no' column. In the example presented, these sums are +9 Rs. and – 5 Rs. respectively. This way, the subject will adopt a liberal criterion and give 'yes' judgement even if she is in doubt.

Lesson 10 : Attention: Selective attention – Kinds of Attention- Barriers of Attention- Concepts and Mechanisms- Selective Attention.

ATTENTION

- Attention was earlier considered as a mental power but psychological experiments have now proved that it is simply a mental process. We cannot do anything without attention. Whatever work we do needs attention. Attention is related to consciousness. Paying attention to anything is to concentrate our consciousness on it. During all the time that we are awake our attention centres round something or the other. Attention is the process whereby we can

select from among the many competing stimuli present in our environment, facilitating processing of some while inhibiting processing of others. Attention is the behavioral and cognitive process of selectively concentrating on a discrete aspect of information, whether deemed subjective or objective, while ignoring other perceivable information. Attention has also been referred to as the allocation of limited processing resources.

Attention refers to the concentration and focusing of mental effort

- Selective
- Shiftable
- Divisable.

- This selection can be driven endogenously by our goals (e.g., to find a particular friend, to follow an instruction, to use an arrow to direct attention), or exogenously by a salient or novel stimulus that captures attention away from the task at hand (e.g., bright light, loud noise). Because there is too much information at any given moment for us to cope with, attention is the mechanism by which the most important information is selected for further processing. The type of information that we miss and the conditions under which we miss it are, therefore, the flip side of the cognitive processes involved in attentional selection. Being unaware of the posters on the wall at a party is a failure of selection that is a property of selectively searching for features of a friend. Although we are capable of processing only a limited quantity of information in both space and time, fortunately selection does not occur randomly. Both our goals and the salience of information around us determine where and to what we attend.

- Attention is the ability to choose and concentrate on relevant stimuli. Attention is the cognitive process that makes it possible to position ourselves towards relevant stimuli and consequently respond to it. This cognitive ability is very important and is an essential function in our daily lives. Luckily, attention can be trained and improved with the appropriate cognitive training.
- Attention is the focus of consciousness on a particular object or idea at a particular time , to the exclusion of other objects or ideas.

Meaning:

- Attention is an active part of consciousness.

- The activity of concentrating mind on a particular matter is called attention. ఒక.
- Attention is not possible in the absence of consciousness, but attention and consciousness are not one.
- The field of consciousness is vast and attention is one of its parts. For example, I am reading at this time. Book, note, table, chair, etc., all this can be under my consciousness, but my attention is on the words being read on the paper.

Definitions:

- Attention is the concentration of consciousness upon one object rather than upon another. (Dumyille- 1938) Attention is the cognitive process of selectively concentrating on one aspect of the environment while ignoring other things – John R. Anderson.

Characteristics of attention

- It is a selective mental activity.
- Mental activity or neurological stimulation can be described as **anything that stimulates, activates or enriches the mind**. Stimulation can be provided internally from thought or externally from the environment.
- **The following mind stimulating activities provide ways to keep seniors sharp, helping to improve memory, problem-solving, creativity, and other cognitive functions.**
- Word Games. ...
- Games and Puzzles. ...
- Card Games. ...
- Chess & Strategy Games. ...
- Reading. ...
- Physical Exercise & Healthy Lifestyle.
- It constantly shifts from one object to another or from one aspect to another.
- Attention is focus of the consciousness on one particular object.

Is attention and consciousness the same?

- In the same sense, eye movements select conscious visual events, which are not the same as conscious visual experience. According to common sense, **visual experience is consciousness, and selective processes are labeled as attention.**
- Attention is the state of preparedness or alertness.
- Attention requires some motor and sensory adjustments on the part of the person being attentive.
- Attention is a mental process.
- Mental process or mental function are terms often used interchangeably for all **the things that individuals can do with their minds**. **These include perception, memory, thinking (such as ideation, imagination, belief, reasoning, etc.), volition, and emotion.** Sometimes the term cognitive function is used instead.
- Attention is a state of preparedness or alertness. Attention has narrow range/span.

What is span of attention in psychology?

- Attention span refers to **an individual's ability to attend to a stimulus or object over a period of time**. This ability is also known as sustained attention or vigilance.

Types of Attention

- The most accepted model for the attention sub-components is currently the hierarchical model from Sohlberg and Mateer (1987, 1989), which is based on clinical cases of experimental neuropsychology. According to this model, attention can be divided into the following parts:
- **Arousal:** Refers to our activation level and level of alertness, whether we are tired or energized.
- Arousal is the physiological and psychological state of being awoken or of sense organs stimulated to a point of perception.
- **Focalized (Focused) Attention:** This refers to your ability to focus attention on one stimulus. For example, when you are writing an exam and need to concentrate entirely on your answers.
- **Sustained Attention:** The ability to attend to a stimulus or activity over a long period of time.

- Examples of sustained attention may include **listening to lecture, reading a book, playing a video, or fixing a car**.
- **Selective Attention:** The ability to attend to a specific stimulus or activity in the presence of other distracting stimuli.

  Here are some everyday examples of selective attention:
  Listening to your favorite podcast while driving to work
  Having a conversation with a friend in a crowded place
  Reading your book on a public transport bus.
- **Alternating Attention:** The ability to change focus attention between two or more stimuli.
- It could also be alternating between unrelated tasks **such as cooking while helping your child with her homework.**
- Divided Attention**:** The ability to attend different stimuli or attention at the same time. This refers to your brain's ability to attend to different stimuli at the same time. For example, when you are driving a car while talking to a passenger.
- **Divided attention is the ability to pay attention to two tasks at once such as cooking a meal while talking to a friend or driving a car and talking to a passenger at the same time –** neither activity is stopped in order to carry out the other activity. Impairments of alternating and divided attention may cause the person
- difficulties
- Sustained Attention: This is your ability to attend to an activity or stimulus over prolonged periods of time. For instance, when you are playing a video game.
- Selective Attention: This refers to your ability to focus on a specific stimulus in the presence of other distracting stimuli. For example, in a classroom, this would be a student's ability to maintain focus on the lesson while his peers chatter or pass notes.
- Alternating Attention: Commonly known as multitasking. This is your ability to change focus attention between stimuli that need different cognitive functions. For example, reading a recipe while preparing a meal.

Factors Related to Attention

- Attention depends upon several factors. These factors may be of two types:

A. External.

B. Internal.

External Factors:

- The external factors are concerned with the environment. These are also called Objective Factors.
- **Size:** Size has effect on attention. It is natural an unusual size attracts attention of the people. Very big size or very small size too draws our attention when compared with normal size. For example, a Lilliputian (dwarf man) walking on the road too draws our attention.
- **Intensity:** Loud sounds, strong smells and deep colours are attractive in nature. If a sound is intense then it would attract our attention. The thunder is louder than a car sound. So, our attention is drawn on thunder.
- **Movement**: Moving things draws our attention more than stationary(sthira) one. A moving car attracts faster than a stationary car.
- **Contrast:** Anything that is different from its surrounding is contrast(viruddam). A black dust in the milk drawn attention quickly. A swan among the crows attracts suddenly.
- **Repetition:** If a thing or person or event is repeated several times, then our attention drawn to it.
- **Duration:** attention is drawn to a thing that lasts longer. A salesperson draws attention by lengthening his voice.
- **Change:** Change draws our attention easily. In the midst of continuous noise a slight moment of silence draws our attention.
- **Novelty:** Newness attracts quickly than traditional one. A new teacher attracts the children very much in the school.

Internal Factors

- The internal factors are concerned with the individual. So, these are also called subjective factors.
- **Interest:** we are interested in some things and disinterested in other things. Interesting things draws our attention soon. An engineer and a botanist going down the same path

will attend entirely different things on the way. Engineer attention will be on the buildings and botanist attention will be on the trees.

- **Desire:** A person's desire becomes a cause of paying attention to a thing. For example, a person has to desire of buying a hammer. There are many things available in a market, but when he goes to a shop where hammers are available.
- **Motives:** Basic motives are important in drawing attention. Human motives like hungry, thirst, sex, safety, etc., play a vital role in drawing attention. A thirst person attention always on where water is available.
- **Aim/Goal:** Every man has some immediate aim and ultimate goal in their life. The immediate aim of a student is to pass in the examination while his ultimate goal may be to become a doctor. The student, whose goal is not to pass the examination, will not be concerned with textbooks or note, etc, but who has the aim to pass in the examination, will at once attend to them.
- **Habit:** Habit is also a vital determinant of attention. The kind of habit we found in our life, our attention is drawn to such things. if a person has habit to play cricket, then his attention is always drawn to it, and he will listen to cricket commentaries with attention.
- **Past Experience:** It is also affect attention. If we know by our past experience that a particular person is sincere to us, we shall pay attention to whatever he advises us. If our experience is contrary, we shall not attend even to his most serous advice.
- Apart from these above factors, aptitude, attitude, mental set, disposition and temperament etc., are also an internal factors.

Sources of Distraction

- The sources of distraction can be roughly divided into two - external and internal sources.
- **External Factors:** It is also called environmental factors. These are more common and prominent.
- Noise, music, improper lighting, uncomfortable seats, inadequate ventilation, defective method of teaching, improper use of teaching aids, defective voice of the teacher are the common external distractors in the classrooms.

- **Internal Factors:** Emotional disturbances, ill-health, anger, fear, feeling of insecurity, boredom, lack of motivation, feeling of fatigue, lack of interest, unrelated subject matter are the examples for internal distractor.

Voluntary attention

- Requires conscious effort on the part of the person.
- Eg: problem solving, answering in the examination.
- Furthur types

1. Implicit volitional attention.
2. Explicit volitional attention.

Implicit volitional attention

- A single act of will is responsible for arousing attention.
- Eg. home work given by teacher and the fear of punishment mat arouse attention in the student to complete the given tasks.

Explicit volitional attention

- Attention is obtained by repeated acts of will.
- One has to struggle hard for keeping oneself attentive. it requires strong will power and motivation to keep the attention focused.

  Eg: during the preparatory days of the examiantion.

Involuntary attention

- This type of attention is aroused without the conscious effort .
- Eg: bright lights, loud noise and strong odours, pain etc.
- This type of attention can be aroused by instincts called enforced non volitional attention.
- Attention given due to preformed sentiments, to people objects and things.

Lesson-11. Determinants and Theories -Broadbent, Treisman and Deutsch and Deutsch, Norman Models, Sustained Attention, Theories Expectancy, Signal Detection, Arousal, Habituation; Intensive Properties of Attention, Fluctuations of Attention, Attention Span.

Lesson-12. Perception: Meaning of Sensation- Types of Sensation - Definition of Perception – Factors influencing Perception (Physiological and Psychological) - Perception and Illusion

PERCEPTION

- The process by which we recognise, interpret or give meaning to the information provided by sense organs is called perception.

DEFINITIONS

STEPHEN ROBBINS

- "Perception is a process by which individual's organize and interpret the sensory impressions in order to give meaning to their environment." (Organization In this stage, stimuli are arranged mentally in a meaningful pattern.)

FRED LUTHANS

- "Perception is an important mediating cognitive process through which persons make interpretations of the stimulus or situation they are forced with."
- In general, it can be defined as "a process that involves seeing, receiving, selecting, organizing, interpreting and giving meaning to the environment"
- Perception is the process by which an individual gives meaning to the environment.

2. It is a cognitive and psychological process.

3. People's action, emotions, thoughts and feelings are triggered by their perceptions or events at any particular moment, it occurs whenever stimuli activate the sense organs.

Expectations or Perceptual Sets

- The expectations about what we might perceive in a given situation also influence our perception. This phenomenon of perceptual familiarisation or perceptual generalisation reflects a strong tendency to see what we expect to see even when the results do not accurately reflect external reality.
- For example, if your milkman delivers you milk daily at about 5.30 A.M., any knocking at the door around that time is likely to be perceived as the presence of the milkman even if it is someone else.

MEANING OF PERCEPTION

- The process by which people select, organize, interpret, retrieve, and respond to information.
- Perceptions differ from person to person.
- Each individual perceives the same situation differently.
- Individuals organize and interpret things based on their past experiences and the important values they consider important.

Nature of perception:

- Sensations are simple impressions of sensory qualities. But their meanings are not known . As soon as their meanings are interpreted, they become perception. For instance ,I hear a sound from a distant place. This is sensation. But when I know it to be a sound of bell from a distant church, it becomes perception. Meaningful sensation is perception.
- Often perception is accompanied by feeling. We perceive a rose, and have a feeling of pleasure. But when we perceive aloud noise we feel pain. This feeling of pleasure or pain depends on our past experiences.
- Perception involves affective processes, e.g., a beautiful flower is pleasant but an ugly object is unpleasant.
- Perception is linked with association. In the past, you have seen and touched ice. Now if you see a piece of ice, you at once perceive it as a cold object.
- Perception is determined by mental readiness. For example, a person may continue to sleep while his wife wakes up at her baby's cry. This is because she is set to hear her baby's cry.
- According to Pillsbury, perception is a synthesis of sensation and memory , i.e.,

  Perception =sensation+ memory.

Attributes of perception:

- Perception has four attributes: a] Quality, [b] Intensity, [c] Extensity and [d] Duration.

- **Quality:** There are differences in our visual, auditory, tactual and olfactory perceptions. Perception of colours, tastes, sounds, smells, heat and cold differ from each other in quality. They are produced by different kinds of stimuli.

- **Intensity:** By intensity we mean the intensity of quality . For example, a sound may be faint or loud or a pressure(physical fource) may be light or heavy. Perceptions of these two qualities will differ from one another.

- **Extensity:** Extensity of perception depends on the area of the surface occupied by the object. Put a coin on your skin and then put a book. The two perceptions will not be the same.

- **DURATION:** Perception may be of shorter duration or longer duration. The sound of a bell may last for a longer or shorter duration. Perceptions alike in quality and intensity may differ in duration. **Duration** is defined as the length of time that something lasts. When a film lasts for two hours, this is **an example** of a time when the film has a two hour **duration**.

- Besides those four attributes perception has certain special features:

[1]Perception requires attention. Without attention, the interpretation of a sensation is not possible.

[2] Perception is entirely personal or subjective. There are individual differences in perception of a thing or an incident. Our perception is influenced by our physical or mental condition. It may be affected by our expectation and past experiences.

For example, if your milkman delivers you milk daily at about 5.30 A.M., any knocking at the door around that time is likely to be perceived as the presence of the milkman even if it is someone else.

- In the early 20th century, three German psychologists Max Wertheimer, and his associates Wolfgang Köhler and Kurt Koffka proposed new principles for explaining perception called as Gestalt principle. Gestalt psychology is form psychology.

- The literal meaning of the word Gestalt is form or configuration(arrangement of elements in a particular form,figure). The basic premise(base) of Gestalt psychology is that ‘whole is different from the sum of its part’. This implies that organization (or structured whole, known as Gestalten) gives a different meaning to the perception.

- Gestalt psychologists proposed a number of principles or laws to explain how we perceive smaller units of stimuli as a whole, having a particular pattern. These principles are known as laws of perceptual organization.

Gestalt view of perception:

- Perception is not a bundle of some separate sensations. It is a simple unitary experience.
- Perceptual knowledge is always organised. By the arranging of some dots you can form a circle. but when the dots are quite separate the property of circle is lost.(Organization In this stage, stimuli are arranged mentally in a meaningful pattern.)

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- **Proximity(nearness in space time or relationship):** Proximity of elements favour grouping of them into a pattern. Things which are in close contact with each other form a unit. For example, four lines in the given figure do not appear as four discrete lines. They are perceived as two pairs of line.
- **Similarity:** Similar objects form a group. In the given figure, the triangles form a group and the circles form another group and they are perceived as two different groups of circle and triangle.
- **Familiarity:** pervious knowledge of a thing favours our perception. In the given figure, several dots have been arranged to give the shape of a human face. Previous knowledge of human face helps us to perceive the figure of dots as a human face.
- **Principle of closure**: There are gaps in each of the given figures. But they are perceived as complete figures having no gaps in them. This filling in gaps in stimulation(encouragement of something to make it develop or become more active) is termed closure.
- **Figure ground relation**: In order to have a knowledge about organisation, we should understand the figure ground relation. The figure seems to have some sort of shape, while the ground tends to be formless. The figure appears to be in front and the ground behind.

The figure is always more organised than the field or ground. The given figure appears as either a flower-vase or two profiles.

- **Figure–ground** organization is a type of perceptual grouping that is a vital necessity for recognizing objects through vision. In Gestalt psychology it is known as identifying a *figure* from the back*ground*. For example, black words on a printed paper are seen as the "figure," and the white sheet as the "background".

- The visual cues that determine which elements will be perceived as figure and which as ground are:

• The figure has a definite shape, whereas the ground is shapeless.
• The ground continues behind the figure.
• The figure seems closer with a clear location in space, whereas the ground seems farther away and has no clear location in space.

- **Continuity:** Continuity of patterns is favourable to grouping elements into regular wholes. The dots in the given figure appear to be continuous though they are separate from one another. Dots **scattered(various locations rather than all together, chellacheduruga**) about irregularly are not easily perceived.

- The lens divides the eye into two unequal chambers, namely aqueous chamber and vitreous chamber. The aqueous chamber is located between the **cornea and the lens**. It is smaller in size and is filled with a water-like substance, called aqueous humor.

- The vitreous chamber is located between the **lens and the retina**. It is filled with a jelly like protein, called vitreous humor. These fluids help in holding the lens at its appropriate place and in proper shape.

- They also allow enough flexibility for the occurrence of accommodation a process through which the lens changes its shape in order to focus the objects at varying distances. This process is regulated by ciliary muscles, which are attached to the lens. These muscles flatten the lens to focus the distant objects and thicken it to focus the near objects.

Physiological factors:

- **Convergence:** The nearer the distance of the object from the eye , the smaller is the convergence. Whereas, the greater is the distance of the object , the greater is the convergence of the lines[see the figure].This results in different kinds of sensation in the muscles of the eye and accordingly distance is perceived.

- **Accommodation:** It is done by the ciliary muscles of the eye. The nearer is the distance, the greater is the convexity of the lens. The greater is the distance , the less is the convexity of it. The different degrees of tension of the ciliary muscle and processes suggest different distances.

- **Retinal disparity :** The left eye and the right eye get somewhat different retinal images of the same object . The right eye gets the retinal image of the right side of the object , Whereas the left eye gets the retinal image of its left side . The two retinal images are slightly different . This phenomenon is called retinal disparity.

Psychological factor

- **Super position:** The distant object is partially covered by a near object. When we see the outline of an object being overlapped by another, we judge the former to be farther, and the latter to be nearer. If there are two mountains, one behind the other, the outline of the distant mountain is partially hidden by the nearer mountain. In the given figure B appears to the farther than A.

- **Light and shade:** The lighted portion of a picture, appears nearer and the shaded portion appears further. If a picture is shaded and lighted, we see the objects at different distances.

- **Clearness of outline:** If outlines of an object are clear and distinct, it is near. But if the outlines are vague and hazy, it is distant.

- **Apparent convergence of parallel lines:** The greater is the distance, the greater is the apparent convergence of the lines which are known to be parallel. As the distance become greater, the parallel railway lines appear to converge. Thus the apparent convergence of parallel lines is a visual cue to distance perception.

- **Size of the object:** From our experience we know that distant object appears to be shorter than the near object. This helps our perception of distance.

- **Filled and empty space:** Filled space appears to be farther from empty space. According to the Gestalt psychologists figure and ground, closure and law of pregnance play an important part in the perception of distance.

- **Parallax:** If you move your head, you can observe that the objects in your visual field appear to move. If you observe closely, you will find that objects that are nearest to you,

appear to move in the opposite direction. But the distant objects appear to move in the same direction as your head.

- When we travel in a train we see that the trees nearby move in opposite direction, but the trees at distance appear to move slowly in the direction we move. This apparent movement of the observed objects due to variation of distance is known as parallax. This helps or misprint as misprint of distance.

Perception and illusion

- Perception is the right interpretation of sensations. Illusion is the wrong interpretation of sensation . It is a false perception.
- For instance, when we perceive a rope as a rope , we have perception. But when we perceive a rope as a snake, we have an illusion.
- If you put a pencil into a glass of water, the pencil will appear bent . Your perception of the pencil as bent is a false one an illusion.
- **Causes of illusions**
- **Habit and familiarity:** Some illusions are due to habit and familiarity. Proofreaders are apt to read the word “Preception” as “Perception” due to habit.
- **Expectation and set :**some times expectations and set cause illusion. For example, if your milkman delivers you milk daily at about 5.30 A.M., any knocking at the door around that time is likely to be perceived as the presence of the milkman even if it is someone else.
- **Anxiety or Fear:** anxiety or fear is the causes of some illusions. A timid person hears the sound of house- breaking by a burglar at night when the sound is caused by rustling of the air outside.
- **Disorders in sense organs:** disorders in some of our sense organs may give rise to illusions. A white cat appears yellow to a man whose eyes are affected by jaundice.
- **Movement;** movements causes illusions. When we travel in a train , the nearby trees appear to move in the opposite direction.
- **Similarity:** similarity between two objects may cause illusions. For instance, we mistake a rope for a snake or rays of the sun for water in a desert.

Unit – IV: Motivation and Emotion

Lesson 13 : Motivation-definition and Nature - Instinct, drive, needs, incentive, motive, Classification of motives.

Motivation

❖ Motivation is the very heart of the learning process. Adequate motivation not only sets in motion the activity which results in learning, but also sustains and directs it. It has been stated, "Motivation arouses interest. Interest is the mother of attention and attention is the mother of learning. Thus to secure learning you must first catch the mother, grandmother and great grand-mother." Motivation is an indispensable technique for learning. It energizes and accelerates the behaviour of learner. Desirable changes in learner's behaviour are only possible when a learner is properly motivated. No learning is possible without motivation. The word motivation has been derived from the Latin word movers which means to move.

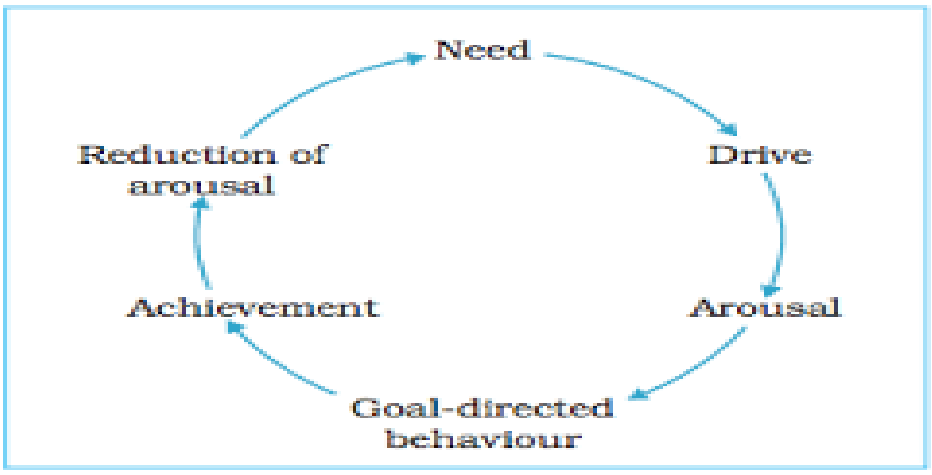
❖ Motivation is an internal force which accelerates a response or behaviour. Some learners learn the same subject-matter or task more efficiently than others, some find it more rewarding and interesting than others; and some enjoy it more than others. At any given time learners vary in the extent to which they are willing to direct their energies to the attainment of goals, due to difference in motivation. Tremendous research has been conducted on motivation in the last five decades and a number of definitions and theories have been given to explain motivation. K B Madson (1975) in his book, Theory of Motivation has given twenty-four definitions and theories of motivations which provide different explanations of learning and human behaviour. Some of the important definitions are given here for having an adequate understanding of the term motivation.

- ❖ The term motivation has been derived from a Latin word 'movere', which means 'to move' or 'to activate'
- ❖ There are various definitions of motivation and some of them have been mentioned as follows:

- Feldman (2015, pg. 287) defined motivation as “the factors that direct and energise the behaviour of humans and other organisms”.
- Feist and Rosenberg (2015, pg. 397) define motivation as “the urge(a strong desire or impulse) to move towards one’s goals, to accomplish tasks”. (achieve or complete successfully).
- Quick, Nelson and Khandelwal (2013, pg. 172) defined motivation as “the process of arousing (evoke or awaken(a feeling ,emotion or response)and sustaining goal directed behaviour”.(strengthen or support physically or mentally)
- Nolen-H oeksema, Fredrickson, Loftus & Lutz (2009, pg 419) describe motivation as “a condition that energizes behaviour and gives it direction”.

MOTIVATION: CONCEPT & NATURE

- Motivation is the need within the individual to do something or to fulfill your desire. There are certain internal and external factors that are responsible for stimulating a desire and providing us with the energy to pursue our goals, needs and desires.
- Motivation involves three aspects that need to be understood. First, it talks about a need state in the organism that drives him/her towards the satisfaction of the bodily need. This need could be due to internal thoughts and feelings or could be due to external factors in the environment. Secondly, this need then induces (give rice to or stimulate) a state of arousal (the action) or an action tendency(type of behaviour).
- Thirdly, the action is directed towards attaining some goal.
- There is a motivational cycle that helps us to understand behaviour (Morgan & King, 1979).This motivational cycle starts with a need, a state of lack or deficit of some necessity. The state of need leads to a driving state. Drives can be stimulated due to internal factors or external factors. Internal factors could be thoughts and memories while external factors could be the environmental factors. This driving state arouses you and pushes you to perform some goal-directed behaviour.



- For example, if we are thirsty there is a need for water, this need will drive us towards attaining water and once we have consumed(drink) water, the need is satisfied and the drive is reduced or diminished(made a smaller or less or reduced). So, once the performed behaviour leads to the achievement of desired goal, it reduces the drive and organism returns to a balanced state. The motivational cycle ends here. Then it arises again after a while when the urge(a strong desire or impulse) for water (or any other need) arises again.
- Before describing further, let us discuss the four terms that are related to motivation. These four terms are needs, drives, incentives, and motives.
- W A Kelly (1955) "Motivation is the central factor in the effective management of the process of learning. Some type of motivation must be present in all learning."
- L D Crow and A Crow (1962) "Motivation is considered with the arousal of the interest in learning and to that extent is basic to learning."
- K Lovell (1964) "Motivation in school learning involves arousing, persisting, sustaining and directing desirable behaviour."
- H W Bernard (1965) "Motivation is the stimulation of actions towards a particular objective where previously there was little or no attraction to that goal."
- T W Atkinson (1966) "The term motivation refers to the arousal of tendency to act to produce one or more effects."
- F G McDonald (1972) "Motivation is an energy change within the person characterized by affective arousal and anticipatory goal relations."

- C W Good (1973) “Motivation is the process of arousing, sustaining and regulating activity.”
- D O Hebb (1975) “The term motivation refers (i) to existence of an organized phase sequence (ii) to its direction and content (iii) to its persistence in given direction or stability of content.”
- Bernard (1980) “Motivation is the stimulation of actions towards a particular objective where previously there was little or no attraction to that goal.”
- Arun Monappa and Mirza S Saiyadain (1985) “Motivation is propensity or the level of desire of an individual to behave in a certain manner at a certain time and in a certain situation.”

Needs: General wants or desires are called needs. Every man **strives** for the satisfaction of his basic needs. (Air,water,food,shelter,safety,sleep).These are related to the biological states of cellular or bodily deficiencies that lead to drives. For example, individuals need water, food and oxygen to survive (Feist and Rosenberg, 2015). Needs may be of two types:[a] Organic and [b] Psycho sociological.

- **Organic needs :** Need for food , water and oxygen; need for rest and sleep. Need for satisfaction of sex urge. Need for protection from natural calamities(a disaster) and wild animals etc.
- **Socio-psychological needs :** Need for freedom,Need for social and economic security, Need for fame(honour) and reputation etc.

DRIVE : Drive is internal stimulus which moves the body. According to Dashiell, drive is an original source of energy-that activates the organism.

- Drive means aroused (evoke or awaken a feeling, emotion or response) awareness. A need gives rise to drive. Drives are of two types:

  [a] Biological drive

  [b] Psycho- sociological drive .
- **Biological drive includes :** hunger, thirst, sex, etc.
- **Psycho-sociological drive includes:** desire for social approval,desire for independence, etc.

- These drives move an organism to act in a particular way.

Incentives :

- An incentive acts as a reinforcing agent. It adds force to a drive. Rewards, praise, appreciation, achievement of the desired goal are a few instances of incentives, **e.g., a piece of bread may act as an incentive for a hungry boy.** There is some distinction between incentives and motives. Incentive refers to the objective appeal, whereas motive to the subjective striving. This is external or is from the environment (as opposed to drive that is internal) and **plays a role in motivating behaviours. It could be an object or an event.** For instance, a trophy won in a game can be termed as an incentive to do well in that game. Reinforcement: A kind of stimulus that aims to increase the strengthens or increases the behavior.

Motives:

- Most often the terms motivation and motives are used interchangeably. However, they are different as motivation can be termed as a general term, whereas motive is a specific term. Motive is something that actually drives an individual to take certain action and motivation is the process within which a motive drives an individual towards certain action.

Extrinsic motivation:

- Extrinsic motivation refers when motivation is external or coming from outside. A person performs an action because it leads to an outcome that is separate from the person (Ryan and Deci, 2000).
- E.g.- giving a child money for every A+ he gets.
- You are motivated to do the activity in order to gain an external reward in return.
- Your goal is focused on an outcome, and does not satisfy your basic psychological needs. Rather, it involves external gains, such as money, fame, power, and avoiding consequences.
- Competing(poti) in sports for trophies.

- completing work for money.
- buy one, get one free sales.

Intrinsic motivation

- Intrinsic motivation refers to motivation that is driven by an interest or enjoyment in the task itself, and exists within the individual rather than relying on external pressures or a desire for reward. These behaviours are interesting and self satisfying. E.g.- playing guitar even when no one is listening.
- You are motivated to do the activity because it is internally rewarding. You choose to do it because it's fun, enjoyable, and satisfying.
- Your goal comes from within, and the outcomes of your goal satisfy your basic psychological needs. for autonomy, competence and relatedness.

Intrinsic	Extrinsic
Participating in a sport because it's fun and you enjoy it.	**Participating in a sport in order to win a reward or get physically fit.**
Learning a new language because you like experiencing new things.	**Learning a new language because your job requires it.**
Cleaning because you enjoy a tidy space.	**Cleaning to avoid making your partner angry.**
Playing cards because you enjoy the challenge.	**Playing cards to win money.**
Exercising because you enjoy physically challenging your body.	**Exercising because you want to lose weight or fit into an outfit.**

Classification:

- There are numerous motives based on the basic needs. For better understanding of the nature and role of these motives we may classify them into two categories:

[a] Primary motives and

[b] Secondary motives.

Primary motives

- Most important of these are: [a] Biological motives, [b] Social motives and [c] personal motives.

Biological motives

- Biological motives originate from organic needs, such as hunger, thirst and sex. want of food gives rise to hunger drive and the individual searches for food thirst motive comes from the sensation of dryness in the mouth and throat.

- **Organic needs are:** Need for food , water and oxygen; need for rest and sleep. Need for satisfaction of sex urge. Need for protection from natural calamities (a disaster) and wild animals etc.

- The source of all kinds of behavior according to Freud is sex motive. Of Course, sex motive in human organism is mainly controlled by social influence. Although not so indispensable like food, air and water,sex motive is a highly powerful factor. It is a complex blend of psycho physiological factors.

- Hunger Motivation When there is a lack of food in the stomach, it causes hunger pangs(a sudden sharp pain or painful emotion). The stomach contracts(decrease in size) and initiates a hunger drive. There are many reasons for hunger. Many studies have shown that after fasting, when the person uses too much of blood sugar or glucose, it causes hunger.

- Friedman and Stricker (1976) suggested that when there is a change in the metabolic functions of the liver, and food is less, then it initiates (cause a process or action to begin) hunger. There are two areas of the hypothalamus that seem to be involved in hunger. Research has shown that when lateral hypothalamus is stimulated, it initiates eating behavior. On the other hand, if this area is damaged, then it causes the animal to stop eating (Epstein, 1960). Ventro-medial area of the hypothalamus is also involved in hunger motivation. It is known to be the inhibitory (preventing an action) area for hunger drive.

- Epstein (1960) showed in his experiments that when the ventro-medial hypothalamus is damaged, then the animal does not know when to stop eating and they become over weight.

- Thirst Motivation It is believed that thirst and drinking are regulated by the body internally. When the mouth is dry, then only one feels thirsty. The Antidiuretic Hormone (ADH) plays an important role in regulating the water level in the body. ADH prevents the loss of water from the kidney. It has been suggested that the thirst drive and drinking behavior are instigated by the loss of water from the cells or the decrease in the volume of the blood. When there is a reduction in the level of water in the body fluids, then the water moves out from the cells into the fluids. This dehydrates the interior part of the cells. The osmo receptors (nerve cells which generate nerve impulses when dehydrated) generate nerve impulses when they are dehydrated. These nerve impulses are mere signals that initiate drinking behavior.

- When thirst is caused by the loss of water from the osmorecpetors, it is known as cellular-dehydration thirst.

- The loss of water from the body also causes a condition known as hypovolemia. In this condition, there is a reduction in the blood volume which causes a decrease in the blood pressure. This reduction in blood pressure causes the kidneys to release renin, an enzyme that helps in the formation of angio tension II. Angio tensin II is a substance that moves about in the blood and tends to stimulate drinking behavior.

- Double-depletion hypothesis explains when both the conditions of hypovolemia and cellular dehydration are at work. Drinking often stops when the mouth and stomach signal that no more water intake is required. Also, when the osmoreceptors are hydrated, the blood plasma level reaches its normal volume.

Sex Motivation

- Sexual drive depends upon the physiological condition of the body. It is also considered as a biological drive. There are various ways in which this drive varies from hunger and thirst drive. The sexual drive does not respond to any deficiency in the body or lack of some substance in the body. Thus, it is not initiated by any homeostatic imbalance or presence of any excessive hormones in the body. In males, the level of the male hormone testosterone is responsible for sexual drive. In females, the female hormone, estrogen is responsible for the sexual behavior. Hypothalamus regulates the hormone release.

- Hunger and thirst drives are necessary for survival but sexual drives are not necessary for survival, though it is important for the species survival. There are various external stimuli that trigger sexual drives. This could be like the odors, smell or sight of the attractive

partner may stimulate this drive. Though, the family upbringing and social norms may also play an important role in the expression of sexual behavior.

Social motives

- The social motives develops from social-interaction. Social motives include gregariousness, acquisitiveness,(excessive interest in acquiring money or material things) aggression,(violent behavior) self-assertion and tendency to seek social approval.
- All animals have strong motivation to live in company of others. When an individual is separated from the group, he feels a strong urge to go back to his group.
- Acquisitive motive is also very strong in human beings. People earn money and love to deposit it in the banks.
- Aggression motive is expressed in violent activities intended to inflict to be suffered by someone or something physical or psychological harm on others. To some, aggression is an essential reaction to frustration.
- Self-assertion another powerful social motive. Showing of excessive desirefor attention, status and prestige can be explained in terms of self-assertion. The impulse to make ourselves the master of the environment is dominant in all of us.
- The desire to seek social approval and earn prestige is one of the important strong social motives. Craving for recognition, fame, power and leadership is the result of achievement or status motive.

Personal motives

- Esteem motive, self-actualisation motive, achievement motive, etc. are included in personal or ego motive. Ego motive drives an individual to strive to gain mastery over challenging situations.
- Every individual desires to unfold his skills and capacities for earning fame and prestige from the members of the society. This egomotive becomes very strong in adolescence.

Secondary motives

- Motives are modified as a result of learning and experience. Such modified motives are attached to different goals or stimuli. These stimuli become learned motives of secondary motives. They are merely reinforcing conditions of motives.

- Punishment, rewards, praise, knowledge of results, interest, etc. work as incentives. Rewards and punishment play a very great role in the motivation of school work.
- Reward is a positive motive. It takes the form of praise, promotion, badges, honorary degree etc. While punishment is a negative motive and based on fear.

Lesson 14 : Theories of Motivation- Instinct theories – Freud & Mc Dougal; Need hierarchy theory – Maslow

McDougall's theory of instinct

- Instincts are the main springs of human action. McDougall defines instincts as certain specific innate modes of behaviour in certain specific situations. which do not have to be learnt and which are common to all members of species.
- Instincts are goal-directed and innate patterns of behavior that are not the result of learning or experience. For example,
- Birds have an innate need to build a nest or migrate during the winter.

In animals example may be

- A dog shaking after it gets wet,
- A sea turtle seeking out the ocean after hatching.

In humans

- The rooting reflex as is the suckling reflex (a reflex in which babies begin sucking when a finger or nipple places pressure on the roof of their mouth,).an action that is performed as aresponse to a stimulus and without conscious thought).
- All of these behaviors occur naturally and automatically. They do not need to be learned in order to be displayed.
- Every living organism is born with Horme, an inner urge to act. This urge to act is not random. It is goal –oriented.

- According to McDougall, all human behaviour can be explained on the basis of instincts.
- McDougall gives a list of 14 important instincts which are appeal, acquisition, Curiosity, Combat, Construction, Sex, Submission, Self-assertion, parental, escape, repulsion, laughter, hunger and gregarious instinct.
- He proposed that each instinct is accompanied by a specific emotional disposition. The instinct off escape, for example, at the sight of a dangerous object, has the emotion of fear in its core.
- Observations About Instinct Theory
- The instinct theory suggests that motivation is primarily biologically based.
- We engage in certain behaviors because they aid in survival.
- Migrating before winter ensures the survival of the flock, so the behavior has become instinctive.
- Birds who migrated were more likely to survive and therefore more likely to pass down their genes to future generations.
- David G. Meyers suggests that in order to be identified as an instinct, the behavior "must have a fixed pattern throughout a species and be unlearned."
- In other words, the behavior must occur naturally and automatically in all organisms of that species.
- For example, infants have an innate rooting reflex that leads them to root for and suck on a nipple.
- This behavior is unlearned and occurs naturally in all human infants.
- McDougall gives a list of 14 important instincts which are appeal, acquisition, Curiosity, Combat, Construction, Sex, Submission, Self-assertion, parental, escape, repulsion, laughter, hunger and gregarious instinct. He proposed that each instinct is accompanied by a specific emotional disposition. The instinct off escape, for example, at the sight of a dangerous object, has the emotion of fear in its core.
- The theory of instincts is not free from criticism. Psychologists have failed to find a limited number of instincts to account for all human behaviour. Again human behaviour cannot be termed innate because most human activities are learned and acquired. Human

behaviour is shaped by the forces of his social and cultural environment. Moreover, instincts are not blind. They are modified by experience, habit and the new environment.

Maslow's Self- actualization theory

- Abraham Maslow[1954] Advocated the hierarchy of five sets of basic needs. The highest need in the hierarchical order is self- actualisation and the lowest need in the system is physiological needs, Such as hunger, thirst etc. A n individual can be motivated towards self- ACTUALISATION ONLY WHEN HIS LOWER ORDER NEEDS SUCH AS PHYSIOLOGICAL NEEDS, Safety needs etc. have been fulfilled. A hungry boy can be motivated to solve a problem in arithmetic only when his hunger has been satisfied. He cannot be motivated to concentrate his attention on study when hungry.
- Abraham Maslow developed the theory of hierarchy of needs, which suggests that individual needs exist in a hierarchy consisting of physiological needs, security needs, belongingness needs, esteem needs, and self-actualisation needs.
- Maslow's hierarchy of needs is most often displayed as a pyramid. The lowest levels of the pyramid are made up of the most basic needs, while the more complex needs are located at the top of the pyramid. Needs at the bottom of the pyramid are basic physical requirements including the need for food, water, sleep and warmth.
- Once these lower level needs have been met, people can move on to the next level of needs, which are for safety and security.
- As people progress up the pyramid, needs become increasingly psychological and social. Soon, the need for love, friendship and intimacy become important. Further up the pyramid, the need for personal esteem and feelings of accomplishment take priority.
- Like Carl Rogers, Maslow emphasised the importance of self-actualisation, which is a process of growing and developing as a person to achieve individual potential.

The Hierarchical Model

- This model was proposed by Maslow, often termed as Maslow's hierarchy of needs. In this model, the needs are placed in a hierarchical order. The model then states that the basic needs are to be fulfilled before the higher order needs are met. Maslow's hierarchy of needs can be explained with the help of a pyramid with basic needs at the bottom of the pyramid and the higher order needs at the top of the pyramid.

Types of Needs

- Maslow believed that these needs are similar to instincts and play a major role in motivating behaviour. Physiological, security, social, and esteem needs are deficiency needs (also known as D-needs), meaning that these needs arise due to deprivation. Satisfying these lower-level needs is important in order to avoid unpleasant feelings or consequences. Maslow termed the highest-level of the pyramid as growth needs (also known as being needs or B-needs). Growth needs do not stem from a lack of something, but rather from a desire to grow as a person.

Five Levels of the Hierarchy of Needs

- There are five different levels in Maslow's hierarchy of needs:

- i) **Physiological Needs:** These include the most basic needs that are vital to survival, such as the need for water, air, food and sleep. Maslow believed that these needs are the most basic and instinctive needs in the hierarchy because all needs become secondary until these physiological needs are met.

- **Meeting a Student's Cognitive Needs**

- Student's cognitive needs can only be met by first fulfilling their basic physiological needs. For example, it will be difficult for a tired and hungry student to focus on learning.

- **Earning For a Living**

- Money is, in fact, the most significant element to satisfy physiological needs. Except for air (oxygen), all the other elements of physiological needs such as food, clothing, and shelter are money-dependent.

- A hungry boy can be motivated to solve a problem in arithmetic only when his hunger has been satisfied. He cannot be motivated to concentrate his attention on study when hungry.

- **Security Needs:** These include needs for safety and security. Security needs are important for survival, but they are not as demanding as the physiological needs.

- Examples of security needs include a desire for steady employment, health insurance, safe neighbourhoods and shelter from the environment.

Insurance Policies

- Today, insurance is considered an essential requirement for physical safety due to the uncertainty of the future and our lives. Different types of insurance facilities like life insurance, health insurance, vehicle insurance, disaster insurance, etc., are daily-life examples of physical safety needs as per Maslow's need hierarchy. A health insurance policy ensures safety against any health issue that may occur in the future. Similarly, life insurance comes handy in case of the sudden death of a person. Besides health and life insurance, there are many more insurance types, such as general insurance, which covers properties, valuables, vehicles, etc. These insurance plans boost mental health so that one can concentrate on other important things.

- **Job Security**

- Job security is also an example of the security needs of Maslow's hierarchy. Everyone wants a safe and secure job. Employees won't or will not be able to work with their full potential if they feel unsafe for their jobs.

A Healthy & Safe Work Environment

- A safe workplace or work environment is related to security and safety in office premises in terms of fire breakdowns, safety instruments or safety measures in factories, security systems like CCTV, emergency exits, etc. Bio-metric scans ensure the entry of employees in the office premises and not strangers.

- **Various Financial Schemes**

- Banks and other financial institutions offer different products and services to provide financial security and stability to their customers. Apart from saving accounts, many more investment and saving schemes like fixed deposits, recurring deposits, mutual funds, and insurance plans are there to boost financial safety and encourage savings for future financial security.

Social Needs (Belonging and love needs): These include needs for belonging, love and affection. Maslow considered these needs to be less basic than physiological and security needs. Relationships such as friendships, romantic attachments and families help fulfill this need for companionship and acceptance, as does involvement in social, community or religious groups.

- **Examples of Belongingness and Love Needs**

- Let's have some real-life examples of Belongingness and Love Needs.

Family get-together

- The family get-together plays an important role in satisfying emotional or belongingness and affection needs of us. On different occasions like birthday parties, marriages, family lunch or dinner, family tours, any misfortunes, etc. family members assemble at one place to share sorrows and happiness with each other. It helps in fulfilling the love and belongingness needs of individuals.

Attractive tour packages by travel agencies

- Nowadays different travel companies or agencies like Make my trip, Cox and Kings, Thomas Cook, Club Mahindra, Goibibo, Yatra.com, etc., offer very attractive tour packages both National and International for married couples, families, group of friends, and corporate employees. These provide discounts on overall tour package, including tickets, hotel stay, etc., so that a large number of families or couples can avail such tour packages and can spend valuable time with each other from their busy life. By doing so, these companies contribute to the love and belongingness needs of humans.

iv) **Self-Esteem Needs:** After the first three needs have been satisfied, esteem needs becomes increasingly important. These include the need for things that reflect on self-esteem, personal worth, social recognition and accomplishment.

Choice of Profession examples

- People choose different professions to gain recognition and fame in their social circle. Different professions like Doctor, Engineer, Professor, Lawyer, Management professionals, etc., that give them a sense of pride, honor, appreciation, and recognition from others.

Social Work or Social Service

- People do social services or social work by contributing as a volunteer in NGOs who help poor and needy people of the society by providing means to live life, like food, shelter, clothes, education, medical assistance, care, etc.

- It satisfies their esteem needs of gaining respect from other people of society and recognition in the society.

v) **Self-actualising Needs:** This is the highest level of Maslow's hierarchy of needs. Self-actualising people are self-aware, concerned with personal growth, less concerned with the opinions of others and interested fulfilling their potential.

- **Artists and Creative People**

Art is considered as a great platform for Self-actualization as it increases the psychological wellness of people. Both artists and audiences interpret art that ultimately moves toward self-realization, which is a form of Self-actualization, like painters are supposed to paint, writers to write, and musicians to play or compose music.

- **Growth and Development of Employees**
  Employees having Self-actualization needs are more concerned about their growth and development in the workplace. They look for more challenging assignments in their work, more independent work with less supervision, leadership roles, participation in decision-making in key issues, an opportunity to enhance their skills and education, etc. So, to meet the Self-actualization needs of employees, the managers promote an environment where employees can get the desired opportunities.

PSYCHOANALYTIC THEORY

- **MAY 6, 1856** – Sigmund Freud was Born in Freyberg Town, Crèche Republic.
- **1881** – He graduated from medical faculty, university of Vienna.
- **1900** – He released 'interpretation of dreams'
- **SEPTEMBER 23, 1939** –Freud passed away in Hampstead house.

STRUCTURE OF PERSONALITY

- Human mind can be divided in to three parts, Id, Ego, Superego.

Id:

- Operates on pleasure principe – to gain pleasure, avoid pain
- Id is conceived as a reservoir of instinctual energy that contains biological urges such as impulses toward survival, sex, and aggression.
- The id is unconscious and operates according to the pleasure principle, the drive to achieve pleasure and avoid pain.
- Driven(operated moved or controlled) by sexual and aggressive urge.

- **pleasure principle** is the driving force of the id that seeks immediate gratification of all needs, wants, and urges. In other words, the **pleasure principle** strives to fulfill our most basic and primitive urges, including hunger, thirst, anger, and sex.
- An **example of pleasure** is enjoying reading books. **Pleasure** means the state of being pleased, delighted or gratified.

Ego:

- Operates on REALITY PRINCIPLES – does realistic and logical thinking.
- Ego is considered as the component that manages the conflict between the id and the constraints of the real world.
- Some parts of the ego are unconscious, while others are preconscious or conscious.
- Source of decision making and rational thought.
- The balance between Id and Superego.
- The ego operates according to the reality principle, the awareness that gratification of impulses has to be delayed in order to accommodate the demands of the real world.
- The ego is characterised by secondary process thinking, which is logical and rational. The ego's role is to prevent the id from gratifying its impulses in socially inappropriate ways.

Superego:

- This is considered to be the moral component of personality.
- It contains all the moral standards learned from parents and society.
- Able to differentiate between good and bad, right and wrong.
- If people follow their superego, they will feel proud but if they don't follow, they will feel guilty and anxious.
- Partially unconscious.
- Operates on MORAL PRINCIPLES
- The superego forces the ego to conform not only to reality but also to its ideals of morality.

- Hence, the superego causes people to feel guilty when they go against society's rules. Like the ego, the superego operates at all three levels of awareness.

LEVEL OF MENTAL LIFE

- Freud believed that most mental processes are unconscious. He proposed that people have three levels of awareness: The conscious the Preconscious, and the unconscious
- **Mental Processes**: an individual's thoughts and feelings that are personal and cannot be directly observed.

THE CONSCIOUS MIND

- It includes everything that we are aware of. This is the aspect of our mental processing that we can think and talk about rationally.
- Events & experiences that are easily remembered or retrieved are considered to be within one's conscious awareness. Example:-Telephone numbers, birthday of self & dates of special holidays.
- The part of the mind we are aware of everyday thoughts and feelings.

THE PRECONSCIOUS MIND

- It includes all memories that may have been forgotten or are not in present awareness but with attention can be readily recalled into consciousness.
- It is thought to be partially under the control of the super-ego, which helps to suppress unacceptable thoughts and behaviors.
- Thoughts and memories not accessible at all the times but easily Recalled.
- The preconscious contains all the information outside of a person's attention but readily available if needed. Example: A close friend's telephone number, the make of one's car, and many of the past experiences are in the preconscious.
- Facts stored in a part of the brain, which are not conscious but are available for possible use in the future (E.g. A person will never think of her home address at that moment but when her friend ask for it, she can easily recall it).

THE UNCONSCIOUS MIND

- It consists of unpleasant & non essential memories that have been repressed & can be retrieved through therapy, hypnosis & with other substances that alter awareness.
- Also contain painful memories, experiences, emotions, guilt feelings and unresolved conflicts which we deliberately suppress.
- The deep dark shameful part. Repressed thoughts, memories and feelings.
- The unconscious contains thoughts, feelings, desires, and memories of which people have no awareness but that which influence every aspect of the individual's day to-day lives.
- Freud believed that information in the unconscious tries to come into the conscious and very often it is seen in slips of the tongue, jokes, dreams, illness symptoms, and the associations people make between ideas.

Life instincts(Eros)

- Sometimes referred to as sexual **instincts**, the **life instincts** are those which deal with basic survival, pleasure, and reproduction.
- These **instincts** are essential for sustaining the **life** of the individual as well as the continuation of the species.
- While they are often called sexual instincts, these drives also include such things as thirst, hunger, and pain avoidance.
- The energy created by the life instincts is known as libido (sexual desire).
- Behaviors commonly associated with the life instinct include love , cooperation, and other pro social actions.
- **Prosocial** behaviors are those intended to help other people. These **actions** are characterized by a concern for the rights, feelings, and welfare of other people. **three different types of prosocial** action: sharing, helping, and comforting.

- Freudian psychology, Eros, also referred to in terms of libido, libidinal energy or love, is the life instinct innate in all humans. It is the desire to create life and favours productivity and construction. Eros battles against Thanatos, what Freud called the destructive death instinct.
- Freud proposed that Eros was opposed by forces of the ego (the organized, realistic part of a person's psyche which mediates between desires).
- In this later view, he maintained that life instincts were opposed by the self-destructive death instincts, known as Thanatos.
- Positive emotions such as love, affection, pro social actions, and social cooperation are also associated with the life instincts. These behaviors support both individual well-being and the harmonious existence of a cooperative and healthy society.

Death (Thanatos)

- The concept of the death instincts was initially described in *Beyond the Pleasure Principle,* in which Freud proposed that "the goal of all life is death." Freud believed that people typically channel their death instincts outwards.[2]
- Aggression,(violent behavior) for example, arises from the death instincts.
- Sometimes these instincts towards destruction can be directed inwards, however, which can result in self-harm or suicide.
- Freud proposed that "the goal of all life is death" (1920).
- He noted that after people experience a traumatic event (such as war), they often reenact the experience.
- He concluded that people hold an unconscious desire to die, but that this wish is largely tempered by the life instincts.
- In Freud's view, self-destructive behavior is an expression of the energy created by the death instincts.
- When this energy is directed outward onto others, it is expressed as aggression and violence.

Lesson 15 : Emotion: Nature, kinds of emotions- physiological basis of emotion.

Emotions

- All of us at some point of time or the other in our everyday life feel happy, sad, angry, love, pity, jealousy, hatred, fear, agony etc. What are these feelings which we experience in our lives? What makes us feel these? Imagine our lives without these feelings or experiences? These are actually emotions. Our life would have been absolutely boring and meaningless without emotions. Our emotions and how we feel about certain things most often influences our thinking.

- Even our perception is not always determined by the outside stimulus but by our internal feelings, emotions, desires, and aversions. Thus feelings and emotions are dynamics of our behaviour, and thus are very important. In our practical life, feelings and outside behaviour are closely related. It is therefore very essential to know the difference between emotions and feelings which are highly inter related and both together and individually affect our behaviours. Emotions and feelings often appear just as different stages of one phenomenon.

- Feelings is a term which is always used for various kinds of experiences. This term is generally used when we are not sure of the answer to a particular question. For instance, when we talk about feelings of pleasure, familiarity, we say I feel fine. These feelings can be put into various categories. Feelings like displeasure, grief, discontent, sadness etc. can be categorised under one category characterised by a tone of unpleasantness and avoidance.

- Similarly feelings like pleasure, happiness, joy, delight etc., will come under one category characterised by desirable pleasant feeling tone. In this manner we can enumerate many more experiences of feelings coming under various categories. Having known a little about what feelings are, now let us look at the dimensions of feelings.

- Try to observe the people around you especially their emotions. Are you able to tell when they are happy or when they are sad or angry& most often we can say whether people are

happy, angry, sad or even surprised based on their facial expressions or even vocal(relating to the human voice) expressions. As a student of psychology, it is important that we discuss the concept of emotion as that may help us understand people in a better way. Motivation and emotion can be termed as two sides of a same coin (Mishra, 2016) and they both play a role in directing(manage or control the operations) behaviours of the individuals.

- Both motivation and emotions are activated when issues are related to the wellbeing of an individual, survival and appetite (Feist and Rosenberg, 2015). For example, an individual may be driven to get involved in activities that are linked to his/ her wellbeing. When the individual is able to achieve certain goals, she/he will experience happiness and if she/he fails, sadness will be experienced.

Definition of Emotion

- Emotion has been derived from a Latin term 'emovere' that means 'stirred-up state'.
- There are various definitions of emotion that are discussed as follows:
- Feist and Rosenberg (2015, pg. 418) defined emotions as "brief, acute changes in consciousness experience and physiology that occur in response to a personally meaningful situation".
- As stated by Gerrig and Zimbardo (2006, pg. 418) emotions are "a complex pattern of bodily and mental changes that includes physiological arousal, feelings, cognitive processes, visible expressions (including face and posture) and specific behavioural reactions made in response to a situation perceived as personally significant".
- Feldman (2015, pg. 312) defined emotion as "feelings that generally have both physiological and cognitive elements and that influence behaviour".
- Mishra (2016, pg. 466) defined emotion as "a state of being moved, stirred up(emotionally arousedor awaked) or behaviourally aroused on experiencing an emotional situation and which involves external and internal physiological changes".
- The physiological changes normally include changes in gestures, muscular movements, facial expression, changes in physiological reactions like blood-pressure, pulse rate, heartbeat, respiration, etc. and psychological changes include blurring(make or become

unclear) or clouding(make or become less clear or transparent) of consciousness, blocking of memory, confusion in perception etc.

- **Facial expressions:** Various muscles of the face including the eye, the lips, the nose and the forehead undergo changes by way of twisting and taking different poses. Most of the emotions have their facial expressions but it may not always be easy to judge the emotion from these expressions. Some facial expressions are universal.
- Internal changes:
- **Change in a Heartbeat:**
- **Change in Blood Pressure:**
- **Change in Rate of Respiration:**
- **Change in Brain Waves:** Brain waves also change during an emotional state. By measuring with the device, it was found that the frequency of brain waves changes in the state of momentum.

- **Muscular adjustment:** Sometimes the entire muscular system of the body undergoes contraction and adjustments. For example, the bodily changes that take place in "startle reaction" when there is quick, unexpected and strong stimulus. The startle response consists of sudden movement of the head, blinking of the eyes, a characteristic facial expression, raising and drawing forward of the shoulders, turning inward of the upper arms, bending of the elbows, turning downward of the forearms, flexion of the fingers, forward movement of the trunk, contraction of the abdomen and bending of the knees.(mokalu)
- Some have questioned considering 'startle reaction' as an emotion as they consider startle reaction as one of the basic reflexes. There is some truth in this claim as the same type of bodily changes also appears in cases of fear response.
- **Gestures and movement of the hands:** Gestures may sometimes involve the movements of the whole body.
- Gestures include movement of the hands, face, or other parts of the body. **Gestures** differ from physical non-verbal communication that does not communicate specific messages, such as purely expressive displays, proxemics, or displays of joint attention.

- **Change in voice:** The voice gets affected in an emotion. When excited the voice becomes loud and high pitched. Under fear the voice takes a sinking(go down) and low tone.
- One of the key points that can be highlighted from the above definitions is that there is a change. This change can be in the conscious experience or could be in terms of physiological arousal and even in cognitive processes.
- Changes can also be in the visible expression that is displayed on the face or in the posture of an individual. This change is as a result of some situation that is important for the individual.

EMOTIONS: NATURE

- Emotions have a great role to play in our lives. It involves feelings of pleasantness and unpleasantness. It can be expressed in various ways, like facial expressions on the face, like gestures, like volume and tone of speech, by the behavioral display or motives of a person.
- There is no moment in our life when we do not experience emotions. They can result from a variety of external or internal stimulations.
- Various events or situations in the environment may produce a variety of emotions as jealousy, guilt and shame. These emotions are more self-conscious in nature as they involve the component of morality.
- **External stimuli** are changes outside the body, or knowledge that is passed to us through our senses. This could be cold or hot weather, light levels, or danger. **Internal stimuli** are changes to conditions inside the body, such as dangerous food in the stomach, viruses in the body, or lack of food or water.
- Ekman (1980) in his seminal work on facial expressions proposed that there are six basic emotions that are universally recognized across cultures. These are fear, anger, surprise, sadness, happiness, and disgust. Though, now ‘contempt’ has also been added.
- Emotions are complex affective states from perception, memory or imagination of a situation. For example, when a mad dog chases us we run away out of fear. Emotions may arise from ideas and thoughts also.
- Emotions are tinged with feelings. Both of them are affective experiences. Some emotions [e.g., joy, elation are pleasant. Some other emotions [e.g., fear, sorrow] are

unpleasant. The core of an emotional experience is feeling. When feelings are violent and the composure of our mind is disturbed we have no longer feelings but emotions.

- McDougall holds that an emotion is a conscious correlate of an instinctive action. He gives the following pairs of instincts and emotions correlated to each other:
- McDougall gives a list of 14 important instincts which are appeal, acquisition, Curiosity, Combat, Construction, Sex, Submission, Self-assertion, parental, escape, repulsion, laughter, hunger and gregarious instinct.

Instinct	Emotion
escape	Fear
Combat	Anger
repulsion	Disgust
Curiosity	wonder
Sex	Lust
parental	Tender emotion
Submission	Inferiority
Self-assertion	uperiority
gregarious	Loneliness
acquisition	Ownership
Construction	Creativeness
laughter	Amusement

Kinds of emotions

- According to Watson, basic emotions are: fear, anger and love. A child gets afraid on hearing a loud sound. He becomes angry when his movement is restricted. On getting a toy he expresses his joy.
- Wundt has divided emotion into three dimensions: pleasant-unpleasant,excitement-calmness and expectation- release. Different kinds of emotions are the product of these three emotions.
- McDougall classifies emotions into three kinds:

1.Primary emotions
2.Secondary emotions
3. Derived emotions.

Primary emotions

- Primary emotions are the elementary correlates of instincts. Fear, anger, disgust, distress,lust(very strong sexual desire), curiosity,(a strong desire to know something) etc., which arise from the instincts of escape, combat, repulsion, appeal, sex, wonder ,etc. are primary emotions.

Secondary emotions

- Secondary emotions are the combinations of two or more primary emotions. For example, scornis a blend of anger and disgust. Similarly jealousy, revenge, gratitude, etc. are blended emotions.

Derived emotions

- Derived emotions are the complex feelings. They are neither primary emotions nor blended emotions. They are related to desires, confidence, hope, anxiety, regret,despairs are derived emotions.

Concept of Emotions

- Emotions expressed by humans can be divided into two broad categories and are called as positive and negative emotions.

- Examples of positive emotions include the following: Interest, enthusiasm(intense and eager enjoyment, interest, or approval.), boredom,(the state of feeling bored.) laughter, empathy,(the ability to understand and share the feelings of another.) action, curiosity.(a strong desire to know or learn something).
- An attempt or intention to exclude is expressed by Negative emotions. Apathy, grief,(intense sorrow, especially caused by someone's death), fear, hatred**,** shame, blame, regret, anger, hostility are examples of negative emotions.
- Emotions are accompanied by activation or an aroused state in the organism. They are normally accompanied by both physiological and psychological changes.
- The physiological changes normally include changes in gestures, muscular movements, facial expression, changes in physiological reactions like blood-pressure, pulse rate, heartbeat, respiration, etc. and psychological changes include blurring or clouding of consciousness, blocking of memory, confusion in perception etc.
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- **TYPES OF EMOTIONS**
- There are various kinds of emotions that we experience and express. There are different sources of these emotional feelings which are experienced for a relatively longer period of time than moods that are more transient in nature.
- **Happiness:** When we are able to achieve something that makes us feel pleasurable and elated. It is a feeling of joy that is experienced when we are content, gratified and satisfied with our life. It contributes to our state of mental health and well-being. It is often expressed in smiles or a positive behaviour as laughing, and being more active and energetic.
- **Fear:** Any situation that induces a feeling of terror or threat to one's wellbeing causes fear. Fear plays an important role in survival as it motivates the person to protect him or herself. There is an increase in heart rate, respiration rate and the body muscles become tense. The person is alert and tries to fight with the fearful situation or take flight from the fearful situation. The fearful situation is perceived as relatively less threatening if one

is exposed to it again and again. Thus, the person is systematically desensitized to the fearful situation.

- **Sadness:** Sadness is another emotion that is characterized by feelings of discontentment, grief, and unhappiness. The person may feel frustrated due to certain failures in life or personal loss and a condition of apathy may occur. The person feels dull, low, disinterested in things around him or her. The person feels hopeless and disappointed. If this continues for a longer period of time then it may also lead to depression. It may be displayed as lowered lips, crying or sulking behavior with drooping shoulders.
- **Anger:** According to McDougall the emotion of anger is the affective aspect of the combat instinct. A child becomes angry if his toy is taken away. A man becomes angry if he is insulted.
- When we are unable to reach our goals, it causes frustration that provokes anger.
- In anger our body assumes an aggressive attitude. It is expressed in a violent motor discharge. It is expressed in grinding the teeth, kicking, striking , howling,(producing a long loud, doleful cry or wailing, etc.
- Men show more anger as compared to women. There are various techniques of therapy that help the person to modulate one's anger, control it and reduce the intensity with which it is experienced or expressed.
- **Surprise:** Surprise is a very brief emotion that is generally very intense. It is a response to a stimulus that may be unexpected, may be pleasant or unpleasant and it leads to a state of startle. The person may display it by raised eye brows, or dilated pupils, loud scream or gasp, jumping behavior or opening of the mouth. The events that cause surprise are the ones that are remembered relatively better than the other events.
- The startle response consists of sudden movement of the head, blinking of the eyes, a characteristic facial expression, raising and drawing forward of the shoulders, turning inward of the upper arms, bending of the elbows, turning downward of the forearms, flexion of the fingers, forward movement of the trunk, contraction of the abdomen and bending of the knees.
- Startle reaction(a complicated involuntary reaction to a sudden unexpected stimulus)

- **Disgust:** When we are faced with a situation or environment that is unhygienic or repulsive in sight or evokes a smell then it produces disgust. Generally, in disgust the person shows a nose crunch and moves away from the annoying stimulus in order to avoid experiencing the unpleasantness. Sometimes people may vomit or move away from a repulsive stimulus.
- Changes in Emotions
- We all have been experiencing various emotions in our day to day life. Have you ever noticed what changes happen in you when you experience an emotion? Let us understand more about the changes which happen in us due to emotions. As mentioned earlier emotions are characterised by definite overt behaviour patterns. Emotions when experienced cause certain overt changes and also disturbances of visceral patterns. Here we enumerate some of the important changes. These changes are produced because of the activities of ductless glands and the autonomic nervous system.
- The Physiological Changes in Emotions
- **Facial expressions:** Various muscles of the face including the eye, the lips, the nose and the forehead undergo changes by way of twisting and taking different poses. Most of the emotions have their facial expressions but it may not always be easy to judge the emotion from these expressions. Some facial expressions are universal. **universal facial expressions** of seven emotions – anger, contempt , disgust, fear, joy, sadness, and surprise
- **Muscular adjustment:** Sometimes the entire muscular system of the body undergoes contraction and adjustments. For example, the bodily changes that take place in "startle reaction" when there is quick, unexpected and strong stimulus. The startle response consists of sudden movement of the head, blinking of the eyes, a characteristic facial expression, raising and drawing forward of the shoulders, turning inward of the upper arms, bending of the elbows, turning downward of the forearms, flexion of the fingers, forward movement of the trunk, contraction of the abdomen and bending of the knees.
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- **Change in voice:** The voice gets affected in an emotion. When excited the voice becomes loud and high pitched. Under fear the voice takes a sinking and low tone.

Components of Emotions

- Most psychologists would agree that an emotion is a complex pattern of changes that include physiological arousal, subjective feelings, cognitive processes and behavioural reactions, all in response to a situation we perceive to be personally significant. Accordingly, an emotion has four components:
- **Physiological arousal:** Emotions involve the brain, nervous system and hormones, so that when you're emotionally aroused the hormone secretion is more to give us instant energy. Each emotion has a specific characteristic of physiological aspects. For example: When angry, the blood rushes(move with urgent haste to our hands in order to fight. When afraid, the blood rushes to our skeletal system and leg to facilitate the fight or flight responses.
- **Subjective feelings:** Emotions also include subjective awareness, or 'feeling' that involves elements of pleasure, liking and disliking. Thus, in studying emotion or knowing another person's feelings, we must rely(depend on with full trust) heavily on that person's own self reports.
- **Cognitive processes:** Emotions also involve cognitive processes such as memorial, perceptions, expectations and interpretations.
- **Cognition** is a term referring to the mental **processes** involved in gaining knowledge and comprehension. These **cognitive processes** include thinking, knowing, remembering, judging, and problem-solving. 1 These are higher-level **functions** of the brain and encompass language, imagination, perception, and planning.

- **Cognition** includes basic mental **processes** such as sensation, attention, and perception. **Cognition** also includes complex mental operations such as memory, learning, language use, problem solving, decision making, reasoning, and intelligence.
- **Behavioural reactions:** Emotions also involve behavioural reactions, both expressive and instrumental. Facial expressions such as smiles and frowns, as well as gestures and Lories of voice, all serve to communicate our feelings that may enhance our chances for survival.
- A specific **response** of a certain organism to a specific stimulus or group of stimuli. 4 the action, **reaction**, or functioning of a system, under normal or specified circumstances.
- a facial expression indicating disapproval, displeasure, or concentration, characterized by a furrowing of one's brows.

Lesson-IV Theories of emotion: The James – Lange theory – The Cannon – Bard theory.

THEORIES OF EMOTIONS

- Psychologists have proposed a number of theories about the origins and function of emotions. The theorists agree on one thing that emotion has a biological basis, which is evidenced by the fact that the amygdala (part of the limbic system of the brain), which plays a large role in emotion, is activated before any direct involvement of the cerebral cortex (where memory, awareness, and conscious "thinking" take place). There are the following theories which explain the complex mental and physical experiences that take place in humans called as "feelings" and these are:

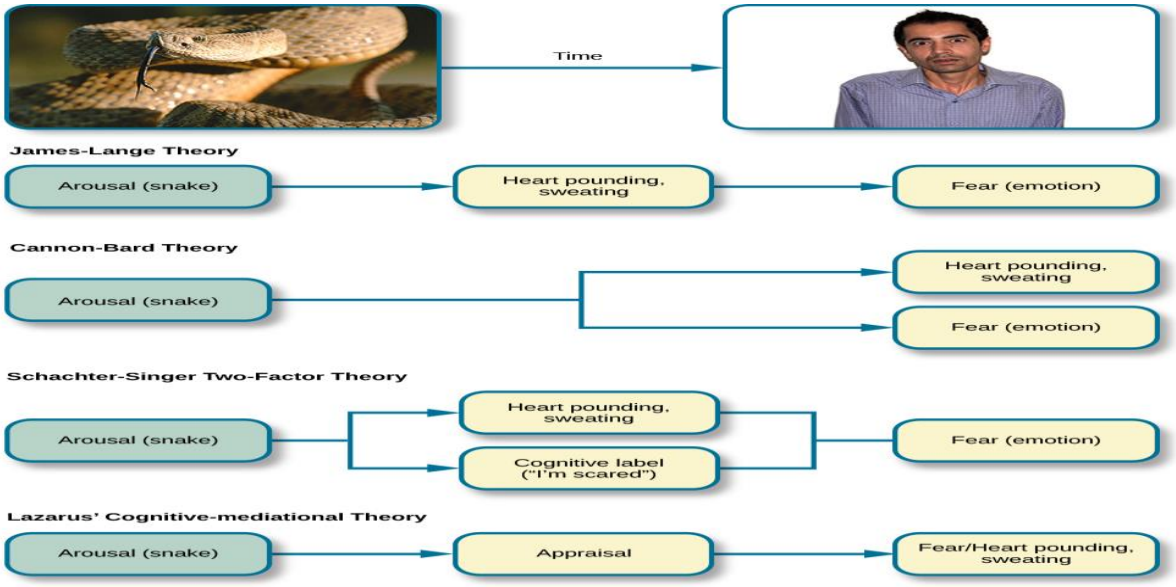
James-Lange Theory of Emotion

- Proposed independently by psychologist William James and physiologist Carl Lange, the James-Lange theory of emotion proposes that emotions occur as a result of physiological reactions to events.
- According to this theory, you see an external stimulus that leads to a physiological reaction. Your emotional reaction is dependent upon how you interpret those physical reactions.

- For example, suppose you are walking in the woods (an area of land, smaller than a forest, that is covered with growing trees). and you come across face to face with a bear. You begin to tremble (to shake involuntarily (as with fear or cold) : shiver)and your heart begins to race. The James-Lange theory proposes that you will interpret your physical reactions and conclude that you are frightened ("I am trembling, therefore I am afraid.")
- William James explained the same thing differently. He said that the bodily changes follow directly the perception of the exciting fact, and that our feeling of the same changes as the facts occur and these feelings are called emotions.
- The James-Lange theory states:
- Environmental influence (event) —> Physiological change —> Psychological experience.
- In other words, James and Lange would say, "I feel afraid because I tremble"(**to shake involuntarily).** If a person sees a bear while walking along in the woods, James and Lange would suggest that the person would tremble **involuntarily** and then realise that, because they are trembling, they are afraid.
- According to James, his theory is that the bodily changes follow directly the perception of the exciting fact. Our feelings of the same changes as they occur and this is called the emotion.
- Common sense says that when we lose our fortune, we are sorry and weep. We encounter a bear, are frightened and run. We are insulted by a rival and we get angry and strike that person.
- This order of the sequence does not seem to be correct. What is actually correct, according to James is that we feel sorry because we cry, angry because we strike, afraid because we tremble.
- Without the bodily states following the perception, the latter would be purely cognitive in form, pale, colourless, and devoid of emotional warmth. We might then see the bear, and judge it best to run, receive the insult and deem it right to strike, but we should not actually feel afraid or angry

Cannon-Bard Theory

- Physiologists Walter Cannon (1927) and Philip Bard (1934) developed another theory which proposed that the emotion that we feel depends upon the activity of the lower brain areas. It suggests that the felt emotion and the bodily changes that occur in response to the emotion are independent of one another, rather both occur simultaneously.(at the same time) This theory proposes that we first perceive the situation that may produce an emotion. This is followed by the activation of the hypothalamus which sends impulses to the internal organs of the body and the muscles that produce the expression of emotion. The impulses are also sent to the cerebral cortex where the emotion is experienced.



- According to Cannon (1927) and Bard (1934), Thalamus is really the seat of emotion. Emotion and physical changes take place simultaneously and one is not dependent on the other. Thalamus (or Hypothalamus) control emotion and physical changes. Hypothalamus works in three ways:

  1. Emotional impulse from receptor goes to the hypothalamus through afferent nerve.

  2. From hypothalamus a part of the impulse passes to the brain. Here the nature of the emotional experiences is determined.

  Again, from hypothalamus, a part of the impulse passes to muscles, viscera and other organs of our body.

- Bard removed the cortex and thalamus of some animals and left only the posterior part of the hypothalamus and found that they still showed rage**.** But when he removed the hypothalamus, they failed to show the integrated rage pattern.

- Thus he proved that the hypothalamus was necessary for the expression of rage.
- **Emotion**-related **impulsivity** is defined as the tendency to react **impulsively** when experiencing heightened **emotional** states (Whiteside & Lynam, 2001). For example, a person high in **emotion**-related **impulsivity** might express anger by abruptly acting aggressively.
- It suggests that the felt emotion and the bodily changes that occur in response to the emotion are independent of one another, rather both occur simultaneously.
- This theory proposes that we first perceive the situation that may produce an emotion. This is followed by the activation of the hypothalamus which sends impulses to the internal organs of the body and the muscles that produce the expression of emotion. The impulses are also sent to the cerebral cortex where the emotion is experienced.

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: To determine the span of attention for single dots or how many dots are reported correctly after a single brief exposure.

Activity 2: To investigate the acquisition of a skill in a motor visual performance (using the step maze

PSY 15142

SOCIAL PSYCHOLOGY

Objectives:

The nature and origins of Social Psychology, the nature of Attitudes and Discrimination, the nature of Social Identity and Interpersonal Attraction, the basic aspects of Social Influence and Pro-social behavior, the meaning and nature of Aggression and Group behavior

Unit – I: Introduction to Social Psychology

Lesson-1 Introduction to Social Psychology- Definition, Historical origins; Relationship with other social sciences-(Sociology, Economics, Political Science and Anthropology).

Lesson-2 Historical Development of Social Psychology- Early Years of Social Psychology, Formational Years of Social Psychology, Expansion of Social Psychology ,Social Psychology in 21st Century ,Social Psychology in India

Lesson-3 Research methods in Social Psychology (Field Research, survey research, Archival Research, Correlational Research, observational research).

Unit – II: Social Perception and Social Cognition

Lesson-4 Person perception; Attribution- Understanding the causes of others beahviour, Kelly's theory; Bem's theory; Regulatory focus theory. Impression formation and impression management.

Lesson-5 Schemas- Types of Schema, Person Schemas, Self Schemas, Group Schemas, Role Schemas, Events Schemas, Impact of Schema.

Lesson-6 Mental frame works- Interpersonal attraction. Communication system in human beings Nonverbal communication; the language of expression

Unit – III: Socialization & Social Influence

Lesson-7 Socialization- Definition, Motive and behavior.
Lesson-8 Agencies of socialization- (Parents, Peergroup, school and media).
Lesson-9 Sexual motives, Sex role identity, Psychological Androgyny, Dependency, Moral development.
Lesson-10 Self-Esteem.
Lesson-11 Self concept.

Unit – IV: Attitudes

Lesson-12 Attitudes -Nature and formation of Attitudes, Functions of Attitudes.
Lesson-13 Theories of attitudes- Reinforcement theory; Balance theory; Cognitive dissonance theory;

Lesson-14 Measurement of attitudes- Self measures; Likert, Thurston Osgood and Bogardus scales: Physiological measures.

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Outcomes:

The nature and origins of Social Psychology
The nature of Attitudes and Discrimination
The nature of Social Identity and Interpersonal Attraction
The basic aspects of Social Influence and Pro-social behavior
The meaning and nature of Aggression and Group behavior

Unit – I: Introduction to Social Psychology

Lesson-1 Introduction to Social Psychology- Definition, Historical origins; Relationship with other social sciences-(Sociology, Economics, Political Science and Anthropology).

In earlier courses you might have got a clear picture about what is psychology, what is the biological basis of our behaviour, how does the human development take place, how research in psychology is done and quantified etc. From this course onwards, we are entering into the 'social' world of an individual because most of our behaviours is a result of our interaction with people and their influence on our behaviour. Also most of these interactions involve other people and it mainly occurs in social context and social situation. The branch of psychology which studies the 'individual behaviour' in social context is called as 'social psychology'. Social psychology is a very important branch of psychology, combining the elements from two strong disciplines – sociology and psychology. Block 1 explains the concept of social psychology and its relevance to individual behaviours. It explains the various definitions of social psychology traces its historical developments and landmarks as well as further explains the relation of social psychology to other discipline. In establishing as a new discipline, social psychology enriched its research, by adopting various research methods – both experimental and non-experimental as well as including modern research through internet. Further the block 1 studies the most important concept of social psychology – social cognition – a process of understanding and predicting the behaviour of self and others i.e. social perception and person perception. It goes on to explore the definition, structure and functional process as well as cognitive and dynamic basis of social perception and person perception.

In this unit, we are going to introduce to you the concepts of social behaviour, social interaction and social influence. Next you see the nature of social psychology in general and the definition of social psychology. Further you will be studying various methods which social psychologists employ in carrying out his / her research about the different social phenomena.

Concept of Social Psychology

It is a truth that human actions do not take place in a vacuum. Every behaviours occur in some 'physical context' and most of what we do take place in the presence of others – that is, in an interpersonal context or social context. Moreover, a considerable portion of the behaviour that individuals perform when alone is based on 'past encounters' with other people.

Psychology studies behaviour which is a result of individuals thoughts and feelings, but individuals thoughts and feelings are inferred from his expressed or overt behaviour. Behaviours which include learning, problem solving, perceiving are the 'overt behaviour', on the other hand, behaviours like dreaming, imaging, memorising represents 'covert behaviour'.
There is another category of behaviour. Behaviour which is result or caused by or occurs because of others presence or influence is called *social behaviour*. This social behaviour is of great significance to social psychology. Whenever our psychological processes of perceiving, learning, motivating, decision making etc. are influenced by or a result of or related to / or occur in a social context, these

processes are called as social perception, social learning, social motivations, group conformity respectively. These behaviours which have a social context and occur in social environment and involve social stimuli are the main subject matter of social psychology.

Concept of Social Behaviour

If I ask you to make a list of all the activities which you perform from the time you get up till you sleep, you will observe that about 20% of these activities are just 'behaviours' but the remaining 80% of these are 'social behaviours'. This would make you understand social behaviour in terms of the significance of 'others' in our life, as most of our time of the day is spent in interacting with other members of our family or society. A careful look at these **'social interaction**' would make it very clear that in many of these interactions, our behaviour is influenced by others and we influence others behaviour. So our life is not only 'social' but our social interactions to a large extent are shaped, influenced and changed as a result of social influence. A major part of our behaviour occurs in a social context, as well as in an environment involving 'others'

Concept of Social Interaction

Unlike animals that come together either for progeny or protecting themselves from danger, human being is a truly social animal. Each one of us is surrounded by and interacts with at least one or two individuals who are close to us. Human being live, grow and strive within the close interpersonal relationship. Many of our needs are satisfied in this social context. In satisfying the needs one establishes contacts, co-operates with other people and adjusts with other members of the society. Actions that are performed by the individual in relation to the members of the society are called 'Social Actions and Interaction' as these actions affect two or more individuals. This results in action, social actions and interaction. This interaction with others is called 'Social Interaction' and this social interaction is the basis for establishing lasting relationship in life. Two mechanisms underlie every social interaction, these are:

1. **Social interaction requires a social contact or social relationship:** Every social situation involves social contact at least between two people without which no interaction can take place. This contact may be direct or indirect and can have negative and positive impact. Positive social contact gives rise to pro-social behaviours like co-operation, organisation as simulation, adjustment, adaptation and accommodation. Negative social contact may retract a person away from entering into social interaction and even if s/he 'has to', it results in unhelpful social behaviours like unhealthy competition, aggression etc.
2. **Social interaction involves communication:** Social interaction between any two individuals involves some form of communication that is verbal or non-verbal without which the social interaction cannot take place. As is known, social interaction takes place at three levels, a) Individual to individual, b) Individual to groups, and c) Group to group level.

Concept of Social Influence

Each one of us depends on others to satisfy, many of our day to day needs – food, clothing, housing, love, security. This interdependence produces interaction between persons. In the course of this interaction an individual is influenced by parents, friends, teachers and they are also in turn influenced by the person. Thus other people affect and influence our outlook, our belief our values and so on marginally or profoundly. Such influence are exerted directly and deliberately, subtly and implied. The individual behaviour is influenced by this social influence, as for example: Sita is newly married and comes to her in-laws house in a new city. Her behaviour is more likely to be influenced by the directions g etc. which slowly do influence her husband and children. Our social life is thus a process of mutual influence and change which enrich our relationship in a social context.

Definition of Social Psychology

Social psychology is defined as the scientific study of "how we feel about, think about and behave toward the people around us and how our feelings, thoughts, and behaviors are influenced by those people in the social context" (Kassin, Fein, & Markus, 2017). In a slightly different expression social psychology is defined as a discipline that uses scientific methods in an attempt to understand and explain how the thought, feeling and behaviour of individuals are influenced by the actual, imagined or implied presence of others (Gordon Allport, 1985, p. 3).

Nature of Social Psychology

Careful explanation of definitions given above reveals that it has three major components that characterise the nature of social psychology. These characteristics have been further elaborated below:

Social Psychology Applies Scientific Methods of Study

Social psychology is scientific in nature. It applies the scientific method of systematic observation, description and measurement to the study of the human behaviour in social context. Social psychological studies refer to the data being collected through direct observation or experiment. Such experiments and observation are carried out carefully and reported in detail so that other investigators can repeat and verify the work. The scientific social psychology carries out three major activities: description, explanation and prediction of social behaviours. Social psychology provides a scientific account of social behaviour based on direct observation rather than on common beliefs. Social psychology also attempts to explain why people behave in a particular way in a particular social situation. Such interconnected explanations of social behaviours lead to formation of theories which help in predicting social behaviours and managing them in a desirable direction.

Social Psychology Studies Individuals' Thought, Feeling and Behaviour

The varied issues of the field of social psychology include- individuals' thought, feeling and behaviour. Cognition is referred as the manner in which people process information. Social

psychology studies cognition that relates to social activities and that helps us in understanding and predicting our social behaviours. Social psychology also studies the feelings we experience as an individual in our social lives. What we think or feel in the social context is finally expressed through our behaviours in social interactions. Social psychology studies these behaviours in the forms of cooperation, helping behaviour, conflict, aggression, etc.

Social Psychology Studies Individuals' Behaviour in Social Contexts

Social psychology studies individuals' thoughts, feelings and behaviours in the social contexts. This component of social psychology refers that our behaviour is influenced by the presence of other people and we also influence other people's behaviour. The social context referred to in the definition of social psychology does not have to be real or present. Even the implied or imagined presence of others can have important effects on individuals (Gordon Allport, 1985). However, in order to establish general principles of human social behaviour, social psychologists sometimes examine non-social factors also. Kurt Lewin (1936), one of the important early figures in social psychology proposed a model for understanding social behaviour, which says that social behaviour is a function of the interaction of the situation and the individual's characteristics

Scope of Social Psychology

Social psychology concentrates on the individual behaviour in the social context, so the subject matter of social psychology is the interactions of the individual with other individuals and society as well. It is the social world, based upon the relations of human to their fellow beings which furnishes the subject matter of social psychology. The scope of social psychology can be broadly outlined in the following ways:

I. People generally express feelings of approval and disapproval, favourability and unfavourability, or likes and dislikes towards different persons, objects or issues that further influence their thought and actions. This phenomenon is termed as attitudes and social psychologists have been emphasizing upon various aspects of attitude, such as formation of attitude, attitude structure, attitude change, function of attitudes and relationship between attitudes and behaviour.
II. One of the emerging areas of social psychology is social cognition, which studies the ways people perceive, ponder and remember information related to social stimuli. Various phenomena studied under social cognition are person perception, attribution process, schema, stereotypes, etc.
III. Social influence is a traditional, core area of study in social psychology which refers to the way people affect the thoughts, feelings, and behaviours of others.
IV. Social psychologists are also interested in the question of why people sometimes act in a pro-social way (helping, liking or loving others), but at other times act in an antisocial way (hostility, aggression or prejudice against others).
V. Social psychologists have vastly studied various phenomena related to social groups

VI. and group dynamics. Groups may be understood in terms of group composition, group structures, group process and the effect this has on individual change and group development, as well as on task performance. Thus, a social psychologist studies almost everything that we do every day in the social context (see box below for detail).

Lesson-2 Historical Development of Social Psychology- Early Years of Social Psychology, Formational Years of Social Psychology, Expansion of Social Psychology ,Social Psychology in 21st Century ,Social Psychology in India

HISTORICAL DEVELOPMENTS or HISTORICAL ORIGIN:

To understand the contribution of social psychology and what is distinct about it,we need to know its historical origins. The formation of pre-conditions of social psychology is as a whole the same as the development of any other scientific discipline. The socio-psychological ideas originally took shape within the realm of philosophy and then gradually branched off from the system of psychological knowledge. First we will briefly discuss the social thought before the advent of social science and then discuss the second stage of the development of social psychology which is deemed to be "more productive.

Social Thought before the Advent of Social Science

Two earlier forms of social thought over the centuries are Platonic and Aristotelian.

Platonic thought emphasised the primacy of state over the individual who had to be educated to become truly social.

Aristotelian thought states that human being is social by nature and nature can be trusted to enable individuals to live together and to enter personal relationships from which families, tribes and ultimately the state will naturally develop.

In modern times these two traditions of social thought have been known as socio centred approach and individual centred approach. Socio centred approach emphasises the determining function of social structures (systems, institutions and groups) for individual experience and behaviour. According to the individual centred approach social systems are said to be explicable in terms of individual processes and functions. For Hegel (1970-1831), the German philosopher, the state is not only the ultimate form of society but the incarnation of the objective social mind of which individual minds are active participants. The notion of Group Mind derived from Hegel's supra-individual nature.

Social psychology from its beginning has been defined as the scientific study of the individual in the social context. Individualism has been an inherent property of the discipline. Individualistic doctrine says that all the explanations of individual social phenomena are to be rejected unless they are expressed in terms of individual. Individualism in psychology has been characterised and criticised

for the self centred denial of the other. Individualism has appeared in two forms in sociological thought:

Hedonism: People act in order to secure and maintain pleasure and to avoid and reduce pain.

Utilitarianism: The doctrine that advocates the pursuit of the greatest happiness of the greatest number.

For most modern theories of conditioning and of motivation, the underlying ideas of individual satisfaction (reinforcement, reward, reduction of stress, of dissonance, uncertainty) are variations of the pleasure or utility principle. Utility and satisfaction are important constructs involved in many social psychological theories

From Machiavelli (1513) and Thomas Hobbes (1651) the concept of power and its role in social relationships returned in social psychology. The concept of social power (social influence) found its proper frame of reference in the field theory and social exchange theory. In Lewinian field theory 'power' became the term for the potential to influence others while control and influence refer to the power of action. Research areas where power has been studied are: aggression, conformity to group pressure and obedience to authority, and power in language.

Modern social psychology was also influenced by the developments in the nineteenth century in two major areas: sociology and theory of evolution. We will discuss these influences later under the titles: social psychology and other disciplines and the influence of evolutionary theory will be discussed in current trends in social psychology.

The Second Stage of Development: Social Psychology Emerges as a Discipline

In the process of branching off from the psychology as a separate discipline, three moments are important to be outlined (Galina Andreyeva 1990):

The requirement concerning the solution of socio-psychological problems which aroused in various related sciences.

The processes involved in the separation of socio-psychological problems within the two parent disciplines: psychology and sociology.

Finally, the description of the first forms of independent socio- psychological knowledge.

In the mid 19th century, the first forms of socio-psychological theories that appeared, three were most important in terms of their influence: *people's psychology, mass psychology and the theory of instincts of social behaviour.* These theories developed in the background of philosophical and descriptive tradition, hence their nature was speculative and abstract.

People's Psychology

People's psychology developed as one of the first forms of socio-psychological theory in Germany in the mid 19th century. Most outstanding creators of people's psychology were Moritz Lazarus (1824-1903) and Heymann Steinthal (1823- 1893). In 1859 the journal *People's Psychology and Linguistics* was founded in which the article by Lazarus and Steinthal entitiled "*Introductory Thoughts on People's Psychology*" was printed. The article expressed the idea that the main force of history is the people, or the 'spirit of the whole', which can be seen in art, religion, language, myths, customs etc. The individual consciousness is only its product, a link in a certain mental connection. The task of social psychology was to perceive the psychological essence of the spirit of the people and discover the laws that guide the spiritual activity of the people.

The views of Wilhem Wundt (1832-1920) furthered the development of People's psychology. Wundt (1900) proposed that psychology consists of two parts: physiological psychology and people's psychology (*Völkerpsychologie*: German word for people's psychology). Physiological psychology, in his views was an experimental discipline, but experimentation is not useful for the study of higher mental processes: thinking and language. For the areas like language, myths, customs and art people's psychology need to adopt other methods.

The views proposed by Wundt were criticised by Vygotsky. People's psychology considered language, myths, customs, art and religion as objectives of study. Vygotsky called these *clots of ideology* or *crystals*. The task of psychology he proposed was not to study these crystals but the solution itself. He opposed the thought that social psychology should study the mentality of collective personality. The personality of the individual, he said, is also social and is therefore an object of study in social psychology. Social psychology focuses on the mentality of the separate individual and collective psychology – on personal psychology under collective manifestation (e.g. army and church). Social psychology is the study of cultural and historical determinati0on of mentality. Lev Vygotsky dealt with two questions directly related to the development of social psychology. The higher mental functions (arbitrary memory, active attention, abstract thinking and volitional act) could not be considered immediate functions of the brain, roots of these functions lie in social condition. He expounded upon the idea of cultural historical determination of all mental processes.

Mass Psychology

This theory emerged in France in the latter half of the 19th century. The creators of mass psychology Italian Lawyer Scipio Sighele (1868-1913) and French sociologist Gustave Le Bon (1841-1931) began with Gabrial Trade's (1843- 1904) basic ideas on the role of irrational movements in social behaviour and the role of imitation. According to Le Bon any accumulation of people represented the idea of the mass with depersonalisation and predominance of emotions over intellect, the general loss of intellect and the loss of the sense personal responsibility. The events like mass movements in the late 19th and early 20th century, rapid social and economic changes due to industrialisation and urbanization wereconducive to mass psychology but like people's psychology it did not develop

within the context of academic psychology. It did not have any significant consequences as regards the future of social psychology.

The First Textbooks of Social Psychology

The year 1908 is considered to the year of final emergence of social psychology as an independent scientific discipline. This year two books appeared with the title social psychology: *An Introduction to Social psychology* by William McDougall and the other *Social Psychology* by sociologist Edward A. Ross. Before these two works in 1897, James Mark Baldwin's work, *Social and Ethical Interpretation in Mental Development* was published in New York which can be considered as one of the first systematic manuals in social psychology.
According to E. A. Ross social psychology deals with uniformities in feeling, belief, volition and action. Uniformities were to be explained by the capacities for imitation and suggestion operating among individuals collectively. McDougall's theory proposed that inborn instincts are the cause of social behaviour. In his views, tendencies of imitation and suggestion are rooted in their biological nature. Imitation supposedly grew out of non-specific innate tendency, whereas suggestion was seen stemming from an instinct to submit to a prestigious person or symbol.

In spite of great popularity of McDougall's ideas, they played a negative role in the history of science. Recognition of instincts as the motive force behind the social behaviour gave importance to the irrational and unconscious motives. Human understanding and thought processes were not given much attention. The overcoming of the theory of instincts is deemed to be an important milestone in the formation of scientific social psychology.

The Beginning of Experimental Research

The early 20th century especially the period after the First World War, is considered the beginning of the metamorphosis of social psychology into an experimental science. But the best known of the early laboratory study was Norman Triplett's 1897 experiment on "the dynamogenic effects of pace making". It was the first study of an individual's performance is affected by the presence of others.

The experimental investigations by Walter Moede in Europe and Floyd Allport in US served as a milestone in this process. Allport compared the performance of individuals working alone with that of the persons working either before an audience or in the presence of others engaged in the same activity. He found that the latter condition often improved performance. The phenomenon is known as the social facilitation effect. In 1924 Allport published the first social psychology textbook making extensive use of experimental research. This work encouraged the growth of an experimentally oriented social psychology.

The next two decades (after Allport's publication) were marked by systemic investigations in the field in the areas such as the development of attitude measures, social norms, aggression, leadership and social influence (conformity). Theodore Newcomb (late 1930s) assessed the social and political attitudes held by college students and demonstrated how their attitudes were modified by the views prevailing

at the college (Peer influence). Muzafer Sherif (1935) studied the nature and impact of social norms-rules indicating how individuals ought to behave. In 1939, Dollard, Doob and Miller demonstrated the relationship between frustration and aggression. They concluded that frustration produces instigation to aggression. Kurt Lewin, Lipitt and White (1939) carried out revealing research on the nature of leadership and related group processes. The main attention began to be focused on the small groups which facilitated experimental method.

From the earlier thoughts of group mind and depersonalization, social psychological theories at this period emphasised the overriding importance of the individual's thought and understanding. In 1948, a revolution got under way in social psychology. Precursor to this change was the establishment of Research Centre for Group Dynamics at MIT by eminent theorist Kurt Lewin. Kurt Lewin is often referred to as the father of applied social psychology. Experimental works due to the efforts of Kurt Lewin put social psychology as a science in a more advantageous position. He believed that significant social problems can be investigated in the laboratory with experimentation. He favoured the analyses based on individual's understanding of the situation surrounding him or her.

All the enthusiasm for experimental orientation stemmed from the need to provide authentic knowledge about the real problems of society. But it was begun to be realised that skillfully conducted laboratory research created a distance between social reality and the topics under investigation. The social content from these researches was taken away in the favour of experimentation. In the mid 20th century, social psychology faced the problem of analysis of the enormous experimental research by theoretical knowledge. The need for the proportional development of two spheres of scientific knowledge – the theoretical and the experimental was felt. 1950s and later, many psychologists contributed significantly by theoretical explanations in the areas such as conformity (Soloman Asch 1956, 1958), cognitive dissonance (Leon Festinger 1950, 1954, 1957) and attribution theory (Fritz Heider, Herold Kelly and E. E. Jones). In this decade, social psychology was brought closer to the cognitive psychology dealing with thought, judgment and decision making.

Middle Range Theories

The need of theoretical explanation for the experimentations in the social psychological areas gave birth to the theories specifically designed to be applied in the areas of social psychological research.

The idea of creating "middle range theory" was first developed by Robert Merton. Middle range theories are the theories that account for a specific aspect of social behaviour and do not try to encompass all of social life. Social psychological theories tend to be specific and focused, rather than global and general. The field theory of Kurt Lewin is often considered a model of middle range theory. The greater part of social psychological theories existing today (theories of frustration aggression, changes of attitudes, cognitive dissonance, cooperation and competition etc.) belong to the bracket of middle range theories. At present, the theories of middle range are mostly concentrated around four trends: behaviourism, psychoanalysis, cognitivism and interactionism. The theories from the vantage point of behaviourism, psychoanalysis and cognitivism are the socio-psychological variants of main

stream psychological thought; and interactionism represents theories mainly contributed by sociological perspective

Lesson-3 Research methods in Social Psychology (Field Research, survey research, Archival Research, Correlational Research, observational research).

Social Psychology and other Related Disciplines

The goal of social psychology is to identify universal properties of human nature that make everyone – regardless of social class or culture susceptible to social influence. By contrast political science, sociology and economics, cultural anthropology have a starting point in the political, societal and cultural systems in which an individual lives while general psychology studies learning processes, motivation and perception, intelligence, social psychology studies the behaviour of individual in his face-to-face interaction with others, anthropology is concerned with the culture of the group, its technology, its beliefs, values and so on, sociology studies the social institutions and social organisations and social systems and attempts to relate the various aspects of social structure to the patterns of human behaviour and regulate the behaviour of individuals through a set of rules. Thus the knowledge of norms, sanctions and roles in the given society will help the sociologist to explain the uniform patterns of human behaviour. Social psychologist, as against this focuses on how individual perceives the norms, sanctions and roles in the group in which he lives and work. Thus social psychology tries to understand the influences which produces regularities and diversities in human social behaviour. Its interest in the individual as a participant in the social relationships and its emphasis is on the understanding of the social influence process underlying relationships.

Levels of Analysis of Behaviour

Social psychology's focus on social behaviour is shared by several other disciplines in the socials sciences including sociology, economics, political science, anthropology, psychology etc. each of these disciplines is concerned with the influence of social and societal factors on human behaviour for example, the effect of recession on the world economy, a typical society with its unique rules and norm, a family which is totally authoritarian, an NGO organisation and its functioning.

In all these examples, the emphasis is on larger social structure like world economy, ethnic group, family, and an NGO and not on individual behaviour. However the important difference between social psychology and the other social sciences is in their level of analysis.

The societal level of analysis is used by sociologists, economists, political scientists, and other social scientists. These scholars attempt to understand general patterns of social behaviour, such as homicide rates, voting behaviour, or consumer spending. According to this viewpoint, social behaviour can be explained by such forces as economic hard times, class conflicts, and clashes between competing ethnic groups, regional crop failures, governmental politics, or technological change. The goal of societal analysis is to identify links between broad social forces and general patterns of social

behaviour. To study violence in urban areas, social scientists might identify relationships between rates of violent crime and such factors as poverty, immigration, or the industrialisation of a society.

The individual level of analysis is typically used by clinical and personality psychologists, who explain behaviour in terms of a person's unique life history and psychological characteristics. According to this viewpoint, personality trait sand motives can explain why individuals behave as they do, and why two people may react quite differently to the same situation. Emphasis is given to individual differences in childhood experiences, in ability and motivation, and in personality or psychological adjustment. The individual approach explains violent crime in terms of the unique histories and characteristics of the criminal. To understand such behaviour, the psychologists using the individual approach would consider the personality and background of the person: was the criminal depressed or suffering from paranoid delusions or using drugs? What kind of life had the bank teller led? For example, was he physically abused as a child?

Social psychologists adopt a different level of analysis: The interpersonal level. Social psychologists typically focus on a person's current social situation. That social situation includes the other people in the environment, their attitudes and behaviours, and their relationship to the individual. The same teenage boy who is outgoing and talkative with his close friends may be shy and withdrawn with unfamiliar adults. The same teenage girl who got. As in math in her supportive all-girl high school may be struggle with math in her competitive, largely male college math classes. Change the social context, and the individual will change.

Social Psychology and General Psychology

Social psychology is so intimately related with general psychology that it may be considered as a branch of psychology itself. This fact is also clear from the matter appearing in the chapter relating to the scope of social psychology in the preceding pages. Many subjects of psychology are studied in the field of social psychology. Besides, social psychology takes within its purview some such subjects which do not form a part of general psychology. Therefore, both of these subjects have been separated.

General psychology, in a way, independently carries out the study of human behaviour whereas social psychology aims at knowing as to how far and in what way the behaviour of an individual affects the behaviour of others. For example: assessing the intelligence of an individual is a job of a psychologist but to study the influence of peer group of intelligence becomes a topic of research of social psychologist. One of the aims of the social psychology is to study changes in human behaviour in the changed modern conditions. It is a fact that the development of various qualities, habits, aptitudes and character of an individual takes place in the society. Conditions in a society are ever-changing and these changing conditions are deeply related to an individual's development. In general psychology, too, we try to study various aspects of man's development. In social psychology while studying these aspects of development, our mind goes particularly towards those social influences which directly or indirectly affect them. In this way we find that general psychology and social psychology are inter-linked. Consequently, it is very difficult to keep them apart. However, the field of activity of both subjects is different and therefore, these are studied independently.

Social Psychology and Sociology

In sociology we study groups, society and institutions, whereas in social psychology our attention is centered round the action and reactions of individuals. Individuals and groups are naturally related to society. For example: studying 'family' as a whole and its structure and function is the job of a sociologist but how family atmosphere influences each child differently becomes the study topic for a social psychologist. As such, relationship between sociology and social psychology is but natural. It is people who form masses and organise society. Many activities of persons take place either in a group or in a society. Therefore, in studying a person's behaviour we have to study those groups, societies or institutions where he shows or had shown his behaviour.

Social customs, religion, marriage and other social traditions develop according to the needs of the people. The sociologist has to take the aid of social psychology in order to find out the necessities that inspire man for creating and organizing his various customs and traditions.

Sociology is the study of human societies and the groups that form those societies. Although both sociologists and social psychologists are interested in how people behave in societies and groups, they differ in what they focus on. Sociologists focus on the group as a single unit, whereas social psychologists focus on the individual members that make up the group. Some sociologists call themselves social psychologists and the exchange of ideas and findings between the two fields that have sometimes been quite fruitful because they bring different perspectives to the same problems.

Social Psychology and Anthropology

In anthropology we study different organisms and developmental and cultural aspects of human life. Anthropology has two branches – physical and cultural. Social psychology appears to be related with both of them. In the field of physical anthropology differences relating to heredity and race are studied. Study of differences relating to heredity and race becomes a subject of social psychological studies. An anthropologist also takes interest in group behaviour, but his interest remains confined to ancient and aborigine races. A study of a racial or cultural group in its detail to observe continuity of a tradition is a subject matter of anthropologist, but how the culture shapes the cognitive and affective tendency of an individual is a matter of interest to a social psychologist. The anthropologist unlike the social psychologist does not study an individual's behaviour in the cultural or social background. However, it is now seen that sociologists have started taking interest in cultural studies along with the study of some very important persons because of the life story of such important persons which plays a vital role in the building of a particular culture. It will have to be admitted that this kind of study conducted by anthropologists will remain incomplete without the help of social psychology. This shows that anthropology and social psychology are inter-dependent upto some extent.

Anthropology is the study of human culture. The field of anthropology and social psychology are inter related. Human culture consists of the shared values, beliefs, and practices of a group of people. These values, beliefs, and practices are passed down from one generation to another. Human beings

are not only social animals, they are also cultural animals. In order to understand human behaviour, social psychology requires understanding the cultural context in which that behaviour occurs

Social Psychology and Ethics

It is evident from the preceding discussion that social psychology is related in some way or other with all those sciences which analyse and define human behaviour. The same holds true in the case of ethics too. Ethics analyses public behaviour and determines its appropriateness or otherwise. In the background of a man's good or bad behaviour lie some of his drives. The study of these drives is one of the main aims of social psychology. It is thus evident that ethics has close relationship with social psychology. For example: ethical appropriateness or inappropriateness of an individual or group behaviour is a study of ethics, anthropologist or a sociologist, but the main motive behind following an ethical code or reasons for doing or not doing so are a subject of concern to social psychologist.

Social Psychology and Political Science

In political science a detailed analysis is made of the mental attitude of rulers and the ruled. It also undertakes the study of such measures and sources that strengthen the state authority and administration. Side by side theories are also propounded to protect the interests of the ruled. How is the public opinion formed? What should the leader do in order to build public opinion in his favour?

What should be done to inspire and encourage the public? How should the ruler deal
with national and international problems? How should he build up organizational? These are some of the issues that come under the purview of political science. To understand the various problems relating to politics, the political scientists have to take the help of social psychology, because it is only through social psychology that political science may succeed in understanding the latest and complex problems and motives of the public. Similarity, social psychology has to lean towards political science for understanding some of its subjects because political science deals with the welfare of masses. Many a studies about leadership, authoritarian personality and group dynamics and group conformity are taken up by social psychologist.

Political science is the study of political organisations and institutions, especially governments. Social psychologists conduct research on political behaviour. They study political issues such as voting, party identification, liberal versus conservative views, and political advertising. They are also interested in what makes some people better leaders than others.

Social Psychology and Economics

Under economics activities relating to earning of money and production are studied. Economics is essentially a subject dealing with money matters, production, consumption and distribution of products and money. In the background of all such matters and activities concerning earning of money lie an individual's natural drives and necessities. As such, economics will certainly need the help of social psychology. An individual's behaviours are motivated due to his monetary considerations.

Consequently, economics, too, has to rely on social psychology in understanding certain matters. A very recent example is the economic recession and its impact on different individuals.

You might know that the field of Economics deals with the trends of production, distribution, and consumption of goods and services of a particular society. This equally diverts the interest of social psychologists. In fact, some social psychological theories are based on economic principles. For example, social exchange theory predicts commitment to relationships by considering factors such as the costs, rewards, investments, and the number of alternatives available. Economics also calls our attention to large social systems (such as the labour market or money system) and to how these systems shape behaviour. Again, a full understanding of human behaviour requires appreciating not just what goes on inside one person's head and what is happening in his or her immediate environment at the time, but also how the person's behaviour fits into the larger social system

Significance of Social Psychology Today

Social psychology is not only related to other social sciences. Since it became an applied field, social psychological concepts of awareness, attitudes, social perceptions, motivational aspects are been used to study – rural problem, organisation problem, health and fertility behaviours and environmental attitudes and so on.

The main utility of social psychology is that through its study we may get enough help in solution of various social problems. Each nation of the world is facing different types of problems. For instance our own India is facing problems of unemployment, caste, communalism, linguism, regionalism, socialism, communism, sectarianism, beggary, prostitution, capitalism, religion, dress, food habits, standard of living, robbery, dacoity, poverty, diseases etc. All these create great difficulties from time to time. Similar problems are found in other countries too. At the root of all these problems are some psychological causes. Social psychology is capable of reaching the roots of these problems and offering some solutions. Therefore, the utility of social psychology for us cannot be dispute. Individuals, institutions, society or the government have equal opportunities to make use of social psychology. In fact, these bodies do take the help of social psychology. Thus, we take the help of social psychology for solving our social problems as and when needed in some form or the other.

RESEARCH METHODS IN SOCIAL PSYCHOLOGY

Goals of Research in Social Psychology

Social psychological research has four goals:

1. **Description:** A major goal is to provide careful and systematic descriptions of social behaviour that permit social psychologists to make reliable generalisations about how people act in various social settings. Example: Are men more aggressive than women's.
2. **Causal analysis:** Much research in psychology seeks to establish cause and effect relationship, because scientific inquiry in the research is to establish cause and effect relations. Example: Does college education make students more liberal in their social attitudes.

3. **Theory building:** Third goal is to develop theories about social behaviour which help social psychologists understand why people behave the way they do. This can further lead to suggest new predictions that can be tested in further research.
4. **Application:** Knowledge gained by the above three attempt can help to solve everyday social problems.

Sources of Research in Social Psychology

Social psychology is the scientific study of social behaviour. These are a diverse range of methods available to social psychologists. Any research begins with a hunch or hypothesis (a tentative solution), that the researcher wants to test. There are two ways in which a researcher chooses the hypothesis.

1. People often generate hypothesis from previous theories and research. Many studies stem from a researcher's dissatisfaction with existing theories and explanation. Example: Leon Festinger was dissatisfied with 'behaviourism' to explain attitude change. He thus formulated a new approach called the dissonance theory that made specific predictions about when and how people would change their attitude. In this way new research is continuously carried out in order to update the existing theory.

2. Theory is not the only way to derive a new hypothesis in social psychology. Researchers often observe a phenomenon in everyday life that they find curious and interesting. The researcher then constructs a theory related to that phenomenon, thus generating new theory. Example: The mere presence of other person that led to better performance lead to the famous phenomena of 'social facilitation'.

Field Method

Experiments are not only confined to the laboratory. They are also conducted in naturalistic settings. For example given a situation; wherein an individual requires help, how many people come forward to help and why? Field experiments have greater external validity (not being artificial but genuine) than laboratory experiment. They are less likely to be influenced by 'demand characteristics' a participants typically have no idea that they are taking part in a study. The Advantage and Disadvantage of Field Method are:

1) The situation is not nearly as controlled as in a laboratory situation and so the impact of external influences cannot be ruled out.

2) It is also not possible to randomly assign participants to conditions

Observation Method

In social psychology, the observers are trained as social scientists who set out to answer questions about a particular social phenomena by observation and coding it according to a prearranged set of criteria. This method varies according to the degree to which the observer actively participates in the scene. At one extreme the observer is a non participant. S/he neither participates nor intervenes in any way as for example: a researcher is interested in children's social behaviour may stand outside a playground to observe. Children at play in some situations, by their very nature, require observer participants**,** who observe, but try not to alter the situation in any way, for example – to get to know the intricacies of certain social phenomena like rituals, cultural way. The observer can be a participant as a friend or relative. Certain behaviors are difficult to observe as they occur rarely or privately. It is confined to one particular group of people, setting, and activity.

Archival Method

Another way that social psychologist can observe social phenomenon without conducting an experiment is to re-analyse existing data. The researcher examines the accumulated document or archives of a culture, for example; diaries, novels, suicide notes, television shows, movies, magazines, newspaper articles, advertising, sexual violence etc. Archival analysis can tell us a great deal about the society's values and beliefs. It has got two advantages. It is inexpensive and it can study the change over time frame. This is particularly useful when researches are interested in the effect of societal events on behaviours, which have occurred in the past. This research comes under 'archival research' first conducted by Hovland and Sears (1940).

Correlation Method

In correlation studies, the researcher carefully observers and records the relationship between (or among) two or more factors technically known as variables. For example: Is physical attractiveness is related to a student's popularity with other students. In a correlation design, the research does not influence the student's behaviour in any way but merely records information. The hallmark of an experimental design is intervention – with putting people in controlled situation or have confederates. Correlation research asks if there is an association between the variables and whether this association is high (+ve) or low (-ve) or neutral (no correlation). Example: whether or not watching violence on television is related to aggressive behaviour.

The *Advantages and Disadvantage* of Correlational method are:

1) Correlation designs enable research to study problems in which intervention is impossible.
2) This design is efficient. It allows research to collect more information and relationship.
3) No clear cut evidence of cause and effect

Survey Method

It is a research method that involves asking participants to respond to a series of questions, through interviews or questionnaires. Poll survey, marketing survey are the best examples. Surveys can be administered to a large, sample with relative ease and at little expense because surveys are gathered from large numbers and researchers can be sure of it is genuinity. The downside to questionnaire is that if they are not very carefully designed they can be misinterpreted by participants. There are also a number of response biases that participants have a tendency to blindly agree with positively worded questions and frequently fail to use the full range of possible responses like 'I don't know' etc

Unit – II: Social Perception and Social Cognition

Lesson-4 Person perception; Attribution- Understanding the causes of others beahviour, Kelly's theory; Bem's theory; Regulatory focus theory. Impression formation and impression management.

SOCIAL COGNITIONA part from the self, the individual is curious to know about the other people. The phenomenon of how people think about other people is discussed in the social cognition. It is a broad term that describe on the way perceiver encode, process, remember and use information in social contexts in order to make sense of other people's behaviour. It becomes an essential element in our understanding of intergroup and interpersonal processes. **SOCIAL PERCEPTION** "We had one girl in our college, she was very popular among boys but was a subject of jealously of girls, because boys use to always follow her and she was constantly interacting with them. One day when I asked the other girls 'what is that this girl has got? The girls unanimously expressed that, she has the knack of talking to boys – the way she looks at them, smiles, her posture, her mannerisms give boys a clue that she is ready for friendship". This example brings us to a very important concept in social psychology, Social Perception, that is the processes through which we seek to understand other persons. Because other people play such important roles in our life, we engage in this process very often and devote a lot of effort in trying to figure out what makes people tick? Why some people is socially popular and others are not? Sometimes their efforts to impress others work but also we often make errors in our efforts to understand them.

ATTRIBUTIONS Humans are rational creatures and always behave in a balanced and coherent and analytic way. This highly influential view was presented by Heider (1958) who said that people are motivated by two primary . The need to form a coherent view of the world and the need to gain control over the environment. Heider believed that this desire for consistency, stability and the ability to predict and control make us. Naïve scientists. In particular this need to attribute causes to effects. (observed behaviours and event) and to create a meaningful, stable world, where things make sense was the basis for a theoretical approach which become highly influential in how social psychologist viewed social cognition. This was referred as the attribution theory. Heider believed that a 'basic need to attribute' make the world a clear, definable and predictable place, thereby reducing uncertainty .

Types of Attributions Attributing causality has generated a great deal of work that attempted to model the way in which humans try to explain the actions of themselves and others. This brings us to the 'Locus of causality'. The main distinction can be made between InternalExternal attributions. The internal attribution locates the cause of behaviour to the personal characteristics of the person such as personality, mood, ability, attitudes and effort; it is also known as personal attribution. On the contrary the external attribution locates the cause to the external events such as actions of others, nature of situation, social pressure or luck. It is also known as situation attribution. A simple example can help us to understand this attribution of causality. A child fails in an examination if he attributes the cause for his failure to his effort then he is showing internal causality but if he attributes the cause for his failure to his luck, then he is showing external causality

IMPRESSION FORMATION The moment we look at a stranger, the process of impression formation starts, although we are not aware of such a process, we end up either forming positive or negative impression about others. We look at a person and immediately a certain impression of his character forms itself before us. A glance, a few spoken words are sufficient to start a story about that person. As we have already seen, some aspects of social perception, such as nonverbal communication, attribution which requires a lot of hard work and it is not always easy to draw inferences about other's motives or traits from their behaviour. In contrast, forming first impressions seems to be relatively effortless.

In forming impression about others, there are six simple and general principles:

➢ People form impressions of others quickly and on the basis of minimal information and go on to infer about general traits about them.

➢ People pay attention to most salient features of a person we notice the qualities that make a person distinctive or unusual, rather than to everything.

➢ In processing information about people involves some coherent meaning in their behaviour and use context of a person's behaviour to infer its meaning, rather than interpret the behaviour in isolation.

➢ We organize our perceptions by categorizing or grouping stimuli i.e. rather than to see each person as a separate individual, we tend to see people as members of a group – for example: people wearing while coat are doctors.

➢ We use cognitive structure to make sense of people's behaviour. If we come across woman doctor, we use our information about doctor more generally to infer her attributes and meaning of her behaviour.

➢ Perceivers own needs and personal goals influence how he or she perceives others. Characteristics, a rather 'warm' person and to others that he was 'cold' person. The lecturer gave an identical talk to each group and when each of class members was asked about their impressions, each group formed

the impression as per earlier instructions that were consistent with their initial description of 'warm' or cold.

Why are our impressions of others sometimes wrong?

Our impressions are sometimes wrong because of the mental shortcuts. We use when forming social judgment people are too ready to attribute other's actions to their personalities rather than to the situation. Another reason our impression can go wrong concerns our use of schemas. People use implicit personality theories to fill in the gaps in their knowledge about other people and use schemas or theories to decide why other people what they do.

Why does it seem like our impressions are accurate? Why does it seem like our impressions are accurate when as we have seen that 'our impressions can go wrong'. These are three reasons:

First – we often see people in a limited number of situations and thus never have the opportunity to see that our impressions are wrong.

Second - we will not realize that our impressions are wrong if we make them come true, this is the case with self-fulfilling prophecies even if an initial impression is incorrect, we often make it come true.

Third – we might not realize we are wrong if a lot of people agree on what someone's is like – even when everyone is wrong.

Impression Management The desire to make a favorable impression on others is a strong one so most of us do our best to look good to others when we meet them for the first time. Social psychologists use the term impression management (or self-presentation) to describe these effort to make a good impression on others and the results of their research on such efforts suggest that they are worthwhile.

The different techniques for boostering their image fall into two categories: self enhancement – effort to increase their appeal to others and other enhancement – effort to make the target person feel good in various ways. With respect to selfenhancement, the strategies include effort to boost one's physical appearance through style of dress, personal grooming and the use of various props. (example: eye glasses). Additional tactics of self enhancement involves efforts to describe oneself in positive terms.

Turning to other enhancement, individuals use many different tactics to induce positive moods and reaction in others example: flattery.

According to William James observation, people often shows different sides of themselves to different groups of people. Youngsters who are demure and obedient before their parents and teachers and may swear and swagger in the company of their friends. Most parents do not show their actual self to their children as they do to their colleagues or intimate friends. We usually talk as if there is a single self who is stable and well defined. But social psychologists believe it is more appropriate to

think of self as multiple selves because people display different aspects of themselves in different situations. Schlenker (1980) has termed this as 'impression management' and defined it as the conscious or unconscious attempt to control images that are projected in real or imagined social interactions. When their images deal with some aspects of self we call the process as self presentation.

PERSON PERCEPTION Person perception is a process of forming impressions of people. Let us understand how do we go about it? What kind of information we use, how accurate it is and what bias affect our impression

Mechanisms of Person Perception Mechanisms of person perception – includes ways of impressions about the other person, for this people use whatever obvious physical cues, their behaviour role which are available to them. Also they further develop mental shortcuts by forming schemas and heuristics and further go and categorize them according to some rules of thumb.

A glance at someone's picture or at an individual passing by on the street gives us an idea about what kind of person s/he is. In fact when two people meet, if only for a moment, they start forming impressions of each other. With more contact they form fuller and richer information of that person. In forming impressions of other person, we depend on the following to help us form impression about a person.

i) Roles/traits
ii) Physical cues and
iii) Salience

I. **Roles/traits:** If roles are informative, rich and well articulated, it can help in summarising a lot of information across a wide range of situations. Roles are more distinctive than traits. Moreover role schemas are more useful than traits for recall ex: try to recall the names of all the people you met in a seminar and next try to recall rude and stubborn people you met, which task is easier, I am sure it is the first task, because people tend to think of others within a 'role context' first and only then according to their personality traits.

II. **Physical cues:** The other person's physical appearance and behaviours are usually taken together to form an impression about that person. The observation that a person is wearing traditional clothes help us infer that the person might be from rural background. We even infer personality traits from person's behaviour – A person who is helping others is inferred as kind, or a chubby faced person is thought to be warm etc.

III. **People's attention** is to drawn to those aspects or characteristics of the persons which stand out. This is termed as 'figure ground' principle. A more outstanding feature or salient feature like fair skinned, handicapped, high pitched voice etc., is used to form the overall impression about the person as a total person. Salient behaviours draw more attention than do less obvious ones. It also influences perception of causality. But a disadvantage here is that salience also can produce 'extreme evaluative judgment'.

We do not stop at just forming impressions from observable information such as appearance, behaviour, but we move on to quickly form inferences about personality traits about the person

from this information. Referring to trait is more economical and general way of describing a person than is referring to behaviour – if someone asked you about what your room mate is like – instead of describing about each and every behaviour of his in detail, it is easier to say that 'he is cool' or 'he is good natured'. This process of inferring also occurs spontaneously, even automatically. These traits can act as bases for predicting future behaviour.

Lesson-5 Schemas- Types of Schema, Person Schemas, Self Schemas, Group Schemas, Role Schemas, Events Schemas, Impact of Schema.

Schemas Social cognition is the process in which people think about themselves and the social world – how they select interpret, remember and use social information to make judgments and decisions. How people think about the social world's people rely on a variety of mental shortcuts that serve them well. They use quite practical, adopt different procedures and rules according to their goals and needs in that situation. Like a scientists, all of us, in our everyday lives, develop theories that help us understand ourselves and the social world. These theories called Schemas are mental structures people use to organise their knowledge around themes or topics or subject. A schema is a way of representing the memory process. In other words, people do not respond to what they see, rather they interpret it on the basis of previous experience that is represented in memory as a schema. Self schema is defined as a generalization about the self based on experience we use to interpret events related to the self. Similarly we have schemata about other people too. The general form of schema can be defined as an organized configuration of knowledge, derived from past experience that is used to interpret our experience. We have schemas about many things – other people, ourselves, social roles and specific events. Schemas effect what information we notice, think about and remember. Schemas are often resistant to change – they show a strong perseverance effect, remaining unchanged even in the face of contradictory information. Perhaps even worse schemas can sometime be self fulfilling People have role schema – these represent the organised abstract concepts we have of people in a particular role, such as cowboy, professor, devoted lover. Other schemas focus on groups. The most familiar is the group stereotype – it is belief about typical characteristics of members of a group or social category, and which attributes specific traits to a particular group of people. Example: certain communities are assumed to be stingy (money wise), absent minded or aggressive etc. Prototype – is an abstract ideal of the schema – for example may be about his body type, mannerisms etc. we assume leaders to be busy – so a well composed and relaxed leader goes contrary to our schema. Schemas about extremely common events are usually called scripts. A script is a standard sequence of behaviour over a period of time. The essence of a script is in its boundedness in time, its causal flow and its being simple, coherent, perceptual unit – ex: a wedding in a community goes through a serious of rituals connected to each other and follow a pattern many schemas contain. Ex: a category that embodies the significant attribute or ideal of the category – for example in India the name of Sachin Tendulkar is an exemplar of the category 'cricket'. **Function or schematic processing Advantages**

Schemas and information processing – schemas help us process an enormous amount of information swiftly and economically. Schemas aid recall – memory often works best when we have a schema representation of past event or people. Schema speed up processing – it is easier to process information relevant to the schema. Schemas aid automatic inference – schematic processing can occur almost automatically, without any conscious effort. When meeting a friendly person, you may automatically attribute to him other characteristics associated with friendliness – kindness wart. Schema aid information – schema can help us fill in missing information where there are gaps in our knowledge. Schemas aid interpretation – schemas help you interpret ambiguous situation. A person having no knowledge about 'mumps' would have no confusion about other inferences about 'mumps' schemas allow confident inferences about matters that would otherwise not be clear. Schemas provide expectations – schemas contain expectations about what should happen expected match lead to unpleasantness. Schemas contain affect – use of a particular schema can produce an emotional response called 'schema-drives-affect' - for example schema for politician. This schema can have 'affective response' attached to it. Some of us feel positively about politician and think of them as helpful law makers, but others may feel negatively thinking of them as power hungry and corrupt.

Lesson-6 Mental frame works- Interpersonal attraction. Communication system in human beings Nonverbal communication; the language of expression

INTERPERSONAL ATTRACTION To the query why it is that a particular person has evoked our positive regard, probably we will reply by making reference to some of the person's "good qualities" such as the person's honesty, sense of humor, or even the particular hair style that the person sports etc. While explanations referring to qualities which seem to "compel" admiration are frequently given, it must be kept in mind that interpersonal attraction is much more complicated than such explanations would imply. In some cases, people are attracted to those persons whom they perceive as similar to themselves. The effect is very small for superficial features like clothes or race but very strong for perceived similarity of attitudes.

In certain other cases, we like people who seem to agree with us because.

a) we think they're smart,

b) we will probably get along, and

c) they'll probably like us too.

To pretend to agree with someone even when you do not really, for the purpose of getting something they can give you, like a job, is called ingratiation. In general this works best if you pretend to agree in about 70%. Research shows that less is not enough, and more is both suspicious and boring. Interpersonal attraction has been an important topic of research in psychology, because humans are social animals, and attraction serves an important function in forming a social network, which in turn provides security and satisfies people's need to belong to a social group (Baumeister & Bushman, 2008). In assessing the nature of attraction, psychologists have used methods such as questionnaires, survey, and rating scale to determine level of one's attraction toward another. Here, the effects of similarity, social reward, familiarity, and physical attractiveness are examined to see how they impact interpersonal attraction. When measuring

interpersonal attraction, one must refer to the qualities of the attracted as well as the qualities of the attractor to achieve predictive accuracy. It is suggested that to determine attraction, personality and situation must be taken into account. Many factors leading to interpersonal attraction have been studied. The most frequently studied are: physical attractiveness, propinquity, familiarity, similarity, complementarily, reciprocal liking, and reinforcement.

Physical Attractiveness Despite the old sayings that "beauty is only skin deep" and "you can not judge a book by its cover", we tend to operate according to Aristotle's 2000-year-old pronouncement that "personal beauty is a greater recommendation than any letter of introduction". One of the most commonly cited factors influencing attraction is physical attractiveness. It is indeed well known that most people show a substantial preference for attractive persons over unattractive others (Baumeister & Bushman, 2008). Perhaps the advantage of good looks and the inferences people make when they see a physically attractive person are the reasons for getting attracted to another person. Studies have shown that when people see an attractive person, they believe that there is more than physical beauty that they see, and they tend to assume certain internal qualities within the person, such as kindness, outgoing, etc. (Barocas & Karoly, 1972). To illustrate this factor, let us take the research illustrating this relationship between physical attractiveness and its stereotypes. For instance, a study on popularity among adolescents was carried out by Cavior & Dokecki in 1973. They found that when physical attractiveness was compared to perceived attitude similarity, physical attractiveness had a stronger effect on popularity. These findings suggest that individuals' perceptions of attitude similarity with those of others may be strongly influenced by more automatic judgments of physical attractiveness. Such demonstrations of preferential treatment may have significant implications at the level of society, as well. For example, in one jury task simulation experiment, more attractive defendants were found to be evaluated more positively and with less certainty of guilt than were other less attractive defendants. Even though physical attractiveness is unrelated to objective measure of internal qualities such as intelligence and personality, many researches indicate that bias for beauty is pervasive in society

Propinquity/Proximity According to Rowland Miller's Intimate Relationships text, the propinquity effect can be defined as: "the more we see and interact with a person, the more likely he or she is to become our friend or intimate partner." This effect is very similar to the mere exposure effect in that the more a person is exposed to a stimulus, the more the person likes it; however, there are a few exceptions to the mere exposure effect.

Similarity The notion of "birds of a feather flock together" points out that similarity is a crucial determinant of interpersonal attraction. According to Morry's attraction similarity model (2007), there is a lay belief that people with actual similarity produce initial attraction. Perceived similarity develops for someone to rate others as similar to themselves in ongoing relationship. Such perception is either self serving (friendship) or relationship serving (romantic relationship). Newcomb (1963) pointed out that people tend to change perceived similarity to obtain balance in a relationship. Additionally, perceived similarity was found to be greater than actual similarity in predicting interpersonal attraction

COMMUNICATION Communication is a process of interpersonal interaction in which thoughts, ideas, emotions and understanding between sender(s) and receiver(s) are exchanged (Guo and Sanchez, 2005). This simple definition of communication directs attention to three important issues:

✓ Communication involves transmission and reception of messages.

✓ Communication involves people, at least two: one to transmit the message (sender) and another to receive the message (receiver).

✓ Communication is best described as a process because it is active, continuous, reciprocal and dynamic.

Types of Communication There are several forms of communication, the most important being verbal communication. Verbal communication can further be divided into two forms: oral and written. Most basic to the oral form of communication involves the spoken word; as it is the quickest and most accurate because messages can be clarified through ongoing dialogue. Written form of communication involves exchange of ideas, thoughts, understanding, etc. with the help of letters, memos, office orders, e-mail, instant messaging, blogs, etc. Another form of communication is non-verbal which consists of unspoken clues that a communicator sends in conjunction with spoken or written messages. Examples may include a person's tone of voice, facial expressions, eye contact, head nodding, posture, way of walking, etc Though verbal and non verbal communications are separate, both operate at the same time. The verbal part of a message conveys content or information. The non verbal component indicates how the verbal message should be interpreted and thus it is miscommunication. Whenever verbal and non verbal messages contradict each other, people are more likely to believe the non verbal.

Communication Style Every individual has his or her own predominant communication style. People with different communications styles use different verbal and non verbal communication patterns. The communication styles can be broadly divided into four types: passive, aggressive, passive-aggressive and assertive. Each communication style conveys certain messages to listeners and leads to a specific outcomes.

➢ **Passive communication:** People with passive communication style are considered as people pleasers, shy and they usually go along with others to avoid conflict. They generally exhibit confused body language and avoid eye contact, place their hands at or over mouth, often play with pen, pencil, clips, paper, etc. during conversation and chew pen or pencil while listening. Because of their body language they are perceived as individual with low selfesteem weak, timid and indecisive.

➢ **Aggressive communication:** Those who use this style of communication often disregard the rights of others and force their own needs and opinions onto other people. This form of communication can lead to shouting and in some cases, physical aggression. Such people often use aggressive tone and gestures to emphasis their point of view.

➢ **Passive-aggressive communication**: In this style of communicating, a person may use passive means of communicating that have an aggressive result, because they often find it difficult to direct their thoughts and feelings. They sometimes use manipulation to get what they want which apparently does not look like manipulation.

➢ **Assertive communication:** People with an assertive style of communicating are able to respectfully express their thoughts and feelings clearly. They consider and value their own needs, but also the needs of others. This form of communicating can help to build strong, balanced, and respectful relationships with others.

Barriers to Effective Communication Anything that filters, blocks or distorts the message or the information during the process of "encoding-sending-decoding" between the sender and the receiver is considered as barriers of communication. Barriers of communication can be broadly classified into two types (Longest, Rakich, & Darr, 2000):

Physical-environmental barriers: Environmental and physical barriers of communication pertain to the physical-environmental components of the communication process. These may include noise in the communication channel, an improper time of communication, significant distance between sender and receiver.

Psychological barriers: Psychological barriers are created due to poor listening skills, biases, prejudices, mistrust, negative attitude, fear of failure, evaluation anxiety, public embarrassment, obsessive thoughts of being judged negatively by others, inferiority complex, etc

Unit – III: Socialization & Social Influence

Lesson-7 Socialization- Definition, Motive and behavior.

Socialization

Socialization is the lifelong process through which people learn the values and norms of a given society. Socialization is not the same as socializing. Socializing is to mix socially with others (i.e., family, friends, neighbors, coworkers), whereas socialization is a process that may include socializing as one element, but is a more complex, multi-faceted and formative set of interactive experiences.

It is also an adaptive lifelong learning experience, because society is constantly changing, and because we may find ourselves in new situations—such as a new job with different norms and values, or in a different familial role—such as that of parent or caregiver to an older relative.

Socialization, the process whereby an individual learns to adjust to a group (or society) and behave in a manner approved by the group (or society). According to most social scientists, socialization essentially represents the whole process of learning throughout the life course and is a central influence on the behavior, beliefs, and actions of adults as well as of children.

Motive and Behavior

A motive is something that causes us to act or behave in order to reach a a goal or desired endpoint. It comes from the Latin word that means 'moving'. For example, a motive for exercise is better health and weight loss.

Psychologists have divided motives into three types—Biological motives, Social motives and Personal motives.

1. Biological Motives

Biological motives are called as physiological motives. These motives are essential for the survival of the organism. Such motives are triggered when there is imbalancement in the body. The body always tends to maintain a state of equilibrium called "Homeostasis"- in many of its internal physiological processes.

2. Social Motives

Physiological motives discussed above pertain to both animals as well as human beings, but the social motives are specific only to human beings. These are called social motives,because they are learnt in social groups as a result of interaction with the family and society. That is why their strength differs from one individual to another.

In addition to the above there are some other social motives like need for self-esteem, social approval, self-actualization, autonomy, master motive, combat, defense, abasement, etc.

3. Personal Motives

In addition to the above said physiological and social motives, there are some other motives which are allied with both of the above said motives. These are highly personalized and very much individualized motives.

Behavior

Behavior (American English) or **Behaviour** (British English; see spelling differences) is the actions and mannerisms made by individuals, organisms, systems or artificial entities in conjunction with themselves or their environment, which includes the other systems or organisms around as well as the (inanimate) physical environment. It is the computed response of the system or organism to various stimuli or inputs, whether internal or external, conscious or subconscious, overt or covert, and voluntary or involuntary.

Behaviors can be either innate or learned from the environment. Behavior can be regarded as any action of an organism that changes its relationship to its environment. Behavior provides outputs from the organism to the environment.

Behavior can be defined as the actions or reactions of a person in response to external or internal stimulus situation." To understand the behavior of a person we have to understand what that person will do if something happens.

Classifications of Human Behavior

Based on the way people act or behave in different situations and in response to different stimuli, human behavior can be divided into different types.
Here are the common types of behaviors human beings can have:

1. Molecular and Moral Behavior

Molecular Behavior**:** It is an unexpected behavior that occurs without thinking. One example is suddenly closing eyes when something is about to this the eyes.
Moler Behavior**:** Unlike molecular behavior, this type of behavior occurs after thinking. For example, a person changes the way when she or he sees a harmful thing.

2. Overt & Covert Behavior

Overt Behavior: It is a visible type of behavior that can occur outside of human beings. Eating food, riding on a bicycle, and playing football are some examples. Covert Behavior: Unlike overt behavior, this type of behavior is not visible. Thinking is agood example of covert behavior because no one can see us thinking.

3. Voluntary and Involuntary Behavior

Voluntary Behavior**:** It is a type of behavior that depends on human want. We can œcharacterize walking, speaking, and writing as voluntary behaviors. Involuntary Behavior: Unlike voluntary behavior, this type occurs naturally and without thinking. Breathing air is a perfect example of involuntary behavior.

Lesson-8 Agencies of socialization- (Parents, Peergroup, school and media)

Agencies of Socialization

In general, it may be said that the total society is the agency for socialization and that each person with whom one comes into contact and interact is in some way an agent of socialization. Socialization is found in all interactions but the most influential interactionoccurs in particular groups which are referred to as agencies of socialization.

The oblivious beginning of the process for the new-born child is-his immediate family group, but this is soon extended to many other groups. Other than the family, the most important are the schools, the peer groups (friends circle) and the mass media.

1. Family

The child's first world is that of his family. It is a world in itself, in which the child learns to live, to move and to have his being. Within it, not only the biological tasks of birth, protection and feeding take place, but also develop those first and intimate associations with persons of different ages and sexes which form the basis of the child's personality development.

The family is the primary agency of socialization. It is here that the child develops an initial sense of self and habit-training—eating, sleeping etc. To a very large extent, the indoctrination ofthe child, whether in primitive or modem complex society, occurs within the circle of the primary family group. The child's first human relationships are with the immediate members of his family—mother or nurse, siblings, father and other close relatives.

Here, he experiences love, cooperation, authority, direction and protection. Language (a particular dialect) is also learnt from family in childhood. People's perceptions of behavior appropriate of their sex are the result of socialization and major part of this is learnt in the family.

As the primary agents of childhood socialization, parents play a critical role in guiding children into their gender roles deemed appropriate in a society. They continue to teach gender role behavior either consciously or unconsciously, throughout childhood. Families also teach childrenvalues they will hold throughout life. They frequently adopt their parents' attitudes not onlyabout work but also about the importance of education, patriotism and religion.

2. School

After family the educational institutions take over the charge of socialisation. In some societies (simple non-literate societies), socialisation takes place almost entirely within the family but in highly complex societies children are also socialised by the educational system. Schools not only teach reading, writing and other basic skills, they also teach students to develop themselves, to

discipline themselves, to cooperate with others, to obey rules and to test their achievements through competition.

Schools teach sets of expectations about the work, profession or occupations they will follow when they mature. Schools have the formal responsibility of imparting knowledge in those disciplines which are most central to adult functioning in our society. It has been said that learning at home is on a personal, emotional level, whereas learning at school is basically intellectual.

3. Peer group

Besides the world of family and school fellows, the peer group (the people of their own age and similar social status) and playmates highly influence the process of socialisation. In the peer group, the young child learns to confirm to the accepted ways of a group and to appreciate the fact that social life is based on rules. Peer group becomes significant others in the terminology of G.H. Mead for the young child. Peer group socialisation has been increasing day by day these days.

Young people today spend considerable time with one another outside home and family. Young people living in cities or suburbs and who have access to automobiles spend a great deal of time together away from their families. Studies show that they create their own unique sub-cultures—the college campus culture, the drug culture, motorcycle cults, athletic group culture etc. Peer groups serve a valuable function by assisting the transition to adult responsibilities.

Teenagers imitate their friends in part because the peer group maintains a meaningful system of rewards and punishments. The group may encourage a young person to follow pursuits that society considers admirable.

On the other hand, the group may encourage someone to violate the culture's norms and values by driving recklessly, shoplifting, stealing automobiles, engaging in acts of vandalism and the like. Some studies of deviant behaviour show that the peer group influence to cultivate behaviour patterns is more than the family.

Why do some youths select peer groups which generally support the socially approved adult values while others choose peer groups which are at war with adult society? The choice seems tobe related to self-image. Perhaps, this dictum works—"seeing- is behaving". How do we see ourselves is how we behave.

The habitual delinquent sees himself as unloved, unworthy, unable, unaccepted and unappreciated. He joins with other such deprived youths in a delinquent peer group which reinforces and sanctions his resentful and aggressive behaviour. The law-abiding youth sees himself as loved, worthy, able, accepted and appreciated. He joins with other such youths in a conforming peer group which reinforces socially approved behaviour.

4. Mass media

From early forms of print technology to electronic communication (radio, TV, etc.), the media is playing a central role in shaping the personality of the individuals. Since the last century, technological innovations such as radio, motion pictures, recorded music and television have become important agents of socialisation.

Television, in particular, is a critical force in the socialisation of children almost all over the new world. According to a study conducted in America, the average young person (between the ages of 6 and 18) spends more time watching the 'tube' (15,000 to 16,000 hours) than studying in school. Apart from sleeping, watching television is the most time-consuming activity of young people.

Relative to other agents of socialisation discussed above, such as family, peer group and school, TV has certain distinctive characteristics. It permits imitation and role playing but does not encourage more complex forms of learning. Watching TV is a passive experience. Psychologist Urie Bronfenbrenner (1970) has expressed concern about the 'insidious influence' of TV in encouraging children to forsake human interaction for passive viewing.

Lesson-9 Sexual motives, Sex role identity, Psychological Androgyny, Dependency, Moral development.

Sexual motives

Sexual motivation also called sex drive, the impulse to gratify sexual needs, either through direct sexual activity or through apparently unrelated activities (sublimation). The term libido was coined by Sigmund Freud and used by him to encompass the seeking of pleasure in general, one of the major motivating forces for human activity. Freud suggested that this drive had a genetic basis as part of the species' need to continue itself. Unlike other basic drives such as hunger, gratification of the libido can be postponed or transferred without endangering the individual's existence. Thus, sexual drives are more likely to be sublimated, or channeled into other avenues of gratification, to achieve socially acceptable goals.
Sexual motivation or libido develops throughout most of the individual's life. Freud identified a series of stages in libido development beginning in infancy, and he suggested that an individual could become fixed in one of these immature stages and not pass on to the mature adult stage of sexuality, in which the chief physical mode of sexual pleasure is genital.

In Freudian psychology, earlier concepts of sexuality were widened to cover all activities from which one derives pleasure. Thus, cultural and intellectual achievements could be seen as the outcome of sublimated sexual drives. As he developed the theory further, Freud saw libido as being in conflict with aggression, and other drives were considered to result from the balance between these two strong motivating forces.

Estrogen, progesterone, and testosterone are hormones that affect a woman's sexual desire and functioning. When it comes to sexual desire, the most influential hormone is testosterone. Though it's often considered a male hormone, testosterone — like estrogen– is present in both men and women, though the proportions differ between the sexes. In women, testosterone is produced through the operation of the adrenals glands — two small glands near the kidneys — and the ovaries.

The hormones that have been regulating your reproductive cycle, sex drive, mood, and more are ebbing, and very often these low levels have a negative effect on your sex life. Reduced levels of estrogen, testosterone, and other hormones are thought to be a primary cause of age-related female sexual dysfunction.

Types of female sex hormones

The two main female sex hormones are estrogen and progesterone. Although testosterone is considered a male hormone, females also produce and need a small amount of this, too.

Estrogen

Estrogen is the major female hormone. The lion's share comes from the ovaries, but small amounts are produced in the adrenal glands and fat cells. During pregnancy, the placenta also makes estrogen.

Estrogen plays a big role in reproductive and sexual development, including:

- puberty
- menstruation
- pregnancy
- menopause

Estrogen also affects the:

- brain
- cardiovascular system
- hair
- musculoskeletal system
- skin
- urinary tract

Estrogen levels can be determined by a blood test. While it can vary from person to person, these are what are considered the normal ranges in picograms per milliliter (pg/mL):

- Adult female, premenopausal: 15-350 pg/mL
- Adult female, postmenopausal: <10 pg/mL
- Adult male: 10-40 pg/mL

Levels will vary greatly throughout the menstrual cycle.

Progesterone

The ovaries produce the female sex hormone progesterone after ovulation. During pregnancy, the placenta also produces some.

The role of progesterone is to:

- prepare the lining of the uterus for a fertilized egg
- support pregnancy
- suppress estrogen production after ovulation

Progesterone levels can be determined by a blood test. Normal ranges are in nanograms per milliliter (ng/mL):

Phase	**Range**
before puberty	0.1–0.3 ng/mL
during first (follicular) stage of menstrual cycle	0.1–0.7 ng/mL
while ovulating (luteal stage of cycle)	2–25 ng/mL
first trimester of pregnancy	10–44 ng/mL
second trimester	19.5–82.5 ng/mL
third trimester	65–290 ng/mL

Testosterone

Small amounts of testosterone come from the adrenal glands and ovaries. This hormone plays a role in several body functions, including:

- sexual desire
- regulation of the menstrual cycle
- bone and muscle strength

A blood test can determine your testosterone level. The normal range for females is 15 to 70 nanograms per deciliter (ng/dL).

Hormones in men

What is testosterone?

Testosterone is the most important androgen (male sex hormone) in men and it is needed for normal reproductive and sexual function, muscle and bone strength. Testosterone is important for the physical changes that happen during male puberty, such as development of the penis and testes, and for the features typical of adult men such as facial and body hair. Testosterone also acts on cells in the testes to make sperm.

What controls the production of testosterone?

The pituitary gland and the hypothalamus, located at the base of the brain, control the production of male hormones and sperm. Hormone messages are sent from these glands to the testes where testosterone is produced and sperm are made. The level of testosterone in the blood feeds back to the pituitary gland and the hypothalamus and when the level is low more hormone messages are sent to increase the production. The Luteinizing hormone (LH) and follicle stimulating hormone (FSH) are the two important messenger hormones made by the pituitary gland that act on the testes. LH is needed for the Leydig cells in the testes to make testosterone that leads to the production of sperm.

Reproduction

The testes hang 'outside' the body in the scrotum because sperm production needs a cooler temperature than the inside of the body. Because the testes are outside the main body cavity they are more vulnerable to injury, but also it is easier for a man to discover lumps or bumps.

Testosterone deficiency

In men with low testosterone levels, the symptoms that may be experienced include low energy levels, little interest in sex, feeling sad, down or depressed, mood, irritability, poor concentration, and reduced muscle strength. Symptoms often overlap with those of other illnesses. The symptoms of androgen deficiency are different depending on the age when testosterone levelsare below the normal range.

Sexual motivation can be measured using a variety of different techniques. Self-report measures, such as the Sexual Desire Inventory, are commonly used to detect levels of sexual motivation in humans. Self-report techniques such as the bogus pipeline can be used to ensure individuals do not falsify their answers to represent socially desirable results. Sexual motivation can also be implicitly examined through frequency of sexual behaviour, including masturbation.

Sex role identity

In the social sciences and humanities, **a sex role or gender role** is a set of behavioral norms of males and females, respectively, in a social group or system. Gender is one component ofthe gender/sex system, which refers to "The set of arrangements by whicha society transforms biological sexuality into products of human activity, and in which these transformed needs are satisfied" (Reiter 1975: 159.). Every known society has a gender/sex system, although the components and workings of this system vary widely from society to society.

Gender role is composed of several elements. A person's gender role can be expressed through clothing, behaviour, choice of work, personal relationships and other factors. Gender roles were traditionally divided into strictly feminine and masculine gender roles, though these roles have diversified today into many different acceptable male or female gender roles. However, gender role norms for women and men can vary significantly from one country or culture to another, even within a country or culture. People express their gender role somewhat uniquely. Gender role can vary according to the social group to which a person belongs or the subculture with which he or she chooses to identify.

A gender role, also known as a sex role, is a social role encompassing a range of behaviors and attitudes that are generally considered acceptable, appropriate, or desirable for a person based on that person's biological or perceived sex.

Gender identity is defined as a personal conception of oneself as male or female (or rarely, both or neither). This concept is intimately related to the concept of gender role, which is defined as the outward manifestations of personality that reflect the gender identity.

Psychological Androgyny

In psychology, androgyny refers to individuals with strong personality traits associated with both sexes, combining toughness and gentleness, assertiveness and nurturing behaviour, as called for

by the situation. Androgynous individuals are more likely to engage in cross-sexual behaviour than those who maintain traditional sex roles.

Androgyny has a long history, although not always by that name. The term, however, gained more prominence in the 1970s when **Bem**, a well known gender scholar, coined the concept of psychological androgyny.

Psychological androgyny referred to the idea that an individual could have both masculine and feminine qualities. More importantly, it reflected the idea that having this mixture of qualities could be a good thing, and was associated with flexibility and better adjustment.

Some researchers argue that there is both positive and negative androgyny. Positive androgyny is where individuals have positive characteristics associated with both masculinity and femininity such as compassion and independence.

In contrast, negative androgyny is associated with having negative masculine and feminine characteristics, such as being temperamental and aggressive. Perhaps unsurprisingly, positive androgyny is associated with better mental health and well-being than negative androgyny.

One criticism of psychological androgyny is that it assumes that certain characteristics are masculine and feminine, rather than seeing masculinity and femininity as being culturally constructed. In that way, psychological androgyny could also be thought of as describing those who are less invested in cultural constructs around gender and its expression.

Dependency

The reliance on others to meet social or psychological needs. Some level of dependency is widely considered normal in human development, however, severe dependency can cause psychological problems. Dependency is sometimes characterized as a personality trait.

Dependency is a need for others to meet one's psychological, social, physical, or financial needs. Any reliance on others can be characterized as a form of dependency, and all people are dependent on others to some extent. Humans are social animals, and without any interactions with others may experience serious psychological consequences.

Things Dependent People Do

1. Dependent people have difficulty making everyday decisions without advice and reassurance.
2. They need others to assume responsibility for many major areas of life.
3. 3. They have difficulty disagreeing with others out of fear.
4. They struggle to start projects or do things on their own.
5. They feel anxious or distressed when alone, or when thinking about being alone.
6. They make themselves responsible when bad things happen.
7. They feel responsible for fulfilling the expectations of others.
8. They have a high need for validation and approval from others.

Moral development

Moral development refers to the ways we distinguish right from wrong as we grow and mature. Very young children generally do not have the same level of moral development as adults.

Lawrence Kohlberg, building upon Jean Piaget's cognitive theory of development, developed three levels with six stages to reflect our progression through moral development. These levels and stages describe how our ideas of right and wrong change as we grow.

Level 1: Preconventional level

At the Preconventional level, morality is externally controlled. Rules imposed by authority figures are conformed to in order to avoid punishment or receive rewards. This perspective involves the idea that what is right is what one can get away with or what is personallysatisfying. Level 1 has two stages.

Stage 1: Punishment/obedience orientation

Behaviour is determined by consequences. The individual will obey in order to avoid punishment.

Stage 2: Instrumental purpose orientation

Behaviour is determined again by consequences. The individual focuses on receiving rewards or satisfying personal needs.

Level 2: Conventional level

At the conventional level, conformity to social rules remains important to the individual. However, the emphasis shifts from self-interest to relationships with other people and social systems. The individual strives to support rules that are set forth by others such as parents, peers, and the government in order to win their approval or to maintain social order.

Stage 3: Good Boy/Nice Girl orientation

Behaviour is determined by social approval. The individual wants to maintain or win the affection and approval of others by being a "good person."

Stage 4: Law and order orientation

Social rules and laws determine behaviour. The individual now takes into consideration a larger perspective, that of societal laws. Moral decision making becomes more than consideration of close ties to others. The individual believes that rules and laws maintain social order that is worth preserving.

Level 3: Post-conventional or principled level

At the post conventional level, the individual moves beyond the perspective of his or her own society. Morality is defined in terms of abstract principles and values that apply to all situations and societies. The individual attempts to take the perspective of all individuals.

Stage 5: Social contract orientation

Individual rights determine behaviour. The individual views laws and rules as flexible tools for improving human purposes. That is, given the right situation, there are exceptions to rules. When laws are not consistent with individual rights and the interests of the majority, they do not bring about good for people and alternatives should be considered.

Stage 6: Universal ethical principle orientation

According to Kohlberg, this is the highest stage of functioning. However, he claimed that some individuals will never reach this level. At this stage, the appropriate action is determined by one's self-chosen ethical principles of conscience. These principles are abstract and universal in application. This type of reasoning involves taking the perspective of every person or group that could potentially be affected by the decision.

Lesson 10: Self-Esteem

Self-esteem is how a person thinks and feels about their own qualities and characteristics. This is described as either high or low that is, a person has either high self-esteem or low self-esteem.

More specifically, a person who has positive thoughts about their qualities and characteristics would have high self-esteem, whereas a person with negative thoughts about their qualities and characteristics would have low self-esteem.

This is important because higher levels of self-esteem translate into improved mental health, and lower levels of self-esteem are related to mental health conditions such as depression. This means self-esteem is an important component of overall health and well-being.

Some common signs or symptoms of low self-esteem include:

- Live in present
- Flexible
- Leaders with a clear vision
- Transparent
- Take care of them
- Goal oriented
- Believe in them
- Problem solver

Some common signs or symptoms of low self-esteem include:

- Lack of confidence
- Negative social comparisons
- Trouble asking for help
- Worry and doubt
- Difficulty accepting compliments
- Negative self-talk
- Fear of failure
- Poor outlook of the future

Lesson 11: Self concept

Self-concept is the image we have of ourselves. It is influenced by many forces, including our interaction with important people in our lives. It is how we perceive our behaviors, abilities, and unique characteristics. For example, beliefs such as "I am a good friend" or "I am a kind person" are part of an overall self-concept.

Our self-perception is important because it affects our motivations, attitudes, and behaviors. It also affects how we feel about the person we think we are, including whether we are competent or have self-worth.

Self-concept tends to be more malleable when we're younger and still going through self-discovery and identity formation. As we age and learn who we are and what's important to us, these self-perceptions become much more detailed and organized. **Carl Rogers** believed that self-concept is made up of three different parts:

- **Ideal self**: The ideal self is the person you want to be. This person has the attributes or qualities you are either working toward or want to possess. It's who you envision yourself to be if you were exactly as you wanted.
- **Self-image**: Self-image refers to how you see yourself at this moment in time. Attributes like physical characteristics, personality traits, and social roles all play a role in your self-image.
- **Self-esteem**: How much you like, accept, and value yourself all contribute to your self-concept. Self-esteem can be affected by a number of factors—including how others see you, how you think you compare to others, and your role in society.

Self-concept develops, in part, through our interaction with others. In addition to family members and close friends, other people in our lives can contribute to our self-identity. For instance, one study found that the more a teacher believes in a high-performing student's abilities, the higher that student's self-concept.

The media plays a role in self-concept development as well—both mass media and social media. When these media promote certain ideals, we're more likely to make those ideals our own.

Unit – IV: Attitudes

Lesson-12 Attitudes -Nature and formation of Attitudes, Functions of Attitudes.

An attitude is a positive, negative, or mixed evaluation of an object expressed at some level of intensity. It is an expression of a favorable or unfavorable evaluation of a person, place, thing, or event. These are fundamental determinants of our perceptions of and actions toward all aspectsof our social environment. Attitudes involve a complex organization of evaluative beliefs, feelings, and tendencies toward certain actions.

Some examples of attitudes are- he has a positive attitude about the changes, she is friendly and has a good attitude, he was showing some attitude during practice today, so the coach benched him, I like my friends that means I am expressing my attitudes towards my friends, etc.

Definition of Attitude

Attitude is the manner, disposition, feeling, and position about a person or thing, tendency, or orientation, especially in mind.

According to **Gordon Allport**, "An attitude is a mental and neural state of readiness, organized through experience, exerting a directive or dynamic influence upon the individual's response to all objects and situations with which it is related."

Frank Freeman said, "An attitude is a dispositional readiness to respond to certain institutions, persons or objects in a consistent manner which has been learned and has become one's typical mode of response."

Thurstone said, "An attitude denotes the total of man's inclinations and feelings, prejudice or bias, preconceived notions, ideas, fears, threats, and other any specific topic."

Anastasi defined attitude as "A tendency to react favorably or unfavorably towards a designated class of stimuli, such as a national or racial group, a custom or an institution."

According to **N.L. Munn**, "Attitudes are learned predispositions towards aspects of our environment. They may be positively or negatively directed towards certain people, service, or institution."

"Attitudes are an 'individual's enduring favorable or unfavorable evaluations, emotional feelings, and action tendencies toward some object or idea." — David Krech, Richard S. Crutchfield, and Egerton L. Ballackey.

"Attitude can be described as a learned predisposition to respond in a consistently favorable or unfavorable manner for a given object." — Martin Fishbein and Icek Ajzen.

"An attitude is a relatively enduring organization of beliefs around an object or situation predisposing one to respond in some preferential manner." — Milton Rokeach.

Characteristics of Attitudes

1. Attitudes are the complex combination of things we call personality, beliefs, values, behaviors, and motivations.
2. It can fall anywhere along a continuum from very favorable to very unfavorable.
3. All people, irrespective of their status or intelligence, hold attitudes.
4. An attitude exists in every person's mind. It helps to define our identity, guide our actions, and influence how we judge people.
5. Although the feeling and belief components of attitude are internal to a person, we can view a person's attitude from their resulting behavior.
6. Attitude helps us define how we see situations and define how we behave toward the situation or object.
7. It provides us with internal cognitions or beliefs and thoughts about people and objects.
8. It can also be explicit and implicit. Explicit attitude is those that we are consciously aware of an implicit attitude is unconscious, but still, affect our behaviors.
9. Attitudes cause us to behave in a particular way toward an object or person.
10. An attitude is a summary of a person's experience; thus, an attitude is grounded in direct experience predicts future behavior more accurately.
11. It includes certain aspects of personality as interests, appreciation, and social conduct.

12. It indicates the total of a man's inclinations and feelings.
13. An attitude is a point of view, substantiated or otherwise, true or false, which one holds towards an idea, object, or person.
14. It has aspects such as direction, intensity, generality, or specificity.
15. It refers to one's readiness for doing Work.
16. It may be positive or negative and may be affected by age, position, and education.

Formation of Attitudes

There are a number of factors that can influence how and why attitudes form. Here is a closer look at how attitudes form.

Experience

Attitudes form directly as a result of experience. They may emerge due to direct personal experience, or they may result from observation.

Social Factors

Social roles and social norms can have a strong influence on attitudes. Social roles relate to how people are expected to behave in a particular role or context. Social norms involve society's rules for what behaviors are considered appropriate.

Learning

Attitudes can be learned in a variety of ways. Consider how advertisers use classical conditioning to influence your attitude toward a particular product. In a television commercial, you see young, beautiful people having fun on a tropical beach while enjoying a sports drink. This attractive and appealing imagery causes you to develop a positive association with this particular beverage.

Conditioning

Operant conditioning can also be used to influence how attitudes develop. Imagine a young man who has just started smoking. Whenever he lights up a cigarette, people complain, chastise him,

and ask him to leave their vicinity. This negative feedback from those around him eventually causes him to develop an unfavorable opinion of smoking and he decides to give up the habit.

Observation

Finally, people also learn attitudes by observing people around them. When someone you admire greatly espouses a particular attitude, you are more likely to develop the same beliefs. For example, children spend a great deal of time observing the attitudes of their parents and usually begin to demonstrate similar outlooks.

Functions of Attitudes

An individual, in his life time, is bound to develop some attitudes. These attitudes may be favourable or unfavourable or both. The personality of a person is coloured by his attitudes. Attitudes determine ones personality.

A person can be called good or bad, sociable or unsociable, acceptable or unacceptable depending upon his attitude. If a person mostly develops all unfavourable attitude, his life becomes miserable. He cannot accept or believe or love anybody. He becomes a social or antisocial. Conversely nobody will accept him as this is a reciprocal process.

From the above stand points it is well evident that attitude determines one's behaviour, one's personality and one's position in the society. While favorable attitude towards others make him pleasant, sociable and acceptable, unfavorable attitudes make many enemies and develop hostile feelings and hatred in his mind.

Attitudes have, therefore, significant functions in moulding, influencing and determining one's behaviour in all contexts. As already discussed, on the basis of attitude personality can be typed.

Attitudes function as a source of motivation which helps in the adjustment to the environment. According to Katz (1960), four different personality functions are served by the maintenance and modifications of social attitudes. They are adjustment, value expression, knowledge and ego defence.

(i) Adjustment Function:

The holding of a particular attitude leads to reward or the avoidance of punishment. It is the utilitarian or instrumental function of attitude which motivates the person to adjust with the environment to gain social approval and support of family, friends and neighbours.

In case of certain social issues like marriage, death, democracy, religion, sacrifice and helping others, he holds opinions similar to his parents and relations and friends. Further favourable attitudes are developed towards those stimuli which satisfy one's needs and unfavourable attitudes towards those which stand on the fulfilment of his needs and motives.

(ii) Value Expression Function:

On the basis of identification with parents and other relatives the child develops certain personal values and self concepts. These values are integrated in the form of different attitudes. Attitudes help in expressing these values. The individual gets satisfaction by expression of attitudes appropriate to his personal values.

Religious, ideological and patriotic beliefs and values normally are based on this function. People get self satisfaction by engaging themselves in social work, care for the aged persons, by helping at the time of flood and famines, by taking care of the orphans or by raising their voice against corruption and social injustice.

(iii) Knowledge Function:

According to Mann this function of attitude is based on the need to understand, make sense and give adequate structure to the universe. Attitudes have a cognitive function in the sense that they help in understanding things properly for the sake of quick adjustment.

Attitudes which prove inadequate dealing with new and changing situations are discarded because, otherwise, they lead to contradictions and inconsistency. The need for cognitive consistency, meaning and clarity is fulfilled by the knowledge function of attitude.

(iv) Ego Defensive Function:

The ego defensive function of attitude provides protection against the knowledge and acceptance of basic unpleasant truths about disease, death, weakness, insecurity, frustration, unemployment, illness and various other harsh realities of life.

By rationalizing and distorting attitudes on the above harsh realities of life the ego tries to defend itself and lead a happy life by avoiding unpleasantness arising out of these unpleasant truths. All these facts lead to believe the tremendous significance of the functions of attitude in human life.

Lesson-13 Theories of attitudes- Reinforcement theory; Balance theory; Cognitive dissonance theory;

Theories of Attitudes

1. **Cognitive Consistency Theories**
2. **Functional Theory**
3. **Social Judgment Theory**

Cognitive Consistency Theories

Research has generally concluded that people seek consistency among their attitudes and between their attitudes and their behaviour. This means that people seek to reconcile divergent attitudes and align their attitudes and behaviour so that they appear rational and consistent. Whenthere is an inconsistency, forces are initiated to return the individual to an equilibrium state where attitudes and behaviour are again consistent. This can be done by either altering the attitude or the behaviour or by developing a rationalization for the discrepancy.

The cognitive consistency theories are concerned with inconsistencies that arise between related beliefs, bits of knowledge and evaluation about an object or an issue. Though various consistency theories differ in several respects, all of them have a common object that is reducing the inconsistency and returning the individual to the equilibrium state.

There are four important theories under this grouping:

- **Balance Theory** (F.Heider)
- **Congruity Theory (**C.E. Os good and P.H. Tannenbaum)
- **Affective Cognitive Consistency Theory** (M.I.Rosenberg)

- **Cognitive Dissonance Theory (**Leon Festinger)

(A) Balance Theory

F. Heider provided the basic model of balance theory. The theory is basically concerned with the consistency in the judgment of people and/or issues that are linked by some form of relationship. According to this theory there are three elements in attitude formation:

Between these three elements two generic types of relationships are considered to exist: Sentiment relations and unit relations.

The sentiment or linking relations include all form of sentiments or effect and the unit relations express the fact that two elements are perceived as belonging together. Both sentiment relations and the unit relations can be positive and negative.

In this three elements system balance exists if all three relations are positive or two relations are negative and one positive. There will be imbalance if all three relations are negative or if two relations are positive and one is negative. People tend to perceive others and objects linked to them so that the system is balanced. This theory assumes that balanced states are stable and imbalanced states are unstable. When imbalanced state occurs, the psychological tension which is created motivates the person to restore the balance cognitively by changing the relations. Thus a person's attitude towards an object depends on his attitudes towards a source who is linked with the object.

Example:

A person consistently argued that the quality of Indian cars is not up to that of the imported cars and that he would never own anything other than an imported car. His father gifts give him a latest model Maruti car. In this situation there are three elements the person, his father and the car. To bring the balance in this situation he will either change his negative relation with the Maruti car, he can ask his father to change his attitude and give him an imported car. Thus if thatperson starts saying that the Maruti car is not so bad, it means he has changed his relation and brought the balance in the situation.

Criticism:

The basic model given by Heider has been criticized on the following grounds:

(1) The theory does not consider the degree of sentimental or unit relationships nor the relevance to the perceiver of the elements and relations.

(2) As a consequence, there are no degrees of balance or imbalance and it is not possible to make quantitative predictions about the degree of attitude change.

(B) Congruity Theory

(C) E. Os good and **P.H. Tannenbaum** have proposed the congruity theory of attitudes which is similar to the balance theory. This theory focuses on the changes in the evaluation of a source and a concept that are linked by an associative or dissociative assertion. Congruity exists when a source and concept that are positively associated have exactly the same evaluations and when a source and concept that are negatively associated have exactly the opposite evaluations attached to them.

Congruity is a stable state and incongruity is an unstable one. As a result, incongruity leads to a change of attitude. This theory states that how much change should be there in the attitudes towards the source and the concept so that incongruity is resolved.

(C) Affective Cognitive Consistency Theory

M.I. Rosenberg has suggested the affective-cognitive consistency theory which is concerned with the consistency between a person's overall attitude or effect towards an object or issue and his beliefs about its relationship to his more general values. This theory is concerned mainly withwhat happens within the individual when an attitude changes. It assumes that the relationship between the affective and cognitive components of the attitude change when an attitude is altered.

The theory postulates the following points

(i) A person's effect towards or evaluation of the attitude object tends to be consistent- with this cognitive structural component.

(ii) When there is inconsistency beyond a certain level of tolerance, the individual is motivated to reduce the inconsistency and thereby to change one or both components to make them more consistent.

(iii) The theory, thus, suggests that changes in the affective component produce changes in the cognitive component in order to bring about consistency between the two.

(iv) The theory also suggests that persuasive communication can also be used to change the attitudes, by revaluating the goals themselves.

(D) Cognitive Dissonance Theory

Leon Festinger, in the late 1950s proposed the theory of cognitive dissonance. Dissonance means an inconsistency. Cognitive dissonance refers to any incompatibility that an individual might perceive between two or more of his attitudes or between his behaviour and attitudes. Festinger argued that any form of inconsistency is uncomfortable and that individuals will attempt to reduce the dissonance and hence the discomfort.
Therefore, individuals will seek a stable state where there is a minimum of dissonance, because an individual cannot completely avoid dissonance.

The desire to reduce dissonance is determined by three factors:

(i) Importance of the elements creating the dissonance

(ii) The degree of influence the individual believes he has over the elements.

(iii) The rewards that may be involved in dissonance.

If the elements creating the dissonance are relatively unimportant, the pressure to correct this imbalance will be low. But if the elements are important then a person will have to correct this imbalance. He can either change his behaviour, or he can change his dissonance, or he can change his attitude. Another choice can be to find out more consonant elements to outweigh the dissonant ones.
The degree of influence the individuals believe they have over the elements will have an impact on how they will react to the dissonance. If they perceive the dissonance to be an uncontrollable

result, something over which they have no choice, they are less likely to be receptive to attitude change. While dissonance exists, it can be rationalized and justified.

Rewards also influence the degree to which individuals are motivated to reduce dissonance. High rewards accompanying high dissonance tend to reduce the discomfort inherent in the dissonance because it increases the consistency side of the individual's balance sheet.

These factors suggest that just because individual experiences dissonance, they will not necessarily move directly toward consistency, that is, toward reduction of this dissonance. This theory helps to predict the propensity of an individual to engage in attitude and behavioural change, if individuals are required.

For example, by the demands of their jobs to do or say things which contradict their personal attitude, they will tend to modify their attitude in order to make it compatible with the cognition of what they have said or done.

Further, the greater the dissonance, after it has been moderated by the above mentioned factors, the greater the pressures to reduce it.

2. Functional Theory

The functional theory considers how attitudes and efforts are related to the motivational structure of the individual.

This theory focuses on two things

(i) The meaning of the influence situation in terms of both the kinds of motives that it arouses and

(ii) The individual's method of coping and achieving his goals.

An understanding of the functions served by attitudes is important for attitude change procedures since a particular method may produce change in individuals whose attitudes serve one particular function, but may produce no change in individuals for whom the attitudes serve a different function.

The most prominent person who visualized functional theory is **Katz** and he suggested four functions of attitudes. However, Katz functional theory has not stimulated much research except for the work on changing ego defensive attitudes.

3. Social Judgment Theory

The social judgment theory was originally formulated by **Sherif** and **Hoveland.** This theory attempts to explain how existing attitudes produce distortions of attitude related objects and how these judgments mediate attitude change. Thus, a person's initial attitude towards an issue, serves as an anchor for the judgment of attitude related stimuli. The person's initial attitude on an issue provides a point of reference against which he evaluates other opinions.

These views can be considered in terms of attitudinal continuum and can be considered as comprised of latitudes. The latitude of acceptance, which is the range of opinions the individual finds acceptable, encompasses the opinion that best characterizes his own stand. The attitude of rejection, which is the range of opinions the individual finds objectionable, encompasses the opinion he finds most objectionable. The attitude of non-commitment is the range of opinions that the person finds neither acceptable nor unacceptable.

Lesson-14 Measurement of attitudes- Self measures; Likert, Thurston Osgood and Bogardus scales: Physiological measures

Attitudes are constructs which are not open to direct observation. Therefore, the common way of measuring attitudes is to study some aspect of behavior and from that makes inferences about the attitudes that may be responsible for behavior.

Many different ways of measuring attitudes have been developed, though the more common techniques make use of self-report questionnaires. In general, an attitude scale or questionnaire will present the individual with a number of statements to which she/he has to respond and from these responses the investigator will arrive at some conclusions about the attitudes of that individual.

1. Thurstone scale

Thurstone (1929) is credited with having first created the attitude-measurement methodology. He is considered to be the **'Father'** of attitude scaling. He developed one of the earliest methods for measuring the attitudes towards religion. It is made up of statements about a particular issue, and each statement has a numerical value, indicating how favorable or unfavorable it is judged to be. People check each of the statements to which they agree, and a mean score is computed, indicating their attitude.

As the developer of the scale needs to make sure, that a wide range of possible views is expressed in the statements used, the Thurstone scales are quite complex and time consuming. This is particularly important, since the only responses allowed are agreement and disagreement. Thus, their main drawback is that they are very time-consuming, and therefore expensive to produce.

2. Likert scale

Likert (1932) proposed a technique, which makes this process simpler by allowing the participant to make a range of possible responses, usually in the form of a five-point scale, ranging from strongly agree, agree, undecided, to disagree, and strongly disagree.

With this method, there is no requirement for the judges to categorize each statement, as the categorization is built into the scale. Assigning the score from 1 to 5 to each of the responses, scores the Likert scales. Then the totaling of the score is done, to give a final measure of the individual's attitude.

3. Bogardus Social distance Scale

This was developed by Bogardus (1926) as a means of measuring racial or ethnic prejudice, a prejudice being seen as an extreme form of an attitude. The respondent is given a series of statements concerning the possible relationships, and is asked to say to what extent he would accept the relationship depicted. The respondents would be asked to tick those relationships they would find acceptable, and overall the scale provides a measure of racial prejudice.

4. Semantic Differential Scale

Osgood et al (1957) suggested that people evaluate attitudes along three dimensions: evaluative (good-bad), potency (strong-weak), and activity (active-passive).

To measure each of these three dimensions, they developed the semantic differential, a scale consisting of a number of different dimensions, against which the attitude object could be evaluated. The dimensions would typically be:

1. Pleasant – Unpleasant
2. Strong – Weak
3. Fast – Slow
4. Active – Passive
5. Hard – Soft
6. Good- Bad

The attitude object is evaluated along each of these dimensions, and the results can then differentiate among the three dimensions of the attitude, i.e. evaluative (1, 6); potency (2, 5) and activity (3, 4). Using this technique, it is possible not only to find out if the attitude objects are judges as favorable or unfavorable, but also why that might be the case in terms of the three dimensions.

Physiological Measures

The emotional reactions are reflected by physiological reactions (skin response, pulse rate, dilation of the pupils etc.). Therefore, it makes sense to look at the physiological reactions, in order to find more objective attitude indicators.

Galvanic Skin Response

The most important objective indicator of attitudes has been the Galvanic Skin Response (GSR). The GSR measure the electrical resistance of the skin, which changes when people are emotionally aroused. Hence, the GSR allows one to assess a subject's emotional response, towards an attitude object.

Facial Electromyogram

The facial Electromyogram (EMG) is an objective attitude indicator, which can be used to measure the quality or direction of attitude (its positiveness/negativity). The facial EMG makes itpossible to detect the differences between the positive and negative attitudes. When people heara message with which they agree rather than disagree, there is a relative increase in the EMG activity in a specific set of muscles, but a relative decrease in another set. Most of these changes cannot be seen with the naked eye.

The physiological measures of attitudes are not very often used in practice. One reason may be the above-mentioned insensitivity of most of the instruments, towards the quality of an attitudinal response. Another important reason, however, is the necessity of using technical devices, which are not easily applied in the field sittings.

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: **To estimate the level of the loneliness of your subject by using suitable test.**

Activity 2:**To assess interpersonal behaviour style by using suitable test**

PSY 15143

ABNORMAL PSYCHOLOGY

Objectives:

It educates the students about abnormality of an individual's both physical and psychological, interventions that are to be followed and to follow the different psychological strategies to overcome the problems.

Unit I: Meaning and definition of abnormality

Lesson-1 Meaning and definition of abnormality- Definition and Criteria of abnormality-

Lesson-2Historicalbackground of abnormal psychology.

Lesson-3 Classification system: Brief overview of classification- DSM IV and ICD – 10, Advantages and disadvantages of classification, Problem of Diagnosis.

Unit II: Different Perspectives of abnormal behaviour

Lesson-4 Psychoanalytical Perspectives- Freud's Psychoanalytical theory-Jung's analytical theory-Adler's Individual psychology.

Lesson-5 Behaviorist perspective- Skinner's operant theory- Bandura social learning theory-

Lesson-6 Humanistic Existential perspective- Maslow needs Hierarchy theory-

Lesson-7 Cognitive perspective – Beck and Ellis

Unit III: Nature of causation of Abnormalities

Lesson-8 Causes of Abnormal Behavior- Biological causes: Constitutional, neurophysiologic and Biochemical Psychological Causes

Lesson-9 Pathogenic causes, Pathogenic Interpersonal relationships.

Lesson-10 Socio– culture causes: Social class, Social roles, and Cultural factors.

Lesson-11 Predisposing, Precipitating reinforcing, Feedback and circularity of Causes.

Unit IV: Classification of Mental Disorders

Lesson-12 Psycho-Neurosis- Causes and Aetiology-Psychoses-Causes and Aetiology Anxiety disorders.

Lesson-13 Somatoform disorders: Somatoform Disorders Somatoform disorders– Hypochondriasis - Somatization Disorder - Pain Disorder - Conversion Disorder - Body Dimorphic Disorder.

Lesson-14 Functional disorders- Mood disorders- Mood disorders Definition & Meaning – Depressive Disorder – Theoretical Perspectives on Depressive Disorder – Major Depressive Disorders – Unipolar and Bipolar disorders – Substance Induced Mood Disorders - Suicide – Classification – Causes - Treatment and Prevention.

Lesson-15 Personality disorders- Personality Disorders Cluster A Personality Disorders – Cluster B Personality Disorders – Personality Disorders – Paranoid – Schizoid – Schizotypal – Histrionic – Narcisstic – Anti-Social Personality Disorders- Addiction.

Lesson- 16 Psychosomatic disorders.

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Outcomes:

- To familiarize with a Concepts and Historical Views on Psychopathology
- To know the Standardized Classification of Mental Disorders
- To understand the Mood and Schizophrenic Disorders
- To learn the Sexual and Personality Disorders
- To learn the Developmental Disorders.

UNIT I: MEANING AND DEFINITION OF ABNORMALITY

Lesson-1 Meaning and definition of abnormality- Definition and Criteria of abnormality-

Defining Abnormality

There are different ways of defining abnormality, but the following characteristics are usually included:

Abnormal behavior occurs infrequently. However, statistical infrequency alone is not a sufficient definition. Some healthy, desirable, and beneficial behaviors also occur infrequently. And other uncommon behaviors or characteristics

- have no bearing on how a person behaves or functions. So just because something is unusual or uncommon does not mean it should be defined as abnormal.
-
- Abnormal behavior creates distress. These behaviors may disturb the individual, or they may be upsetting and disruptive to others.
- Abnormal behavior affects a person's ability to function. People who are displaying these behaviors may struggle to function normally in their daily life, which can affect their relationships, work, school, and home life.
- Abnormal behavi or is socially disruptive. It may violate social norms and make it difficult for people to function in social settings and maintain social relationships.

According to the DSM 5, a mental disorder is defined as a syndrome that ispresent in an individual and that involves clinically significant disturbance in behaviour, emotional regulation, or cognitive functioning. These disturbances are thought to reflect a dysfunction in biological, psychological or developmental processes that are necessary for mental func tioning. Thus, abnormality is associated with significant decrease in social, occupational and other activities that are important for human functioning.

Lesson-2 Historicalbackground of abnormal psychology.

Throughout history the dominant social, economic and religious views have had a great influence on how people perceived abnormal behaviour.In the ancient world superstitious explanations for mental disorders were popular. In the fifteenth and sixteenth centuries it was widely believed that mental disorders were attributed by demonic possessions.

However, Hippocrates denied that Gods and demons caused illnesses and insisted that mental disorders had natural causes and required proper treatment. He also believed that dreams are important to understand the patients' personality. He also recognised the importance of environment formental health and thus removed some patients from their families.

The Greek and Roman were among the few to treat people with mental disorders with appropriate care. They provided pleasant surroundings with constant activities like parties, dances and massages. Theyalso followed the principle of contrary is contraries (opposite by opposite). For example, having their patient drink chilled wine while taking a warm bath. Chinese physician, Chung Ching conducted treatments that were similar to Hippocrates. Other references to mental health disorders were also made inthe Indian texts of Charaka Samhitha and Sushruta Samhita.

In Europe during the Middleage and Renaissance period, there was a general movement away from

superstitions and toward reasoned scientific studies. Mental asylums were created in the 16th century, however, it lead to the isolation and maltreatment of the patients. Some patients were displayed to the public in return for money. Slowly by the eighteenth century efforts were made for the better care of patients by providing them withbetter living conditions and humane treatment.

In the 19th and 20th century rapid growth of abnormal psychology was observed. This can be because of the growth of technology and scientific advancements. Thus the treatment of individuals with mental illnesses was advanced. Because of the works of several renowned psychologists like Pavlov and Freud, the gradual acceptance of patients with mental illnesses as people who need care and attention, was possible. Successful applicationof biomedical methods to disorders and the growth of scientific research into the biological, psychological and sociocultural roots ofabnormal behaviour were observed.

Approaches to the Classification of Psychopathology

Psychologists use three approaches or strategies to classify disorders:

i) Categorical approach: It was Kraepelin, the first psychiatrist to classify psychological disorders from a biological or medical point of view. For Kraepelinin term of physical disorders, we haveone set of causative factors which do notoverlap with other disorders. We have one defining criteria, which everybodyin the category or in the group should meet, e. g. Schizophrenia. After a categoryhas been defined, an object is either a member of the category or it is not. Acategorical approach to classification assumes that distinctions among membersof different categories are qualitative. In other words, the differences reflect a difference in kind (quality) rather than a difference in amount (quantity).

i) Dimensional approach: A second strategy is a dimensional approach, in whichwe note the variety of cognitions, moods and behaviours with which the patientpresents and quantify them on a scale. For example, on a scale of1 to 10, apatient might be rated as severely anxious (10), moderately depressed (5), andmildly manic (2) to create a profile of emotional functioning (10, 5, 2). Althoughdimensional approaches have been applied to psychopathology, they are relatively unsatisfactory. Dimensional approach to classification describes the objects of classification in terms of continuous dimensions. Rather than assuming that anobject either has or does not have a particular property, it may be useful to focus on a specific characteristic and determine how much of that characteristicthe object exhibits.

iii) Prototypical approach: A third approach, for organising and classifying behaviouraldisorders which is an alternative to the first two. It is called a prototypical approach. It identifies some essential characteristics of a disorderand it also allows for certain non-essential variations that do not necessarily change the classification. With this approach classifying the disorder by different possible features or properties anycandidate must meet (but not all) of them tofall in that category. In depression, there are five important symptoms such as:depressed mood all the day, weight loss, insomnia, fatigue and feeling of worthlessness. For a person might have three or four of the characteristics ofthe depression but not all five of them. Yet we still diagnos e the person as depressed.

most ancient classification of psychopathology was of senile deterioration, melancholia and hysteria. The oldest systematic classification finds its mention in the Ayurveda, an Indian body of thought. The Greek philosopher Hippocrates (460–370 B.C.) classified mental illness into delirium, mania, paranoia, hysteria, melancholia resulting from 4 basic temperaments. Philippe Pinel's (1745–1826) father of modernpsychiatry, classification system was based on functional disorders of nervous system.He describedfour functional disorders: dementia, mania, melancholia and idiotism. Karl Ludwig Kahlbaum (1828-1899) distinguished organic and non organic mentaldisorders. Emil Kraepelin's (1856- 1926) classification system was based on clinicalfeatures of disorders: cause, course and outcomes. His primary classifications weremanic depressive psychosis and dementia praecox. Eugen Bleuler combined Kraepelinand Meyerian approaches and classified mental disorders on the basis of psycho-pathological processes.

Lesson-3 Classification system: Brief overview of classification- DSM IV and ICD – 10, Advantages and disadvantages of classification, Problem of Diagnosis.

Development of ICD

In 1893 1st international list of causes of death was published. This stimulated worldwide organised effort for classification of diseases which resulted in the publicationof International Statistical Classification of Diseases and Related Health Problems-1 (ICD-1) by the World Health Organisation in 1900. However, it wasonly ICD-6 which was published with a separate section on mental disorder in 1949.ICD-8 was published in 1972 with a comprehensive glossary of mental disorders.ICD-9 was published in 1977 with greater clinical modification. Vol. 1 and 2 ofICD-9 described diagnostic codes, while vol. 3 explained procedure codes for theMental and Behavioural Disorders.

ICD-10: In 1978, WHO entered into a long-term collaborative project with the Alcohol, Drug Abuse and Mental Health Administration (ADAMHA) in the USA, aiming to facilitate further improvements in the classification and diagnosis of mentaldisorders, and alcohol and drug related problems. A series of workshops brought together researchers and practitioners from a number of different psychiatric traditions and cultures, reviewed knowledge in specified areas, and developed recommendations

Disorders in ICD-10

A brief description of classification of mental disorders as per ICD-10 is given below:

- F00-F09: Organic, including symptomatic, mental disorders: Dementia, delirium, organic.
- F10-F19: Mental and behavioural disorders due to use of psychoactive substances: Alcohol, cocaine and tobacco.
- F20-F29: Schizophrenia, schizotypal and delusional disorders.
- F30-F39: Mood (affective) disorders: Manic, bipolar, depressive.
- F40-F48: Neurotic, stress-related and somatoform disorders: Phobia, OCD, adjustment, dissociative.
- F50-F59: Behavioural syndromes associated with physiological disturbances and physical factors: Eating, sleep, sexual disorders.
- F60-F69: Disorders of personality and behaviour in adult persons: Specific, impulse disorder, gender identity.
- F70-F79: Mental retardation
- F80-F89: Disorders of psychological development: Speech and language, pervasive development.
- F90-F98: Behavioural and emotional disorders with onset usually occurring in childhood and adolescence: Hyperkinetic, conduct, tic.
- F-99: Unspecified mental disorders.

A Brief History of DSM

Need of statistical information regarding mental disorders stimulated revolution in the efforts of development of a classification system in the United States. The National census of 1840 used a single category, “idiocy/insanity”. Seven categories of mental disorders were mentioned in the 1880 census: mania, melancholia, monomania, paresis, dementia, dipsomania, and epilepsy. The American Psychiatric Association (APA), then known as Committee on Statistics, together with the National Commission on Mental Hygiene,

developed a new guide for mental hospitals called the "Statistical Manual for the Use of Institutions for the Insane" in 1917, which included 22 diagnoses. Subsequently, this was revised several times by APA over the years.

for future research. A major international conference on classification and diagnosiswas held in Copenhagen, Denmark, in 1982 to review the recommendations thatemerged from these workshops and to outline a research agenda and guidelines forfuture work. Several major research efforts were undertaken to implement the recommendations of the Copenhagen conference. All these efforts resulted in thepublication of ICD-10 in 1992 in which in chapter V (F) pertained to the classificationof mental disorders explaining their inclusion and exclusion terms.

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Revisions of DSM

In view of needs arisen from World War II, a committee headed by psychiatrist andbrigadier general William C. Menninger with the help of US psychiatrists developeda new classification scheme called Medical 203 in 1943.

DSM-I (1952): In 1949, the World Health Organisation published the sixth revisionof the International Statistical Classification of Diseases (ICD-10) which included asection on mental disorders for the first time. Consequently, an APA Committee onNomenclature and Statistics was empowered to develop a classification system foruse in the United States. In 1950 the APA committee undertook a review and consultation to circulated the Diagnostic and Statistical Manual of Mental Disorders-1 (DSM-I) which was approved in 1951 and published in 1952. The structure and conceptual framework were the same as in Medical 203, and many

passages of text identical. The manual was 130 pages long and listed 106 mentaldisorders. DSM-I made 94% changes in nomenclature from the prior system andseventy disorder terms used "Reaction," e.g., schizophrenic reaction. This includedreaction to internal conflict. DSM-I is criticised on the basis that it was psychoanalyticin its theoretical orientation. The term "Unconscious" was mentioned a few times in describing psychoneurotic disorders in DSM-I.

DSM-II (1968): Despite APA's involvement in the revision of the mental disordersection of the ICD-8 in 1968), it also published a revision of the DSM in 1968, listed182 disorders, and was 134 pages long. It was quite similar to the DSM-I. The term"reaction" was dropped but the term "neurosis" was retained. Both the DSM-I andthe DSM-II reflected the predominant psychodynamic psychiatry, although they alsoincluded biological perspectives and concepts from Kraepelin's system of classification. Symptoms were not specified in detail for specific disorders. Many of the disorderswere seen as reflections of broad underlying conflicts or maladaptive reactions to life problems, rooted in a distinction between neurosis and psychosis (roughly, anxiety/depression broadly in touch with reality, or hallucinations/delusions appearing disconnected from reality). Sociological and biological knowledge was also incorporated, in a model that did not emphasise a clear boundary between normalityand abnormality. After wide post publication criticism of DSM-II, the term "Homosexuality" was replaced with "Ego-dystonic Homosexuality" in 1973.

DSM-III (1980): To make the DSM nomenclature consistent with the InternationalStatistical Classification of Diseases and Related Health Problems (ICD) the decisionto create a new revision of the DSM was made in 1974 with Robert Spitzer as chairman of the task force. This revision was aimed at improving the uniformity andvalidity of psychiatric diagnosis, standardising diagnostic practices across the worldand to facilitate the pharmaceutical regulatory process to avoid criticisms levelled onDSM-II. The criteria for many of the mental disorders were taken from the ResearchDiagnostic Criteria and Feighner Criteria. New categories of disorder and their criteria were established by consensus during meetings of the committee. The psychodynamic or physiologic view was abandoned, in favour of a regulatory orlegislative model. A new "multi-axial" system attempted to yield a picture more amenable to a statistical population census, rather than just a simple diagnosis. DSMconceptualised each of the mental disorders as a clinically significant behavioural orpsychological syndrome.

Finally published in 1980, the DSM-III was 494 pages long and listed 265 diagnosticcategories. It rapidly came into widespread international use. DSM-III was publishedwith 93% changes in nomenclature from the earlier version of DSM with diagnostic

criteria for each of the disorders mentioned. There was a multi-axial classificationwith five axes. DSM-III provided a vast increase in background information abouteach disorder, adding diagnostic features, associated features, cultural and genderfeatures; prevalence, course, familiar patterns, differential diagnosis, decision trees and glossary. However, DSM-III was later criticized on the ground that 20-30 percent of the population would have been diagnosed as having behavioural disorderswithout having any serious mental problems.

DSM-III-R (1987): The DSM-III-R was published as a revision of DSM-III in 1987. Categories were renamed, reorganised, and significant changes in criteria weremade. Six categories were deleted, while some new categories were added. Controversial diagnoses such as pre-menstrual dysphoric disorder and Masochistic Personality Disorder were discarded. "Sexual orientation disturbance" was alsoremoved and was largely subsumed under "sexual disorder not otherwise specified"which can include "persistent and marked distress about one's sexual orientation."DSM-III-R contained 292 diagnoses and was 567 pages long.

DSM-IV (1994): DSM-IV was published in 1994 listing 297 disorders in 886 pages. Process of development of DSM-IV included extensive literature review ofdiagnoses, analyses to determine required change in criteria and multicenter fieldtrials relating diagnoses to clinical practice. A major change from previous versionswas the inclusion of a clinical significance criterion to almost half of all the categories,which required symptoms cause "clinically significant distress or impairment in social,occupational, or other important areas of functioning".

DSM-IV (TR) (2000): A "Text Revision" of the DSM-IV, known as the DSM-IV-TR, was published in 2000.

The diagnostic categories and the vast majority of thespecific criteria for diagnosis were unchanged. The text sections giving extra informationon each diagnosis were updated, as were some of the diagnostic codes in order tomaintain consistency with the ICD.

DSM-TR

The Diagnostic and Statistical Manual of Mental Disorders (DSM) is the standardfor classifying mental disorders that are used by mental health professionals in theUnited States. It is intended to be applicable in a wide variety of contexts and usedby clinicians and researchers of many different orientations, for example; biological,psychodynamic, cognitive, behavioural, interpersonal, family/systems.

The DSM IV (Text Revision) has been designed for use across settings such as inpatient, outpatient, partial hospitalisation, consultation-liaison, clinic, private practiceand primary care. Professionals that use the DSM diagnosis are psychiatrists, psychologists, social workers, nurses, occupational and rehabilitation therapists, counsellors and other health and mental health professionals. The DSM is also anecessary tool for collecting and communicating accurate public health statistics. TheDSM consists of three major components/features: the diagnostic classification, thediagnostic criteria sets and the descriptive text.

i) The Diagnostic Classification: The DSM-IV (TR) is a categorical classificationsystem. The categories are prototypes, and a patient with a close approximation tothe prototype is said to have that disorder with qualifiers, for example mild, moderateor severe forms of a disorder. For nearly half the disorders, symptoms must be sufficient to cause clinically significant distress or impairment in social, occupational,

or other important areas of functioning, although DSM-IV-TR removed the distresscriterion from tic disorders and several of the paraphilias. Making a DSM diagnosisconsists of selecting those disorders from the classification that best reflects the signsand symptoms that are afflicting the individual being evaluated. Associated with eachdiagnostic label is a diagnostic code, which is used primarily by institutions andagencies for data collection. These diagnostic codes are derived from the coding system used by all health care professionals in the United States, known as the ICD-9-CM.

ii) The Diagnostic Criteria: Each disorder included in the DSM-IV (TR) includesa set of diagnostic criteria including symptoms that are present and for how long.These criteria called inclusion criteria as well as those symptoms that must not bepresent called exclusion criteria qualify an individual for a particular diagnosis. Manyusers of the DSM-IV (TR) find these diagnostic criteria useful because they provideа compact description of each disorder. Use of this diagnostic criterion has increaseddiagnostic reliability and the likelihood that different individuals will assign the samediagnosis. It is important to remember that these criteria are meant to be used as aguideline by an informed clinician.

iii) Descriptive Text: The third component of the DSM-IV (TR) is the descriptive text that accompanies each disorder. The text of the DSM-IV (TR) systematicallydescribes each disorder under the following headings: Diagnostic Features; Subtypesand/or Specifies; Recording Procedures; Associated Features and Disorders; SpecificCulture, Age, and Gender Features; Prevalence; Course; Familial Pattern; and Differential Diagnosis.

Multi-Axial Classifications

The DSM-IV (TR) recommends clinicians to assess an individual's mental stateacross five factors or axes. Together the five axes provide a broad range of informationabout the individual's functioning, not just a diagnosis. The system contains thefollowing axes.

1) Axis I: *Clinical Disorders and Other Conditions That May Be a Focus of Clinical Attention:* This axis incorporates a wide range of clinical syndromes,including anxiety disorders, mood disorders, schizophrenia and other psychotic disorders, adjustment disorders, and disorders usually first diagnosed during infancy, childhood, or adolescence (except for mental retardation, which iscoded on Axis II). Axis I also includes relationship problems, academic or occupational problems, and bereavement, conditions that may be the focus ofdiagnosis and treatment but that do not in themselves constitute definable psychological disorders. Also coded on Axis I are psychological factors thataffect medical conditions, such as anxiety that exacerbates an asthmatic conditionor depressive symptoms that delay recovery from surgery. The Axis I clinicaldisorder categories are as follows:
    1) Disorders Usually First Diagnosed in Infancy, Childhood, or Adolescence
    2) Delirium, Dementia, and Amnestic and Other Cognitive Disorders
    3) Mental Disorders Due to a General Medical Condition Not ElsewhereClassified
    4) Substance Related Disorders
    5) Schizophrenia and Other Psychotic Disorders
    6) Mood Disorders
    7) Anxiety Disorders

8) Somatoform Disorders
9) Factitious Disorders
10) Dissociative Disorders
11) Sexual and Gender Identity Disorders
12) Eating Disorders
13) Sleep Disorders
14) Impulse Control Disorders Not Elsewhere Classified
15) Adjustment Disorders
16) Other Conditions That May Be a Focus of Clinical Attention

2) Axis II: Personality Disorders and Mental Retardation: Personality disorders are enduring and rigid patterns of maladaptive behaviour that typicallyimpair relationships with others and social functioning. These include antisocial,paranoid, narcissistic, and borderline personality disorders. Mental retardation, which is also coded on Axis II, involves pervasive intellectual impairment. Peoplemay be given either Axis I or Axis II diagnoses or a combination of the twowhen both apply. For example, a person may receive a diagnosis of an anxietydisorder (Axis I) and a second diagnosis of a personality disorder (Axis II).This axis includes following disorders:
    1) Paranoid Personality Disorder
    2) Narcissistic Personality Disorder
    3) Schizoid Personality Disorder
    4) Avoidant Personality Disorder
    5) Schizotypal Personality Disorder
    6) Dependent Personality Disorder
    7) Antisocial Personality Disorder
    8) Obsessive-Compulsive Personality disorder
    9) Borderline Personality Disorder
    10) Personality Disorder Not Otherwise Specified
    11) Histrionic Personality Disorder
    12) Mental Retardation
3) Axis III: General Medical Conditions: All medical conditions and diseases that may be important to the understanding or treatment of an individual's mentaldisorders are coded on Axis III. For example, if hypothyroidism were a directcause of an individual's mood disorder (such as major depression), it would be coded under Axis III. Medical conditions that affect the understanding or treatment of a mental disorder (but that are not direct causes of the disorder) are also listed on Axis III. For instance, the presence of a heart condition may determinewhether a particular course of drug therapy should be used with a depressedperson.
4) Axis IV: Psychosocial and Environmental Problems: The psychosocial and environmental problems that affect the diagnosis, treatment, or outcome of amental disorder are placed on

Axis IV. These include job loss, marital separationor divorce, homelessness or inadequate housing, lack of social support, thedeath or loss of a friend, orexposure to war or other disasters. Some positivelife events, such as a job promotion, may also be listed on Axis IV, but only when they create problems for the individual, such as difficulties adapting to anew job. Table 1 lists examples from this axis.

5) Axis V: Global Assessment of Relational Functioning (GARF): The clinicianrates the client's current level of psychological, social, and occupational functioningusing a 0-100 scale. The clinician may also indicate the highest level of functioningachieved for at least a few months during the preceding year. The level of current functioning indicates the current need for treatment or intensity of care.The level of highest functioning is suggestive of the level of functioning that might be restored. The GARF Scale can be used to indicate an overall judgment ofthe functioning of a family or other ongoing relationship on a hypothetical continuum ranging from competent, optimal relational functioning to a disrupted,dysfunctional relationship (APA, 2000).

Pros and cons

the Diagnostic and Statistical Manual (DSM) is sometimes referred to as psychiatry's bible.[1] Within its covers are specific diagnostic criteria for mental disorders, as well as a series of codes that allow therapists to easily summarize often complex conditions for clinical research and insurance purposes.

This method offers a number of advantages, such as standardization of diagnoses across different treatment providers. But increasingly, mental health professionals are considering the drawbacks of the DSM, including the possibility of over-diagnosis. In order to understand the debate, it is first necessary to understand what the DSM is and is not.

In 1949, the World Health Organization released its sixth edition of the International Classification of Diseases (ICD), which for the first time included mental illnesses.[3]

Although this work represented some of the earliest standards for mental health diagnosis, it was far from complete.

DSM-I and DSM-II

In 1952, the American Psychiatric Association (APA) published the DSM-I, an adaptation of a classification system developed by the armed forces during WW2. It was designed for use by doctors and other treatment providers.

The DSM-I was the first of its kind, but experts agreed that it still needed work. The DSM-II, released in 1968, attempted to incorporate the psychiatric knowledge of the day. It was heavily influenced by psychoanalytic concepts that were prominent at that time.

DSM-III

Published in 1980, DSM-III represented a radical change in the DSM structure. Its goal was to improve the validity and standardization of psychiatric diagnoses.

It was the first version to introduce such elements as the multi-axis system and explicit diagnostic criteria. It also removed much of the earlier versions' bias toward psychodynamic, or Freudian, thought in favor of a more descriptive and categorical approach.

Although the DSM-III was a pioneering work, real-world usage soon revealed its flaws and limitations. Confusing diagnostic criteria and inconsistencies led the APA to develop a revision.

Some of these changes were based on changing societal norms. For example, in the DSM-III, homosexuality was diagnosed as ego-dystonic homosexuality. In the revision, this specific classification was removed.

By the late 1980s, however, homosexuality was no longer seen as a disorder, although anxiety and distress about sexual orientation were. The DSM-III-R, released in 1987, fixed many of the internal difficulties of the earlier work.

DSM-IV and DSM-5

Published in 1994, the DSM-IV reflected numerous changes in the understanding of mental health disorders. Some diagnoses were added, others subtracted or reclassified. In addition, the diagnostic system was further refined in an effort to make it more user-friendly.

The DSM-5, published in May 2013, represented another radical shift in thinking in the mental health community. Diagnoses have been changed, removed or added, and the organizational structure underwent a major reworking.

Previous editions went decades between revisions. The DSM-5 is expected to be revised more regularly with mini additions (such as the DSM-5.1, DSM-5.2, etc.) in an effort to be more responsive to research.

DSM-5-TR

The most recent edition of the DSM is the DSM, fifth edition, text revision, known as the DSM-5-TR. The American Psychiatric Association published this update in 2022.

This version of the manual includes text revisions intended to add clarity, use more precise language, and minimize reader confusion. Notable changes include:[4]

- Significant updates regarding the language used to describe gender dysphoria
- Updates for the wording of autism spectrum disorder for criterion A
- Language changes to reduce racial and cultural biases
- Descriptions of how symptoms can manifest differently in people of different demographic backgrounds
- New codes for non-suicidal self-injury and suicidal behavior

- Revised diagnostic criteria for 70 disorders
- Added a new diagnosis for prolonged grief disorder

Clinical Uses of the DSM

Every mental health professional uses the DSM in his or her own way. Main clinical uses for the DSM include:

- A diagnostic and treatment blueprint: Some practitioners rigidly stick to the manual, developing treatment plans for each client based solely on the book's diagnoses.
- A guideline: Others use the DSM as a guideline—a tool to help them conceptualize cases while focusing on each client's unique set of circumstances.
- Billing purposes: In the modern world, virtually every mental health professional must refer to the DSM's codes to bill treatment to insurance companies.

Advantages of the DSM

Despite its flaws, the DSM is uniquely helpful for several reasons.

Standardization

Beyond billing and coding, standardization provides a number of important benefits to the clinician and the client. Standardization of diagnoses helps ensure that clients receive appropriate, helpful treatment regardless of location, social class, or ability to pay.

It provides a concrete assessment of issues and helps therapists develop specific goals of therapy, as well as assess the effectiveness of treatment.[5]

Research Guidance

In addition, the DSM helps guide research in the mental health field. The diagnostic checklists help ensure that different groups of researchers are studying the same disorder—although this may be more theoretical than practical, as so many disorders have such widely varying symptoms.

Therapeutic Guidance

For the mental health professional, the DSM eliminates a lot of guesswork. Proper diagnosis and treatment of mental illness remains an art, but the DSM diagnostic criteria serve as a sort of map.

In the age of brief therapy, a clinician may see a specific client only a handful of times, which may not be long enough to delve fully into the client's background and issues.

Using the diagnostic criteria contained in the DSM, the therapist can develop a quick frame of reference, which is then refined during individual sessions.[6]

Advantages

- Standardizes billing and coding
- Standardizes diagnoses and treatment
- Guides research
- Guides treatment

Disadvantages

- Oversimplifies human behavior
- Increases risk of misdiagnosis or over-diagnosis
- Provides labels, which can be stigmatizing

Disadvantages of the DSM

No tool is perfect, and the DSM is no exception. Being aware of its drawbacks is important for both patients and therapists.

Oversimplification

The latest round of criticism echoes a long-running debate on the nature of mental health. Many critics of the DSM see it as an oversimplification of the vast continuum of human behavior.[7]

Some worry that by reducing complex problems to labels and numbers, the scientific community risks losing track of the unique human element.

Misdiagnoses and Over-Diagnoses

Possible risks include misdiagnosis or even over-diagnosis, in which vast groups of people are labeled as having a disorder simply because their behavior does not always line up with the current ideal.[8]

Childhood attention deficit/hyperactivity disorder (ADHD) is a common example. Shifts in terminology and diagnostic criteria in DSM-IV coincided with a massive upturn in the number of children on Ritalin or other medications.

Labeling and Stigmatization

Other risks involve the possibility of stigmatization. Although mental health disorders are not viewed in the negative light that they once were, specific disorders can be perceived as labels.

Some therapists take great care to avoid attaching labels to their clients. But for a variety of reasons, a specific diagnosis may be required.

Criticisms of the DSM

The DSM is not without criticism. Such critiques often center on the oversimplification of mental health, the potential for misdiagnosis, and the risks of labeling and stigma.[9]

Other critics have suggested that the pharmaceutical industry has had too much influence on revisions to the manual. Critics note that 69% of DSM-5 task force members had direct ties to the pharmaceutical industry.[10]

This suggests that the manual may overly medicalize conditions, implying that even relatively normal or non-serious behavior and mood patterns require treatment, often in the form of medication.

Limitations of ICD-10

The review of the literature establishes the fact that the implementation of ICD-10 will become an initial turmoil for the for-profit physicians. Hirsch et al. (2015) remark that ICD-10 is a rather expensive proposition. The authors note that the current regulatory difficulties will be aggravated since the cost of ICD-10 implementation will become the responsibility of healthcare providers.

Southern et al. (2017) consider the need to implement significant technology changes as a complication of ICD-10. According to the authors, the new classification system will enhance the arrangement of patient data safety, but it demands financial support. Machikanti et al. (2014) argue that patient outcomes and the arrangement of treatment require more details than diagnosis codes and thus consider ICD-10 an insufficient measure to improve the system of healthcare.

Benefits of ICD-10

Despite some challenges presented by the implementation of ICD-10, there will also be substantial benefits associated with the system available for physicians. Wing (2016) remarks that the codification of data collection was "long overdue," and emphasizes that with the help of ICD-10, it will be possible to improve this issue (p. 179). The author mentions that the unique structure of coding suggested by ICD-10 will provide better possibilities of data collection (Wing, 2016). The main benefit, as outlined by Wing (2016), is that ICD-10 will establish opportunities to enhance patient outcomes and care.

Sanders et al. (2012) identify three significant advantages of ICD-10 implementation: cost benefits, improved quality of care, and better access to collecting data. Caskey et al. (2014) analyze the positive implications of ICD-10 for pediatric specialists. Researchers note that some of the codes in ICD-9 were outdated, and the introduction of ICD-10 was inevitable and necessary. Caskey et al. (2014) remark that ICD-9 codes concerned with pregnancy and postnatal complications required supplementary clinical information that was added to ICD-10.

Advantages ICD

Easier comparison of mortality and morbidity data
Improved quality data
New codes includes liability severity and complexity of the diseases conditions

Terminology of diseases classification are now consistent with new technology and current clinical practice.
Injuries causes and complexity can known and better health record maintence.
Flexibility more additional emerging procedures and diagnosis.
Disadvantages
Complex and hard to teach
New technology is required, number of codes 150000.
No benefits from large codes

UNIT II: DIFFERENT PERSPECTIVES OF ABNORMAL BEHAVIOUR

Lesson-4 Psychoanalytical Perspectives- Freud's Psychoanalytical theory-Jung's analytical theory- Adler's Individual psychology.

Psycholo analytical theory –freud.

In simple terms, Sigmund Freud's theory suggests that human behavior is influenced by unconscious memories, thoughts, and urges. This theory also proposes that the psyche comprises three aspects: the id, ego, and superego. The id is entirely unconscious, while the ego operates in the conscious mind

The superego operates both unconsciously and consciously.

Psychology students spend a fair amount of time learning about Freud's theory. Even those outside the psychology field often have some awareness of Freudian psychoanalysis, the school of thought created by Sigmund Freud.

Knowing more about Freudian psychology, along with the key concepts in psychoanalysis—like the unconscious, fixations, defense mechanisms, and dream symbols—can help you understand the influence Freud's theories

The psychodynamic perspective

This perspective emphasizes the role of unconscious motives and thoughts that govern behaviour of human beings. According to this perspective abnormal behaviour is because of the hurtful memories, forbidden desires and repressed experiences in the unconscious mind. The unconscious continues to express itself in dreams, fantasies and slips of tongue. When such unconscious material is brought to the consciousness it can lead to irrational and maladaptive behaviour. Sigmund Freud is considered the founder of the psychoanalytic school of thought.

Jung's Psychodynamic Psychology

Jung believed that the psyche, or the soul, was driven toward individuation. His psychodynamic psychology revolved around the archetypes within the collective unconscious, as well as the personal unconscious and the ego.

Analytical psychologist Carl Jung is well known for his theory of personality. This theory suggests that personality is inherited and part of the collective unconscious—which includes both conscious and unconscious aspects.

According to Jung, personality appears in the form of archetypes, or universal patterns of thought and behavior that affect what we focus on and how we interact with the world. Jungian archetypes categorize people in terms of various personality patterns.

The psychological types identified by Jung are based on factors such as general attitude and psychological functions. The four basic psychological functions or dimensions that are the focus of Jung's theory are:

1. Extraversion vs. introversion
2. Sensation vs. intuition
3. Thinking vs. feeling
4. Judging vs. perceiving

Jung's model of the psyche

Jung posited that there are three components that make up the human psyche:

- The ego
- The personal unconscious
- The collective unconcscious

The ego represents the conscious mind that contains the awareness of existing and the sense of personal identity. This is where your personality exists and where your thoughts, intuitions, feelings and sensations are organized. It's the door between the inner and outer worlds of the psyche.

The personal unconscious is made up of the memories that are subliminal, forgotten and/or repressed. Some of the personal unconscious can be recalled to the conscious mind. Jung believed that for individuation to occur, the personal unconscious and the conscious ego have to be fully integrated.

The collective unconscious, also known as the transpersonal unconscious, is one of Jung's more unique and controversial additions to personality theory. The idea proposes that there is a universal version of the personal unconscious, which is shared with all other members of the human species. These shared ancestral memories, born from evolution, are called archetypes by Jung and are represented by universal themes that appear in various cultures.

Some of these innate characteristics include being scared of the dark or spiders, for example. However, Jung posited that more than these isolated examples, archetypes have developed into separate subsystems of the personality through ancestral memories and images.

Individuation

According to the American Psychological Association, individuation is “the gradual development of a unified, integrated personality that incorporates greater and greater amounts of the unconscious, both personal and collective, and resolves any conflicts that exist, such as those between introverted and extraverted tendencies.”

In essence, the therapeutic goal of individuation within analytical psychology is the process through which a person becomes a whole psychological individual. The person recognizes their own self-worth and uniqueness and embraces both the consciousness and the unconscious.

Archetypes

Derived from the collective unconscious, Jungian archetypes are depicted as images and themes with universal meanings across a wide range of cultures. These archetypes may appear in dreams, literature, religion or art.

There are numerous archetypes that Jung explored within Analytical Psychology, but below we will explore the main four that are discussed most often:

1. The Persona

The persona, also referred to as the “mask,” signifies the outward face that each human being presents to the world. It’s not our true self but represents our “conformity” within society. Like acting, the persona is the performance we put on for others, which isn’t who we really are.

2. The Shadow

The shadow archetype represents the animal side of our personality. From the shadow, individuals gain both creative and destructive energies, which influence their predispositions. This includes all the things that individuals do not want to know about themselves or do not like. It is the part of the unconscious that is most accessible by the conscious.

3. The Anima/Animus

Jung described the anima/animus as the mirror images of our biological sex. In other words, the unconscious feminine side in men and, vice versa, the unconscious masculine side in women. The anima represents the feminine aspects, while the animus represents the masculine aspects.

4. The Self

The self is an intrinsically important part of Jung’s Analytical Psychology. The self is what proves a sense of unity in the human experience. Every individual should be aiming to achieve a state of selfhood to create a balance between the conscious and unconscious.

Discover more about the complex world of Analytical Psychology from one of Jung's most important and famous books, Psychological Types.

Adlers individual theory

Individual psychology, the theory of the Austrian psychiatrist Alfred Adler, held that the main motives of human thought and behavior are particular man's striving for superiority and power, partly in compensation for his inferiority. Every individual, in this view, is unique, and his

personality structure, including his great goal and ways of striving for it, finds expression in his style of life, this lifestyle being the product of his creativity. Besides, the individual cannot be considered apart from society; all crucial problems, including problems of general human relations, occupation, and love, are social.

According to Adler, every person has a sense of inferiority, and people "strive for supremacy" from an early age to overcome this inadequacy. Adler believed this drive was the motivation behind human behaviors, emotions, and thoughts. Adler called his theory individual psychology as it expressed his belief that every individual personality is unique and indivisible. His emphasis did not preclude the social.

Individual Psychology's Basic Theory

According to Adler, people are born with weak, inferior bodies, which leads to feelings of inadequacy and, as a result, reliance on others. As a result, people have a social interest from birth, the most reliable indicator of psychological wellness. The essential elements of Adlerian philosophy, mainly, can be sketched forth—in the following excerpt from a list that exhibits individual psychology's basic theory (Adler, 1964).

- The desire for achievement or dominance is the driving factor behind people's activities.
- People's actions and personalities are shaped by their subjective perceptions.
- The personality is coherent and self-contained.
- The worth of all human action must be viewed through the lens of social interest.
- A self-consistent personality structure forms a person's living style.
- People's creative power shapes their way of life.

Pursuing Success or Superiority

The first principle of Adlerian philosophy states that the only dynamic force driving people's conduct is a desire for achievement or supremacy. Adler used the phrase masculine protest, which connoted a desire for power or dominance over others. However, he quickly abandoned masculine protest as a universal urge while maintaining a restricted role for it in his theory of abnormal development. Adler referred to the one dynamic force vying for superiority. However, his ultimate theory limited striving for supremacy to those who aspire for personal superiority over others.

Subjective Beliefs

According to Adler's second postulate, people's subjective impressions impact their conduct and personality. People strive for superiority or success to compensate for emotions of inadequacy, but the method they strive for is formed not by reality but by their subjective conceptions of reality, that is, by their future fiction.

Personality Unity and Self-Consistency

Adlerian theory's third postulate is that personality is unified and self-consistent. Adler wanted to emphasize his idea that each individual is unique and indivisible. Individual psychology thus insists on the essential unity of personality and the absence of contradictory conduct. Thoughts, thoughts, and actions are all directed toward and serve a single aim. When people act erratically or unpredictably, it forces others to be on the defensive, wary, and not confused by arbitrary

actions. Adler (1956) identified various ways the whole person functions with unity and consistency.

Social Issues

According to Adler's fourth precept, the value of all human effort must be evaluated through the prism of social interest. The English translation of Adler's original German word, "gemeinsgefühl," is somewhat misleading. Although "social feeling" or "community feeling" may be a better translation, Gemeinschaftsgefühl has a deeper meaning that no English word or phrase can fully convey. It generally translates to a feeling of unity with all of humanity and signifies inclusion in the global social structure.

According to Adler, the early social milieu's consequences are tremendously essential. The bond between a child and their parents is so strong that it suffocates the consequences of genetics. Adler felt that beyond the age of five, the effects of inheritance are masked by the tremendous influence of the child's social environment. By then, environmental impacts had altered or sculpted practically every facet of a child's personality.

Style of Lifestyle

Adler developed the term "style of lifestyle" to describe the flavor of a person's life. It involves a person's objective, self-concept, feelings towards others, and worldview. It results from the inheritance, environment, and an individual's creative ability. The age of four or five firmly defines a person's living style. After that, everything we do revolves around our unified way of life. Although the end aim is unique, the way of life does not have to be limited or inflexible. Psychologically sick people frequently lead very rigid lifestyles characterized by an incapacity to find new ways of reacting to their surroundings.

Creative Ability

Adler thought that each person had the opportunity to construct their way of life. Ultimately, everyone is responsible for who they are and how they behave. Their creative power gives them control over their lives, is accountable for their ultimate objective, defines how they will achieve that aim, and contributes to the growth of societal interest. In short, creative power liberates each individual. Creative ability is a multidimensional construct that implies movement, and mobility is the most noticeable feature of existence. All psychological existence entails movement toward a goal, movement in a specific direction.

The behavioural perspective

The behavioural perspective emerged as a reaction against the unscientific methods of the psychodynamic approach. Behavioural scientists believed that the study of subjective experiences cannot be tested by other investigators. They resorted to laboratory research rather than clinical practice. Behaviorists focus on the effects of environmental conditionswhen subjected to various stimuli. The central theme of this perspective is learning- the modification of behaviour based on its consequences.

The Cognitive-Behavioural Perspective

This approach focused on cognitive processes and their impact on behaviour. It involved the study of information processing mechanisms like attention, memory, thinking, planning and decision making. Thus, the cognitive behavioural perspective on abnormal behaviour generally focuseson how thoughts and information processing can become

distorted and leads to maladaptive emotions and behaviour.

Lesson-5 Behaviorist perspective- Skinner's operant theory- Bandura social learning theory

Skinner argued that the goal of a science of psychology was to predict and control an organism's behavior from its current stimulus situation and its history of reinforcement.

Skinner was influenced by John B. Watson's philosophy of psychology called behaviorism, which rejected not just the introspective method and the elaborate psychoanalytic theories of Freud and Jung, but any psychological explanation based on mental states or internal representations such as beliefs, desires, memories, and plans. The very idea of "mind" was dismissed as a pre-scientific superstition, not amenable to empirical investigation. Skinner argued that the goal of a science of psychology was to predict and control an organism's behavior from its current stimulus situation and its history of reinforcement. In a utopian novel called Walden Two and a 1971 bestseller called Beyond Freedom and Dignity, he argued that human behavior was always controlled by its environment. According to Skinner, the future of humanity depended on abandoning the concepts of individual freedom and dignity and engineering the human environment so that behavior was controlled systematically and to desirable ends rather than haphazardly.

In the laboratory, Skinner refined the concept of operant conditioning and the Law of Effect. Among his contributions were a systematic exploration of intermittent schedules of reinforcement, the shaping of novel behavior through successive approximations, the chaining of complex behavioral sequences via secondary (learned) reinforcers, and "superstitious" (accidentally reinforced) behavior.

Skinner was also an inveterate inventor. Among his gadgets were the "Skinner box" for shaping and counting lever-pressing in rats and key-pecking in pigeons; the cumulative recorder, a mechanism for recording rates of behavior as a pen tracing; a World-War II-era missile guidance system (never deployed) in which a trained pigeon in the missile's transparent nose cone continually pecked at the target; and "teaching machines" for "programmed learning," in which students were presented a sentence at a time and then filled in the blank in a similar sentence, shown in a small window. He achieved notoriety for a mid-1950s Life magazine article showcasing his "air crib," a temperature-controlled glass box in which his infant daughter would play. This led to the urban legend, occasionally heard to this day, that Skinner "experimented on his daughter" or "raised her in a box" and that she grew up embittered and maladjusted, all of which are false.

B. F. Skinner's Theory of Learning: Operant Conditioning

According to B. F. Skinner's theory of learning, our behaviors are developed or conditioned through reinforcements. He referred to this process as operant conditioning, with operant referring to any behavior that acts on the environment and leads to consequences.

Operant behaviors (actions under our control) differ from respondent behaviors. Skinner described respondent behaviors as anything that occurs reflexively or automatically—such as jerking your hand back when you accidentally touch a hot pan.

Skinner's idea of operant conditioning influenced thoughts about child development, or that a child's behavior could be impacted through positive and negative reinforcements. It also contributed to the behavioral theory of personality, explaining that we respond in certain ways based on our learned experiences.

Skinner's ABCs of Behaviorism

B. F. Skinner's theory of learning says that a person is first exposed to a stimulus, which elicits a response, and the response is then reinforced (stimulus, response, reinforcement). This, ultimately, is what conditions our behaviors.

To make this process easier to remember, the ABCs of behaviorism were developed. The ABCs are antecedent (stimulus), behavior (response), and consequence (reinforcement).

Reinforcement

In B. F. Skinner's theory of learning, reinforcement plays a critical role in behavior development. Reinforcement is any event that strengthens a certain behavior and it can be positive or negative in nature.

Positive reinforcement includes actions or events that strengthen a response by providing a stimulus for certain behaviors, such as giving a child a reward or praise for cleaning their room. Negative reinforcement also strengthens a response, but by the removal of an unfavorable outcome, such as the child cleaning their room to avoid being grounded.

Schedules of Reinforcement

Skinner further described schedules of reinforcement in his operant conditioning research. These schedules determine when specific behaviors are reinforced (either based on the number of responses or time) and impact how strong a learned behavior becomes.

Skinner's four schedules of reinforcement are:

- Fixed-ratio schedules: Responses are reinforced after a specific number of responses
- Variable-ratio schedules: Responses are reinforced after an unspecified or unpredictable number of responses
- Fixed-interval schedules: Responses are reinforced after a specific amount of time
- Variable-interval schedules: Responses are reinforced after an unspecified or unpredictable amount of time

Punishment

Punishment can also play a role in the learning process, according to B. F. Skinner. Punishment is described as the application of an adverse outcome in an effort to decrease or weaken a specific behavior.

Punishment may involve presenting a negative reinforcer—such as a spanking, scolding, or the imposition of a prison sentence—which some refer to as positive punishment. Or it can involve the removal of a positive reinforcer, such as taking away a favorite toy, also known as a negative punishment.

Bandura social learning theory.

Social learning theory, proposed by Albert Bandura, emphasizes the importance of observing, modeling, and imitating the behaviors, attitudes, and emotional reactions of others. Social learning theory considers how both environmental and cognitive factors interact to influence human learning and behavior.

In social learning theory, Albert Bandura (1977) agrees with the behaviorist learning theories of classical conditioning and operant conditioning. However, he adds two important ideas:

1. Mediating processes occur between stimuli & responses.

2. Behavior is learned from the environment through the process of observational learning.

Observational Learning

Observational Learning

Children observe the people around them behaving in various ways. This is illustrated during the famous Bobo doll experiment (Bandura, 1961).

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Individuals that are observed are called models. In society, children are surrounded by many influential models, such as parents within the family, characters on children's TV, friends within their peer group, and teachers at school.

These models provide examples of behavior to observe and imitate, e.g., masculine and feminine, pro and anti-social, etc.

Children pay attention to some of these people (models) and encode their behavior. At a later time, they may imitate (i.e., copy) the behavior they have observed.

They may do this regardless of whether the behavior is 'gender appropriate' or not, but there are a number of processes that make it more likely that a child will reproduce the behavior that society deems appropriate for its gender.

First, the child is more likely to attend to and imitate those people it perceives as similar to itself. Consequently, it is more likely to imitate behavior modeled by people of the same gender.

Second, the people around the child will respond to the behavior it imitates with either reinforcement or punishment. If a child imitates a model's behavior and the consequences are rewarding, the child is likely to continue performing the behavior.

If a parent sees a little girl consoling her teddy bear and says, "what a kind girl you are," this is rewarding for the child and makes it more likely that she will repeat the behavior. Her behavior has been reinforced (i.e., strengthened).

Reinforcement can be external or internal and can be positive or negative. If a child wants approval from parents or peers, this approval is an external reinforcement, but feeling happy about being approved of is an internal reinforcement. A child will behave in a way that it believes will earn approval because it desires approval.

Positive (or negative) reinforcement will have little impact if the reinforcement offered externally does not match with an individual's needs. Reinforcement can be positive or negative, but the important factor is that it will usually lead to a change in a person's behavior.

Third, the child will also take into account what happens to other people when deciding whether or not to copy someone's actions. A person learns by observing the consequences of another person's (i.e., models) behavior, e.g., a younger sister observing an older sister being rewarded for a particular behavior is more likely to repeat that behavior herself. This is known as vicarious reinforcement.

This relates to an attachment to specific models that possess qualities seen as rewarding. Children will have a number of models with whom they identify. These may be people in their immediate world, such as parents or older siblings, or they could be fantasy characters or people in the media. The motivation to identify with a particular model is that they have a quality that the individual would like to possess.

Identification occurs with another person (the model) and involves taking on (or adopting) observed behaviors, values, beliefs, and attitudes of the person with whom you are identifying.

The term identification as used by Social Learning Theory, is similar to the Freudian term related to the Oedipus complex. For example, they both involve internalizing or adopting another person's behavior.

However, during the Oedipus complex, the child can only identify with the same-sex parent, whereas with Social Learning Theory, the person (child or adult) can potentially identify with any other person.

Identification is different from imitation as it may involve a number of behaviors being adopted, whereas imitation usually involves copying a single behavior.

Mediational Processes

SLT is often described as the 'bridge' between traditional learning theory (i.e., behaviorism) and the cognitive approach. This is because it focuses on how mental (cognitive) factors are involved in learning.
Unlike Skinner, Bandura (1977) believes that humans are active information processors and think about the relationship between their behavior and its consequences.

Observational learning could not occur unless cognitive processes were at work. These mental factors mediate (i.e., intervene) in the learning process to determine whether a new response is acquired.

Therefore, individuals do not automatically observe the behavior of a model and imitate it. There is some thought prior to imitation, and this consideration is called the mediational process. This occurs between observing the behavior (stimulus) and imitating it or not (response).

There are four mediational processes proposed by Bandura:

Attention

The individual needs to pay attention to the behavior and its consequences and form a mental representation of the behavior. For a behavior to be imitated, it has to grab our attention. We observe many behaviors on a daily basis, and many of these are not noteworthy. Attention is, therefore, extremely important in whether a behavior influences others to imitate it.

Retention

How well the behavior is remembered. The behavior may be noticed, but it is not always remembered, which obviously prevents imitation. It is important, therefore, that a memory of the behavior is formed to be performed later by the observer.

Much of social learning is not immediate, so this process is especially vital in those cases. Even if the behavior is reproduced shortly after seeing it, there needs to be a memory to refer to.

Reproduction

This is the ability to perform the behavior that the model has just demonstrated. We see much behavior daily that we would like to be able to imitate, but this is not always possible. We are limited by our physical ability, and for that reason, even if we wish to reproduce the behavior, we cannot.

This influences our decisions whether to try and imitate it or not. Imagine the scenario of a 90-year-old lady who struggles to walk while watching Dancing on Ice. She may appreciate that the skill is a desirable one, but she will not attempt to imitate it because she physically cannot do it.

Motivation

The will to perform the behavior. The rewards and punishments that follow a behavior will be considered by the observer. If the perceived rewards outweigh the perceived costs (if there are any), then the behavior will be more likely to be imitated by the observer.

If the vicarious reinforcement is not seen to be important enough to the observer, then they will not imitate the behavior.

Critical Evaluation

The social learning approach takes thought processes into account and acknowledges the role that they play in deciding if a behavior is to be imitated or not.
As such, SLT provides a more comprehensive explanation of human learning by recognizing the role of mediational processes.

For example, Social Learning Theory can explain many more complex social behaviors (such as gender roles and moral behavior) than models of learning based on simple reinforcement.

However, although it can explain some quite complex behavior, it cannot adequately account for how we develop a whole range of behavior, including thoughts and feelings.

We have a lot of cognitive control over our behavior, and just because we have had experiences of violence does not mean we have to reproduce such behavior.

It is for this reason that Bandura modified his theory and in 1986 renamed his Social Learning Theory, Social Cognitive Theory (SCT), as a better description of how we learn from our social experiences.

Some criticisms of social learning theory arise from their commitment to the environment as the chief influence on behavior. It is limiting to describe behavior solely in terms of either nature or nurture, and attempts to do this underestimate the complexity of human behavior.

It is more likely that behavior is due to an interaction between nature (biology) and nurture (environment).
Social learning theory is not a full explanation for all behavior. This is particularly the case when there is no apparent role model in the person's life to imitate for a given behavior.

The discovery of mirror neurons has lent biological support to the social learning theory. Although research is in its infancy, the recent discovery of "mirror neurons" in primates may constitute a neurological basis for imitation. These are neurons that fire both if the animal does something itself and if it observes the action being done by another.

Bobo doll experiment (Bandura, 1961).

Stage 1: Modeling

In the experimental conditions, children were individually shown into a room containing toys and played with some potato prints and pictures in a corner for 10 minutes while either:

- 24 children (12 boys and 12 girls) watched a male or female model behaving aggressively towards a toy called a "Bobo doll". The adults attacked the Bobo doll in a distinctive manner – they used a hammer in some cases, and in others threw the doll in the air and shouted "Pow, Boom."

- Another 24 children (12 boys and 12 girls) were exposed to a non-aggressive model who played in a quiet and subdued manner for 10 minutes (playing with a tinker toy set and ignoring the bobo-doll).

- The final 24 children (12 boys and 12 girls) were used as a control group and not exposed to any model at all.

Stage 2: Aggression Arousal

All the children (including the control group) were subjected to "mild aggression arousal." Each child was (separately) taken to a room with relatively attractive toys.

As soon as the child started to play with the toys, the experimenter told the child that these were the experimenter's very best toys and she had decided to reserve them for the other children.

Stage 3: Test for Delayed Imitation

- The next room contained some aggressive toys and some non-aggressive toys. The non-aggressive toys included a tea set, crayons, three bears and plastic farm animals. The aggressive toys included a mallet and peg board, dart guns, and a 3 foot Bobo doll.

- The child was in the room for 20 minutes, and their behavior was observed and rated though a one-way mirror. Observations were made at 5-second intervals, therefore, giving 240 response units for each child.

- Other behaviors that didn't imitate that of the model were also recorded e.g., punching the Bobo doll on the nose.

Results

- Children who observed the aggressive model made far more imitative aggressive responses than those who were in the non-aggressive or control groups.

- There was more partial and non-imitative aggression among those children who had observed aggressive behavior, although the difference for non-imitative aggression was small.

- The girls in the aggressive model condition also showed more physically aggressive responses if the model was male, but more verbally aggressive responses if the model was female. However, the exception to this general pattern was the observation of how often they punched Bobo, and in this case the effects of gender were reversed.

- Boys were more likely to imitate same-sex models than girls. The evidence for girls imitating same-sex models is not strong.

- Boys imitated more physically aggressive acts than girls. There was little difference in verbal aggression between boys and girls.

Lesson-6 Humanistic Existential perspective- Maslow needs Hierarchy theory

Humanistic perspective –Maslow theory

The basic beliefs of humanism include that people are generally good, have almost limitless potential, and all persons are unique individuals.

Maslow's humanistic theory came about as an alternative to psychoanalysis and behaviorism beliefs. The former espouses that people are mostly controlled by various urges they experience, while the latter believes that punishments and rewards are the root cause of most human deportment.

Maslow believed human beings were destined for greater ideals, and in fact, the word "potential" was one of his main points when referring to the human condition. He also spoke of the concept of "peak experiences," and in a well-known interview in 1968; he even used the word "meta motivation" to describe top human achievers.

Abraham Maslow founded the Brandeis Psychology Department in Waltham, Massachusetts, and taught there through most of the 1950s and 1960s.

Hierarchy of Needs

In Maslow's hierarchy of needs, persons attempt to fulfill bottom-level needs before moving up through the other levels.



The Famous Hierarchy of Needs

1) At the bottom of Maslow's pyramid are physiological needs, which entail:

- Air
- Food
- Shelter
- Warmth
- Water

A human being can only survive without air, specifically oxygen, for about three minutes, three days without water, and three weeks without food. Thus, according to the hierarchy of needs, these basic life requirements must first be met before moving up the pyramid to "higher-level needs." These needs include sleep, excretion, and staying mostly in a state of homeostasis as well.

2) The next level up is security needs, which include:

- Insurance
- Safety
- Steady job

Above is delineated how a human being meets the basic requirements of day-to-day living in this world. These include not only the security of the physical body but also of employment and property.

3) We see that the bottom two pyramid levels deal a lot with merely surviving on this harsh planet. At the next level are the social needs of:

- Belonging
- Family
- Love

Above are deemed needs, but technically, a human being does not need them to survive. These include meaningful friendships and close family bonds which last for many years.

4) The second level from the top includes the two esteem needs of:

- Accomplishment
- Self-worth

Above are ideas such as gaining the respect of other human beings, giving them respect in return, having the self-confidence to complete a task such as running a marathon, setting a goal such as graduating from college and then achieving it, and looking in the mirror and liking oneself.

5) The top level of the pyramid is reserved for self-actualizing needs that involve:

- Self-aware personal growth

Lesson-7 Cognitive perspective – Beck and Ellis

Cognitive perspective Beck and Ellis

The cognitive model posits that the way people perceive their experiences influences their emotional, behavioral, and physiological reactions. Correcting misperceptions and modifying unhelpful thinking and behavior brings about improved reactions (Beck, 1964).

RATIONAL EMOTIVE BEHAVIOR THERAPY

In order to understand Ellis' perspective on therapy, one must first understand his perspective on the basis of psychological disturbances, whether they are minor problems with personal adjustment or more serious forms of mental illness. Individuals with problems typically have a long history related to the disorder. Using the same example as Ellis (1957), suppose a woman is chronically depressed because she has been rejected by men she really liked and with whom she wanted to have long-term relationships. She understandably concludes that the activating event of being rejected leads to the consequence of being depressed. However, this conclusion is wrong! According to Ellis, it is not the rejection that causes her depression, it is the belief system that arises within her, particularly irrational beliefs, that cause her depression. For example, if she rationally believed that it was unfortunate that someone she liked rejected her, or that it was frustrating that she was rejected, she might not become depressed. However, if she irrationally believes that it is *awful* that she has been rejected, or that she *should* have been more beautiful so that he wouldn't have rejected her, then she is quite likely to become depressed. This is what Ellis referred to as the A-B-C's of emotional disturbance or self-defeating behaviors and attitudes. *A* refers to the activating event (being rejected), *B* refers to the person's beliefs (this is awful, I'm not pretty), and *C* is the consequence of the beliefs: depression. The basis for therapy in such situations can be found by extending the A-B-C's to the D-E's: disputing (D) the irrational beliefs, which hopefully leads to the cognitive effect (E -effective new philosophies, emotions, and behaviors) of disrupting the self-defeating patterns of behavior (Ellis, 1957, 1973, 1996). More recently, Ellis has proposed one more letter for his ABC theory of personality, the letter *G* for goals (*note: there is no "F"). Goals consist of a person's purposes, values, standards, and hopes. When these goals are thwarted by an activating event, the person can respond by choosing healthy or unhealthy alternatives, and the nature of that choice is based on one's beliefs (Ellis, 1994). Ellis considered the ABC theory to be so straightforward that it could prove helpful to anyone.

When an individual has become trapped by unhealthy belief systems and the corresponding self-defeating behavior patterns that accompany them, the potential need for psychotherapy arises. The therapy that Ellis developed has come to be known as rational emotive behavior therapy, or REBT (the name went through several permutations over Ellis' career, and he finally settled on REBT). The primary task of the therapist using REBT is to challenge the client's irrational beliefs and, in so doing, to help the client change their belief systems. In essence, when the client believes that it would be catastrophic for a certain negative outcome to occur, the

therapist tries to help them by disputing the irrational belief with questions such as: "Why would a certain outcome be catastrophic?" While trying to dispute the irrational beliefs, the therapist also searches for underlying philosophies that support the irrational belief system, philosophies that can then also be disputed. In addition to the cognitive aspect of REBT, the therapist often encourages the client to act against their irrational fears. If the client is willing, they have the opportunity to experience anxiety-ridden situations without the catastrophic consequences they have feared (although the help of the therapist may prove necessary along the way). Throughout this process, REBT does not intend to ignore the person's feelings. However, when a client is suffering from unhealthy and self-defeating feelings, such as anxiety, depression, or anger, REBT can help to minimize those unhealthy feelings. In addition, REBT encourages healthy, positive emotions, and recognizes that sometimes a strong negative response, such as sadness or grief, to a tragic activating event may be healthy or constructive. When utilized effectively, REBT offered what Ellis believed was a better, deeper, and more enduring therapy, which could achieve those results in a fairly brief amount of time (Ellis, 1962, 1973, 1994, 1995, 1996). Ellis also believed that REBT was applicable to a wider range of clients than any other psychotherapy:

> RET [later known as REBT], on the contrary, seems to be almost the only major kind of psychotherapy (aside, perhaps, from Zen Buddhism, if this is conceptualized as psychotherapy…) that holds that the individual does not need *any* trait, characteristic, achievement, purpose, or social approval in order to accept himself. In fact, he does not have to rate himself, esteem himself, or have any self-measurement or self-concept whatever. (pg. 65; Ellis, 1973)

First and foremost, Ellis focused on practical applications of psychotherapy, and he considered his approach to be humanistic in its emphasis on the whole person. He acknowledged that REBT shared important elements with the approaches of other classic theorists who had emphasized the value of individuals, including Alfred Adler, Viktor Frankl, Rollo May, Carl Rogers, Abraham Maslow, and Karen Horney (Ellis, 1973, 1995). Ellis' discussion of the practice of REBT seems to focus on what Horney addressed in her concepts of neurotic needs and the tyranny of "the should." Taken together, the desire to focus on practical applications and helping individuals has led to a wide variety of self-help books based on REBT (Yankura & Dryden, 1994). REBT has also been applied to a wide variety of other problems that have been covered in Ellis' books, such as marriage counseling, personality disorders, depression, and even schizophrenia, and REBT has proved successful in both individual and group settings (Ellis, 1962, 1973, 2001; Ellis & Dryden, 1987).

Ellis, as well as other therapists using REBT, have also addressed addictive disorders, including alcoholism (Ellis, 2001; Ellis, McInerney, DiGuiseppe, & Yeager, 1988; Trimpey, Velten, & Dain, 1993; Yeager, Yeager, & Shillingford, 1993). In the early 1990s, Rational Recovery (RR) was developed by Jack Trimpey as an alternative to the Alcoholics Anonymous (AA) approach to treating alcohol abuse. Both Ellis and Trimpey challenge the basic principles of AA: that the addict has no control over their alcohol cravings, that they must turn over control to a higher power (such as God), that they can never drink alcohol again, and that only AA works for alcoholics. Objective research simply does not support these assertions, and practitioners of REBT and RR have great faith in the ability of individuals to take control of their own lives (though they may need some help from a therapist to get on the right path). RR also does not include the strong religious overtones of AA, which may be an impediment to recovery for anyone required to attend AA meetings but who does not believe in God.

Aaron Beck cognitive behavioral therapy

Beck developed cognitive therapy with the belief that a person's experiences result in cognitions or thoughts. These cognitions are connected with schemas, which are core beliefs developed from early life, to create our view of the world and determine our emotional states and behaviors.

Aaron Beck was an American psychiatrist who analyzed how a person's thought process affects their daily life. He felt these thoughts, also known as cognitions, were a direct reflection of a person's lived experiences and could shape the way a person perceived the world. If a person had negative lived experiences and those became daily cognitions, those negative narratives would, in turn, cause issues such as depression and anxiety. These narratives ultimately transform into a set of beliefs or expectations called schemas, which dictate a person's quality of life in various ways.

Beck believed these schemas are generally created early in life, paving the way for distorted views of the self and reality. When a person can only see their life through a pessimistic lens, their understanding and interpretation of reality become distorted, which ultimately leads a person into a negative emotional state. Aaron Beck's theory became known as the cognitive triad, each aspect playing off the next. This triad includes negative self-image, negative world image, and negative future images. He felt a person's depression would start at one of these triad points. For example, a negative perspective of oneself would lead to a negative view of the world. This perception would ultimately lead to a negative view about their future. The triad becomes a vicious cycle that keeps a person stuck in a loop of depression.

Through his research with depressed patients, Beck developed what is now known as cognitive therapy. Cognitive therapy is one of Aaron Beck's contributions to psychology that is rooted in the idea that what one thinks becomes their reality. Therefore, the practice aims to weed out distorted and negative beliefs so a person can view the world with optimism and hope. While the creation of Aaron Beck's cognitive therapy was founded through work with depressed patients, it has since been used to help with mental health issues of all kinds, such as PTSD, anorexia, and anxiety, to name just a few.

Negative Self-Schemas

Beck's cognitive model centers on the idea of schemas, such as negative self-schemas. These beliefs or expectations focus on how a person perceives themselves within the world. He believed that depressed patients have a tendency to focus solely on negative and pessimistic ideas about themselves, which leads to the triad's never-ending loop. For example, say a student studies all night for a math test, using practice tests and online tools to ensure they understand the formulas that will be presented on the test. The following day, they take the test and end up failing. While failing the first test does not necessarily result in a negative self-schema, failing another test in the same or a different subject on the same day may result in a negative self-schema. The student may become disheartened about their intelligence level, leading to a pessimistic view of their abilities and skill sets.

Aaron Beck's Cognitive Therapy

Based on his work with depressed patients, Aaron Beck created what is known as cognitive therapy, a psychological modality to help people work through and change their cognitive distortions and ways of thinking. The goal of Beck's cognitive therapy is to help identify and alter these schemas, or distorted beliefs, to improve behavior and experiences, helping clients live happier and healthier lives. To start the process, a therapist will assess the client's distortions and automatic thoughts. Automatic thoughts are typically strong and negative narratives that a person is generally unaware of, yet these patterns of pessimism rule a person's perspective. From here, the real work can begin.

This is where cognitive therapy can help. The therapist will identify the distortions and the automatic thoughts through talk therapy and help the client remove and replace them with positive and/or rational thinking so they can begin to see the world through a clearer and more positive lens. This type of therapy helps reframe negative thinking and builds self-worth and self-esteem.

Therefore, negative and unrealistic thoughts can cause us distress and result in problems. When a person suffers from psychological distress, how they interpret situations becomes skewed, which, in turn, has a negative impact on the actions they take.

CBT aims to help people become aware of when they make negative interpretations and of behavioral patterns which reinforce distorted thinking. Cognitive therapy helps people develop alternative ways of thinking and behaving to reduce their psychological distress.

Cognitive behavioral therapy is, in fact, an umbrella term for many different therapies that share some common elements. Two of the earliest forms of Cognitive Behavioral Therapy were Rational Emotive Behavior Therapy (REBT), developed by Albert Ellis in the 1950s, and Cognitive Therapy, developed by Aaron T. Beck in the 1960s.

General Assumptions

• The cognitive approach believes that mental illness stems from faulty cognitions about others, our world, and us. This faulty thinking may be through cognitive deficiencies (lack of planning) or cognitive distortions (processing information inaccurately).

• These cognitions cause distortions in how we see things; Ellis suggested it is through irrational thinking, while Beck proposed the cognitive triad.

• We interact with the world through our mental representation of it. If our mental representations are inaccurate or our ways of reasoning are inadequate, our emotions and behavior may become disordered.

The cognitive therapist teaches clients how to identify distorted cognitions through a process of evaluation. The clients learn to discriminate between their own thoughts and reality. They learn the influence that cognition has on their feelings, and they are taught to recognize, observe and monitor their own thoughts.

The behavior part of the therapy involves setting homework for the client to do (e.g., keeping a diary of thoughts). The therapist gives the client tasks to help them challenge their irrational beliefs.

The idea is that the client identifies their own unhelpful beliefs and then proves them wrong. As a result, their beliefs begin to change. For example, someone who is anxious in social situations may be set a homework assignment to meet a friend at the pub for a drink.

Albert Ellis – REBT

Rational Emotive Behavior Therapy (REBT) is a type of cognitive therapy first used by Albert Ellis, focusing on resolving emotional and behavioral problems.

The goal of this therapy is to change irrational beliefs to more rational ones.
REBT encourages people to identify their general and irrational beliefs (e.g., 'I must be perfect') and subsequently persuades them to challenge these false beliefs through reality testing.

Albert Ellis (1957, 1962) proposes that each of us holds a unique set of assumptions about ourselves and our world that guide us through life and determine our reactions to the various situations we encounter.

Unfortunately, some people's assumptions are largely irrational, guiding them to act and react in inappropriate ways that prejudice their chances of happiness and success. Albert Ellis calls these basic irrational assumptions.

Some people irrationally assume they are failures if they are not loved by everyone they know – they constantly seek approval and repeatedly feel rejected. All their interactions are affected by this assumption so that a great party can leave them dissatisfied because they don't get enough compliments.

According to Ellis, these are other common irrational assumptions :

• The idea that one should be thoroughly competent at everything.

• The idea that it is catastrophic when things are not the way you want them to be.

• The idea that people have no control over their happiness.

• The idea that you need someone stronger than yourself to depend on.

• The idea that your history greatly influences your present life.

• The idea that there is a perfect solution to human problems, and it's a disaster if you don't find it.

Ellis believes that people often forcefully hold on to this illogical way of thinking and therefore employ highly emotive techniques to help them vigorously and forcefully change this irrational thinking.

The ABC Model

A major aid in cognitive therapy is what Albert Ellis (1957) called the ABC Technique of Irrational Beliefs.

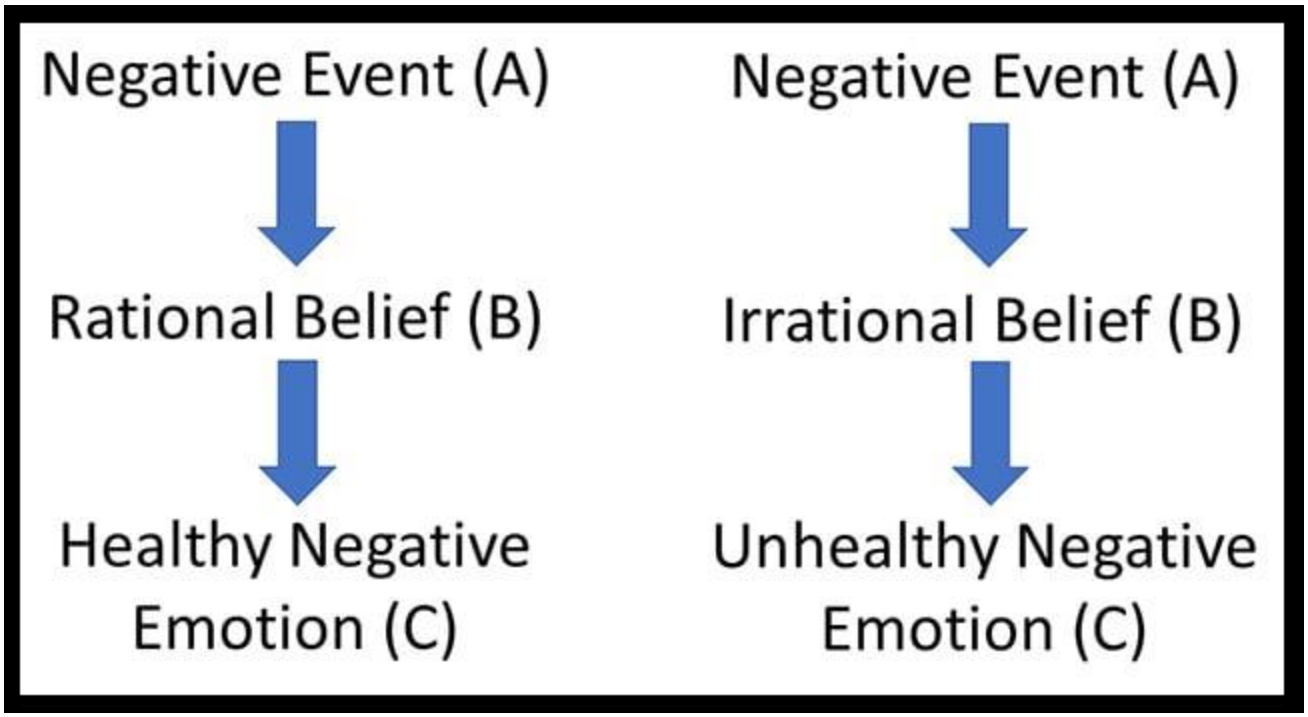
The first three steps analyze the process by which a person has developed irrational beliefs and may be recorded in a three-column table.

* A – Activating Event or objective situation. The first column records the objective situation, that is, an event that ultimately leads to some type of high emotional response or negative dysfunctional thinking.

* B – Beliefs. In the second column, the client writes down the negative thoughts that occurred to them.

* C – Consequence. The third column is for the negative feelings and dysfunctional behaviors that ensued. The negative thoughts of the second column are seen as a connecting bridge between the situation and the distressing feelings. The third column, C, is next explained by describing emotions or negative thoughts that the client thinks are caused by A. This could be anger, sorrow, anxiety, etc.

Ellis believes that it is not the activating event (A) that causes negative emotional and behavioral consequences (C) but rather that a person interprets these events unrealistically and therefore has an irrational belief system (B) that helps cause the consequences (C).



REBT Example

Gina is upset because she got a low mark on a math test. The Activating event, A, is that she failed her test. The Belief, B, is that she must have good grades or she is worthless. The Consequence, C, is that Gina feels depressed.

After identifying irrational beliefs, the therapist will often work with the client in challenging the negative thoughts based on evidence from the client's experience by reframing it, meaning to re-interpret it in a more realistic light. This helps the client to develop more rational beliefs and healthy coping strategies.

A therapist would help Gina realize that there is no evidence that she must have good grades to be worthwhile or that getting bad grades is awful. She desires good grades, and it would be good to have them, but it hardly makes her worthless.

If she realizes that getting bad grades is disappointing but not awful and that it means she is currently bad at math or studying but not as a person, she will feel sad or frustrated but not depressed. The sadness and frustration are likely healthy negative emotions and may lead her to study harder from then on.

Critical Evaluation

Rational emotive behavior therapists have cited many studies in support of this approach. Most early studies were conducted on people with experimentally induced anxieties or non-clinical problems such as mild fear of snakes (Kendall & Kriss, 1983).

However, several recent studies have been done on actual clinical subjects and have also found that rational emotive behavior therapy (REBT) is often helpful (Lyons & Woods 1991).

Aaron Beck – Cognitive Therapy

Beck's (1967) system of therapy is similar to Ellis's but has been most widely used in cases of depression. Cognitive therapists help clients to recognize the negative thoughts and errors in logic that cause them to be depressed.

The therapist also guides clients to question and challenge their dysfunctional thoughts, try out new interpretations, and ultimately apply alternative ways of thinking in their daily lives. Aaron Beck believes that a person's reaction to specific upsetting thoughts may contribute to abnormality. As we confront the many situations that arise in life, both comforting and upsetting thoughts come into our heads. Beck calls these unbidden cognitions automatic thoughts.

When a person's stream of automatic thoughts is very negative, you would expect a person to become depressed (e.g., 'I'm never going to get this essay finished, my girlfriend fancies my best friend, I'm getting fat, I have no money, my parents hate me – have you ever felt like this?').

Quite often, these negative thoughts will persist despite contrary evidence.

Beck (1967) identified three mechanisms that he thought were responsible for depression:

1. The cognitive triad (of automatic negative thinking)
2. Negative self-schemas
3. Errors in Logic (i.e., faulty information processing)

The Cognitive Triad

The cognitive triad is three forms of negative (i.e., helpless and critical) thinking that are typical of individuals with depression: namely, negative thoughts about the self, the world, and the future. These thoughts tended to be automatic in depressed people as they occurred spontaneously.

As these three components interact, they interfere with normal cognitive processing, leading to impairments in perception, memory, and problem-solving, with the person becoming obsessed with negative thoughts.

Negative Self-Schemas

Beck believed that depression-prone individuals develop a negative self-schema. They possess a set of beliefs and expectations about themselves that are essentially negative and pessimistic.

Beck claimed that negative schemas might be acquired in childhood due to a traumatic event. Experiences that might contribute to negative schemas include:

- Death of a parent or sibling.
- Parental rejection, criticism, overprotection, neglect, or abuse.
- Bullying at school or exclusion from a peer group.

People with negative self-schemas become prone to making logical errors in their thinking, and they tend to focus selectively on certain aspects of a situation while ignoring equally relevant information.

Cognitive Distortions

Beck (1967) identifies several illogical thinking processes (i.e., distortions of thought processes). These illogical thought patterns are self-defeating and can cause great anxiety or depression for the individual.

• Arbitrary interference: Drawing conclusions on the basis of sufficient or irrelevant evidence: for example, thinking you are worthless because an open-air concert you were going to see has been rained off.

• Selective abstraction: Focusing on a single aspect of a situation and ignoring others: E.g., you feel responsible for your team losing a football match even though you are just one of the players on the field.

• Magnification: exaggerating the importance of undesirable events. E.g., if you scrape a bit of paintwork on your car and, therefore, see yourself as a totally awful driver.

• Minimisation: underplaying the significance of an event. E.g., you get praised by your teachers for an excellent term’s work, but you see this as trivial.

• Overgeneralization: drawing broad negative conclusions on the basis of a single insignificant event. E.g., you get a D for an exam when you normally get straight As and you, therefore, think you are stupid.

• Personalisation: Attributing the negative feelings of others to yourself. E.g., your teacher looks really cross when he comes into the room, so he must be cross with you.

UNIT III: NATURE OF CAUSATION OF ABNORMALITIES

LESSON-8 Causes of abnormal behavior- biological causes: constitutional, neurophysiologic and biochemical psychological causes

The major factors involved are biological, psychological, social, and socio-cultural. It is called integrative because no one factor in isolation impacts the behavior, but multiple factors interact to influence abnormal behavior. Other than genes, further biological factors that contribute efficiently to the abnormality of one's behaviour are neurotransmitters and hormones. Imbalance in neurotransmitters like GABA, norepinephrine, etc., causes abnormal neural plasticity or brain dysfunction.

Biological factors such as faulty genes, endocrine imbalances, malnutrition, injuries and other conditions may interfere with normal development and functioning of the human body. These factors may be potential causes of abnormal behaviour. causes of abnormal Behaviours?

Cause

- Genetics.
- Biological causal factors.
- Socio-cultural factors.
- Systemic factors.
- Biopsychosocial factors.

The biological model of abnormality says that psychological disorders are a result of biological issues. According to this model, Percy's depression might be caused by a chemical imbalance in his brain. Let's look closer at the strengths and limitations of the biological model of abnormal psychology.

Constitutional neurophysilogicalcauses.

Neurogological system ,ANS and CNS brain functional system. Legal Mentalillnessmental retardation. legal protections.

Biochemical causes.

Biochemical diseases as a group are not recognizable by a common clinical trait or a common pattern of organ involvement. Depending on the particular defect, biochemical diseases may affect different organs and organ systems (liver, heart, kidney, muscle, brain and others) as single organs or as multi-organ disease.

Throughout life, human beings undergo several hormonal changes responsible for growth and maturation. These alterations in hormone secretion include enhanced or decreased production, the latter of which is mainly observed during aging. These processes are intrinsic to human development, but may vary from individual to individual. Thus, experienced metabolic changes can alter the state of health and even trigger the development of certain pathologies.

The main metabolic differences observed in newborns and children when compared to adults result from the fact that the organism is not yet fully developed. During adolescence, changes in hormone secretion occur that lead to sexual maturation. In the same way, during pregnancy, women suffer alterations in the secretion of certain hormones which allow the adaptation of their bodies to that physiological state and the normal development of the foetus. As for the elderly, a general decline of health is observed during ageing, and hormonal dysfunctions, such as the development of insulin resistance and thyroid dysfunction, frequently occur.

This volume focuses on metabolic and hormonal changes during the human lifetime. Screenings best suited for each life stage, the reasons for doing them, and the diseases they allow the diagnosis of are also presented.

Lesson-9 Pathogenic causes, Pathogenic Interpersonal relationships.

Diseases in humans that are caused by infectious agents are known as pathogenic diseases. Not all diseases are caused by pathogens, such as black lung from exposure to the pollutant coal dust, genetic disorders like sickle cell disease, and autoimmune diseases like lupus.

Your body is naturally full of microbes. However, these microbes only cause a problem if your immune system is weakened or if they manage to enter a normally sterile part of your body.

Pathogens are different and can cause disease upon entering the body.

All a pathogen needs to thrive and survive is a host. Once the pathogen sets itself up in a host's body, it manages to avoid the body's immune responses and uses the body's resources to replicate before exiting and spreading to a new host.

Pathogens can be transmitted a few ways depending on the type. They can be spread through skin contact, bodily fluids, airborne particles, contact with feces, and touching a surface touched by an infected person.

Pathogen types

There are different types of pathogens, but we're going to focus on the four most common types: viruses, bacteria, fungi, and parasites.

Viruses

Viruses are made up of a piece of genetic code, such as DNA or RNA, and protected by a coating of protein. Once you're infected, viruses invade host cells within your body. They then use the components of the host cell to replicate, producing more viruses.

After the replication cycle is complete, these new viruses are released from the host cell. This usually damages or destroys the infected cells.

Some viruses can remain dormant for a time before multiplying again. When this happens, a person appears to have recovered from the viral infection, but gets sick again.

Antibiotics do not kill viruses and therefore are ineffective as a treatment for viral infections. Antiviral medications can sometimes be used, depending on the virus.

Bacteria

Bacteria are microorganisms made of a single cell. They are very diverse, have a variety of shapes and features, and have the ability to live in just about any environment, including in and on your body. Not all bacteria cause infections. Those that can are called pathogenic bacteria.

Your body can be more prone to bacterial infections when your immune system is compromised by a virus. The disease state caused by a virus enables normally harmless bacteria to become pathogenic.

Antibiotics are used to treat bacterial infections. Some strains of bacteria have become resistant to antibiotics, making them difficult to treat. This can happen naturally, but also happens because of the overuse of antibiotics, according to the World Health Organization (WHO)Trusted Source.

Fungi

There are millions of different fungal species on Earth. Just 300Trusted Source or so are known to cause sickness. Fungi can be found just about everywhere in the environment, including indoors, outdoors, and on human skin. They cause infection when they overgrow.

Fungi cells contain a nucleus and other components protected by a membrane and a thick cell wall. Their structure can make them harder to kill.

Some new strains of fungal infections are proving to be especially dangerous, such as Candida aurus, and have prompted more research into fungal infections.

Parasites

Parasites are organisms that behave like tiny animals, living in or on a host and feeding from or at the expense of the host. Though parasitic infections are more common in tropical and subtropical regions, they can occur anywhere.

Three main types of parasites can cause disease in humans. These include:

- protozoa, which are single-celled organisms that can live and multiply in your body
- helminths, which are larger, multi-celled organisms that can live inside or outside your body and are commonly known as worms
- ectoparasites, which are multi-celled organisms that live on or feed off your skin, including some insects, such as ticks and mosquitos

They can be spread several ways, including through contaminated soil, water, food, and blood, as well as through sexual contact and via insect bites.

Diseases caused by pathogens

Pathogens can cause a number of diseases that range in severity and how they're transmitted. Let's look at some of the diseases caused by the different types of pathogens

Viruses

Viruses can cause a number of infections, many of which are contagious. Examples of viral diseases include:

- common cold
- flu
- meningitis
- warts, including genital warts
- oral and genital herpes
- chickenpox/shingles
- measles
- viral gastroenteritis, including norovirus and rotavirus
- hepatitis A, B, C, D, E
- yellow fever
- dengue fever
- HIV and AIDS

Bacteria

Here are some examples of bacterial infections:

- strep throat
- urinary tract infection (UTI)
- bacterial gastroenteritis, such as salmonella food poisoning or E.coli infection
- bacterial meningitis
- Lyme disease
- tuberculosis
- gonorrhea
- cellulitis

Fungi

Some examples of common fungal infections are:

- vaginal yeast infections
- thrush
- ringworm
- athlete’s foot
- jock itch
- fungal nail infections (onychomycosis)

Parasites

Some examples of diseases caused by parasites include:

- giardiasis. trichomoniasis. malaria toxoplasmosis.intestinal worms
- pubic lice

Protecting against pathogens

The following are ways that you can protect yourself and others against pathogens.

- Wash your hands often.
- Get vaccinated and ensure vaccinations are up to date.
- Prepare, cook, and store meat and other foods properly.
- Stay home when you're sick, especially if you have a fever or diarrhea, or are vomiting.
- Don't share personal items, such as razors or toothbrushes.
- Don't share drinking glasses or utensils.
- Protect against insect bites.
- Practice safe sex.
- Travel wisely by getting informed about health risks and special vaccinations.

Lesson-10 Socio– culture causes: Social class, Social roles, and Cultural factors.

According to this viewpoint our life experiences and interaction with the society help us face challenges resourcefully and may lead to resilience during stress in the future. Unfortunately some of our experiences as a child may be unhelpful and may influence us later in life. Social factors areinfluences in the environment that consists of unpredictable and uncontrollable negative events.

Different social factors that can have a detrimental effect on a child's socioemotional development are listed below.

Early Deprivation and trauma

Children who do not receive adequate food, shelter, love and attention maybe left with deep and irreversible psychological scars. This kind of treatment is usually observed in foster homes and other institutions for children. Sometimes deprivation can occur in families where the parents suffer from mental disorders themselves thus are unwilling to provide care to the child.

Problems in parenting style

Deviations in parenting can also have profound impacts on a child's ability to cope with life's difficulties. This may cause the child to be vulnerable to various forms of psychopathology. For example, children who are anxious, irritable and impulsive may cause the parents to become anxious and irritable thus further worsening the condition of the child. Parenting styles like authoritarian and neglectful parents may result in aggressive behaviourof children and cause them to resort to drug and alcohol abuse.

Marital Discord and Divorce

A disturbed family structure serves as a high risk factor to psychopathology. Marital discord can affect the offspring's marriage and may lead to negative interaction styles. Divorce of parents and have traumatic effects on the child. It can lead to a feeling of insecurity, disloyalty and delinquency.

Low socioeconomic status and unemployment

The lower economic class have a higher incidence of mental and physical disorders. For example antisocial disorder occurs in the lower socioeconomic backgrounds thrice as often as it occurs in better economic conditions. People with mental disorders are usually prejudiced and slide down the economic ladder because of the lack of opportunities. Unemployment, financial hardships, self-devaluation and emotional distress is associated with enhanced chance of psychopathology.

Other social factors that can cause abnormal behaviors include prejudice, discrimination and strained relationship among peers.

Cultural variables such as over and under controlled behaviour can alsocontribute to mental disorders. Although many serious mental disorders arefairly universal, the form some mental disorders take varies widely among different cultures.

Lesson-11 Predisposing, Precipitating reinforcing, Feedback and circularity of Causes.

Predisposing factors are those that put a child at risk of developing a problem (in this case, high anticipatory distress). These may include genetics, life events, or temperament. Precipitating factors refer to a specific event or trigger to the onset of the current problem.

What are examples of precipitating factors?

Examples of Precipitating Factors

- Fear, anxiety, stress.
- Unmet physical needs (hunger, silence) or emotional needs (recognition, love)
- Traumatic experiences.
- Pain.
- Impaired cognitive ability (e.g., a result of intellectual disabilities, mental illness, or dementia)
- Impaired communication skills.
- Frustration.

common symptoms of circulatory system diseases?

- Angina (chest pain with exertion).
- Dizziness.
- Dyspnea (shortness of breath).
- Edema (swelling, most commonly in your legs).
- Fainting.
- Fatigue.

- Heart palpitations (an irregular or forceful heartbeat).
  Circulatory System Diseases: What You Should Know
- High blood pressure.
- CAD.
- Heart attacks.
- Heart failure.
- Strokes.
- Aneurism.
- PAD.
- Risk factors.

Unit IV: Classification of Mental Disorders

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- Lesson-12 Psycho-Neurosis- Causes and Aetiology-Psychoses-Causes and Aetiology Anxiety disorders.

- A functional personality disorder characterized by disturbing emotional symptoms such as morbid fears, obsessive thoughts, or depressive states, but without gross personality disorganization or loss of contact with reality. The terms neurosis and psychoneurosis are interchangeable.

  Neurotic symptoms are generally viewed as exaggerated defense mechanisms arising out of unconscious attempts to cope with internal conflicts and the anxiety they produce—for example, one person screens out his difficulties by becoming preoccupied with repetitive thoughts (obsessions), while another may escape from an unbearable situation by forgetting who he is (amnesia).

  These faulty reactions weaken the individual's effectiveness and generate intense feelings of distress and unhappiness. The psychoneurotic therefore needs help in changing his basic reaction patterns so that he can deal more successfully with the problems and demands of life.Many authorities of today believe it is theoretically possible to draw a continuous line from normal behavior through psychoneurosis to psychosis.

  A strong external stimulation, which even people with a "strong psyche" find difficult to cope with, is the second cause of psychogenic neurosis. Such irritants include chronic stress at work, disputes at home, household troubles, health problems, loss of loved ones, financial issues, and so on.

  Anxiety disorder

  Occasional anxiety is a normal part of life. Many people worry about things such as health, money, or family problems. But anxiety disorders involve more than temporary worry or fear. For people with an anxiety disorder, the anxiety does not go away and can get worse over time.

The symptoms can interfere with daily activities such as job performance, schoolwork, and relationships.

There are several types of anxiety disorders, including generalized anxiety disorder, panic disorder, social anxiety disorder, and various phobia-related disorders.

Generalized anxiety disorder

Generalized anxiety disorder (GAD) usually involves a persistent feeling of anxiety or dread, which can interfere with daily life. It is not the same as occasionally worrying about things or experiencing anxiety due to stressful life events. People living with GAD experience frequent anxiety for months, if not years.

Symptoms of GAD include:

- Feeling restless, wound-up, or on-edge
- Being easily fatigued
- Having difficulty concentrating
- Being irritable
- Having headaches, muscle aches, stomachaches, or unexplained pains
- Difficulty controlling feelings of worry
- Having sleep problems, such as difficulty falling or staying asleep

Panic disorder

People with panic disorder have frequent and unexpected panic attacks. Panic attacks are sudden periods of intense fear, discomfort, or sense of losing control even when there is no clear danger or trigger. Not everyone who experiences a panic attack will develop panic disorder.

During a panic attack, a person may experience:

- Pounding or racing heart
- Sweating
- Trembling or tingling
- Chest pain
- Feelings of impending doom
- Feelings of being out of control

People with panic disorder often worry about when the next attack will happen and actively try to prevent future attacks by avoiding places, situations, or behaviors they associate with panic attacks. Panic attacks can occur as frequently as several times a day or as rarely as a few times a year.

Social anxiety disorder

Social anxiety disorder is an intense, persistent fear of being watched and judged by others. For people with social anxiety disorder, the fear of social situations may feel so intense that it seems

beyond their control. For some people, this fear may get in the way of going to work, attending school, or doing everyday things.

People with social anxiety disorder may experience:

- Blushing, sweating, or trembling
- Pounding or racing heart
- Stomachaches
- Rigid body posture or speaking with an overly soft voice
- Difficulty making eye contact or being around people they don't know
- Feelings of self-consciousness or fear that people will judge them negatively

Phobia-related disorders

A *phobia* is an intense fear of—or aversion to—specific objects or situations. Although it can be realistic to be anxious in some circumstances, the fear people with phobias feel is out of proportion to the actual danger caused by the situation or object.

People with a phobia:

- May have an irrational or excessive worry about encountering the feared object or situation
- Take active steps to avoid the feared object or situation
- Experience immediate intense anxiety upon encountering the feared object or situation
- Endure unavoidable objects and situations with intense anxiety

There are several types of phobias and phobia-related disorders:

Specific Phobias (sometimes called simple phobias): As the name suggests, people who have a specific phobia have an intense fear of, or feel intense anxiety about, specific types of objects or situations. Some examples of specific phobias include the fear of:

- Flying
- Heights
- Specific animals, such as spiders, dogs, or snakes
- Receiving injections
- Blood

Social anxiety disorder (previously called social phobia): People with social anxiety disorder have a general intense fear of, or anxiety toward, social or performance situations. They worry that actions or behaviors associated with their anxiety will be negatively evaluated by others, leading them to feel embarrassed. This worry often causes people with social anxiety to avoid social situations. Social anxiety disorder can manifest in a range of situations, such as within the workplace or the school environment.

Agoraphobia: People with agoraphobia have an intense fear of two or more of the following situations:

- Using public transportation
- Being in open spaces
- Being in enclosed spaces
- Standing in line or being in a crowd
- Being outside of the home alone

People with agoraphobia often avoid these situations, in part, because they think being able to leave might be difficult or impossible in the event they have panic-like reactions or other embarrassing symptoms. In the most severe form of agoraphobia, an individual can become housebound.

Separation anxiety disorder: Separation anxiety is often thought of as something that only children deal with. However, adults can also be diagnosed with separation anxiety disorder. People with separation anxiety disorder fear being away from the people they are close to. They often worry that something bad might happen to their loved ones while they are not together. This fear makes them avoid being alone or away from their loved ones. They may have bad dreams about being separated or feel unwell when separation is about to happen.

Selective mutism: A somewhat rare disorder associated with anxiety is *selective mutism*. Selective mutism occurs when people fail to speak in specific social situations despite having normal language skills. Selective mutism usually occurs before the age of 5 and is often associated with extreme shyness, fear of social embarrassment, compulsive traits, withdrawal, clinging behavior, and temper tantrums. People diagnosed with selective mutism are often also diagnosed with other anxiety disorders.

What are the risk factors for anxiety?

Researchers are finding that both genetic and environmental factors contribute to the risk of developing an anxiety disorder.

The risk factors for each type of anxiety disorder vary. However, some general risk factors include:

- Shyness or feeling distressed or nervous in new situations in childhood
- Exposure to stressful and negative life or environmental events
- A history of anxiety or other mental disorders in biological relatives

Anxiety symptoms can be produced or aggravated by:

- Some physical health conditions, such as thyroid problems or heart arrhythmia
- Caffeine or other substances/medications

If you think you may have an anxiety disorder, getting a physical examination from a health care provider may help them diagnose your symptoms and find the right treatment.

How is anxiety treated?

Anxiety disorders are generally treated with psychotherapy, medication, or both. There are many ways to treat anxiety, and you should work with a health care provider to choose the best treatment for you.

Psychotherapy

Psychotherapy or “talk therapy” can help people with anxiety disorders. To be effective, psychotherapy must be directed at your specific anxieties and tailored to your needs.

Cognitive behavioral therapy

Cognitive Behavioral Therapy (CBT) is an example of one type of psychotherapy that can help people with anxiety disorders. It teaches people different ways of thinking, behaving, and reacting to situations to help you feel less anxious and fearful. CBT has been well studied and is the gold standard for psychotherapy.

Exposure therapy is a CBT method that is used to treat anxiety disorders. Exposure therapy focuses on confronting the fears underlying an anxiety disorder to help people engage in activities they have been avoiding. Exposure therapy is sometimes used along with relaxation exercises.

Lesson-13 Somatoform disorders: Somatoform Disorders Somatoform disorders– Hypochondriasis - Somatization Disorder - Pain Disorder - Conversion Disorder - Body Dimorphic Disorder.

Somatoform disorder, also known as somatic symptom disorder (SSD) or psychosomatic disorder, is a mental health condition that causes an individual to experience physical bodily symptoms in response to psychological distress.

The somatoform disorders are a group of psychiatric disorders in which patients present with a myriad of clinically significant but unexplained physical symptoms

When physical symptoms are caused by mental (psychological) or emotional factors it is called somatisation. For example, many people have occasional headaches caused by mental stress. But, stress and other mental health problems can cause many other physical and bodily symptoms such as:

- Chest pains.
- Tiredness.
- Dizziness.
- Back pain.
- Feeling sick (nauseated

Hypochondrisis

- Illness anxiety disorder, sometimes called hypochondriasis or health anxiety, is worrying excessively that you are or may become seriously ill. Excessive and persistent fear of, or preoccupation with, having or developing a severe illness; excessive health care seeking · Early childhood. You may have no physical symptoms. reoccupation with a serious illness for at least 6 months.
- Inability to control fears and worries.
- Misinterpreting normal body sensations as symptoms of a disease.
- Constant fear of illness despite reassurance of health status by health care providers.

Someone with hypochondria might have a stomach ache, for example, and rather than understand it as a reaction to something they ate or a bug, fear that they have stomach cancer or appendicitis. They tend to fear severe illnesses like cancer or HIV, rather than more common health ailments like strep throat or a cold.
Paindisorder..
Pain disorder is chronic pain experienced by a patient in one or more areas, and is thought to be caused by psychological stress. The pain is often so severe that it disables the patient from proper functioning. Duration may be as short as a few days or as long as many years.
Causes

causes of pain disorder.

- Psychodynamic theory: unconscious conflicts or desires are converted into somatic symptom to protect the person from conscious awareness of it
- Emotions and communication: children show distress in what may be the only way they can, physical symptoms, when they lack the ability to speak or express their thoughts in any way
- Social influences: where psychological disorders are frowned upon, whether in families or cultures, distress may be expressed in physical terms
- Learning theory: children learn to imitate a family member or pick up on possible gains of being "sick"
- Family systems theory: a child's role in a family may be the sick one as part of the family dynamics. Reasons why fall under four possibilities: enmeshment, overprotection, rigidity or lack of conflict resolution
- Trauma and abuse: this includes physical,[7] psychological, or both combined with somatization. It is a common combination.[1] People who have a history of physical or sexual abuse are more likely to have this disorder. However, not every person with pain disorder has a history of abuse. Early intervention when pain first occurs or begins to become chronic offers the best opportunity for prevention of pain disorder

Conversional disorder

Conversion disorder (also known as functional neurological system disorder) is a condition in which a person experiences physical and sensory problems, such as paralysis, numbness, blindness, deafness or seizures, with no underlying neurologic pathology.

Conversion disorder is a psychiatric disorder characterized by signs and symptoms affecting sensory or motor function inconsistent with patterns of known neurologic diseases or other medical conditions and significantly impact the patient's ability to function.

The symptoms of conversion disorder can vary widely depending on the part of the brain involved. Some of the most common symptoms include:

- Psychogenic non-epileptic seizures (PNES). Seizures typically happen because of conditions or issues with your brain's structure or how it's working. Psychogenic non-epileptic seizures happen because of mental health conditions (the word "psychogenic" means "of mental health origin").
- Sense-related disruptions. Trouble with senses of vision, hearing, smell, taste and touch are all possible with conversion disorder. Some examples include double or tunnel vision, hearing loss or numbness, and the inability to feel something touching your skin.
- Pain. People with conversion disorder often feel pain, sometimes with other symptoms and sometimes on its own.
- Unusual muscle tension, spasms, twitches and tremors. These all happen because of a disruption in how your brain controls your muscles.
- Muscle weakness or paralysis.
- Trouble swallowing (dysphagia).
- Dizziness.
- Fainting or passing out (syncope).
- Chronic fatigue or lack of energy.

People with conversion disorder often seem like they're not worried about their symptoms. This phenomenon, "la belle indifference" (which is French for "beautiful ignorance"), happens most commonly with conversion disorder. However, it isn't necessarily a symptom, and it doesn't happen with every case of conversion disorder.

What causes conversion disorder?

Experts don't know exactly why conversion disorder happens. However, they do know that it's more likely to happen along with certain circumstances and some medical conditions.

Common circumstances seen in people with conversion disorder include:

- A history of childhood abuse.
- Having other mental health conditions, especially depression or anxiety.
- A recent stressful or traumatic event.
- A recent health condition or event acting as a trigger for conversion disorder.

Body dysmorphic disorder

Body dysmorphic disorder (BDD), or body dysmorphia, is a mental health condition where a person spends a lot of time worrying about flaws in their appearance. These flaws are often unnoticeable to others. People of any age can have BDD, but it's most common in teenagers and young adults.

Like many other mental health conditions, body dysmorphic disorder may result from a combination of issues, such as a family history of the disorder, negative evaluations or experiences about your body or self-image, and abnormal brain function or abnormal levels of the brain chemical called serotonin. Body dysmorphic disorder typically starts in the early teenage years and it affects both males and females.

Certain factors seem to increase the risk of developing or triggering body dysmorphic disorder, including:

- Having blood relatives with body dysmorphic disorder or obsessive-compulsive disorder
- Negative life experiences, such as childhood teasing, neglect or abuse
- Certain personality traits, such as perfectionism
- Societal pressure or expectations of beauty
- Having another mental health condition, such as anxiety or depression

Complications

Complications that may be caused by or associated with body dysmorphic disorder include, for example:

- Low self-esteem
- Social isolation
- Major depression or other mood disorders
- Suicidal thoughts or behavior
- Anxiety disorders, including social anxiety disorder (social phobia)
- Obsessive-compulsive disorder
- Eating disorders
- Substance misuse

- Health problems from behaviors such as skin picking
- Physical pain or risk of disfigurement due to repeated surgical interventions

Prevention

There's no known way to prevent body dysmorphic disorder. However, because body dysmorphic disorder often starts in the early teenage years, identifying the disorder early and starting treatment may be of some benefit.

Long-term maintenance treatment also may help prevent a relapse of body dysmorphic disorder symptoms.

Lesson-14 Functional disorders- Mood disorders- Mood disorders Definition & Meaning – Depressive Disorder – Theoretical Perspectives on Depressive Disorder – Major Depressive Disorders – Unipolar and Bipolar disorders – Substance Induced Mood Disorders - Suicide – Classification – Causes - Treatment and Prevention.

Functional neurological disorder (FND) describes a problem with how the brain receives and sends information to the rest of the body. It's often helpful to think of your brain as a computer. In someone who has FND, there's no damage to the hardware, or structure, of the brain.

Symptoms

- Weakness or paralysis.
- Abnormal movement, such as tremors or difficulty walking.
- Loss of balance.
- Difficulty swallowing or feeling "a lump in the throat"
- Seizures or episodes of shaking and apparent loss of consciousness (nonepileptic seizures)
- Episodes of unresponsiveness.

Common associated symptoms or conditions

There are other symptoms or conditions that are commonly associated with FND. These include:

- chronic pain, including fibromyalgia, back and neck pain, and complex regional pain syndrome
- persistent fatigue
- sleep problems including insomnia (not sleeping enough) and hypersomnia (sleeping too much)
- migraines and other types of headache and facial pain
- irritable bowel syndrome and other problems with the function of your stomach and bowel

- anxiety and panic attacks
- depression
- post-traumatic stress disorder
- chronic urinary retention despite all tests being normal (which may be diagnosed as Fowlers syndrome)
- dysfunctional breathing

What causes FND?

We know that the symptoms of FND happen because there's a problem with how the brain is sending and receiving messages to itself and other parts of the body. Using research tools, scientists can see that certain circuits in the brain are not working properly in people with FND.

However, there's still a lot of research to be done to understand how and why FND happens.

Why does FND happen?

FND can happen for a wide range of reasons. There's often more than one reason, and the reasons can vary hugely from person to person.

Some of the reasons why the brain stops working properly in FND include:

- the brain trying to get rid of a painful sensation
- a migraine or other neurological symptom
- the brain shutting down a part or all of the body in response to a situation it thinks is threatening

In some people, stressful events in the past or present can be relevant to FND. In others, stress is not relevant.

The risk of developing FND increases if you have another neurological condition.

Mood disorder.

your general emotional state or mood is distorted or inconsistent with your circumstances and interferes with your ability to function. You may be extremely sad, empty or irritable (depressed), or you may have periods of depression alternating with being excessively happy (mania).

Anxiety disorders can also affect your mood and often occur along with depression. Mood disorders may increase your risk of suicide.

Some examples of mood disorders include:

- Major depressive disorder — prolonged and persistent periods of extreme sadness
- Bipolar disorder — also called manic depression or bipolar affective disorder, depression that includes alternating times of depression and mania
- Seasonal affective disorder (SAD) — a form of depression most often associated with fewer hours of daylight in the far northern and southern latitudes from late fall to early spring
- Cyclothymic disorder — a disorder that causes emotional ups and downs that are less extreme than bipolar disorder
- Premenstrual dysphoric disorder — mood changes and irritability that occur during the premenstrual phase of a woman's cycle and go away with the onset of menses
- Persistent depressive disorder (dysthymia) — a long-term (chronic) form of depression
- Disruptive mood dysregulation disorder — a disorder of chronic, severe and persistent irritability in children that often includes frequent temper outbursts that are inconsistent with the child's developmental age
- Depression related to medical illness — a persistent depressed mood and a significant loss of pleasure in most or all activities that's directly related to the physical effects of another medical condition
- Depression induced by substance use or medication — depression symptoms that develop during or soon after substance use or withdrawal or after exposure to a medication

Depressive disorder.

Depression (major depressive disorder) is a common and serious medical illness that negatively affects how you feel, the way you think and how you act. Fortunately, it is also treatable. Depression causes feelings of sadness and/or a loss of interest in activities you once enjoyed.

Depression is a mood disorder that causes a persistent feeling of sadness and loss of interest. Also called major depressive disorder or clinical depression, it affects how you feel, think and behave and can lead to a variety of emotional and physical problems. You may have trouble doing normal day-to-day activities, and sometimes you may feel as if life isn't worth living.

More than just a bout of the blues, depression isn't a weakness and you can't simply "snap out" of it. Depression may require long-term treatment. But don't get discouraged. Most people with depression feel better with medication, psychotherapy or both.

Symptoms

Although depression may occur only once during your life, people typically have multiple episodes. During these episodes, symptoms occur most of the day, nearly every day and may include:

- Feelings of sadness, tearfulness, emptiness or hopelessness
- Angry outbursts, irritability or frustration, even over small matters
- Loss of interest or pleasure in most or all normal activities, such as sex, hobbies or sports
- Sleep disturbances, including insomnia or sleeping too much
- Tiredness and lack of energy, so even small tasks take extra effort
- Reduced appetite and weight loss or increased cravings for food and weight gain
- Anxiety, agitation or restlessness
- Slowed thinking, speaking or body movements
- Feelings of worthlessness or guilt, fixating on past failures or self-blame
- Trouble thinking, concentrating, making decisions and remembering things
- Frequent or recurrent thoughts of death, suicidal thoughts, suicide attempts or suicide
- Unexplained physical problems, such as back pain or headaches

For many people with depression, symptoms usually are severe enough to cause noticeable problems in day-to-day activities, such as work, school, social activities or relationships with others. Some people may feel generally miserable or unhappy without really knowing why.

Depression symptoms in children and teens

Common signs and symptoms of depression in children and teenagers are similar to those of adults, but there can be some differences.

- In younger children, symptoms of depression may include sadness, irritability, clinginess, worry, aches and pains, refusing to go to school, or being underweight.
- In teens, symptoms may include sadness, irritability, feeling negative and worthless, anger, poor performance or poor attendance at school, feeling misunderstood and extremely

sensitive, using recreational drugs or alcohol, eating or sleeping too much, self-harm, loss of interest in normal activities, and avoidance of social interaction.

Depression symptoms in older adults

Depression is not a normal part of growing older, and it should never be taken lightly. Unfortunately, depression often goes undiagnosed and untreated in older adults, and they may feel reluctant to seek help. Symptoms of depression may be different or less obvious in older adults, such as:

- Memory difficulties or personality changes
- Physical aches or pain
- Fatigue, loss of appetite, sleep problems or loss of interest in sex — not caused by a medical condition or medication
- Often wanting to stay at home, rather than going out to socialize or doing new things
- Suicidal thinking or feelings, especially in older men

Uni and bipolar

Unipolar depression is another name for major depressive disorder. The term “unipolar” means that this form of depression does not cycle through other mental states, such as mania. In contrast, bipolar conditions cause periods of both depression and mania. A mental health disorder characterised by persistently depressed mood or loss of interest in activities, causing significant impairment in daily life.

Possible causes include a combination of biological, psychological and social sources of distress. Increasingly, research suggests that these factors may cause changes in brain function, including altered activity of certain neural circuits in the brain.
The persistent feeling of sadness or loss of interest that characterises major depression can lead to a range of behavioural and physical symptoms. These may include changes in sleep, appetite, energy level, concentration, daily behaviour or self-esteem. Depression can also be associated with thoughts of suicide.
The mainstay of treatment is usually medication, talk therapy or a combination of the two. Increasingly, research suggests that these treatments may normalise brain changes associated with depression

One of the widely accepted theories is that unipolar depression is caused by lack of balance in the naturally occurring chemicals known as neurotransmitters, that are present in the brain as well as the spinal cord.

Characteristics and Symptoms of Unipolar Depression

Experiencing unipolar depression, or major depressive disorder, is different from feeling sad and down from time to time. Unipolar depression is characterized by feelings of depression that are persistent, intense, and that make it challenging for you to function normally or relate to others.

Some of the main characteristics of unipolar depression include:[2]

- Feeling sad most days or having a chronic low mood
- Not enjoying activities that used to bring you pleasure
- Finding it difficult to experience joy or happiness
- Feeling emotionally numb
- Having trouble concentrating
- Lack of energy
- Noticeable changes in hunger
- Trouble sleeping
- Feeling agitated
- Having trouble completing tasks
- Feeling like you don't matter
- Feeling guilty
- Experiencing thoughts of suicide or self-harm

Bipolar disorder

Bipolar disorder (formerly called manic-depressive illness or manic depression) is a mental illness that causes unusual shifts in a person's mood, energy, activity levels, and concentration. These shifts can make it difficult to carry out day-to-day tasks.

There are three types of bipolar disorder. All three types involve clear changes in mood, energy, and activity levels. These moods range from periods of extremely "up," elated, irritable, or energized behavior (known as manic episodes) to very "down," sad, indifferent, or hopeless periods (known as depressive episodes). Less severe manic periods are known as hypomanic episodes.

- Bipolar I disorder is defined by manic episodes that last for at least 7 days (nearly every day for most of the day) or by manic symptoms that are so severe that the person needs immediate medical care. Usually, depressive episodes occur as well, typically lasting at least 2 weeks. Episodes of depression with mixed features (having depressive symptoms and manic symptoms at the same time) are also possible. Experiencing four or more episodes of mania or depression within 1 year is called "rapid cycling."
- Bipolar II disorder is defined by a pattern of depressive episodes and hypomanic episodes. The hypomanic episodes are less severe than the manic episodes in bipolar I disorder.
- Cyclothymic disorder (also called cyclothymia) is defined by recurring hypomanic and depressive symptoms that are not intense enough or do not last long enough to qualify as hypomanic or depressive episodes.

Sometimes a person might experience symptoms of bipolar disorder that do not match the three categories listed above, and this is referred to as “other specified and unspecified bipolar and related disorders.”

Bipolar disorder is often diagnosed during late adolescence (teen years) or early adulthood. Sometimes, bipolar symptoms can appear in children. Although the symptoms may vary over time, bipolar disorder usually requires lifelong treatment. Following a prescribed treatment plan can help people manage their symptoms and improve their quality of life.

Symptoms - Bipolar disorder

- feeling sad, hopeless or irritable most of the time.
- lacking energy.
- difficulty concentrating and remembering things.
- loss of interest in everyday activities.
- feelings of emptiness or worthlessness.
- feelings of guilt and despair.
- feeling pessimistic about everything.
- self-doubt.

Substance induced mood disorder

Substance/medication-induced mental disorders refer to depressive, anxiety, psychotic, or manic symptoms that occur as a physiological consequence of the use of substances of abuse or medications. It may occur during active use, intoxication or withdrawal.

Substance-induced psychotic disorder is typically caused by the misuse of alcohol or drugs. Some other risk factors may make you more prone to developing this type of disorder, like: A traumatic brain injury or stroke. A family or personal history of schizophrenia, mood disorders, or psychotic disorders.

Although, both illicit substances and iatrogenic medications are ingested with the impetus to alleviate mood, a substantial proportion of patients experience paradoxical affective disorders following the ingestion of said substances. Instead of the prosaic euphoria experienced while intoxicated or the subsequent day 'hangover,' some individuals will become manic or enter into a state of depression. Mood disorders that precipitate only in association with substance use are specified as 'substance-induced.' Affective disorders that can precipitate in the setting of substance use include both bipolar and related disorders and depressive disorders. These disorders were previously found within the nosological category of substance-induced mood disorders, in DSM-IV. However, in the current DSM 'substance-induced' is now a specifier for mood disorders.

Causes

Depression and bipolar disorder frequently co-occur with substance use disorders (SUDs) and are prevalent in the general population.[1][2] This review will reflect on both substance-induced

depressive disorder and substance-induced bipolar and related disorders, and further elaborate on how they can be distinguished from mood disorders that are comorbid with SUDs.

For the purpose of this review, the designation of substance-induced mood disorders - although no longer a distinct category in DSM-V - will represent both substance-induced depression and bipolar disorders.

A wide variety of psychoactive substances can cause substance-induced depression. The following disorders are recognized:[3]

- Alcohol-induced depressive disorder
- Phencyclidine-induced depressive disorder
- Other hallucinogen-induced depressive disorder
- Inhalant-induced depressive disorder
- Opioid-induced depressive disorder
- Sedative-induced depressive disorder
- Hypnotic-induced depressive disorder
- Anxiolytic-induced depressive disorder
- Amphetamine-induced depressive disorder
- Other stimulant-induced depressive disorder
- Cocaine-induced depressive disorder
- Other substance-induced depressive disorder
- Unknown substance-induced depressive disorder

Many medications are known to cause substance-induced depression. The following disorders are recognized:

- Steroid-induced depressive disorder
- L-dopa-induced depressive disorder
- Antibiotic-induced depressive disorder
- Central nervous system drug-induced depressive disorder
- Dermatological agent-induced depressive disorder
- Chemotherapeutic drug-induced depressive disorder
- Immunological agent-induced depressive disorder

Specific medications that have been implicated in medication-induced depression through surveillance studies, retrospective observational studies, or case reports, which are prone to difficulty in determining the actual cause, include antiviral agents (such as efavirenz), cardiovascular agents (such as clonidine, guanethidine, methyldopa, reserpine), retinoic acid derivatives (such as isotretinoin), antidepressants, anticonvulsants, anti-migraine agents (triptans), antipsychotics, hormonal agents (corticosteroids, oral contraceptives, gonadotropin-releasing hormone agonists, tamoxifen), smoking cessation agents (varenicline), and immunological agents (interferon).

Suicide- causes and prevention.

Suicide is when people harm themselves with the goal of ending their life, and they die as a result.

A suicide attempt is when people harm themselves with the goal of ending their life, but they do not die.

Avoid using terms such as “committing suicide,” “successful suicide,” or “failed suicide” when referring to suicide and suicide attempts, as these terms often carry negative meanings.

Warning Signs

Warning signs that someone may be at immediate risk for attempting suicide include:

- Talking about wanting to die or wanting to kill themselves
- Talking about feeling empty or hopeless or having no reason to live
- Talking about feeling trapped or feeling that there are no solutions
- Feeling unbearable emotional or physical pain
- Talking about being a burden to others
- Withdrawing from family and friends
- Giving away important possessions
- Saying goodbye to friends and family
- Putting affairs in order, such as making a will
- Taking great risks that could lead to death, such as driving extremely fast
- Talking or thinking about death often

Other serious warning signs that someone may be at risk for attempting suicide include:

- Displaying extreme mood swings, suddenly changing from very sad to very calm or happy
- Making a plan or looking for ways to kill themselves, such as searching for lethal methods online, stockpiling pills, or buying a gun
- Talking about feeling great guilt or shame
- Using alcohol or drugs more often
- Acting anxious or agitated
- Changing eating or sleeping habits
- Showing rage or talking about seeking revenge

It is important to note that suicide is not a normal response to stress. Suicidal thoughts or actions are a sign of extreme distress and should not be ignored. If these warning signs apply to you or someone you know, get help as soon as possible, particularly if the behavior is new or has increased recently.

Risk Factors

Suicide does not discriminate. People of all genders, ages, and ethnicities can be at risk. Suicidal behavior is complex, and there is no single cause. The main risk factors for suicide are:

- Depression, other mental disorders, or substance use disorder

- Chronic pain
- A history of suicide attempts
- Family history of a mental disorder or substance use
- Family history of suicide
- Exposure to family violence, including physical or sexual abuse
- Presence of guns or other firearms in the home
- Having recently been released from prison or jail
- Exposure, either directly or indirectly, to others' suicidal behavior, such as that of family members, peers, or celebrities

Most people who have risk factors will not attempt suicide, and it is difficult to tell who will act on suicidal thoughts. Although risk factors for suicide are important to keep in mind, someone who is showing *warning signs* of suicide may be at higher risk for danger and need immediate attention.

Stressful life events (such as the loss of a loved one, legal troubles, or financial difficulties) and interpersonal stressors (such as shame, harassment, bullying, discrimination, or relationship troubles) may contribute to suicide risk, especially when they occur along with suicide risk factors.

Family and friends are often the first to recognize the warning signs of suicide, and they can take the first step toward helping a loved one find mental health treatment. See the resources on NIMH's Find Help for Mental Illnesses page if you're not sure where to start.

Identifying People at Risk for Suicide

- Universal Screening: Research has shown that a three-question screening tool helps emergency room personnel identify adults at risk for suicide. Researchers found that screening all patients – regardless of the reason for their emergency room visit – doubled the number of patients identified as being at risk for suicide. The researchers estimated that suicide-risk screening tools could identify more than three million additional adults at risk for suicide each year.

- Predicting Suicide Risk Using Electronic Health Records: Researchers from NIMH partnered with the VA and others to develop computer programs that could help predict suicide risk among veterans receiving VA health care. Other healthcare systems are beginning to use data from electronic health records to help identify people with suicide risk as well.

Suicide, taking your own life, is a tragic reaction to stressful life situations — and all the more tragic because suicide can be prevented. Whether you're considering suicide or know someone who feels suicidal, learn suicide warning signs and how to reach out for immediate help and professional treatment. You may save a life — your own or someone else's.

Children and teenagers

Suicide in children and teenagers can follow stressful life events. What a young person sees as serious and insurmountable may seem minor to an adult — such as problems in school or the loss of a friendship. In some cases, a child or teen may feel suicidal due to certain life circumstances that he or she may not want to talk about, such as:

- Having a psychiatric disorder, including depression
- Loss or conflict with close friends or family members
- History of physical or sexual abuse
- Problems with alcohol or drugs
- Physical or medical issues, for example, becoming pregnant or having a sexually transmitted infection
- Being the victim of bullying
- Being uncertain of sexual orientation
- Reading or hearing an account of suicide or knowing a peer who died by suicide

If you have concerns about a friend or family member, asking about suicidal thoughts and intentions is the best way to identify risk.

Murder and suicide

In rare cases, people who are suicidal are at risk of killing others and then themselves. Known as a homicide-suicide or murder-suicide, some risk factors include:

- History of conflict with a spouse or romantic partner
- Current family legal or financial problems
- History of mental health problems, particularly depression
- Alcohol or drug abuse
- Having access to a firearm

Starting antidepressants and increased suicide risk

Most antidepressants are generally safe, but the Food and Drug Administration requires that all antidepressants carry black box warnings, the strictest warnings for prescriptions. In some cases,

children, teenagers and young adults under 25 may have an increase in suicidal thoughts or behavior when taking antidepressants, especially in the first few weeks after starting or when the dose is changed.

However, keep in mind that antidepressants are more likely to reduce suicide risk in the long run by improving mood.

Prevention

To help keep yourself from feeling suicidal:

- Get the treatment you need. If you don't treat the underlying cause, your suicidal thoughts are likely to return. You may feel embarrassed to seek treatment for mental health problems, but getting the right treatment for depression, substance misuse or another underlying problem will make you feel better about life — and help keep you safe.
- Establish your support network. It may be hard to talk about suicidal feelings, and your friends and family may not fully understand why you feel the way you do. Reach out anyway, and make sure the people who care about you know what's going on and are there when you need them. You may also want to get help from your place of worship, support groups or other community resources. Feeling connected and supported can help reduce suicide risk.
- Remember, suicidal feelings are temporary. If you feel hopeless or that life's not worth living anymore, remember that treatment can help you regain your perspective — and life will get better. Take one step at a time and don't act impulsively.

Lesson-15 Personality disorders- Personality Disorders Cluster A Personality Disorders – Cluster B Personality Disorders – Personality Disorders – Paranoid – Schizoid – Schizotypal – Histrionic – Narcisstic – Anti-Social Personality Disorders- Addiction.

Personality disorders

Personality disorders are conditions in which an individual differs significantly from an average person, in terms of how they think, perceive, feel or relate to others. Changes in how a person feels and distorted beliefs about other people can lead to odd behaviour, which can be distressing and may upset others

Personality disorders are conditions in which an individual differs significantly from an average person, in terms of how they think, perceive, feel or relate to others.

Changes in how a person feels and distorted beliefs about other people can lead to odd behaviour, which can be distressing and may upset others.

Personality disorders typically emerge in adolescence and continue into adulthood.

They may be mild, moderate or severe, and people may have periods of "remission" where they function well.

Personality disorders may be associated with genetic and family factors. Experiences of distress or fear during childhood, such as neglect or abuse, are common.

Types of personality disorder

Several different types of personality disorder are recognised. They have been broadly grouped into one of three clusters – A, B or C – which are summarised below.

Read about the symptoms of personality disorders for a full list of the main types and signs of personality disorders.

Cluster A personality disorders

A person with a cluster A personality disorder tends to have difficulty relating to others and usually shows patterns of behaviour most people would regard as odd and eccentric. Others may describe them as living in a fantasy world of their own.

An example is paranoid personality disorder, where the person is extremely distrustful and suspicious.

Cluster B personality disorders

A person with a cluster B personality disorder struggles to regulate their feelings and often swings between positive and negative views of others. This can lead to patterns of behaviour others describe as dramatic, unpredictable and disturbing.

An example is borderline personality disorder, where the person is emotionally unstable, has impulses to self-harm, and has intense and unstable relationships with others.

Cluster C personality disorders

A person with a cluster C personality disorder struggles with persistent and overwhelming feelings of fear and anxiety. They may show patterns of behaviour most people would regard as antisocial and withdrawn.

An example is avoidant personality disorder, where the person appears painfully shy, socially inhibited, feels inadequate and is extremely sensitive to rejection. The person may want to be close to others, but lacks confidence to form a close relationship.

Signs and symptoms of personality disorders

The different types of personality disorder that might need treatment can be broadly grouped into one of three clusters, called A, B or C.

Cluster A personality disorders

A person with a cluster A personality disorder tends to have difficulty relating to others and usually shows patterns of behaviour most people would regard as odd and eccentric.

The main personality disorders in this category are listed below.

Paranoid personality disorder

A person with a paranoid personality disorder is extremely distrustful and suspicious. Other features include:

- thinking other people are lying to them or trying to manipulate them
- feeling they cannot really trust their friends and associates
- worrying that any confidential information shared with others will be used against them
- thinking there are hidden meanings in remarks most would regard as innocent
- worrying that their spouse or partner is unfaithful, despite a lack of evidence

Schizoid personality disorder

Someone with a schizoid personality disorder may appear cold and detached, and avoid making close social contact with others. Other features include:

- preferring to take part in activities that do not require interaction with others
- having little desire to form close relationships, including sexual relationships
- being uninterested when receiving criticism or praise
- having a limited ability to experience pleasure or joy

Schizotypal personality disorder

A person with a schizotypal personality disorder is likely to have poor social skills and delusional thoughts, and behave in unusual ways. Other features include:

- attaching undue and misguided significance to everyday events – such as thinking newspaper headlines are secret messages to them
- believing in special powers – such as telepathy or the ability to influence other people's emotions and actions
- having unusual ways of speaking – such as long, rambling vague sentences or going off on a tangent
- experiencing excessive anxiety in social situations – even if they have known a particular person or group of people a long time

Cluster B personality disorders

Someone with a cluster B personality disorder struggles to relate to others. As a result, they show patterns of behaviour most would regard as dramatic, erratic and threatening or disturbing.

The main personality disorders in this category are listed below.

Antisocial personality disorder

A person with an antisocial personality disorder sees other people as vulnerable and may intimidate or bully others without remorse. They lack concern about the consequences of their actions.

Symptoms include:

- lack of concern, regret or remorse about other people's distress
- irresponsibility and disregard for normal social behaviour
- difficulty in sustaining long-term relationships
- little ability to tolerate frustration and to control their anger
- lack of guilt, or not learning from their mistakes
- blaming others for problems in their lives

The National Institute for Health and Care Excellence (NICE) has published guidelines on how people with antisocial personality disorders should be treated. For more information, see NICE: antisocial personality disorder (PDF, 250kb).

Borderline personality disorder

A person with borderline personality disorder is emotionally unstable, has impulses to self-harm, and has very intense and unstable relationships with others.

Histrionic personality disorder

A person with histrionic personality disorder is anxious about being ignored. As a result, they feel a compulsion (overwhelming urge) to be noticed and the centre of everyone's attention. Features include:

- displaying excessive emotion, yet appearing to lack real emotional sincerity
- dressing provocatively and engaging in inappropriate flirting or sexually seductive behaviour
- moving quickly from one emotional state to another
- being self-centred and caring little about other people
- constantly seeking reassurance and approval from other people

Symptoms and signs may co-exist with borderline and narcissistic personality disorders.

Narcissistic personality disorder

A person with narcissistic personality disorder swings between seeing themselves as special and fearing they are worthless. They may act as if they have an inflated sense of their own importance and show an intense need for other people to look up to them.

Other symptoms include:

- exaggerating their own achievements and abilities
- thinking they are entitled to be treated better than other people
- exploiting other people for their own personal gain
- lacking empathy for other people's weaknesses
- looking down on people they feel are "beneath" them, while feeling deeply envious of people they see as being "above" them

Cluster C personality disorders

Someone with a cluster C personality disorder fears personal relationships and shows patterns of anxious and fearful behaviour around other people. Others may be withdrawn and reluctant to socialise.

The main personality disorders in this category are listed below.

Avoidant personality disorder

A person with avoidant personality disorder appears painfully shy, is socially inhibited, feels inadequate and is extremely sensitive to rejection.

Unlike people with schizoid personality disorders, they desire close relationships with others, but lack the confidence and ability to form them.

Dependent personality disorder

A person with dependent personality disorder feels they have no ability to be independent. They may show an excessive need for others to look after them and are "clingy". Other features include:

- finding it difficult to make decisions without other people's guidance
- needing others to take responsibility over what should be their own important life choices
- not being able to express disagreement with other people
- finding it difficult to start new activities due to a lack of confidence
- going to extremes to obtain support and comfort
- feeling helpless and uncomfortable when alone
- urgently needing to start a new relationship once a previous relationship comes to an end
- having an unrealistic and constant fear they will be left alone to fend for themselves

Obsessive compulsive personality disorder

A person with obsessive compulsive personality disorder is anxious about issues that seem out of control or "messy". They are preoccupied with orderliness and ways to control their environment, and may come across to others as a "control freak".

Other features include:

- having an excessive interest in lists, timetables and rules
- being so concerned with completing a task perfectly that they have problems completing it (perfectionism)
- being a workaholic
- having very rigid views about issues such as morality, ethics and how a person should behave in daily life
- hoarding items that seem to have no monetary or sentimental value
- being unable to delegate tasks to other people
- disliking spending money, as they think it is always better to save for a "rainy day"

This personality disorder differs from obsessive compulsive disorder (OCD), a related mental health condition, in several important ways:

- People with OCD are aware that their behaviour is abnormal and are anxious about it. Most people with obsessive compulsive personality disorder think their behaviour is perfectly acceptable and have no desire to change it.

- Some people with OCD are compelled to carry out rituals, such as having to touch every second lamppost as they walk down the street. This is not usually the case with people with obsessive compulsive personality disorder.
- People with OCD may feel compelled to make lists or organise items in their house, but feel anxious about doing so. People with obsessive compulsive personality disorder find relief from anxiety when doing such tasks and may become irritated when prevented from doing so.

Lesson- 16 Psychosomatic disorders.

Psychosomatic disorders

Psychosomatic means mind (psyche) and body (soma). A psychosomatic disorder is a disease which involves both mind and body. Some physical diseases are thought to be particularly prone to being made worse by mental factors such as stress and anxiety. Your mental state can affect how bad a physical disease is at any given time.

Psychosomatic disorder is a psychological condition involving the occurrence of physical symptoms, usually lacking a medical explanation. People with this condition may have excessive thoughts, feelings or concerns about the symptoms — which affects their ability to function well.

Psychosomatic disorder is a psychological condition involving the occurrence of physical symptoms, usually lacking a medical explanation. People with this condition may have excessive thoughts, feelings or concerns about the symptoms — which affects their ability to function well.

People with psychosomatic disorder usually don’t report overt symptoms of psychiatric distress. Instead, they believe their problems are caused by medical conditions. They tend to visit healthcare providers frequently to get tests and treatments, often not receiving a diagnosis, which may lead to frustration and distress.

Psychosomatic disorder is sometimes called somatic symptom disorder, somatic symptoms or somatic pain.

Scientists aren’t sure. Some believe that stress releases hormones and chemicals in the body that cause damage or dysfunction.

Psychosomatic disorder can affect almost any part of the body. Common examples include:

- Fatigue
- Insomnia
- Aches and pains, such as muscle pain or back pain.
- High blood pressure (hypertension).
- Trouble breathing (dyspnea, or shortness of breath).
- Indigestion (upset stomach).

- Headaches and migraines.
- Erectile dysfunction (impotence).
- Skin rash (dermatitis).
- Stomach ulcers (peptic ulcer disease).

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: Introduction of ICDS

Activity 2: To get awareness about Projective test like TAT , Rorschach

PSY15144

PSYCHOLOGICAL MEASUREMENT AND TESTING

Objectives:

To Demonstrate the knowledge and understanding of basic psychometric principles.
Learning Psychophysical methods and their relevance to psychological testing and assessment.

Unit-I: Nature of Psychological Measurement

Lesson-1 Nature of Psychological Measurement -Definition and Advantages of Measurement - History of Measurement.

Lesson-2 Scales of Measurement-The Nominal Scale-The ordinal scale-The interval scale-The ratio scale.

Lesson-3 Nature of Psychological Variable- Mental test tradition-Psycho Physical tradition

Unit-II: Psycho- Physical Methods and Psychological scaling methods

Lesson-4 Method of Average Error-Constant and Variable Errors-Evaluation.

Lesson-5 Methods of Minimal Changes-Evaluation.

Lesson-6 Method of Constant Stimuli- -Evaluation.

Lesson-7 Psychological scaling methods- Method of Pair comparison and rank order - evaluation.

Unit-III: Test Development and Test Standardization Procedures

Lesson-8 Test Development- Uses of Psychological test- Criteria of a Psychological Test. Test Standardization Procedures- -Construction of psychological test-

Lesson-9 Reliability-Types of reliability- Test -retest reliability- Interrater reliability- Parallel forms - reliability-Internal Consistency

Lesson-10 Validity- Types of validity- Construct Validity- Content Validity- Face validity- Criterion Validity -Concurrent Validity.

Lesson-11 Test Construction- Item writing - Meaning and types of items and their differences, guidelines for item writing. Response sets – Meaning, types and implications.

Lesson-12 Item analysis – Meaning and purpose. Item indices ad Item discrimination index.

Lesson-13 Test score - interpretation- Types of scores.

Lesson-14 Ethical issues in test use and interpretation.

Unit-IV: Standard Psychological tests &miscellaneous techniques:

Lesson-15 Classification of Psychological Tests - Intelligence- Aptitude Tests-Tests of Interest

Lesson-16 The semantic differential technique-Attitude scales- Personality Testing- MMPI.

Lesson-17 Projective Techniques -TAT, Rorschach etc.

Lesson-18 Culture and Psychological tests.

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UNIT-I: NATURE OF PSYCHOLOGICAL MEASUREMENT

Lesson-1 Nature of Psychological Measurement -Definition and Advantages of Measurement History of Measurement.

Psychological Measurement

Researchers Tara MacDonald and Alanna Martineau were interested in the effect of female college students' moods on their intentions to have unprotected sexual intercourse (MacDonald & Martineau, 2002).MacDonald, T. K., & Martineau, A. M. (2002). Self-esteem, mood, and intentions to use condoms: When does low self-esteem lead to risky health behaviors? *Journal of Experimental Social Psychology*, *38*, 299–306. In a carefully designed empirical study, they found that being in a negative mood increased intentions to have unprotected sex—but only for students who were low in self-esteem. Although there are many challenges involved in conducting a study like this, one of the primary ones is the measurement of the relevant variables. In this study, the researchers needed to know whether each of their participants had high or low self-esteem, which of course required measuring their self-esteem. They also needed to be sure that their attempt to put people into a negative mood (by having them think negative thoughts) was successful, which required measuring their moods. Finally, they needed to see whether self-esteem and mood were related to participants' intentions to have unprotected sexual intercourse, which required measuring these intentions. To students who are just getting started in psychological research, the challenge of measuring such variables might seem insurmountable. Is it really possible to measure things as intangible as self-esteem, mood, or an intention to do something? The answer is a resounding yes, and in this chapter we look closely at the nature of the variables that psychologists study and how they can be measured. We also look at some practical issues in psychological measurement.

Measurement is the assignment of scores to individuals so that the scores represent some characteristic of the individuals. This very general definition is consistent with the kinds of measurement that everyone is familiar with—for example, weighing oneself by stepping onto a bathroom scale, or checking the internal temperature of a roasting turkey by inserting a meat thermometer. It is also consistent with measurement throughout the sciences. In physics, for example, one might measure the potential energy of an object in Earth's gravitational field by finding its mass and height (which of course requires measuring *those* variables) and then multiplying them together along with the gravitational acceleration of Earth (9.8 m/s^2). The result of this procedure is a score that represents the object's potential energy.

Of course this general definition of measurement is consistent with measurement in psychology too. (Psychological measurement is often referred to as psychometrics.) Imagine, for example, that a cognitive psychologist wants to measure a person's working memory capacity—his or her ability to hold in mind and think about several pieces of information all at the same time. To do this, she might use a backward digit span task, where she reads a list of two digits to the person and asks him or her to repeat them in reverse order. She then repeats this several times, increasing the length of the list by one digit each time, until the person makes an error. The length of the longest list for which the person responds correctly is the score and represents his or her working memory capacity. Or imagine a clinical psychologist who is interested in how

depressed a person is. He administers the Beck Depression Inventory, which is a 21-item self-report questionnaire in which the person rates the extent to which he or she has felt sad, lost energy, and experienced other symptoms of depression over the past 2 weeks. The sum of these 21 ratings is the score and represents his or her current level of depression.

The important point here is that measurement does not require any particular instruments or procedures. It does not require placing individuals or objects on bathroom scales, holding rulers up to them, or inserting thermometers into them. What it *does* require is *some* systematic procedure for assigning scores to individuals or objects so that those scores represent the characteristic of interest.

Levels of Measurement

The psychologist S. S. Stevens suggested that scores can be assigned to individuals so that they communicate more or less quantitative information about the variable of interest (Stevens, 1946).Stevens, S. S. (1946). On the theory of scales of measurement. *Science, 103,* 677–680. For example, the officials at a 100-m race could simply rank order the runners as they crossed the finish line (first, second, etc.), or they could time each runner to the nearest tenth of a second using a stopwatch (11.5 s, 12.1 s, etc.). In either case, they would be measuring the runners' times by systematically assigning scores to represent those times. But while the rank ordering procedure communicates the fact that the second-place runner took longer to finish than the first-place finisher, the stopwatch procedure also communicates *how much* longer the second-place finisher took. Stevens actually suggested four different levels of measurement (which he called "scales of measurement") that correspond to four different levels of quantitative information that can be communicated by a set of scores.

The nominal level of measurement is used for categorical variables and involves assigning scores that are category labels. Category labels communicate whether any two individuals are the same or different in terms of the variable being measured. For example, if you look at your research participants as they enter the room, decide whether each one is male or female, and type this information into a spreadsheet, you are engaged in nominal-level measurement. Or if you ask your participants to indicate which of several ethnicities they identify themselves with, you are again engaged in nominal-level measurement.

The remaining three levels of measurement are used for quantitative variables. The ordinal level of measurement involves assigning scores so that they represent the rank order of the individuals. Ranks communicate not only whether any two individuals are the same or different in terms of the variable being measured but also whether one individual is higher or lower on that variable. The interval level of measurement involves assigning scores so that they represent the precise magnitude of the difference between individuals, but a score of zero does not actually represent the complete absence of the characteristic. A classic example is the measurement of heat using the Celsius or Fahrenheit scale. The difference between temperatures of 20°C and 25°C is precisely 5°, but a temperature of 0°C does not mean that there is a complete absence of heat. In psychology, the intelligence quotient (IQ) is often considered to be measured at the interval level. Finally, the ratio level of measurement involves assigning scores in such a way that there is a true zero point that represents the complete absence of the quantity. Height measured in meters and weight measured in kilograms are good

examples. So are counts of discrete objects or events such as the number of siblings one has or the number of questions a student answers correctly on an exam.

Stevens's levels of measurement are important for at least two reasons. First, they emphasize the generality of the concept of measurement. Although people do not normally think of categorizing or ranking individuals as measurement, in fact they are as long as they are done so that they represent some characteristic of the individuals. Second, the levels of measurement can serve as a rough guide to the statistical procedures that can be used with the data and the conclusions that can be drawn from them. With nominal-level measurement, for example, the only available measure of central tendency is the mode. Also, ratio-level measurement is the only level that allows meaningful statements about ratios of scores. One cannot say that someone with an IQ of 140 is twice as intelligent as someone with an IQ of 70 because IQ is measured at the interval level, but one can say that someone with six siblings has twice as many as someone with three because number of siblings is measured at the ratio level.

Characteristics of Psychological / Psychometric tests

Every standard Psychological test will have following characteristics-

1. Uniformity
2. Objectivity measurement
3. Comprehensiveness
4. Psychometric Properties

1. Uniformity:

- It refers to uniformity of administration and scoring.
- This feature is mainly required for intelligence, aptitude and personality tests.
- To establish uniformity researcher try to form an objective scoring key.
- This particular characteristic is a limitation for Projective Psychometric tests.
- Objective administration and scoring results in minimal errors hence results of the test are more valid.
- Personal biases and limitation are taken care of to make results more accurate.
- Examples .Intelligence test ,Aptitude test

2. Objective Measurement:

- It refers to the objective measurement of discrimination and/or difficulty index.
- Items in the tests should be having moderate level of difficulty.
- For example, In a Maths test; if all the problems are difficult it would be difficult to judge the students because nobody will get good marks, vice a versa if all the problems are easy everybody will get good marks. Hence in both the cases judgement of the students and discriminating them into categories will be difficult.
- ***Discrimination index*** is used in personality tests where there is no difficulty but individuals are discriminated on the basis of the their attributes and characteristics.

3. Comprehensiveness:

- Comprehensiveness refers to inclusion of various functions related to the topic of the test.
- Hence a test should comprehended the data and should have a relevant content in it.
- For example- A primary school maths test should include all the mathematical functions which children may have already learnt and not only the selected functions.

4. Psychometric Properties:

- A sound psychological test must have 3 psychometric properties including all of the following:
- Reliability – Obtaining the same result after multiple testing.
- Validity- must measure what it is intended to measure.
- Norms- The average test score , to compared to the results of others by establishing a point of comparison or frame of reference.

Definition of Measurement

Measurement is defined as assignment of numerals to objects or events according to some rule (Stevens, 1946).

It consists of rules for assigning numbers to objects in such a way as to represent qualities of attributes (Nunnally,1970).

Definition of Test

In general test is a critical examination, observation or evaluation.

It use a set of questions or problems to measure someone's skills, knowledge, or ability.

Kaplan defined Test as a measurement device or technique used to quantify behavior or aid in the understanding and prediction of behavior.

For example- A spelling test measures how well someone learned to spell.

Test is a systematic procedure for observing a person's behaviour and describing it with the aid of a numerical scale or a category system (L.J. Cronbach, 1949).

What is Psychological Measurement

It is assignment of numerals to characteristics or attributes of individual, according to rules (Brown, 1987).

What is Psychological/ Psychometric Test

It is a standardised and objective technique / tools of eliciting and assessing a sample of behaviour.

It is the test that assesses person's abilities, aptitudes, interests or personality based on systematically obtained sample of behaviour (Rea,2001).

According to Gregory (2015) Psychological test is a standardized procedure for sampling behavior and describing it with categories or scores.

Anastasi (1982), "A psychological test is essentially an objective and standardized measure of a sample behaviour."

Kaplan (2005) defined psychological test as a set of items that are designed to measure characteristics of human beings that pertain to behavior.

Behavior can be overt (Observable) or Covert (Hidden) like feelings, thoughts.

According to American Psychological Association (APA), Psychological Test is any standardized instrument, including scales and self-report inventories, used to measure behavior or mental attributes, such as attitudes, emotional functioning, intelligence and cognitive abilities (reasoning, comprehension, abstraction, etc.), aptitudes, values, interests, and personality characteristics.

For example, a researcher might use a psychological test of emotional intelligence to examine whether some managers make better decisions in conflict situations than do others.

Psychological tests are also called Psychometric test.

Psychological tests measure past or current or predict future behavior.

History of Psychological measurement

Psychological testing in its modern form originated little more than 100 years ago in laboratory studies of sensory discrimination, motor skills, and reaction time. Francis Galton invented the first battery of tests.

Rudimentary forms of testing in China in 2200 b.c

Rudimentary forms of testing date back to at least 2200 b.c when the Chinese emperor has his officials examined every third year to determine their fitness for office. Such testing was modified and refined over the centuries until written exams were introduced in the Han dynasty. The testing practices were unnecessarily gruelling, and the Chinese also failed to validate their selection procedures. But the examination program incorporated relevant selection criteria. Physiognomy, phrenology, and the psychograph Physiognomy: based n the notion that we can judge the inner character of people from their outward appearance, especially the face. It represents an early form of psychological testing.

Interest in physiognomy can be dated back to the fourth century. Physiognomy remained popular for centuries and laid the foundation for the more specialized form of quackery, phrenology.

Phrenology: reading 'bumps' on the head. The founding of phrenology is usually attributed to Franz Joseph Gall. He argued that the brain is the organ of sentiments and faculties and that these capacities are localized. To the extent that a faculty was well developed, the corresponding component of the brain would be enlarged and in turn form a bump because the skill conforms the shape of the brain. (This is incorrect). Johann Spurzheim popularized phrenology. The psychograph was a machine that measured phrenoloy. Made by Henry C. Lavery in 1931.

The brass instruments era of testing

Experimental psychology flourished in the late 1800s in continental Europe and Great Britain. Human abilities were tested in laboratories with objective procedures that were capable of replication. The problem with experimental psychology was that it mistook simple sensory processes for intelligence. They used assorted brass instruments to measure sensory thresholds and reaction times, thinking that such abilities were at the heart of intelligence.

Wilhelm Wundt founded the first psychological laboratory in 1879 in Leipzig. He believed that the speed of thought might differ from one person to the next.

Galton and the first battery of mental tests

Francis Galton (1822-1911) pioneered the new experimental psychology in Great Britain. He was obsessed with measurements. Galton demonstrates that individual differences not only exist but also are objectively measurable. Galton continued the tradition of brass instruments, but with an important difference: his procedures were much more amenable to the timely collection of data from hundreds if not thousands of subjects. The tests and measures involved both the physical and behavioural domains. He demonstrated that objective tests could be devised and that meaningful scores could be obtianed through standardized procedures.

Cattell imports brass instruments to the United States

James McKeen Cattell studied the new experimental psychology with both Wundt and Galton before settling at Columbia University where, for 26 years, he was the undisputed dean of American psychology. Cattell wanted to study individual differences (and did). He invented

the term ‘mental test’. Wissler made experimental psychology turn away from the brass instruments approach. In 1905, Binet introduced his scale of intelligence.

Rating scales and their origins

Rating scales are widely used in psychology as a means of quantifying subjective psychological variables of many kinds. A crude form of rating scale can be traced back to Galen, the second century Greco-Roman physician. The first person to devise and apply rating scales for psychological variables was Christian Thomasius (1655-1728).

Changing conceptions of mental retardation in the 1800s

By the early 1800s medical practitioners realized that some of those with psychiatric impairment had reversible illnesses that did not necessarily imply diminished intellect, whereas those with mental retardation showed a greater developmental continuity and invariably had impaired intellect. A newfound humanism began to influence social practices toward individuals with psychological and mental disabilities. With this humanism there arose a greater interest in the diagnosis and remediation of mental retardation.squirol and diagnosis in mental retardationAround the beginning of the nineteenth century, many physicians had begun to perceive the difference between mental retardation and mental illness. Esquirol (1772-1840) was the first to formalize the difference in writing. He proposed the first classification system in mental retardation and language skills were the main diagnostic criteria.Seguin and education of individuals with mental retardationEdouard Seguin helped establish a new humanism toward those with mental retardation in the late 1800s.

Influence of Binet’s early research on his test

Alfred Binet (1857-1911) invented the first modern intelligence test in 1905. Binet was a prolific researcher and author long before he turned his attentions to intelligence testing.

Binet and testing for higher mental processes

Binet argued that intelligence could be better measured by means of the higher psychological processes rather than elementary sensory processes such as reaction time.Binet and Simon were called to develop a practical tool for selecting children for which special placement in schools was needed. Thus arose the first formal scale for assessing the intelligence of children. The scale was appropriate for assessing the entire gamut of intelligence. The purpose was classification, not measurement.

The revised scales and the advent of IQ

In 1908, Binet and Simon published a revision of he 1905 scale. The major innovation of he 1908 scale was the introduction of the concept of mental level. The idea of deriving a mental level was a monumental development that was to influence the character of intelligence testing throughout the twentieth century (later mental age).

Testing form the early 1900s to the present

With the successful application of Binet’s mental test, psychologists realized that their inventions could have pragmatic significance for many different segments of society. The profusion of tests developed early in the twentieth century helped shape the character of contemporary tests.

Early used and abuses of tests in the United States

First translation of the Binet-Simon scale

Goddhard gained a reputation as one of the leading experts on the use of intelligence tests to identify persons with impaired intellect. He believed that the impaired children should be segregated so that they would be prevented from 'contaminating society'. The Binet-Simon immigration When Goddard visits Ellis Island, he became convinced that the rates of feeblemindedness were much higher than estimated by the physicians who staffed the immigration service. He became a apostle for the use of intelligence tests to identify feebleminded immigrants.Goddard's schlarly views were influenced by the social ideologies of his time.

Testing for giftedness:

Leta Stetter Hollingworth One of the earliest uses of IQ tests was testing for giftedness. Hollingworth demonstrated that children with high genius showed significantly greater school achievement than those of mere ordinary genius. She dispelled the belief that gifted children should not be moved ahead in school because they would lag behind older children in penmanship and other motor skills. She advanced the science of IQ testing. She proposed a revolving fund from which gifted children could draw for their development. The Stanford-Binet: The early mainstay of IQ It was Stanford professor Lewis M. Terman (1857-1956) who popularized IQ testing with his revision of the Binet scales in 1916. The Stanford-Bined was a substantial revision. It was the standard of intelligence testing for decades.

Group tests and the classification of WWI army recruits

Researchers sought group mental test to supplement tests imported from France. Group test were slow to catch on, partly because the early versions still had to be scored laboriously by hand.

The slow pace of developments in group testing picked up dramatically as the United States entered WWI.

Robert M. Yerkes convinced the U.S. government and the army that all of its recruits should be given intelligence tests for purposes of classification and assignment. It gave rise to the army Alpha and the army Beta.

The army Alpha and Beta examinations

The alpha consisted of eight verbally loaded tests for average and high-functioning recruits. The army beta was a nonverbal group test designed for the use with illiterates and recruits whose first language was not English.

The army testing was intended to help segregate and eliminate the mentally incompetent, to classify men according to their mental ability, and to assist in the placement of competent men in responsible positions.

Early educational testing

Yerkes's grand scheme for testing army recruits helped to usher in the era of group tests. Machine scoring was introduced in the 1930s, making objective group tests even more efficient than before.

The development of the aptitude tests

Aptitude tests measure more specific and delimited abilities than intelligence tests. A single aptitude test will measure just one ability domain. The development of aptitude tests lagged behind of that of intelligence tests for two reasons

- Statistical A new technique, factor analysis, was often needed to discern which aptitudes were primary and, therefore, distinct from each other. Research on this was not refined until the 1930s
- The absence of a practical application for such refined instruments. It was not until WWII that a pressing need arose to select candidates who were highly qualified for very difficult and specialized tasks.

Personality and vocational testing after WWI

It was not until WWI that personality tests emerged in a form resembling their contemporary appearance. It was needed to detect which army recruits were susceptible for psychoneurosis. Made by Woodworth. In addition, the MMPI introduced the use of validity scales to determine fake bad, fake good, and random response patterns.

The origins of projective testing

The projective approach originated with the word association method pioneered by Galton. Also the Rorschach ink bolds.

The development of interest inventories

Psychologists were devising measures for guidance and counselling of the masses of more normal persons.

The interest inventory, which has roots going back to Thorndike.

The emergence of structured personality tests

Beginning in the 1940s, personality tests began to flourish as useful tools for clinical evaluation and also for assessment of the normal spectrum of functioning.

The expansion and proliferation of testing

In the twenty-first century, the reach of testing continues to increase. Both in one-to-one clinical uses and in group testing fro societal applications.

Evidence-based practice and outcomes assessment

Evidence-based practice: the integration of best research evidence with clinical expertise and patient values.

The advance of evidence-based practice is part of a worldwide trend to require proof that treatments and interventions yield measurable positive outcomes.

Psychological Evaluation in 20th Century

Intelligence geared for assisting in student classification in Paris's public schools. It is generally agreed that the creation of the Binet-Simon Intelligence Scale marked the beginning of a new era in the history of I.Q. testing. The following sequence of test development was seen.

- In 1908, A.D. Frank Parsons establishes the Vocational Bureau of Boston to serve as a center for vocational counseling for young adults.
- In 1914 A.D., when thousands of American recruits were screened for intelligence and mental stability, the field of psychological testing experienced a boom.
- In 1919, Robert Woodworth created the Personal Data Sheet to aid in screening recruits for susceptibility to "shell shock."
- In 1921, the Rorschach Inkblot Test was developed after Swiss psychiatrist Hermann Rorschach published his seminal book, Psychodiagnostics.
- The first SAT, or Scholastic Aptitude Test, was created and given in 1926.

- In 1927, A.D. Carl Spearman published a two-factor theory of intelligence in which he hypothesized the presence of a general intellectual capacity factor and particular components of that general ability. It is 1938 A.D., and intelligence testing has become a multimillion-dollar industry. As of 1938, there were over 4,000 psychological tests published in the Mental Measurements Yearbook.
- The Wechsler-Bellevue Intelligence Scale, developed by David Wechsler and colleagues, was administered in 1939 A.D. to evaluate adult IQ. The most common instruments used to gauge a person's I.Q. today are variants of these tests, many of which have been published.
- The Minnesota Multiphasic Personality Inventory first appeared in print in 1943.
- The original Wechsler Intelligence Scale for Children was released in 1949.
- The 16PF Questionnaire, First Edition, was made available to the public in 1949.
- When the original Wechsler Adult Intelligence Scales were released in 1955, the world was introduced to a new way of measuring cognitive ability.
- 1962 A.D. The Meyers Briggs Type Indicator is released with the help of Katherine Briggs and Isabel Briggs Meyers (MBTI).
- 1962 A.D. In his debut article on the topic, psychologist Warren T. Norman presents his Big Five model of personality.

The original version of the Self Directed Search (SDS) was released to the public by John L. Holland in 1970. We hoped that this checklist would aid people in zeroing down on occupations that are a good fit for their unique character traits.

Lesson-2 Scales of Measurement-The Nominal Scale-The ordinal scale-The interval scale-The ratio scale.

SCALES OF MEASUREMENT

Level of measurement or scale of measure is a classification that describes the nature of information within the numbers assigned to variables. Psychologist Stanley Smith Stevens developed the best known classification with four levels or scales of measurement: nominal, ordinal, interval, and ratio.

Stevens proposed his typology in a 1946 Science article titled "On the theory of scales of measurement". In this article, Stevens claimed that all measurement in science was conducted using four different types of scales that he called "nominal" "ordinal" "interval" and "ratio" unifying both "qualitative" (which are described by his "nominal" type) and "quantitative" (all the rest of his scales). The concept of scale types later received the mathematical rigor that it lacked at its inception with the work of mathematical psychologists Theodore Alper (1985, 1987), Louis Narens (1981a, b), and R. Duncan Luce (1986, 1987, 2001). As Luce (1997, p. 395) wrote; S.

S. Stevens (1946, 1951, and 1975) claimed that what counted was having an interval or ratio scale. Subsequent research has given meaning to this assertion, but given his attempts to invoke scale type ideas it is doubtful if he understood it himself ... no measurement theorist I know accepts Stevens' broad definition of measurement ... in our view, the only sensible meaning for 'rule' is empirically testable laws about the attribute.

1. Nominal Scale: A nominal scale of measurement deals with variables that are non-numeric or where the numbers have no value. The lowest measurement level you can use, from a

statistical point of view, is a nominal scale. A nominal scale, as the name implies, is simply some placing of data into categories, without any order or structure. A physical example of a nominal scale is the terms we use for colours. The underlying spectrum is ordered but the names are nominal. In research activities a YES/NO scale is nominal. It has no order and there is no distance between YES and NO.

2. Ordinal Scale: The ordinal type allows for rank order (1st, 2nd, 3rd, etc.) by which data can be sorted, but still does not allow for relative degree of difference between them. An ordinal scale of measurement looks at variables where the order matters but the differences do not matter. When you think of 'ordinal,' think of the word 'order.' In the case of letter grades, we don't really know how much better an A is than a D. We know that A is better than B, which is better than C, and so on. But is A four times better than D? Is it two times better? In this case, the order is important but not the differences. Examples include, on one hand, dichotomous data with dichotomous (or dichotomized) values such as 'sick' vs. 'healthy' when measuring health, 'guilty' vs. 'innocent' when making judgments in courts, 'wrong/false' vs. 'right/true' when measuring truth value, and, on the other hand, non-dichotomous data consisting of a spectrum of values, such as 'completely agree', 'mostly agree', 'mostly disagree', 'completely disagree' when measuring opinion.

3. Interval Scale: Interval scales are numeric scales in which we know not only the order, but also the exact differences between the values. The classic example of an interval scale is Celsius temperature because the difference between each value is the same. For example, the difference between 60 and 50 degrees is a measurable 10 degrees, as is the difference between 80 and 70 degrees. Time is another good example of an interval scale in which the increments are known, consistent, and measurable.

4. Ratio Scale: The ratio scale of measurement is the most informative scale. It is an interval scale with the additional property that its zero position indicates the absence of the quantity being measured. You can think of a ratio scale as the three earlier scales rolled up in one. Like a nominal scale, it provides a name or category for each object (the numbers serve as labels). Like an ordinal scale, the objects are ordered (in terms of the ordering of the numbers). Like an interval scale, the same difference at two places on the scale has the same meaning. And in addition, the same ratio at two places on the scale also carries the same meaning. Ratio scales are the ultimate nirvana when it comes to measurement scales because they tell us about the order, they tell us the exact value between units and they also have an absolute zero–which allows for a wide range of both descriptive and inferential statistics to be applied. Everything above about interval data applies to ratio scales i.e. ratio scales have a clear definition of zero. Good examples of ratio variables include height and weight. Ratio scales provide a wealth of possibilities when it comes to statistical analysis. These variables can be meaningfully added, subtracted, multiplied, divided (ratios). Central tendency can be measured by mode, median, or mean; measures of dispersion, such as standard deviation and coefficient of variation can also be calculated from ratio scales

Lesson-3 Nature of Psychological Variable- Mental test tradition-Psycho Physical tradition.

Mental test tradition.

Nature of Psychological Variable

A variable, as the name implies, is something that varies. This is the simplest way of defining a variable. Webster says that a variable is “a thing that is changeable” or “a quantity that may

have a number of different values." True, a variable is something that has at least two values: however, it is also important that the values of the variable be observable. Thus, if what is being studied is a variable, it has more than one value and each value can be observed. For example, the outcome of throwing a dice is a variable. That variable has six possible values (each side of the dice having one to six dots on it), each of which can be observed. However, a behavioural scientist attempts to define a variable more precisely and specifically. Kerlinger (1986) defined variable 'a property that taken as different values'. According to D'Amato (1970) variables may be defined as those attributes of objects, events, things and beings, which can be measured. According to Postman and Egan (1949) a variable is a characteristic or attribute that can take on a number of values, for example, number of items that an individual solves on a particular test, the speed with which we respond to a signal, IQ, sex, level of anxiety, and different degree of illumination are the examples of variables that are commonly employed in psychological research. Before discussing the types of variables, it is important to know how the variables of study related to theoretical concepts. Because the variables exist in the world but the theory is an idea, researcher makes certain assumption to relate the two. These assumptions are guide ropes that tie a theory to the real world. The variables are tangible: duration, frequency, rate, or intensity of bar presses; items checked on a questionnaire; murders committed; books written. The theoretical concept is intangible: hunger, motivation, anxiety. The variables are related to the theoretical concepts by means of the operational definitions used to measure the concepts.

Suppose a theory reveals that increasing anxiety will increase the affiliation motive. To test out this theory, you may take the theoretical concepts of anxiety and affiliation motive and relate them to variables by means of operational definitions. The theory is an abstract statement. .For example, anxiety can be measured by the Anxiety Scale and affiliation by how close people sit to each other in the experiment. These two measures constitute the variables of the study. The scores on the variables of anxiety and distance apart are related to one another as test of the hypothesis. The relationship between the variable is taken as providing support for or against the particular theory that generated the experiment.

TYPES OF VARIABLES

To understand how variables are used and discussed in psychological researches, you would like to understand several distinctions that are made among the type of variables. The descriptions of different types of variables are given below:

Stimulus, Organism and Response Variables

Psychologists are interested in studying the behaviour or causes of behaviour as variables. Many psychologists have adopted a theoretical viewpoint or model called the S-O-R model to explain all behaviour. The symbols S, O, and R represent different categories of variables. S is the symbol of stimuli, and the category may be referred to in general as stimulus variables. A stimulus variable is some form of energy in the environment, such as light, to which the organism is sensitive. O is the symbol for organism variables , that is the changeable physiological and psychological characteristics of the organisms being observed. Examples of such variables are anxiety level, age and heart rate etc. Finally, R is the symbol for response

and, in general, response variables, which refer to some behaviour or action of the organism like pressing a lever, and reaction to any stimulus, are the examples of responses variables. You can understand an application of S-O-R model through the following example.

Suppose that an experiment is conducted in which a rat is placed on a metal grid floor, the grid is electrified, and the length of time it takes the rat to jump from the grid to a platform is measured. Using the S-O-R model, the electrical shock would be called a stimulus variable. The intensity of shock would be the value of the variable. The particular state of the organisms would be measured by the organismic variables. For example, the skin resistant of the rat at the time of shock was introduced would be an organismic variables. A response variable would be the latency (i.e. the elapsed time between the onsets of the shock and when the rat reaches the platform).

Independent and Dependent Variables An independent variable or stimulus variable (as Underwood calls it) is that factor manipulated or selected by the experimenter in his attempt to ascertain its relationship to an observed phenomenon. Dependent upon the mode of manipulation, some experts have tried to divide the independent variable into 'Type E' independent variable and 'Type S' independent variable (D'Amato, 1970). Type E independent variable is one of which is directly or experimentally manipulated by the experimental and type S.

independent variable is one which is manipulated through the process of selection only. For example the experimenter wants to study the effect of noise upon the task performance in an industry. Here the IV (Independent Variable) is the noise and the DV(Dependent Variable) is the task performance. He may manipulate the noise by dividing into three categories — continuous noise, intermittent noise and no noise and examine its effect upon the task performance. Here the noise is being directly manipulated by the experimenter and hence, it constitutes the example of Type-E independent variable. Suppose, for the time being, that the experimenter is interested in answering the question: Is the rate of production dependent upon the age of the workers? Age is here the independent variable. For investigating this problem, the experimenter will have to select groups of workers on the basis of their age in a way by which he can get an appropriate representation from different age groups ranging from say, 16 to 55 years. Subsequently, he will compare the rate of production obtained by each age group and finally, conclude whether or not age is a factor in enhancement of the performance .Hence this constitutes the examples of S-independent variables.

A dependent variable is the factor that appears, disappears, or varies as the experimenter introduces, removes or varies the independent variable. (Townsend,1953). The dependent variable is a measure of the behaviour of the subject. The dependent variable is the response that the person or animal makes. This response is generally measured using at least one of several different dimensions (Alberto & Troutman 2006). The dimensions are – (a) frequency – Number of times that a particular behaviour occurs, (b) duration - the amount of time that a behaviour lasts. (c) latency –the amount of time between and when the behaviour is actually performed (d) force – the intensity or strength of a behaviour. Here, you can examine the relationship between independent and dependent variables. The relationship is that of dependence. One variable depends upon the other. Suppose you find a relationship between

meaningfulness of the learning material and speed of learning. Speed of learning then depends upon meaningfulness; the greater the meaningfulness, the faster the learning. The speed of learning is, therefore, called dependent variable; meaningfulness is independent variable. Similarly, rest between work periods is independent variables; output of work is dependent variable. Sudden noise is independent variable; change in breathing is dependent variable. In an experiment one discovers and confirms a relationship between an independent variable and a dependent variable.

Extraneous and Confounded Variables

Any and all other variables that may 'mask' the relationship between independent variable and dependent variable are known as extraneous variables. Extraneous variables may directly affect the dependent variable or may combine with the independent variable to produce an affect. Therefore, extraneous variables must be controlled so that the experimenter can determine whether the dependent variable changes in relation to variation in the independent variable. Several others factors operating in a real life situation may affect changes in the dependent variable. These factors, not measured in the study, may increase or decrease the magnitude or strength of the relationship between independent and dependent variables. Extraneous variables are relevant in nature, and in experimental studies, they belong to three major types i.e., organismic variables, situational variables and sequential variables. The subject related variables include age, sex, intelligence, personality etc. are organismic variables. The situational variables include environmental variables operating in the experimental setting (e.g. noise, temperature, humidity) and variables related to the experimental task. The sequence related variables deal with sequence effects. They arise when participants in experiments are required to be tested in several conditions. Exposure to many conditions may result in adaptation, fatigue or practice effects which, if allowed to operate, may make the results difficult to interpret.

Confounding variables Variables and Constructs is one that varies with the independent variable. While doing a study if we are not careful then two variables may get combined so that the effect of one cannot be separated from the effect of other. This is known as confounding. For instance, if you conducted a study of the effect of television viewing on perception of violence and the experimental group contained only adolescents, whereas the control group only adults, the age of participants would be confounded with the independent variable under study. Confounding makes the conclusions of the study doubtful. It is ,therefore, necessary that effort should be made to unconfound the variables. To explain these variables let us take one example. Suppose you want to study the relationship between smoking and coronary heart disease. You assume that affecting this relationship, such as a number of cigarettes or the amount of tobacco smoked every day; the duration of smoking; the age of the smoker; dietary habits; and the amount of exercise undertaken by the individuals. All of these factors may affect the extent to which smoking might cause coronary heart disease. These variables may either increase or decrease the magnitude of the relationship. In this example, the extent of smoking is the independent variable, coronary heart disease is the dependent variable and all the variables that might affect this relationship, either positively or negatively, are extraneous variables.

Active and Attribute Variables Any variable that is manipulated is called active variables. Examples of active variables are reward, punishment, methods of teaching, creating anxiety

through instructions and so on. Attribute variable isthat variable which is not manipulated but measured by the experimenter. Variables that are human characteristics like intelligence, Aptitudes, sex, socio economic status, education, field dependence and need for achievement are the example of attributes variables. The word 'attribute' is more accurate enough when used within animated objects or references.

Quantitative and Categorical Variables

Quantitative variables is one that varies in amount whereas categorical variables varies in kind. Speed of response, intensity of sound, level of Illumination, intelligence etc. are the example of quantitative variables and gender, race, religion are the example of categorical variables. Precise and accurate measurement are possible with the quantitative variables because they can be easily ordered in terms of increasing and decreasing magnitude categorical variables can be of three types: Constant, dichotomous and polytomous. When a variable can have only one value or category, for example taxi, tree and water, it is known as a constant variables. When a variable can have only two categories as in yes/no, good/bad and rich/poor, it is known as dichotomous variables. When variables can be divided into more than two categories, for example: religion (Christian, Muslim, Hindu); political parties (Labor, Liberal, Democrat); and attitudes (strongly favorable, favorable, uncertain, unfavorable, strongly unfavorable), it s called a polytomous variable.

Continuous Variables and Discrete Variables

Quantitative variables are further divided into two categories, namely, continuous variables and discrete variables. A distinction between continuous and discrete variables is especially useful in planning of research and analysis of data. A continuous variable is one which is capable of being measured in any arbitrary degree of fineness or exactness. Age, height, intelligence, reaction time, etc., are some of the examples of a continuous variable. The age of the person can be measured in years, month and days. Thus, all such variables which can be measured in the smallest degree of fineness are called continuous variable. The discrete variables are those variables which are not capable of being measured in any arbitrary degree of fineness or exactness because the variables contain a clear gap. For example, the number of members in a family, no. of females in particular group, no of books in library and so on constitutes the examples of a discrete variable

psychological testing, also called **psychometrics**, the systematic use of tests to quantify psychophysical behaviour, abilities, and problems and to make predictions about psychological performance.

The word "test" refers to any means (often formally contrived) used to elicit responses to which human behaviour in other contexts can be related. When intended to predict relatively distant future behaviour (*e.g.*, success in school), such a device is called an aptitude test. When used to evaluate the individual's present academic or vocational skill, it may be called an achievement test. In such settings as guidance offices, mental-health clinics, and psychiatric hospitals, tests of ability and personality may be helpful in the diagnosis and detection of troublesome behaviour. Industry and government alike have been prodigious users of tests for

selecting workers. Research workers often rely on tests to translate theoretical concepts (*e.g.*, intelligence) into experimentally useful measures.

Psychophysics quantitatively investigates the relationship between physical stimuli and the sensations and perceptions they produce. Psychophysics has been described as "the scientific study of the relation between stimulus and sensation"[1] or, more completely, as "the analysis of perceptual processes by studying the effect on a subject's experience or behaviour of systematically varying the properties of a stimulus along one or more physical dimensions".[2]

Psychophysics also refers to a general class of methods that can be applied to study a perceptual system. Modern applications rely heavily on threshold measurement,[3] ideal observer analysis, and signal detection theory.[4]

Psychophysics has widespread and important practical applications. For example, in the study of digital signal processing, psychophysics has informed the development of models and methods of lossy compression. These models explain why humans perceive very little loss of signal quality when audio and video signals are formatted using lossy compression.

Thresholds

Psychophysicists usually employ experimental stimuli that can be objectively measured, such as pure tones varying in intensity, or lights varying in luminance. All the senses have been studied: vision, hearing, touch (including skin and enteric perception), taste, smell and the sense of time. Regardless of the sensory domain, there are three main areas of investigation: absolute thresholds, discrimination thresholds and scaling.

A threshold (or limen) is the point of intensity at which the participant can just detect the presence of a stimulus (absolute threshold[15]) or the presence of a difference between two stimuli (difference threshold[7]). Stimuli with intensities below the threshold are considered not detectable (hence: sub-liminal). Stimuli at values close enough to a threshold will often be detectable some proportion of occasions; therefore, a threshold is considered to be the point at which a stimulus, or change in a stimulus, is detected some proportion p of occasions.

Detection

An absolute threshold is the level of intensity of a stimulus at which the subject is able to detect the presence of the stimulus some proportion of the time (a p level of 50% is often used).[16] An example of an absolute threshold is the number of hairs on the back of one's hand that must be touched before it can be felt – a participant may be unable to feel a single hair being touched, but may be able to feel two or three as this exceeds the threshold. Absolute threshold is also often referred to as detection threshold. Several different methods are used for measuring absolute thresholds (as with discrimination thresholds; see below).

Discrimination

A difference threshold (or just-noticeable difference, JND) is the magnitude of the smallest difference between two stimuli of differing intensities that the participant is able to detect some proportion of the time (the percentage depending on the kind of task). To test this threshold, several different methods are used. The subject may be asked to adjust one stimulus until it is

perceived as the same as the other (method of adjustment), may be asked to describe the direction and magnitude of the difference between two stimuli, or may be asked to decide whether intensities in a pair of stimuli are the same or not (forced choice). The just-noticeable difference (JND) is not a fixed quantity; rather, it depends on how intense the stimuli being measured are and the particular sense being measured.[17] Weber's Law states that the just-noticeable difference of a stimulus is a constant proportion despite variation in intensity.[18]

In discrimination experiments, the experimenter seeks to determine at what point the difference between two stimuli, such as two weights or two sounds, is detectable. The subject is presented with one stimulus, for example a weight, and is asked to say whether another weight is heavier or lighter (in some experiments, the subject may also say the two weights are the same). At the point of subjective equality (PSE), the subject perceives the two weights to be the same. The just-noticeable difference,[19] or difference limen (DL), is the magnitude of the difference in stimuli that the subject notices some proportion p of the time (50% is usually used for p in the comparison task). In addition, a two-alternative forced choice (2-afc) paradigm can be used to assess the point at which performance reduces to chance on a discrimination between two alternatives (p will then typically be 75% since p=50% corresponds to chance in the 2-afc task).

Absolute and difference thresholds are sometimes considered similar in principle because there is always background noise interfering with our ability to detect stimuli.[6][20]

Experimentation:

In psychophysics, experiments seek to determine whether the subject can detect a stimulus, identify it, differentiate between it and another stimulus, or describe the magnitude or nature of this difference.[6][7] Software for psychophysical experimentation is overviewed by Strasburger.[21]

Classical psychophysical methods

Psychophysical experiments have traditionally used three methods for testing subjects' perception in stimulus detection and difference detection experiments: the method of limits, the method of constant stimuli and the method of adjustment.[22]

Method of limits

In the ascending method of limits, some property of the stimulus starts out at a level so low that the stimulus could not be detected, then this level is gradually increased until the participant reports that they are aware of it. For example, if the experiment is testing the minimum amplitude of sound that can be detected, the sound begins too quietly to be perceived, and is made gradually louder. In the descending method of limits, this is reversed. In each case, the threshold is considered to be the level of the stimulus property at which the stimuli are just detected.[22]

In experiments, the ascending and descending methods are used alternately and the thresholds are averaged. A possible disadvantage of these methods is that the subject may become accustomed to reporting that they perceive a stimulus and may continue reporting the same way even beyond the threshold (the error of habituation). Conversely, the subject may also

anticipate that the stimulus is about to become detectable or undetectable and may make a premature judgment (the error of anticipation).

To avoid these potential pitfalls, Georg von Békésy introduced the staircase procedure in 1960 in his study of auditory perception. In this method, the sound starts out audible and gets quieter after each of the subject's responses, until the subject does not report hearing it. At that point, the sound is made louder at each step, until the subject reports hearing it, at which point it is made quieter in steps again. This way the experimenter is able to "zero in" on the threshold.[22]

Method of constant stimuli

Instead of being presented in ascending or descending order, in the method of constant stimuli the levels of a certain property of the stimulus are not related from one trial to the next, but presented randomly. This prevents the subject from being able to predict the level of the next stimulus, and therefore reduces errors of habituation and expectation. For 'absolute thresholds' again the subject reports whether they are able to detect the stimulus.[22] For 'difference thresholds' there has to be a constant comparison stimulus with each of the varied levels. Friedrich Hegelmaier described the method of constant stimuli in an 1852 paper.[23] This method allows for full sampling of the psychometric function, but can result in a lot of trials when several conditions are interleaved.

Method of adjustment

In the method of adjustment, the subject is asked to control the level of the stimulus and to alter it until it is just barely detectable against the background noise, or is the same as the level of another stimulus. The adjustment is repeated many times. This is also called the method of average error.[22] In this method, the observers themselves control the magnitude of the variable stimulus, beginning with a level that is distinctly greater or lesser than a standard one and vary it until they are satisfied by the subjective equality of the two. The difference between the variable stimuli and the standard one is recorded after each adjustment, and the error is tabulated for a considerable series. At the end, the mean is calculated giving the average error which can be taken as a measure of sensitivity.

Adaptive psychophysical methods

The classic methods of experimentation are often argued to be inefficient. This is because, in advance of testing, the psychometric threshold is usually unknown and most of the data are collected at points on the psychometric function that provide little information about the parameter of interest, usually the threshold. Adaptive staircase procedures (or the classical method of adjustment) can be used such that the points sampled are clustered around the psychometric threshold. Data points can also be spread in a slightly wider range, if the psychometric function's slope is also of interest. Adaptive methods can thus be optimized for estimating the threshold only, or both threshold and slope. Adaptive methods are classified into staircase procedures (see below) and Bayesian, or maximum-likelihood, methods. Staircase methods rely on the previous response only, and are easier to implement. Bayesian methods take the whole set of previous stimulus-response pairs into account and are generally more robust against lapses in attention.[24] Practical examples are found here.[21]

Staircase procedures

Staircases usually begin with a high intensity stimulus, which is easy to detect. The intensity is then reduced until the observer makes a mistake, at which point the staircase 'reverses' and intensity is increased until the observer responds correctly, triggering another reversal. The values for the last of these 'reversals' are then averaged. There are many different types of staircase procedures, using different decision and termination rules. Step-size, up/down rules and the spread of the underlying psychometric function dictate where on the psychometric function they converge.[24] Threshold values obtained from staircases can fluctuate wildly, so care must be taken in their design. Many different staircase algorithms have been modeled and some practical recommendations suggested by Garcia-Perez.[25]

One of the more common staircase designs (with fixed-step sizes) is the 1-up-N-down staircase. If the participant makes the correct response N times in a row, the stimulus intensity is reduced by one step size. If the participant makes an incorrect response the stimulus intensity is increased by the one size. A threshold is estimated from the mean midpoint of all runs. This estimate approaches, asymptotically, the correct threshold

Bayesian and maximum-likelihood procedures

Bayesian and maximum-likelihood (ML) adaptive procedures behave, from the observer's perspective, similar to the staircase procedures. The choice of the next intensity level works differently, however: After each observer response, from the set of this and all previous stimulus/response pairs the likelihood is calculated of where the threshold lies. The point of maximum likelihood is then chosen as the best estimate for the threshold, and the next stimulus is presented at that level (since a decision at that level will add the most information). In a Bayesian procedure, a prior likelihood is further included in the calculation.[24] Compared to staircase procedures, Bayesian and ML procedures are more time-consuming to implement but are considered to be more robust. Well-known procedures of this kind are Quest,[26] ML-PEST,[27] and Kontsevich& Tyler's method.[28]

Magnitude estimation

In the prototypical case, people are asked to assign numbers in proportion to the magnitude of the stimulus. This psychometric function of the geometric means of their numbers is often a power law with stable, replicable exponent. Although contexts can change the law & exponent, that change too is stable and replicable. Instead of numbers, other sensory or cognitive dimensions can be used to match a stimulus and the method then becomes "magnitude production" or "cross-modality matching". The exponents of those dimensions found in numerical magnitude estimation predict the exponents found in magnitude production. Magnitude estimation generally finds lower exponents for the psychophysical function than multiple-category responses, because of the restricted range of the categorical anchors, such as those used by Likert as items in attitude scales.

UNIT-II: PSYCHO- PHYSICAL METHODS AND PSYCHOLOGICAL SCALING METHODS

Lesson-4 Method of Average Error-Constant and Variable Errors-Evaluation.

Introduction:

Psychophysical methods

The term psychophysical methods is sometimes extended to include certain scaling technique s which are most often usedwith subjective dimensions to which there correspond no simple p hysical dimensions, for example, food preferences.In 1860, G. Fechner designed psychophysi cal methods to measure the absolute threshold, defined as the minimum stimulusenergy that a n organism can detect, and the differential threshold, defined as the minimum detectable chan ge in a stimulus.Both quantities had to be defined as statistical averages. To obtain reliable m easurements for these averages, Fechner devisedthe method of limits (also called the method of minimal changes) and the method of constant stimuli.In the method of limits, the experime nter begins with a stimulus which is too weak for the subject to detect. In successivepresentati ons, the stimulus intensity is increased in small, equal steps, the subject reporting after each p resentation whetherthe stimulus was perceived until it has been detected. The descending seri es is then begun, the stimulus intensity beginning atan abovethreshold value and decreasing i n steps until the subject signals the disappearance of the stimulus. Many such seriesare given measuring the difference threshold, essentially the same procedure is involved, except that th e subject now signals therelation of a comparison stimulus to a standard stimulus. After a larg e number of such trials, the average of each of these fourthreshold values is computed.To me asure the absolute threshold by the method of constant stimuli, the experimenter selects a sma ll number of stimulusvalues in the neighborhood of the absolute threshold (previously roughl y located by informal use of the method of limits) andpresents them to the subject a large nu mber of times each, in an irregular order unknown to the subject. Each time a stimulusis prese nted, the subject reports the presence or absence of sensation.

The method of average error, the third of the three methods devised by Fechner, is a special a pplication of directmatchingmethods to cases in which the point of interest is in discrepancies between perception and stimulation. The subject adjusts acomparison stimulus to match a stan dard stimulus; the average of a number of such settings gives the point of subjectiveequality, a nd the difference between this point and the standard stimulus is the average error. Two illustr ative uses of themethod are the measurement of accuracy of distance perception and the meas urement of the magnitude of so-called opticalillusions.

Weber's law, also called Weber–Fechner law, historically important psychological law quantifying the perception of change in a given stimulus. The law states that the change in a stimulus that will be just noticeable is a constant ratio of the original stimulus. It has been shown not to hold for extremes of stimulation.

The law was originally postulated to describe researches on weight lifting by the German physiologist Ernst Heinrich Weber in 1834 and was later applied to the measurement of sensation by Weber's student Gustav Theodor Fechner, who went on to develop from the law the science of psychophysics. By stating a relationship between the spiritual and physical worlds, the law indicated to Fechner that there is really only one world, the spiritual. To others, the law meant the possibility of a scientific, quantitative psychology. The combined work of Weber and Fechner has been useful, especially in hearing and vision research, and has had an impact on attitude scaling and other testing and theoretical developments.

Fechner's law

The magnitude of a stimulus can be estimated by the formula S=k log R, where S = sensation, R = stimulus, and k = a constant that differs for every sensory modality (sight, touch, temperature, etc)

It refers to a psychology-based term formulated by the German psychologist Gustav Theodor Fechner, that focused on the relationship between physical stimulus and psychological experiences. It asserts the proportionality that exists between the intensity of a subjective sensation and the logarithm of the physical stimulus.

Steven's power law

A law of magnitude estimation that is more accurate than Fechner's law and covers a wider variety of stimuli. It is represented by the formula S=kl^a, where S = sensation, K= a constant, l = stimulus intensity, and a = a power exponent that depends on the sense being measured.

Signal detection theory (SDT)

Explains how he detect "signals" consisting of stimulation affectiing our eyes, ears, nose, skin, and other sense organs. Signal detection theory says that sensation is a judgment the sensory system makes about incoming stimulation. Often, it occurs outside of consciousness. In contrast to older theories from psychophysics, signal detection theory takes observer characteristics into account.

Classical psychophysics

Psychophysics is commonly defined as the quantitative branch of the study of perception, examining the relations between observed stimuli and responses and the reasons for those relations.

Psychophysics was established by German scientist and philosopher Gustav Theodor Fechner. He coined the word, developed the fundamental methods, conducted elaborate psychophysical experiments, and began a line of investigation that still persists in experimental psychology. Fechner's classic book Elemente der Psychophysik (1860) may be looked upon as the beginning not only of psychophysics but also of experimental psychology.

Psychophysics quantitatively investigates the relationship between physical stimuli and the sensations and perceptions they effect. Psychophysics has been described as "the scientific study of the relation between stimulus and sensation" or, more completely, as "the analysis of perceptual processes by studying the effect on a subject's experience or behaviour of systematically varying the properties of a stimulus along one or more physical dimensions".

Psychophysics also refers to a general class of methods that can be applied to study a perceptual system. Modern applications rely heavily on threshold measurement, ideal observer analysis, and signal detection theory.

Absolute threshold

Definition: The amount of stimulation necessary for a stimulus to be detected. In practice, this means that the presence or absence of a stimulus is detected correctly half the time over many trails.

- The lowest level of stimulation that any one person can detect.
- A person's ability to sense a stimulus at it lowest intensity.

- The least amount of stimuli need to be detected or noticed.

Eg. If I can hear a very high pitched noise, but someone else cannot, then the sound is above my own absolute threshold and below the other person's absolute threshold. A good example of this would be the "Mosquito" ringtone:

A threshold (or limen), is the point of intensity at which the participant can just detect the presence of a stimulus or presence of a difference between two stimuli. There are two kinds of thresholds: absolute and difference.

An absolute threshold is the level of intensity of a stimulus at which the subject is able to detect the presence of the stimulus some proportion of the time (a *p* level of 50% is often used). Absolute threshold is also often referred to as detection threshold.

Difference threshold

A difference threshold (or just-noticeable difference) is the magnitude of the smallest difference between two stimuli of differing intensities that the participant is able to detect some proportion of the time (again, 50% is often used). To test this threshold, several different methods are used. The subject may be asked to adjust one stimulus until it is perceived as the same as the other, may be asked to describe the magnitude of the difference between two stimuli, or may be asked to detect a stimulus against a background.

JND

Just noticeable difference (JND) describes the marginal difference in a stimulus (compared to a similar stimulus) needed for that difference to be perceived. Also mathematically represented.

Method of Average Error

The method of average error is one of the oldest and most fundamentalmethods of Psychophysics. In this method, 'O' (subject) is provided with a standardstimulus (Ss) and with a variable stimulus (Sv). The objective of this method is toproduce equal stimuli by active adjustment on the part of the observer. 'O' attempts toadjust the Sv until it seems to be equal to Ss. Since 'O' attempts to reproduce a givenstimulus it is often called as the "method of Reproduction" and also called as" Method ofAdjustment".AccordingtoTitchner,themethodofAverageerrorisfreegifttopsychophysics from the exact science of physics and astronomy. Fechner introducedthismethodintopsychophysics andmentioneditsusewithvisualand tactualstimuli. The method of average error will be illustrated with the measurement of the amount ofdistortion of a line of standard length bounded by arrow heads or feather heads in thewell-known Muller-Lyer illusion apparatus .Visual illusion refers to misinterpretation ofthe stimulus or its erroneous perception. The lines in the Muller-Lyer illusion apparatus appear that arrow heads constrict the space and feather heads expand the space between them.Hence it leads to undere stimation or over estimation of lines. The main interest is to determine the discrepancy between the standardand the average of 'O's judgements .Some of the possible errors in this method areconstant error due to illusion, space error ,movement error and other variable errors.The constant error is produced by a uniform condition resulting on a deviation in a fixeddirection and in a given extent. The difference between the mean judgment (Mj) and thestandard stimulus (Ss) value gives us an estimate of the size and direction of theillusion. Therefore PSE minus Ss (PSE-Ss) gives us both the directions and magnitudeof illusion. This method has the distinct practical advantage of economizing the time ofboth E and O when compared to the other Psychophysical methods.

Every trial gives ameasurement, whereas in other methods several trials or judgements are required toobtain what is equivalent to a single measurement. It is especially useful wherever anumber of measurements must be made in a limited time. This method has been foundtobemostusefulinapplicationtothestudyofvisualintensitiesandalsointhemeasurementofchromasofcolor;studyofto0nal attributes pitchandintensity.

The aim of the test To estimate the extent of distortion in perceived length of line of standard length inMuller-Layer illusionbyusingtheMethodof AverageError. Materials used in the test:Muller-Layerillusionapparatus,paperandpencil.

Description of the apparatus: the muller-Lyer illusion apparatus consists of a standardline (S) and a variable line (V). the standard line is having two arrow heads facinginwards. The variable line (V) is having two feather heads facing outwards. The variableline is adjustable to increase or decreasethelength between the two feather heads.The subject is required to adjust the variable line so that the line with feather heads willappearequaltothelinewitharrowheads.

Procedure: The Subjectis seated comfortably at a fixed place in front of the muller-Lyer illusion apparatus kept at eye level and is instructed as follows: "look at this lineenclosedbytwo arrowheads calledas standardlineandlookatthe otherportionofline enclosed by feather heads called as variable line (which can be varied in length). Yourtask is to adjust the length of the variable line (feather headed line) until you find it equalto line with arrow heads (standard line).once you have adjusted, you should not readjustit".

The experimental is given under 4 conditions- 2space and 2 movement condition. in halfof the trials the variable is placed to the left side of the subject and in the another half ofthetrialsthevariableisplacedtotherightsideofthesubject;thesetwospatialarrangement are denoted as L and R. in half of each of their seats, R and L, the trialsshouldstartwith thevariableobviouslytoogreatsothatthemovementofterminalarrowis inward (Q). In another half of the trials the variable is set too small so that themovement is outward (Q). These two movement conditions are denoted as I and Orespectively.Tocounterbalancetheeffectofspaceandmovementerrors,theexperimental conditions are arranged in the sequence of RO, LI,LO, RL, RO, RI, LO.Five trials are given in each of the experimental conditions and within each set of trialsthe starting length of variable is varied. The length of variable line adjusted by thesubject(ineachtrial)isrecorded.

Table – I: 40 observations of the subject for the method of Average error experiment.

Trial no	RO	LI	LO	RI	LI	RO	RI	LO
1	11.5	12.5	12.0	11.8	12.0	11.8	12.0	11.8
2	11.7	12.1	11.8	11.8	12.0	11.8	12.0	11.8
3	11.4	12.0	12.1	11.8	12.0	11.8	12.0	12.2
4	12.2	12.0	12.0	12.1	12.0	12.0	11.8	12.2

5	11.9	12.0	11.8	12.2	11.8	11.8	12.0	12.0
	58.7/5	60.6/5	59.6/5	59.7/5	59.8/5	59.2/5	59.8/5	60.0
	11.7	12.1	11.9	11.9	11.9	11.8	11.9	12.0

Table – II summary of Means for the muller-Lyer experiment by space and movement conditions and for the total data

condition					
	Space		**movement**		**total**
	R	**L**	**O**	**I**	
Mean	11.82	11.97	11.95	11.85	11.89

Constant error:

Constant error is computed as the average positive or negative difference between the observed and actual values along a dimension of interest. For example, if a weight of 1 kg is judged on average to be 1.5 kg, and a weight of 2 kg is judged to be 2.5 kg, the constant error is 500 g.

Variable error:

The difference between the observed and true values of the variable is called as measurement error or. errors-in-variables.

Lesson-5 Methods of Minimal Changes-Evaluation.

General Introduction: The method of minimal changes originated from the Method of Just Noticeable Differences. The presumption in this method is that the human observer c an recognize a just noticeable difference (jnd) when he comes across such an experience. Weber for eg., would set the variable stimulus (Sv) equal to standard stimulus (Ss) and then increase or decrease (Sv) by small steps until the observer reported that he perceived a jnd. Later Fechner introduced a change in this method which has been generally accepted. He recommended the approach to the jnd not only from the position of physical equality but also from the position of inequality. This change in the procedure led to the concept of "just not noticeable Difference" (jnnd). The jnd and jnnd are the two different measurements of the difference limens. In these two the jnnd is usually smaller than the jnd. Muller concluded that either of these two by itself is a true measure of the limen. But the true limen lies in between these two. Further, he asserted that difference limen is not an observable psychological quantity as Weber and others had believed. It is rather an ideal, calculated statistical value to be reached by averaging jnd and jnnd.

This method is known by several other names such as Method with complete knowledge, Method Serial Reproduction, Method of Least Noticeable Stimulus differences, Method of limits etc. Since the procedure involves ascending and descending series, this method is known as serial reproduction. This is the direct method to locate a threshold of given sensory modality.

The comparison will be made by moving the stimulus forward and backward from the standard by small steps to sense the particular kind of change that is to occur. The subject can be relaxed at first but this attention increases to a maximum extent as the limen is approached. In each trial the point at which the subject's response shifts from one category to another category would be noted down as transition point. Whereas in the estimation of difference limen, in each trial two stimuli would be presented and the subject asked to compare the standard (Ss) and the variable stimulus (Sv) in succession. The responses of the subjects would be categorized in three ways: greater, smaller and equal or doubtful. These were symbolized as +,- and = respectively. The common errors in this method are constant errors of habituation and expectation. Other possible errors are stimulus error, accidental error and variable errors.

ABSOLUTE LIMEN

Some lights are too dim to be seen, some sounds are too soft to be heard. To perceive the presence of stimulus a specific amount of stimulus Quantity is necessary. The absolute Limen or absolute threshold refers to the least amount of a stimulus that is responded to or recognized fifty percent of the trials. Since on any one trial many factors may determine if a minimal stimulus is perceived, the concept of AL is statistically computed. Therefore the absolute limen is defined as the stimulus value a person responds or perceives on 50 percent of the trials.

DIFFERENCE LIMEN

Generally one fails to notice every increment or decrement to the value of a stimulus. A stimulus may increase in size or intensity without the subject being able to appreciate or notice the change. Only when the amount of the increase/decrease reaches a certain limit, one can notice the difference. The value of noticeable difference is called Difference Limen or the DL. Aim of the test is to learn to use the method of Minimal Changes to determine the Difference Limen(DL) and to estimate Weber's ratio for the given visual stimulus.

DESCRIPTION OF THE APPARATUS: Aubert's diaphragm is an apparatus which has a variable white square in the Center against a white background. The area of the square can be increased or decreased by manipulating the knob situated below the diaphragm. At the back of the diaphragm there is a pointer on a linear scale to indicate the size of the square. The standard stimulus is a four inches square which is mounted on a rod against the black back ground.

PROCEDURE: The standard stimulus and variable stimulus were kept close in line with each other. The subject is seated comfortably in front of the apparatus of 2.5 meters (approximately). The variable stimulus should be kept by the side of the standard stimulus in such a way that the two white squares appear at the same height (at eye level) to the subject. The subject is instructed as follows:

'You see there are two white square. The size of square one(1) is fixed (S) and the size of the other square (2) wil be increased or decreased in small steps. Your task is to observe the two squares and report whether the size of the square two (2) appears to be greater than or less than or equal to the square one (1). If it appears larger say 'greater', if it appears smaller say 'lesser' and if it appears equal, say 'equal'. The responses are recorded in the table.

Table-I: Responses given by the subjects in the estimation of DL by using the method of Minimal changes.

Value of stimulus	VR				VL				VR				VL			
	d	a	d	a	d	a	d	a	d	a	d	a	d	a	d	a
16.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
16.5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
17.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
17.5	=	=	=	-	=	=	-	-	-	-	=	-	=	=	-	=
18.0	+	=	=	=	=	=	=	=	=	=	=	=	=	+	=	+
18.5	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
19.0	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
19.5	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
20.0	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

Table-II Summary of the means for the estimation of series in the different Limen experiment

Series Order			Space Arrangement	
Mean	ASC	DES	R	L
	14.68	14.12	17.46	18.15

Table-III Summary of IU, PSE, Constant error, DL & WR for the visual stimulus in the method of Minimal changes.

UTV	LTV	IU (UTV-LTV)	PSE UTV+LTV/2	DL= IU/2	EC=PSE-Ss	WR=DL/PSE

18.15	17.46	0.69	17.80	0.34	1.8	0.019

Lesson-6 Method of Constant Stimuli- -Evaluation.

- **General Introduction:** Of all the psychophysical methods, the constant methods are considered as the most accurate and most widely applicable. These methods are convenient to measure stimulus limens, differential limens, equal sense distances and in the estimation of many other psychosocial constants not only in the field of psychophysics but outside the sphere of psychophysics as well. In the procedure of this method, after initial (preliminary) trials, the experimenter (E) selects a limited number of stimuli (usually 4 to 7) which are presented to (O) subject a large number of times usually from 50 to 200 times each, in a pre arranged order unknown to O. In finding an S0 (absolute limen) each stimulus is present alone and O judges either the presence or absence of desired experience, for eg., "two" when assessing the two-point threshold. The procedure is known as the "Method of constant stimuli". Each stimulus is presented simultaneously with or in temporal sequence with a single standard and 'O' reports whether one stimulus is apparently "greater" or "lesser" than the other.Aim of the test is to determine the two point tactual threshold by the method of constant stimuli.

Materials: Two point Aesthesiometer (caliper type). Scale, blindfold, paper and pencil.

Description of the apparatus: Aesthesiometer is a caliper type of metallic apparatus with three (pyramid shaped) needles. One needle is horizontal (for single touch) which the other two are vertical (for use as two points). Out of the two vertical needles, one is a fixed where as the other is movable and slides on a graduated (in cms.) metallic strip enabling the reading of the distance of separation of the two vertical points.

Procedure: The subject is seated comfortably and should be blindfolded and is asked to stretch and rest his fore hand on the table. The ventral side (of the edge) of his elbow is taken as the skin area on which the limen is to be determined i.e., an area free of hair. A line is marked lightly in that area for presenting stimuli. Then the following instructions are given to the subject. "You are blind folded. Now I will press gently on your forearm sometimes with one needle point and sometimes with two needle points. Carefully observe. You have to respond saying whether you experienced "one" or "two" points. If you experience, one say, "one" and if it is two say "two". In this experiment sometimes the subject should be presented with the single needle instead of two. This will be like a check to know whether the subject is attentive and careful in judging the stimulus.

Some preliminary trials are administered very roughly the transition zone within which some judgments are "two" and some are "one". Five stimuli are selected such that the middle one is probably close to the two point limen. The stimulus should be presented at a constant pressure for 0.5 seconds. The smallest point separation is likely to elicit responses of two about five percent of the times and the largest separation about 95 percent of the times. The stimulus

intervals are kept equal. Then the stimuli in a set are applied in a prearranged random order until each stimulus in the set has been applied ten trials (each trial consists of applying all stimuli in a set). The data from which the limen is to be computed is tabulated.

Table-I The judgements of "two" for each stimulus applications in the Tactual Threshold experiment.

STIMULUS VALUES (in mms)						
TRAILS	0.5	0.3	0.1	0.4	0.2	0.6
1	2	1	1	1	2	1
2	1	2	2	1	1	2
3	2	2	1	2	2	2
4	2	2	1	2	2	2
5	2	2	1	2	2	2
6	2	2	2	2	2	2
7	2	2	1	2	2	2
8	2	2	1	2	2	2
9	2	2	1	2	2	2
10	2	2	1	2	2	2

Table-II Shows the Propoportion of 2s by the subject.

NAME OF THE SUBJECT		**Stimulus values**					
B.B		0.1	0.2	0.3	0.4	0.5	0.6
	f	2	3	9	8	9	9
	p	0.2	0.3	0.9	0.8	0.9	0.9

Lesson-7 Psychological scaling methods- Method of Pair comparison and rank order - **evaluation.**

Method of Pair comparison

Introduction The method of paired comparisons has been used in the cross-moderation strand of inter-board comparability studies for around a decade. The first studies to use it were reported in D'Arcy (1997). It replaced the cross-moderation techniques of ratification (where judges decide whether a script is above, at, or below a particular grade boundary) and identification (where judges examine a set of scripts in a range around the presumed boundary

and identify the mark that best represents the grade boundary). The cross-moderation strand is the part of a comparability study that requires expert judgements about examinee performance. The use of paired comparisons has not diminished the role of the specification review strand, which requires expert judgements about the level of demand of syllabuses and question papers (see Chapter 5) – this work is as important a pre-requisite for paired comparison judgements as it is for ratification or identification. Section 2 of this chapter reviews some of the work of the American psychologist L. L. Thurstone (1887–1955), who established and developed the theoretical approach to psychological measurement that underpins the use of paired comparisons and related methods. Section 3 shows how the paired comparison method has been used in practice in comparability research in British examinations. Section 4 describes a rank-ordering method – a recent development based on the same underlying theory and using similar techniques of analysis. Finally, section 5 contains more general discussion about the nature and validity of cross-moderation exercises.

The method of paired comparisons is a simple and direct way of collecting judgement data. The judge is presented with two objects (or 'stimuli') and has to decide which object is 'x-er' – in other words which object possesses more of a specified attribute, 'x'. In early psychophysical research 'x' was often a physical attribute such as weight, loudness or brightness. One aim of such research might be to discover or verify a mathematical function linking the perceived magnitude of the objects to their actual measured physical magnitude. A second aim might be to discover the 'limen' – the smallest increment in stimulus magnitude that could be discriminated at an arbitrary level of accuracy. Two well-known results from this field of research are Fechner's law and Weber's law. Fechner's law states that the perceived magnitude of a stimulus is proportional to the logarithm of its physical magnitude, and Weber's law states that the smallest perceptible difference in stimulus magnitude is proportional to the absolute magnitude. Thurstone liked to quote the psychologist William James's view of psychophysics: William James said that psychophysics was the dullest part of psychology. He was right. But if we extend and adapt the psychophysical concepts to the theory of discriminatory and selective judgment, the subject takes a different colour and it is no longer dull One of Thurstone's main achievements was to liberate psychophysical judgements from being restricted to attributes of stimuli that had a measurable physical magnitude and to develop a theory of psychological or 'subjective' measurement for non-physical attributes such as 'seriousness of crimes', 'attitude towards gambling', 'excellence of handwriting' and so on. This theory was first expounded in a series of articles published between 1925 and 1935, and Thurstone was still refining his theory in the years shortly before his death in 1955.

Method of rank order - evaluation

In every psychophysical experiment it is necessary for the investigator to select a psychophysical method and in doing so he is guided by two principal types of consideration, namely experimental convenience and theoretical propriety. Other things being equal, he will choose that experimental procedure which is simplest for the subject and most convenient. He will also be constrained to select a method which is legitimate for his problem. One of the simplest of experimental procedures ,is to ask the subject to arrange a series of stimuli in absolute rank order according to length, weight, brightness, beauty or whatever the psychological continuum may happen to be. This experimental procedure does not ordinarily lend itself to psychological measurement because the subjective increments represented by pie

successive rank orders are ordinarily entirely unknown. Occasionally it is legitimate to assume that the entire distribution of subjective values is Gaussian and then it is possible to translate rank orders into legitimate psychological measurement. We are dealing now with the situation in which a small number of stimuli, such as ten or twenty, are to be dealt with and in which no assumption can be made legitimately regarding their distribution in subjective value.
For theoretical purposes the constant method is probably the best in most experimental problems. The ideal form of the constant method is the method of paired comparison in which not one but all of the stimuli serve as standards, but 188) the constant method is also one of the most laborious experimental methods. If there are twenty stimuli and if the method is to be complete there would be required 190 judgments in order to compare each stimulus with every other stimulus. This assumes that the constant method is used in complete form with every stimulus serving in turn as a standard. The method has serious limitations when only one or two of the stimuli are used as standards. We shall also assume that the intermediate category is excluded. If the reader insists on using the intermediate category of judgment in the constant method, this paper is of no interest to him.

Our present problem is to devise a plan whereby simple absolute rank order may be used as the experimental procedure with the advantages of the much more laborious constant method. Given the data for absolute rank order we shall extract the proportion of judgments "A is greater than B" for every possible pair of stimuli in the given series. These derived proportions will be used instead of the proportions that are obtained directly in the constant method. From these derived proportions the subjective separations between any pair of stimuli can then be readily calculated by the equation of comparative judgment. The method will be derived first theoretically and then we shall describe its empirical verification.

If a subject has placed four stimuli *A B C D* in the rank order *B D A C* it is possible to tabulate his various comparisons as though he had made them separately. If each of the four stimuli were to be compared with every other one in the series it would require six separate judgments, namely *AB AC AD BC BD CD.* If there are *n* stimuli in the series it would require *n(n — 1)* such judgments with counterbalanced order of presentation or half that many if counterbalanced order is disregarded. This would give only one judgment for each of the possible pairs of stimuli. Now if the four stimuli have been placed in the rank order *B D A* C by one subject, it is clear that six judgments may be extracted from this one rank order. Evidently the above rank order series is equivalent to the judgments *B > D, B > A, B > C, D > A, D > C, A > C.* If a large number of subjects have arranged fifteen or twenty stimuli in rank order it is an almost prohibitive task to tabulate the separate judgments to which the single rank order is equivalent. However, it can be done by a shorter procedure.

Let there be *n* specimens in the series to be arranged in rank order by *N* subjects. Let *A* and *B* be two of these specimens and let a_1= frequency with which specimen *A is* placed in rank 1 by the *N* subjects, b_1= frequency with which specimen *B is* placed in rank i by the *N* subjects, pa_1= proportion of the *N* subjects who place specimen A in rank 1, p_{b1} = proportion who place specimen *B* in rank 1, and similarly for the other specimens and the other rank orders.

UNIT-III: TEST DEVELOPMENT AND TEST STANDARDIZATION PROCEDURES

Lesson-8 Test Development- Uses of Psychological test- Criteria of a Psychological Test. Test Standardization Procedures- -Construction of psychological test-

Definition: - A Psychological test is a standardized measure of a sample of a person's behaviour that is used to measure the individual differences that exist among people.

A psychological test is an objective and standardized measure of an individual's mental and/or behavioural characteristics.

A Psychological test is a systematic procedure or observing person's behaviour or performing, describing it with the aid of a numerical scale or category system. Mostly tests are used as a way of measuring differences between people or differences in the same person over time. Psychological tests are written, visual, or verbal evaluations administered to assess the cognitive and emotional functioning of children and adults.

Need for psychological test: - 1. Identifies weaknesses and strength 2. Supports individualized lesson plans 3. Enables placement decisions 4. Monitors progress 5. Identifying disabilities 6. Helping the administrative and academic works 7. Vocational ability. For children, academic achievement, ability psychological tests may be used as tools in school placement, in determining the presence of a learning disability or a developmental delay in identifying giftedness, or in tracking intellectual development. It may also be used with teens and young adults to determine vocational ability (e.g., in career counselling). Tests are administered for a wide variety of reasons, from diagnosing psychopathology (e.g., personality disorder, depressive disorder) to screening job candidates. They may be used in an educational setting to determine personality strengths and weaknesses.

Major Uses of Tests: - The basic use of tests is to provide information for decision makers. • Selection or placement • Diagnosis • Accountability evaluations • Judging progress and following trends • Self- discovery

Main Characteristics of a Good Psychological Test Five main characteristics of a good psychological test are as follows: 1. Objectivity 2. Reliability 3. Validity 4. Norms 5. Practicability! 1.

Objectivity: The test should be free from subjective—judgement regarding the ability, skill, knowledge, trait or potentiality to be measured and evaluated.

2. Reliability: This refers to the extent to which they obtained results are consistent or reliable. When the test is administered on the same sample for more than once with a reasonable gap of time, a reliable test will yield same scores. It means the test is trustworthy. There are many methods of testing reliability of a test.

3. Validity: It refers to extent to which the test measures what it intends to measure. For example, when an intelligent test is developed to assess the level of intelligence, it should assess

the intelligence of the person, not other factors. Validity explains us whether the test fulfils the objective of its development. There are many methods to assess validity of a test.

4. Norms: Norms refer to the average performance of a representative sample on a given test. It gives a picture of average standard of a particular sample in a particular aspect. Norms are the standard scores, developed by the person who develops test. The future users of the test can compare their scores with norms to know the level of their sample.

5. Practicability: The test must be practicable in- time required for completion, the length, number of items or questions, scoring, etc. The test should not be too lengthy and difficult to answer as well as scoring.

Test anxiety:

Test anxiety is a type of performance anxiety. In situations where the pressure is on and a good performance counts, people can become so anxious that they are actually unable to do their best.

Test anxiety is a psychological condition in which people experience extreme distress and anxiety in testing situations. While many people experience some degree of stress and anxiety before and during exams, test anxiety can actually impair learning and hurt test performance.

Physical symptoms may include: excessive sweating;;nausea, vomiting, or diarrhea; stomach pain; rapid heartbeat; shortness of breath; headaches; feeling lightheaded or faint
Emotional symptoms of test anxiety can include feelings of: self-doubt; fear; stress; hopelessness; inadequacy; anger

Anxiety can also cause difficulty concentrating. You may feel like your thoughts are jumbled and you've forgotten everything that you've learned. You can also become more indecisive, and you may struggle to choose between two different answers.

There are some different techniques you can use to cope with test anxiety, both before the test and during it.

To cope with test anxiety before a test, the best thing you can do is prepare as much as possible. Learn the best study methods that work for you, and spend plenty of time studying before each test. When possibl, take practice tests beforehand.

Lesson-9: Reliability-Types of reliability- Test -retest reliability- Interrater reliability- Parallel forms -reliability-Internal Consistency

Reliability' of any research is the degree to which it gives an accurate score across a range of measurement. It can thus be viewed as being 'repeatability' or 'consistency'. In summary:

- Inter-rater: Different people, same test.
- Test-retest: Same people, different times.
- Parallel-forms: Different people, same time, different test.

- Internal consistency: Different questions, same construct.

Inter-Rater Reliability

When multiple people are giving assessments of some kind or are the subjects of some test, then similar people should lead to the same resulting scores. It can be used to calibrate people, for example those being used as observers in an experiment.

Inter-rater reliability thus evaluates reliability *across different people*.

Two major ways in which inter-rater reliability is used are (a) testing how similarly people *categorize* items, and (b) how similarly people *score* items.

This is the best way of assessing reliability when you are using observation, as observer bias very easily creeps in. It does, however, assume you have multiple observers, which is not always the case. Inter-rater reliability is also known as *inter-observer reliability* or *inter-coder reliability*.

Examples

Two people may be asked to categorize pictures of animals as being dogs or cats. A perfectly reliable result would be that they both classify the same pictures in the same way.

Observers being used in assessing prisoner stress are asked to assess several 'dummy' people who are briefed to respond in a programmed and consistent way. The variation in results from a standard gives a measure of their reliability.

In a test scenario, an IQ test applied to several people with a true score of 120 should result in a score of 120 for everyone. In practice, there will be usually be some variation between people.

Test-Retest Reliability

An assessment or test of a person should give the same results whenever you apply the test.

Test-retest reliability evaluates reliability across *time*.

Reliability can vary with the many factors that affect how a person responds to the test, including their mood, interruptions, time of day, etc. A good test will largely cope with such factors and give relatively little variation. An unreliable test is highly sensitive to such factors and will give widely varying results, even if the person re-takes the same test half an hour later.

Generally speaking, the longer the delay between tests, the greater the likely variation. Better tests will give less retest variation with longer delays.

Of course the problem with test-retest is that people may have learned and that the second test is likely to give different results.

This method is particularly used in experiments that use a no-treatment control group that is measure pre-test and post-test.

Examples

Various questions for a personality test are tried out with a class of students over several years. This helps the researcher determine those questions and combinations that have better reliability.

In the development of national school tests, a class of children are given several tests that are intended to assess the same abilities. A week and a month later, they are given the same tests. With allowances for learning, the variation in the test and retest results are used to assess which tests have better test-retest reliability.

Parallel-Forms Reliability

One problem with questions or assessments is knowing what questions are the best ones to ask. A way of discovering this is do two tests in parallel, using different questions.

> Parallel-forms reliability evaluates different questions and question sets that seek to assess the same construct.

Parallel-Forms evaluation may be done in combination with other methods, such as *Split-half*, which divides items that measure the same construct into two tests and applies them to the same group of people.

Examples

An experimenter develops a large set of questions. They split these into two and administer them each to a randomly-selected half of a target sample.

In development of national tests, two different tests are simultaneously used in trials. The test that gives the most consistent results is used, whilst the other (provided it is sufficiently consistent) is used as a backup.

Internal Consistency Reliability

When asking questions in research, the purpose is to assess the response against a given construct or idea. Different questions that test the same construct should give consistent results.

> Internal consistency reliability evaluates individual questions in comparison with one another for their ability to give consistently appropriate results.

Average inter-item correlation compares correlations between all pairs of questions that test the same construct by calculating the mean of all paired correlations.

Average item total correlation takes the average inter-item correlations and calculates a total score for each item, then averages these.

Split-half correlation divides items that measure the same construct into two tests, which are applied to the same group of people, then calculates the correlation between the two total scores.

Cronbach's alpha calculates an equivalent to the average of all possible split-half correlations and is calculated thus:

$$a = (N \cdot \bar{r}) / (1 + (N-1) \cdot \bar{r})$$

Where N is the number of components,
and r-bar is the average of all Pearson correlation coefficients

Lesson-10 Validity- Types of validity- Construct Validity- Content Validity- Face validity- Criterion

Validity is the extent to which a test measures what it claims to measure.[1] It is vital for a test to be valid in order for the results to be accurately applied and interpreted.

Psychological assessment is an important part of both experimental research and clinical treatment. One of the greatest concerns when creating a psychological test is whether or not it actually measures what we think it is measuring.

For example, a test might be designed to measure a stable personality trait but instead, it measures transitory emotions generated by situational or environmental conditions. A valid test ensures that the results are an accurate reflection of the dimension undergoing assessment.[2]

Validity isn't determined by a single statistic, but by a body of research that demonstrates the relationship between the test and the behavior it is intended to measure. There are four types of validity: content validity, criterion-related validity, construct validity, and face validity.

This article discusses what each of these four types of validity is and how they are used in psychological tests. It also explores how validity compares with reliability, which is another important measure of a test's accuracy and usefulness.

Content Validity

When a test has content validity, the items on the test represent the entire range of possible items the test should cover.[3] Individual test questions may be drawn from a large pool of items that cover a broad range of topics.

In some instances where a test measures a trait that is difficult to define, an expert judge may rate each item's relevance. Because each judge bases their rating on opinion, two independent judges rate the test separately. Items that are rated as strongly relevant by both judges will be included in the final test.

Internal and External Validity

Internal and external validity are used to determine whether or not the results of an experiment are meaningful. Internal validity relates to the way a test is performed, while external validity examines how well the findings may apply in other settings.

Criterion-Related Validity

A test is said to have criterion-related validity when it has demonstrated its effectiveness in predicting criteria, or indicators, of a construct.[4]

For example, when an employer hires new employees, they will examine different criteria that could predict whether or not a prospective hire will be a good fit for a job. People who do well on a test may be more likely to do well at a job, while people with a low score on a test will do poorly at that job.

There are two different types of criterion validity: concurrent and predictive.

Concurrent Validity

Concurrent validity occurs when criterion measures are obtained at the same time as test scores, indicating the ability of test scores to estimate an individual's current state.[5] For example, on a test that measures levels of depression, the test would be said to have concurrent validity if it measured the current levels of depression experienced by the test taker.

Predictive Validity

Predictive validity is when the criterion measures are obtained at a time after the test.[6] Examples of tests with predictive validity are career or aptitude tests, which are helpful in determining who is likely to succeed or fail in certain subjects or occupations.

Construct Validity

A test has construct validity if it demonstrates an association between the test scores and the prediction of a theoretical trait.[7] Intelligence tests are one example of measurement instruments that should have construct validity. A valid intelligence test should be able to accurately measure the construct of intelligence rather than other characteristics, such as memory or education level.

Essentially, construct validity looks at whether a test covers the full range of behaviors that make up the construct being measured. The procedure here is to identify necessary tasks to perform a job like typing, design, or physical ability.

In order to demonstrate the construct validity of a selection procedure, the behaviors

demonstrated in the selection should be a representative sample of the behaviors of the job.

Face Validity

Face validity is one of the most basic measures of validity. Essentially, researchers are simply taking the validity of the test at face value by looking at whether it *appears* to measure the target variable.[8] On a measure of happiness, for example, the test would be said to have face validity if it appeared to actually measure levels of happiness.

Obviously, face validity only means that the test *looks* like it works. It does not mean that the test has been proven to work. However, if the measure seems to be valid at this point, researchers may investigate further in order to determine whether the test is valid and should be used in the future.

A survey asking people which political candidate they plan to vote for would be said to have high face validity, while a complex test used as part of a psychological experiment that looks at a variety of values, characteristics, and behaviors might be said to have low face validity because the exact purpose of the test is not immediately clear, particularly to the participants.

Reliability vs. Validity

While validity examines how well a test measures what it is intended to measure, reliability refers to how *consistent* the results are. There are four ways to assess reliability:[3]

- **Internal consistency**: Internal consistency examines the consistency of different items within the same test.
- **Inter-rater**: In this method, multiple independent judges score the test on its reliability.
- **Parallel or alternate forms**: This approach uses different forms of the same test and compares the results.
- **Test-retest**: This measures the reliability of results by administering the same test at different points in time.

It's important to remember that a test can be reliable without being valid. Consistent results do not always indicate that a test is measuring what researchers designed it to.

Lesson-11 Test Construction- Item writing - Meaning and types of items and their differences, guidelines for item writing. Response sets – Meaning, types and implications.

Attention must be given to the below mentioned points while constructing a potent, constructive and relevant questionnaire/schedule:

• The researcher must first define the problem that s/he wants to examine, as it will lay the foundation of the questionnaire. There must be a complete clarity about the various facets of the research problem that will be encountered as the research progresses.

• The correct formulation of questions is dependent on the kind of information the researcher seeks, the objective of analysis and the respondents of the schedule/questionnaire. Whether to use open ended or close ended questions should be decided by the researcher. They should be uncomplicated and made with such a view that there will be an objective part of a calculated tabulation plan.

• A researcher must prepare a rough draft of the schedule while giving ample thought to the sequence in which s/he wants to place the questions. Previous examples of such questionnaires can also be observed at this stage.

• A researcher by default should recheck and if required make changes in the rough draft to improve the same. Technical discrepancies should be examined in detail and changed

accordingly. • There should be a pre-testing done through a pilot study and changes should be made to the questionnaire if required.

• The questions should be easy to understand the directions to fill up the questionnaire clearly mentioned; this should be done to avoid any confusion. The primary objective of developing a tool is obtaining a set of data that is accurate, trustworthy and authentic so as to enable the researcher in gauging the current situation correctly and reaching conclusions that can provide executable suggestions. But, no tool is absolutely accurate and valid, thus, it should carry a declaration that clearly mentions its reliability and validity

Standardization of Psychological Tests: Standardization refers to the consistency of processes and procedures that are used for conducting and scoring of a test. To compare the scores of different individuals the conditions should be the same. In case of a new step the first and major step in standardization is formulating the directions. This also includes the type of materials to be used, verbal instructions, 123 Introduction to Test Construction time to be taken, the way to handle questions by test takers and all other minute details of a testing environment. Establishing the norms is also a key step for standardization. Norm refers to the average performance. To standardize a test, we administer it to a big, representative sample of the kind of individuals it was designed for. The aforementioned group sets the norms and is called the standardization sample. The norms for personality tests are set in the same way as those set for aptitude tests. For both, the norm would refer to the performance of average individuals. To construct and administer a test, standardization is a very important. The test is administered on a large set number of the people (the conditions and guidelines need to be the same for all). After which the scores are modified using Percentile rank, Z-score, T-score and Stanine, etc. The standardization of a test can be established from this modified score. Hence, "standardization is a process of ensuring that a test is standardized, (Osadebe, 2001)". There are lots of advantages when a test is standardized. A standard test is usually produced by experts and it is better than teacher made test. The standardized test is highly valid, reliable and normalized with Percentile rank, Z-score, T-score among scores derived from others to produce age norm, sex norm, location norm and school-type norm. Generally, a standardized test could be used to assess, and compare students in the same norming group.

The normal process for administering standardization includes: 1) A calm, quiet and disturbance free setting 2) Accurately understanding the written instructions, and 3) Provisioning of required stimuli. This makes the normative data applicable to the individuals being evaluated. 9.6.2 Classification of Standardized Testing Norm-referenced Testing: It is used to measure the result or performance in relation to all other individuals being administered the same test. It can be used to compare an individual to the others. Criterion referenced Testing: It is used for measuring the real knowledge of a certain topic. For example: Multiple choice questions in a geography quiz. 9.6.3 Steps for Constructing Standardized Tests A carefully constructed test where the scoring, administration and interpretation of result follows a uniform process can be termed as a standardized test. Following are the steps that can be followed to construct a standardised test: Steps 1) Plan for the test. 2) Preparation of the test. 3) Trial run of the test. 4) Checking the Reliability and Validity of the test. 5) Prepare the norms for the test. 6) Prepare the manual of the test and reproducing the test. 124 Methods of Data Collection 1) Planning – There needs to be a systematic planning in order to formulate a standardized test. Its objectives should be carefully defined. The type of content should be determined for example using short/long/very short answers or using multiple type questions, etc. A blue print must be ready with instructions to the method to be used for sampling, making the necessary requirements for preliminary and final administration. The length, time for completing the test and number of questions should be fixed.

Detailed and precise instructions should be given for administration of the test and also it's scoring. 2) Writing the items of the test – This requires a lot of creativity and is dependent on the imagination, expertise and knowledge. Its requirements are: • In-Depth knowledge of the subject • Awareness about the aptitude and ability of the individuals to be tested. • Large vocabulary to avoid confusion in writing. Words should be simple and descriptive enough for everybody to understand. • Assembly and arrangement of items in a test must be proper, generally done in ascending order of difficulty. • Detailed instructions of the objective, time limit and the steps of recording the answers must be given. • Help from experts should be taken to crosscheck for subject and language errors. 3) Preliminary Administration – After modifying the items as per the advise of the experts the test can be tried out on experimental basis, which is done to prune out any inadequacy or weakness of the item. It highlights ambiguous items, irrelevant choices in multiple choice questions, items that are very difficult or easy to answer. Also the time duration of the test and number of items that are to be kept in the final test can be ascertained, this avoids repetition and vagueness in the instructions. This is done in following three stages: a) Preliminary try-out – This is performed individually and it helps in improving and modifying the linguistic difficulty and vagueness of items. It is administered to around hundred people and modifications are done after observing the workability of the items. b) The proper try-out – It is administered to approximately four hundred people wherein the sample is kept same as the final intended participants of the test. This test is done to remove the poor or less significant items and choose the good items and includes two activities: • Item analysis – The difficulty of the test should be moderate with each item discriminating the validity between high and low achievers. Item analysis is the process to judge the quality of an item. • Post item analysis: The final test is framed by retaining good items that have a balanced level of difficulty and satisfactory discrimination. The blue print is used to guide in selection of number of items and then arranging them as per difficulty. Time limit is set.

Final try-out – It is administered on a large sample in order to estimate the reliability and validity. It provides an indication to the effectiveness of the test when the intended sample is subjected to it. 4) Reliability and Validity of the test – When test is finally composed, the final test is again administered on a fresh sample in order to compute the reliability coefficient. This time also sample should not be less than 100. Reliability is calculated through test-retest method, split-half method and the equivalent -form method. Reliability shows the consistency of test scores. Validity refers to what the test measures and how well it measures. If a test measures a trait that it intends to measure well then the test can be said to be a valid one. It is correlation of test with some outside independent criterion. 5) Norms of the final test – Test constructor also prepares norms of the test. Norms are defined as average performance scores. They are prepared to meaningfully interpret the scores obtained on the test. The obtained scores on test themselves convey no meaning regarding the ability or trait being measured. But when these are compared with norms, a meaningful inference can be immediately drawn. . The norms may be age norms, grade norms etc. as discussed earlier. Similar norms cannot be used for all tests. 6) Preparation of manual and reproduction of the test – The manual is prepared as the last step and the psychometric properties of the test norms and references are reported. It provides in detail the process to administer the test, its duration and scoring technique. It also contains all instructions for the test.

Lesson-12 Item analysis – Meaning and purpose. Item indices ad Item discrimination index.

Item analysis is a process which examines student responses to individual test items (questions) in order to assess the quality of those items and of the test as a whole. • Item analysis is especially valuable in improving items which will be used again in later tests, but it can also

be used to eliminate ambiguous or misleading items in a single test administration. • In addition, item analysis is valuable for increasing instructors' skills in test construction, and it is an important tool to uphold test effectiveness and fairness. • Item analysis is likely something educators do both consciously and unconsciously on a regular basis. • In fact, grading literally involves studying student responses and the pattern of student errors, whether to a particular question or particular types of questions. • But when the process is formalized, item analysis becomes a scientific method through which tests can be improved, and academic integrity upheld.

Need of the Item Analysis: Following are the needs of a test: • To select the candidates • To classify the candidates • To provide the ranking to the candidates • To promote the candidates • To frame the statements about the future behavior of the candidates • To establish the individual differences among the candidates To achieve the above objectives a test need to have the appropriate items so that the test can differentiate the individuals in different categories like superior, average and inferior. Therefore for selecting the appropriate items for the final form of the test following may be the objectives of the item analysis: Therefore we can say that the following may be the functions of Item Analysis: • Item analysis can increase the efficacy of the exams by testing knowledge accurately. • Item analysis not only can drive exam design, but it can also inform course content and curriculum. • When it comes to item difficulty, it's important to note whether errors indicate a misunderstanding of the question or of the concept the item addresses. • When a large number of students answer an item incorrectly, it's notable. It may be a matter of fine-tuning a question for clarity; is the wording of the question confusing? Are the answers clear? • It could be that the material may have to be reviewed in class, possibly with a different learning approach.

Following three functions are the main functions of Item Analysis: 1. Selecting the appropriate items for the test 2. Rejecting in appropriate items, and 3. Modification in the structure of the items Characteristics of an item: There are two main characteristics of an item 1. Difficulty Value or Pass Percentage "The difficulty value of an item is defined as the proportion or percentage of the examinee's who answer the item correctly" 2. Discriminating Power Discriminating power further can be divided into two parts (i) Item Reliability- "Item reliability may be defined as the degree to which an item differentiate high and low groups on the basis of the same test scores" (ii) Item Validity- "Item validity may be defined as the degree to which the item differentiate between high and low groups on the basis of some criterion test score" Item analysis can be performed in the following ways: • Item Difficulty: It is about the exam question too easy or too hard? When an item is one that every student either gets wrong or correct, it decreases an exam's reliability. If everyone gets a particular answer correct, there's less of a way to tell who really understands the material with deep knowledge. Conversely, if everyone gets a particular answer incorrect, then there's no way to differentiate those who've learned the material deeply.

• Item Discrimination -- Does the exam question discriminate between students who understand the material and those who do not? Exam questions should evaluate the varying degrees of knowledge students have on the material, reflected by the percentage correct on exam questions. Desirable discrimination can be shown by comparing the correct answers to the total test scores of students--i.e., do students who scored high overall have a higher rate of correct answers on the item than those who scored low overall? If you separate top scorers from bottom scorers, which group is getting which answer correct? The data from item analysis can drive the way in which one can design his future tests. As noted previously, if student knowledge assessment is the bridge between teaching and learning--then exams ought to measure the

student learning gap as accurately as possible. Item analysis should bring to light both questions and answers as one revise or omit items from his test. • Is the item difficulty level appropriate? • Does the item discriminate appropriately? Procedural steps for performing item analysis: Dichotomization: It is the process of forming higher score group and lower score group as per the Kelly's Method. On the basis of obtained scores we have to consider only the top and bottom percentage in any of the following manner: 1. Considering Top 25% and Bottom 25%, discarding the middle 50% cases. 2. Considering Top 33% and Bottom 33%, discarding the middle 34% cases. 3. Considering Top 27% and Bottom 27%, discarding the middle 46% cases. From the above, the third one (Kelly's Method of dichotomization) is the most promising one. Difficulty Value: Guilford defined Difficulty Value as: The formula for calculating the difficulty value of an item is The formula for calculating the Difficulty Value if there are some examinees are such that they are not giving any response to an item Discrimination Index: Blood and Budd defined the Discrimination Index as: Following are the types of Discrimination Index: The formula for the determination of the value of Discrimination Index is given as: According to Ebel the following criterion are looking important for the selection of an appropriate item in the light of its Discriminating Index and Difficulty Index: How are the Difficulty Value and Discrimination Power related? Criterion for the selection of appropriate item for the final form of the test

Lesson-13 Test score - interpretation- Types of scores.

Definition: Placing measurement data in a context, or making sense of test scores.

Description. Interpretation of test scores depends upon all the steps that came before it. That is, the test construction process must have produced a valid test if the interpretation is to be valid; the test must have been administered and scored with a minimum of error during those processes. Since tests are never perfectly valid, interpretation should include statements about the limits of the test as influenced by demonstrated and likely sources of error. Without such statements of limitations, you may misinterpret the scores of the measurement methods you employ.

Test interpretation, particularly in educational settings, traditionally has focused on norms. In *norm-referenced tests* a test score is interpreted by comparing it to a group of scores. I can say, for example, that a 3rd grade student's score on an achievement test places her or him at the 90th percentile of performance. Norm-referenced interpretations are typically contrasted with criterion-referenced test interpretations (i.e., comparison to a standard, instead of other persons). That same 3rd grade student may have correctly answered 35 of 40 test items that assessed previously taught material; the teacher may have set of criterion of 30 correct answers for students to pass the course.

Other types of interpretations are also useful. With *formative* tests, interpretation focuses on an individual's performance on the *components* of a course. In a mathematics class, for example, a formative test might provide information about the particular types of addition or subtraction problems a particular student answered correctly and incorrectly. During a course, formative tests provide feedback to the teacher and student that reveal progress and guide adjustment of the teaching. In education, **Cross and Angelo (1988)** described this process as a loop "from teaching technique to feedback on student learning to revision of the technique" (p. 2).

Summative tests provide an overall evaluation of an individual's performance in a course (e.g., a course grade). Summative tests provide data convenient for administrative decision-making. Summative tests can suggest initial hypotheses relevant to teaching: A standardized achievement test, for example, can describe a student's strengths and weaknesss (compared to other students) across subject areas. This information might be relevant to inclusion or exclusion from an educational program (e.g., a remedial course or repeating a grade). More sensitive measures will be needed to develop and test those hypotheses, however, and it is here that formative tests can be useful (**Bloom, Hastings, &Madaus, 1971**; **Cross & Angelo, 1988**). The interpretation of summative tests focus on an aggregate score (of items and components), while administrators of formative tests tend to examine item response patterns (Bloom et al., 1971).

Much more attention has been paid in the literature to how the test administrator or researcher interprets test scores than how test-takers make sense of them. One exception to this is research on the *Barnum effect*. Gauging the validity of a particular test interpretation depends upon making comparisons with other types of test interpretation. The Barnum effect occurs when individuals take a test and receive test interpretations based not on their test data, but simple generic statements that might apply to anyone, such as the statements that appear in horoscopes ("Work hard today and your efforts will pay off"). Test-takers usually find such bogus feedback as accurate as actual test interpretations. **Guastello and Rieke (1990)** evaluated the accuracy of real computer-based test interpretations (CBTIs) based on 16PF scores (a personality inventory) with bogus reports. A sample of 54 college students rated the real reports as 76% accurate and the bogus reports as 71% accurate. Computer-based reports are likely to increase the Barnum effect because many people ascribe increased credibility to computer operations.

4.1 Norms

Definition: Data about a distribution of scores for a particular test.

Description. As described previously, in norm-referenced interpretations the purpose of testing is to compare scores among individuals. Thus, the test is intended to detect individual differences on a construct of interest. **Gronlund (1988)** indicated that developers of norm-referenced tests seek items with the greatest possible variability. With achievement tests, these items are pursued through a selection process that retains items of average difficulty; easy and difficult items, that everyone passes or fails, are likely to be discarded. Aggregation of such items increases the possibility of making valid distinctions among individuals.

Norm-referenced testing has been the predominant approach in selection testing (**Murphy &Davidshofer, 1994**). Besides their lower cost, norm-referenced tests also seem more applicable when the test administrator desires to select some portion of a group (e.g., the top 10% of applicants) as compared to all applicants who could successfully perform a function. Thus, norm-referenced tests are useful in selection situations where individuals are chosen partially on the basis of scarce resources. Suppose you conduct a research study and find that 95% of all graduate students who score 600 or above on the GRE Verbal scale are able to pass all required graduate school courses. From the perspective of criterion-referenced testing, everyone scoring 600 or above should be admitted. In many graduate departments, however, that would mean admitting more students than available courses, instructors, or financial supports. Such a situation certainly occurs in other educational, occupational, and clinical

settings with fixed quotas. Norm-referenced testing, then, provides a solution: identify the top-scoring number who match the available resources.

If a test is intended to function as a selection device, its items should be developed on a sample representative of the population for whom the test is intended. Thus, the selection of a norm group for test development has potentially serious consequences for the interpretation of future scores compared to that group. Much controversy has occurred over the widespread use of intelligence tests or vocational interest inventories, for example, that were developed and normed on predominantly white, middle class individuals.

Measurement-related statistics

Definition: Statistics employed to facilitate the interpretation of test scores.

Description. Making sense of test scores often depends at least partially on understanding a number of statistical indices normally computed with tests. For example, test developers usually examine (and present information about) the frequency distribution of all test scores to determine if it is normally distributed. Similarly, developers may present information about the range and standard deviation of scores to examine whether sufficient individual differences exist. Below I describe statistics commonly used during the test interpretation process. Students who wish to review or learn about statistics may consult this **website** or try this **video**.

A *mean* or average is a measure of central tendency; that is, in a group of scores, where is the middle or most representative value? The mean is found by summing the scores in a group and dividing the number of scores. Other measures of central tendency are the *median* and the *mode*. These measures provide a typical score that characterizes the performance of the entire sample. A mean, along with the other measures of central tendency, is particularly useful for comparing different groups (such as children of different ages) who take the same test as well as describing individuals in relation to a group's set of scores (where does one individual score on a course quiz in relation to the whole class).

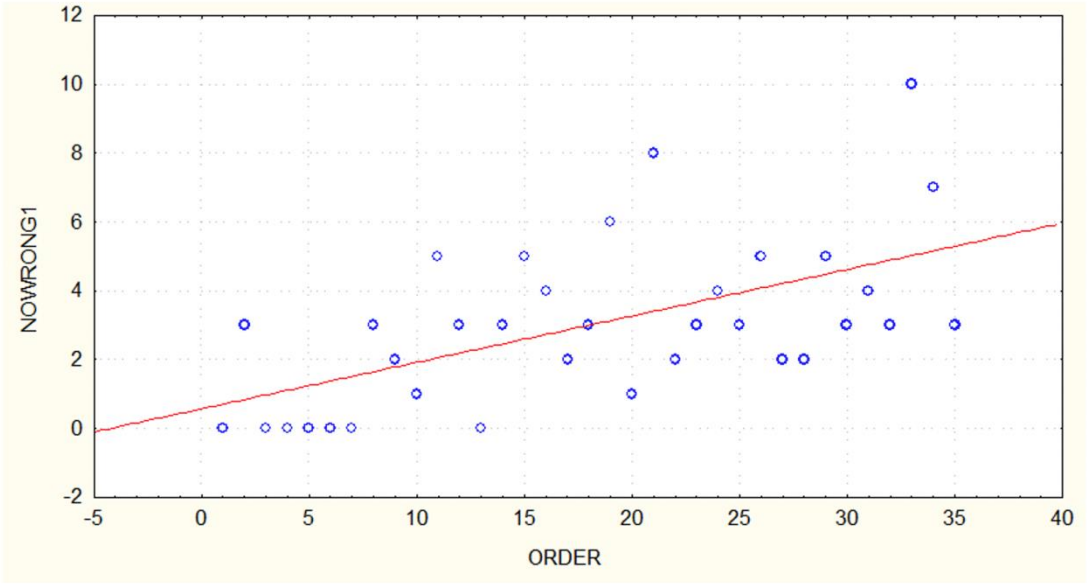
Besides knowing the central tendency in a group of scores, it is often useful to know how dispersed the scores are. One such index of dispersion, the *standard deviation*, refers to the average deviation of scores from the mean. The larger the standard deviation, the more widely spread the distribution of scores.

A *correlation* refers to the extent to which two variables covary. A *correlation coefficient* expresses the degree of relationship between two sets of scores. For example, if the highest scoring individual on Test 1 has also obtained the top score on Test 2, and the second-best individual on Test 1 is also second-best on Test 2, as so on down to the lowest scoring individual on each test, a perfect positive correlation would exist (+1.00). If there is a complete reversal of scores, so that the highest scoring individual on variable 1 obtains the lowest score on variable 2 and so forth, there would be a perfect negative correlation (-1.00). A zero correlation indicates the absence of a relationship between two variables, such as might occur by chance. Thus, correlation coefficients fall between the range of -1.00 and +1.00.

The data that form the basis of a correlation coefficient can also be graphed. The graph below shows the relation between the number of quiz questions students answered incorrectly in relation to the order in which they turned in their quiz:

Figure 12

Scatterplot Between Number of Incorrect Responses and Order of Quiz Completion



As the scatterplot shows (**Figure 3-4**), students who completed the quiz sooner generally had a fewer number of incorrect answers. However, the relationship is not perfect; for example, the second student to turn in a quiz had 3 incorrect answers. The correlation computed for these data is .51, with a mean of 1.66 and a standard deviation of 2.29. Although the reason to present these (actual) data is to explain the idea of correlation, do you have any substantive idea about why this relation should exist? In other words, how would you explain why students who finished a quiz faster generally had better grades?

A *standard score* or *z score* is a transformation of a raw score to show how many deviations from the mean that score lies. The formula is:

z = (Raw score – mean) / Standard deviation)

Thus z equals the person's raw score minus the mean of the group of scores, divided by the standard deviation of the group of scores. Frequently the best information that a test score can give us is the degree to which a person scores in the high or low portion of the distribution of scores. The z score is a quick summary of the person's standing: positive z scores indicate that the person was above the mean, while negative scores indicate the person scored below the mean.

Other types of standard scores have also been developed, including stanines, deviation IQs, sten scores, and T-scores. *T-scores*, for example, allow us to translate scores on a test to a distribution of scores of our choice. T-scores use arbitrarily fixed means and standard deviations and eliminate decimal points and signs. The formula is:

T = (SD * z) + M

where SD is the chosen standard deviation, M is the chosen mean, and z is the standard score for a person's score on a test. For example, I might find it simpler to give feedback using a distribution of scores whose mean is 50 and whose standard deviation is 10. If a person had a score on a test whose z equaled -.5, the T-score would be:

(10 -.5) + 50 = 45

Tests such as the **Analysis of Learning Potential** use a fixed mean of 50 and a standard deviation of 20, while the **Scholastic Aptitude Test (SAT)** and **Graduate Record Exam (GRE)** historically have employed 500 as the mean and 100 as the standard deviation. Again, the T-score provides a convenient translation of scores so that they might be more understandable during test interpretation.

Acknowledging that error influences any particular testing occasion, the *standard error of measurement* (SEM) is the standard deviation that would be obtained for a series of measurements of the same individual if the individual did not change on the measured construct over that time period. For example, assume that I administer a test measuring a stable trait 10 times to a particular person. If that person received the same score for each test occasion, there would be no error of measurement. In reality, however, the test score would vary for each testing, and SEM is a statistic designed to summarize the amount of variation. If you have an estimate of a test's reliability, SEM can be calculated as follows:

SEM = Standard deviation * SqRt (1 – r)

Thus, SEM equals the standard deviation of the group of scores times the square root of 1 minus the reliability estimate. SEMs help us know the extent to which an individual's particular test score can be trusted as indicative of the person's true score on the test.

Finally, the *standard error of estimate* (SEE) helps us know the trustworthiness of a test score's ability to predict a criterion of some sort. Just as no test produces the same score when administered repeatedly to a person, no single score will be associated with the identical score on a criterion. Thus, the SEE refers to the spread of scores around a criterion, or more precisely, the standard deviation of criterion scores for individuals who all have the same score on the predictor test. The formula for SEE is:

SEE = Standard deviation * SqRt ($1 – v^2$)

SEE equals the standard deviation for the group of criterion scores times the square root of 1 minus the squared validity coefficient (v). The validity coefficient is simply the correlation between the predictor test and the criterion that is attempted to be predicted. For example, graduate schools frequently screen candidates on the basis of their GRE scores because GRE scores (the predictor test) have been shown to have a modest correlation with first year GPA (the criterion). SEE helps us gain a sense of how large the variation is likely to be around the criterion given an individual's particular test score.

Let's walk through simple computations of the standard score, SEM, and SEE. Start with the z or standard score. Assume that the following represents a group of test scores. To compute a z score, I need the mean (which equals 87.95) and standard deviation (6.82) for this group of scores.

Figure 13

Sample Dataset of Test Scores

78	90	95	70	85
88	85	85	90	83
94	95	88	91	99
93	81	94	91	84

If your score on this test was 90, your z score would be:

(90 – 87.95) / 6.82 = .30

A z of .30 indicates you scored slightly above the mean in this group of scores.

On the other hand, if your score was 70, your z score would be:

(70 – 87.95) / 6.82 = -2.63

This z indicates your score was well below the mean.

SEM depends upon the standard deviation and the reliability of the particular test. If I have a test with a reliability estimate of .90 (high) and a standard deviation of 15, then SEM equals:

15 * SqRt (1-.9) = 4.7

Thus, 4.7 represents 1 standard deviation unit for the distribution of scores around the individual's true score. However, if the test's reliability estimate was .7, SEM increases:

15 * SqRT (1-.7) = 8.21

Thus, the lower the reliability of the test, the less confidence I have that an individual's true score is close to the actual score obtained.

Finally, with SEE, I need the correlation between the test and criterion as well as the standard deviation for the group of criterion scores. If the correlation between test and criterion equaled .61, and the standard deviation for the criterion scores equaled 100, then SEE would be:

100 * SqRt (1-[.61*.61]) = 79

Thus, 79 represents 1 standard deviation unit around the criterion score. However, if the correlation between predictor and criterion dropped to .30, the SEE would increase:

100 * SqRT (1-[.30*.30]) = 95

Thus, the lower the correlation, the less confidence I have that the predicted criterion score is the true score the individual would actually obtain.

4.3 Criterion-referenced interpretations

Definition: Interpreting a test score in relation to a criterion or pre-established level instead of other persons.

Description. Suppose an individual received a score of 95% on a classroom test. What does that mean? In a norm-referenced interpretation, that would indicate that the student scored higher than 94% of the rest of the class. A criterion-referenced statement would be "correctly completed 95 of 100 questions." Criterion-referenced interpretations simply describe performance in relation to a standard other than other persons.

With criterion-referenced tests items are retained during test development because of their relation to a criterion, regardless of the frequencies of correct or incorrect responses. However, criterion-referenced tests cost more than norm-referenced tests because they (a) require considerable effort in the analysis and definition of the performance criteria to be measured and (b) may necessitate special facilities and equipment beyond self-report materials. If one is interested in predicting performance on a criterion–the major purpose of selection testing–then criterion-referenced approaches would seem a logical choice. If one is interested in knowing whether a person can shoot a basketball, it usually makes more sense to give her or him 20 shots than a test of eye-hand coordination.

During item development of criterion-referenced tests, **Swezey (1981)** emphasized the importance of precisely specifying test objectives. Criteria can be described in terms of variables such as product or process, quality, quantity, time to complete, number of errors, precision, and rate (**Gronlund, 1988**). A criterion may be a product such as "student correctly completes 10 mathematics problems"; a process criterion would be "student completes division problems in the proper sequence." Process measurement is useful when diagnostic information is required, when the product always follows from the process, and when product data are difficult to obtain.

Criterion-referenced tests should be reliable and valid to the extent that performances, testing conditions, and standards are precisely specified in relation to the criteria. Swezey (1981) preferred "within 5 minutes" to "under normal time conditions" as a precise testing standard. In some respects, the criterion-referenced approach represents a move away from a search for general laws and toward a specification of the meaning of test scores in terms of important measurement facets. Discussing test validity, **Wiley (1991)** presented a similar theme when he wrote that the labeling of a test ought to be "sufficiently precise to allow the separation of components of invalidity from valid variations in performance" (p. 86). Swezey's and Wiley's statements indicate the field's increasing emphasis on construct explication.

Gentile and Murnyack (1989) described a set of criteria for grading students performing art criticism assignments. They noted that art criticism is a complex analytic skill requiring students to evaluate and interpret their and others' art work. Gentile and Murnyack suggested a 50-point rating system for evaluating students' assignments:

1. Applies critical thinking criteria (0-10).

2. Employs technical vocabulary (0-10).

3. Provides feedback according to criteria (0-10).

4. Presents the criticism (0-10).

Gentile and Murnyack (1989) suggested a possible passing grade of 35 points. Students who scored lower would revise and resubmit their paper based on the instructor's feedback on these criteria.

Lesson-14 Ethical issues in test use and interpretation.

Ethics in Psychological testing: It must be remembered that there are certain important fallacies in psychological assessment. These have to be kept in mind while doing assessment. In addition, we have many ethical considerations that have to be considered which are given in the sections below. There are 10 common fallacies and pitfalls that plague psychological testing and assessment. They are: 1) mismatched validity; 2) confirmation bias; 3) confusing retrospective and prospective accuracy (switching conditional probabilities) 4) unstandardising standardised tests; 5) ignoring the effects of low base rates; 6) misinterpreting dual high base rates; 7) perfect conditions fallacy; 8) fmancial bias; 9) ignoring the effects of audio-recording, video-recording, or the presence of third-party observers; and 10) uncertain gate keeping. These assessment fallacies and pitfalls are discussed in more detail below.

The Ethics Code contains six general principles:

i) Competence: Psychologists maintain high standards of competence, including knowing their own limits of expertise. Applied to testing, this might suggest that it is unethical for the psychologist to use a test with which he or she is not familiar to make decisions about clients.

ii) Integrity: Psychologists seek to act with integrity in all aspects of their professional roles. As a test author for example, a psychologist should not make unwarranted claims about a particular test.

iii) Professional and scientific responsibility: Psychologists uphold professional standards of conduct. In psychological testing this might require knowing when test data can be useful and when it cannot. This means, in effect, that a practitioner using a test needs to be familiar with the research literature on that test.

iv) Respect for people's rights and dignity: Psychologists respect the privacy and confidentiality of clients and have an awareness of cultural, religious, and other sources of individual differences. In psychological testing, this might include an awareness of when a test is appropriate for use with individuals who are from different cultures.

v) Concern for others' welfare: Psychologists are aware of situations where specific tests (for example, ordered by the courts) may be detrimental to a particular client. How can

these situations be resolved so that both the needs of society and the welfare of the individual are protected?

vi) Social responsibility: Psychologists have professional and scientific responsibilities to community and society. With regard to psychological testing, this might cover counseling against the misuse of tests by the local school.

UNIT-IV: STANDARD PSYCHOLOGICAL TESTS &MISCELLANEOUS TECHNIQUES

Lesson-15 Classification of Psychological Tests - Intelligence- Aptitude Tests-Tests of Interest

Psychological tests are tests that are conducted to know about the abilities, attitudes, motivations and needs of individuals. According to Anne Anastasi, a famous psychologist, 'Psychological test can be defined as a sample of an individual's, behaviour, obtaining under standard conditions and scored according to a fixed set of rules that provide a numeric score.' In a standardized psychological test, the scores of the individual are compared with test scores obtained from samples of tests that were conducted when the test was first developed. Psychological tests can be conducted in various ways. These may require the individual to answer in Yes/ No or Right/ Wrong or in some cases these tests may require the individual to give detailed information about their tendencies and dispositions. Generally, psychological tests are conducted and interpreted on the basis of a manual that gives detailed information regarding the test, its accuracy and reliability.

Each of these tests has a defined purpose and assesses different traits and attitudes of the individual. Any psychological test to be of use to the counsellor as well as the client must have the following characteristics: The test must be reliable in the sense that it must give accurate and consistentresults every time it is used. The test must be valid in the sense that it measures exactly what it is intended to measure. The test must be practical and usable in the sense that it can be easily administered in a cost- effective manner. The test must also be conducted in such a manner that it takes very less time. According to Lee Joseph Cronbach, an American educational psychologist, psychological tests are undertaken for the following purposes:

Prediction:The various psychological tests that the individual undertakes provide a basis for predicting the decisions that the individual will take at a later stage in order to overcome the problem.

Selection: Psychological tests are often administered by various organizations and institutions to know the skills and abilities that an individual possesses. On the basis of these skills and abilities, the individual may be selected for a job or a course.

Classification: : Psychological tests are used to classify individuals on the basis of their aptitude and attitude. For example, students may need to undertake psychological tests to classify those with high aptitude levels as one group.

Evaluation: Psychological tests are also undertaken to assess and evaluate programmes, methods and techniques to know whether these are effective in application or not.

Intelligence Tests

Intelligence is one of the most commonly used psychological term that has been used to distinguish people. Intelligence is basically an understanding of how well the individual understands a complex process and learns from it and other experiences. On this basis, a person

who is able to understand the complex processes well and use them equally well is considered to be an intelligent person. Intelligence, however, cannot be measures only on the basis of a single entity. In fact, intelligence tests measure a variety of mental functions and abilities of the individual. Intelligence tests are, thus, based on nine different and independent intelligences. These multiple intelligences have been proposed by Howard Gardener, an American developmental psychologist, in his Theory of Multiple Intelligences. According to Gardener, these nine different and independent intelligences interact and work together to enable the individual to find a solution to his problems. Thus, the Theory of Multiple Intelligences states that intelligence is not a single entity and a person's intelligence cannot be measured on the basis of excellence at a specific task.

The following are the nine types of intelligences as described by Gardener in his Theory of Multiple Intelligences:

Linguistic intelligence: Linguistic intelligence refers to the ability of the individual to make use of language he knows in expressing his thoughts and in understanding others.

Logical mathematical intelligence: Logical mathematical intelligence refers to the ability of the individual to think logically and critically to manipulate symbols and solve mathematical problems.

Spatial intelligence: Spatial intelligence refers to the ability of a person to form, transform and use mental images. It is spatial intelligence that enables a person to understand relationships of objects in space.

Musical intelligence: Musical intelligence is the capability of the individual to create, produce and manipulate musical patterns.

Bodily-kinaesthetic intelligence: Bodily intelligence refers to the individual's ability to make use of his body structure and muscles in a coordinated manner to facilitate movement and to do various tasks in a routine manner.

Interpersonal intelligence: Interpersonal intelligence is the ability of the individual to relate with other people and understand the behaviour, attitudes and emotions of other people. This type of intelligence enables a person to find out why the other person behaves in a specific manner.

Intrapersonal intelligence: Intrapersonal intelligence is the individual's ability to understand his inner self. It enables a person to identify his strengths and weaknesses so that the person can relate with others on this basis.

Naturalistic intelligence: Naturalistic intelligence refers to the ability of a person to understand the nature and the environment that surrounds the person. It enables a person to identify the beauty of the nature and also makes a person aware of the importance of environment to lead a healthy life.

Existential intelligence: This intelligence refers to the ability of a person to understand the basics of human existence and answer several questions about the existence of mankind.

With the emergence of the concept of multiple intelligences, the intelligence tests today require a person to perform several tasks. There are four basic intelligence tests that are used to measure the intelligence of a person. These are as follows: Binet-Simon scale: The Binet-Simon intelligence test was first devised in

Individual and Group Intelligence Tests

A. Individual Tests

An individual test is one that is administered to one individual at a time. There are many standardised individual tests such as The Kaufman Scales, Stanford-Binet Scale and, Wechsler Intelligence Scales. We will limit our discussion with two most famous intelligence tests i.e., Stanford - Binet Test and Wechsler Intelligence Tests.

Stanford-Binet Scale of Intelligence Binet-Simon intelligence test was the first intelligence test, developed by Binet and Simon (1905). It was one of the popular intelligence tests among psychologists. Later this test was revised and adapted by an American psychologist Lewis M. Terman who was working at Stanford University. After validating it on the American population, he renamed the original scale as "Stanford-Binet Scale". In 2003, the fifth version of the Stanford-Binet Scale, Fifth Edition (SB5) was introduced with 10 subtests measuring following five factors:

Fluid reasoning
Knowledge
Quantitative Reasoning
Visual-Spatial Processing
Working Memory

Other than scores with reference to these five factors, SB5 gives two distinct scores, namely Verbal IQ, and Nonverbal IQ. The instrument has 10 subtests. The SB5 can be used for people within the age range from two years to 85 years old individuals.

The Wechsler scales were developed by Dr. David Wechsler. He developed three scales; for adults, for school-age children, and one for preschool children.

All three of his tests contain several subtests from verbal as well as nonverbal domain and they can measure intelligence and cognitive abilities.

He developed his first test (Wechsler- Bellevue scale) in 1939 when he was working in Bellevue hospital.

He devised a new formula for calculating IQ from his scales. As we know, the usual formula of IQ is, IQ = Mental Age/Chronological Age × 100 According to Wechsler, IQ = Attained or Actual Score / Expected Mean score for Age.

There are three versions of Wechsler's intelligence scale:

i) **WPPSI – Wechsler Pre-School & Primary Scale of Intelligence:** This scale can be administered on children from 2 years and 6 months to 7 years and 7 months.
   It was introduced in 1967 and originally designed for children between 4 years and 6.5 years old.
   It consists of 14 subtests measuring three indices, viz., verbal, performance and full-scale IQ. Currently, it is in fourth revision known as WPPSI-IV.

ii) **WISC – Wechsler Intelligence Scale for Children:** This test can be administered on children from 6 to 16 years old.
   This test was developed from the Wechsler-Bellevue Intelligence Scale and was first introduced in 1949.
   This test is often used in schools and other educational setups with the aim to identify gifted children as well as children with learning difficulties.
   The most recent version of the test is the WISC-V, which was released in 2014.

iii) **WAIS – Wechsler Adult Intelligence Scale:** This test is used for adolescents from 16 years of age through adulthood to measure general intelligence, by administering many subtests. Each of the test is an indicator and estimator of intelligence.

The current version of the test is the WAIS-IV which was launched in 2008.

B) Group Tests

A group test is one that can be administered to more than one person at the same time. Thus, making the tests quick in administration.

There are many intelligence tests which can be considered as group tests such as

Multidimensional Aptitude Battery (MAB; Jackson, 1984),

Cognitive Abilities Test (Lohman & Hagen, 2001),

Culture Fair Intelligence Test (1940) and, Raven's Progressive Matrices (1938,1992). As an example, we will discuss only Raven's Progressive Matrices briefly. Raven's Progressive Matrices (RPM) Raven's Progressive Matrices (RPM) was developed by John C. Raven in 1983. It is a non-verbal test of inductive reasoning, designed to measure Spearman's g factor or general intelligence. It consists of 60 multiple choice items and can be administered on children from 5 years-old to the older adults. The test contains visual geometric designs with a missing piece and the task of the test taker is to choose the missing part of the matrix from six to eight given alternatives. Raven constructed three different forms of tests: Standard Progressive Matrices (SPM), Colored Progressive Matrices (CPM) and, Advanced Progressive Matrices (APM).

SPM is suitable for average individual between the ages of 6 and 80 years. CPM is availabe for younger children and for special groups who cannot be tested on SPM.

APM is available for above average adolescents and adults.

Verbal and Nonverbal Tests

A) **Verbal Tests** Verbal intelligence is the ability to use and solve problems using language-based reasoning. Verbal tests are those which require the use of language for successful performance in it. Verbal intelligence is the ability to comprehend and solve languagebased problems. Initially, approximately all intelligence tests were based on language only but later it was realised that such tests are of no use for people who were illiterate, young children who have not acquired the language abilities fully and people with speech difficulties. To overcome the limitation of these verbal tests, many psychologists came up with a number of non-verbal intelligence tests. Moreover, many verbal standardised tests such as Wechsler scales and Kaufman scales now also have some non-verbal test components.

B) **Non-Verbal Tests** A nonverbal test of intelligence measures one's ability to analyze visual information and solve problems without necessarily using words. Nonverbal tests are also known as performance tests as they generally require a construction of certain patterns.Some of the famous nonverbal tests are Koh's Block Design Test, Cube Construction Tests, and Pass along Tests. Raven's Progressive Matrices (1938, 1986, 1992, 1995) is also a well-known nonverbal intelligence test which has been discussed in the previous section.Picture making test, Part missing test.

Aptitude tests

In general words Aptitude is the ability/capability or talent of a person either inner or outer or both to perform any work or duty. It is directly related to the mind and the creativity that person use while doing any task. The tests which are conducted to measure or judge these capabilities and talents are known as Aptitude tests.

Types of Aptitude tests

Aptitude tests are broadly divided into three categories.

1. General Aptitude test
2. Special Aptitude test
3. Manual Dexterity Aptitude test

General Aptitude test:

These are most common types of tests majorly conducted for checking different reasoning abilities, Numerical abilities, language, spelling etc.

THEY ARE FURTHER DIVIDED INTO 2 TWO CATEGORIES

A. General Aptitude test battery

B. Differential Aptitude test (DAT)

A. General Aptitude test Battery

It is developed in U.S.A for employment service bureau. It have total 12 tests 8 were paper/pencil based and 4 were activity based.

B. Differential Aptitude test (DAT)

It is developed by psychological corporation of U.S.A. It consists of some tests i.e.

- Verbal reasoning
- Numerical Ability
- Abstract Reasoning
- Space Relations
- Mechanical Reasoning
- Clerical Speed and Accuracy
- Language Usage, Grammar and Spelling

2. Spatial Aptitude Test

These tests are related to judge the Aesthetic and musical sense of an individual. They are also divided into four categories.

A. Mechanical Aptitude Test

B. Clerical Aptitude Test.

C. Aesthetic Aptitude Test

D. Musical Aptitude test.

A. Mechanical Aptitude Test

These tests involve other tests like

a) Performance Test: These tests Include Assembling of things.

a) Paper pencil Test: These tests includes Question & answers.

B. Clerical Aptitude Test.

This test is also known as Minnesota Clerical Test. Meier art judgment test is an example of CAT. This test has two parts.

Part 1: It contains 200 pair of numbers containing of 3-12 digits.

Part2: It contains 200 pair of names to be remind.

C. Aesthetic Aptitude Test

This test is related to how a person is appreciating the beauty. It contains a 100 pair of pictures and the subject should choose the best pictures from them.

D. Musical Aptitude test.

This tests helps in determining the sense of hearing that how the subject is detecting the different sounds. It contains 6 tests for different aspect of the music.

- Pitch
- Rhythm
- Loudness
- Time
- Tone
- Timbre

3. Manual Dexterity Aptitude test

These tests are manual activity-based tests. They are further classified into two categories.

A. Minnesota Manual Dexterity Test

B. O' Conner Finger & Tweezers dexterity Test.

A. Minnesota Manual Dexterity Test

In this test the subject has to arrange a list in the same order after observing it carefully.

B. O' Conner Finger & Tweezers dexterity Test

In this test there are small holes on the board and all the wholes should be filled by pins. This test helps in determining the coordination of fingers and hand, wrist and steadiness.

Tests of Interest

Interest tests are tests that assess the interests of the individual. It has been proved that interests are related to academic success and also job satisfaction. Therefore, it is essential to know the interests of the individual or a student. Interests basically refer to activities that a person enjoys and likes to indulge in. Interest tests are classified into formal assessment techniques and informal assessment techniques.

Formal assessment techniques: Interest inventories are the formal tools that are used to identify the interests of the individual. The Strong Vocational Interest Bank is an interest inventory that is based on the assumption that the individual with interest in an occupation will find satisfaction and will also enjoy success in that specific occupation. In this interest inventory, a test taker has to differentiate between different activities as L(Like), I (Indifferent) and D (Dislike). Thus, this interest inventory is termed as category-response type assessment. Kuder Preference Record is an interest inventory developed by G. Frederick Kuder, a pioneer in vocational guidance. The Kuder Preference Record enables one to assess preferences for

specific activities. Each activity listed in the records has three choices and the test taker has to select one of the three choices as the first choice and another as the third choice. This type of assessment is also called forced choice pattern type as the test taker is required to specify which activity he likes the best and which he likes the least and the reason for disliking that activity. The most common type of interest test that is conducted is career test. This test helps people find careers that match their interests. The career test typically asks a few questions about the career interests of the individual and the results of the test point out, as to whether the career is right for the person or not. The test is based on the premise that a person should opt for a career that interests him because only then can one enjoy the work and achieve success in that field. Interest tests are typically carried out for the following reasons:

This test provides teachers, parents and counsellors information related to the preference and aversions of the students so that they can be understood better. It enables the parents, teachers and counsellors to assist the student in a better manner in terms of his educational and vocational interests. This test also helps in selecting the right person for the right task and, hence, increasing the productivity.

Interest tests simply show what interests a person and not the strengths of a person. As such, interest test results should not be used alone to decide which vocation or educational field is best suited for a person. Interest tests on their own have very less significance and so their results must be combined with other tests like achievement and aptitude tests to help a person to make the right educational and career choice.

HOPE test of interests

Online HOPE Interest Tests are field-tested, reliable and valid. These tests have been developed by a trained psychometrician of Boston University. These tests are duly copyrighted with the Government of India. These tests are now available online, not only in English but also in Hindi and Tamil and can be administered anywhere in the world. Students, therefore, have the option to take the online tests in their preferred language. These tests are designed for individual students who want to know about their level of intelligence, vocational aptitude and personality. These tests are specially recommended for students in the X class to help them select proper course groups in XI class and for students in the XII class to select appropriate courses in higher studies. The tests are also recommended for undergraduate and graduate students and students pursuing professional degrees. These tests help these students in identifying careers in which they will be successful. These tests help educational and vocational institutions, to identify careers/vocations in which the students would excel.

Lesson-16 The semantic differential technique-Attitude scales- Personality Testing- MMPI.

Attitude Tests

Attitude tests are the tests that assess the attitudes of people with respect to other people and society. These tests assess how well the individual adjusts to the social norms and how he manages interpersonal relations. Basically, attitude is an expression of how much one likes or dislikes a thing or a person. Based on the attitude, a person approaches things or people he likes and rejects things or people that are not liked. Attitudes are basically learnt and adopted and

are not natural capacities of a person. To test the attitude of a person, various scaling techniques have been developed. One such technique is known as equal-appearing intervals. This method was developed by Thurstone and Chave (scaling theorists). The scale consists of several statements. The score of a person is calculated based on the psychological continuum of the scale value. If the score falls in the middle range of the psychological continuum, the person's attitude to a statement and, thus, a concept is regarded as neutral. If the score falls towards the favourable end of the psychological continuum, the attitude is considered favourable and unfavourable otherwise.

The other method is that of summative ratings. As per this method, each statement has arbitrary weights associated with it, which are as follows:

- 5 strongly agree (SA)
- 4 agree (A)
- 3 undecided (U)
- 2 disagree (D)
- 1 strongly disagree (SD)

The individual who shows favourable attitude for the statement uses this weight series. On the other hand, the weights 1, 2, 3, 4, and 5 are used by people who oppose the statement or who show unfavourable attitude. The score is then determined by summing up the ratings of all items. Attitude tests are used to assess the level of adjustment of the individual with the people around him. These tests are also used to assess how good a person is at developing and maintaining interpersonal relationships.

Personality Tests

Personality can be defined as a unique pattern of traits that the individual possesses. It is these unique traits that distinguish the person from others and provide a consistent way in which the person interacts with the environment. Personality is the sum total of emotional, motivational, interpersonal, attitudinal and even moral aspects of a person. Personality tests are tests which are conducted to assess the personality of the individual. It is important to conduct personality tests due to the following reasons:

Personality plays a vital role in determining the vocational and educational success of a person. It is, therefore, essential to know the personality traits so that the individual can make educational and occupational adjustments as required.

Personality tests help the individual to overcome emotional conflicts. When personality tests are conducted, the individual gets to know what traits are becoming a problem in managing emotional issues and, so can work on improving them thereby resolving emotional issues.

Personality tests are also important for social counselling.

There are several tools and techniques that are used for testing the personality. Some of them are as follows:

Self-report measures: Self-report measures are personality tests that require the individual to answer a series of questions regarding his characteristic behaviour. During these tests, a person answers several questions about himself in a true/ false manner. These tests can be used to measure several personality traits including attitude, emotions, feelings and so on.

Projective techniques: The projective techniques help to focus on the personality of the individual as a whole. When taking these tests, a person needs to project his personality in the

test exercise. The idea behind projection of personality in these tests is to unearth the inner motives and personality traits that are otherwise hidden. These tests, therefore, make the individual aware of whatever he does not know about himself.

To take the test, the individual has to respond to several stimuli like a series of pictures, ink-blots, or other such stimuli. The interpretation of such results is carried out by professionals who are trained to do so. The interpretation also takes a long time.

Behavioural assessment: Behavioural assessment is a personality test tool that assesses the way the individual behaves under certain conditions. This behaviour is an indicator of the personality of a person.

Advantage of personality tests:

- These tests help to understand the individual and acquire information about him in detail.
- The disadvantages of personality tests include the following:
- Personality tests are not based on adequate norms.
- The information and results of the test can be interpreted only by skilled and trained professionals.

Personality questionnaires are used to assess the personal behavioural preferences. The questionnaire helps a person to know how he likes to work. The test is not aimed at assessing the abilities of a person but the aim of the test is to enable a person to see how he relates to others, when it comes to solving problems and dealing with emotions and feelings. In a personality questionnaire, there are no right or wrong answers. The only thing he has to do is to choose the option that best describes the person's feeling or his liking and disliking for a particular thing or concept. Personality questionnaires can be in two formats. In one type, a person is required to rate himself and in the other format, the person is required to choose statements that he feels relate to his personality.

Some well-known personality questionnaires are: 1) Minnesota multiphasic personality inventory (MMPI):It is the most widely used personality inventory for both clinical and non-clinical populations and commonly used to help with the diagnosis of personality disorder.

It was first published in 1943 with 504 true/false questions, An updated version including 567 questions was released in 1989,and is known as MMPI-2.A version for adolescents MMPI-A was published in 1992, An alternative version of the test, the MMPI-2 restructured form MMPI-2-RF, published in 2008.This test takes approximately 60 to 90 minute to complete. This test is useful in measuring the anxiety, hostility, hallucinations, phobias and suicidal impulses. Responses are scored to produce a clinical profile composed of 10 scales: Hypochondriasis, Depression, Hysteria, Psychopathic Deviance, Masculinity versus femininity, Paranoia, Psychathenia, Schizophrenia, Hypomania and social Introversion. It is used for occupational screening for career like law enforcement, college and marital counselling.

NEO PI-R:

It is psychological personality inventory with240 items used to measure the big 5 personality traits: Neuroticism Extraversion Openness to experience Agreeableness and Conscientiousness. It has been widely used in research.

THE 16 PF: It measures the 16 sources traits as outlined by Cattell. It has been translated into over 20 languages): It yields scores on 16 primary traits such as social boldness, dominance, vigilance, emotional stability and rule consciousness. It is used for selection, recruitment, marital counselling and for therapy planning. 4) The Eysenck Personality Questionnaire (Epq). It measures the 3 major dimensions: Neuroticism, extraversion, and Psychoticism. It used as a clinical test and research.

Lesson-17 Projective Techniques -TAT, Rorschach etc.

Projective techniques: The projective techniques help to focus on the personality of the individual as a whole. When taking these tests, a person needs to project his personality in the test exercise. The idea behind projection of personality in these tests is to unearth the inner motives and personality traits that are otherwise hidden. These tests, therefore, make the individual aware of whatever he does not know about himself.

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RORSCHACH INK BLOT TEST:

- It was developed by Herman Rorschach in 1920.
- This test contains 10 cards with ink blots.
- The subject is asked to look at the cards and say what it reminds him or her of.
- The responses are interpreted based on a scoring scheme and the structure of the personality is analyzed.
- used for Degree of intelligence, emotional aspect, personality.

THEMATIC APPRECEPTION TEST (TAT)

- In this test cards with pictures of human beings are shown one by one.
- The person being tested is asked to write or tell stories about "what is happening in the picture?" "what are these people thinking and how they are feeling" "How does it turn out in the end" .
- TAT brings out the dynamic of personality.
- Areas of TAT are family relationships, motivation, inner fantasies, Level of aspiration, social relationship etc
- used to assess the locus of problems, nature of needs, quality of interpersonal relationship.
- It requires the examiner to call out a word to which the subject respond with the first word that comes to his mind.
- Reaction time and any other behavioural responses, as well as the response, are noted down.

Lesson-18 Culture and Psychological tests.

Culture-Fair Tests and Culture-Biased Tests

Culture-Fair: Tests Every culture is unique in terms of their values, language, expectations, demands and environmental experiences.

A child reared in the US will be very different in many respects with a child been brought-up in Indian sub-urban area. Due to this reason, for assessing individual belonging to different cultures, psychologists came up with tests which are free from any cultural biases.

Some of the famous culturefair tests are The Culture Fair Test (Cattell, 1940), Raven's Progressive Matrices (Raven, 1938, 1986, 1995), The Leiter International Performance Scale-Revised (Roid& Miller, 1997) and, Draw-a-Man Test (Goodenough, 1926). All these and other culturally fair tests are non-verbal in nature. Now, we will discuss an example of culture-fair intelligence test-Draw-a-Man test (Goodenough, 1926).

Draw-a-Man Test This test was initially developed by Goodenough (1926). Later, it was revised by Goodenough and Harris in 1963, known as Goodenough-Harris Drawing Test. Based on the projective technique, this test requires a test taker (children only) to make three pictures on three separate papers. They are asked to draw a man, women and themselves without giving any further instructions. Interestingly, instead of artistic skill, emphasis is given on the child's ability to observe accurately and think conceptually.

Cultural-Biased Tests: Many psychologists have attempted to develop culture-fair intelligence tests by making it non-verbal in nature. However, it was realized that the impact of culture cannot be eliminated completely from these tests even after making it nonverbal completely. Due to this reason, only the term 'culture fair' is used in place of 'culture free' tests.

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: To estimate the extent of distortion in perceived length of line of standard length inMuller-Layer illusion by using the Method of Average Error.

Activity 2: To learn to use the method of Minimal Changes to determine the Difference Limen(DL) and to estimate Weber's ratio for the given visual stimulus

PSY15145 ENVIRONMENTAL PSYCHOLOGY

Objectives:

To improve the relationship between mankind and the environment using theory, research and practice. It also endeavors to find ways to improve our relationship with the surrounding world. To discuss both theoretical and applied perspectives of environmental issues to help reduce stress, pollution and crowding concerns; To provide knowledge and understanding of well-established theories in environmental psychology.

Unit – I: Introduction

Lesson-1 Introduction -Nature, scope, Origin of environmental psychology;

Lesson-2 Fields of Environmental psychology- Relationship with other fields-

Lesson-3 Basic principles of environmental psychology-

Lesson-4 Major research methods in environmental psychology-Questionnaire studies, Field studies, Case studies.

Unit – II: Environmental Risk and Perception

Lesson-5 Natural Disasters: Characteristics of Natural Disasters, Perception and Psychological Effects of Natural Disasters;

Lesson-6 Noise and Air Pollution: Source and effects of Noise in the Work Place, Noise and Social Behaviour.

Lesson-7 Perception, Health, Performance.

Lesson-8 Social Behaviour and Air Pollution.

Lesson-9 Environmental Perception and Cognition; Environmental Cognition, Cognitive Maps, Way finding.

Unit – III: Environment and Behaviour- Theories

Lesson-10 Changing Behaviour to save the Environment: The Commons Dilemma as an Environment-Behaviour Problem.

Lesson-11 Strategies to Encourage Environmentally Responsible Behavior-Contemporary Values and Environmental Attitudes.

Lesson-12 Functions of theories; the arousal perspective: Environmental load perspective; Adaptation level theory;

Lesson-13 Environmental stress perspective.

Unit – IV: Psycho-educational aspects of Environmental Protection.

Lesson-14 Environmental education and laws of environmental Protection

Lesson-15 Disaster Management-Social Forestry and Ecological Aesthetics

Lesson-16 Managing scarce resources.

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Outcomes:

- To improve the relationship between mankind and the environment using theory, research and practice. It also endeavors to find ways to improve our relationship with the surrounding world.
- To discuss both theoretical and applied perspectives of environmental issues to help reduce stress, pollution and crowding concerns.
- To provide knowledge and understanding of well-established theories in environmental psychology

Unit – I: Introduction

Lesson-1 Introduction -Nature, scope, Origin of environmental psychology

Introduction:
Environmental psychology is the interdisciplinary field of study that is concerned with the interplay between individuals and their surroundings. Environment plays a major role in the well-being and development of mental faculty of human beings. Environmental psychology also deals with socio-architecture, ecological psychology, Eco psychology, behavioural geography, person environment studies, social ecology, and environmental design research. This unit will discuss the scope of environmental psychology and the role played by environmental psychologists. The psychological benefits of nature will also be discussed.

NATURE AND SCOPE OF ENVIRONMENT PSYCHOLOGY
Environment is derived from the French word 'Environ' which means 'around', 'round about', 'to surround', and 'to encompass'. Environment includes land, water, vegetation, air and social system. It also includes the physical and ecological environment.
Environment is thus a broad concept which includes a wide range of surroundings in which human beings perceive, experience and react to events and changes. It is quite dynamic in nature and is continuously changing with or without any human interference. Despite these changes, there is a dynamic equilibrium in the environment of a specific region. This is called balance of nature.

Environment mainly comprises the natural environment (such as places, geographical features like rivers, mountains, valleys; environmental conditions like temperature, rainfall, flora and fauna) and the built up environment (such as houses, buildings, cities, communities, etc.). Our environment encompasses a wide range of diverse surroundings in which humans experience and react to various events.
A close relationship between physical and socio-cultural environment and human being's psychological processes is seen. The environmental factors have a significant effect on both mental and behavioural activities of human beings. Mental activities refer to the way humans perceive, think, reason, attribute, believe, interpret and develop attitudes about different aspects of their environment. On the other hand, behavioural activities refer to the way human beings respond or react or behave in their environment based on their perceptions, thinking, attitudes and beliefs.
The social, economic, psychological, mental and physical well-being of a human being is also greatly affected by occurrence of different environmental hazards such as cyclones, floods, draughts, tsunami, etc.
According to Garling and Golledge (1993), and Kaplan and Kaplan (1982), environment can have a significant effect on our overall functioning.
1. **Attention:** For understanding human behaviour, we need to first understand how humans notice and perceive environment. The key component while maintaining human effectiveness in environment is to re-establish and enhance an individual's competence to willingly express their attention.
2. **Perception and cognitive maps:** How individuals perceive the nature and built of environment has been an important aspect of environmental psychology. Cognitive maps are spatial networks through which we memorise information in our brain. This information links experiences with an individual's perception of current actions, ideas and emotions. Humans recognize and perceive environment, plan and conduct these plans through these spatial networks.
3. **Ideal environments:** Everyone has a tendency to look for a place, where they can feel self-assured and competent. Humans thus tend to look for a place which seems familiar to them and where they can engage with their surroundings. Researchers have extended the concept of environmental psychology to embrace unity and legibility. This acts as a contributor to environmental understanding. To be able to better engage and investigate within an environment, the environment should have some degree of complexity (enough information and diversity to make it worth learning about) and mystery (expectation of acquiring more information about an environment) attached to it. After all, the sense of human wellbeing and behavioural effectiveness is enhanced by maintaining, reestablishing and developing an ideal environment.
4.**Environmental stress and managing:** Environment in which we live can have a significant influence on our behaviours and cognitions. An unhealthy environment can lead to poor physical health, reduced selflessness and weaknesses. One needs to thus make necessary adjustments in our physical and social surroundings by establishing our privacy, having a sense of proper personal space etc., so that we are able to create a more supportive environment for ourselves. Also, each environment inherits a certain degree of stressors and only when we can make a better sense of these

stressors and the circumstances around can we reduce their stressful effect on our psyche and existence.
5. **Involvement:** The main aim of Environmental psychology is to improve human participation in environmental design. It focuses on promoting and ensuring an individual understands of environmental issues. They also focus on human-environment participation, adjustment and organization.
6. **Protective behaviour:** Environmental psychology also aims to develop an ecologically protracting society by promoting environmentally appropriate behaviour. It also investigates environmental attitudes, perceptions and principles that people carry.
Environmental Psychology Environmental psychology studies the intricate and complex relationship humans share with their natural environment. It basically studies the impact environment can have on human experiences, behaviour, well-being, and the reciprocal influence individuals can have on their environment. It also studies the various factors that influence environmental behaviour, and looks into the key ways to encourage proenvironmental behaviour. It for this reason that environmental psychology is also referred to as 'human factor science'. It also deals with socio-architecture, ecological psychology, Eco psychology, behavioural geography, person environment studies, social ecology, and environmental design research.

The keys aspects of environmental psychology are as follows: 1) Socio-architecture: The term socio-architecture was coined by psychologist Humphry Osmond and architect Kyo Izumi. They were involved in the research to find the best architectural form for the mental hospital, the OSMONDS Weyburn in 1951. Over the years, Osmond became one of the best research centres for the treatment of schizophrenia, and focused on gaining insight into mystical experiences one may have when experimenting with psychedelic drugs. Whereas, Weyburn hospital became a design and research lab to study the functional aspects of the architecture and the impact it can have on people with mental illness. Osmond based his ideas of hospital design on the species-habitat work of German zoologist Fritz Hediger.

To describe the stimulating or limiting effect seating can have on social interactions, Osmond coined the terms 'sociopetal' and 'sociofugal'. His work on architecture was continued by his colleague Robert Sommer.

2. Ecological Psychology: Ecological psychology is the scientific study of perception-action from a non-functionalism approach. Ecological psychology is largely based on the works of Roger Barker and James J. Gibson, who tried to study the relationship between perception, action, and dynamical systems and largely believed that perception and action are inseparable.
Roger Barker argued that one cannot make accurate predictions about human behaviour unless one knows the situation or context or environment in which that human being lives. Human behaviour can be very different in different contexts; for instance, the behaviour that may seem appropriate in the setting of church may be quite different from behaviour seen when working in a factory. Also, the behaviour of different people in the same setting may be quite similar. James J Gibson stated that perception of the environment also influences the way humans behave in that setting. He even believed that perception is based more on information than sensation.

3) **Eco Psychology:** The term 'eco psychology' was coined by Theodore Roszak, in his book The Voice of the Earth in 1992. It is a therapeutic technique and ideology that tries to treat people psychologically by bringing them spiritually closer to nature as it strongly believes that human beings are born with an instinct to emotionally connect with nature instantly. It is based on the idea that although mind is shaped by the modern world but its underlying structure is created in a natural non-human environment. One of its primary goal is to expand and remedy the emotional connection between humans and nature, and promote sustainability.
Roszak also believed that connection of human beings to nature can improve their interpersonal relationships and emotional wellbeing. Therefore, he believed that patients can improve better when treated outdoors amidst nature. He also claimed that a bad connection of
human beings with their environment seems to lie behind psychological problems. The more we become disconnected from nature, the more we become negative and our risk of developing pathology increases.
4) **Behavioural Geography:** Behavioural Geography focuses on the cognitive processes, spatial reasoning, decision making and behaviour to understand the role played by the cognitive processes in an individual's perception of or response and reaction to their environment, using the behavioural approach. It tries to study the relationship between environmental perception and cognition and examines topics like the construction of cognitive maps, attachment of a place or an environment,

the development of attitudes about space and place, human decisions and behaviours based on imperfect knowledge of one's environment, etc. It studies the mental maps of the individual concerning his/her environment. It researches on environmental values, meanings and preferences. Behaviour maps are prepared for exploratory behaviour, neighbourhood feelings, etc.

5**) Person Environment studies:** It makes use of psychological science to improve the interactions of people with the world around us. They study human responses to natural and technological hazards and examine the influence of different environments, such as offices, homes and urban areas, on loneliness and stress, for example. It tries to answer questions like the effect of sun on our mood or the effect of colours of the walls of your office on productivity of the employees.

6) **Social Ecology**: It studies the relationship between people and their environment and aims to study the interdependence of people, collectives and institutions. It also aims to study the social structure and organization as influenced by an organism's environment.

7) **Environmental research design:** It is a field that tries to study the interrelationship between environments, society, culture, and human behavior. The knowledge so obtained is used by architects and designers to make houses, offices and buildings.

Different perspectives of Environmental Psychology Environmental psychology consists of many theories about how and why we act the way we do in our environment, but they tend to fall into one of the following key perspectives:

(i) Geographical Determinism
(ii) Ecological Biology
(iii) Behaviourism
(iv) Gestalt Psychology.
(v) Ecological theory
(vi) Behavioural constraint theory
(vii) Adaptation level theory
(viii) Arousal theory
(ix) Environmental stress theory
(x) Environmental load theory.

Let us discuss these perspectives in detail.

(i) **Geographical Determinism**: It tries to study how geographic and ecological forces influence state-building, economic development and institutions. Theorists believe that too many environmental challenges can lead to destruction of civilizations and very little challenge can result in a stagnation of culture. These environmental factors have a huge impact on what we know about our societies and the way individuals live and work together.

(ii)**Ecological biology:** Ecological biology relies on theories of biological and sociological interdependence between organisms and their environment. It studies how organisms interact with one another and with their physical environment at many levels such as organism, population, community, ecosystem, and biosphere level. It assumes that the presence of both living and non-living factors together shape and determine the population of organisms at a place. Organismal ecologists study morphological, physiological, or behavioral adaptations that makes it possible for an organism to live in specific habitats. They define population as a group of same species that live in the same area at the same time and they are largely interested in the study of size, density, and structure of populations and how they change over time. Community ecologists are more interested in studying the way different populations interact with each other and its effect on the community. Ecosystem ecologists try to focus on flow of energy and recycling of nutrients among different species that live together at a place. But, ecologists working at the biosphere level are more interested in the study of how different ecosystems interact with each other and affect the entire globe in ways such as bringing a change in the climate etc

(iii) **Behaviourism:** It tries to study the effect of environment on person's personality, disposition, attitude, views, and experience.

(iv) Gestalt psychologists are more interested in studying how humans perceive, think and make sense of the environment around them.

(v) Ecological theorists believe that there is an inseparable connection between the person and the environment in which they exist. They believe that the two mutually benefit each other and share a symbiotic relationship. Roger Barker studied the various ways in which the number and variety of behavior settings remained remarkably the same even as institutions increased in size. For example, no major difference in the behaviour of students was seen whether they studied in a large or a small school.

(vi) **Behavioural Constraint Theory:** This theory states that when particular aspects of our environment are perceived to be threatening in nature, we react in certain ways to either escape or cope with the stressor at hand. We continue to behave in this way irrespective of whether the stressor

is real or perceived. We do so, as it helps us to prepare in advance for certain stressors that we may have to face in future, thus enhancing our coping capacities and survival. When we are unable to cope with a stressor, it results in learned helplessness, which may make one feel powerful and thus reduce the motivation to take some active steps to better deal with the situation at hand.

(vii) **Adaptation Level Theory:** This theory postulates that all human beings tend to adapt to a certain level of stimulation present in their environment, which allows human beings to function at their optimum level. It also believed that too much or very little stimulation from the environment can have an undesirable effect. For instance, an ideal teacher-student ratio is 1:25. A class that has very few children may seem dull and a class that is too crowded may add to distraction level of each child and hence both the conditions are likely to have an undesirable effect. Similarly, being with few people in life may add to our loneliness and being with many people may make the place crowded and add to our irritability and anger.

(viii) **Arousal Theory:** This theory states that our behaviour and experiences get significantly affected and influenced by the degree, manner and the extent to which we feel physiologically aroused by various stimuli in our environment. Human beings are likely to perform the best, when they are optimally aroused. This physiological arousal can have a significant effect on our behaviours. For instance, people who tend to engage in high risk behaviours have a higher optimum level of arousal compared to others. Similarly, people who tend to get satisfied easily and have a lesser need for excitement tend to have lower optimum level of arousal compared to others. In addition, the optimum level of each individual is not a static figure as it may vary depending on the task at hand. For instance, we need high optimum level of arousal to perform a simple task but a low level of arousal to perform more complex and difficult tasks. For example, a high level of arousal in our exams, can add to anxiety and impair our performance, whereas a small level of arousal may not motivate us to study hard for our exams.

(iv) **Environmental Stress Theory:**This theory proposes that humans respond to environmental stimuli based on their cognitive and autonomic evaluation of the stressor. It is these evaluations that help us determine whether an environmental stimuli is stressful or not for us. If a stimulus is perceived as threatening in nature, it results in a stressful reaction. If we perceive a stranger standing at the corner of a street at night alone as a burglar, we are likely to react in a stressful manner by either avoiding going through that street or running back home as quickly as possible.

(x)**Environmental Load Theory:** This theory states that human beings have a certain capacity to deal with a specific amount of information at a given point in time. To deal with the problem of information over load, human beings try to focus on relevant information and tend to ignore other pieces of information, as this allows them to process this information better. That is why, when we are engaged in an interesting interaction with someone, we tend to remember the conversation well, but may not remember the colour of his dress or the brand of his shoes.

Characteristics of environmental psychology

Environmental psychology, according to Gifford (2007), can be defined as the study of transactions between individuals and their physical settings. Some of its main characteristics are as follows:

(i) A holistic molar perspective: This means that environmental psychology examines behaviour in its respective context and tries to not only study complex psychological processes and environmental factors but also the dynamic relationship between them.

(ii) An applied problem-solving perspective:Environmental psychology tries to study the basic principles of behaviour in order to find out solutions of social problems involving the physical environment.

(iii) A broad and eclectic methodology: It refers to various methods environmental psychologists use to study this complex multi-factorial relationship between human beings and the environment in which they live such as laboratory experiments, field experiments, survey studies and natural observations, etc. (iv)

(iv) A wide range of levels of analysis: Environmental psychologists do Micronveleles, which is the study of the effect of noise on the performance analysis at various levels such as Large units of study such as the design of communities and cities.

- Moderate scales of analysis of the design and domestic use; of a task; Important psychological processes like perception of the environment,
- They also study the following:

spatial cognition, and personality as they filter and structure human experience and behaviour, and The management of social space: personal space, territoriality, crowding.

(v) wide range of theoretical approaches: Environmental psychologists borrow different theoretical ideas from other areas of psychology and other social sciences rather than developing their own theories to study human environment interactions. Barker was the only environmental psychologist who developed his theory of ecological psychology.

Goals of environmental psychology

To use this knowledge to solve several problems faced by humans in the

- To study human-environmental interactions
- The key goals of environmental psychology are as follows: To create, manage, protect, and restore environments that promote proper environment in which they live behavior.

Lesson-2 Fields of Environmental psychology- Relationship with other fields

Relationship of Environmental Psychology to other Disciplines

Environmental Psychology and Architectural Psychology

There is relationship between environmental psychology and architectural psychology in many respects. The attempt by Ittelson and Proshansky's group in the United States studied the relationship between architectural design and the behaviour of the patients in psychiatry department. At the same time, other psychologists and psychiatrists were involved in similar research projects both in the United States and in other countries. Besides this: Sommer developed the concepts of human "territoriality" and "personal space" destined to find a remarkable following in psychology in general and environmental psychology in particular and in the sciences of architectural design.

Environmental Psychology and Biological Sciences

In 1956, the Research Commission of the American lnsti,tute of Architects (AIA) presented a proposal to the NationalFscience Foundation (NSF) to hold a conference, which took place in 1959 at the University of Michigan, the main topic was the relationship between the physical, biological and social sciences with regard to the problems of creating optimal environments of human activities. Prior to this, at the 1958 convention of the AIA in Cleveland, a preliminary seminar was held on the topic, in which the participants included architects, civil engineers, urban planners, psychologists and sociologists. Other disciplines outside of psychology, which contributed greatly towards the emergence of environmental psychology, are represented by the sciences of the physical geographical environment and those of the ecological naturalistic one, by the end of the 1960's they showed a growing "human" or "anthropic" factor. Each man's perceptions are his own, unique and personal.

Perception is the process by which a particular person, from his particular behaviowl centre, attributes significance to his immediate environmental situation. So, it can be said that there exists relationship between environmental psychology and other disciplines in many respects. One can easily observe it in one's daily life.

Environmental Psychology and General Psychology Environmental Psychology emphasises on the studies of the effect of physical environment such as: the heat, noise, crowding, upon behavior, feeling and health General Psychology is the broad branch of psychology which seeks to discover the laws and principles that apply to persons in general, which is opposed to differential psychology which finds the way in which people differ.

Enviromental Psychology and Human Engineering

Human engineering is a multidisciplinary field incorporating contributions from various fields including psychology, engineering, industrial design and anthropometry. It deals with the properties of human capability. The human factor in human engineering is a physical property of the individual which is specific to the person and influences the human environment equilibriums. Human factors refer to all aspects of the ways in which humans relate to the world around them with the aim to improving operational performance , safety etc . In one way the human engineering contributes to making the environemtn for the humans better. Thus it contributes to the tasks of environmental psychology.

Environmental Psychology and Applied Psychology

Applied Environmental psychology aims at better management of the environment for better life and psychological growth. It studies effective ways of conserving the natural environment, better ways of designing towns and cities and means of promoting environmental awareness among people. Psychology has a great deal of application in town planning. Studies on how the community works, the psychological needs of the people and their likes and dislikes should be considered while planning the growth of towns. Since the environment shapes and limits behaviour, proper planning to ensure maximum satisfaction, efficiency and growth is essential.

Environmental Psychology and Community Psychology

Community psychology focuses on the influence of the social environment, whereas environmental psychology focuses on the influence of the physical environment. The overlap of community and

environmental psychology involves the study of social processes in physical settings the psychology of social settings like the home, the workplace, and the school.

Lesson-3 Basic principles of environmental psychology

SALIENT FEATURES

Environmental psychology studies the interactions between people and the physicalenvironment as well as the built environment. Environmental psychologists rather thandefining discipline have been in charge of describing some of its main features:

1. A holistic molar perspective: The researcher examines behaviour in its contextor surroundings and tries to delve into complex psychological processes andenvironmental factors.
2. An applied problem-solving perspective: here the research is designed todiscuss the basic principles of behaviour and find solutions to social problemsinvolving a physical environment.
3. A broad and eclectic methodology: environmental psychology uses laboratory experiments, field experiments, surveys research, observations, on diverseenvironments, and diverse methodology to study problems.
4. A wide range of levels of analysis: It employs micro and macro-level approaches to study human problems concerning everyday life problems. It applies a 360-degree approach.
5. Environmental psychology is a multi-disciplinary subject that has adapted theoretical ideas not only from core psychology but also other social sciences in developing its theories.

Bell et. al has precisely stated that "environmental psychology is characterized by the following:

1. Study of the environment- behaviour, relationships as a unit
2. Study of interrelationships of environment and behaviour.
3. A relative lack of distinction between applied and theoretical research.
4. An interdisciplinary and international appeal.
5. An eclectic methodology (a rich mixture of methods).

Characteristic Features of Environmental Psychology

The four key features that characterize the field of environmental psychology are brieflydescribed below (Steg and Groot, 2019):

- **Interactive Approach** Environmental psychology is mainly interested in theinteractions between people and their built and natural environment, it alsotakes into account the way the environment influences behaviour as well aswhich factors have an impact on human behaviour that can be helpful inimproving the quality of the environment.
- **Interdisciplinary Approach** Environmental psychologists often work ininterdisciplinary settings and collaborate with other experts from other disciplines as this can help them in having a more holistic approach. Each andevery discipline brings in a different point of view to the given phenomenon,when all perspectives are integrated, it helps in providing a comprehensive scenario to solve the problem at hand and answer questions accordingly.

  Environmental psychologists often work with architects, geography experts,cognitive psychologists, and social psychologists.
- **Problem-focusedApproach** Environmental psychologyintends to solve real-life problems and often carries

out studies in a manner that can be helpful insolving a problem but it also works relentlessly to study the various phenomenon, test theories, and understand and explain various aspects ofhuman behaviour in the context of their environment and their interaction withthe environment. The approach can be centered on the immediate surroundingof an individual and can be extended to a global level as well to deal withissues like global warming, climate change, ecological restoration, etc.

- **Diversity of Methods Approach** Environmental psychology also utilizes qualitative and quantitative research methodologies as used in other disciplinesof psychology. The distinctive feature of this field, in particular, is the way ituses various methods that are very diverse in nature and work through theproblem or research work. The optimal method is often chosen by weighingthe strengths and weaknesses of a particular type of method and then takinginternal and external validity into consideration.

Lesson-4 Major research methods in environmental psychology-Questionnaire studies, Field studies, Case studies.

RECENT TRENDS AND FUTURE DIRECTIONSIN ENVIRONMENTAL PSYCHOLOGY

Environmental issues such as climate change, pollution, and deforestation have beenrecognized as serious difficulties impacting people's health, economic prospects, andfood and water supplies since the beginning of the twenty-first century (IPCC 2013).It is also widely acknowledged that human behaviour is a major contributor to theseenvironmental issues. Finding strategies to influence people's behaviour to reverse environmental problems while preserving human well-being and quality of life is an ongoing and developing concern of environmental psychology. To this goal, a comprehensive concept of sustainability has been widely adopted, which incorporates environmental, social, and economic factors (World Commission on Environment andDevelopment 1987). This broad concept of sustainability has grown in importance asa guiding and unifying element for environmental psychology research (Giuliani andScopelliti, 2009). Indeed, it has been suggested that, over the past decades, the fieldof environmental psychologyhas graduallyevolved into a 'psychologyof sustainability' (Gifford 2007).

Environment influences human behaviour and well-being. Environmental psychology thus studies the environmental risk perceptions and environmental stress. It also studies the positive influence of the natural environment on humans. Researches have been inthis field to create a sustainable environment that supports human well-being andincreases the quality of life.

Environmental psychology also studied factors influencing environmental behaviour like values, social norms, affect, and conflicts between individual and collective interests.An integrative and multi-disciplinary approach is undertaken by environmental psychologists. The best example to depict this is the *Chipko movement*. A social event happened in the western Himalayan region in the 1970s, when a small-scale social invention (the Chipko movement) to rescue the forests spread far and wide.This is regarded as a significant rethinking of the significance of man's relationship withnature. 'Grassroots activists like those seen in India's Chipko Movement have suggested a new alternative ecological approach in which insight into ecosystem interrelationships is gained through folk knowledge rather than scientific observation,' argues Hannigan, a social scientist (Hannigan 1995: 119). Although Chipko's success story has many social dimensions and interpretations, its articulation as a native cognitive model is an insider's socio-cultural fabrication of the wisdom of those involved in the movement (Pirta, 2007). They emulated Gandhi's nonviolent protest, drew on Hindu wisdom,made naturalistic observations, and synthesized data from a variety of sources. However, the leadership and organization of these community natural resource management initiatives are crucial for their success (Kellert et al. 2000).

The other topics of research by environmental psychologists include encouraging pro- environmental behaviour concerning informational strategies, reinforcements, punishments, and persuasive technology and strategies for behaviour change. It also tried to develop various models that integrate interactions between man and environment. The future directions include challenges for environmental psychologists towards a greater engagement that are studied embracingsustainabilityas a central focus of researchdelivering many valuable insights and tools for promoting sustainability at local andglobal levels.

Environmental psychologists are struggling to save the planet from environmental andsocial degradation and increase the well-being of the inhabitants. The further action ofenvironmental psychologists is to increase the political relevance and applicability oftheir work. Finally, the initiative should be taken at the interface of science, politics, and society such as the establishment of international forums and bringing together knowledge and behavioural determinants of global environmental problems (Ehrlichand Kennedy, 2005) Such initiatives can take a significant role in engaging citizens, policymakers, globally to explore the moral elements and the consequences of choicesabout environmental psychologists can engage and contribute to interdisciplinary andtrans-governmental solutions that make a difference from present to future generations.

Environmental psychologists thus, should explore studying and researching oncommunities that are residing in areas that are tremendously growing and dislocated bythe government for various projects and industrial developments

MAJOR RESEARCH METHODS INENVIRONMENTAL PSYCHOLOGY

The main research methods used in environmental psychology-related research arequestionnaire studies, laboratory experiments, simulation experimental methods, fieldstudy methods, and case studies. All the major research methods are discussed belowbriefly:

Questionnaire Studies – These studies intend to provide a description ofbehaviours and also gather information about people's opinions, attitudes, beliefs, and perceptions about various issues. These are also used in order to establish relationships between two or more variables. These became popular in environmental psychology for various reasons, such as these tend to have higher external validity, manipulation of environmental circumstances is a little easier, and these methods often incur lower monetary cost than the othermethods.

a. **Laboratory Experimental Methods** – These include studies conductedunder artificial and controlled environmental conditions. Laboratory experiments are often helpful in establishing and understanding the causal relationships between variables. These methods are characterized by two main features: random assignment or randomization and manipulation. When the independent variable (IV) is only manipulated while keeping the other variables controlled or constant, any changes in the responses between theconditions will be attributed to the manipulation of the independent variable. Randomization refers to the technique used by researchers in which all theparticipants of the experiment have an equal chance of getting assigned toeach experimental condition. Random assignment mainlydecreases the chance that differences between the different experimental groups are caused byconfounding individual factors like personality, socio-demographics, etc. These methods are often low in external validity and the result can not be easilygeneralized to real-world settings.

b. **Computer Simulation Studies -** Real people or authentic environmentalstimuli aresometimes impossible to useinthe study. Studies involving thousandsof people to learn about complicated systems or studies on how people judge future environmental scenarios are two examples. For this reason, environmental psychologists are increasingly turning to environmentalsimulations. Environments and/or humans are reproduced as exactly andrealistically as feasible in this style of research. Immersive virtual environments created with computers to give participants a realistic impression of what itwould be like to experience specific environments or events (De Kort et al. 2003), 3D visualization of data in Geographical Information Systems, or agent-based models of land use or resource use are examples of simulations. Ingeneral, simulations allow you to maintain some control over the environment, boosting internal validity while not jeopardizing external validity.

c. **Field Studies** – Many environmental psychologists use field investigationsand experiments to obtain high external validity without sacrificing too muchinternal validity. Field experiments have a high level of external validitybecause they are conducted in real-life settings. Internal validity, on the other hand, isrelatively high because the experimenter attempts to control the situation bysystematically manipulating independent variables (e.g., placing or removing a bin from the environment) and/or by randomly assigning participants to different study conditions (e.g., environments with and without bins). Internal validity is ensured when researchers can be relatively certain that any variations between conditions are due to the manipulations (rather than, for example, personality differences). However, because field studies take place in real- world situations, it's impossible to account for confoundingfactors like changing weather or unexpected interruptions. Furthermore, random assignment is notachievable in many instances.

Case Studies - A case study is an in depth study of a specific situation. it's away used when a researcher really wants to narrow down a really broadtopic of research into one single case, i.e., a

person, setting, situation, or event. for instance, the broad topic of urban environmental quality could also be studied in one particular neighbourhood where the municipality has recentlyinstalled garbage bins to tackle littering. instead of employing a strict protocol and close ended inquiries to study a limited number of variables, case studymethods involve an exploratory, qualitative examination of one situation orevent: a case. Qualitative research uses words or other non numerical indicators(such as images or drawings) as data. the most purpose of case studies andother sorts of qualitative research is to explore and understand the meaningthat individuals or groups ascribe to a phenomenon. during a case study, peopleor events are studied in their own context, within present settings, such as thehome, playing fields, the university, and therefore the street. These settingsare 'open systems' where conditions are continuously suffering from interactions with the social, physical, historical, and cultural context to offerrise to a process of ongoing change, including ethnography, grounded theory,and phenomenology (Wolcott 2001). Although there will not be one objective truth of the interpretation of the phenomenon (Willig 2001).

UNIT – II: ENVIRONMENTAL RISK AND PERCEPTION

Lesson-5 Natural Disasters: Characteristics of Natural Disasters, Perception and Psychological Effects of Natural Disasters

NATURALDISASTERS

Definition and Characteristics

We are all familiar with environmental calamities that result in excessive propertydestruction, human life loss, or both. Untold human agony has been brought byearthquakes, tornadoes, floods, fires, storms, and avalanches.Although disasters caused by naturally occurring phenomena usually have minimal human involvement, there aremoments when natural events and our impact on the environment intersect. Adisaster is anything that has extremely bad implications for individuals who are impacted by itand is caused by the interaction of natural and societal events (Veitch and Arkkelin, 1995) This definition recognizes that a natural event is onlya disaster if it has an impacton humans.Anatural disaster is a major calamitycaused by the earth's natural processes.

Characteristics of natural disasters Natural disasters are distinguished by the following characteristics:

- They occur suddenly.
- They are usually unpredictable, and even when we have some advance notice,we donot always know when the event will occur or what will happen.
- They are out of control
- They are incredibly powerful
- They are frequently acute, meaning they last only a few minutes rather than days (Oliver, 2008).

Some examples of natural disastersare tornadoes, severe storms, hurricane, tropical storms, floods, wildfires, earthquakes, droughts, volcanic eruption, tsunami, and landslide.

Categories of Natural Disasters

The natural disaster category is generally divided into six groups of disasters, namely, biological, geophysical, metrological, hydrological, climatological, and extra-terrestrial.

a. Biological disaster – Biological disasters are natural events that result in widespread disease, disability, or death among humans, animals, and plants as a result of microorganisms such as bacteria, viruses, or poisons. Examples are, dengue, malaria, meningitis, HIV, Bird flu, Swine flu, Cholera, etc.

b. Geophysical disasters– They are potentially damaging natural events and/or phenomenathat may cause the loss of life or injury, property damage, socialand economic disruption, or natural degradation, extreme natural hazards are pre-manifestation of the complex, hierarchal and non-linear earth systems. Examplesare, Volcanic eruption, earthquakes, landslides, mudflows, etc.

c. Metrological disasters – These are caused by extreme weather such asrain, drought, snow, extreme

heat or cold, ice or wind.Examples of weatherdisasters embrace blizzards, cyclones, hailstorms, heat waves, hurricanes,floods, and tornadoes.

d. Hydrological disasters – A hydrological disaster is a violent, sudden, anddestructive change either in the quality of earth's water or in the distribution ofthe movement of water on land, below the sea-level or in the atmosphere.Examples are, floods, limnic eruptions (eruption of carbon dioxide present inthe deep waters of a lake that creates gas cloud), puddle, landslides.

e. Climatological disasters – These are events that are brought about by drasticfluctuations of the climate states and variabilities. Or, these are a phenomenonthat is a result of climate fluctuations that shift outside of their normal state todifferent factors that impact the climate. Examples areElNino SouthernOscillation, La Nina, enhanced greenhouse effect (climate change), aciddeposition, ozone depletion.

f. Extra-terrestrial disasters – These are hazards caused by asteroids, meteorites, and comets as they pass near Earth, enter the earth's atmosphere,and/or strike the earth. They cause changes in interplanetary conditions that affect the earth magnetosphere, ionosphere, and thermosphere. Examples are, meteorites, interstellar dust, explosion from comets, nuclear accidents in space.

Causes of Natural Disasters

Earthquakes, flooding, volcanic eruptions, landslides, hurricanes, and other natural calamities can all contribute to a natural disaster. To be classified as a disaster, it must have a significant environmental impact, as well as human loss and, in many cases,financial loss

Natural disasters are caused by natural events such as extreme weather conditions ormovements of the earth's surface that tend to be localized in certain areas of the world,and they are remembered vividly because of their dramatic nature. For example,Hurricanes form over moisture-rich oceans and coastlines, such as the Atlantic Ocean,Caribbean Sea, and the Gulf of Mexico, whereas earthquakes occur along fault lines inthe earth's crust.

Avalanches of snow or excessive rainfall leading to flooding can also cause disasters;the magnitude of the disaster may be exacerbated by human intervention in some areas,but humans are not directly responsible for the events(Oliver, 2008). Soil erosion,seismic activity, tectonic movements, air pressure, and ocean currents, among other factors, all contribute to natural disasters. The main causes of these disasters are natural activities occurring in the earth's crust as well as on the surface.

PERCEPTION AND EFFECTS OFNATURALDISASTERS

Perception of natural hazards and disasters involve intuitive judgments, beliefs, andattitudes adopted by individuals and groups of people about the likelihood and occurrence, and mechanism of development of such phenomenon. The subjective nature of the understanding of natural hazards influences people's decisions.

Risk perception is influenced by many factors which are firstly related to the nature ofthe hazard itself and secondly to psychological, social, and cultural components andtheir mutual interactions. In the case of natural hazards, risk perception analysis has totake into account that they vary immensely from certain events such as flash floods,avalanches, or earthquakes to long-lasting phenomena such as heatwaves and droughts.Most comparative studies focus on human-induced dangers whereas risk perceptionof natural hazards has been researched less.

Some limitations of human risk perceptions are related to psychological mechanisms.Natural disasters may cause unpleasant feelings of anxiety and lack of sense ofsecurity.For this reason, each individual's emotional and psychological traits have animpact on their perception.

Effects of Natural Disasters

Natural disasters generally affect the physical infrastructural facilities, agriculturalproductivity, and even lead to loss of life and cause property damage. Various factorsinfluence the effects that are disastrous for a country, like the magnitude of the disaster and geographic location of the affected area and recovery efforts directed towards reducing immediate effects of a disaster.Failure to mitigate various short term economiceffects of a disaster affects the long-term economy of the country. The primary effectsare direct results of a natural disaster such as collapsed buildings and waste damage.Secondary effects are the result

of primary effects. The effect of natural disasters isboth short-term including loss of life and damage to propertyand long-term affectingthe economic stability of a region or a country.

ENVIRONMENTAL THEORIES AND DISASTERS

Over the centuries there have been four fundamental theories about the disaster, whichhave conceived disasters as acts of fate/acts of God, acts of nature, joint effects of nature and society, and social construction, respectively. Let us explore them.

Acts of fate/acts of God For millennia, disasters were considered to arise fromimpersonal and uncontrollable sources either from unfortunate alignments of stars andplanets or acts of God that were beyond human understanding. Both forms of thistheory viewed disasters as predetermined and thus completely beyond the victim'scontrol. Avariation on this theory was that disasters were cosmic or divine retributionfor human failings- personal disasters for personal failings, and collective disasters forsocietal failings.

Acts of Nature Overtime increased scientific knowledge led many people to substitutenatural causes for supernatural ones.Thus floods occur because of a large amount ofrainfall from a serious form exceeded capacity to absorb it.Accordingly the term naturaldisaster came to refer to an outside attack upon social systems that broke down in theface of such assaults(Quarantelli, 1998).

Joint effects of Nature and Society It was proposed that hazards arise from theinteraction of a physical event system and human use system.Thus, it takes both thehazardous event system and the vulnerable human use system to produce disaster

According to this view, human societies adapt to the prevailing environmental conditions, e.g.,temperature, wind speed, precipitation, seismic activities, at a given location.Unfortunately, they fail to anticipate the variations in those environmental conditions;consequently, their adaptation to normal conditions usually is inadequate for extremeevents – blizzards, heat waves, tornadoes, hurricanes, and floods.

Social Constructions Most recentlyresearchers have recognized that disasters arequite systematic in the types of people they harm as well as the type of geographicallocation and use systems they impact. While the interactive theory is concerned with hazard-exposure at specific locations and physical vulnerability of specific structures,the social construct theory calls attention to the social vulnerability of specific population segments. To say that hazard vulnerability is socially constructed does not mean thatpeople are vulnerable because they think the wrong thoughts, as most people wouldcategorize the belief that floods are caused by alignments of planets and stars. Rather,socially vulnerable population segments emerge because their psychological, demographical, economical, and political processes tend to produce them.

Lesson-6 Noise and Air Pollution: Source and effects of Noise in the Work Place, Noise and Social Behaviour.

Noise pollution, unwanted or excessive sound that can have deleterious effects on human health, wildlife, and environmental quality. Noise pollution is commonly generated inside many industrial facilities and some other workplaces, but it also comes from highway, railway, and airplane traffic and from outdoor construction activities.

Measuring and perceiving loudness

Sound waves are vibrations of air molecules carried from a noise source to the ear. Sound is typically described in terms of the loudness (amplitude) and the pitch (frequency) of the wave. Loudness (also called sound pressure level, or SPL) is measured in logarithmic units called decibels (dB). The normal human ear can detect sounds that range between 0 dB (hearing threshold) and about 140 dB, with sounds between 120dB and 140 dB causing pain (pain threshold). The ambient SPL in a library is about 35 dB, while that inside a moving bus or subway train is roughly 85 dB; building construction activities can generate SPLs as high as 105 dB at the source. SPLs decrease with distance from the source.

The rate at which sound energy is transmitted, called sound intensity, is proportional to the square of the SPL. Because of the logarithmic nature of the decibel scale, an increase of 10 dB represents a 10-fold increase in sound intensity, an increase of 20 dB represents a 100-fold increase in intensity, a 30-dB increase represents a 1,000-fold increase in intensity, and so on. When sound intensity is doubled, on the other hand, the SPL increases by only 3 dB. For example, if a construction drill causes a noise level of about 90 dB, then two identical drills operating side by side will cause a noise

level of 93 dB. On the other hand, when two sounds that differ by more than 15 dB in SPL are combined, the weaker sound is masked (or drowned out) by the louder sound. For example, if an 80-dB drill is operating next to a 95-dB dozer at a construction site, the combined SPL of those two sources will be measured as 95 dB; the less intense sound from the compressor will not be noticeable.

Frequency of a sound wave is expressed in cycles per second (cps), but hertz (Hz) is more commonly used (1 cps = 1 Hz). The human eardrum is a very sensitive organ with a large dynamic range, being able to detect sounds at frequencies as low as 20 Hz (a very low pitch) up to about 20,000 Hz (a very high pitch). The pitch of a human voice in normal conversation occurs at frequencies between 250 Hz and 2,000 Hz.

Precise measurement and scientific description of sound levels differ from most subjective human perceptions and opinions about sound. Subjective human responses to noise depend on both pitch and loudness. People with normal hearing generally perceive high-frequency sounds to be louder than low-frequency sounds of the same amplitude. For this reason, electronic sound-level meters used to measure noise levels take into account the variations of perceived loudness with pitch. Frequency filters in the meters serve to match meter readings with the sensitivity of the human ear and the relative loudness of various sounds. The so-called A-weighted filter, for example, is commonly used for measuring ambient community noise. SPL measurements made with this filter are expressed as A-weighted decibels, or dBA. Most people perceive and describe a 6- to 10-dBA increase in an SPL reading to be a doubling of "loudness." Another system, the C-weighted (dBC) scale, is sometimes used for impact noise levels, such as gunfire, and tends to be more accurate than dBA for the perceived loudness of sounds with low frequency components.

Noise levels generally vary with time, so noise measurement data are reported as time-averaged values to express overall noise levels. There are several ways to do this. For example, the results of a set of repeated sound-level measurements may be reported as L_{90} = 75 dBA, meaning that the levels were equal to or higher than 75 dBA for 90 percent of the time. Another unit, called equivalent sound levels (L_{eq}), can be used to express an average SPL over any period of interest, such as an eight-hour workday. (L_{eq} is a logarithmic average rather than an arithmetic average, so loud events prevail in the overall result.) A unit called day-night sound level (DNL or L_{dn}) accounts for the fact that people are more sensitive to noise during the night, so a 10-dBA penalty is added to SPL values that are measured between 10 PM and 7 AM. DNL measurements are very useful for describing overall community exposure to aircraft noise.

Effects

Noise is more than a mere nuisance. At certain levels and durations of exposure, it can cause physical damage to the eardrum and the sensitive hair cells of the inner ear and result in temporary or permanent hearing loss, known as noise-induced hearing loss. Hearing loss does not usually occur at SPLs below 80 dBA (eight-hour exposure levels are best kept below 85 dBA), but most people repeatedly exposed to more than 105 dBA will have permanent hearing loss to some extent. In addition to causing hearing loss, excessive noise exposure can raise blood pressure and pulse rates, cause irritability, anxiety, and mental fatigue, and interfere with sleep, recreation, and personal communication. Children living in areas with high levels of noise pollution may suffer from stress and other problems, such as impairments in memory and attention span. Noise pollution control is therefore important in the workplace and in the community.

Noise regulation and mitigation

Noise-control ordinances and laws enacted at the local, regional, and national levels can be effective in mitigating the adverse effects of noise pollution. Environmental and industrial noise is regulated in the United States under the Occupational Safety and Health Act of 1970 and the Noise Control Act of 1972. Under these acts, the Occupational Safety and Health Administration set up industrial noise criteria in order to provide limits on the intensity of sound exposure and on the time duration for which that intensity may be allowed.

If an individual is exposed to various levels of noise for different time intervals during the day, the total exposure or dose (D) of noise is obtained from the relation$D = (C_1/T_1) + (C_2/T_2) + (C_3/T_3) +\ldots$,where C is the actual time of exposure and T is the allowable time of exposure at any level. Using this formula, the maximum allowable daily noise dose will be 1, and any daily exposure over 1 is unacceptable.

Criteria for indoor noise are summarized in three sets of specifications that have been derived by collecting subjective judgments from a large sampling of people in a variety of specific situations. These have developed into the noise criteria (NC) and preferred noise criteria (PNC) curves, which provide limits on the level of noise introduced into the environment. The NC curves, developed in 1957, aim to provide a comfortable working or living environment by specifying the maximum allowable level of noise in octave bands over the entire audio spectrum. The complete set of 11 curves specifies noise criteria for a broad range of situations. The PNC curves, developed in 1971, add limits on low-frequency rumble and high-frequency hiss; hence, they are preferred over the older NC standard. Summarized in the curves, these criteria provide design goals for noise levels for a variety of different purposes. Part of the specification of a work or living environment is the appropriate PNC curve; in the event that the sound level exceeds PNC limits, sound-absorptive materials can be introduced into the environment as necessary to meet the appropriate standards.

Low levels of noise may be overcome using additional absorbing material, such as heavy drapery or sound-absorbent tiles in enclosed rooms. Where low levels of identifiable noise may be distracting or where privacy of conversations in adjacent offices and reception areas may be important, the undesirable sounds may be masked. A small white-noise source such as static or rushing air, placed in the room, can mask the sounds of conversation from adjacent rooms without being offensive or dangerous to the ears of people working nearby. This type of device is often used in offices of doctors and other professionals. Another technique for reducing personal noise levels is through the use of hearing protectors, which are held over the ears in the same manner as an earmuff. By using commercially available earmuff-type hearing protectors, a decrease in sound level can be attained ranging typically from about 10 dB at 100 Hz to more than 30 dB for frequencies above 1,000 Hz.

Outdoor noise limits are also important for human comfort. Standard house construction will provide some shielding from external sounds if the house meets minimum standards of construction and if the outside noise level falls within acceptable limits. These limits are generally specified for particular periods of the day—for example, during daylight hours, during evening hours, and at night during sleeping hours. Because of refraction in the atmosphere owing to the nighttime temperature inversion, relatively loud sounds can be introduced into an area from a rather distant highway, airport, or railroad. One interesting technique for control of highway noise is the erection of noise barriers alongside the highway, separating the highway from adjacent residential areas. The effectiveness of such barriers is limited by the diffraction of sound, which is greater at the lower frequencies that often predominate in road noise, especially from large vehicles. In order to be effective, they must be as close as possible to either the source or the observer of the noise (preferably to the source), thus maximizing the diffraction that would be necessary for the sound to reach the observer. Another requirement for this type of barrier is that it must also limit the amount of transmitted sound in order to bring about significant noise reduction.

What is Air Pollution?

The air pollution definition says that when any physical, chemical, or biological change takes place in the air and contaminates it, then it is called air pollution. The contamination of air can be caused due to many factors such as poisonous or harmful gases, smoke, fog, smog, dust, etc. air pollution affects both plants as well as animals

Types of Air Pollutants

The air pollutants are divided into primary and secondary pollutants. Pollutants are those substances that cause air pollution.

- **Primary Pollutants:**

  The primary pollutants responsible for air pollution are the ones that directly cause air pollution. These include harmful gases such as sulfur dioxide coming from the factories. Primary pollutants are those that are produced as a direct result of the process. Sulfur dioxide, generated by factories, is a classic example of a primary pollutant.

- **Secondary Pollutants:**

  The secondary pollutants are formed by the process of intermixing or intermingling of primary pollutants. Smog, which is a combination of fog and smoke, is a secondary pollutant.

Causes of Air Pollution:

To prevent the pollution of air around, you have to understand the causes of air pollution at first. The main causes are –

- **Burning of Fossil Fuels:**

  Fossil fuel emits harmful gases such as sulfur dioxide and carbon monoxide into the air. One of the biggest causes of air pollution is sulfur dioxide, which is emitted through the combustion of fossil fuels such as coal, petroleum for energy in power plants, and other industry combustibles.

- **Automobiles:**

  The emission of harmful gases is caused by the excessive use of automobiles.

- **Agricultural Activities:**

  Various processes take place during agricultural activities such as the emission of ammonia, overuse of insecticides, pesticides, and fertilizers. Ammonia is a typical byproduct of agriculture and one of the most dangerous gases in the atmosphere. Insecticides, pesticides, and fertilizers have all become increasingly common in agricultural practices. They release hazardous chemicals into the atmosphere and can pollute water.

  Farmers also set fire to the fields and old crops to clear them up for the new cycle of sowing. According to reports, burning to clean up fields pollutes the air by emitting toxic pollutants.

- **Factories and Industries:**

  Emission of harmful gases and chemicals into the air by the increasing industrial activities. Manufacturing companies emit a significant amount of carbon monoxide, hydrocarbons, organic compounds, and chemicals into the air, lowering air quality.

  Manufacturing industries may be found in every corner of the globe, and no region has escaped their influence. Petroleum refineries also emit hydrocarbons and a variety of other pollutants, which damage the air and soil.

- **Mining Activities:**

  Increasing emission of harmful substances through mining activities. Mining is the extraction of minerals from under the earth's surface utilizing heavy machinery. Dust and chemicals are released into the air throughout the process, resulting in significant air pollution.

  This is one of the factors contributing to the deteriorating health of workers and inhabitants in the area.

- **Domestic Resources:**

  Effects of domestic sources such as the use of chemical paints and overuse of air conditioners. Household cleaning products and painting supplies release hazardous chemicals into the air, polluting the environment. Have you ever observed that when you paint your house's walls, it emits a noxious odor that makes it nearly impossible to breathe?

  Another source of pollution is suspended particle matter, sometimes known as SPM. SPM refers to the particles that float in the air and is typically caused by dust, combustion, and other factors.

Effects of Air Pollution:

The air pollution information shows that increasing air pollution can have an adverse effect on plants, animals, and humans.

- **Global warming**

  Air Pollution can increase the amount of global warming as the temperature of the earth will keep rising with the emission of harmful gases. With rising global temperatures, rising sea levels, melting ice from colder places and icebergs, relocation, and habitat loss, an imminent crisis has already been signaled if preservation and normalization measures are not done soon.

- **Acid rain**

  When water droplets combine with harmful chemicals and pollutants, it will lead to acid rain. When fossil fuels are burned, harmful chemicals such as nitrogen oxides and sulfur oxides are emitted into the environment. When it rains, the water droplets interact with the contaminants in the air, becoming acidic and falling to the earth as acid rain. Acid rain has the potential to harm humans, animals, and agriculture.

- **Ozone layer Depletion**

  All this will eventually lead to depletion of the ozone layer that protects us from harmful UV sun rays. The presence of chlorofluorocarbons and hydrochlorofluorocarbons in the atmosphere is degrading the ozone layer on Earth.

  As the ozone layer thins, damaging rays are emitted back to Earth, potentially causing skin and eye problems. UV rays have the power to harm crops as well.
  Thus, we have to work on the prevention of air pollution.

Lesson-7 Perception, Health, Performance.

Environmental perception is the way in which an individual perceives the environment, and the process of evaluating and storing information received about the environment. It is the perception of the environment which is important because individuals base their judgements on the environment as they perceive it, not as it is. The nature of such perception includes warm feelings for an environment, an ordering of information, and an understanding, however subjective, of the environment.
Environmental perception is not simply perception of the natural environment but includes built environments, other people, values, cognition and aesthetics. Environmental perception studies have made an important contribution to geographical knowledge. For instance, behavioural geographers, who drew heavily upon environmental psychology and cognition, developed an elaborate array of general categories which can describe and explain environmental perception in a variety of contexts. This was part of their commitment to making geography a spatial science, but it also paved the way for less analytical investigatioils into huinan perception and into hcnneneutics (science and methodology of interpreting texts) which are associated with humanistic geography. TTouch is a sense that we use all day without typically thinking a second thought about it. We use our sense of touch to respond to various elements of the environment with every movc. We do not only experience touch through just our fingertips, but through every inch of our bodies. We rcspond to the environment through touch in three major ways, namely (i) Thermoreceptors (ii) Noceptors and (iii) Mechanoreceptors. A thermoreceptor is a sensory receptor that responds to heat and cold. Mechanoreceptors are those which we and other animals have, that is, several types of receptors of mechanical stimuli. Each initiates nerve impulses in sensory neurons when it is physically deformed by an outside force such as touch, pressure, strctching, sound waves and motion. Mechanoreceptors enable us to dctcct touch, monitor the position of our muscles, bones, and joints, which is called as the sense of proprioception We respond to the elements of temperature changes of space through our Thennoreceptors and to the changes in pressure in our environment through both Noceptors and Mechanoreceptors. These three ways allow us to continually evaluate our environments almost instantly. The concept of the 'perceived environment' has been used to challenge the concept of economic marl, which lies at the heart of neoclassical economics (see bounded rationality satisficer) and to explain supposedly irrational behaviour, such as moving to a flood- or earthquake-prone location (see cognitive dissonance). It is suggested that environmental perception can be seen in the following stages: An emotional response. An orientative response with the construction of mental maps. . : A classibing response as the individual sorts out the incoming information. 4) An organising response as the individual sees causes and effects in the information.
Environment is taken to refer to anything external to the perceiver which influences or might influence the perception process. Generally physical environment is composed of (a) the natural environment and (b) the built environment. Natural environment refers to places, such as valley, mountains, environmental conditions such as temperature and rainfall, flora and fauna. Built environment refers to the results of people's alterations of environments, e.g., houses and buildings, cities, communities etc. Mental activities include things that they see, hear and their interpretation about the physical environment. Mental activities also include beliefs, attitudes which can be positive or negative. All persons receive the same information in a given real life environmental

display and they process it in different fashion. How we behave in a setting is a function of how we perceive it. People who live in different environments develop distinctive ways of recording, processing and acting on this information. The concept of mental or cognitive maps illustrates the complex character of environmental perception. All of us carry a mental map of our surroundings around us. A cognitive map of an environment, thus, is our internalised image of that environment. It is not necessarily be correct and accurate representation. Like all perceptual processes environmental perception plays a dual role in our lives. 0 First, it is the source of our phenomenal experience of the world. All its sights, sounds and smells, all its simple and complex things, all its ugliness and its beauty, all its sense of value comes to us through the process of perception. n Second, perception provides us with a guide to action in the environment. It gives us both arena within which actions take place and the ability to register and record the consequences of these actions.

INDIVIDUAL'S RESPONSE TO ENVIRONMENT

In order to have an extensive view of human behavoiur in large ecological contexts, it is very essential to know the environmental perception and cognition of individuals that is, the manner in which individuals respond to their environment, viz., their cognitive perceptual capacities and the components of the information processing system are hnctioning uniquely as a whole interactive Physical and socio-cultural environment are linked with people and various psychological environments. It means that man moves through various settings with clear expectations of how others will behave as well. Whereas the conventional approach to perception examines the way the brain interprets messages from the sensory organs concerning specific elements in the environment, environmental perception views the perceptual experience as more encompassing, including cognitive, affective, interpretive and evaluative responses. Also environmental perception is likely to consider the systems or transactional perspectives. Environmental perception involves activity on our part, especially in terms of exploring the environment to determine what needs to meet. Exposure to a particular environment may result in adaptation or habituation. In other words, how the weakening of responses follow repeated exposure to a stimulus. Individual differences in the ways people think about and relate to the everyday physical environment are assumed to be understood by environmental disposition including conservation, recreation and leisure activities, architecture and geography, science and technology, urban life and culture, aesthetic preferences, privacy and adaptation etc.

Lesson-8 Social Behaviour and Air Pollution

ENVIRONMENTAL EFFECTS ON HUMAN BEHAVIOUR The human-environment relationship can be appreciated fully by understanding that the two influence each other, and depend on each other for their survival and maintenance. When we focus our attention on the effects of the natural environment on human beings, we find a variety of environmental influences, ranging from physical effects such as a change of clothing in response to climate changes to severe psychological effects such as serious depression after a natural disaster. Some of the effects pointed out by psychologists are described below. • Environmental influences on perception : Some aspects of the environment influence human perception. For example, as you have studied in Class XI, a tribal society of Africa lives in circular huts, that is, in houses without angular walls. They show less error in a geometric illusion (the Muller-Lyer illusion) than people from cities, who live in houses with angular walls. • Environmental influences on emotions : The environment affects our emotional reactions as well. Watching nature in any form, whether it is a quietly flowing river, a smiling flower, or a tranquil mountain top, provides a kind of joy that cannot be matched by any other experience. Natural disasters, such as floods, droughts, landslides, quakes on the earth or under the ocean, can affect people’s emotions to such an extent that they experience deep depression and sorrow, a sense of complete helplessness and lack of control over their lives. Such an influence on human emotions is a traumatic experience that changes people’s lives forever, and can last for a very long time after the actual event in the form of post-traumatic stress disorder (PTSD). • Ecological influences on occupation, living style and attitudes : The natural environment of a particular region determines whether people living in that region rely on agriculture (as in the plains), or on other occupations such as hunting and gathering (as in forest, mountainous or desert regions), or on industries (as in areas that are not fertile enough for agriculture). In turn, the occupation determines the lifestyle and

attitudes of the residents of a particular geographical region. Compare the daily routine followed by a person living in a desert with one living in a mountainous region, and with one living on the plains. It has been observed that an agricultural society has to depend on the collective efforts of its members. Therefore, the members of an agricultural society develop an attitude of cooperativeness, and consider group interests more important than the individual's wishes. They are also closer to nature, more dependent on natural events such as the monsoon, and may face situations in which necessary natural resources such as water are in limited supply. Accordingly, the members of an agricultural society may become more fatalistic in their beliefs. On the other hand, highly industrialised societies feel less close to and less dependent on nature. Members of industrialised societies may value independent thinking, develop an attitude of competitiveness, and cultivate a belief of personal control over what happens to them.

Human Influence on the Environment Human beings also exert their influence on the natural environment for fulfilling their physical needs and other purposes. All the examples of the built environment express human influence over the environment. For example, the human being started building something called 'houses' by changing the natural environment in order to provide shelter for herself/himself. Some of these human actions harm and damage the environment, and ultimately harm themselves, in numerous ways. For example, human beings use appliances such as refrigerators and air conditioners that generate chemical substances (e.g., CFCs, or chlorofluorocarbons) that pollute the air and, in the long run, may be responsible for physical diseases such as some forms of cancer. Smoking is known to pollute the air around us, and the burning of plastic or metal articles has disastrous polluting effects on the environment. Cutting down trees, or deforestation, may disrupt the carbon cycle and the water cycle. This may eventually affect the pattern of rainfall in a region, and increase soil erosion and desertification. Industries that discharge effluents, and pump this untreated sewage into rivers, seem to be unconcerned about the dangerous physical and psychological consequences of this kind of pollution. What is the psychological message in all of these examples? The message is that although most of the effects mentioned above are physical in nature, human beings have brought about these effects in order to exhibit their control and power over the natural environment. It is somewhat paradoxical that human beings are using technology to change the natural environment in order to improve the quality of their life but, in reality, they may actually be worsening the quality of life! Noise, pollution, crowding, and natural disasters are some examples of environmental stressors, which are stimuli or conditions in the environment that create a stress for human beings.

Pollution

Environmental pollution may be in the form of air, water, and soil pollution. Waste or garbage that comes from households or from industries are a big source of air, water, and soil pollution. Scientists know it very well that any of these forms of pollution is hazardous to physical health. However, there are some research studies that have shown direct or indirect psychological effects of these forms of pollution as well. It has to be understood that in general, any form of environmental pollution may affect the nervous system because of the presence of toxic substances and, to that extent, influence psychological processes in some way. Another form of influence is seen in the emotional reactions to pollution which, in turn, create discomfort, and have consequences such as decreased work efficiency, lowered interest in the job, and increase in anxiety level. People do not like to live or work in a locality that is littered with garbage, or has a constant foul smell. Similarly, the presence of dust particles, or other suspended particles, may give a feeling of suffocation and difficulty in breathing, and may actually lead to respiratory disorders. People who experience this kind of discomfort cannot pay full attention to their work, or be in a pleasant mood. Specific psychological effects of air pollution have been reported by some researchers. For example, in one part of Kolkata, the psychological reactions to air pollution were compared between a group living in an industrial area, and a group living in a non-industrial residential area. Those living in the industrial area reported greater tension and anxiety than those living in a non-industrial residential area. In a study conducted in Germany, the presence of pollutants such as sulphur dioxide in the air was found to decrease the ability to concentrate on a task, and lowering performance efficiency. Pollution caused by leaks of dangerous chemical substances can cause other kinds of harm. The infamous Bhopal gas tragedy of December 1984 that claimed many lives, also left behind psychological effects because of the gas. Many of those who had inhaled the poisonous gas, methylisocyanate (MIC) along with other substances, showed disturbances in memory, attention and alertness. There can be harmful air pollution in the home and office environment (indoor environments) also. For example, tobacco smoke pollution, that is, pollution through cigarette, cigar or beedi smoking, can also cause psychological effects. Such effects are supposed to be more dangerous for the smoker; however, those who inhale tobacco smoke (passive smoking) can also suffer the negative effects. One researcher found that inhaling tobacco smoke can increase the

aggression level of individuals. The presence of polluting substances in water and soil are hazardous for physical health. Some of these chemicals can also have damaging psychological effects. The presence of specific chemicals such as lead can cause mental retardation by affecting brain development. Such toxic substances affect human beings through various routes, i.e. through water, or through soil by being absorbed by vegetables grown in polluted soil. Another source of toxicity is household and industrial waste, or garbage, which are non-biodegradable. Common examples of such waste are plastics, tin or any metal container. This kind of waste material should be destroyed or burned through special techniques, and the smoke should not be allowed to escape into the air that people breathe. In general, there is sufficient evidence to show that toxic chemicals in the air, water and soil may lead to harmful effects not only on normal psychological functioning, but may also cause serious mental disorders. Therefore, there is absolutely no doubt that all forms of pollution need to be curbed.

Lesson-9 Environmental Perception and Cognition; Environmental Cognition, Cognitive Maps, Way finding.

Environmental Perception and Cognition;

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Mental Maps or Cognitive Maps

Cognitive maps also known as mental maps, mind maps, cognitive models, or mental models are a type of mental processing composed of a series of psychological transformations by which an individual can acquire, code, store, recall, and decode information about the relative locations and attributes of phenomena in their everyday or metaphorical spatial environment.

The credit for the creation of this term is given to Edward Tolman. Cognitive maps have been studied in various fields;-such as psychology, education, archaeology, planning, geography, cartography, architecturc, landscape architecture, Cognition and Perception urban planning and management. As a consequence. these mental models are often referred to, variously, as cognitive maps, mental maps, scripts, schemata, and frames of rqference.

Putting it into simpler terms, cognitive maps are a method we use to construct and accumulate spatial knowledge, allowing the "mindb qe" to visualize images in order to reduce cognitive load, enhance recall and learning of information. This type of spatial thinking can also be used as a metaphor for non-spatial tasks involving mernon, and imaging. The oldest known formal method of using spatial locations to remember data is the "method of loci". This method was originally used by students of rhetoric in ancient Rome when memorising speeches. 'I'he neural correlate^ of a cognitive map have been speculated to be theplace cell system in the hippocampus and the recently discovered grid cells in the entorhinal cortex.

There is a popular view that people's mental representations of environments are embodied in 'cognitive maps'. Like many useful concepts the term cognitive map has many meanings, leading to inevitable misunderstandings. One view is that cognitive maps are map like mental constructs that can be mentally inspected. They are presumed to be learned by gradually acquiring elements of the world, first landmarks (point like elements), then routes (line like elements), and finally unifying the landmarks and routes with metric suniey information. As maps they are presumed to be like real maps availalsle to real inspection.

In many instances especially for environments not known in dctail, the information relevant to memory or judgement inay be in different forms some of them not map like at all. In these cases, rather than resembling maps, people's internal representations seein to be more like collages, which are thematic overlays from different points of view. In other situations, where environments are simple and well learned, people seen1 to have quite accurate mental representations of spatial layouts. These are relations that are easily comprehended from language as well as from direct

experience. If an environment is familiar, most of us will find our ways most of the time. Or we use maps or instructions or environmental cues or all of them. For some well learned environments large scale or small, people's knowledge can be well organised and systematic. In such cases the knowledge often has (i) the form of locating elements relative to one another from a point of view or of (ii) locating an element relative to a higher order environmental feature or reference frame. These two simple relations can form a foundation for spatial knowledge from which memory and judgement are connected. These relations also form the basis for spatial language used in descriptions of environments. .

Perception has always been conceptualised as one of the most relevant processes within General Psychology. Environmental perception plays a similar role within the flamework of Environmental Psychology. Research and theories in environmental perception have primarily focused on the development and processing of mental representations by subjects to better interpret and understand their surroundings.

One of the first and principal concepts developed to investigate the mental representations of an environment is the cognitive map. As mentioned above, this term was coined by Tolman and the concept of cognitive map has evolved to integrate many different 5ources which can influence such mental representations. The first of these sources is the environment itself. Studies on spatial orientation and spatial perception primarily focus on the way that physical characteristics of the environment influence knowledge or the way finding ability or orientation capabilities of the subjects.

Environmental Perception and Man Built Environment

Environmental perception is not simply perception of the natural environment but includes built environments, other people, values, cognition and aesthetics.

Environmental perception studies have made an important contribution to geographical knowledge. For instance, behavioural geographers, who drew heavily upon environmental psychology and cognition, developed an elaborate array of general categories which can describe and explain environmental perception in a variety of contexts. This was part of their commitment to making geography a spatial science, but it also paved the way for less analytical investigatioils into huinan perception and into hcnneneutics (science and methodology of interpreting texts) which are associated with humanistic geography.

Touch is a sense that we use all day without typically thinking a second thought about it. We use our sense of touch to respond to various elements of the environment with every movc. We do not only experience touch through just our fingertips, but through every inch of our bodies. We rcspond to the environment through touch in three major ways, namely (i) Thermoreceptors (ii) Noceptors and (iii) Mechanoreceptors.

A thermoreceptor is a sensory receptor that responds to heat and cold. Mechanoreceptors are those which we and other animals have, that is, several types of receptors of mechanical stimuli. Each initiates nerve impulses in sensory neurons when it is physically deformed by an outside force such as touch, pressure, strctching, sound waves and motion. Mechanoreceptors enable us to dctcct touch, monitor the position of our muscles, bones, and joints, which is called as the sense of proprioception. We respond to the elements of temperature changes of space through our Thennoreceptors and to the changes in pressure in our environment through both Noceptors and Mechanoreceptors. These three ways allow us to continually evaluate our environments almost instantly.

The concept of the 'perceived environment' has been used to challenge the concept of economic marl, which lies at the heart of neoclassical economics (see bounded rationality satisficer) and to explain supposedly irrational behaviour, such as moving to a flood- or earthquake-prone location (see cognitive dissonance).

It is suggested that environmental perception can be seen in the following stages:

1. An emotional response.
2. An orientative response with the construction of mental maps.
3. A classibing response as the individual sorts out the incoming information.
4. An organising response as the individual sees causes and effects in the information.

Environment is taken to refer to anything external to the perceiver which Cognition and Perception influences or might influence the perception process. Generally physical environment is composed of (a) the natural environment and (b) the built environment. Natural environment refers to places,

such as valley, mountains, environmental conditions such as temperature and rainfall, flora and fauna. Built environment refers to the results of people's alterations of environments, e.g., houses and buildings, cities, communities etc. Mental activities include things that they see, hear and their interpretation about the physical environment. Mental activities also include beliefs, attitudes which can be positive or negative.

All persons receive the same information in a given real life environmental display and they process it in different fashion. How we behave in a setting is a function of how we perceive it. People who live in different environments develop distinctive ways of recording, processing and acting on this information. The concept of mental or cognitive maps illustrates the complex character of environmental perception. All of us carry a mental map of our surroundings around us.

A cognitive map of an environment, thus, is our internalised image of that environment. It is not necessarily be correct and accurate representation. Like all perceptual processes environmental perception plays a dual role in our lives.

i.First, it is the source of our phenomenal experience of the world. All its sights, sounds and smells, all its simple and complex things, all its ugliness and its beauty, all its sense of value comes to us through the process of perception.

ii.Second, perception provides us with a guide to action in the environment. It gives us both arena within which actions take place and the ability to register and record the consequences of these actions.

Relationship between Environmental Perception and Environmental Situation

The environment is taken to refer to anything external to the perceiver which influences or might influence the perception process. There is a close relationship between environmental situation and environmental perception.

The significant information's are:

- Environments have no fixed or given boundaries to space and time.
- Environments provide information through all the senses.
- Environments include peripheral as well as central information.
- Environments include more information than can adequately be handled.
- Environments are defined by and experienced through action.
- Environments have symbolic meaning.
- Environmental experience always takes on the systematic quality of a coherent and predictable value.
- Environmental ptrception involves activity on our part, especially in terms Environmental Perception and Cognition of exploring the environment to determine what needs it meets.
- Environmental perception is likely to consider the person environment relationship from a holistic system or transactional perspective.
- Exposure to a particular environment may result in a adaptation or habitation, with the weakening of a response following repeated exposure to a stimulus.

We typically perceive our environment as bounded, yet the environment as such does not provide any information as to its spatial or temporal boundaries. It is the person, who sets his own boundaries for the various settings he experiences. His purposes and actions constantly interact with environmental information. The person is able to shift the boundaries one after which he perceives. There is a continuous interplay between environmental information and individual along with culture. Environments are multimodal, the senses function always. A blind person learns to get around through his sense by touch.

A kind of private radar closely related to his sense of hearing. Environments almost always provide more information than can possibly be processed. Environments influence the behaviours which take place in them is through their symbolic meaning. The symbolic meanings and motivational messages emitted by an environment are integral to our perception of it. Every setting induces feelings, associations and attitudes in the perceiver that can be described as its ambience. Whereas symbolic meaning, generally cany cognitive information, ambience is related to the way we feel about the environment.

A setting can be exotic, pleasant, gloomy or restful. Such feelings can be general across many individuals or unique to one person. Another important characteristic of environment is their aesthetic quality. The experience of the aesthetic quality of an environment reflects most dramatically the complex interrelationship between the perceiver and the situation of which he is a part. It will vary from culture to culture but the perception of all environments necessarily involves a degree of aesthetic awareness.

The various components of an environment relate to each other in ways that, characterise and define the particular environment that is being perceived. The environment provides no direct information as to its boundaries in space or time, it provides information through all the sense modalities and this information is of a very complex nature. Environmental information is always acquired through action on the part of the perceiver and the information conveys among other things symbolic meanings which have cognitive, affective arid aesthetic aspects and finally environments have a systematic quality of orderly interrelatedness.

ENVIRONMENTAL PERCEPTION AND OTHER PSYCHOLOGICAL PROCESS –

The concept of mental or cognitive maps illustrates the complex character of rh~: environmental perception. It is one way of approaching the perception of room, home or city. We all carry a mental map of our surroundings and ahis mapis often more important in determining our orientation toward an environment than any particular feature or group of features which rnay exist in that situation.
A cognitive map of an environment is our internalized image of that environment. It need not necessarily be a correct and accurate representation. Language ability is not always visual. Sounds also contribute to our sense of space and our orientation to it. It is true that the cognitive map is not pure perccption nor pure cognition, nor pure memory but it is in fact what comes to mind when we ask oursclves to visualise any particular environmental situation. A mental image of the environment which we anticipate for our action is a necessary part of this ability to plan our behaviour.

ENVIRONMENTAL PERCEPTION AND ITS FUNCTIONAL ASPECTS

Environmental perception actually orients us in our surroundings. It provides us with information about the systematic relationships among the components of our world and also the means whereby we can relate our own 'goal striving' to the environment within which we live.
The overall function of environmental perception is neither to reveal the present reality nor to recall past reality; rather it is to predict the future. Perception is anticipatory. This anticipation is not possible without our experience with the recording of past environments. Environmental perceptions reconcile the world we experience a new each day with the world wc have already come to know.
We can refer our previous experiences, our previous perception and also our previous assumption to create and maintain a stable environment.

Cognitive map:

Cognitive map is a mental iiamework that holds some representation for the spatial representation of the physical environn-~ent. Cognitive map may be sketchy, incomplete, distorted, simplified and so on. It mainly consists of three elements, viz., places, the spatial relations between places and trahel plans. Place may be a room, building, city, nation etc. Cognitive inap reflects spatial characteristics, such as the distance and direction between places and the inclusion of one place within another. Cognitive maps are very personal representation of the familiar environmcnt that wc all experience. Lynch found that there are five categories in cognitive map, viz., path, edges, districts, nodes and landmarks.

Paths: Paths are shared traveled corridors viz.. streets, river ways etc.

Edges: Edges are limiting or enclosing features that tend to be linear but are not functioning as paths, viz., wall, seashore etc.

Districts: Districts are larger spaces of cognitive maps that have some common character viz., "China town" found in many cities.

Nodes: Nodes are major points where behaviour is focused. It is typically associated with the intersections of major paths or places, viz., a traffic circle.

Landmarks: Landmarks are distinctive features that people use for reference points.

UNIT – III: ENVIRONMENT AND BEHAVIOUR- THEORIES

Lesson-10 Changing Behaviour to save the Environment: The Common Dilemma as an Environment-Behaviour Problem.

The environmental challenges facing us are striking. Whether it is the threat of the sixth mass extinction or global climate change, these challenges can seem fundamentally intractable. What links nearly all present environmental problems is their root cause: human behavior (Foley et al., 2005; IPCC, 2018). Yet this cause also presents a solution: to address these challenges, humans must act

differently (Schultz, 2011). In other words, environmental problems are behavioral problems, and environmental solutions must also be behavioral solutions. Whenever one approaches developing an environmental program, what they are doing is developing a behavior change program (Cowling, 2014).

Behavior Change Levers for the Environment

Even when not explicitly identified, changing behaviors have long been at the core of delivering environmental programs. Historically, there have been three main levers pulled for changing behavior: shifting material incentives, promulgating rules and regulations, and providing information to actors. Shifting material incentives involves increasing or decreasing the costs, time, or effort for doing a behavior. This lever has its roots in neoclassical economics, where an actor is assumed to respond to only the material incentives for engaging or not engaging in a specific behavior. Standard methods for shifting incentives include enforcing penalties for non-compliance with rules, providing rewards for positive behavior, or making a target behavior materially easier, such as removing time friction or promoting substitute actions.

Passing rules and regulations that promote or restrict a behavior is perhaps the most commonly used strategy for achieving environmental outcomes. Rules and material incentives often work together, but each can exist without the other. For example, a seller might offer an incentive to purchase a product without any legal requirement. Similarly, laws and rules can be passed without their enforcement shifting the material incentives. Even without enforcement, rules can shift behavior due to people having a general preference to conform to rules even without positive or negative sanctions (Funk, 2007) or where rules convey factual or social information (Sunstein, 1996).

Providing actors with information has also been a common tactic in traditional environmental programming, including explaining what the desired behavior is, why it is important, and how to engage in it. Informational programs implicitly assume something similar to the information deficit model; the lack of change in someone's behavior is assumed to be because they do not know key information, rather than psychological or socio-contextual factors (Burgess et al., 1998).

While these levers can be successful at changing behavior, they have also been well-documented as generally insufficient for changing behavior on their own (Cinner, 2018). Environmental behavior change program designers have recently expanded their toolkit to include a more comprehensive set of levers for shifting behavior and achieving environmental outcomes. These levers are choice architecture, emotional appeals, and social influences. These three novel levers, along with the three traditional levers, represent the Behavioral Lever Framework for categorizing behavioral interventions in the environmental field (Rare, 2020).

Using choice architecture means constructing an actor's choice environment without changing the value of said actor's underlying options. This lever deviates from the more traditional levers by not assuming that actors are solely influenced by their rational deliberation, but also how a choice is presented to them. There are many ways in which a designer might construct the choice environment. These include prominent strategies such as directing attention by increasing salient features or changing what outcome occurs by default, using timely moments to prompt action, and providing decision aids that encourage short- or long-term decision making.

Emotional appeals function differently by changing how an actor feels about a set of options. Humans like to believe that they deliberate over all of their decisions, yet emotions often drive our decisions. Emotional appeals can include messaging that makes the behavior feel consistent with the target actor's core identities and values or encourage the actor to experience a particular emotion known to result in a particular behavioral pattern.

Finally, leveraging an actor's social networks and influences is an effective behavior change strategy. Social influence strategies involve understanding how an actor relates to others in their social system, including those with power and prestige, and leveraging these dynamics to support changes in the actor's behavior. Changing behavior in this way often includes social learning, making behavior more observable, or shifting social norms by changing an actor's expectations for what others in their reference network are doing or think is right or wrong. These novel strategies complete the six levers of the Behavior Levers framework. These levers provide a typology for categorizing the majority of existing behavior change interventions, often delivered in combination rather than isolation.

The logic, ethics, and effectiveness of behavior change programming across these levers have been an intense subject of research. This work has mainly been conducted from the behavioral science perspective, which focuses on the cognitive processes affecting how individuals make decisions, and the social science perspective, which focuses on how social structures shape an actor's capacity and interest in adopting a behavior.

The Behavioral Science Perspective

While there are many different definitions of behavioral science, we focus on the systematic study of human judgment and decision making. This research has been conducted by those working in several fields but is most commonly associated with psychology and behavioral economics. This perspective tends to take the individual actor as the central unit for analysis and understanding behavior.

The roots of what is now commonly known as behavioral science can be traced to rational choice models in neoclassical economics and the inability of those models to account for the decisions people often make. These systematic deviations from rational choice models are known as biases, which result from people applying cognitive heuristics to solve real-world decision problems (Tversky & Kahneman, 1974).

Research in this field focuses on the decision processes that affect how an actor is making a particular decision. These processes are often described as falling into two broad and simplified categories. The first mode is quick and automatic and is more likely to be driven by an emotional reaction. The second mode of thinking more closely approximates rational choice models. This way of thinking is often slow and deliberate, and the decision-maker is generally conscious of this mode. These two groups of processes are often labeled as System 1 and System 2 (Stanovich & West, 2000). Research in the behavioral sciences primarily focuses on documenting the mechanisms underpinning System 1.

Researchers have documented a host of deviations from rational choice models in decision making and the cognitive processes underpinning them. The most extensive set of this work has been conducted in contexts where people face some risky decision, where an outcome could end up going better or worse than their current state.

One pattern is loss aversion, where people feel a loss more strongly than a similarly sized gain. Another is risk aversion, where people prefer a sure thing over a risky proposition, even when the risky proposition is likely to return even more. A third is ambiguity aversion, where people prefer to choose options where they know the likelihood of the different outcomes, even when they are guaranteed to do worse. Many of these findings have been replicated frequently and cross-culturally (Ruggeri et al., 2020).

This research has also documented an effect called status quo bias, a general tendency for people to keep doing what they have previously done, even when not in their best interest (Kahneman et al., 1991). This bias describes how habitual behaviors persist but also why it is difficult to form new habits that are inconsistent with one's previous status-quo.

While behavioral science researchers generally take the individual as their unit of analysis, this does not mean researchers ignore social influences. A large body of work on social preferences has documented how people— unlike what would be predicted by a selfish economic model—care deeply about what those in their social network do, believe, and receive. While early research attempted to identify universal social preferences (Fehr & Schmidt, 1999), these social influences differ dramatically across cultural contexts (Henrich et al., 2005). Behavioral scientists now primarily focus on the cognitive mechanisms that result in a particular pattern of behavior within a social context. For example, social norms describe where an individual's actions are influenced by their beliefs of what others are doing and what others think they should be doing (Bicchieri, 2016). The fact that these expectations may be different for different social groups, and different for individuals having different reference networks within a social group, allows for the varied social preferences we see among people of different social groups.

Behavioral science insights have recently been deliberately incorporated into behavior change program design, including at the bilateral, national, and regional levels of government and non-government entities (Whitehead et al., 2019). Many applications of behavioral science have been to design a choice environment to nudge people to perform behaviors in their interest (Thaler &

Sunstein, 2009). Nudges are intended to be consistent with libertarian paternalism, where each person's actual choices are not restricted, but their environment is designed to encourage a particular behavior. Nudges are often subtle changes, such as shifting the default offering or making one choice more salient. However, nudges represent only one area of the application of behavioral science to behavior change. Other applications of behavioral science incorporate rich insights from the program's target actors. They also often involve shifting entrenched social norms, such as encouraging the adoption of toilets (Ashraf et al., 2020), reducing female genital cutting (Evans et al., 2019), or encouraging treatment adherence to painful drug regimens like those used to treat tuberculosis (Yoeli et al., 2019). This latter set of interventions differs from traditional uses of nudges by addressing actors as members of a community rather than narrowly as individuals, being more overt about the intervention itself, and often targeting socially constructed practices.

In summary, the behavioral science perspective has studied how individuals make decisions, concentrating on the ways human behavior deviates from the predictions of rational choice models. The field has documented various biases that result from people relying on cognitive heuristics for making decisions, many of which are the result of quick, implicit, and sometimes emotional processes rather than slow deliberation. While this work analyzes decisions from the perspective of the individual, it also investigates social influences, showing how people process their social environment and then apply it to their choices. This work has recently been adopted into behavior change program design across various institutions and levels of decision-makers, sometimes within the framework of nudges and larger-scale behavior change campaigns that often target more entrenched behaviors.

The Ethics of Behavior Change

Behavior change practitioners seeking to realize a social good must pay attention to ethics, as ethics are particularly relevant to a program that uses persuasion to influence a target audience and achieve its goals. Therefore, we encourage practitioners to adhere to the following three principles:

Principle of well-being Particular intervention strategies may target different kinds of well-being, and we cannot expect all forms of well-being to be equally pursued at all times. Even so, following the principle of well-being means that our work should always entail the goal of helping humans and nature flourish. **Principle of integrity** At no time should conservation advocates misrepresent facts or people's ideas, even if they do not agree with them. Nor should advocates misrepresent the intentions and consequences of particular environmental policies or instances of management. Finally, conversation practitioners should not deceive or manipulate people in ways that cause them to do behavior they do not want to do or removes their agency or free will**. Principle of empowerment** Whenever possible, behavior change interventions should empower citizens to make democratic decisions about conservation, such as through environmental policy and management. To do this, interventions should strive to make the best scientific, political, economic, and moral arguments available to the public. There is a rich body of literature on the ethics of behavior change, and nudging strategies in particular, which often embrace non-conscious forms of influence. We cannot do this literature justice here and direct interested readers to resources such as Cass Sunstein's The Ethics of Influence. Such literature demonstrates how behavioral science sheds new light on the nature of agency and of personal choice. In practice, influence in unavoidable, as we are constantly being nudged towards certain choices by aspects of our environment, whether done with intent or not. The ethical influencer, therefore, seeks to ensure their influence is for good rather than for profit or personal agenda; that their influence leads to improved decisions on the target audience's own terms; and freedom of choice is not restricted, unless justified to reduce harms committed to others (as with restricted freedoms through environmental laws, for example). We encourage you to explore what ethical influence means to you in your work and use our principles as helpful guidelines.

MOTIVATE THE CHANGE

Many things motivate us to take action, and these may be in tension with each other. For instance, we may have ambitions to be more sustainable, healthier, more successful, or more charitable. But these aspirations may be in conflict with our deepest desires for pleasure, wealth, security, a sense of purpose, comfort, convenience, happiness, belonging, self-expression, and so on. Often our self-interested side poses a barrier to conservation efforts, since the most enjoyable, convenient, or profitable option may not be the most sustainable. But this part of ourselves also presents an opportunity: Tesla sells sustainable cars partly on status and luxury, and researchers have found that healthy and sustainable food is better sold on the basis of indulgence and pleasure. This raises the

obvious but often overlooked insight that we don't need to promote conservation behaviors solely on the merits of conservation, if we can harness more powerful self-interests such as profit, convenience, security, or enjoyment.

Other motivations lie beneath the surface of our day-to-day awareness but are equally profound. For instance, related to the tension between our self-interests and our more virtuous aspirations, is the desire to make sense of our lives in a consistent, ego-enhancing, and positive light.

This is a task requiring constant psychological trickery to overcome our own hypocrisies: the restaurant-goer opts for the steak despite her concern for the planet; the conservationist still flies on vacation; the fisherman uses the cheaper nets despite knowing they increase by-catch, and so on. As noted previously, we're good at rationalizing these inconsistencies while doing just enough to maintain our sense of integrity. Naturally, we don't like being called out on our shortcomings, revealing another important insight: guilt-based messaging, common in environmental campaigns, causes defensiveness and disengagement. Positive emotions, such as pride, can be more effective at eliciting engagement.

The broader point is that our beliefs and attitudes are not shaped solely by facts or logic, but through a lens of sense-making and self-enhancement: the extent to which something concords with our prior worldview, with our identity, and with the flattering autobiography we hold in our minds. This is a process of story-telling and narrative filtered by what feels personally relevant, and so we must appeal to these ways of thinking when we tell stories of conservation.

All of these motivations, conscious and non-conscious, manifest through myriad decision-making processes which themselves are complex and subject to the influence of bias and emotion. This reveals yet another key insight: campaigns can more effectively motivate change when they harness or address the biases, heuristics, and emotions that dominate our decision-making. A few examples are given in the adjacent box. For instance, we might harness present bias by front-loading the benefits and delaying the costs associated with the adoption of a new conservation program. Many such biases exist and provide novel tools for boosting the impact of conservation campaigns and policies.

Finally, we must recognize the distinction between intrinsic motivations – doing something for its own inherent appeal – versus extrinsic motivation – doing something to achieve some other reward or benefit. The most common way to harness extrinsic motivations is to use financial incentives including taxes, fines, payments for ecosystem services or new pricing schemes. Behavioral insights have a lot to contribute to the design of financial incentives and disincentives, because they invariably embody psychological mechanisms as well as economic ones.

First, extrinsic motivations (e.g., in response to payments or fines) can undermine or 'crowd out' the intrinsic motivation to do the right thing. For example, paying someone to adopt a certain agricultural practice may imply that it is an otherwise undesirable thing to do since it 'deserves' compensation.60 Precisely this effect was found among a Swiss community who were less likely to support the construction of nuclear facility when offered compensation, since payment implied risk.61 For similar reasons, payments to volunteers have been found to undermine their efforts,62 since 'doing it for money' cheapens the act and undercut other motives for doing it: to feel good about themselves, to contribute to a worthy cause, or to gain social kudos by signaling their virtues. This is why symbolic, non-financial rewards can sometimes be more effective at promoting pro-social behaviors. For instance, public recognition amplifies rather than crowds-out these intrinsic motivations for prosocial behavior. This was found to be so among Zambian hairdressers encouraged to hand out condoms to their community: gold stars in the window for each pack they gave out were more effective rewards than a generous financial incentive.

But incentives don't always go wrong. Often, they work as we would intuitively expect, and well-designed extrinsic rewards can 'crowd-in' (strengthen) intrinsic motivations and achieve a stronger effect than would be expected from the financial element alone.

Finally, the biases and heuristics which shape our perceptions and decision-making can also be used to boost the power of conventional financial incentives. For instance, we tend to overweight very small probabilities, and focus on the size of a potential windfall rather than the odds of it happening.

THIRTY BEHAVIORAL SOLUTIONS FOR CLIMATE MITIGATION

Sector	Solution (Optimum scenario ranking)	Description
Food	Reduced food waste	Minimizing food loss and wastage throughout the food supply chain from harvest to consumption
	Plant-rich diets	Eating more plant-based foods and fewer animal proteins and products (e.g., meat, dairy)
	Clean cookstoves	Using cookstoves that burn fuel more efficiently
	Composting	Converting biodegradable waste into a useful soil fertilizer instead of sending it to the landfill
AGRICULTURE AND LAND MANAGEMENT	Silvopasture	Adding trees to pastures to increase productivity
	Tropical staple trees	Growing trees and other perennial crops for staple protein, fats, and starch
	Tree intercropping	Growing trees together with annual crops in a given area at the same time
	Regenerative agriculture	Adopting at least four of the following six agricultural practices: compost application, cover crops, crop rotation, green manures, no-till or reduced tillage, and/or organic production
	Farmland restoration	Restoring degraded, abandoned farmland to grow crops or native vegetation
	Managed grazing	Adjusting stocking rates, timing, and intensity of grazing in grassland soils
	System of rice intensification and improved rice cultivation	Adopting low-methane rice production methods for small or large operations
	Conservation agriculture	Adopting crop rotation, cover crops, and reduced tillage practices on agricultural land
	Nutrient management	Reducing the use of fertilizer use on farmland
	Farmland irrigation	Installing water and energy saving irrigation systems, such as drip irrigation
	Electric vehicles	Driving battery and plug-in vehicles instead of conventional vehicles

TRANSPORTATION	. Ridesharing	Using ride-sharing services and/or carpooling
	Mass transit	Using public transportation for commuting in cities instead of individual vehicles
	Telepresence	Using video-conferencing technologies in place of commercial flights
	Hybrid cars	Driving hybrid cars instead of conventional cars
	Bicycle infrastructure	Biking to destinations in cities instead of using cars
	Walkable cities	Walking to destinations in cities instead of using cars
	Electric bicycles	Using electric bikes for urban transport instead of using cars
ENERGY AND MATERIALS	Rooftop solar	Installing rooftop photovoltaic systems under one megawatt
	Solar water	Using solar radiation to pre-heat or heat water for building use
	Methane digesters	Adopting technologies that produce biogas for household heating through anaerobic digestion of organic waste
	LED lighting	Using energy efficient lighting in households
	Household water saving	Using water saving devices in homes such as low-flow showerheads
	Smart thermostats	Using devices that reduce heating and cooling demand through sensors and settings in the home
	Household recycling and recycled paper	Recycling paper, metal, plastic, and glass materials
	Micro wind	Installing small wind turbines to provide household electricity needs

The urgency of the climate crisis cannot be overstated. Impacts are already being felt around the world from longer heat waves, reductions in crop yields, depleted freshwater reserves, melting ice caps, and irreversible damage to much of the world's ecosystems. If swift action is not taken to reduce emissions over the next several decades, the planet will be locked in to increasingly higher temperatures, the impacts of which will gain in intensity and severity, threatening much of the world's people and natural systems. Most of the world's governments are aware of the existential threat posed by climate change, but their current ambition to mitigate this threat is not nearly enough.

If all nations follow through on climate targets pledged in the Paris Agreement, this would limit the global average temperature rise to 2.6-3.2˚C above preindustrial levels. Existing policies do not yet enable us to reach these levels.200 With temperature changes in this range over the 21st century, the climate impacts are likely to be catastrophic for Earth's systems and much of the global population.201 Though parties to the Paris Agreement will likely increase ambition in coming years, and though the 1.5˚C target is not yet a geophysical impossibility,202 rapid decarbonization is needed across the entire economy to peak emissions in the next several years and reach net-zero by the mid-point of the century.

Global political efforts to achieve these rapid changes will benefit considerably from increased ambition from individuals and communities around the world to reduce overconsumption of natural resources and foster sustainable lifestyles that do not surpass the ecological limits of the planet. But changing behavior to prevent environmental harm and slow global warming is notably difficult because of the often-distant link between individual behaviors and these challenges, many of which develop over decades.

Emerging evidence from new and innovative approaches to changing human behavior give reason for hope. These approaches leverage insights from behavioral science, among other disciplines, to reshape unsustainable patterns of behavior, such as through the 30 solutions presented here. With this list of solutions comes the challenge to encourage adoption using what we know from behavioral insights in order to maximize emissions mitigation. This presents an opportunity to draw on new conceptions and models of why humans behave the way they do and use this knowledge to direct humans toward more sustainable decisions that have numerous additional benefits. We need more evidence on the applicability and effectiveness of these approaches to continue building the case.

As such, a systematic review is needed to identify organizations and their approaches to apply behavioral science for the purpose of changing unsustainable patterns of consumption. Developing this knowledge base and leader network is the best opportunity to support ongoing efforts to rapidly reduce global emissions and thus limit the devastating impacts that will result from global temperature rise. It is imperative that we act now to preserve the planet’s rich natural resources for generations to come. Behavioral science offers us a promising path forward

Lesson-11 Strategies to Encourage Environmentally Responsible Behavior-Contemporary Values and Environmental Attitudes.

For several centuries, the environment has provided habitation for humans and numerous organism but the insatiable needs of humans have driven them to devise strategies for survival and adaptation. Several of these strategies, especially technology, have had direct and indirect negative consequences on the immediate environment, resulting in the degradation of the latter. Many of today's environmental problems are increasingly the outcomes of individual actions, personal consumer decisions, and the activities of small and large businesses. Nevertheless, the fact remains that the healthiness of the world’s economy and people is inextricably bound to the wellbeing of the environment. This implies that now, much more than previously, there is a greater need to understand patterns, connections, systems and root causes of the degrading environment. A very strong tool for nipping the 21st century environmental problems, which are becoming lot more alarming in the bud, is environmental education.

Motivating Action

The likelihood that individuals will behave in environmentally responsible ways reflects a number of factors. It depends on the effectiveness of information they receive, the credibility of the information source, and how the problem and proposed solutions relate to their values and self image. In addition, individuals will be influenced by how they perceive others responding to the challenge and by how costly or inconvenient the problem and the solutions will be.

Getting people to care deeply about adverse climate change is a task that faces major obstacles. Conventional media coverage can get such matters onto the public agenda, but it is less effective in producing strong commitments and prompting effective action. Such change requires that media presentations be a part of a broader, coordinated public education and influence strategy—one that emphasizes not only information, but also motivation and the challenge of converting positive motives and intentions into individual and collective action. Focusing on proven tactics for sustaining behavioral change should be part of the process.

Behavioral research suggests that:

• **Information should be available in multiple forms from multiple sources, targeted to particular audiences' circumstances and concerns.** Given the diversity in consumer circumstances and motivations, multiple appeals from multiple sources are necessary. In the context of residential home improvements, for example, different households have different levels of concern about energy bills, uncomfortable drafts, environmental preservation, and related factors. Knowledge about specific customer priorities is critical in shaping effective communication.

• **Sources should be credible and include celebrities, local officials, community organizations, and social networks of friends, neighbors, and colleagues.** To overcome complacency, lack of interest, or ignorance, the most effective communication strategy is to present easily understandable material from credible sources in inviting, vivid formats. Celebrity and familiar spokespersons can attract popular attention, donor support, and access to policy circles. The most important role models for most people, however, may lie closer to home. In residential improvements, people are most persuaded by personal stories: seeing what friends and neighbors have done, finding out exactly what was involved in doing it, and hearing them recount the benefits and sources of satisfaction they derived from the experience. Such personalized accounts are apt to prove more effective than information available from more official sources.

• **Visual images should be used to enlist media and public attention and evoke concerns about environmental preservation and stewardship for future generations.** Compelling visual images shape individual values directly and indirectly through replication by the media, and do so more effectively than text. They convey, in uniquely powerful form, the global, local and personal implications of climate change.

Messages should be concrete, vivid, and dramatic; they should also be personalized and responsive to the core audience's values and experiences. Individuals tend to be more responsive to concrete, vivid, and dramatic information than to general warnings or appeals. Changes in monthly energy bills or in the cost of filling one's gas tank heighten motivation and provide timely occasions to step up campaigns. Values also play an important role in encouraging conservation behaviors. Research suggests that ethical commitments are more likely to promote sustained, environmentally responsible behavior than short-term financial incentives and that certain measures, such as solar panels on roofs and hybrid purchases, are often adopted out of commitment to environmental preservation and the sense of personal environmentalist identity, even when they yield little or no financial savings.

• **The risks of global climate change are best conveyed in ways that underscore urgency but that do not evoke doomsday scenarios that may result in denial and perceptions of futility**. Research suggests that people generally overestimate the likelihood and seriousness of threat posed by vivid, readily visible or imaginable, catastrophic threats that have low probability while underestimating the likelihood and seriousness of less visible, cumulative, long-term risks. Problems like global warming that lack incontrovertible villains and are partly attributable to policy and lifestyle choices provoke more resignation than indignation. Individuals are also apt to feel that it will be fruitless and quixotic to change their own behavior if others do not change.

• **Highlighting environmentally responsible behavior by peer groups is especially important.** Research suggests that perceptions about the behavior of one's peer group may be the most critical factor in determining whether those beliefs and values get translated into action. In one study, utility customers reduced energy consumption most when they learned that the vast majority of their neighbors were conserving, rather than when they were told about environmental benefits or personal cost savings. As a result, opt-out programs are more effective than opt-in ones. They convey the sense that other citizens are also doing their part to deal with the problem.

• **Messages that emphasize opportunities to avoid losses or negative changes are particularly important.** Individuals are more willing to act to prevent losses than to achieve gains. Research suggests they are more likely to be persuaded if choices are presented in a way that emphasizes unnecessary costs and risks and the savings and other opportunities that will be lost if they fail to act.

• **The focus of any campaign should be on behavior change rather than information, attitudes, or values alone.** Specific behaviors should be targeted, and, where possible, specific opportunities

should be created for individuals to commit themselves to those behaviors. Giving people concrete, convenient strategies to reduce risks is the best way to reduce feelings of anxiety and helplessness.

• **The behavior change initially sought should be made as easy, convenient, and inexpensive as possible. More significant behavior change can be achieved more readily after individuals have made climate concerns a source of personal identity and self-esteem.** Once individuals have agreed to small behavioral changes and taken those steps, they tend to identify with the relevant goals and become more likely to undertake more substantial changes in behavior.

• **Getting people to act on positive environmental values and intentions can be best achieved by face-to-face encounters and requests for personal commitments.** A Canadian study of strategies to reduce summer use of water on residential lawns found that simply providing information on efficient use actually increased consumption by 15 percent. However, when residents in the study received a home visit and were asked to commit to reduced usage, the results were dramatic. Three-quarters of those visited made the commitment in question, and overall water consumption was reduced by more than 50 percent.

To achieve a global sustainable future, individuals must make climate change a priority in both their personal and political behavior. For that to occur, policy makers - government, business and individuals – will need a coordinated strategy informed by the type of behavior research summarized. Careful planning with involvement from researchers and target audiences is necessary to achieve the fundamental behavior and policy changes that reduction of global warming requires.

ENVIRONMENTAL ATTITUDE

Traditional personality theory and assessment tend to focus on various aspects of the self and its relationship to the human environment. They tend to regard human behavior as a fiction of social environment. Environment is the total of the things or circumstances around an organism, including humans. It is not limited to the surrounding neighborhood of a specific place or the neighborhood or vicinity.

At the core of such physical environment related' cognitions is the "environmental past" of the persons. This is a past consisting of places, spaces and their properties which have served instrumentality in the satisfaction of the person's biological, psychological, social and cultural needs. Similarly, modem environment as an objective reality has not yet been proved as truly contributive to well being of the community. It is well known that due to high 'place identity', permanent dwellers of a place, as a community, prefer to accommodate stress in avoiding their rehabilitation in a modernized place.

The personality system refers to the skills, cognitive capacities, values and psychodynamics of the individual agent, while the social system consists of the institutional structures, regulatory principles, technological capacities and adjustment ~mechanisms of a society, It is required to look deeper into environmental needs, traits dispositions, values, attitudes, opinions etc. of professionals who are Environmental Attitudes, Assessments and Preferences engaged in environmental planning, designing and decision-making.

One can assume that psychological factors play an important role in creating and shaping the environment, as do technical, cultural and sociological factors. More research in this direction might help us to understand the development of today's environmental deterioration. Furthermore, this endeavor could clarify better man's environmental needs and thus contribute to improved environmental design. Thus, the following objectives have been considered in this investigation.

Environmental attitudes are conceptualized as an indicator and component of environmental behavior. There are many empirical and theoretical approaches to investigate attitude towards environment. Most studies have been conducted since 1970 onwards.

Dimensionality in Environmental Attitude

Dimensionality was one of the most critical factors of environmental attitude Many studies explored the multidimensionality of environmental attitudes.

Albrecht conceptualized environmental attitudes as a three dimensional concept consisting of the following:

i Environment world view
ii Environmental concerns
iii Environmental commitment

The Strongest and the most consistent predictor of environmental behavior is age. As for gender it is stated that women as compared to men are more concerned with environment. Antecedent factors such as social structural variables have been associated with value orientation, attitudes and environmental behaviors. Women, high level education, youngster with liberal political orientation~ support the principles of sustainable resource management.

Psychology and Environmental Attitudes

The science of psychology can play an important role in understanding and solving these problems. Environmental problems are caused by human behavior, and solving these problems will require changes in behavior.

Psychology, as the science of behavior, can make a substantial contribution to this change. Psychologists can work to develop theories of human cognition and behavior that lead to environmental problems develop and assess program that move us toward sustainability, or study the impact of culture on environmental attitudes, beliefs, or behavior.

At the very core of environmental problems is an individual's understanding of his or her relationship with the natural environment. All cultures that exist today, and throughout history, have addressed the issue of the relationship between individuals and the natural environment (Ponting, 199 1).

Just as every culture has prescribed a set of rules for social interaction, culture also flames the relationship between the individual and nature. As environmental problems have become more apparent, our understanding of the problems, and the solutions we develop to address the problems, occur through the filter of culture. This chapter examines the influence of culture on the beliefs, attitudes, and behaviors of individuals relevant to environmental issues.

Attitudes about Environmental Issues

i) **Attitudes about Environmental Issues in the United States of America**: As the extent and severity of human caused environmental problems became apparent, people in the United States began to take notice. Survey research on attitudes about environmental issues showed a slow but steady increase in recognition and concern about environmental issues.

Beginning in the 1960s and lasting through the 1990s, people in the States expressed increasing levels of concern. Sparked by the first Earth Day in 1970, environmental problems began to emerge as one of the most pressing set of social issues. Although shadowed by the threat of global nuclear war that dominated public concern throughout the 1960s, 70s, and 80s, recognition of the importance and severity of environmental problems continued to grow. Public concern continued to rise fairly steadily throughout the 1970s and 1980s, and reached an all-time peak in the mid-1990s. An example of the high level of environmental concern in the U.S. is a 1990 nationwide survey which found that 71% of respondents believed the U.S. was spending "too little" on "improving and protecting the environment," while only 4% stated "too much" (Dunlap, 1991).

ii) **Attitudes about Environmental Issues in Other Countries**: Concern for environmental issues is not solely a U.S. or Western phenomenon. Recently researchers have found high levels of support for environmental protection from countries around the world. In the most ambitious multi-national study of environmental issues conducted to date, kley Dunlap and the Gallup Organization surveyed approximately 1,000 people in each of 24 countries, which ranged widely in economic development, political structure, and geographic region. The counties were divided into industrialized nations (Denmark, Germany, Finland, Norway, Canada, Ireland, Switzerland, USA, Netherlands, Japan, Great Britain, and Portugal) and developing nations (Mexico, Brazil, IJruguay, Chile, Korea, Philippines, Poland, Russia, Hungary, India, Turkey, and Nigeria). The first question in the survey asked people "What do you think is the most important problem facing our nation?" At this point, respondents did not know anything about the survey and they were not prompted to think about environmental issues. Yet in 16 of the 24 countries, environmental issues were among the top three problems identified; only economic issues were mentioned more often. When asked about their "personal concern about the environment," more than 50% of respondents in all but three of the countries (Switzerland, Turkey, and Poland) stated "a great deal" or a "fair amount." Taken together, these results suggest that concern for environmental issues is high in countries around the world. Even in poorer countries, people are concerned about environmental issues and even favour protecting the environment over economic growth.

Different Types of Attitudes (Egoistic versus Biospheric)

When we think about attitudes toward environmental issues, we often assume that these attitudes fall along a continuum from low (not concerned) to high (very concerned). These attitudes might reflect general concern for environmental issues, or attitudes about specific issues like recycling,

energy conservation, or public transportation. However, research suggests that there are different types of attitudes, and that two people can be equally concerned about environmental problems, but for very different reasons. For psychologists, the more interesting question is not whether or not a person is concerned about environmental Environmental problems (most people are), but why they are concerned.

Research examining the different types of environmental attitudes has attempted to identify the values associated with different concerns. In a survey, many of the respondents indicated that their greatest concern was pollution. Among these, some respondents indicated that their concern was based on the fact that air pollution has been shown to cause cancer, and they (the individuals) didn't want to breath polluted air. Another set of respondents indicated that polluted air will increase the rate of sickness in people around the world and reduce the quality of life for everyone. Finally, a third group of respondents indicated that pollution is killing plants and animals and threatens to disrupt the balance of nature.

Each of these sets of concerns reflects different underlying values. We refer to these as egoistic, altruistic, and biospheric attitudes (Schultz, 2000).

Egoistic concerns are focused on the individual, and reflect a concern about environmental problems for self. These concerns include personal health, financial well-being, quality of life, and availability of resources.

Altruistic concerns focus on people other than self, including friends, family, community, future generations, or humanity.

Finally, biospheric concerns focus on all living things, including plants, animals, ecosystems, and the biosphere.

Lesson-12 Functions of theories; the arousal perspective: Environmental load perspective; Adaptation level theory

THEORIES IN ENVIRONMENTAL PSYCHOLOGY

On the one hand, the interdisciplinary nature of environmental psychology is laudable in that they brought various perspectives to bear on a single phenomenon, leading, it is hoped, to fewer "tunnel-vision" theories, and applicable solutions. It is this very multiplicity that leads to difficulties in the systematic application of knowledge and the development of well-articulated unifying theories.

Historical Influences

Theories have influenced thinking among environmental psychologists both within and outside of the discipline of psychology. Some of these theories are very broad in scope, whereas others are more focused; some are lacking in empirical basis and others are more data-based. We will review several of them to provide the context for a consideration of current theories of environment-behaviour relationships. These perspectives include: *Geographical Determinism, Ecological Biology, Behaviourism* and *Gestalt Psychology*.

Some historians and some geographers have attempted to account for the rise and fall

of entire civilizations based on environmental characteristics. For example, Toynbee (1962) theorized that the environment (specifically, topography, climate, vegetation, availability of water, etc.) presents challenges to its inhabitants. Extreme environmental challenge leads to the destruction of a civilization, whereas too minor challenge leads to stagnation of culture. Thus, Toynbee proposed that an intermediate level of environmental challenge enhances the development of civilizations, and extremely diminished or excessive levels are debilitating. The notion of environmental challenge and behavioural response, although rooted in the thinking of such geographical determinists, appears often in one form or another in various theories in environmental psychology.

As one example, Barry, Child, and Bacon (1959) have suggested that agricultural, non-nomadic cultures seem to emphasize responsibility, obedience, and compliance in child-rearing practices, whereas nomadic cultures often emphasize independence and resourcefulness. These differences, they suggest, result from the fact that people who live and work together in organized non-mobile communities require a more structured organization and therefore stress the importance of obedience and compliance. Nomads inculcate independence and resourcefulness in preparation for meeting the changing and unpredictable demands of an

environment confronted by "roaming" people. Thus, the argument goes, the environment sets the stage for the development of cultures having the best chance of surviving it.

This does not differ from saying that ghetto culture cultivates a set of skills in its inhabitants that are best suited to the ghetto. Someone not skilled in the craft of street fighting, perhaps, is not likely to survive the environment of the streets. We will see later in this Unit that some have carried this argument to the point, and other institutional environments develop in their inhabitants that are ostensibly survival certified to these environments, yet to be maladaptive in the larger environment.

The development of ecological theories, theories concerned with biological and sociological interdependence between organisms and their environment, have also significantly influenced thinking in environmental psychology. With the development of ecological science, organisms were no longer viewed to be separate from their environment but were integral to it. This notion of organism-environment reciprocity now appears in many current environment-behaviour theories. The environment and its inhabitants are still often studied as separate components, but no one doubts their interdependency. These various components constitute a total system and interactions between the components are assumed to bring the change.

Arousal Theories

Arousal theories have typically been concerned with the influence of arousal on performance. They maximize performance at intermediate levels of arousal but falls off as arousal is increased or decreased. This relationship sometimes referred to as an inverted-U relationship, has been shown to differ slightly depending on whether performance is measured on simple or complex tasks (see Figure 3.1) and is often referred to as the Yerkes-Dodson law. These relationships are consistent with other findings that humans seek intermediate levels of stimulation (Berlyne, 1974), and are reminiscent of Toynbee's assertion, referred to earlier, that cultures only develop in environments that provide intermediate environmental challenges.

It has also shown performance changes that are very curvilinear with temperature increases. One explanation of these findings is that increase in ambient temperature leads to increase in arousal levels. Initially, the higher arousal leads to performance enhancement, but as it increases further, over-arousal occurs, causing performance decrements. Similarly, it has been shown that personal space invasions lead to increases in arousal (Middlemist, Knowles, & Matter, 1976) and performance decrements (Evans & Howard, 1972; McBride, King, & James; 1965). Increases in noise level have been associated with changes in arousal and performance (Evans & Cohen, 1987). Thus, several variables associated with arousal changes are related to performance changes, and performance has consistently been curvilinear related to arousal.

Other theorists using an arousal perspective have featured physiological responses to environmental stimulation. Changes in heart rate, blood pressure, respiration rate, galvanic skin response, and adrenaline secretion, among others, have been shown to occur with changes in the environment. Increased ambient temperature leads to blood vessel dilation, perspiration, increased heart rate, and, in extreme conditions, lowered blood pressure and insufficient oxygen supply to the brain. It has linked personal space invasion to delayed onset and shorter duration of maturation for males. And noise exposure alters blood pressure, heart rhythm, and the flow of gastric juices to the stomach.

Neurologists, such as Hebb (1972) have linked arousal with increased activity of the reticular activating system of the brain. Still, other theorists have equated arousal with changes in motor activity or with self-report of arousal. Berlyne (1974) has, for example, characterized arousal as lying on a continuum anchored at one end by sleep and the other by excitement, and Mehrabian and Russell (1974) have identified arousal as a major component in people's affective responses to their environment.

Independent of the orientation taken concerning arousal, several consistencies are apparent:

1. The associated changes in arousal with changes in the environment
2. Pleasant as well as unpleasant stimulation increases arousal — that is, room temperatures above 100 degrees Fahrenheit and obnoxious noises influence arousal in ways similar to a roller coaster rides and such other stimulating situations.

3. Changes in arousal lead people to seek information about their internal states(Schacter & Singer, 1962) as well as to seek information from others (Festinger,1954).

4. People evaluate moderate levels of arousal positively.

5. Often, great expenditures of energy are used by individuals to bring the environment to alevel of moderate stimulation.

Performance is predicted to be optimal for both simple and complex tasks at intermediate levels of arousal. Arousal above the level, leads to decrements in performance.

Factors affecting Arousal Level

Furthering the point, various researchers have tried to identify the impact of different environmental factors on the arousal level:

i. **Temperature**An increase in ambient temperatures leads to increases inarousal levels. Usually, higher arousal leads to performance enhancement, but as it increases, further over-arousal occurs, causing performance decrements.Increased ambient temperature leads to the following physiologicalchanges: blood vessel dilation, pupil dilation, perspiration, increased heartrate. Extreme conditions also lead to lowered blood pressure and insufficientoxygen reaching to the brain.

ii. **Personal Space Invasion**An increase in personal space invasion leads to an increase in arousal and performance decrements (Middlemist, Knowels&Matter 1976).

iii. **Noise Level**An increase in noise level leads to an increase in arousal and performance decrements(Evans & Kohen 1987). It alters blood pressure,heart rhythm, and the flow of gastric juices to the stomach.

Arousal and Nervous System

It linked arousal to increased activity of the reticular activating system of the brain andthe associated changes in arousal with changes in the environment. Pleasant as well asunpleasant stimulation increases arousal, i.e. room temperatures above 100 degrees Fahrenheit. Changes in arousal lead people to seek information about their internalstates. People associate moderate levels of arousal positively. Individuals use great expenditure of energy to bring the environment to a moderate level of stimulation.

Stimulus Load Theories

Central to stimulus load theories is the notion that humans have a limited capacity toprocess information. When inputs exceed that capacity, people ignore some inputsand devote more attention to others (Cohen, 1978). These theories account for responses to environmental stimulation in terms of the organism's momentary capacityto attend to and deal with salient features of its milieu.The stimuli most important to thetask are in the focus of attention whereas the others are in the margin.For example,while driving during heavy traffic, we invest attention in the vehicles and road signsaround us, and we pay less attention to the commentator on the car radio, people inthe back seat, or the clouds in the sky. If the less important stimuli interfere with thetask at hand, then ignoring them will enhance performance; for instance, ignoring the children's fighting will make you a better and safer rush-hour driver. If, however, the less important stimuli are important to the task at hand, then performance will not beoptimal; for instance, ignoring the road signs because you are attending to the moreimportant trucks, cars, etc., may lead you thirty miles out of your way in getting home.

Sometimes the organism's capacity to deal with the environment is overtaxed or evendepleted. When this occurs, we attend only the most important information, with all other information filtered out.Once attentional capacities have been depleted, even small demands for attention can be draining. Thus, behavioural aftereffects, includingerrors in judgment, decreased tolerance for frustration, ignoring others in need of help, can be accounted for by these theories. For example, the exhausted rush-hour driver eventually might reach the point where he or she doesn't notice the traffic light turnfrom red to green (or worse yet, from green to yellow to red), even though this is averyimportant stimulus.

Decreased tolerance for frustration may lead to "laying on the horn" or "lane hopping"and motorists in the break-down lane may be ignored, if not looked upon with disdain.

Wohlwill (1966) has argued that environments should be depicted in terms of

measurements applied to the dimensions of intensity, novelty, complexity, temporal variation, surprisingness, and incongruity, all of which contribute to stimulus load. Subsequent behaviours can then be related to the stimulus properties of environments in systematic and comparable ways.

Behaviour Constraint Theories

Behaviour constraint theories focus on the real, or perceived, limitations imposed on the organism by the environment. According to these theories, the environment can prevent, interfere with, or limit the behaviours of its inhabitants (Rodin & Baum, 1978; Stokols, 1978).

The development of the behavioural constraint model comes from Brehm's (1966) reactance theory which states that there is a basic desire among human beings to maintain their behavioural freedom. An environment is constraining when something is limiting or prevents an individual from achieving his intentions. Certain environmental conditions like noise, crowding, temperature, location or specific features such as bad weather, barriers, objective experiences such as control may be perceived as constraining by the individual. In such a situation, the person feels as being out of his/her control. This feeling of not being able to control the situation produces psychological reactance (Brehm, 1966). Wortman and Brehm (1975) suggest that if failures or loss of control persist for a longer period of time without any chances of restoration, the individual will cease to make any instrumental efforts and thus enter into a helplessness state.

Allen and Greenberger (1980) worked extensively in this area and suggested that before entering into a state of helplessness, an individual will try to alter his/her physical environment, if the previous effects of modifying the social environment did not prove successful. For instance, while trying to teach in a noisy classroom environment, the teacher first tries to request the children to remain quiet before asking the naughty ones to move out of the class. This modification if expressed in the form of personalization or any other social change will be in a socially approved and benign manner, else could also resort to drastic causes like intentional damage to the environment. Breaking of school furniture, switchboards, vandalizing or looting of public property during riots are certain examples of these negative form of perceived control.

It has also been suggested that boundary mechanisms are crucial to an individual and one sets up certain regulatory systems to regain certain amount of control. So for instance, when someone invades our personal space during the course of a conversation, we try to regain it by moving a step further away or displaying some other behaviour like crossing our arms and the like. We may sometimes invent systems to prevent or reduce the chances of invasion. Sitting on one edge of an empty park bench or trying to get the aisle seat while travelling are examples to prevent encroachment. Very few of us will sit in the centre, or even if we do we spread out personal belongings on either side of us

An important aspect of everyday life of an individual is having a sense of controlling the environment which induces a sense of well being. When people perceive control in a noisy situation, like by putting on ear plugs (Glass & Singer, 1972); they are less aggressive (Donnerstein & Wilson, 1976; Moser & Levy-Leboyer, 1985); and they are more often helpful (Sherrod & Doves, 1974). On the contrary, the perception of loss of control leads to manifestation of negative consequences on behaviour, well-being and health (Barnes, 1981).

Averill (1973) has discussed three categories of control: (a) the behavioural control which provides the subject with the opportunity to change the disturbing environmental events, (b) through cognitive control the individual adopts reappraisal technique to perceive the environment as less threatening and (c) the decision control provides the subject with an opportunity to choose the best possible alternative.

The behavioural constraint model is very effective in explaining human behaviour in terms of control component. However, this approach alone cannot explain the environment-behaviour relationship in an effective manner.

Adaptation-Level Theories

Adaptation level theories assume that each of us has a minimum threshold for a certain level of environmental stimulus. A slight rise or fall in this threshold is a focus of arousal, overload, under load or stress. Variations in behaviour will therefore occur if the stimulus load is outside our experienced adaptation levels. Major proponents of this position include: Helson (1964) and Wohlwill (1974). While all environmental psychologists emphasize the interrelationship of humans to their environment, adaptation-level theorists speak specifically of two processes that make up this relationship-the processes of adaptation and change. Organisms either adapt (i.e., change their response to the environment) or they adjust (i.e., change the

environment with which they are interacting). Adaptationto decreases in ambient temperature includes piloerection (hair on the body standingup or what is commonly called getting "goose pimples"), muscle rigidity, increasedmotor activity, vasoconstriction; adjustments include throwing another log on the fireor turning up the thermostat. Either process brings the organism back to equilibriumwith its environment.

Wohlwill(1974) distinguished three categories of stimulation: sensory, social andmovement; along the three dimensions of intensity, diversity and patterning. Amongstthese; optimal or intermediate level of arousal or stimulation is desirable for positiveresponses and task performance. For instance, if aloofness and gregariousness are thetwo poles of a social interaction scale then either of them can be desired by a subject at some span of time depending on the level of exposure he/she is facing. The urge tospend some time alone or nowadays known as "My time" either in a quiet resort or inan aloof place by the working young generation of today is probably getting its ideasfrom this theory only.

Another value of this approach is that it recognizes individual differences in adaptationlevel (i.e., the level of stimulation/arousal that the individual has become accustomed to and expects or desires in an environment). Thus, this approach can explain the different responses of two individuals to the same environment. For example, we may perceive a boisterous partyas pleasant to a person high in need for sensation, but as overwhelmingfor the person who prefers a low level of sensation. By the same token, some peoplerevel in the crowded atmosphere of last-minute festival shopping, while others abhorthe inconvenience of having two or more shoppers in the same store with them. Theseindividual differences in adaptation level lead to quite different behaviours. The personhigh in need for sensation will seek boisterous parties, whereas the person preferringlow levels of sensation would avoid them or seek havens of solitude within them. Wecan ascribe some differences in their behaviours to differences in their adaptation level.Inyet another study, it was observed that when concentrating on important tasks, thesubjects were perturbed both by the presence of a persisting noise as well as by thecomplete lack of it. Thus an optimal level of stimulation is desirable for an appreciabletask performance. For instance some students like to hear some soothing music ormantras while studying or concentrating on important tasks.

However there may be some variations in the optimal level in different individualsdepending on their experience and proximity with a given environment. For example: the heat and temperature of a bangle factory can be too overpowering for a person onentering for the first time; while the workers of the place are quite immune to it. Similarly, people of hilly areas when settled in lowlands have a high resistance to cold temperatureas compared to the native residents of that place. This variation in optimal level istermed as adaptation. According to Wohlwill(1974), adaptation is a quantitative shiftin the distribution of judgmental or affective responses along a stimulus continuum as a function of continued exposure to a stimulus.Dubos(1980) defines adaptation as theprocess which allows people to adjust to their environment, variations in the perception of the environment and coping mechanisms undertaken to overcome the problems it creates. Adaptation involves physiological as well as psychological processes. The phenomenon of light and dark adaptation accomplished by the human eye is a classic example of physiological adaptation to intense stimuli of light. An individual previouslyresiding in an area close to railway station may perceive his/her new house in the centreof the city as much quieter, is an example for setting a psychological benchmark ofadaptation for evaluating new conditions.

The process of defensive re-appraisal or defensive re-evaluation can be very useful indealing with stress emancipating from the extreme environmental stimuli. In thisreappraisal, the environmental conditions which were once perceived as overwhelming or threatening ; over a period of time maybe re-evaluated as bearable or less threatening.Defensive re-appraisals are self generated; they arise from needs within the personrather than from environmental pressures. In a study conducted on a flock of sheep, itwas observed that when the paddock of the sheep was near the highway, initially thesheep used to run hither-thither on the arrival of any motor vehicle. However with thepassage of time the approaching vehicles were perceived as less threatening and theyused to continue with their grazing. Appraisal focused strategies occur when personsmodify the way they think, for example, employing denial or distancing oneself fromthe problem. People may also alter the way they think about a problem by altering theirgoals, values such as by seeing humour in a situation. So, defensive reappraisal consists of any effort made to interpret the past more effectively or to deal with present harmsand threats by viewing them in a less damaging and threateningway.

Sometimes subjects are seen to resort to behavioural adaptations. This kind of adaptation can be defined as the response a person gives to environmental stimulusand a change in that corresponding behaviour. In India, for instance, adjusting to therituals and customs by the bride (being different from her

family) after her marriage. Similarly, when an individual emigrates to a new country, he/she has to cut down on his traditions to follow the traditions of the new country.

The theory of adaptation presents an active and dynamic relation of the individual with his/her environment. However, the postulates of this theory cannot be generalized as it does not explain variation in adaptation levels across individuals, so precise prediction of behaviour with respect to environment becomes difficult.

1.

Lesson-13 Environmental stress perspective.

Environmental Stressors

Environmental stressors (stressors) are factors whose influence is to constrain productivity, reproductive success, and ecosystem development (see Chapter 9). To some degree, stressors affect all organisms as well as their populations, communities, and ecoscapes (landscapes and seascapes). Stressors may be natural in origin, being associated with such environmental influences as:

- competition, predation, disease, and other interactions among organisms
- constraints related to climate or to inadequate or excessive nutrients, moisture, or space
- disturbances such as wildfire and windstorms

The effects of natural stressors are not always negative. Some individuals, populations, and communities may benefit from the effects of natural stress, even while others suffer a degree of damage.

Increasingly, however, stressors associated with human activities are the most critical influence on species and ecosystems. In too many cases, anthropogenic stressors are causing important damage to resources that are needed to sustain people and their economy, and also to natural biodiversity and ecosystems.

Environmental stressors may occur as an intense, short-lived event of destruction, also known as a disturbance. Alternatively, stressors may exert their influence over an extended period of time – that is, in a chronic manner. The interaction of organisms with a stressor at a particular place and time is called exposure. Exposure can be instantaneous or it may accumulate over time. If an exposure is intense enough, it will cause some sort of biological or ecological change, called a response, to occur. It is important to understand, however, that individuals, populations, and communities are capable of tolerating a range of intensity of stressors without suffering significant damage. In other words, certain thresholds of biological or ecological tolerance must be exceeded before damage is caused.

Stressors are environmental factors that affect organisms and ecosystems. They may exist at varying intensities of exposure, as is suggested by the metaphor of a water faucet, whose dial if turned to the right will increase the flow of water, or decrease or stop it if turned the other way. For actual stressors, if a threshold of biological or ecological tolerance is exceeded, then a response will occur.

Damage occurs when one or more stressors elicit responses that can be interpreted as a degradation of environmental quality. Such responses may include illness or death caused by an exposure of wild animals to pesticides, or as a reduction of the productivity of ecosystems, or the endangerment of vulnerable elements of biodiversity. In this chapter, we examine a conceptual framework for the study of damage caused by stressors. In the following 11 chapters, we deal with specific kinds of stressors and examine case studies of the kind of damage they may cause to occur.

Kinds of Stressors

The diverse kinds of environmental stressors are grouped into classes, although they are not entirely exclusive.

- Physical stress is a disturbance in which there is an intense exposure to kinetic energy, which causes damage to habitats and ecosystems. Examples include such disruptive events as a hurricane or tornado, a seismic sea wave (tsunami), the blast of a volcanic eruption, an explosion, or trampling by heavy machinery or hikers.
- Wildfire is another disturbance, which involves the uncontrolled combustion of the biomass of an ecosystem. A wildfire can be ignited by people, or naturally by lightning. A severe fire consumes much of the biomass of an ecosystem, but even a less-severe wildfire may kill many organisms by scorching and poisoning by toxic gases.
- Chemical pollution occurs when one or more substances occur in a concentration high enough to elicit physiological responses in organisms, potentially causing toxicity and ecological change. Chemical stressors include pesticides, gases such as ozone and sulphur dioxide, and toxic elements

such as arsenic and mercury. Pollution may also be caused by excessive nutrients, which can distort productivity and other ecological functions. Note that the mere presence of a potentially toxic agent does not necessarily cause pollution. (The distinction between contamination and pollution is examined later in this chapter.)

- Thermal pollution is caused by the release of heat (thermal energy) into the environment, which results in ecological stress because species vary in their tolerance of temperature extremes. Thermal stress may occur at natural springs and submarine vents where geologically heated water is emitted. It is also associated with discharges of hot water from power plants.
- Radiation stress is caused by excessive exposure to ionizing energy. The radiation may be emitted by nuclear waste or explosions, or it can be diagnostic X-rays or solar ultraviolet energy.
- Climatic stress is associated with insufficient or excessive regimes of temperature, moisture, solar radiation, wind, or combinations of these.
- Biological stressors are associated with interactions occurring among organisms, such as competition, herbivory, predation, parasitism, and disease. For example, individuals of the same or different species may compete for essential resources that are limited in supply. Herbivory, predation, parasitism, and disease are trophic interactions, in which one species exploits another. Exploitation can be anthropogenic, as when humans harvest wild animals or trees, or it can be natural, perhaps associated with defoliating insects or disease-causing pathogens.
- Biological pollution occurs when people release organisms beyond their natural range. This might involve the introduction of alien species that invade and alter natural habitats, or it may be the release of pathogens into the environment through discharges of raw sewage.

Biological "pollution" is caused when species are introduced into habitats beyond their natural range, where they may cause ecological damage. This non-native lupine (Lupinus polyphyllus) has been introduced to eastern Canada, where it thrives in gardens and along roadsides. Although an attractive wildflower, it displaces indigenous plants.

Ecological Responses

An ecosystem that has been affected by a disturbance typically suffers mortality among its species, along with damage to its structural properties (such as species composition and biomass distribution) and functional attributes (such as productivity and nutrient cycling). Once the disturbance event is over, a process of recovery through succession begins. If the succession proceeds for a long enough time, it will restore another mature ecosystem, perhaps one similar to that existing before the disturbance.

Chronic stressors operate over longer periods of time (rather than as events), and they include climatic factors and many kinds of chemical and thermal pollution. Depending on the intensity of exposure, organisms may suffer acute toxicity resulting in tissue damage or even death, or a less-obvious chronic damage that results in decreased productivity.

Exposure to a higher intensity of environmental stressors can result in evolutionary changes if individual organisms vary in their tolerance and those differences are genetically based. Under such conditions, natural selection in favour of tolerant individuals will eventually result in increased tolerance at the population level. At the community level, relatively vulnerable species will be reduced or eliminated from the habitat if the intensity of stress increases markedly. The niches of those species may then be occupied by more tolerant members of the community, or by invading species that are capable of exploiting a stressful but weakly competitive habitat.

A prolonged intensification of stress will cause longer-term ecological change to occur. Consider, for example, a case in which a new metal smelter is constructed in a forested landscape. If the smelter emits toxic sulphur dioxide gas, the toxic stress will damage the tree-sized plants of the forest and eventually cause them to give way to shrub-sized and herbaceous vegetation. If the long-term stress is extremely severe, the landscape could entirely lose its vegetation. This kind of damage has actually occurred around a number of Canadian smelters, such as those near Sudbury (Chapter 16).

This kind of ecological damage involve changes in the composition and dominance of species in communities, in the spatial distribution of biomass, and in functions such as productivity, litter decomposition, and nutrient cycling. Because a smelter is a discrete point source of environmental stress, the ecological responses eventually stabilize as gradients of community change that radiate outward, in a downwind or downstream direction from the source of pollution.

The intensity of a stressor may also decrease in time and space. When this happens, the ecological responses are, in many respects, the reverse of the damage that occurs when the stress intensifies. These changes represent a process of recovery through succession. In the case of the Sudbury smelters, emissions of pollutants have decreased greatly because of the installation of pollution-control technologies. This has resulted in much less toxic stress in the surrounding environment, which has allowed some ecological recovery to occur (Chapters 16 and 18).

Ecologists have described the general attributes of ecosystems that have been subjected to severe stress for a period of time. In general, as environmental stress intensifies significantly (such as by increasing pollution), the following changes are observed:

- mortality increases, especially of the most vulnerable species
- species richness decreases
- the stocks of nutrients and biomass become depleted
- the rate of community respiration exceeds that of production, so the net production becomes negative.
- sensitive species are replaced by more-tolerant ones
- top predators and large-bodied species may be lost from the ecosystem
- previously self-maintaining ecosystems may require active management to sustain their desirable attributes, for example, to maintain declining populations of rare or economically valuable species that have become threatened

Ecosystems that are chronically exposed to intense stress (such as climate-stressed tundra) eventually stabilize. Typically, the stable ecosystems are low in species richness, simple in structure and function, and dominated by relatively small, long-lived species. As well, they have low rates of productivity, decomposition, and nutrient cycling.

If an increase in environmental stress has an anthropogenic causation, then the resulting ecological changes are often considered to represent damage and are viewed as a degradation of environmental quality and ecological integrity

UNIT – IV: PSYCHO-EDUCATIONAL ASPECTS OF ENVIRONMENTAL PROTECTION

Lesson-14 Environmental education and laws of environmental Protection

Environmental education and legislation are crucial in promoting sustainability and protecting the natural environment. Environmental education refers to the process of educating individuals and communities about environmental issues and promoting sustainable practices. Environmental legislation refers to the regulations, laws, and guidelines established by governments and organizations to address environmental issues and promote sustainability. Understanding the importance, types, and issues of environmental education and legislation is essential in promoting a sustainable future.

Meaning and History of Environmental Education and Legislation

Environmental education refers to the process of educating individuals and communities about environmental issues and promoting sustainable practices. The concept of environmental education has been recognized for centuries, with historical accounts of environmental education dating back to ancient civilizations. The modern environmental education movement began in the 1960s and 1970s, with the passage of major environmental laws such as the Clean Air Act and the Clean Water Act. Environmental legislation refers to the regulations, laws, and guidelines established by governments and organizations to address environmental issues and promote sustainability.

Types of Environmental Education

- **Formal Education:** Formal education refers to environmental education that occurs in formal educational settings, such as schools and universities. This type of education often includes courses and programs focused on environmental issues and sustainability.
- **Non-formal Education:** Non-formal education refers to environmental education that occurs outside of formal educational settings, such as community programs and workshops. This type of education often includes hands-on learning and practical skills development.

- **Informal Education**: Informal education refers to environmental education that occurs through everyday experiences, such as learning about recycling or energy conservation at home or in the workplace.

Examples of Environmental Education

- **Environmental Education in Schools:** Environmental education programs in schools can include courses and extracurricular activities focused on environmental issues and sustainability. These programs can help to promote environmental awareness and sustainable practices among students.
- **Community Environmental Programs:** Community environmental programs, such as recycling and composting programs, can help to promote sustainable practices among community members.
- **Workshops and Conferences:** Workshops and conferences focused on environmental issues and sustainability can provide opportunities for individuals to learn about new technologies and approaches to environmental protection.

Environmental Education and Legislation Issues

- **Funding:** Funding for environmental education and legislation can be a significant issue, with limited resources available for these programs.
- **Political Polarization:** Environmental education and legislation can be politicized, with some political groups opposed to environmental regulations and others advocating for more stringent regulations. This political polarization can lead to delays in policy development and implementation.
- **Limited Resources:** Governments and organizations may not have the financial or technical resources necessary to develop and implement effective environmental education and legislation programs.

Types of Environmental Legislation

- **Command-and-Control Regulations**: Command-and-control regulations involve setting specific limits on pollution levels and enforcing those limits through fines and other penalties. This type of policy is often criticized for being inflexible and not allowing for innovation in pollution reduction.
- **Economic Incentives:** Economic incentives involve using financial incentives, such as taxes or subsidies, to encourage environmentally friendly behavior. This type of policy is often more flexible than command-and-control regulations and can lead to more innovative solutions.
- **Voluntary Agreements:** Voluntary agreements involve agreements between industry and government to reduce pollution levels voluntarily. This type of policy is often criticized for not being enforceable and not providing sufficient incentives for compliance.

Examples of Environmental Legislation

- **The Clean Air Act:** The Clean Air Act is a major environmental law in the United States that sets standards for air quality and regulates emissions from industry and transportation. The Clean Air Act has been credited with significant improvements in air quality in the United States.
- **The Clean Water Act**: The Clean Water Act is a major environmental law in the United States that regulates discharges of pollutants into navigable waters and sets water quality standards for surface waters. The Clean Water Act has been credited with significant improvements in water quality in the United States.
- **The Endangered Species Act:** The Endangered Species Act is a major environmental law in the United States that aims to protect endangered and threatened species and their habitats. The Endangered Species Act has been credited with saving many species from extinction.

Environmental Legislation Issues

- **Implementation and Enforcement:** Implementation and enforcement are significant issues in environmental legislation. Many environmental regulations are not effectively enforced, leading to noncompliance and continued environmental degradation.
- **Political Polarization:** Environmental legislation has become increasingly politicized, with some political groups opposed to environmental regulations and others advocating for more stringent regulations. This political polarization can lead to delays in policy development and implementation.
- **Limited Resources:** Limited resources can be a significant issue in environmental legislation. Governments and organizations may not have the financial or technical resources necessary to develop and implement effective environmental policies.

Strategies for Effective Environmental Education and Legislation

- **Collaboration:** Collaboration between governments, organizations, and industry is essential in developing and implementing effective environmental education and legislation. Collaboration can lead to more innovative solutions and greater support for environmental regulations.

- **Innovation:** Innovation is crucial in developing effective environmental education and legislation. New technologies and approaches can lead to more efficient and effective environmental regulations.
- **Education and Awareness:** Education and awareness are crucial in promoting environmental education and legislation. Educating the public on the importance of environmental protection and sustainability can lead to greater support for environmental regulations.
- **Inclusivity:** Inclusivity is essential in developing effective environmental education and legislation. Ensuring that all stakeholders have a voice in the policy development process can lead to greater support for environmental regulations.

Environmental Laws in India

Environmental laws are an important part of any governance body. It comprises a set of laws and regulations concerning air quality, water quality, and other aspects of the environment.

The environmental laws in India are guided by environmental legal principles and focus on the management of specific natural resources, such as forests, minerals, or fisheries.

The environmental laws in India are a direct reflection of what was envisaged in the constitution. The need for protection and conservation of the environment and sustainable use of natural resources is reflected in the constitutional framework of India and also in the international commitments of India.

History of environmental laws in India:

The detailed and developed framework for environmental protection came after the UN conference on Human Environment in Stockholm, in 1972.

This led to the formation of the National Council for Environmental Policy and Planning in 1972 within the science and technology department.

This was set up to establish a regulatory body for the overview of the environmental-related issues and concerns.

This council was later converted to the Ministry of Environment and Forests.

The government of India has made numerous acts to protect the environment and biodiversity. The important and impactful environmental laws and acts are listed and explained below.

(1) The Wildlife (Protection) Act, 1972

The Act provides for the protection of wild animals, birds, and plants; and for matters connected therewith or ancillary or incidental thereto. It extends to the whole of India.

It has six schedules that give varying degrees of protection:

- **Schedule I and part II of Schedule provide absolute protection**, offences under these are prescribed the highest penalties.
- Species listed in **Schedule III and Schedule IV** are also protected, but the **penalties are much lower**.
- Animals under **Schedule V**, e.g. common crows, fruit bats, rats, and mice, are legally considered **vermin** and may be hunted freely.
- The specified **endemic plants in Schedule VI are prohibited from cultivation and planting.**

Statutory bodies under WPA:

1. National Board for Wildlife and state wildlife advisory boards

2. Central Zoo Authority
3. Wildlife Crime Control Bureau
4. National Tiger Conservation Authority

(2) The Water (Prevention and Control of Pollution) Act, 1974

Objective: To provide prevention and control of water pollution. Maintaining or restoring of wholesomeness and purity of water in the various sources of water.

It vests regulatory authority in **Centre Pollution Control Boards (CPCB)** and **State Pollution Control Board (SPCB).**

CPCB and SPSB are statutory bodies created under the Water Act, 1974. It empowers CPCB and SPCB to establish and enforce effluent standards for factories discharging pollutants into water bodies.

CPCB performs these same functions for union territories along with formulating policies related to the prevention of water pollution and coordinating activities of different SPSBs.

SPCB controls sewage and industrial effluent discharge by approving, rejecting, and granting consent to discharge.

(3) The Air (prevention and control of pollution) act, 1981

The act targets to control and prevent air pollution in India and its main objectives are:

- To provide for prevention, control, and abatement of air pollution.
- To provide for the establishment of the boards at the central and state levels to implement the act**.**

CPCB and SPCB were given the responsibility.

It states that the sources of air pollution such as internal combustion engines, industry, vehicles, power plants, etc., are not permitted to release particulate matter, lead, carbon monoxide, sulfur dioxide, nitrogen oxide, volatile organic compounds (VOCs), or other toxic substances beyond the predetermined limit.

It empowers the state government to designate air pollution areas.

(4) The Environment (Protection) Act, 1986

This act was passed under **article 253 (legislation for giving effect to international agreements)**

This was passed in the wake of the **Bhopal gas tragedy** in December 1984.

It was enacted to achieve the UN conference on the human environment, 1972- Stockholm declaration**.**

Eco-sensitive zones or ecologically fragile areas are notified by MoEFCC under EPA, 1986 – 10 km buffer zones around protected areas.

Statutory bodies under the EPA, 1986:

1. **Genetic Engineering Appraisal Committee**
2. **National Coastal Zone Management Authority** (later converted to **National Ganga Council** under Ministry of Jal Sakthi)

The ozone-depleting substances (regulation and control) rules, 2000.

It set deadlines for phasing out of various Ozone Depleting Substances (ODSs) and regulating production, trade import, and export of the product containing ODS.

These rules prohibit the use of CFCs, halons, ODSs such as carbon tetrachloride and methyl chloroform, and SFC except in metered-dose inhalers and for other medical purposes.

Coastal Regulation zone notification 2018:

It was notified based on the recommendations of the **Shailesh Nayak Committee.**

To promote sustainable development while taking into account the natural hazards such as increasing sea levels due to global warming.

To conserve and protect biodiversity besides livelihood security to local communities including the fishermen.

CRZs have been classified into 4 zones for regulation:

1. **CRZ I**– **ecologically sensitive areas such** as mangroves, coral reefs, salt marshes, turtle nesting ground, and the inter-tidal zone.
2. **CRZ II**– **areas close to the shoreline**, and which have been developed.
3. **CRZ III- Coastal areas that are not substantially built up**, including rural coastal areas.
4. **CRZ IV- water area from Low Tide Line (LTL) to the limit of territorial waters of India.**

(5) The energy conservation act, 2001

It was enacted as a step towards improving energy efficiency and reducing wastage. It specifies the energy consumption standards for equipment and appliances.

It prescribes energy consumptions norms and standards for consumers. It prescribes energy conservation building codes for commercial buildings.

Bureau of energy efficiency (BEE) is a statutory body established under the act.

(6) Biological diversity act 2002

It was implemented to give effect to **CBD, Nagoya Protocol**.

To check biopiracy, protect biological diversity, and local growers through a three-tier structure of central and state boards and local committees.

To set up National Biodiversity Authority (NBA), State Biodiversity Boards (SBBS), and Biodiversity Management Committees (BMCS).

(7) Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 (FRA)

The act recognizes and vests the forest rights and occupation in forest land in **Forest Dwelling Scheduled Tribes (FDST)** and Other **Traditional Forest Dwellers (OTFD)**who have been residing in such forests for generations. This act comes under the aegis of the Ministry of Tribal Affairs.

The act also establishes the responsibilities and authority for sustainable use, conservation of biodiversity, and maintenance of the ecological balance of FDST and OTFD.

It strengthens the conservation regime of the forests while ensuring the livelihood and food security of the FDST and OTFD.

It seeks to rectify colonial injustice to the FDST and OTFD who are integral to the very survival and sustainability of the forest ecosystem.

The act identifies four types of rights:

1. Title rights

It gives FDST and OTFD the right to ownership of land farmed by tribals or forest dwellers subject to a maximum of 4 hectares.

Ownership is only for land that is being cultivated by the concerned family and no new lands will be granted.

2. Use rights

The rights of the dwellers extend to extracting Minor Forest Produce, grazing areas, pastoralist routes, etc.

3. Relief and development rights

To rehabilitation in case of illegal eviction or forced displacement and basic amenities, subject to restrictions for forest protection

4. Forest management rights

It includes the right to protect, regenerate or conserve or manage any community forest resource which they have been traditionally protecting and conserving for sustainable use.

(8) The National Green Tribunal Act, 2010

It was established in concurrence to **Rio Summit 1992** to provide judicial and administrative remedies for the victims of the pollutants and other environmental damage.

It also agrees with **article 21, the Right to a healthy environment to its citizens of the constitution.**

The NGT has to dispose of the cases presented to it within 6 months of their appeals.

NGT has original jurisdiction on matters related to substantial questions of the environment.

NGT deals with the civil cases under the 7 acts related to the environment:

1. Water (Prevention And Control Of Pollution) Act, 1974
2. Water (Prevention And Control Of Pollution) Cess Act, 1974
3. Air (Prevention And Control Of Pollution) Act, 1977
4. Forest Conservation Act, 1980
5. Environmental Protection Act, 1986
6. Public Liability Insurance Act 1991
7. Biological Diversity Act, 2002

2 acts have been kept out of the jurisdiction of NGT:

1. Wildlife Protection Act, 1972
2. Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 (FRA)

The decisions of the NGT can be challenged in High Courts and the Supreme Court.

(9) Compensatory Afforestation Fund Act, 2016

The CAF Act was enacted to manage the funds collected for compensatory afforestation which till then was managed by **ad hoc Compensatory Afforestation Fund Management and Planning Authority (CAMPA).**

- **Compensatory afforestation** means that every time forest land is diverted for non-forest purposes such as mining or industry, the user agency pays for planting forests over an equal area of non-forest land, or when such land is not available, twice the area of degraded forest land.

As per the rules, **90% of the CAF money is to be given to the states while 10% is to be retained by the Centre.**

The funds can be used for the treatment of catchment areas, assisted natural generation, forest management, wildlife protection and management, relocation of villages from protected areas, managing human-wildlife conflicts, training and awareness generation, supply of wood saving devices, and allied activities.

Disaster Management

Disaster management refers to effective management of counter measures that are taken in order to mitigate the effect natural calamities that lead to desperate situations after calamities such as earthquakes, floods, landslides, tsunamis, etc. Although these sudden calamities are natural geographical processes that have been taking place from beginning and have played important role in shaping of earth, these geographical activities are wreck havoc and bring misfortune to people in region affected. Among the 36 states and Union territories in the country, 22 are prone to disasters. Among all the disasters that occur in the country, floods are the most frequently occurring natural disasters, due to the irregularities of the Indian monsoon. About 75 percent of the annual rainfall in India is concentrated in three to four months of the monsoon season. As a result there is a very heavy discharge from the rivers during this period causing widespread floods. Approximately 40 million hectares of land in the country has been identified as being prone to floods. Major floods are mainly caused in the Ganga-Brahmaputra-Meghna basin which carries 60 percent of the total river flow of our country.These processes inflict huge losses to life and property and it can take years for life to take normal shape.

Every region of the world will confront disaster in some way or other. As these disasters are sudden and rarely predictable, best way to mitigate their effect is to be prepared to them. This requires preplanning and professional approach.

Disaster management pivots around preplanning, which includes

- Organizing general disaster management teams to respond to any general disaster and in any terrain.
- Organizing special quick response teams that are highly specific to nature and region of disaster.
- Most important part is indentifying threats that a particular region is most venerable to. This involves setting up of research stations that study the terrain, climate and underground seismic activities of the region.

21.2 Major Causes for Disasters

1. Anthropogenic activities such as Impoundment of huge quantities of water in the lake behind a big dam e.g. Koyna Dam in Maharashtra have created few incidence of minor and major earthquakes., under ground nuclear testing e.g.Pokharan II testing at desert of Rajasthan, Deep well disposal of liquid waste.

2. Due to heavy rainfalls or sudden snow melt can swell the rivers disproportionately- causes a great economic loss and health related problems.
3. Landslides occur when coherent rock of soil masses move down slope due to gravitational pull. Water and vegetation influence landslides. Chemical action of water gradually causes chemical weathering of rocks making them prone to landslides.

Here are some control measures that are disaster-type specific.

21.3 Earthquake

In order to abate the effect of earthquakes, these precautionary measures must be taken.

- People should evacuate buildings and stay in open until the time, tremors have ceased.
- In case people are unable to get out of the buildings, they should try and stay in corners of the rooms.
- People using transport should stop vehicles and wait for tremors to subside.
- Buildings should be made by using construction material that is recommended by authorities.
- Design of the houses and buildings must be approved by authorities. Rectangular building design is most effective design that can withstand earthquake.
- People should help each other and provide first aid to the victims and not just wait for disaster management teams to arrive.
- Temporary relief camps and rehabilitation centres should be provided to people who have been affected.
- Compensation should be given to people who lost their house and livelihood.
- People should be made aware and trained through campaigns to tackle adversities as it is not possible for disaster management teams to reach everywhere.

21.4 Cyclones

In order to abate the effect of cyclones, following measures are advised.

- With help of technology, advent and paths cyclones can be predicted to some extent. First and foremost measure is to vacate the region that is predicted to be affected.
- People should be warned about cyclones through weather news, internet, newspapers, radio broadcast, etc.
- People should take to shelter in safe buildings during cyclones. Storm shelter should be constructed by authorities.
- Fisherman should be warned not to go to sea.
- Electricity supply should be cut off to the region that is affected.
- Temporary relief camps and rehabilitation centres should be provided to people who have been affected.
- Compensation should be given to people who lost their house and livelihood.

21.5 Floods

In order to pacify the effects of flood disaster, following steps must be taken.

- Floods in general are caused by heavy and concentrated rains. Therefore best defence is to study and predict weather developments and issue early warnings through broadcast and print media.
- People should be evacuated to safer places and relief camps should be provided.
- People who could not be evacuated should move to relatively higher places.
- Dams and embankments must be constructed by the government to check the flow in regions frequently affected by flood disasters.
- Floods often result in breaching of canal embankments and river embankments. Strength of these embankments must be periodically evaluated by authorities so that they can withstand deluge.
- Sand bangs must be used to repair temporary breaches in canals during floods.

21.6 Landslides

In general the chief mitigatory measures to be adopted for landslide areas are

- Drainage correction,
- Proper land use measures,
- Reforestation for the areas occupied by degraded vegetation
- Creation of awareness among local population.

The most important triggering mechanism for mass movements is the water infiltrating into the overburden during heavy rains and consequent increase in pore pressure within that overburden. When this happens in steep slopes, the safety factor of the slope material gets considerably reduced causing it to move down. Hence, the natural way of preventing this situation is by reducing infiltration and allowing excess water to move down without hindrance. As such, the first and foremost mitigation measure is drainage correction. This involves maintenance of natural drainage channels both micro and macro in vulnerable slopes.

The universal use of contour bunding for all types of terrain without consideration of the slope, overburden thickness and texture or drainage set-up needs to be controlled especially in the plateau edge regions. It is time to think about alternative and innovations, which are suitable for the terrain, to be set up. It need not be emphasized the governmental agencies have a lot to contribute in this field.

2Agencies Working on Disaster Management

21.7.1 National disaster management authority (NDMA)

NMDA is headed by the Prime Minister of India, is the Apex Body for Disaster Management in India. The setting up of the NDMA and the creation of an enabling environment for institutional mechanisms at the State and District levels is mandated by the Disaster Management Act, 2005. NDMA as the apex body is mandated to lay down the policies, plans and guidelines for Disaster Management to ensure timely and effective response to disasters.

21.7.2 International association of emergency managers (IAEM)

IAEM is a non-profit educational organization dedicated to promoting the goals of saving lives and protecting property during emergencies and disasters. The mission of IAEM is to serve its members by providing information, networking and professional opportunities, and to advance the emergency management profession. It currently has seven Councils around the World: Asia, Canada, Europa,International, Oceania, Student and USA.

21.7.3 Red cross/Red crescent

National Red Cross/Red Crescent societies often have pivotal roles in responding to emergencies. Additionally, the International Federation of Red Cross and Red Crescent Societies (IFRC, or "The Federation") may deploy assessment teams, e.g.Field Assessment and Coordination Team – (FACT) to the affected country if requested by the national Red Cross or Red Crescent Society. After having assessed the needs Emergency Response Units (ERUs) may be deployed to the affected country or region. They are specialized in the response component of the emergency management framework.

21.7.4 United nations

Within the United Nations system responsibility for emergency response rests with the Resident Coordinator within the affected country. However, in practice international response will be coordinated, if requested by the affected country's government, by the UN Office for the Coordination of Humanitarian Affairs (UN-OCHA), by deploying a UN Disaster Assessment and Coordination (UNDAC) team.

SOCIAL FORESTRY: PERSPECTIVE

The National Commission on Agriculture, considering the need to expand the tree resource base of the country, suggested the adoption of social forestry in rural areas in 1976. The Sixth Five Year Plan period (1980-85) gave a boost to the afforestation activities with the Centrally Sponsored "Social Forestry including Rural Fuel Wood Plantations" scheme. A series of social forestry projects followed with the support of international agencies including the World Bank, SIDA, DANIDA, US AID and others. Social forestry projects aimed at taking the pressure away from the forests by afforestation of all the unused and fallow lands including village common lands, the Government wasteland and Panchayat lands. Tree planting in and around agricultural fields, along railway lines, roadsides, river and canal banks were carried out to meet the growing demand for timber, fuel wood, fodder, etc., thereby reducing the pressure on the traditional forest area. It was also considered as a means of putting to the optimum use the fallow and degraded lands. Community Development

schemes like NREP and RLEGP too had the components of social forestry, but the implementation through the Government establishment was aimed at planting trees as mentioned above. People's participation was envisaged through awareness creating campaigns and extension services.

TYPES OF SOCIAL FORESTRY

Social forestry activities may be categorized into three groups: farm forestry, agroforestry and community and extension forestry

3.3.1 Farm Forestry Individual farmers are encouraged to plant trees in their own farmlands to meet the market requirements of wood products. Forest departments as well as industries promoted forestry as a diversification option to conventional agriculture in suitable farmlands. In the initial stages, planting material was made available by the state forest departments. Subsequently, industries established buy-back agreements with tree growers and financial institutions also facilitated investments. These systems developed in the states like Punjab, Haryana, Gujarat, Karnataka, Uttar Pradesh, parts of Himachal Pradesh, etc. Now industries have their own nursery development programmes and quality seedlings are provided to the farmers with whom they have buy-back arrangements. WIMCO, Mysore Paper Mills, some units of Hindustan Paper Corporation and ITC Bhadrachalam are some of the pioneers in this field. The programme experienced a setback in the wake of smaller holdings coming to the fore and difficulties of the industries to organise these arrangements with smaller farmers and also the relaxed import of pulp. Relatively longer gestation period, demand driven market and emphasis on exotics and monoculture also contributed to the slow progress of the programme. However, economic considerations ensured that the system survived in the private sector and is still prevalent in many parts of the country. 3.3.2 Agro-Forestry Agro-forestry is the system of land use that combines growing of forest trees along with agricultural crops. The farmlands are used for agricultural crops and trees are grown along the margins for various needs of the farmers. In fact, most farmers in India grow agricultural crops, rear animals, and plant certain trees on their land, often on the boundary areas. Growing of trees helps in conserving soil, enhancing soil fertility, and providing shelterbelts for crops and fruit trees, apart from wood and other produce. Agro-forestry reduces farmers' dependence on forests even as it provides them economic benefits resulting in a healthy and sustainable land-use system. There have been several traditional agro-forestry systems in vogue in different regions of the country. Usually multipurpose trees are grown along the farmlands. Different versions of shifting agriculture are also termed as agro-forestry systems, which are now considered unsustainable because of their short cycles and high frequency. The agro-forestry programmes in India were started in the late 1970s as a component of social forestry programmes. The Government promoted agro- forestry systems included fast growing species like eucalyptus, poplar, subabul, siris, etc. These systems though still prevalent in some areas, have not become very popular because of the market situation, which led to the slump in farm forestry systems. In the Tenth Five Year Plan, it is proposed to popularize agro-forestry as a strategy to increase the green cover of the country in areas outside the government forests. 3.3.3 Community Forestry Another scheme taken up under the social forestry programme is the raising of trees on community land and not on private land as in farm forestry. All these programmes aim at creating tree resources for the entire community and not for any individual. The government takes the responsibility of providing seedlings, fertilizers, etc., but the community has to take the responsibility of protecting the trees. Some communities have managed the plantations sensibly and in a sustainable manner and the public continues to benefit. Planting of trees on the sides of roads, canals and railways, along with planting on wastelands came to be known as 'extension forestry', as it increases the domain of forests. Planting of trees in vacant lands in urban areas, mainly along roadsides for beautification and for reducing noise and dust levels, came to be known as 'urban forestry'. Developing beautiful landscapes for recreation like picnics, jogging, relaxation, education, etc. has been named as 'recreation forestry'. Under this system wood lots have been created in village common lands, government wastelands, panchayat lands and urban areas

ASSESSMENT OF SOCIAL FORESTRY PROGRAMMES

Social forestry was originally conceived by the government as a response to the forestry crisis and to the accelerating deforestation in India. The original objectives of social forestry projects were to assist rural communities and the landless people to meet their needs for fodder, fuel wood, small timber, fruits, and minor forest produce through tree plantations planned and managed by the community. Most social forestry projects, however, came under increasing criticism because they failed to adequately involve the local communities and the rural poor, the main beneficiaries. Instead, these projects catered mainly to urban and commercial interests through the widespread promotion of fast growing tree species for pulp and paper manufacture, rayon production, urban fuel wood supply and other commercial uses. Private farmlands, wastelands and community lands were converted for these uses, and in a number of cases the access of the poorer rural population to fodder,

fuel wood and other forest products actually reduced. The only benefit was the availability of seasonal employment to the rural people in tree planting operations, which actually reduced in net terms in the case of farm forestry. On the management front, the main drawbacks included greater emphasis on silviculture, lack of the required extension work, involvement of people and Panchayati Raj Institutions, commensurate modifications in the forest regulations to motivate people to grow trees and that of orientation for the grass roots functionaries towards the needs of social forestry. Focus on economic benefits gave greater emphasis to fast growing species, which could not cater to the subsistence needs of the poor. Although eucalyptus trees grow more quickly, and remained at the focus of the projects, they could not provide the community with basic subsistence items. A 1988 midterm review of social forestry cited the lack of the participation of local communities in project design and implementation and over reliance on industry biased management as the main causes of the shortcomings of the project.

THE SOCIAL FORESTRY PROGRAMME: SOME LESSONS

In the social forestry programmes, emphasis on planting trees without appreciating the requirements of the communities impacted its sustainability negatively. Planting of tree is still taken as a good option for degraded soils and wastelands instead of a diversification crop. Farm forestry became popular more for its long gestation period, for low labour requirement and absentee landlordism, rather than its remunerative aspects. Agro-forestry seems to be acceptable when a balance is struck between agriculture and the planting of tree. However, it works only for farmers with bigger holdings, because for marginal farmers and the landless, livelihood issues compel the use of land mainly for conventional agriculture. Some aspects of agro- and farm forestry, namely private enterprises in tree planting, have suffered not only from the indifferent forest related regulations but also from the fiscal policies related to them. The control on harvest and the movement of forest produce as provided in forest laws works as a disincentive for the farmer to grow trees. Permissions are required from the forest department for felling trees as well as transporting the forest produce in almost all the states of the country. Even where these regulations are relaxed, adequate evidence is required to prove (during transportation) that the produce has been collected from private lands and not government forests. In the initial stage of agricultural development, expansion of agriculture took place at the expense of forests. These sectors competed for space in the private sector and plantation of trees lagged because of the lack of incentives, policy back up and the resultant economic considerations. The concessionary supply of the forest raw material from state governments and the relaxed conditions for the import of pulp have acted as additional disincentives for growing of trees because of the availability of cheaper raw material to industries. Thus, social forestry in its initial form was faced with the problems of low viability. Today, the social forestry establishments within the state governments continue with the work of extension and awareness campaigns, afforestation on non-forest lands, urban forestry and the related activities. Community participation in social forestry is pursued through government interventions in the preservation of sacred groves, planting of institutional areas, other vacant lands and urban areas with resident associations, etc. Though by law social forestry is a Panchayat subject, Panchayats have seldom played any active role in the process of afforestation in the case of common lands. It is necessary that the establishment of social forestry operates in collaboration with Panchayati Raj Institutions, as these institutions have adequate allocation of resources for afforestation and the management of existing tree lands as common property resources.

Lesson-16 Managing scarce resources

some natural resources like soil, air and water and how various components are cycled over and over again in nature. In the previous chapter we also learnt about the pollution of these resources because of some of our activities. In this chapter, we shall look at some of our resources and how we are using them. May be we should also think about how we ought to be using our resources so as to sustain the resources and conserve our environment. We shall be looking at our natural resources like forests, wild-life, water, coal and petroleum and see what are the issues at stake in deciding how these resources are to be managed for sustainable development. We often hear or read about environmental problems. These are often global-level problems and we feel helpless to make any changes. There are international laws and regulations, and then there are our own national laws and acts for environmental protection. There are also national and international organisations working towards protecting our environment.

There are some measurable factors which are used to quantify pollution or the quality of the water that we use for various activities. Some of the pollutants are harmful even when present in very small quantities and we require sophisticated equipment to measure them

But we need not feel powerless or overwhelmed by the scale of the problems because there are many things we can do to make a difference. You must have come across the three R’s to save the

environment: Reduce, Recycle and Reuse. What do they refer to? **Reduce:** This means that you use less. You save electricity by switching off unnecessary lights and fans. You save water by repairing leaky taps. You do not waste food. Can you think of other things that you can reduce the usage of ? **Recycle**: This means that you collect plastic, paper, glass and metal items and recycle these materials to make required things instead of synthesising or extracting fresh plastic, paper, glass or metal. In order to recycle, we first need to segregate our wastes so that the material that can be recycled is not dumped along with other wastes. Does your village/town/city have a mechanism in place for recycling these materials? **Reuse:** This is actually even better than recycling because the process of recycling uses some energy. In the 'reuse' strategy, you simply use things again and again. Instead of throwing away used envelopes, you can reverse it and use it again. The plastic bottles in which you buy various food-items like jam or pickle can be used for storing things in the kitchen. What other items can we reuse? But even while making everyday choices, we can make environment friendly decisions. For doing this, we need to know more about how our choices affect the environment, these effects may be immediate or longterm or long-ranging. The concept of sustainable development encourages forms of growth that meet current basic human needs, while preserving the resources for the needs of future generations. Economic development is linked to environmental conservation. Thus sustainable development implies a change in all aspects of life. It depends upon the willingness of the people to change their perceptions of the socio-economic and environmental conditions around them, and the readiness of each individual to alter their present use of natural resources.

WHY DO WE NEED TO MANAGE OUR RESOURCES? 16.1 WHY DO WE NEED TO MANAGE OUR RESOURCES? Not just roads and buildings, but all the things we use or consume – food, clothes, books, toys, furniture, tools and vehicles – are obtained from resources on this earth. The only thing we get from outside is energy which we receive from the Sun. Even this energy is processed by living organisms and various physical and chemical processes on the earth before we make use of it. Why do we need to use our resources carefully? Because these are not unlimited and with the human population increasing at a tremendous rate due to improvement in health-care, the demand for all resources is increasing at an exponential rate. The management of natural resources requires a long-term perspective so that these will last for the generations to come and will not merely be exploited to the hilt for short term gains. This management should also ensure equitable distribution of resources so that all, and not just a handful of rich and powerful people, benefit from the development of these resources. Another factor to be considered while we exploit these natural resources is the damage we cause to the environment while these resources are either extracted or used. For example, mining causes pollution because of the large amount of slag which is discarded for every tonne of metal extracted. Hence, sustainable natural resource management demands that we plan for the safe disposal of these wastes too.

FORESTS AND WILD LIFE Forests are 'biodiversity hot spots'. One measure of the biodiversity of an area is the number of species found there. However, the range of different life forms (bacteria, fungi, ferns, flowering plants, nematodes, insects, birds, reptiles and so on) is also important. One of the main aims of conservation is to try and preserve the biodiversity we have inherited. Experiments and field studies suggest that a loss of diversity may lead to a loss of ecological stability We all use various forest produce. But our dependency on forest resources varies. Some of us have access to alternatives, some do not. When we consider the conservation of forests, we need to look at the stakeholders who are – (i) the people who live in or around forests are dependent on forest produce for various aspects of their life (see Fig. 16.2). (ii) the Forest Department of the Government which owns the land and controls the resources from forests. (iii) the industrialists – from those who use 'tendu' leaves to make bidis to the ones with paper mills – who use various forest produce, but are not dependent on the forests in any one area. (iv) the wild life and nature enthusiasts who want to conserve nature in its pristine form. Let us take a look at what each of these groups needs/gets out of the forests. The local people need large quantities of firewood, small timber and thatch. Bamboo is used to make slats for huts, and baskets for collecting and storing food materials. Implements for agriculture, fishing and hunting are largely made of wood, also forests are sites for fishing and hunting. In addition to people gathering fruits, nuts and medicines from the forests, their cattle also graze in forest areas or feed on other fodder which is collected from forests. Do you think such use of forest resources would lead to the exhaustion of these resources? Do not forget that before the British came and took over most of our forest areas, people had been living in these forests for centuries. They had developed practices to ensure that the resources were used in a sustainable manner. After the British took control of the forests (which they exploited ruthlessly for their own purposes), these people were forced to depend on much smaller areas and forest resources started becoming over-exploited to some extent. The Forest Department in independent India took over from the British but local knowledge and local needs continued to be ignored in the management practices. Thus vast tracts of forests have been converted to monocultures of pine, teak or eucalyptus.

In order to plant these trees, huge areas are first cleared of all vegetation. This destroys a large amount of biodiversity in the area. Not only this, the varied needs of the local people – leaves for fodder, herbs for medicines, fruits and nuts for food – can no longer be met from such forests. Such plantations are useful for the industries to access specific products and are an important source of revenue for the Forest Department. Do you know how many industries are based on forest produce? A short count reveals timber, paper, lac and sports equipment.

Industries would consider the forest as merely a source of raw material for its factories. And huge interest-groups lobby the government for access to these raw materials at artificially low rates. Since these industries have a greater reach than the local people, they are not interested in the sustainability of the forest in one particular area. For example, after cutting down all the teak trees in one area, they will get their teak from a forest farther away. They do not have any stake in ensuring that one particular area should yield on optimal amount of some produce for all generations to come. What do you think will stop the local people in behaving in a similar manner? Lastly, we come to the nature and wildlife enthusiasts who are in no way dependent on the forests, but who may have considerable say in their management. The conservationists were initially taken up with large animals like lions, tigers, elephants and rhinoceros. They now recognise the need to preserve biodiversity as a whole. But shouldn't we recognise people as forming part of the forest system? There have been enough instances of local people working traditionally for conservation of forests. For example, the case of the Bishnoi community in Rajasthan, for whom conservation of forest and wildlife has been a religious tenet. The Government of India has recently instituted an 'Amrita Devi Bishnoi National Award for Wildlife Conservation' in the memory of Amrita Devi Bishnoi, who in 1731 sacrificed her life along with 363 others for the protection of 'khejri' trees in Khejrali village near Jodhpur in Rajasthan. Studies have shown that the prejudice against the traditional use of forest areas has no basis. Here is an example – the great Himalayan National Park contains, within its reserved area, alpine meadows which were grazed by sheep in summer. Nomadic shepherds drove their flock up from the valleys every summer. When this national park was formed, this practice was put to an end. Now it is seen that without the regular grazing by sheep the grass first grows very tall, and then falls over preventing fresh growth. Management of protected areas, by keeping the local people out, by using force cannot possibly be successful in the long run. In any case, the damage caused to forests cannot be attributed to only the local people – one cannot turn a blind eye to the deforestation caused by industrial needs or development projects like building roads or dams. The damage caused in these reserves by tourists or the arrangements made for their convenience is also to be considered. We need to accept that human intervention has been very much a part of the forest landscape. What has to be managed in the nature and extent of this intervention. Forest resources ought to be used in a manner that is both environmentally and developmentally sound – in other words, while the environment is preserved, the benefits of the controlled exploitation go to the local people, a process in which decentralised economic growth and ecological conservation go hand in hand. The kind of economic and social development we want will ultimately determine whether the environment will be conserved or further destroyed. The environment must not be regarded as a pristine collection of plants and animals. It is a vast and complex entity that offers a range of natural resources for our use. We need to use these resources with due caution for our economic and social growth, and to meet our material aspirations. 16.2.2 Sustainable Management We need to consider if the goals of all the above stakeholders with regard to the management of the forests are the same. Forest resources are often made available for industrial use at rates far below the market value while these are denied to the local people. The Chipko Andolan ('Hug the Trees Movement') was the result of a grassroot level effort to end the alienation of people from their forests. The movement originated from an incident in a remote village called Reni in Garhwal, high-up in the Himalayas during the early 1970s. There was a dispute between the local villagers and a logging contractor who had been allowed to fell trees in a forest close to the village. On a particular day, the contractor's workers appeared in the forest to cut the trees while the men folk were absent. Undeterred, the women of the village reached the forest quickly and clasped the tree trunks thus preventing the workers from felling the trees. Thus thwarted, the contractor had to withdraw. Inherent in such a competition to control a natural resource is the conservation of a replenishable resource. Specifically the method of use was being called into question. The contractor would have felled the trees, destroying them forever. The communities traditionally lop the branches and pluck the leaves, allowing the resource to replenish over time. The Chipko movement quickly spread across communities and media, and forced the government, to whom the forest belongs, to rethink their priorities in the use of forest produce. Experience has taught people that the destruction of forests affected not just the availability of forest products, but also the quality of soil and the sources of water. Participation of the local people can indeed lead to the efficient management of forests.

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: To assess the steadiness of hands of a given subject.

Activity 2:To investigate the phenomenon of transfer of training.

PSY 15146

RESEARCH METHODOLOGY

Objectives: To orient students to the different stages of research. To give insight into the various research methods. To identify and apply appropriate research tools. To acquire the skill of reporting the research.

Unit I: Introduction to Research Methodology

Lesson 1 Basic concepts – meaning and characteristics of scientific research.
Lesson 2 factors affecting, steps or stages in research and research process
Lesson 3 Types of Research – Experimental and Non experimental, Laboratory experiments and Field experiments and quasi experiment,
Lesson 4 Quantitative and Qualitative Research, Ex-post Facto research, Survey research and Types of experiments.

Unit II: Identification of Problem

Lesson 5 Problem: Types of problems, identifying a Research Problem –
Lesson 6 Variables: Independent, Dependent and Intervening Variable –
Lesson 7 Experimental group and control group.
Lesson 8 Hypotheses: The nature and types of Hypotheses, stating a hypothesis, criteria of Hypothesis and functions of hypothesis.

Unit III: Sampling & Data Collection.

Lesson 9 Concept of Population, sample and sampling
Lesson 10 Types of sampling - Random, Stratified, Purposive, Incidental and Multi stage Sampling.
Lesson 11 issues related to selection of sample size
Lesson 12 Data collection & presentation: Primary data (observation Method/ Personal interview, survey method, questionnaire) secondary data.

Unit IV: Ethical Consideration and Report Writing

Lesson 13 Ethics in human research: APA guidelines,
Lesson 14 scientific fraud and monitoring ethical practice.Presenting research results:
Lesson-15Report Writing: Meaning of Interpretation - Techniques of Interpretation - Precaution in Interpretation - Significance of Report Writing - Different Steps in Writing Report
Lesson-16 Layout of the Research Report - Types of Reports - Mechanics of Writing a Research Report - Precautions for Writing Research Reports

Outcomes :

- To understand and comprehend the basics in research methodology and applying them in research/ project work.
- To select an appropriate research design.
- To take up and implement a research project/ study. • The course will also enable them to collect the data, edit it properly and analyse it accordingly. Thus, it will facilitate students' prosperity in higher education.
- To develop skills in qualitative and quantitative data analysis and presentation.
- To demonstrate the ability to choose methods appropriate to research objectives.

Unit I: Introduction to Research Methodology

Lesson 1 Basic concepts – meaning and characteristics of scientific research.

Meaning of scientific research

Scientific research is the systematic and empirical investigation of phenomena, theories, or hypotheses, using various methods and techniques in order to acquire new knowledge or to validate existing knowledge.

It involves the collection, analysis, interpretation, and presentation of data, as well as the formulation and testing of hypotheses. Scientific research can be conducted in various fields, such as natural sciences, social sciences, and engineering, and may involve experiments, observations, surveys, or other forms of data collection. The goal of scientific research is to advance knowledge, improve understanding, and contribute to the development of solutions to practical problems.

Scientific research follows a systematic approach. It aims to acquire new information that adds to the existing knowledge in the research field. The consensus of scientific research is that researchers should plan their investigation before executing it.

Scientific research aims to identify and build scientific knowledge that discovers and explains laws or principles of natural or social phenomena.

Characteristics of scientific research

Scientific research is basically about the search for knowledge in an orderly way, structured and attached to certain rules. This type of study has several defining characteristics.

1- Systematic

The systematization of scientific research is linked to the need for it to be rigorous in procedures. This is not a random observation, but is the result of a well-structured plan, with specific objectives.

The processes must be standardized, always be sought to execute the actions in the same way, so that the result can be reliable as a result of having always followed the same guidelines.

The systematic plan that must guide a scientific investigation must consider all the aspects and moments of this research: from the objects of study and the variables to be taken into account, to the rhythm of work that must be followed in order to arrive at conclusions in time expected.

2- Controlled

Scientific research must avoid chance, and the process must be supported by control mechanisms that allow it to obtain truthful results.

Chance has no place in scientific research: all actions and observations are controlled, according to the researcher's criteria and according to the object investigated, through well-defined methods and rules.

3- Empirical

The results of a scientific investigation must deal with the aspects of reality related to the subject under investigation. The aspects that characterize a particular research must be observable in the real world.

Scientific research refers to issues that can be measured and identified as facts.

Is about Experiment with evidence . In this way it is possible to test the research hypothesis, and thus be able to affirm, deny or supplement it, as the case may be.

4- Rational

Science in general is characterized by being rational and logical. In a scientific investigation must emphasize the rationality on the subjectivity.

Its empirical characteristic makes it necessary to be based on real and verifiable facts, and demands from the researcher a critical attitude and a dispossession of his personal conceptions or judgments of value.

Some scientists and philosophers maintain that it is precisely the rational and critical character of an investigation that generates progress in the intellectual field and an important development of knowledge.

5- Reproducible

The findings obtained through scientific research should be able to be reproduced under the same conditions established in the study.

Given the systematized nature of scientific research, it must be verifiable. The fact of having controlled the variables that were part of the process, allows to be able to reproduce the results achieved.

6- Consider everyday problems

In a scientific investigation, the hypotheses constitute the nucleus of the study, and must be generated of problems and situations of the daily life, that affect the people of habitual form.

It is hoped that scientific research will solve a problem that ideally affects several groups of people. By critically observing this problem and making it an object of study, it is possible to find an answer that, hopefully, can improve the quality of life of many people in different areas.

7- Objective

Just as rationality and critical character must be emphasized in scientific research, it must also be objective.

The goal of the investigator is not to justify own postures, but to expose the facts in the purest way possible.

The explanation arising from scientific research must be legitimate for people with different inclinations of thought. The results of scientific research must be universal.

8- Provisional

Science is constantly expanding. Scientific research is considered provisional because it must be open to further studies that reaffirm, refute or complement the findings obtained in that research.

The debate is a fundamental part of the scientific field. Therefore, a scientific investigation must be able to be questioned and, if there is some subsequent research proving contrary hypotheses, it must be able to rectify.

9- Original

There is no sense in focusing scientific research on proven facts. A scientific investigation must treat new or little studied aspects, so that the result of the study implies a true contribution to the science and the humanity.

If it is based on an existing research, the researcher should focus on a different area of the problem, look for alternative results to those presented in the first place, or refute the research hypothesis as mistaken.
In any case, it is essential that scientific research brings something new and useful for people.

10- Ordered

Scientific research needs rigorous planning so that it can yield true results. This planning must have a specific order, which responds to the interests of the study.
In a scientific investigation it is necessary that the processes are designed and ordered in such a way that they reach secondary objectives that, in the last instance, can help to verify the main objectives raised by the researcher.
In turn, the whole exploratory process of a scientific investigation must be based on an orderly structure that allows to develop a veracious, empirical and verifiable study.

Lesson 2 factors affecting, steps or stages in research and research process

Steps or stages in research and research process

The research process consists of a series of systematic procedures that a researcher must go through in order to generate knowledge that will be considered valuable by the project and focus on the relevant topic.
To conduct effective research, you must understand the research process steps and follow them.
Here are a few steps in the research process to make it easier for any researcher.

Step 1: Identify the Problem

Finding an issue or formulating a research question is the first step. A well-defined research problem will guide the researcher through all stages of the research process, from setting objectives to choosing a technique. There are a number of approaches to get insight into a topic and gain a better understanding of it. Such as:

- A preliminary survey
- Case studies
- Interviews with a small group of people
- Observational survey

Step 2: Evaluate the Literature

A thorough examination of the relevant studies is essential to the research process. It enables the researcher to identify the precise aspects of the problem. Once a problem has been found, the investigator or researcher needs to find out more about it.
This stage gives problem-zone background. It teaches the investigator about previous research, how they were conducted, and its conclusions. The researcher can build consistency between his work and others through a literature review. Such a review exposes the researcher to a more significant body of knowledge and helps him follow the research process efficiently.

Step 3: Create Hypotheses

Formulating an original hypothesis is the next logical step after narrowing down the research topic and defining it. A belief solves logical relationships between variables. In order to establish a hypothesis, a researcher must have a certain amount of expertise in the field. It is important for researchers to keep in mind while formulating a hypothesis that it must be based on the research topic. Researchers are able to concentrate their efforts and stay committed to their objectives when they develop theories to guide their work.

Step 4: The Research Design

Research design is the plan for achieving objectives and answering research questions. It outlines how to get the relevant information. Its goal is to design research to test hypotheses, address the research questions, and provide decision-making insights.
The research design aims to minimize the time, money, and effort required to acquire meaningful evidence. This plan fits into four categories:

- Exploration and Surveys
- Experiment
- Data Analysis
- Observation

Step 5: Describe Population
Research projects usually look at a specific group of people, facilities, or how technology is used in the business. In research, the term population refers to this study group. The research topic and purpose help determine the study group.
Suppose a researcher wishes to investigate a certain group of people in the community. In that case, the research could target a specific age group, males or females, a geographic location, or an ethnic group. A final step in a study's design is to specify its sample or population so that the results may be generalized.

Step 6: Data Collection
Data collection is important in obtaining the knowledge or information required to answer the research issue. Every research collected data, either from the literature or the people being studied. Data must be collected from the two categories of researchers. These sources may provide primary data.

- Experiment
- Questionnaire
- Observation
- Interview

Secondary data categories are:

- Literature survey
- Official, unofficial reports
- An approach based on library resources

Step 7: Data Analysis
During research design, the researcher plans data analysis. After collecting data, the researcher analyzes it. The data is examined based on the approach in this step. The research findings are reviewed and reported.
Data analysis involves a number of closely related stages, such as setting up categories, applying these categories to raw data through coding and tabulation, and then drawing statistical conclusions. The researcher can examine the acquired data using a variety of statistical methods.

Step 8: The Report-writing
After completing these steps, the researcher must prepare a report detailing his findings. The report must be carefully composed with the following in mind:

The Layout: On the first page, the title, date, acknowledgments, and preface should be on the report. A table of contents should be followed by a list of tables, graphs, and charts if any.

Introduction: It should state the research's purpose and methods. This section should include the study's scope and limits.

Summary of Findings: A non-technical summary of findings and recommendations will follow the introduction. The findings should be summarized if they're lengthy.

Principal Report: The main body of the report should make sense and be broken up into sections that are easy to understand.

Lesson 3 Types of Research – Experimental and Non experimental, Laboratory experiments and Field experiments and quasi experiment.

Types of Research – Experimental

Experimental research is a form of comparative analysis in which you study two or more variables and observe a group under a certain condition or groups experiencing different conditions. By assessing the results of this type of study, you can determine correlations between the variables applied and their effects on each group. Experimental research uses the scientific method to find preferable ways of accomplishing a task for providing a service. The three main types of experimental research design are:

1. Pre-Experimental Design:

A group, or various groups, is kept under observation after implementing cause and effect factors. You'll conduct this research to understand whether further investigation is necessary for these particular groups.

You can break down pre-experimental research further in three types:

- One-shot Case Study Research Design
- One-group Pretest-posttest Research Design
- Static-group Comparison

2. True Experimental Design:

It relies on statistical analysis to prove or disprove a hypothesis, making it the most accurate form of research. Of the types of experimental design, only true design can establish a cause-effect relationship within a group. In a true experiment, three factors need to be satisfied:

- There is a Control Group, which won't be subject to changes, and an Experimental Group, which will experience the changed variables.
- A variable that can be manipulated by the researcher
- Random distribution

This experimental research method commonly occurs in the physical sciences.

3. Quasi-Experimental Design:

The word "Quasi" indicates similarity. A quasi-experimental design is similar to an experimental one, but it is not the same. The difference between the two is the assignment of a control group. In this research, an independent variable is manipulated, but the participants of a group are not randomly assigned. Quasi-research is used in field settings where random assignment is either irrelevant or not required.

Types of Research –Non experimental

Non-experimental research is research that lacks the manipulation of an independent variable, control of extraneous variables through random assignment, or both. Non-experimental research can take the following forms:

Cross-sectional research:
Cross-sectional research is used to observe and analyze the exact time of the research to cover various study groups or samples. This type of research is divided into:

- Descriptive: When values are observed where one or more variables are presented.
- Causal: It is responsible for explaining the reasons and relationship that exists between variables in a given time.

Longitudinal research:
In a longitudinal study, researchers aim to analyze the changes and development of the relationships between variables over time. Longitudinal research can be divided into:

- Trend: When they study the changes faced by the study group in general.
- Group evolution: When the study group is a smaller sample.
- Panel: It is in charge of analyzing individual and group changes to discover the factor that produces them.

Field experiments:

Field Experiments Field experiments on the other hand refer to experiments conducted in real life situations. Here the control of extraneous factors is not possible as it is a natural setting and there is no way to control any factor so absolutely as one does in the laboratory experiments. Hence in field experiments we take two groups matched for a number of factors such as age, sex, education, socio-economic status etc. Both these groups are in real life setting and thus are subjected to similar extraneous variables and thus the experimenter can observe the effects of his manipulation on one group and compare with the other group which is not subjected to any intervention.

Lesson 4 Quantitative and Qualitative Research, Ex-post Facto research, Survey research and Types of experiments.

Qualitative Research
Qualitative research is based on quality, and it looks in-depth at non-numerical data. It enables us to understand the comprehensive details of the problem. The researcher prepares open-ended questions to gather as much information as possible.
Examples:

- Stress level among men and women.

Quantitative Research
Quantitative research is associated with the aspects of measurement, quantity, and extent. It follows the statistical, mathematical, and computational techniques in the form of numerical data such as percentages and statistics. The research is conducted on a large group of population.

Example:

- Find out the weight of students of the fifth standard Studying in government schools.

Ex-post Facto research
The ex-post facto research is a kind of research in which the researcher predicts the possible causes behind an effect that has already occurred. For example, if a child is delinquent (that is, one who indulges in criminal activities), then in order to find the basic reason behind such delinquency, the researcher would try to find out the various events that have occurred and the many possibilities

that could have contributed to the concerned delinquent behaviour. The expected possibilities may be lack of discipline at school/ family history/ peer effect/ neighborhood or socialization.
It is an interesting point to note that, the researcher predicts a cause on the basis of a controlled effect (since no variation can be done on the effect which has already taken place on the basis of the independent variable or the cause).
Thus, an ex-post facto research can be defined as an empirically based investigation which does not involve the researchers' direct control over the 21 independent variables because they have already led to effects which can no Ex-Post Facto Research more be manipulated. The conclusions regarding the relationship between the variables are inferred without intervening or varying the independent or dependent variable.
The term ex-post facto according to Landman (1988: 62) is used to refer to an experiment in which a researcher, instead of finding a treatment, examines the effect of a naturally occurring treatment after it has occurred. In other words it is a study that attempts to discover the pre-existing causal conditions between groups.
It should, however, be pointed out that the most serious danger of ex-post factoresearch is the conclusion that because two factors go together, one is the cause and the other is the effect.

Survey research

Survey research is a **quantitative approach** that features the use of self-report measures on carefully selected samples. It is a flexible approach that can be used to study a wide variety of basic and applied research questions. Survey research has its roots in applied social research, market research, and election polling.
Survey Research is defined as the process of conducting research using surveys that researchers send to survey respondents. The data collected from surveys is then statistically analyzed to draw Meaningful research conclusions.

In the 21st century, every organization's eager to understand what their customers think about their products or services and make better business decisions. Researchers can conduct research in multiple ways, but surveys are proven to be one of the most effective and trustworthy research methods. An online survey is a method for extracting information about a significant business matter from an individual or a group of individuals. It consists of structured survey questions that motivate the participants to respond.

Stages of survey research

There are five stages of survey research design:

Decide an aim of the research:

There can be multiple reasons for a researcher to conduct a survey, but they need to decide a purpose for research. This is the primary stage of survey research as it can mold the entire path of a survey, impacting its results.

Filter the sample from target population:

Who to target? is an essential question that a researcher should answer and keep in mind while conducting research. The precision of the results is driven by who the members of a sample are and how useful their opinions are. The quality of respondents in a sample is essential for the results received for research and not the quantity. If a researcher seeks to understand whether a product

feature will work well with their target market, he/she can conduct survey research with a group of market experts for that product or technology.

Zero-in on a survey method:

Many qualitative and quantitative research methods can be discussed and decided. Focus groups, online interviews, surveys, polls, questionnaires, etc. can be carried out with the pre-decided sample of individuals.

Design the questionnaire:

What will the content of the survey be? A researcher is required to answer this question to be able to design it effectively. What will the content of the cover letter be? Or what are the survey questions of this questionnaire? Understand the target market thoroughly to create a questionnaire that targets a sample to gain insights about a survey research topic.

Send out surveys and analyze results:

Once the researcher decides on which questions to include in a study, they can send it across to the selected sample. Answers obtained for this survey can be analyzed to make product-related or marketing-related decisions.

Unit II: Identification of Problem

Lesson 5 Problem: Types of problems, identifying a Research Problem

Research problem is a specific and well-defined issue or question that a researcher seeks to investigate through research. It is the starting point of any research project, as it sets the direction, scope, and purpose of the study.

Types of Research Problems

Types of Research Problems are as follows:

Descriptive problems

These problems involve describing or documenting a particular phenomenon, event, or situation. For example, a researcher might investigate the demographics of a particular population, such as their age, gender, income, and education.

Exploratory problems

These problems are designed to explore a particular topic or issue in depth, often with the goal of generating new ideas or hypotheses. For example, a researcher might explore the factors that contribute to job satisfaction among employees in a particular industry.

Explanatory Problems

These problems seek to explain why a particular phenomenon or event occurs, and they typically involve testing hypotheses or theories. For example, a researcher might investigate the relationship between exercise and mental health, with the goal of determining whether exercise has a causal effect on mental health.

Predictive Problems

These problems involve making predictions or forecasts about future events or trends. For example, a researcher might investigate the factors that predict future success in a particular field or industry.

Evaluative Problems

These problems involve assessing the effectiveness of a particular intervention, program, or policy. For example, a researcher might evaluate the impact of a new teaching method on student learning outcomes.

Research Problem

The research problem is the heart of a study. It is a clear, definite statement of the area of concern or investigation and is backed by evidence (Bryman, 2007). It drives the research questions and processes and provides the framework for understanding the research findings. To begin, you will need to know where to look for your research problem and how to evaluate when a research problem for success.

As well as outlining the topic and/or opportunity, a research problem will explain:

1. why the area/issue needs to be addressed,
2. why the area/issue is of importance,
3. the parameters of the research study
4. the research objective
5. the reporting framework for the results and
6. What the overall benefit of doing so will provide (whether to society as a whole or other researchers and projects).

Identification of Research Problem:

It is the foremost and initial step in a scientific method for doing research. To conduct a research, there should be exiting ideas that developed the necessity for the research to be conceded out. The thoughts or ideas constructed while going through different discourses like literature with specialized and continuation of activities concerning to the major matter. These ideas come up into particular topics that will be motivating or gratifying if searched out. These queries or topics are called research problems.

Problems are recognized by means of group involvement. A group of knowledge persons are recognized and their suggestions in negative sense are incorporated into dissimilar groups. Afterwards, from each group, a statement that comes most delegate of the group is chosen. Statements are organized in an order as they appear to the group of specialized in the major problem or matter. These statements are known as problems.

Lesson 6 Variables: Independent, Dependent and Intervening Variable

Variables

Researchers and statisticians use variables to describe and measure the items, places, people or ideas they're studying. Many types of variables exist, and you must choose the right variable to measure when designing studies, selecting tests and interpreting results. A strong understanding of variables can lead to more accurate statistical analyses and results.

Variables in psychology are things that can be changed or altered, such as a characteristic or value. Variables are generally used in psychology experiments to determine if changes to one thing result in changes to another.

Variables in psychology play a critical role in the research process. By systematically changing some variables in an experiment and measuring what happens as a result, researchers are able to learn more about cause-and-effect relationships.

The two main types of variables in psychology are the independent variable and the dependent variable. Both variables are important in the process of collecting data about psychological phenomena.

Independent variable

Independent variable is the variable that is manipulated by the experimenter. An example of an independent variable in psychology: In an experiment on the impact of sleep deprivation on test performance, sleep deprivation would be the independent variable. The experimenters would have some of the study participants be sleep-deprived while others would be fully rested.

Dependent variable

Dependent variable is the variable that is measured by the experimenter. In the previous example, the scores on the test performance measure would be the dependent variable.

Intervening Variables

Intervening variables, also sometimes called intermediate or mediator variables, are factors that play a role in the relationship between two other variables.

An intervening variable, also known as a mediator or mediating variable, explains the process through which two variables are related, while a moderating, or moderator, variable affects the strength and direction of that relationship.

In the previous example, sleep problems in university students are often influenced by factors such as stress. As a result, stress might be an intervening variable that plays a role in how much sleep people get, which may then influence how well they perform on exams.

Lesson 7 Experimental group and control group.

Experimental group

An experimental group is a test sample or the group that receives an experimental procedure. This group is exposed to changes in the independent variable being tested. The values of the

independent variable and the impact on the dependent variable are recorded. An experiment may include multiple experimental groups at one time.

In a comparative experiment, the experimental group (aka the treatment group) is the group being tested for a reaction to a change in the variable. There may be experimental groups in a study, each testing a different level or amount of the variable.

Control group

A control group is a group separated from the rest of the experiment such that the independent variable being tested cannot influence the results. This isolates the independent variable's effects on the experiment and can help rule out alternative explanations of the experimental results.

Using a control group means that any change in the dependent variable can be attributed to the independent variable. This helps avoid extraneous variables or confounding variables from impacting your work, as well as a few types of research bias, like omitted variable bias.

Lesson 8 Hypotheses: The nature and types of Hypotheses, stating a hypothesis, criteria of Hypothesis and functions of hypothesis.

Nature and types of Hypotheses

Hypothesis is a predictive statement which is unproven or a presumption to be proved or disproved about any factor. This statement can be tested and verified by scientific methods and is related to the independent factor of a dependent factor.

Example - Consumption of fast food everyday leads to obesity.

A hypothesis statement is an assumption that is made based on some evidence. Hypothesis is the starting point of investigation which translates the research questions into assuming predictions. Components of hypotheses are variables, variable relations, population. A research hypothesis is used to test the relationship between two or more variables.

Examples of Hypothesis based on their types-

- **Simple hypothesis-** Consumption of coke every day leads to obesity.
- **Null hypothesis-** All flower lilies have the same number of petals.
- **Directional hypothesis-** A person gets 7 hours of sleep, will make him feel less fatigue than if he sleeps less.

Nature of Hypotheses

The following are the main features of a Hypothesis –

1. Its conceptual in nature.
2. It is a verbal statement in a declarative form.
3. It has the empirical referent.
4. It indicates the tentative relationship between two or more variables.
5. It is a powerful tool of advancement of knowledge, consistent with existing knowledge and conducive to further enquiry.
6. It can be tested, verifiable or falsifiable.
7. It is not moral or ethical questions.

8. It is neither too specific nor to general.
9. It is a prediction of consequences.
10. It is considered valuable even if proven false.

Types of Hypotheses

Alternative Hypothesis
The alternative hypothesis states that there is a relationship between the two variables being studied (one variable has an effect on the other).
An experimental hypothesis predicts what change(s) will take place in the dependent variable when the independent variable is manipulated.
It states that the results are not due to chance and that they are significant in terms of supporting the theory being investigated.

Null Hypothesis
The null hypothesis states that there is no relationship between the two variables being studied (one variable does not affect the other). There will be no changes in the dependent variable due to the manipulation of the independent variable.
It states results are due to chance and are not significant in terms of supporting the idea being investigated.

Non-directional Hypothesis
A non-directional (two-tailed) hypothesis predicts that the independent variable will have an effect on the dependent variable, but the direction of the effect is not specified. It just states that there will be a difference.
E.g., there will be a difference in how many numbers are correctly recalled by children and adults.

Directional Hypothesis
A directional (one-tailed) hypothesis predicts the nature of the effect of the independent variable on the dependent variable. It predicts in which direction the change will take place. (i.e. greater, smaller, less, more)
E.g., adults will correctly recall more words than children.

Stating a hypothesis
Research hypothesis is a statement that introduces a research question and proposes an expected result. It is an integral part of the scientific method that forms the basis of scientific experiments. Therefore, you need to be careful and thorough when building your research hypothesis. A minor flaw in the construction of your hypothesis could have an adverse effect on your experiment. In research, there is a convention that the hypothesis is written in two forms, the null hypothesis, and the alternative hypothesis (called the experimental hypothesis when the method of investigation is an experiment).
A testable hypothesis is not a simple statement. It is rather an intricate statement that needs to offer a clear introduction to a scientific experiment, its intentions, and the possible outcomes. However, there are some important things to consider when building a compelling hypothesis.

1. State the problem that you are trying to solve.
Make sure that the hypothesis clearly defines the topic and the focus of the experiment.

2. Try to write the hypothesis as an if-then statement.
Follow this template: If a specific action is taken, then a certain outcome is expected.

3. Define the variables
Independent variables are the ones that are manipulated, controlled, or changed. *Independent variables* are isolated from other factors of the study.
Dependent variables, as the name suggests are dependent on other factors of the study. They are influenced by the change in independent variable.

4. Scrutinize the hypothesis
The types of research hypothesis are stated below:
1. Simple Hypothesis
It predicts the relationship between a single dependent variable and a single independent variable.
2. Complex Hypothesis
It predicts the relationship between two or more independent and dependent variables.

3. Directional Hypothesis
It specifies the expected direction to be followed to determine the relationship between variables and is derived from theory. Furthermore, it implies the researcher's intellectual commitment to a particular outcome.
4. Non-directional Hypothesis
It does not predict the exact direction or nature of the relationship between the two variables. The non-directional hypothesis is used when there is no theory involved or when findings contradict previous research.
5. Associative and Causal Hypothesis
The associative hypothesis defines interdependency between variables. A change in one variable results in the change of the other variable. On the other hand, the causal hypothesis proposes an effect on the dependent due to manipulation of the independent variable.

6. Null Hypothesis
Null hypothesis states a negative statement to support the researcher's findings that there is no relationship between two variables. There will be no changes in the dependent variable due the manipulation of the independent variable. Furthermore, it states results are due to chance and are not significant in terms of supporting the idea being investigated.
7. Alternative Hypothesis
It states that there is a relationship between the two variables of the study and that the results are significant to the research topic. An experimental hypothesis predicts what changes will take place in the dependent variable when the independent variable is manipulated. Also, it states that the results are not due to chance and that they are significant in terms of supporting the theory being investigated.

Criteria of Hypothesis
There exist two criteria for formulation of a good hypothesis. First, it is a statement about the relations between variables. Secondly it carries clear implications for testing the stated relations. Thus, these couple of criteria implies that the hypotheses comprise two or more variables which are measurable or potentially measurable and that they specify the way in which they are related. A statement which fails to meet these criteria is no scientific hypothesis in the true sense of the term. However, there are legitimate hypotheses, formulated in factor analytic studies.

Functions of hypothesis.

A hypothesis functions as an answer to the research question and guides data collection and interpretation. A hypothesis enables researchers not only to discover a relationship between variables, but also to predict a relationship based on theoretical guidelines and/or empirical evidence.
Following are the functions performed by the hypothesis statement-

- Hypothesis helps in making an observation made by the researcher.
- Hypothesis becomes the initial point of an investigation.
- Hypothesis helps in verifying the observations made.
- Hypothesis helps in directing the inquiries in the right direction.

UNIT III: SAMPLING & DATA COLLECTION

Lesson 9 Concept of Population, sample and sampling

Sampling design refers to the process of selecting samples from a population. There are two types of sampling designs—probability sampling design and non-probability sampling design. Probability sampling designs are used in conclusive research. In a probability sampling design, each and every element of the population has a known chance of being selected in the sample. The known chance does not mean equal chance. Simple random sampling is a special case of probability sampling design where every element of the population has both known and equal chance of being selected in the sample. In case of non-probability sampling design, the elements of the population do not have any known chance of being selected in the sample. These sampling designs are used in exploratory research. Probability Sampling Design Under this, the following sampling designs would be covered—simple random sampling with replacement (SRSWR), simple random sampling without replacement (SRSWOR), systematic sampling, stratified random sampling and cluster sampling. Simple random sampling with replacement Under thisscheme, a list of all the elements of the population from where the samplesto be drawn is prepared. If there are 1000 elementsin the population, we write the identification number or the name of all the 1000 elements on 1000 different slips. These are put in a box and shuffled properly. If there are 20 elements to be selected from the population, the simple random sampling procedure involves selecting a slip from the box and reading of the identification number. Once this is done, the chosen slip is put back to the box and again a slip is picked up and the identification number is read from that slip. This process continues till a sample of 20 is selected. Please note that the first element is chosen with a probability of 1/1000, the second one is also selected with the same probability and so are all the subsequent elements of the population. An alternative way of selecting the samples from the population is by using random number tables. Table 7.1 gives an illustrative example of random numbers.

Probability and Non-Probability Sampling Methods, Sample Size and Sampling and Non-sampling Errors to identify a representative sample. For example, the shoppers at a shopping centre may serve to represent the residents of a city or some of the cities may be selected to represent a country. Judgemental sampling design is used when the required information is possessed by a limited number/category of people. This approach may not empirically produce satisfactory results and, may, therefore, curtail generalizability of the findings due to the fact that we are using a sample of experts (respondents) that are usually conveniently available to us. Further, there is no objective way to evaluate the precision of the results. Accompany wanting to launch a new product may use judgemental sampling for selecting ‘experts’ who have prior knowledge or experience of similar

products. A focus group of such experts may be conducted to get valuable insights. Opinion leaders who are knowledgeable are included in the organizational context. Enlightened opinions (views and knowledge) constitute a rich data source. A very special effort is needed to locate and have access to individuals who possess the required information. The most common application of judgemental sampling is in business to-business (B to B) marketing. Here, a very small sample of lead users, key accounts or technologically sophisticated firms or individuals is regularly used to test new product concepts, producing programmes, etc. Snowball sampling Snowball sampling is generally used when it is difficult to identify the members of the desired population, e.g., deep-sea divers, families with triplets, people using walking sticks, doctors specializing in a particular ailment, etc. Under this design each respondent, after being interviewed, is asked to identify one or more in the field. This could result in a very useful sample. The main problem is in making the initial contact. Once this is done, these cases identify more members of the population, who then identify further members and so on. It may be difficult to get a representative sample. One plausible reason for this could be that the initial respondents may identify other potential respondents who are similar to themselves. The next problem is to identify new cases. Quota sampling In quota sampling, the sample includes a minimum number from each specified subgroup in the population. The sample is selected on the basis of certain demographic characteristics such as age, gender, occupation, education, income, etc. The investigator is asked to choose a sample that conforms to these parameters. Field workers are assigned quotas of the sample to be selected satisfying these characteristics. A researcher wants to measure the job satisfaction level among the employees of a large organization and believes that the job satisfaction level varies across different types of employees. The organization is having 10 percent, 15 per cent, 35 per cent and 40 per cent, classI, classII, classIII and class IV, employees, respectively. If a sample of 200 employees is to be selected from the organization, then 20, 30, 70 and 80 employees from class I, class II, class III and class IV respectively should be selected from the population. Now, various investigators may be assigned quotas from each class in such a way that a sample of 200 employees is selected from various classes in the same proportion as mentioned in the population. For example, the first field worker may be assigned a quota of 10 employees from class I, 15 from class II, 20 from class III and 30 from class IV. Similarly, a second investigator may be assigned a different quota such that a total sample of 200 is selected in the same proportion as the population is distributed. Please note that the investigators may choose the employees from each class as conveniently available to them. Therefore, the sample may not be totally representative of the population, hence the findings of the research cannot be generalized. However, the reason for choosing this sampling design is the convenience it offers in terms of effort, cost and time. In the example given above, it may be argued that job satisfaction is also influenced by education level, categorized as higher secondary or below, graduation, and postgraduation and above. By incorporating this variable, the distribution of population may look as given in Table 7.2. From the table, we may note that there are 8 per cent class I employees who are postgraduate and above, there are 35 per cent class IV employees with a higher secondary education and below and so on. Now, suppose a sample of size 200 is again proposed. In this case, the distribution of sample satisfying these two conditions in the same proportion in the population is given in Table 7.3. Table 7.2 Distribution of Population (percentage) Table 7.3 Distribution of Sample (numbers) Table 7.3 indicates that a sample of 20 class II employees who are graduates should be selected. Likewise, a sample of 10 employees who possess postgraduate and above education should be selected. In the above table, the sample to be taken from each of the 12 cells has been specified. Having done so, each of the investigators is assigned a quota to collect information from the employees conforming to the above norms so that a sample of 200 is selected. Quota sampling design may look similar to the stratified random sampling design. However, there are differences between the two. In the stratified sampling design, the selection of sample from each stratum is random but in the quota sampling, the respondents may be chosen at the convenience or judgement of the researchers. Further, as already stated, the results of stratified random sampling

could be generalized, whereas it may not be possible in the case of quota sampling. Quota sampling has some advantages over the probabilistic techniques. This design is very economical and it does not take too much time to set it up. Also, the use of this design does not require a sampling frame. However, quota sampling also has certain weaknesses like: • The total number of cells depends upon the number of control characteristics associated with the objectives of the study. If the control characteristics are large, the total number of cells increases, which may result in making the task of the investigator difficult. • The chosen control characteristics should be related to the objectives of the study. The findings of the study could be misleading if any relevant parameter is omitted for one reason or the other. • The investigator may visit those places where the chances of getting the respondents with the required control characteristics are high. The investigator could also avoid some responses that appear to be unfriendly. All this could result in making the findings of the study less reliable.

Lesson 10 Types of sampling - Random, Stratified, Purposive, Incidental and Multi stage Sampling.

Probability and Non-Probability

Sampling Methods,

Sample Size and Sampling and Non-sampling Errors Table 7.1 Select Four-digit Random Numbers I II III IV V 2807 0495 6183 7871 9559 8016 5732 3448 0164 2367 1322 4678 8034 1139 1474 0843 4625 7407 9987 5734 2364 1187 4565 2343 9786 4885 8755 4355 5465 0575 3406 4678 5950 7222 8494 5927 6010 7545 8979 1041 4447 3476 9140 0736 2332 4968 7553 1073 2493 4251 7489 1630 2330 4250 6170 4010 2707 3925 6007 8089 6531 9784 5520 7764 0008 7052 3861 7115 9521 2192 6573 2793 8710 2127 3846 8094 3205 2030 3035 5765 8615 6092 1900 4792 7684 9136 4016 3495 6549 9603 9656 5246 5090 8306 1522 2017 8323 1685 3006 3441 Table 7.1 givesfour-digit random numbers arranged in 20 rows and five columns. These random numbers can be generated by a computer programmed to scramble numbers. The logic for generating random number is that any number can be constructed from numbers 0 to 9. The probability that any one digit from 0 through 9 will appear is the same as that for any other digit and the appearance of the numbers is statistically independent. Further, the probability of one sequence of digits occurring is the same as that for any other sequence of the same length. The use of random number table for selecting samples could be illustrated through an example. Suppose there are 75 students in a class and it is decided to select 15 out of the 75 students. These students can be numbered from 01 to 75. Now, to pick up 15 students using random numbers and following the scheme of simple random sampling with replacement, we proceed as follows: • With eyes closed, we place our finger on a number on the random number table. Suppose it is on the first row and the first column of our table. Now, we go down the first two columns and choose two-digit random numbers running from 01 to 75. If any number greater than 75 appears, it gets rejected. This way, the first number to be selected would be 28. The second number is 80, which would be rejected as we are choosing numbers from 01 to 75. The next selected number would be 13, followed by 08, 23, 48, 34, 59, 44, 49, 74, 40, 65, 70 and 65. Note that 65 has appeared twice. Since we are using the scheme of simple Probability and Non-Probability Sampling Methods, Sample Size and Sampling and Non-sampling Errors random sampling with replacement, we would retain it. This way we have selected 14 samples. The 15th number selected would be 20. In brief, the scheme explained above states that any number greater than the population size (in this case 75) is rejected and only the numbers from 01 to 75 are selected. A number may get repeated because simple random sampling scheme is done with replacement. Simple random sampling without replacement In the case of simple random sample without replacement, the procedure is identical to what was explained in

the case of simple random sampling with replacement. The only difference here is that the chosen slip is not placed back in the box. This way, the first unit would be selected with the probability of 1/1000, second unit with the probability of 1/999, the third will be selected with a probability of 1/998 and so on, till we select the required number of elements (in this case, 15) in our sample. The simple random sampling (with or without replacement) is not used in a consumer research. This is because in a consumer research the population size is usually very large, which creates problems in the preparation of a sampling frame. For example, there is a large number of consumers of soft drinks, pizza, shampoo, soap, chocolate and so on. However, these (SRSWR and SRSWOR) designs could be useful when the population size is very small, for example, the number of steel/aluminium-producing companies in India and the number of banks in India. Since the population size is quite small, the preparation of a sampling frame does not create any problem. Another problem with these (SRSWR and SRSWOR) designs is that we may not get a representative sample using such a scheme. Consider an example of a locality having 10,000 households, out of which 5,000 belong to low-income group, 3,500 belong to middle income group and the remaining 1,500 belong to high-income group. Suppose it is decided to take a sample of 100 households using the simple random sampling. The selected sample may not contain even a single household belonging to the high- and middleincome group and only the low-income households may get selected, thus, resulting in a non-representative sample. Systematic sampling Systematic sampling takes care of the limitation of the simple random sampling that the sample may not be a representative one. In this design, the entire population is arranged in a particular order. The order could be the calendar dates or the elements of a population arranged in an ascending or a descending order of the magnitude which may be assumed as random. List of subjects arranged in the alphabetical order could also be used and they are usually assumed to be random in order. Once this is done, the steps followed in the systematic sampling design are as follows:

Probability and Non-Probability Sampling Methods, Sample Size and Sampling and Non-sampling Errors

• First of all, a sampling interval given by K = N/n is calculated, where N = the size of the population and n = the size of the sample. It is seen that the sampling interval K should be an integer. If it is not, it is rounded off to make it an integer.

• A random number is selected from 1 to K. Let us call it C.

• The first element to be selected from the ordered population would be C, the next element would be C + K and the subsequent one would be C + 2K and so on till a sample of size n is selected. This way we can get representation from all the classesin the population and overcome the limitations of the simple random sampling. To take an example, assume that there are 1,000 grocery shops in a small town. These shops could be arranged in an ascending order of their sales, with the first shop having the smallest sales and the last shop having the highest sales. If it is decided to take a sample of 50 shops, then our sampling interval K will be equal to 1000 ÷ 50 = 20. Now we select a random number from 1 to 20. Suppose the chosen number is 10. This means that the shop number 10 will be selected first and then shop number 10 + 20 = 30 and the next one would be 10 + 2 × 20 = 50 and so on till all the 50 shops are selected. This way we can get a representative sample in the sense that it will contain small, medium and large shops. It may be noted that in a systematic sampling the first unit of the sample is selected at random (probability sampling design) and having chosen this, we have no control over the subsequent units of sample (non-probability sampling).

Because of this, this design at times is called mixed sampling. The main advantage of systematic sampling design is its simplicity. When sampling from a list of population arranged in a particular order, one can easily choose a random start as described earlier. After having chosen a random start, every Kth item can be selected instead of going for a simple random selection. This design is statistically more efficient than a simple random sampling, provided the condition of ordering of the population is satisfied. The use of systematic sampling is quite common asit is easy and cheap to select a systematic sample. In systematic sampling one does not have to jump back and forth all over the sampling frame wherever random number leads, and neither does one have to check for duplication of elements as compared to simple random sampling. Another advantage of a systematic sampling over simple random sampling is that one does not require a complete sampling frame to draw a systematic sample. The investigator may be instructed to interview every 10th customer entering a mall without a list of all customers. There may be situations where it may not be possible to get a representative sample. The design can create problems if the sampling Probability and Non-Probability Sampling Methods, Sample Size and Sampling and Non-sampling Errors interval is a whole number multiple of some cycle related to the problem.

On this design there may be a problem that there is a high probability of systematic bias creeping into the sample resulting in a non-representative sample. Consider, for example, the case of a certain PVR cinema hall where there may be a couple of snack bars. We may be interested in estimating the average daily sales of a particular snack bar in that PVR. Now, using the daily data with the population and sample size known, we compute a sampling interval which may be a multiple of seven. Using this, we may select our first element which would reflect one of the seven days of the week, say Friday. The next element would also be Friday, as our sampling interval is a multiple of seven and so the subsequent elements of the population. Therefore, our sample would comprise only Fridays and the sample would not reflect day of the week variation in the sales data, which could result in a non-representative sample. Therefore, while using daily data, care should be taken that our sampling interval is not a multiple of seven.

Stratified random sampling: Under this sampling design, the entire population (universe) is divided into strata (groups), which are mutually exclusive and collectively exhaustive. By mutually exclusive, it is meant that if an element belongs to one stratum, it cannot belong to any other stratum. Strata are collectively exhaustive if all the elements of various strata put together completely cover all the elements of the population. The elements are selected using a simple random sampling independently from each group. There are two reasons for using a stratified random sampling rather than simple random sampling. One is that the researchers are often interested in obtaining data about the component parts of a universe. For example, the researcher may be interested in knowing the average monthly sales of cell phones in ‘large’, ‘medium’ and ‘small’stores. In such a case, separate sampling from within each stratum would be called for. The second reason for using a stratified random sampling is that it is more efficient as compared to a simple random sampling. This is because dividing the population into various strata increases the representativeness of the sampling as the elements of each stratum are homogeneous to each other. There are certain issues that may be of interest while setting up a stratified random sample. These are: What criteria should be used for stratifying the universe (population)? The criteria for stratification should be related to the objectives of the study. The entire population should be stratified in such a way that the elements are homogeneous within the strata, whereas there should be heterogeneity between strata. As an example, if the interest is to estimate the expenditure of households on entertainment, the appropriate criteria for stratification would be the household income. This is because the expenditure on entertainment and household income are highly correlated. As another example, if the Probability and Non-Probability

Sampling Methods, Sample Size and Sampling and Non-sampling Errors objective of the study is to estimate the amount of money spent on cosmetics, then, gender could be used as an appropriate criteria for stratification. This is because it is known that though both men and women use cosmetics, the expenditure by women is much more than that of their male counterparts. Someone may argue out that gender may no longer remain the appropriate criteria if it is not backed by income. Therefore, the researcher might have to use two or more criteria for stratification depending upon the problem in hand. This would only increase the number of strata thereby making the sampling difficult. Generally stratification is done on the basis of demographic variables like age, income, education and gender. Customers are usually stratified on the basis of life stages and income levels to study their buying patterns. Companies may be stratified according to size, industry, profits for analysing the stock market reactions. How many strata should be constructed? Going by common sense, as many strata as possible should be used so that the elements of each stratum will be as homogeneous as possible. However, it may not be practical to increase the number of strata and, therefore, the number may have to be limited. Too many strata may complicate the survey and make preparation and tabulation difficult. Costs of adding more strata may be more than the benefit obtained. Further, the researcher may end up the practical difficulty of preparing a separate sampling frame as the simple random samples are to be drawn from each stratum. What should be appropriate number of samples size to be taken in each stratum? This question pertains to the number of observations to be taken out from each stratum. At the outset, one needs to determine the total sample size for the universe and then allocate it between each stratum. This may be explained as follows: Let there be a population of size N. Let this population be divided into three strata based on a certain criterion. Let N1 , N2 and N3 denote the size of strata 1, 2 and 3 respectively, such that N = N1 + N2 + N3 . These strata are mutually exclusive and collectively exhaustive. Each of these three strata could be treated as three populations. Now, if a total sample of size n is to be taken from the population, the question arises that how much of the sample should be taken from strata 1, 2 and 3 respectively, so that the sum total of sample sizes from each strata adds up to n. Let the size of the sample from first, second and third strata be n1 , n2 , and n3 respectively such that n = n1 + n2 + n3 . Then, there are two schemes that may be used to determine the values of ni, (i = 1, 2, 3) from each strata. These are proportionate and disproportionate allocation schemes. Proportionate allocation scheme: In this scheme, the size of the sample in each stratum is proportional to the size of the population of the Probability and Non-Probability Sampling Methods, Sample Size and Sampling and Non-sampling Errors strata. As an example, if a bank wants to conduct a survey to understand the problems that its customers are facing, it may be appropriate to divide them into three strata based upon the size of their deposits with the bank. If we have 10,000 customers of a bank in such a way that 1,500 of them are big account holders (having deposits more than ` 10 lakh), 3,500 of them are medium sized account holders (having deposits of more than ` 2 lakh but less than ` 10 lakh), the remaining 5,000 are small account holders (having deposits of less than ` 2 lakh). Suppose the total budget for sampling is fixed at ` 20,000 and the cost of sampling a unit (customer) is ` 20. If a sample of 100 is to be chosen from all the three strata, the size of the sample from strata 1 would be: 1 1 N 1500 n n 100 15 N 10000 =× = × = The size of sample from strata 3 would be: 2 2 N 3500 n n 100 35 N 10000 =× = × = The size of sample from strata 3 would be: 3 3 N 5000 n n 100 50 N 10000 =× = × = This way the size ofthe sample chosen from each stratum is proportional to the size of the stratum. Once we have determined the sample size from each stratum, one may use the simple random sampling or the systematic sampling or any other sampling design to take out samples from each of the strata. Disproportionate allocation: As per the proportionate allocation explained above, the sizes of the samples from strata 1, 2 and 3 are 15, 35 and 50 respectively. As it is known that the cost of sampling of a unit is ` 20 irrespective of the strata from where the sample is drawn, the bank would naturally be more

interested in drawing a large sample from stratum 1, which has the big customers, as it gets most of its business from strata 1. In other words, the bank may follow a disproportionate allocation of sample as the importance of each stratum is not the same from the point of view of the bank. The bank may like to take a sample of 45 from strata 1 and 40 and 15 from strata 2 and 3 respectively. Also, a large sample may be desired from the strata having more variability. Cluster sampling In the cluster sampling, the entire population is divided into various clustersin such a way that the elements within the clusters are heterogeneous. However, there is homogeneity between the clusters. This design, therefore, is just the opposite of the stratified sampling design, where there was homogeneity within the strata and heterogeneity between the strata. To illustrate the example of a cluster sampling, one may assume that there is a company Probability and Non-Probability Sampling Methods, Sample Size and Sampling and Non-sampling Errors having its corporate office in a multi-storey building. In the first floor, we may assume that there is a marketing department where the offices of the president (marketing), vice president (marketing) and so on to the level of management trainee (marketing) are there. Naturally, there would be a lot of variation (heterogeneity) in the amount of salaries they draw and hence a high amount of variation in the amount of money spent on entertainment. Similarly, if the finance department is housed on the second floor, we may find almost a similar pattern. Same could be assumed for third, fourth and other floors. Now, if each of the floors could be treated as a cluster, we find that there is homogeneity between the clusters but there is a lot of heterogeneity within the clusters. Now, a sample of,say, 2 to 3 clustersis chosen at random and once having done so, each of the cluster is enumerated completely to be able to make an estimate of the amount of money the entire population spends on entertainment. Examples of cluster sampling could include ad-hoc organizational committees drawn from various departments to advise the CEO of a company on product development, new product ideas, evaluating alternative advertising programmes, budget allocations and marketing strategies. Each of the clusters comprises a heterogeneous collection of members with different interests, background, experience, value system and philosophy. The CEO of the company may be able to take strategic decisions based upon their combined advice. Although the per unit costs of cluster sampling are much lower than those of other probability sampling, the applicability of cluster sampling to an organizational context may be questioned as a cluster may not contain heterogeneous elements. The condition of heterogeneity within the cluster and homogeneity between the clusters may not be met. As another example, the households in a block are to be similar rather than dissimilar and as a result, it may be difficult to form heterogeneous clusters. Although the per unit costs of cluster sampling are much lower than those of other probability sampling, the applicability of cluster sampling to an organizational context may be questioned as a cluster may not contain heterogeneous elements. The condition of heterogeneity within the cluster and homogeneity between the clusters may not be met. As another example, the households in a block are to be similar rather than dissimilar and as a result, it may be difficult to form heterogeneous clusters. Cluster sampling is useful when populations under a survey are widely dispersed and drawing a simple random sample may be impractical. Non-probability Sampling Designs Under the non-probability sampling, the following designs would be considered—convenience sampling, purposive sampling, snowball sampling and quota sampling. Probability and Non-Probability Sampling Methods, Sample Size and Sampling and Non-sampling Errors

Convenience sampling Convenience sampling is used to obtain information quickly and inexpensively. The only criterion for selecting sampling unitsin this scheme is the convenience of the researcher or the investigator. Mostly, the convenience samples used are neighbours,

friends, family members, colleagues and 'passers-by'. This sampling design is often used in the pre-test phase of a research study such as the pre-testing of a questionnaire. Some of the examples of convenience sampling are: • People interviewed in a shopping centre for their political opinion for a TV programme. • Monitoring the price level in a grocery shop with the objective of inferring the trends in inflation in the economy. • Requesting people to volunteer to test products. • Using students or employees of an organization for conducting an experiment. • Interviews conducted by a TV channel of people coming out of a cinema hall, to seek their opinion about the movie. • A researcher visiting a few shops near his residence to observe which brand of a particular product people are buying, so as to draw a rough estimate of the market share of the brand. In all the above situations, the sampling unit may either be self-selected or selected because of ease of availability. No effort is made to choose a representative sample. Therefore, in this design the difference between the population value (parameters) of interest and the sample value (statistic) is unknown both in terms of the magnitude and direction. Therefore, it is not possible to make an estimate of the sampling error and researchers won't be able to make a conclusive statement about the results from such a sample. It is because of this, convenience sampling should not be used in conclusive research (descriptive and causal research). Convenience sampling is commonly used in exploratory research. This is because the purpose of an exploratory research is to gain an insight into the problem and generate a set of hypotheses which could be tested with the help of a conclusive research. When very little is known about a subject, a small-scale convenience sampling can be of use in the exploratory work to help understand the range of variability of responses in a subject area. Judgemental sampling Under judgemental sampling, experts in a particular field choose what they believe to be the best sample for the study in question. The judgement sampling calls for special efforts to locate and gain access to the individuals who have the required information. Here, the judgement of an expert is used

Lesson 11 Issues related to selection of sample size

The sample size for a study needs to be estimated at the time the study is proposed; too large a sample is unnecessary and unethical, and too small a sample is unscientific and also unethical. The necessary sample size can be calculated, using statistical software, based on certain assumptions. There are many factors that are intertwined in the calculation of sample sizes or power of a study. Some factors depend on the design of the study, others on the investigator's choices, and others on the data themselves. The first consideration is the type of response variable. The study design and the response variable will determine the type of statistical method used in the data analysis. For example, if the data are continuous, and two groups are being compared for their means, a t-test may be the appropriate statistical method of analysis. The t-test defines the formula for the sample size or power calculation. The second set of factors depend on the investigator's choices. He/she needs to define the acceptable levels of significance and power of the study. The sample size may be more a consideration of availability and/or resources than what is necessary to achieve a certain power. The third factor is the variation of the data in the population of interest. It is intuitive to realize that the higher the variation of the data (this may include measures of variance and correlation among observations), the larger the sample size will have to be to give us enough confidence that we have a good estimate of the mean, for example. The last five items in the slide above are related to each other in the formula defined by specific statistical test used in the study. If one knows

four of those values, the fifth will be determined. Therefore, some values will have to either come from previous studies and/or knowledge, or will have to be assumed. Variance of outcome measure (cannot be controlled by researcher) Characteristics of the study design Quantities related to the research question (defined by the researcher).

Sample size for two proportions: example
Example: The efficacy of 'treatment A ' is expected to be 70%, and for 'treatment B' to be 60%. A study is planned to show the difference at the significance level of 1% and power of 90%.The sample size can be calculated as follows:

p1 = 0.6; q1= =0.4; p2 = 0.7;

q2 =1-0.7=0.3;Z0.01 = 2.58;

Z1-0.9 = 1.28.

The sample size required for each group should be:

n = ( )2[(0.6)(0.4)+(0.7)(0.30] /( )2 = 670.5

Total sample size = 1342 (consider for drop outs & lost to followup)

Lesson 12 Data collection & presentation: Primary data (observation Method/ Personal

interview, survey method, questionnaire) secondary data.

UNIT IV: ETHICAL CONSIDERATION AND REPORT WRITING

Lesson 13 Ethics in human research: APA guidelines,

Ethics in Human research as per APA guidelines

Summarized below are the Ethical guidelines for research with human subjects, which were published by the American Psychological Association in 1981. These regulations have been incorporated into broader ethical guidelines published more recently. Those broader guidelines can be found on the APA website. However, we believe that the simple guidelines published in 1981 still capture the broad principles of ethics in research well and so have chosen to summarize them here for you.

Preamble: Scientific research contributes to human welfare. Every researcher automatically assumes responsibility to conduct research with concern and sensitivity for the welfare and dignity of all human participants and to conform to all professional standards and all state and federal regulations regarding research with human participants. This general principle is clarified in the following ten principles.

1. In planning research, it is the researcher's responsibility to evaluate and ensure its ethical acceptability, to obtain ethical advice when needed, and to apply careful safeguards to protect participants' rights.
2. A primary concern of the investigator is to determine whether the procedures will place participant at risk.

3. It is the investigator's responsibility to ensure ethical practice, including that of all collaborators, assistants, employees, and students.
4. Informed consent must be obtained from all participants prior to the research. If potential participants cannot give consent, such as children, informed consent must be obtained from a responsible surrogate, such as a parent.
5. If deception is necessary, full disclosure must be made as soon as possible after completion of the research.
6. Participants' freedom to decline or to withdraw at any time must be fully respected by the investigator.
7. Participants must be protected from physical and mental harm. The investigator must provide participants with ways to contact the investigator at any time after completion of the research, should questions arise.
8. After data collection, full disclosure. explanation, and answers to any questions raised by participants must be provided.
9. It is the researcher's responsibility to detect and to remove any negative effects of the research, including long-term consequences.
10. All information obtained must be held in confidentiality.

Lesson 14 scientific fraud and monitoring ethical practice. Presenting research results:

Ethical reporting is a process of transparency. Foundational to proper, professional, and ethical conduct is the intent to be honest, accurate, and complete when providing required information. Many clinical, research, and teaching scenarios can potentially foster a natural tension between professional obligation and personal preference/gain. In reporting, misrepresentation by both errors of omission and commission, as well as by conflicts of interest, may create ethical violations. Adherence to professional obligations and standards in the Code is essential to minimize a possible ethical violation.

Reporting ethically requires you to proactively and deliberately provide honest and complete information in all areas of professional practice (e.g., credentials, services, relationships, payments, and advertising).

Lesson-15 Report Writing: Meaning of Interpretation - Techniques of Interpretation - Precaution in Interpretation - Significance of Report Writing - Different Steps in Writing Report

An important point to remember in report writing is that the document compiled is meant for specific readers. Thus, one needs to design the same according to the needs of the reader. Listed below are some features of a good research study that should be kept in mind while documenting and preparing the report. Clear report mandate: While writing the research problem statement and study background, the writer needs to be focused, precise and very explicit in terms of the problem under study, the background that provided the impetus to conduct the research and the study domain. This is prepared on the assumption that the writer at no point in time needs to be physically present in order to clarify the research mandate. One cannot make an assumption that the reader has earlier insights into the problem situation. The writer needs to be absolutely clear on the need for lucidity of thought and dissemination of this knowledge to the reader. Clearly designed methodology: Any research study has its unique orientation and scope and thus has a specific and customized research design, sampling and data collection plan. The writer, thus, needs to be explicit in terms of the logical justification for having used the study methods and techniques. However, as stated earlier, the language should be non-technical and reader friendly and any

technical explanations or details must be provided in the appendix. In researches, that are not completely transparent on the set of procedures, one cannot be absolutely confident of the findings and resulting conclusions. Clear representation of findings: The sample size for each analysis, any special conditions or data treatment must be clearly mentioned either as a footnote or as an endnote, so that the reader takes this into account while interpreting and understanding the study results. The sample base is very important in justifying a trend or taking a strategic decision; for example, if amongst a sample of bachelors we say that 100 per cent young bachelors want to buy grocery online or on the telephone and the recommended strategy is to suggest this as the delivery channel, one might be making an error if the size of the bachelors was four out of a total sample of 100 grocery buyers considered. Thus, complete honesty and transparency in stating the treatment and editing of missing or contrary data is extremely critical. Representativeness of study finding: A good research report is also explicit in terms of extent and scope of the results obtained, and in terms of the applicability of findings. This is also dependent on whether the assumptions and preconditions made for formulating the conclusions and recommendations of the study have been explicitly stated. Report Writing In order to ensure that one has been able to achieve the above stated objective, the reader must ensure a standardization of procedures in writing the document as well as follow standard protocols for preparing graphs and tables. In the following section we will briefly discuss some simple rulesthat the researcher can use as guidelines for this. Guidelines for Effective Documentation The guidelines for an effective documentation may be discussed under the following heads: Command over the medium: Even though one may have done an extremely rigorous and significant research study, the fundamental test still remains as to how the learning has been disseminated. Regardless of how effective the graphs and figures are in showcasing the findings, the verbal description and explanation—in terms of why it was done, how it was done, and what was the outcome, still remain the acid test. Thus, a correct and effective language of communication is critical in putting ideas and objectives in the vernacular of the reader/decision-maker. The writer may, thus, be advised to read professionally written reports and, if necessary, seek assistance from those proficient in preparing business reports. Phrasing protocol: There is a debate about whether or not one makes use of personal pronoun while reporting. To understand this, one needs to revisit the responsibility of the researcher, which is to present the findings of his/ her study, with complete objectivity and precision. The use of personal pronoun such as 'I think…..' or 'in my opinion…..' lends a subjectivity and personalization of judgement. Thus, the tone of the reporting should be neutral. For example: 'Given the nature of the forecasted growth and the opinion of the respondents, it is likely that the……' Whenever the writer is reproducing the verbatim information from another document or comment of an expert or published source, it must be in inverted commas or italics and the author or source should be duly acknowledged. For example: Sarah Churchman, Head of Diversity, Price water house Coopers, states 'At PricewaterhouseCoopers we firmly believe that promoting work– life balance is a 'business-critical' issue and not simply the 'right thing to do'. Profitable growth and sustainable business depend on attracting and retaining top talent and we know, from our own research and experience that work–life policies are an essential ingredient of successful recruitment and retention strategies.' The writer should avoid long sentences and break up the information in clear chunks, so that the reader can process it with ease. Similar is the case in structuring of the chapters or sections of the report that can be logically Report Writing broken down into smaller sections that are comprehensive and complete and yet maintain a strong but logical link with the flow of reporting. With the onset of the use of abbreviated communications in SMS and emails, most people tend to use shortened form as 'cd.'for could and 'u'for you, etc. Also the use of colloquial language and slangs must be avoided, as this is a formal document and one must maintain the sanctity of the formal documentation required in a research report. Simplicity of approach: Along

with grammatically and structurally correct language, care must be taken to avoid technical jargon as far as possible. The business manager, might have been a business student who had prepared a research report in his academic pursuits but now understands simple common terms and does not have the time or inclination to juggle the dictionary and the report together. In case it is imperative to use certain terminology, then, as stated earlier, the definition of these terms can be provided in the glossary of terms at the end of the report. Sometimes the writer may prepare different research reports for the same study to suit the need of diverse readers, for example, the business report needs to be crisp and simple with definable and workable recommendations. On the other hand, an academic report could discuss extensively the literature review section, as well as the statistical analysis and interpretation.

Lesson-16 Layout of the Research Report - Types of Reports - Mechanics of Writing a Research - Report - Precautions for Writing Research Reports

Report formatting and presentation:In terms of paper quality, page margins and font style and size, a professional standard should be maintained. The font style must be uniform throughout the report. The topics, subtopics, headings and subheadings must be construed in the same manner throughout the report. Sometimes certain academic reports have a mandated format for presentation which the writers need to follow, in which case there is no choice in presentation. However, when this is not clear, it is advisable that the writer creates his/her own formatting rules and saves it on a notepad so that they can be implemented in a standardized and professional manner. The researcher can provide data relief and variation by adequately supplementing the text with graphs and figures. Pictorial representations are simple to comprehend and also break the monotony and fatigue of reading. They should be used effectively whenever possible in the report.

Principles of a Good Report Writing Based on the above description, it may be concluded that report writing should be based on the following principles: • Principle of purpose: A report must have a clear and meaningful purpose that can be converted into an effective management. A clear statement of purpose helps prepare a well-focused report on which the management can work. Specification of purpose is important because: o Reports are analyses of Report Writing facts and proposals. o They are records of particular business activities. • Principles of organization: A written report should be well-designed and well-ordered. The managerial plan of a report must include the following: o Purpose of the report o Information required to be included in the report o Method used to collect report data o Summary of the report o Problems and solutions of the subject mentioned in the report o An appendix that describes and confirms the content and conclusion of the report • Principles of brevity: Reports should be concise. It is essential because long reports: o Are costly. o Are difficult to examine. o Are prone to disapproval, as they seem insufficient. o Focus on irrelevant minor details that may lead to the ignorance of major points. • Principles of clarity: Reports should be clear. Clarity can be maintained by using simple language for writing the report. New terms, if any in the report, should be properly explained to avoid confusion. • Principle of scheduling: Reports should be prepared at that time when there is no undue burden on the staff or when the staff has sufficient time to prepare reports. However, the time period between the gathering of data and generating finished reports should not be long; otherwise, the report may become outdated and useless if it is not completed in time. • Principle of cost: While preparing reports, it is necessary that their cost–benefit analysis should be done. A report should be minimum at costs and maximum at benefits. If the cost of preparation of the report is high but its benefit is low, then it is not advisable to prepare that report.

Features of a Good Research Report Research reports must be absolutely efficient and well formatted and the matter should be clear, analytical and directive. The actual facts need to be explained clearly. Data and results should be furnished in graphical or tabular format as it would create a substantially good impression and would be unambiguous to understand.

Report Writing The characteristic features of a good research may be listed as follows:
• Information collected in the report should be relevant and focused to derive desired results.
• The report should strictly adhere to predefined goals and objectives.
• The report should provide the description of the questionnaires used in analysis and the means adopted in their preparation.
• The report should elaborate the methodology used in the interviews.
• There must be an executive summary of the work in the report.
• The report should not only present the actual analysis but also the reasons of making this report. It should also highlight the advantages and profit it can provide after successful implementation of business plans described inside the report.
• It should also mention the methodology of the research presenting the overall process adopted to create the report.
• The report needs to be flexible enough so that it may be changed according to requirements.

Criteria for Evaluating Research Reports/Research Findings Research reports/findings are evaluated on the basis of following criteria: • Clarity: The report should be clear in terms of representation of data. It should be easy to understand. • Statement of objective: The objective of the report should be stated in the beginning of a report. While evaluating, it is important to check whether the research achieved the stated objective. • Relevance of data: The data of the report should be relevant to the research topic. In addition, it is important to check whether the recent data was used in report. • Analysis of data: The data should be properly analyzed. Thus, the evaluator checks whether all the findings are supported by analysis. • Unbiased: The report should not be biased towards a particular interpretation because biasness affects the complete process of research.

RESEARCH REPORT: LANGUAGE FLOW AND GRAMMATICAL QUALITY A report must have a clear and logicalstructure with clear indication of where the ideas are leading. It should be able to make a good first impression. The presentation of the report is very important. All reports must be written in a good language, using short sentences and correct grammar and spellings. The main points to be kept in mind in this light are as follows: Report Writing
• Context and style: o Appropriate, informative title for the report o Crisp, specific, unbiased writing with minimal jargon o Adequate analysis of prior relevant research
• Questions/hypotheses: o Clearly stated questions or hypotheses o Thorough operational definitions of key concepts along with the exact wording or measurement of the key variables
• Research procedures: o Full and clear description of the research design o Demographic profile of the participants/subjects o Specific data gathering procedures
• Data analysis: o Appropriate inferential statistics for sample or experimental data and appropriate use of descriptive statistics o Clear and reasonable interpretation of the statistical findings, accompanied by effective tables and figures
• Summary: Fair assessment of the implications and limitations of the findings o Effective commentary on the overall implications of the findings for theory and/or policy

Clarity and Brevity of Expressions There is a famoussaying that ‘words are like mirror that reflect the personality of the person from whose mouth they come out’. Thus, if a research wants to absorb the attention of the reader, then it must have clarity in its expression asthis will also tell a lot about the clarity of the researcher’sthought. Experts emphasize the importance of using as few words as possible to deliver your message. However, sometimes messages that are very brief sacrifice clarity and leave out vital information. Thus, while crafting hisreport, the researcher should choose clarity over brevity, and include all relevant information and be sure it is logically organized. Three rules for bringing clarity in reports In business writing, you get points for clarity, not style. Instead of trying to wax poetic about your division’s plans for the next 60 days, just make your point. Here are three ways to do that (i) Limit one idea to one paragraph: The researcher should limit his thoughtsto one per paragraph.When he has anothersuggestion, thought or idea, he should start a new paragraph. (ii) Make it scannable: The report should be so prepared that audience is able quickly scan the researcher’s message and understand his point. (iii) Put your point in the first sentence. The researchershould not entice hisreaders with background information and build-up. He should make his primary point first. Then he should go into supporting detail.

References and Annotations Several report types like scientific, engineering, technical and census reports contain either original writing or text adopted from a previous work. As such, a report writer should be careful and should avoid any violation of copyright laws and plagiarism. The necessary rule of thumb in this regard can be stated as follows:

• Citations and referencing: o A citation is the acknowledgement in your writing of the work of other authors and includes paraphrasing and making direct quotes. o Unless citation is very necessary, you should write the material in your own words. This shows that you understand what you have read and know how to apply it, to your own context. o Direct quotes should be used sparingly.

• Direct quotes:

o Short direct quotes: These need to be placed between quotation marks. For example, Rosenfield defines a cluster as a ‘geographically bounded concentration of similar, related or complementary businesses, with active channels for business transactions, communications and dialogue that share specialized infrastructure, common opportunities and threats.’ This shows clearly that the words being used are not your own words.

o Long direct quotes: There are occasions when it is useful to include long direct quotes. If you are quoting more than forty words, you should again use quotation marks but also indent the text. For example: The sustainability of higher value-added industry is grounded in the diminishing significance of cost structures. At the level of the European Union, a weak capacity to innovate has been identified as an innovation, in the sense of product, process and organizational innovation, accounts for a very large amount, perhaps 80–90 per cent of the growth in productivity in advanced economies.

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: To investigate the status of short term memory for digits of an individual.

Activity 2: To study the interference of one habit system with another, i.e to observe negative transfer of learning.

SEMESTER - II

PSY 25141 **COGNITIVE PSYCHOLOGY**

Objectives:

Cognitive Psychology is to study how humans acquire and put to use the acquired knowledge and information mentally just like a computer processor

Unit-1The Study of Cognition

Lesson -1 Definition and Historical origins of Cognitive Psychology

Lesson -2 The Philosophical Background: Early Greek Philosopher – Descartes; Locke;

Lesson -3 British Associationism; Kant. Associationism in Psychology: Contributions of Ebbinghaus.

Lesson -4 The twentieth century Associationism. The Gestalt view, Bartlett's outlook.

Unit-2Learning & Memory

Lesson -5 Definition and Nature - Learning theories, - Pavlov, Thorndike and Skinner

Lesson -6 Types of Learning – Conditioning, Habitual, Insight- Transfer of Training- Motor and verbal learning - Motor skills, distribution of practice, warm up, patterns of movement & knowledge of results

Lesson -7 Verbal learning – process in verbal learning. Organization, practice, individual differences

Lesson -8 Memory: Definition, Types of Memory- Working memory, Semantic memory, Picture memory and Verbal memory, Remote Memory, Procedural memory etc.

Unit-3Concept Formations and Problem Solving

Lesson -9 Concept formation- Process in concept formation – Analytical and non analytical Strategies

Lesson -10 Types, functions and Determinants of concept formation

Lesson -11 Problem Solving: Functional fixedness, Organization and insight Syllogistic Reasoning; Strategies for problem solving- Role of reasoning

Unit-4Creativity and Intelligence

Lesson -12 Creativity; Creative process, creativity and functional fixity, Investment theory and Creativity-Judging creativity'

Lesson -13 Intelligence: Definition and nature, Differences between Creativity and intelligence

Lesson -14 Factors influencing intelligence: Spearman, Thurston and Guilford's Gardner's Theory of multiple intelligences

Lesson 15 Stenberg's Triarchic theory –Measurement of Intelligence

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Outcomes:

To explain the mediatory role of cognition in behavior

To explain the process and function of attention

To describe sensational, perceptual phenomena and its different scientific explanations

To elucidate how the memory system functions
To explain the process and function of Neuropsychology
To describe Plasticity and Restoration of brain function

UNIT-I- STUDY OF COGNITION

PSY25141

Lesson 1 : Definition and Historical origins of Cognitive Psychology Introduction to Cognitive Psychology

Cognitive psychology is the study of mental processes such as perceiving, remembering, and reasoning. Since the beginning of recorded history, people have expressed curiosity about the operation of the mind, largely because they believed that behaviour is the result of mental processes. For example, how are we to understand the very behaviour in which you are engaged at this moment, reading this course book? At one level, we are interested in explaining your ability to comprehend what you are reading, and in so doing, we are likely to appeal to processes of perception of words and computation of meaning. At another level, we might explain our motivation for reading in terms of our goal to complete this course, which in turn is motivated by our goal of obtaining a degree in order to follow some plan that you have for a career. The point is that our behaviour of reading this book is determined in part by our intent to meet some goal and fulfill some plan. Intentionality, goals, and plans are mental phenomena that affect behaviour. Further, the specific behaviour, in this case, reading, is understood by appeal to the specific mental processes involved in perception and comprehension of text. In short, the study of mental processes is important because these processes are responsible for much of our behaviours. In this unit you will be studying the definition and description of cognitive psychology, distinctive research methods, domains of cognitive psychology, metaphor of cognition etc.

- **Cognitive psychology** is the scientific investigation of human cognition, that is, all our mental abilities – perceiving, learning, remembering, thinking, reasoning, and understanding.

- The word ‘cognition’ is derived from the Latin word cognoscere, meaning “to know” or “to come to know”. Thus, cognition includes the activities and processes concerned with the acquisition, storage, retrieval and processing of knowledge. In other words, it might include the processes that help us to perceive, attend, remember, think, categorize, reason, decide, and soon.
- Learning define“relatively permanent change in behaviour (or behaviour potential) resulting from experience” (Baron, 2001).
- Fundamentally, cognitive psychology studies how people acquire and apply knowledge or information. It is closely related to the highly interdisciplinary (relating to more than one branch of knowledge) cognitive science and influenced by artificial intelligence, computer science, philosophy, anthropology, linguistics, biology, physics, and neuroscience.
- Cognitive psychology, as the name suggests, is that branch of psychology that deals with cognitive mental processes.
- Cognitive Psychology deals with how people perceive, learn, remember and think about information (Sternberg, 2003) and how do they use this information (Matlin, 2002).
- Cognitive psychology holds a major domain in human psychology.
- It studies the mind and how it works.
- And it has a widespread influence on other areas of psychology.

Artificial Intelligence (AI)

- Machines or programs using intelligence to solve complex problems, the way humans solve problems.
- Application will include computers doing medical diagnoses, flying jet planes, being the senses of the disabled.
- Linguistics is the scientific study of language. It encompasses the analysis of every aspect of language, as well as the methods for studying and modeling them. The traditional areas of linguistic analysis include phonetics, phonology, morphology, syntax, semantics, and pragmatics.

- Neuroscience is the scientific study of the nervous system. It is a multidisciplinary science that combines physiology, anatomy, molecular biology, developmental biology, cytology, computer science and mathematical modeling to understand the fundamental and emergent properties of neurons and neural circuits.

Definitions:

- Sternberg (1999) defined Cognitive psychology as that which deals with how people perceive, learn, remember, and think about information."
- In 2005, Solso gave another definition of Cognitive psychology as the study of processes underlying mental events.
- A mental event is any event that happens within the mind of a conscious individual. Examples include thoughts, feelings, decisions, dreams, and realizations. Some believe that mental events are not limited to human thought but can be associated with animals and artificial intelligence as well.
- In general, Cognitive psychology can thus be defines as that branch of psychology that is concerned with how people acquire, store, transform, use and communicate language.
- The cognitive psychologists study the various cognitive processes that make up this branch. These processes include attention, the process through which we focus on some stimulus; perception, the process through which we interpret sensory information; pattern recognition, the process through which we classify stimuli into known categories; and memory, the process through which information is stored for later retrieval,(the process of getting something back from some were. thirigi pondadam), and soon. Thus, the work of cognitive psychologists is extended to a number of areas, which can be depicted(SHOW) as follows.

Cognitive Psychologist

- Cognitive psychologists study internal processes that include perception, attention, language, memory, and thinking.
- While many cognitive psychologists specialize in research and are employed by universities or government agencies, others take a clinical focus and work directly with individuals who are experiencing challenges related to different mental processes. They may work in hospitals, mental health clinics, or private practices.

Reasons to Consult a Cognitive Psychologist

- Alzheimer's disease, dementia, or memory loss
- Brain trauma treatment
- Cognitive therapy for a psychological illness
- Interventions for learning disabilities
- Perceptual or sensory issues
- Therapy for a speech or language disorder
- The work of cognitive psychologists is essential for helping people who have experienced issues with mental processes. While we tend to take abilities such as attention and problem solving for granted, perhaps because they are so woven into the fabric of our everyday existence, cognitive disruptions can create havoc in multiple areas of an individual's life.
- Dementia is a term used to describe a group of symptoms affecting memory, thinking and social abilities severely enough to interfere with your daily life. It isn't a specific disease, but several diseases can cause dementia. Though dementia generally involves memory loss, memory loss has different causes.

Although the early signs vary, common early symptoms of dementia include:

- memory problems, particularly remembering recent events.
- increasing confusion.
- reduced concentration.
- loss of ability to do everyday tasks.
- According to the American Psychological Association (APA), cognitive psychology is the "study of higher mental processes such as attention, language use, memory, perception, problem solving, and thinking." As a scientific study of mind and mental functioning, the core focus of modern cognitive psychology is **on studying** .
- Attention is the ability to choose and concentrate on relevant stimuli. Attention is the cognitive process that makes it possible to position ourselves towards relevant stimuli and consequently respond to it. This cognitive ability is very important and is an essential

function in our daily lives. Luckily, attention can be trained and improved with the appropriate cognitive training.

Types of Attention

- The most accepted model for the attention sub-components is currently the hierarchical model from Sohlberg and Mateer (1987, 1989), which is based on clinical cases of experimental neuropsychology. According to this model, attention can be divided into the following parts:

- **Arousal:** Refers to our activation level and level of alertness, whether we are tired or energized.

- **Focalized (Focused) Attention:** This refers to your ability to focus attention on one stimulus. For example, when you are writing an exam and need to concentrate entirely on your answers.

- **Sustained Attention:** The ability to attend to a stimulus or activity over a long period of time.

Examples of sustained attention may include **listening to lecture, reading a book, playing a video.**

- **Selective Attention:** The ability to attend to a specific stimulus or activity in the presence of other distracting stimuli.

- Having a conversation with a friend in a crowded place

- Reading your book on a public transport bus.

- **Alternating Attention:** The ability to change focus attention between two or more stimuli.

- It could also be alternating between unrelated tasks such as cooking while helping your child with her homework.

- Divided Attention**:** The ability to attend different stimuli or attention at the same time. This refers to your brain's ability to attend to different stimuli at the same time. For example, when you are driving a car while talking to a passenger.

- Divided attention is the ability to pay attention to two tasks at once such as cooking a meal while talking to a friend or driving a car and talking to a passenger at the same time

– neither activity is stopped in order to carry out the other activity. Impairments of alternating and divided attention may cause the person difficulties:

- In 1885, Hermann Ebbinghaus (1850-1909), a German psychologist was the first person to study memory in a scientific way.
- Ebbinghaus conducted memory experiments on himself and the findings were found to be valid. Ebbinghaus taught himself nonsense syllables (consonant-vowelconsonant trigrams, like TOB, etc.). They can be pronounced but are meaningless.
- Ebbinghaus created 20,000 such trigrams(a group of three consecutive written units such as letters, syllables, or words), in his experimental study. He assumed that meaningful stimuli would be more memorable than non-meaningful stimuli.
- Based on the experimental findings, Ebbinghaus also concluded that distributed practice (the efforts to learn the material is spread overtime) is often better than massed practice (the attempt to memorize the material at once).

Definition of Memory:

- The power or process of reproducing or recalling what has been learned and retained especially through associative mechanisms.
- Memory refers to the ability of retaining information and reproducing it over a period of time when required to perform a cognitive task. It has been conceptualized as a process comprising of three stages.
- **Memory** is the faculty of the brain by which data or information is encoded, stored, and retrieved when needed.
- Memory refers to the ability of retaining information and reproducing it over a period of time when required to perform a cognitive task. It has been conceptualized as a process comprising of three stages; (i) encoding, (ii) storage, and (iii) retrieval. All information received by our senses goes through these stages.

i) **Encoding:** It is the process of converting sensory information into a form that can be processed further by the memory systems.

ii) **Storage:** In this second stage, received information by memory systems are stored so that it can be used at later time also.

iii) **Retrieval:** It refers to locating and bringing the stored material to one's awareness when required to complete a task

- **Ulric Richard Gustav Neisser** (December 8, 1928 – February 17, 2012) was a German-American psychologist and member of the US National Academy of Sciences. He has been referred to as the "father of cognitive psychology". Neisser researched and wrote about perception and memory. He posited that a person's mental processes could be measured and subsequently analyzed.
- In 1967, Neisser published *Cognitive Psychology*.
- **Died:** February 17, 2012 (aged 83); Ithaca, New ...
- **Born:** December 8, 1928; Kiel, Schleswig-Hols...
- In 1986, while teaching at Emory, Neisser conducted a famous experiment that centered on the space shuttle Challenger explosion. The day following the **tragedy**, he asked students to write down their impressions immediately after hearing the news; then, almost three years later, he had them perform the same exercise again. Most of the accounts were remarkably different—supporting Neisser's theory that the mind distorts and reshapes the past, drawing on layered memories rather than actual events.

Cognitive Psychology and the Brain

- Introduction Imagine the following situation:
- A young man, let's call him Knut, is sitting at his desk reading some papers which he needs to complete a psychology assignment. In his right hand he holds a cup of coffee. With his left one he reaches for a bag of sweets without removing the focus of his eyes from the paper. Suddenly he stares up to the ceiling of his room and asks himself: "What is happening here?" Probably everybody has had experiences like the one described above.
- Even though at first sight there is nothing exciting happening in this everyday situation, a lot of what is going on here is highly interesting particularly for researchers and students in the field of Cognitive Psychology. They are involved in the study of lots of incredibly fascinating processes which we are not aware of in this situation. Roughly speaking, an analysis of Knut's situation by Cognitive Psychologists would look like this: Knut has a problem, he really needs to do his assignment.

- To solve this problem, he has to perform loads of cognition. The light reaching his eyes is transduced(prasaram) into electrical signals traveling through several stations to his visual cortex. Meanwhile, complex nets of neurons filter the information flow and compute contrast(differ strikingly), colour, patterns, positions in space, motion(kadalika) of the objects in Knut's environment. Stains(marakalu) and lines on the screen become words; words get meaning, the meaning is put into context, analyzed on its relevance for Knut's problem and finally maybe stored in some part of his memory.
- At the same time an appetite for sweets is creeping from Knut's hypothalamus, a region in the brain responsible for controlling the needs of an organism. This appetite finally causes Knut to reach out for his sweets.

History of Cognitive Psychology

- In 1885, Hermann Ebbinghaus (1850-1909), a German psychologist was the first person to study memory in a scientific way.
- Ebbinghaus conducted memory experiments on himself and the findings were found to be valid. Ebbinghaus taught himself nonsense syllables (consonant-vowel consonant trigrams, like TOB, etc.). They can be pronounced (very noticeable or marked; conspicuous)but are meaningless.
- **Consonant Trigram** (ACT) a memory test in which three letters consisting only of consonants, such as DCJ, are presented verbally and the participant.
- a memory test in which three letters consisting only of consonants, such as *DCJ*, are presented verbally and the participant is asked to recall the sequence following delays of varying lengths (e.g., 9, 18, and 36 seconds). During the delay intervals, the participant performs a distractor task (e.g., counting backwards from a specified number by threes)
- Ebbinghaus created 20,000 such trigrams(a group of three consecutive written units such as letters, syllables, or words), in his experimental study. He assumed(Guessed) that meaningful stimuli would be more memorable than non-meaningful stimuli.
- Based on the experimental findings, Ebbinghaus also concluded that distributed practice (the efforts to learn the material is spread overtime) is often better than massed practice (the attempt to memorize the material at once).

- Psychology has a short history but a long past', this remark was made by Hermann Ebbinghaus over a century ago, which is a kind of reflection on Greek philosophers who wrote on human nature. Thus, Greek philosophers like Socrates (428-348 BCE), Plato (428-347 BCE) and Aristotle (384-322 BCE) tried to explain human mind and its relationship to the physical body as early as 4 BCE. Socrates famous thought 'know thyself' emphasized on the importance of self and personal reflection. Later on, French philosopher René Descartes (1596-1650) considered pineal gland (body physiology) as the 'seat of the soul', the place where all thoughts are formed.

- The success of experimental methods in physics motivated some scientists to use experiments to study mind and behavior. One of the first scientists to study psychological processes was the German physiologist Gustav Theodor Fechner (1801-1887), who studied psychophysics, a branch of psychology. The first laboratory was established in 1879 in Leipzig, Germany by Wilhelm Wundt(1832-1920).In 1883 in the USA, the first formal psychology laboratory was set up in John Hopkins University by Wundt's student, G. Stanley Hall (1844-1924).

- There was much progress, but moving away from philosophy was quite difficult. American Psychological Association (APA) started in 1892 and Hall became its first president.

- William James (1842-1910) wrote the first textbook on psychology known as "The Principles of Psychology". The psychologists, in the beginning, had an empirical approach to understand thinking, attention, imagery, etc. The experiments were done to study the mind and mental experiences. Gradually, different schools of thought came up with different viewpoints about the nature of the study of psychology. They **emphasized** that the focus should be on the study of behavior rather than the mind.

Structuralism

Founders of Structuralism

Wilhelm Wundt (1832-1920)

- **Wilhelm Wundt:-** Wilhelm Wundt was German physiologist, Philosopher and Professor, know today as one of the founders of modern psychology.

- **Born:-** 16 August 1832.

- **Died:-** 31 August 1920.

- ❖ In 1879 at university of Leipzig ,Wundt founded the first psychological laboratory for psychological research and also called as father of experimental psychology.
- Elements of mind: sensation, feelings, and images.
- The main focus is on analyzing the basic structures of the human mind. This view is known as structuralism. It is associated with Wilhelm Wundt and Edward Titchener (Wundt's student). Wundt believed that consciousness could be broken into thoughts, experiences, emotions and other basic elements.
- To analyze the structure of the mind. Through the method of introspection.
- They believed that the mind is made up of building blocks in the various types of sensation and perception and that these building blocks could be discovered through introspection or looking into one's own mind. Introspection, which required subjects to look inward and observe and report on the working of their mind.
- Aimed to study identify the building blocks of conscious experience.
- Sensations, feelings and images as building blocks.
- ❖ Edward B. Tichener (1867-1927)Professor of Psychology at Cornell University.
- ❖ According to him psychology may be regarded as the science of consciousness or study of experience.
- ❖ Tichener's analyzed consciousness there basic elements.

(1). Sensations.
(2). Images.
(3). Affections.

- With structuralism came an active interest in knowing the elements of consciousness. Structuralism was very strong in the beginning, but faded (gradually grow faint and disappear)out in 1900s.
- **Sensations:-** A sensation is simplest form of mental activity and originate from the sense organs. Basic elements of perception sight, sound, taste, smell, and touch.
- ❖ **Images:-** Elements of Ideas- reflects of experiences not actually present at the time. Example- Memory of a sight.

- **Affections:-** Feelings- the elements of emotions- Love, hate, anger, sadness etc….

What are the Five Element's emotions

- Traditionally, the five emotions are sadness, fear, anger, happiness, and worry. ...
- Knowledge of these five emotions is important for both diagnosis and treatment.

Functionalism

- William James's focus on how mental activities help a person adapt to his or her environment is known as functionalism.
- As James saw it, psychology's purpose was to study the function of behavior in the world, and as such, his perspective was known as **functionalism**.
- Mind or consciousness cannot be divided into elements.
- Introspection cannot yield measures about the mind.
- Functionalism focused on how mental activities helped an organism fit into its environment.
- Functionalism school of psychology concerned with How the mind helps us adopt to our environment.
- This view was strongly influenced by Darwin's theory of natural selection (functionalists believed that the theory could be applied to psychological characteristics) and the focus of study was functions of mind and behavior (like learning, memory, problem-solving and motivation).
- John Dewey (1859-1952) developed functionalism, focusing on what the mind and behavior do and how they enable a person to adapt to new and complex situations.
- William James (1842- 1910), the main contributor, was concerned on how the mind allows people to function in the real world. This approach came to be known as functionalism. James established psychology laboratory at Harvard University.

What difference between Structuralism and Functionalism

- Structuralism focus on introspection and breaking down mental events to the smallest elements.

- A mental event is any event that happens within the mind of a conscious individual. Examples include thoughts, feelings, decisions, dreams, and realizations. Some believe that mental events are not limited to human thought but can be associated with animals and artificial intelligence as well.
- Functionalism focused on the wholeness of an mental events, taking in to the impact of the environment on behavior.

Behaviorism

- This school originated with John B.Watson (1879-1958) and Burrhus Fredrick Skinner (1904-1990), who disregarded mind as the matter of study and stressed that psychology should study only observable forms of behavior, ignoring the covert processes. They gave more importance to the study of behavior and what activities animals or humans engage in. Watson emphasized on conditioned responses, learned behavior and animal behavior.
- Ivan Petrovich Pavlov (1849-1936), a physiologist whose studies of digestion won him the Nobel Prize in 1904, concluded that stimuli other than food, may produce salivation and this led him to study conditioned reflex in detail.
- Behavioral psychology is the study of external behavior.
- Behavior is the response of an organism to stimuli.
- Behavior is the out word organic of mental processes or behavior as an act which can be Observed and measured Objectively- **J.B. Watson.**
- Behavior is any Overt (observable) response or activity by an organism. From the word behavior one can extract two meaning, because there are two kind of behavior Overt and Covert behavior.
- **Overt behavior** are those which is visible to other in other word seen behaviors, Example: The way we Talk, Walk, Dress, Eat and so on… in short these are the action we see and observe in individual and group around us..
- **Covert behavior** are those which is not obvious to the people around us. Simply they are unseen behavior. For example, Feeling such as anger, jealousy, kindness, pity, happiness and sad.

Gestalt Psychology

- Opposed behaviorism and structuralism.

- S-R components or mental elements could not explain behavior or mind of an individual.
- Holistic approach to understanding behavior or mind.
- Later psychologists conducted experiments to study sensations in order to know how the mind functions. In 1912, Max Wertheimer (1880-1943), Kurt Koffka (1886-1941) and Wolfgang Kohler (1887-1967) founded gestalt psychology in Germany. They emphasized upon the whole notion of sensory experience, connecting the sensations to relations and organization as a whole. They studied the mind by applying the principles of organization to human behavior.

Cognitive Perspective:

- The gestalt approach and the development of computers stimulated interest in studying higher-order mental processes. This perspective gained prominence in 1960s and focused on memory, intelligence, language, problem-solving, and decision-making. Main contributors are German psychologist Hermann Ebbinghaus (1850–1909), who started the study on memory by investigating the formation of associations by learning serial lists of nonsense syllables, and the English psychologist, Sir Frederic Bartlett (1886–1969), who studied the cognitive and social processes of remembering.

History of Cognitive Psychology

- Generally speaking two approaches have been used to study the mind.
- Historically, philosophical approaches have rendered an understanding to the workings of the mind using introspection.
- And biology, has used observational (and empirical(by experimental) methods to study the mind.

Lesson 2 : The Philosophical Background: Early Greek Philosopher – Descartes; Locke.

Mind & Reality

- Plato was an Athenian philosopher during the Classical period in Ancient Greece, founder of the Platonist school of thought and the Academy, the first institution of higher learning in the Western world.
- Two part : 1) Basic Education First Stage: (Birth – 6 years) Study their own Language using Pictures

- Second Stage : (7-18 years) Music & Gymnastics (for soul & mind
- Third Stage : 18-19 Compulsory Military Training
- Higher Education Two Part: 20-30 years (a selective test) 31- 35 years Aim : To make people Wise Subject: Logic, Maths, Astrology.

Plato (ca. 447-327 BC): Reality does not reside in concrete objects but in abstract forms represented in our mind.

- Plato was an idealist.
- What is Plato's view of reality?
- Plato believed that **true reality is not found through the senses**. Phenomenon is that perception of an object which we recognize through our senses. Plato believed that phenomena are fragile and weak forms of reality. They do not represent an object's true essence.
- **Aristotle (384-322 BC):**
- Aristotle was one of the greatest philosophers who ever lived and **the first genuine scientist in history**. He made pioneering contributions to all fields of philosophy and science, he invented the field of formal logic, and he identified the various scientific disciplines and explored their relationships to each other.
- Reality lies only in the concrete world of objects accessed through the senses.
- Aristotle was a realist.

Aristotle philosophy?

- Aristotle is a towering figure in ancient Greek philosophy, who made important contributions to **logic, criticism**, rhetoric, physics, biology, psychology, mathematics, metaphysics, ethics, and politics. ... As the father of western logic, Aristotle was the first to develop a formal system for reasoning.

Seventeenth Century

- **Rene Descartes (1596-1650):** Descartes agreed with Plato and emphasized reflective methods (rationalism) over observation to study mind.
- **John Locke (1632-1704):** Locke favored Aristotle and emphasized observation as a method (empiricism) to study mental processes.

- **Immanuel Kant** In eighteenth century German philosopher Kant addressed the issue of rationalism and empiricism and said that both approaches must be used to decipher truth and about reality of mind.

- **Rene Descartes (1596-1650):**

- René Descartes was a French philosopher, mathematician, and scientist who invented analytical geometry, linking the previously separate fields of geometry and algebra.

- Descartes agreed with Plato and emphasized reflective methods (rationalism) over observation to study mind.

- He was born on March 31 1596 in La Haye En Touraine inside the kingdom of France.

Carrier:

- Descartes was the first to provide a framework in philosophy for the nature sciences and mathematics as they developed France.

- René Descartes (1596–1650) was a creative mathematician of the first order, an important scientific thinker, and an original metaphysician. During the course of his life, he was a mathematician first, a natural scientist or "natural philosopher" second, and a metaphysician third. In mathematics, he developed the techniques that made possible algebraic (or "analytic") geometry. In natural philosophy, he can be credited with several specific achievements: co-framer of the sine law of refraction, developer of an important empirical account of the rainbow, and proposer of a naturalistic account of the formation of the earth and planets (a precursor to the nebular hypothesis).

- Metaphysics is the branch of philosophy that studies the first principles of being, identity and change, space and time, causality, necessity and possibility. It includes questions about the nature of consciousness and the relationship between mind and matter.

- More importantly, he offered a new vision of the natural world that continues to shape our thought today: a world of matter possessing a few fundamental properties and interacting according to a few universal laws. This natural world included an immaterial mind that, in human beings, was directly related to the brain; in this way, Descartes formulated the modern version of the mind–body problem. In metaphysics, he provided arguments for the existence of God, to show that the essence of matter is extension, and that the essence of mind is thought. Descartes claimed early on to possess a special

method, which was variously exhibited in mathematics, natural philosophy, and metaphysics, and which, in the latter part of his life, included, or was supplemented by, a method of doubt.

- Descartes presented his results in major works published during his lifetime: the *Discourse on the Method* (in French, 1637), with its essays, the *Dioptrics*, *Meteorology*, and *Geometry*; the *Meditations on First Philosophy* (i.e., on metaphysics), with its Objections and Replies (in Latin, 1641, 2nd edn. 1642); the *Principles of Philosophy*, covering his metaphysics and much of his natural philosophy (in Latin, 1644); and the *Passions of the Soul*, on the emotions (in French, 1649).

- Important works published posthumously included his *Letters* (in Latin and French, 1657–67); *World, or Treatise on Light*, containing the core of his natural philosophy (in French, 1664); *Treatise on Man* (in French, 1664), containing his physiology and mechanistic psychology; and the *Rules for the Direction of the Mind* (in Latin, 1701), an early, unfinished work attempting to set out his method.

Early life and education

- Descartes was born on 31 March 1596 in his maternal grandmother's house in La Haye, in the Touraine region of France. His father Joachim, a lawyer who lived in Châtellerault (22 kilometers southwest of La Haye, across the Creuse River in the Poitou region), was away at the Parliament of Brittany in Rennes. The town of La Haye, which lies 47 kilometers south of Tours, has subsequently been renamed Descartes.

- When Descartes was thirteen and one-half months old, his mother, Jeanne Brochard, died in childbirth. The young René spent his first years with his grandmother, Jeanne Sain Brochard, in La Haye, together with his older brother Pierre and older sister Jeanne. It is likely that he then moved to the house of his great uncle, Michel Ferrand, who, like many of René's male relatives, was a lawyer, and who was Counselor to the King in Châtellerault.

- In 1606 or 1607, Descartes entered the newly founded Jesuit College of La Flèche, where he remained until 1614 or 1615. He followed the usual course of studies, which included five or six years of grammar school, including Latin and Greek grammar, classical poets, and Cicero, followed by three years of philosophy curriculum. By rule, the Jesuit philosophy curriculum followed Aristotle; it was divided into the then-standard topics of

logic, morals, physics, and metaphysics. The Jesuits also included mathematics in the final three years of study.

- Descartes died on February 11 1650 age 53 in Stockholm, Sweden while he was invited as a philosophy teacher for the Queen Christina of Sweden.
- Comparison
- Rationalism
- Understanding of mind (or knowledge) through logical analysis and introspection (rationalism).
- Innateness of mind or mental faculties.
- Theory building in psychology today.
- Rationalism is the view that reasoning by itself is a source of knowledge or proof. It is a philosophical movement that began in the 17th century, but the same ideas were also talked about before then.
- **René Descartes** (1596-1650), the first of the modern rationalists, laid the groundwork for debates developed during the Enlightenment. He thought that the knowledge of eternal(without end or beginning) truths could be attained by reason alone (no experience was necessary**French philosopher René Descartes**, who wrote "I think therefore I am," is considered the father of rationalism. He believed that eternal truths can only be discovered and tested through reason.
- Rationalism is an epistemological position in which reason is said to be the primary source of all knowledge, superior to the senses.
- Epistemology is defined as a branch of philosophy that is defined as the study of knowledge. An example of epistemology is a thesis paper on the source of knowledge.
- In general, rationalists believe that abstract reasoning can produce undeniable, absolutely certain truths about nature, existence, and the whole of reality. These truths are called a priori, or innate, ideas – because they are discovered independently of experience, without empirical observation or experimentation.
- Descartes stands not only as the “father of modern philosophy,” but as the original archetype of the modern rationalist.

- What is Philosophy of Rationalism?
- Encompasses several strands of thought all of which usually share the conviction that REALITY is actually rationale in nature and that making the proper deductions is essential in achieving knowledge.
- Latin word ratio, meaning reason-point of view that states that reason plays the main role in understanding the world and obtaining knowledge.
- is “any view appealing to reason as a source of knowledge or justification.” in which the criterion of the truth is not sensory but intellectual and deductive.
- The philosophical view that regards reason as the chief source and test of knowledge. Holding the reality itself has an inherent logical structure. **Rationalism and Empiricism**
- Empiricism: All knowledge of the world comes from experience.
- Rationalism: Some knowledge of the world is independent of experience— that is, some knowledge is inborn (innate).
- **Innate ideas**
- “Some ideas are present from birth.”
- Ideas that do not require the proof or suggestion of sense experience, are concept which are present from birth.
- It could be theoretically be discovered or brought out from within the mind of each individual.
- Example is Descartes arguments for the existence of God

John Locke:

- John Locke FRS (29 August 1632 – 28 October 1704) was an **English philosopher and physician**, widely regarded as one of the most influential of Enlightenment thinkers and commonly known as the "Father of Liberalism". ... Locke was the first to define the self through a continuity of consciousness.
- **Born:** John Locke; 29 August 1632; Wrington, ...
- **Died:** 28 October 1704 (aged 72); High Laver, ...
- **School:** Empiricism; Foundationalism; Concept...

- **Main interests:** Metaphysics, epistemology
- **Liberalism** is a political and moral philosophy based on liberty, consent of the governed and equality before the law.

Some examples of liberalism?

- Modern liberalism includes issues such as same-sex marriage, reproductive and other women's rights, voting rights for all adult citizens, civil rights, environmental justice and government protection of the right to an adequate standard of living.

Empiricism

- **John Locke (1632-1704):** Locke favored Aristotle and emphasized observation as a method (empiricism) to study mental processes.
- All knowledge of the world comes from experience.
- Observations of the external world are the only means to arrive at truth (empiricism).
- Acquired experiences through interaction with the environment.
- Experimentation in psychology today.
- Believed that at birth human mind is a blank slate; a tabula rasa (empty of ideas).
- Tabula rasa is a Latin word that originates from the Roman tabula or wax tablet used for notes, which was blanked by heating the wax and then smoothing it.
- The mind in it's hypothetical primary empty state before receiving outside
- All ideas are based on sensation – arriving at explanation by observing phenomena.
- Learning should be a gradual process ,slow & cumulative.
- Locke believed that we gain knowledge by experience, this is called empiricism
- Locke agreed with Aristotle that we are not born with innate ideas, and that we learn through experience.
- Locke's theory was that the only way we learn is by tasting, smelling, touching, and hearing the external world
- Locke believed that there were two types of ideas, Simple and Complex

- Simple ideas are easy sensations like feeling a hot stove
- Complex ideas are a grouping of simple ideas like when you look at a banana, you see it is yellow (color) it is mushy (texture) it is sweet (taste) these ideas form the idea of a banana
- Ideas that are nonmaterial were also explained by simple ideas
- If you imagine a griffin, you have put together the parts of different animals the body of a lion, the head, wings, and talons of an eagle
- Locke stated that all things have primary and secondary qualities
- Primary: height and weight (undeniable(kadanaleni) truths)
- Secondary: rely on subjective or personal judgments (color, taste and sound).
- Locke disagreed with Rene Descartes, he believed that humans are born without any knowledge of the world
- Descartes borrowed ideas from Plato, he believed that we are all born with innate ideas
- Locke agreed with Aristotle's theory that humans are born with a clean slate, and that we learn while we grow, experiencing the world as we go
- Aristotle believed that we gain knowledge by experiencing the world with our senses
- SENSE: Sight, hearing, smell, taste, or touch
- SENSATION: The operation or function of the senses; perception or awareness of stimuli through the senses
- REFLECTION: Thinking about what you experienced, processing the information
- IMPRESSION: The first and immediate effect of an experience or perception upon the mind
- QUALITIES: Essential or distinctive characteristic, property, or attribute.

Classical Empiricism According to Classical Empiricism:

- The mind is a blank slate (tabula rasa)
- Knowledge of existing things is 'written' onto this blank slate through sense experience.
- All Knowledge, concepts, universals, and general ideas come from experience Classical Empiricism.

Modern Empiricism Locke's Theory of Knowledge

- There are two forms of experience:
- Inner Experience
- Reflection Processes of cognition
- Reflection is a **means of**. **processing thoughts and feelings about an incident**, or a difficult day…and gives us a chance to come to terms with our thoughts and feelings about it. Reflection can be particularly useful in dealing with a difficult or challenging. situation.
- External Experience
- Sensations of objects in the external world
- Modern Empiricism
- All knowledge is derived from external and internal Experience.
- Your external experience includes **anything that you can sense via your five senses**, so anything that you can see, hear, taste, smell and touch. This exercise will help you to develop two important skills for managing your mental health.
- Your internal experiences include **your thoughts, emotions, physical sensations (including your breath) and any urges that you notice**. Your external experience includes anything that you can sense via your five senses, so anything that you can see, hear, taste, smell and touch.
- **Locke's Arguments against Rationalism (Negative arguments)**
  Locke argues that the theory of innate ideas employed by the rationalists is incorrect for the following reasons (pp):

- Rationalists claim that there are certain principles agreed upon by everyone
- Universal consent proves nothing innate
- There is no universal consent
- If there were universal ideas children and 'idiots' would possess them
- Rationalists claim that innate ideas can only be known through the use of reason
- Knowledge derived from reason is not necessarily innate
- Reason is only a faculty of deduction
- Modern Empiricism
- **Locke's Arguments for Empiricism (Positive arguments)**
  (pp)Idea is the object of thinking
- All ideas come from Sensation or Reflection
- The objects of sensation one source of ideas
- The operations of our mind the other source of them
- All ideas are either from sensation or reflection
- Observable in children
- Uncompounded appearance
- The mind cannot make or destroy ideas
- Complex ideas are created (by the mind) from simple ones
- Made voluntary

Immanuel Kant

- Immanuel Kant was a German philosopher and one of the central Enlightenment thinkers. Kant's comprehensive(samagra) and systematic works in epistemology, metaphysics, ethics, and aesthetics have made him one of the most influential figures in modern Western philosophy.
- Born: 22 April 1724, Königsberg
- Died: 12 February 1804, Königsberg

- Influenced: Georg Wilhelm Friedrich Hegel, Friedrich Nietzsche, MORE
- Influenced by: Aristotle, René Descartes, Plato, David Hume, MORE.
- Parents: Regina Dorothea Reuter, Johann Georg Kant.
- What is Kant's philosophy.
- His moral philosophy is a **philosophy of freedom**. Without human freedom, thought Kant, moral appraisal and moral responsibility would be impossible. Kant believes that if a person could not act otherwise, then his or her act can have no moral worth.

Lesson 3 : British Associationism; Kant. Associationism in Psychology: Contributions of Ebbinghaus.

Associationism

- Studied “higher mental processes”, like memory and forgetting.
- Used systematic experimental introspection.
- Followed Aristotelian law of association. Repetition improves memory.
- Associationism is the idea that mental processes operate by the association of one mental state with its success or states. It holds that all mental processes are made up of discrete psychological elements and their combinations, which are believed to be made up of sensations or simple feelings.
- For instance, you hear the word "summer." This word may bring up memories of the beach, swimming, hot weather, or going camping. Associationism is the concept that mental processes occur as a result of whatever the individual may associate (either good or bad) with the stimulus word.
- Aristotle's analysis of learning memory includes four laws of association: **the laws of contiguity, contrast, frequency and similarity**. Accounting for learning and memory in terms of such laws of association is called as

laws of contiguity:

- A principle of association stating that forming connections between ideas, events (e.g., stimuli and responses), or other items depends on their proximity in space or time.

- In cognitive science, association by contiguity is the principle that ideas, memories, and experiences are linked when one is frequently experienced with the other. For example, if you constantly see a knife and a fork together they become linked (associated).

- The Law of Contiguity states that we associate things that occur close to each other in time or space. For example, if we think of thunder, we immediately think of lightning, since the two often occur one after the other. ... For example, when we think of coffee, we often think of tea as well.

Law of contrast:

- A principle of association stating that opposites are reminders of one another: encountering or thinking about one (e.g., a snow-covered field) tends to bring to mind the other (e.g., a sunny beach).

- For example, when we think of coffee, we often think of tea as well. The Law of Contrast states that the thought of something is likely to trigger the thought of its direct opposite. For example, when we hear the word "hot," we often think of the word "cold."

Law of frequency:

- The law of frequency suggests that the more a person practices the correct and desired behavior, the higher the probability there is to use the desired behavior. This principle is often called "drill and practice".

- The more often two things or events are linked, the more powerful will be that association. If you have an eclair with your coffee every day, and have done so for the last twenty years, the association will be strong indeed -- and you will be fat.

The Law of Similarity:

- The Law of Similarity is the gestalt grouping law that **states that elements that are similar to each other tend to be perceived as a unified group**. Similarity can refer to any number of features, including color, orientation, size, or indeed motion.

- **Similarity (**also known as **Invariance):** The human eye tends to build a relationship between similar elements within a design. Similarity can be achieved using basic elements such as shapes, colors, and size.

British associationism:

- David Hume, John Locke, **David Hartley**, George Berkeley, James Mill, and John Stuart Mill are the six British philosophers most famous as starting the school of associationism. The general theory of associationism says that simple processes, associated with one another, will over time yield more complex processes.

What is British associationism?

- Associationism is the theory that complex mental processes, such as learning and thinking, occur because external stimuli are linked together in the mind. The idea became a hallmark of British philosophers, but it is attributed to Aristotle and other Greek philosophers.
- Hume's associationism was, first and foremost, a theory connecting how perceptions ("Impressions") determined trains of thought (successions of "Ideas"). Hume's empiricism, as enshrined in the Copy Principle, demanded that there were no Ideas in the mind that were not first given in experience.
- David Hume presented arguably the first attempt to understand thought generally in associative terms. He first lays out these views in A Treatise of Human Nature (1739) and then reiterates them in An Enquiry Concerning Human Understanding (1748). According to Hume, the trains of thought are made up of ideas, which are basically images in the mind. Simple ideas, such as a specific color, taste, or smell, are copies of sensory impressions. Thoughts in general are built from these simple ideas by association.

John Locke (1632-1704)

Empiricism

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External Experience

- Sensations of objects in the external world

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David Hartle:

- Hartley's Observations on Man (1749) was published just after Hume's Enquiry, though he claimed to have been thinking about the power of association for about 18 years. Hartley's discussion of association is more focused and sustained than Hume's because of his explicitly programmatic goals. Following Newton's axiomatization of physics, Hartley sought to axiomatize psychology on the twin principles of association and vibration. Vibrations, in Hartley's system, are the physiological counterpart of associations.

- **Axiomatisation definition**: the process of reducing down to a system of basic truths , or axioms

- As association carries the mind from idea to idea, vibrations in the nerves carry sensations to the brain and through it. He references physical vibrations as causing mental associations but then expresses dissatisfaction with this framing and uncertainty on the exact association-vibration relation.

- The idea is that external stimuli act on nerves, inciting(preraipinchadam) infinitesimally small vibrations in invisible particles of the nerve. These vibrations travel up the nerves, and upon reaching the brain, cause our experience of sensations. If a particular frequency or pattern of vibration is repeated, the brain gains the ability to incite new vibrations like them. This is, effectively, storing a copy of the idea for later thought.

- These 'ideas of sensation' are the elements from which all others are built. Ideas become associated when they are presented at the same time or in immediate succession, meaning that the first idea will bring the second to mind, and, correspondingly, their vibrations in the brain will follow in sequence.

- Hartley, like Hume, viewed association as both the principle by which ideas came to follow one another and by which simple ideas were combined into complex ideas: A complex idea is the end point of the process of strengthening associations between simple

ideas. Unlike Hume, though, Hartley only posited association by contiguity and did not allow for any other laws of association.

- Mill presents his views in his Analysis of the Phenomena of the Human Mind. Like Hartley, James Mill argues that contiguity is the only law of association. Specifically, James Mill argues that similarity is just a kind of contiguity. The claim is that we are used to seeing similar objects together, as sheep tend to be found in a flock, and trees in a forest. In his editorial comments in the 1869 edition, John Stuart Mill calls this "perhaps the least successful attempt at a generalisation and simplification of the laws of mental phenomena, to be found in the work". For his part, James Mill does not attribute much significance to the question, saying: Whether the reader supposes that resemblance is, or is not, an original principle of association, will not affect our future investigations.

- James Mill reduces the various "active" and "intellectual" powers of the mind to association. He limits his discussion of association to mental phenomena, though he recognizes the significance of physiology for motor movements and reflexes. For instance, conception, consciousness, and reflection simply refer to the train of conscious ideas itself. Memory and imagination are particular segments of the trains. Motives are associations between actions and positive or negative sensations which they produce.

- John Stuart Mill argues, against his father, that complex ideas are new entities, not mere aggregates of simple ideas, and that intermediate ideas can drop out of sequences like that above. In general, John Stuart Mill analogizes the association of ideas to a kind of chemistry, where a new compound has new properties separate from its constituent elements. In James Mill's view of association, ideas retain their identity in combination, like bricks in a wall.

Hermann Ebbinghaus contributions

- Hermann Ebbinghaus was a German psychologist who pioneered(margadarsakudu) the experimental study of memory, and is known for his discovery of the forgetting curve and the spacing effect. He was also the first person to describe the learning curve. He was the father of the neo-Kantian philosopher Julius Ebbinghaus.

- Born: 24 January 1850, Barmen, Germany

- Died: 26 February 1909, Halle (Saale), Germany

- Known for: Serial position effect, Über das Gedächtnis

- Children: Julius Ebbinghaus
- Parents: Julie Ebbinghaus, Carl Ebbinghaus
- Grandchild: Ernst A. Ebbinghaus.
- In 1885, Hermann Ebbinghaus (1850-1909), a German psychologist was the first person to study memory in a scientific way.
- Ebbinghaus conducted memory experiments on himself and the findings were found to be valid. Ebbinghaus taught himself nonsense syllables (consonant-vowelconsonant trigrams, like TOB, etc.). They can be pronounced but are meaningless.
- Ebbinghaus created 20,000 such trigrams(a group of three consecutive written units such as letters, syllables, or words), in his experimental study. He assumed that meaningful stimuli would be more memorable than non-meaningful stimuli.
- Based on the experimental findings, Ebbinghaus also concluded that distributed practice (the efforts to learn the material is spread overtime) is often better than massed practice (the attempt to memorize the material at once.
- Ebbinghaus has discovery of the forgetting curve and the spacing effect. He was also the first person to describe the learning curve.

Forgetting:

- The inability to retrieve previously stored information is called forgetting. Anything stored in the memory is subject to forgetting. forgetting may be of slow or fast depending upon the individual, the situation and the nature of the information.

Definition of Forgetting.

Psychologists define forgetting scientifically. Some of the definitions are as follows:

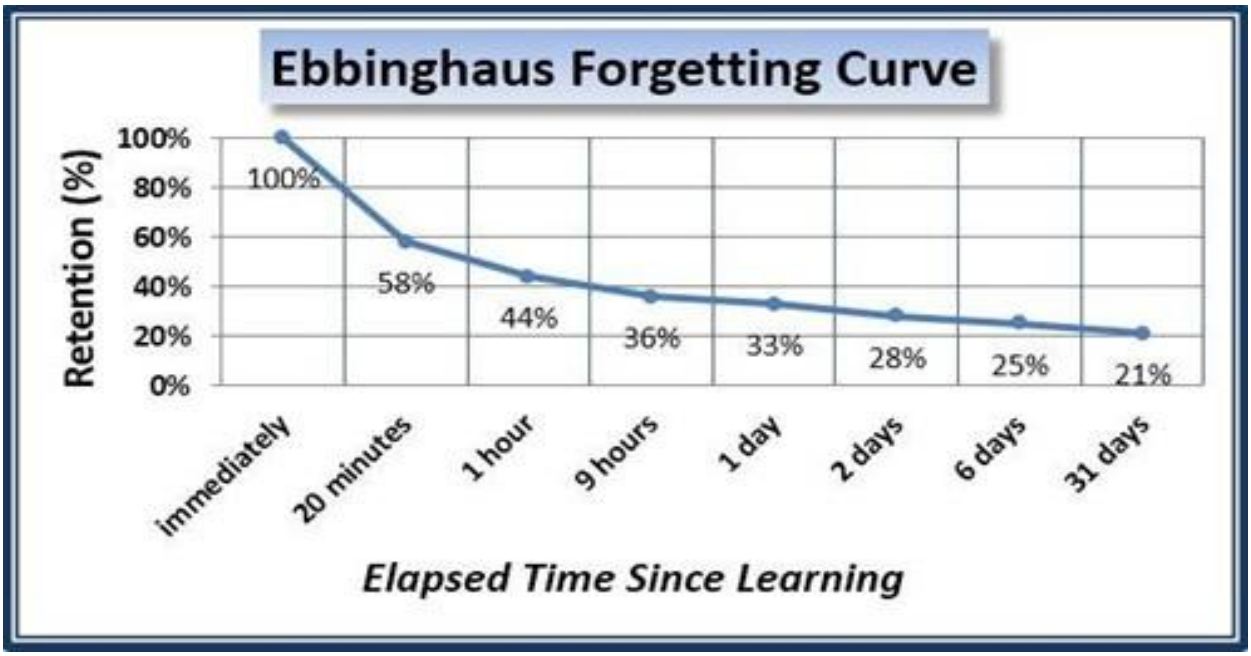
- **Drever (1952):** "Forgetting means failure at any time to recall an experience. when attempting to do so, or to perform an action previously learned."
- **Munn (1967):** "Forgetting is the loss, permanent or temporary, of the ability to recall or recognize something learned earlier.**“**
- **Bhatia (1968):** "Forgetting is the failure of the individual to revive in consciousness an idea or group of ideas without the help of the original stimulus**.“**

- **Fernald and Fernald (2005):** "Forgetting is an absence of memory; specifically, the inability to recall, to recognize, or to relearn at an improved rate the something which, the individual was exposed earlier."

- Thus, we understand that forgetting is failure in retrieving the past experiences either temporarily or permanently.

Ebbinghaus's Curve of Forgetting.

- The first formal research on forgetting was conducted by German Psychologist Ebbinghaus in 1885 by using non-sense syllables. He himself worked as a subject for his research and described his results by plotting a curve of forgetting. He memorized a list of non-sense syllables and then tested himself at various intervals. The results of material forgetton were as follows.



- The forgetting curve is a mathematical formula by Hermann Ebbinghaus that originated in 1885. The curve demonstrated the rate at which information is forgotten over time if we don't attempt to retain it.

- Learners will rapidly lose their memory of learned knowledge in a matter of days or weeks unless the information is consciously reviewed. Some studies suggest that humans forget approx 50% of new information within an hour of learning it. That goes up to an average of 70% within 24 hours.

- However, the rate at which a person forgets depends on several factors including memory strength, how meaningful the material is, and physiological factors such as stress. The good news is that there are a number of methods you can use in your courses to help your learners challenge the forgetting curve.

Spaced learning

- To thoroughly understand what is learned, there are two important elements to consider. One is time. And the other is the application of repetition.
- This can be achieved with spaced learning, which is why it's considered one of the best methods for combating the learning curve. Spaced learning helps learners manage what learned information is retained, enabling them to reshape the forgetting curve. In turn, spaced learning benefits organizations as it supports the retention of skills and increased productivity in the long-term.
- Spaced learning is a learning method in which highly condensed learning content is repeated three times, with two 10-minute breaks during which distractor activities such as physical activities are performed by the students.
- So, what exactly is spaced learning? Well, it's a learning methodology where learners are presented with material they have to learn in a timed session, with a short break provided after they've completed it. **Spaced learning strengthens memory retention because the learner studies the information, and periodically returns to review it in order to retain the knowledge**.
- The learner also practices retrieving the learned information using different formats such as solving problems, completing exams, and so on. This reinforces what has been learned, battling the forgetting curve.

- Hermann Ebbinghaus conclusion of forgetting, Ebbinghaus found **the forgetting curve to be exponential in nature**. Memory retention is 100% at the time of learning any particular piece of information. However, it drops rapidly to 40% within the first few days. After which, the declination of memory retention slows down again.
- In the early 20th century, three German psychologists Max Wertheimer, and his associates Wolfgang Köhler and Kurt Koffka proposed new principles for explaining perception called as Gestalt principle. Gestalt psychology is form psychology.

Gestalt view:

- The literal meaning of the word Gestalt is form or configuration(arrangement of elements in a particular form,figure). The basic premise(base) of Gestalt psychology is that **'whole is different from the sum of its part'.** This implies that organization (or structured whole, known as Gestalten) gives a different meaning to the perception.
- Gestalt psychologists proposed a number of principles or laws to explain how we perceive smaller units of stimuli as a whole, having a particular pattern. These principles are known as laws of perceptual organization.

Gestalt view of perception:

- Perception is not a bundle of some separate sensations. It is a simple unitary experience.
- Perceptual knowledge is always organised. By the arranging of some dots you can form a circle. but when the dots are quite separate the property of circle is lost.(Organization In this stage, stimuli are arranged mentally in a meaningful pattern)
- **Proximity(nearness in space time or relationship):** Proximity of elements favour grouping of them into a pattern. Things which are in close contact with each other form a unit. For example, four lines in the given figure do not appear as four discrete lines. They are perceived as two pairs of line.
- **Similarity:** Similar objects form a group. In the given figure, the triangles form a group and the circles form another group and they are perceived as two different groups of circle and triangle.
- **Familiarity:** pervious knowledge of a thing favours our perception. In the given figure, several dots have been arranged to give the shape of a human face. Previous knowledge of human face helps us to perceive the figure of dots as a human face.
- **Principle of closure**: There are gaps in each of the given figures. But they are perceived as complete figures having no gaps in them. This filling in gaps in stimulation is termed closure.
- **Figure ground relation**: In order to have a knowledge about organisation, we should understand the figure ground relation. The figure seems to have some sort of shape, while the ground tends to be formless. The figure appears to be in front and the ground behind.

The figure is always more organised than the field or ground. The given figure appears as either a flower-vase or two profiles.

- **Continuity:** Continuity of patterns is favourable to grouping elements into regular wholes. The dots in the given figure appear to be continuous though they are separate from one another. Dots **scattered(various locations rather than all together, chellacheduruga**) about irregularly are not easily perceived.

Lesson -4 The twentieth century Associationism. The Gestalt view, Bartlett's outlook.

Cognitive Psychology in Early Twentieth Century

- The representation of knowledge took a radical turn with the advent of twentieth century behaviourism and Gestalt psychology. The behaviourist views of human and animal psychology were cast in a framework of stimulus-response (S-R) psychology, and Gestalt theorists built elaborate conceptualisations of internal representation within the context of isomorphism–one-to-one relationship between representation and reality.
- Psychological studies of mental processes as conceptualised in the late nineteenth century suddenly became unfashionable, displaced by behaviourism. Studies of internal mental operations and structures such as attention, consciousness, memory, and thinking were laid to rest and remained so for about fifty years.
- To the behaviourists, internal states were subsumed under the label of "intervening variable," that mediated the effects of stimuli on responses and were neglected in favor of making observations on behaviour rather than on the mental processes.
- In 1932, some years before the cognitive revolution swept across psychology, learning psychologist Edward Tolman from the University of California at Berkeley published Purposive Behavior in Animals and Men. In this seminal work, Tolman observed that what rats learn in a maze is the layout of the land rather than simply a series of S-R connections.
- The animal, according to Tolman's interpretation, gradually developed a "picture" of his environment that was later used to find the goal. This picture was called a cognitive map.

- Tolman’s postulate about cognitive maps in animals did anticipate the contemporary preoccupation with how knowledge is represented in a cognitive structure.
- Also in 1932 Sir Frederick Bartlett from Cambridge University wrote Remembering in which he rejected the then popular view that memory and forgetting can be studied by means of nonsense syllables, as had been advocated by Ebbinghaus in Germany during the previous century. In the study of human memory, Bartlett argued, the use of rich and meaningful material under naturalistic conditions would yield far more significant conclusions.
- Bartlett introduced the concept of schema as a unifying theme that describes the essence of an experience. Schema theory plays a central role in modern theories of memory. The fecund ideas of Tolman in America and Bartlett in England highly influenced the thinking of future cognitive psychologists.

Bartlett's Schema Theory

- Schemata were initially introduced into psychology and education through the work of the British psychologist Sir Frederic Bartlett (1886–1969). In carrying out a series of studies on the recall of Native American folktales, Bartlett noticed that many of the recalls were not accurate, but involved the replacement of unfamiliar information with something more familiar.
- They also included many inferences that went beyond the information given in the original text. In order to account for these findings, Bartlett proposed that people have *schemata*, or unconscious mental structures, that represent an individual's generic knowledge about the world. It is through schemata that old knowledge influences new information.

Reconstructive Memory (Bartlett)

- Reconstructive memory suggests that in the absence of all information, we fill in the gaps to make more sense of what happened. According to Bartlett, we do this using schemas. These are our previous knowledge and experience of a situation and we use this process to complete the memory.
- In his major work, Remembering: A Study in Experimental and Social Psychology (1932), Bartlett advanced the concept that memories of past events and experiences are

actually mental reconstructions that are coloured by cultural attitudes and personal habits, rather than being direct recollections of observations made at.

- For example, one of Bartlett's participants read the phrase "something black came out of his mouth" and later recalled it as "he foamed at the mouth." This finding could be accounted for by assuming that the input information was not consistent with any schema held by the participant, and so the original information was reconstructed in a form that was consistent with one of the participant's schemata. The schema construct was developed during the period when psychology was strongly influenced by behaviorist and associationistic approaches; because the schema construct was not compatible with these worldviews, it eventually faded from view.

Unit – II Learning & Memory Definition and Nature - Learning theories, Pavlov, Thorndike, Skinner, Types of Learning – Conditioning, Habitual, Insight- Transfer of Training- Motor and verbal learning - Motor skills, distribution of practice, warm up, patterns of movement & knowledge of results. Verbal learning – process in verbal learning. Organization, practice, individual differences

Lesson 5 : Definition and Nature - Learning theories, - Pavlov, Thorndike and Skinner

LEARNING

- Learning is the key factor involved in behavioural change of an organism. Through learning, we make changes in our behaviour. There are many processes from which we get experience in life.
- The term learning has been defined by psychologists in many ways. According to the most acceptable definition, it is a “relatively permanent change in behaviour (or behaviour potential) resulting from experience” (Baron, 2001).
- According to R S Woodworth (1945), ‘Any activity can be called learning so far as it develops the individual (in any respect, good or bad) and makes his behaviour and experiences different from what that would otherwise have been.’
- 2. H L Kingsley and R Garry, (1946) said, ‘Learning is the process by which behaviour (in the broader sense) originates or changes through practice and training.’

- 3. Gates and Others (1946) observed, ‘Learning is the modification in behaviour to meet environmental requirements.’
- 4. F S Freeman (1958) defined, ‘Learning is the process of developing the ability to respond adequately to a situation which may or may not have been properly encountered.’
- 5. B L Hilgard (1958) was of the view, ‘Learning is the process by which an activity originates or is changed through reacting to an encountered situation, provided that the characteristics of the change in activity cannot be explained on the basis of native responses, tendencies, maturation or temporary states of the organism (e.g., fatigue or drugs, etc.)’

Nature & Characteristics

- Learning is the key factor involved in behavioural change of an organism. Through learning, we make changes in our behaviour. There are many processes from which we get experience in life. Psychologists have found out such processes and are constantly engaged in conducting research in such areas. The term learning has been defined by psychologists in many ways. According to the most acceptable definition, it is a “relatively permanent change in behaviour (or behaviour potential) resulting from experience” (Baron, 2001). Three apects in this definition that need attention are, first, ‘relatively permanent change’. It is important to mention here that any temporary change in behaviour can not be termed as learning. Such as, feeling sleepy after taking drugs or heavy meals or feeling tired due to illness. Second, permanent change due to ageing or maturation (as a person grows and develops), will not be considered as learning. For example, change in height takes place because of maturation. Third, here ‘experience’ does not mean our own experience only. Learning can also occur through direct experiences as well as vicarious, i.e., observing events and participating in them (Bandura, 1986). From the above explanation, it is obvious that all modifications of behaviours are not learned. Some modifications do take place due to physical maturity. In most of the cases the distinction between learning and maturation is very clear but in some places this distinction is less obvious. You take an example of infant’s walking. Normally, infant does not walk before the age about 12-15 months. They walk when they are physically fit and ready, perhaps, without learning. So, walking here does not have the role of learning. But in children, recognition of colour is the outcome of learning. This way, the impact of learning and maturation on modification of behaviour are

different. Now, the question is why we 'learn'? What is the importance of the process called 'learning'. Learning is an important process in human behavior. The reason being, it helps us in adaptation and in survival. Learning may be good as well as bad, and learning can be of many types. Some of the most important types of learning have been discussed in the following section.

- Learning is a process.
- It involves all the experiences from birth to death: produce change in behavior.
- Learning makes change in behavior either +ve or –ve.
- Learning prepares an individual for adjustment and adaptation.
- All learning is purposeful and goal oriented.
- Learning is universal and is continuous.

Trial and Error Method of Learning – Thorndike (1874-1949).

- In Thorndike's experiment, the cat was put in a box . This box was called puzzle box. A cat was kept hungry in the box. All round the box there were iron rods. There is only one door from which the cat can come out. The door can be opened by a lever. Outside the box a fish was kept in a plate .The sight of the fish and smell work as stimuli. The hungry cat was activated. It made a number of trials like biting the rods, trying to squeeze between the rods, pushing the door etc. None of the trials opened the door. By chance, the cat pressed the lever, the door was opened and cat came out and ate the fish. This experiment is repeated several times. Whenever the cat successfully pressed the lever, the stimulus (food) was reinforced . Gradually the number of unsuccessful trails (errors) were reduced and successful attempts increased. Thorndike says that, after several trials the cat learnt the connection or bond between pressing a lever and the food.
- Thorndike and his followers conducted extensive research on humans and animals. This research helped them to formulate the laws of learning.
- **Law of Readiness:** the first law of learning states that unless the learner is ready to learn, learning process does not take place. That is unless there is an inner drive that facilitates or serves as a moving force, the organism does not learn. You can take the horse to the pond you cannot make it drink water if it is not thirsty.

- **Law of Effect:** the second law states that a bond is fixed if it satisfies, extinguished if it is painful. Rewards lead to pleasant and satisfying experiences that are learnt and remembered. Punishment is unpleasent and develops a sense of annoyance and hence forgotten. All those experiences which are pleasant and satisfying are remembered and all those annoying are forgotten.
- **Law of Exercise:** the third law of learning highlights the importance of exercise or the use or disuse of a particular behavior or skill. The law of exercise stated in two parts. 1. the law of use 2. the law of disuse. The law of use states that more and more use makes the newly learnt S-R connection stronger and stronger. The law of disuse states that are the bond weaking and disappears if it is not used or practiced. Practice makes man perfect.

Pavlov's Classical Conditioning

- Ivan Pavlov was a famous Russian psychologist.
- **Ivan Pavlov**, in full **Ivan Petrovich Pavlov**, (born September 14, 1849, Ryazan, Russia.
- Died February 27, 1936, Leningrad [now St. Petersburg].
- He lived from 1849 to 1936.
- He made many discoveries in the psychology field.

He was awarded the Nobel Prize for Physiology or Medicine in 1904 for his work on digestive secretions.

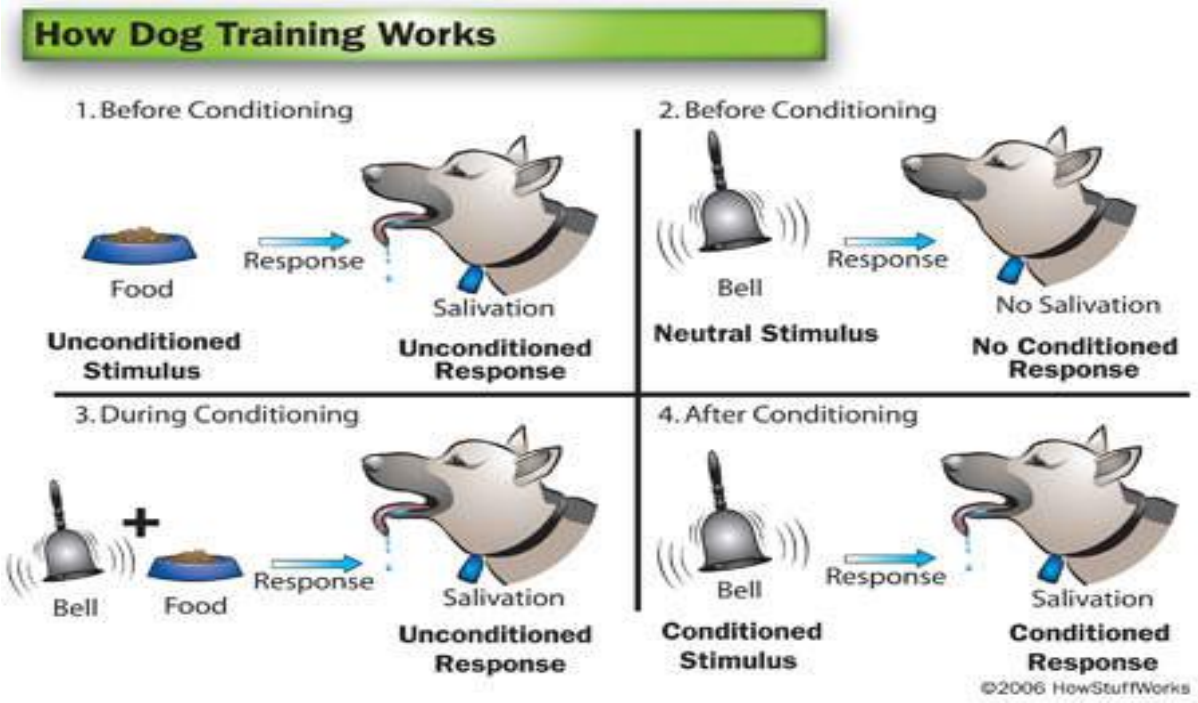
- Pavlov's Classical Conditioning Ivan Petrovich Pavlov (1849-1936), a famous Russian physiologist, was the pioneer in the study of learning. He was awarded Nobel Prize in 1904 in the field of medicine, for his work on physiology of digestion. He used classical experiments in learning and established many basic principles of learning. Classical conditioning is a kind of learning that is based on the classical experiments by Pavlov. This kind of conditioning is also known as respondent conditioning or Pavlovian conditioning.
- This kind of conditioning involves pairing of two stimuli. One is called as the conditioned stimulus (CS) and the other is unconditioned stimulus (UCS).
- The CS is also known as the neutral stimulus. This is so because when it is presented initially for a first few times, it merely serves to alert the organism and not evoke any response.

Pavlov's experiment:

- Pavlov made experiments with a hungry dog and evoked his salivary reflex by conditioning stimuli. He kept a dog hungry for a few days and before giving it food , he rang the bell . When the food was given and the bell was rung, there was automatic secretion of saliva from the mouth of the dog. This process of presenting the food accompanied with the bell was repeated several times. The amount of saliva secreted was measured.

- After several trial, the bell was rung but no food was given. The amount of saliva secreted was recorded. It was found that even in the absence of food [the unconditioned stimulus, UCS], the sound of the bell [conditioned stimulus CS] caused the dog to secrete the saliva[natural response, established by connecting UCS]. A conditioned response was the natural stimulus with an artificial stimulus.

- The UCS is a stimulus that produces a reflex response consistently known as unconditioned response (UCR). The CS and UCS are paired repeatedly a number of times so that conditioning can be done. The CS is presented a little before the UCS. The time gap between the CS and UCS is known as the inter-stimulus interval and it ranges from about half a second to a few seconds. When the neutral stimulus is repeatedly paired with the UCS, it acquires the properties of the UCS and hence produces the unconditioned response,UCR.



Some phenomena of classical conditioning:

- **Extinction:** If the conditioned stimulus CS[sound of the bell] is repeatedly applied without the natural stimulus UCS[food], it gradually fails to evoke the conditioned reflex[salivation]. This is called extinction. In the case of the dog , this would mean ringing the bell without giving food. After a while, the dog would no longer salivate at the sound of the bell. The response would have become extinct.

- The inhibition of behaviour due to lack of reinforcement is called extinction. if there is no reinforcement or there is no food after the bell rang, then the dog slowly does not salivate it, as there is no reward associated with it. thus there is extinction of behaviour.

- **spontaneous recovery :**In the context of Pavlov's dog, if the bell was not followed by any food for many presentations, then after some time, the dog will stop salivating in response to the bell. **But, when CS (bell) is again followed by UCS (food) after extinction has taken place, conditioned response (salivation) will return very quickly-a process known as reconditioning.** The reappearance of the conditioned response after a time interval due to UCS-CS pairing is known as **spontaneous recovery.**

- **Stimulus generalisation :** An animal conditioned to salivate to a bell, would also salivate to sound of a buzzer. Thus the animal tended to generalise the conditioned response to stimuli that were different from, but somewhat similar to the one to which it was

specifically conditioned. It was found that the greater the similarity between stimuli, the greater the generalisation between them.

Stimulus generalization :

- In this type of conditioning, pavlov shows that in a situation there are various ways to respond. **There is also a chance to respond in a similar manner to several stimuli. That means that we can be conditioned to respond to a number of stimuli and this is called stimulus generalization.**
- **Unconditioned Stimulus (UCS):** A stimulus that can produce response unconditionally and naturally, whenever it is presented. For example, cutting an onion can make you cry or pollen from flower can make you sneeze. Here, onion and pollen are two examples of UCS.
- **Conditioned Stimulus (CS):** A stimulus that was initially neutral in nature i.e., was not capable of producing any response. Later, with repeated paring with UCS, it is able to predict a UCS and thus can elicit the response originally meant for UCS only.
- **Unconditioned Response (UCR):** Response for which we do not need any previous learning i.e., which occurs automatically without any condition. As discussed already, 'crying' while cutting onion and 'sneezing' due to pollen, are two examples of UCR. Conditioned Response (CR): The response we give to a conditioned stimulus because of some experience is known as CR.
- **Generalization and Discrimination:** Responding to similarities and differences Pavlov also found that if the animal could be trained to respond to a bell, he could also be trained to respond to a buzzer with a similar response. This is known as generalization of the CS to other stimuli that were somewhere similar to the original CS where the learning took place.

Pavlov's contribution: The principles of classical conditioning has a great influence in the following areas:

Language development:

- Most of the child's language development is associated with the process of conditioning. A child learns to call his mother mummy or his father daddy through S-R association and substitution. When mother repeats the word ball while giving it to the child learns to say the word ball . Thus the principles of conditioning are used to teach alphabets.

Habit development:

- Children learn some good habits such as cleanliness , punctuality, early rising etc. through the principles of conditioning. They also learn some good manners, such as how to behave with elders and classmates through the process of conditioning.

Breaking bad habits:

- Bad habits, such as nail biting, carelessness, errors in writing etc. can be eliminated by using the principles of deconditioning .

Operant conditioning

- Burrhus Frederic Skinner was an American psychologist, behaviourist, author, inventor, and social philosopher. He was a professor of psychology at Harvard University from 1958 until his retirement in 1974.
- Born**:** 20 March 1904, Susquehanna, Pennsylvania, United States
- Died**:** 18 August 1990, Cambridge, Massachusetts, United States.
- Prof. B F Skinner started his research work on behaviour while he was a graduate in the Department of Psychology at Harvard University. In 1931, he wrote his thesis entitled, The Concept of the Reflex in the Description of the Behaviour. Thereafter, in the middle of forties, Skinner conducted a good deal of research at the Minnesota and Indiana Universities, on the theory of operant conditioning. Skinner was a practical psychologist who conducted several experiments on rats and pigeons. He popularized 'teaching machines' in learning in 1954. His important publications are: The Behaviour of Organism (1938); Science and Human Behaviour (1953); Verbal Behaviour (1957); Cumulative Record (1959); Beyond Freedom and Dignity (1971) and About Behaviourism (1974).
- Meaning of Operant Conditioning: Skinner called his theory as operant conditioning, as it is based on certain 'operations or actions' which an organism has to carry out. The term 'operant' stresses that behaviour is carried out in the environment to generate its own

consequences. An operant is a set of acts which conditions an organism in doing something. In the process of operant conditioning, operant responses are modified or changed by reinforcement. Reinforcement is a special kind or aspect of conditioning within which the tendency for a stimulus to evoke a response on subsequent occasions is increased by reduction of a need.

- Most SR theorists have assumed the existence of a stimulus as a prerequisite for evoking a response. In the absence of any external stimulus, they have assumed some internal stimuli for evoking the response. Skinner was against this 'No stimulus—no response' theory and believed that most of the responses could not be attributed to the known stimuli. He defined two kinds of responses—the one elicited by the known stimuli, which he called as respondent or reflexive behaviour, and the other emitted by the unknown stimuli, which he called as operant behaviour.

- Respondent behaviour is learnt according to Pavlovian model of conditioning. Since it is concerned with the stimuli, it is known as S-type conditioning. Skinner attached greater importance to operant behaviour which is primarily concerned with response rather than stimuli, it is known as R-type conditioning. Out of many responses which an organism is capable of giving, the problem with the experimenter is to evoke only the appropriate responses and fix them properly. Thus Skinner changed the usual SR formula into an RS formula.

- Instrumental conditioning is also known as operant conditioning. This kind of learning involves some kind of behavior or action that is instrumental in producing reinforcement (or reward) when it operates upon the environment. Operant conditioning is based on the research findings of Edward L. Thorndike and B.F.Skinner. In this learning, the most important thing to understand is reinforcement. Reinforcement is any stimulus or event which increases the likelihood that the response will occur again. It could be a reward or a punishment.

- For example, when the rabbit is given a carrot after running a distance, he will run faster to eat more carrots. Here, carrot is reinforcement in the form of reward. Thus, we can say that the nature or occurrence of a particular response depends on the extent to which the reinforcement is given. Hence, the term operant is used.That is the way the response operates on the environment, its desirability or non- desirability will strengthen or weaken the occurrence of the response.

- Skinner used a simple experiment to explain this concept. He placed a rat in a box. There was a lever which if pressed would activate the food or water delivery mechanism. This was called as the Skinner box. The rat is placed in a box and the experimenter delivers the food pellets by pressing the lever from outside. The rat eats the pellet one at a time. After some training, the rat is left alone in the box and the experimenter does not release the food pellets for him. After some time, when the rat is hungry again, he moves about exploring the box and by chance tends to press the lever that releases the food pellet. Thus, food pellet is the reinforcement that depends upon the pressing of the lever.
- Then the rat moves about in the box and again tends to press the lever by chance. This enables him to eat another food pellet. Thus, after that he does not explore much but tends to press the lever again and again quickly to eat the food pellets that are released.
- Hence, the operant behavior is very evident now. The number of times the rat presses the bar in a particular time are counted and recorded as rate of responding per unit. A cumulative recorder is used to record the rate of responding, that is the number of responses recorded per unit time.
- Operant conditioning helps in shaping a response. The reinforcement can be used to shape the behavior in a relatively shorter period of time. The learner learns the desired behavior when the chain of simple behaviors leading to the desired behavior is reinforced step by step. As each step is learned and the desired response is given, reinforcement is given. This makes the occurrence of the response more likely to occur again. This method of shaping is known as the method of successive approximations. This method of shaping is often used to shape the behavior of children, attitude, language, beliefs, etc.
- For example, when a child learns to walk, each step she/he takes, is reinforced to encourage her/him to walk

Shaping:

- Step by step attempts in a planned systematic manner to make the learner elicit appropriate responses and to move in the right direction is a process called shaping. Preparing and making the organism to make the first step in learning is crucial in this technique of learning.

- Operant conditioning can be described as a process that attempts to modify behaviour through the use of positive and negative reinforcement. Through Operant conditioning, an individual makes a association between a particular behaviour and a consequence.
- Example- parents rewarding a child's excellent grade with candy or some other prize.
- Reinforcement is any event that strengthens or increases the behaviour it follows.

There are two kinds of reinforcers:

- **Positive reinforcers** are favourable events or outcomes that are presented after the behaviour.(Rewarding learning). In situations that reflect positive reinforcement, a response or behaviour is strengthened by the addition of something, such as praise or a direct reward.
- **Negative reinforcers** involve the removal of an unfavourable events or outcomes after the display of a behaviour. In these situations, a response is strengthened by the removal of something considered unpleasant In both of these cases of reinforcement, the behaviour increases.
- Punishment, on the other hand, is the presentation of an adverse event or outcome that causes a decrease in the behaviour it follows.
- Any event that causes a decrease in the behaviour.

There are two kinds of punishment:

- **Positive punishment,** sometimes referred to as punishment by application, involves the presentation of an unfavourable event or outcome in order to weaken the response it follows.
- **Negative punishment,** also known as punishment by removal, occurs when a favourable event or outcome is removed after a behaviour occurs. In both of these cases of punishment, the behaviour decreases.

Lesson 6 : Types of Learning – Conditioning, Habitual, Insight- Transfer of Training- Motor and verbal learning - Motor skills, distribution of practice, warm up, patterns of movement & knowledge of results

Definitions

- Learning is the key factor involved in behavioural change of an organism. Through learning, we make changes in our behaviour. There are many processes from which we get experience in life.
- The term learning has been defined by psychologists in many ways. According to the most acceptable definition, it is a "relatively permanent change in behaviour (or behaviour potential) resulting from experience" **(Baron, 2001).**
- It is acquiring of knowledge.
- Learning is the acquisition of new behavior or the strengthening or weakening of old behavior as the result of experience. – **Henry P Smith**
- Learning is the acquisition of habits, knowledge and attitude. – **Crow and Crow**
- Learning is an important process in human behavior. The reason being, it helps us in adaptation and in survival. Learning may be good as well as bad, and learning can be of many types. Some of the most important types of learning have been discussed in the following section.

Insight Learning

- Insight Learning Given by one of the founders of Gestalt psychology, Wolfgang Kohler, insight learning refers to the sudden realization of a problem's solution. Kohler proposed that not all kind of learning depends on trial-error or conditioning. We also use our cognitive processes to learn. Using cognitive processes, we visualize the problem and solution for it internally. Even though this learning takes place **implicitly** but the change in the behaviour is long lasting.
- In the early 20th century, three German psychologists Max Wertheimer, and his associates Wolfgang Köhler and Kurt Koffka proposed new principles for explaining perception called as Gestalt principle. Gestalt psychology is form psychology.
- The literal meaning of the word Gestalt is form or configuration. The basic premise of Gestalt psychology is that 'whole is different from the sum of its part'. This implies that organization (or structured whole, known as Gestalten) gives a different meaning to the perception.

Kohler's experiment:

- Kohler conducted a number of experiments on chimpanzees during the period 1913-1917 and recorded his findings in his book, Mentality of Apes (1925). These Experiments Demonstrated that Chimpanzees do not learn by Trial and Error but by Insight, Provided the path is clearly visible to the animal. Some of Kohler's experiments are:

- **Experiment 1 :** A chimpanzee, called Sultan, was put in a cage. Outside the cage some bananas were placed. On seeing the bananas, Sultan showed restlessness and tried to reach the bananas outside but failed. All of a sudden he saw a stick inside the cage and pulled the bananas inside by means of the stick.

- **Experiment 2** : In the second experiment, the chimpanzee was given two sticks, one of which could be fitted into the other end and made a long stick. The chimpanzee first tried to get the bananas with the help of a single stick but he failed. He gave up the job and began playing with the two sticks. By chance, he succeeded in fitting both the sticks and pulled in the bananas. Next day, sultan made a few useless movements, but in a few seconds joined the sticks and got the bananas. This shows the animal learned the movement through insight.

- **Experiment 3:** A banana was hung from the ceiling of a cage. A box was put in the corner of the cage. The animal could not reach the fruit by stretching his hands. He suddenly dragged the box under the banana, climbed on it and got the banana.

- **Experiment 4:** In another experiment, kohler made the problem more difficult and a few boxes were put on the floor. The animal tried to reach the banana by jumping but failed. Suddenly he got an idea and put two or three boxes one on top of the other in order to reach it. He learned the trick by insight.

- The famous story behind Archimedes "Eureka" moment and sudden realization of gravitational force after seeing a falling apple by Isaac Newton, are some of the most famous examples of insight learning.

- **Motor learning:** Motor learning involves acquiring of a new motor skill or functions as a result of practice or experience. This learning helps us in executing motor functions, for example walking, running, skating, driving, climbing, etc.

- **Verbal learning:** It involves acquiring skills to communicate with others by using words, sounds, pictures, symbols, figures etc. The earlier studies were conducted with nonsense syllables which differed according to meaningfulness. A list of syllables of high

association value, such as LUV, LOS, RUF, were compared to syllables which are comparatively meaningless, like XUY, ZER, XUT.

- This type of learning involves the language we speak, the communication devices we use. Signs, pictures, symbols, words, figures, sounds, etc, are the tools used in such activities. We use words for communication.
- **Concept learning:** It is the form of learning which requires higher order mental processes like thinking, reasoning, intelligence, etc. we learn different concepts from childhood. For example, when we see a dog and attach the term 'dog', we learn that the word dog refers to a particular animal. Concept learning involves two processes, viz. abstraction and generalisation. This learning is very useful in recognising, identifying things.
- The type of learning in which we learn to classify stimuli based on its characteristics and features. For example, our ability to identify a barking, four legged animal with a tail as a 'dog', is the part of concept learning. We have learned that the word dog refers to this particular type of described animal.
- **Problem solving:** This is a higher order learning process. This learning requires the use of cognitive abilities-such as thinking, reasoning, observation, imagination, generalization, etc. This is very useful to overcome difficult problems encountered by the people.
- **Discrimination learning:** Learning to differentiate between stimuli and showing an appropriate response to these stimuli is called discrimination learning. Example, sound horns of different vehicles like bus, car, ambulance, etc
- **Discrimination learning** is defined in psychology as the ability to respond differently to different stimuli. This type of learning is used in studies regarding operant and classical conditioning. Operant conditioning involves the modification of a behavior by means of reinforcement or punishment. In this way, a discriminative stimulus will act as an indicator to when a behavior will persist and when it will not. Classical conditioning involves learning through association when two stimuli are paired together repeatedly.

Lesson 7 : Verbal learning – process in verbal learning. Organization, practice, individual

Differences.

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- This type of learning involves the language we speak, the communication devices we use. Signs, pictures, symbols, words, figures, sounds, etc, are the tools used in such activities. We use words for communication.
- There are three main processes that characterize how memory works. These processes are encoding, storage, and retrieval (or recall).
- Memory is conceptualised as a process consisting of three independent, though interrelated stages. These are encoding, storage, and retrieval. Any information received by us necessarily goes through these stages.
- (a) Encoding is the first stage which refers to a process by which information is recorded and registered for the first time so that it becomes usable by our memory system. Whenever an external stimulus impinges on our sensory organs, it generates neural impulses. These are received in different areas of our brain for further processing. In encoding, incoming information is received and some meaning is derived. It is then represented in a way so that it can be processed further.
- (b) Storage is the second stage of memory. Information which was encoded must also be stored so that it can be put to use later. Storage, therefore, refers to the process through which information is retained and held over a period of time.
- (c) Retrieval is the third stage of memory. Information can be used only when one is able to recover it from her/his memory. Retrieval refers to bringing the stored information to her/his awareness so that it can be used for performing various cognitive tasks such as problem solving or decision-making. It may be interesting to note that memory failure can occur at any of these stages. You may fail to recall an information because you did not encode it properly, or the storage was weak so you could not access or retrieve it when required.

Lesson 8 : Memory: Definition, Types of Memory- Working memory, Semantic memory, Picture memory and Verbal memory, Remote Memory, Procedural memory etc

INFORMATION PROCESSING APPROACH:

THE STAGE MODEL

- Initially, it was thought that memory is the capacity to store all information that we acquire through learning and experience. It was seen as a vast storehouse where all information that we knew was kept so that we could retrieve and use it as and when needed. But with the advent of the computer, human memory came to be seen as a system that processes information in the same way as a computer does. Both register, store, and manipulate large amount of information and act on the basis of the outcome of such manipulations. If you have worked on a computer then you would know that it has a temporary memory (random access memory or RAM) and a permanent memory (e.g., a hard disk). Based on the programme commands, the computer manipulates the contents of its memories and displays the output on the screen. In the same way human beings too register information, store and manipulate the stored information depending on the task that they need to perform. For example, when you are required to solve a mathematical problem, the memory relating to mathematical operations, such as division or subtraction are carried out, activated and put to use, and receive the output (the problem solution). This analogy led to the development of the first model of memory, which was proposed by Atkinson and Shiffrin in 1968. It is known as Stage Model.

NATURE OF MEMORY

Memory refers to retaining and recalling information over a period of time, depending upon the nature of cognitive task you are required to perform. It might be necessary to hold an information for a few seconds. For example, you use your memory to retain an unfamiliar telephone number till you have reached the telephone instrument to dial, or for many years you still remember the techniques of addition and subtraction which you perhaps learned during your early schooling. Memory is conceptualised as a process consisting of three independent, though interrelated

stages. These are encoding, storage, and retrieval. Any information received by us necessarily goes through these stages.

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subtraction are carried out, activated and put to use, and receive the output (the problem solution). This analogy led to the development of the first model of memory, which was proposed by Atkinson and Shiffrin in 1968. It is known as Stage Model.

MEMORY SYSTEMS: SENSORY, SHORT-TERM AND LONG-TERM MEMORIES

According to the Stage Model, there are three memory systems : the Sensory Memory, the Short-term Memory and the Long-term Memory. Each of these systems has different features and performs different functions with respect to the sensory inputs (see Fig.7.1). Let us examine what these systems are:

Sensory Memory

The incoming information first enters the sensory memory. Sensory memory has a large capacity. However, it is of very short duration, i.e. less than a second. It is a memory system that registers information from each of the senses with reasonable accuracy. Often this system is referred to as sensory memories or sensory registers because informations from all the senses are registered here as exact replica of the stimulus. If you have experienced visual after-images (the trail of light that stays after the bulb is switched off) or when you hear reverberations of a sound when the sound has ceased, then you are familiar with iconic (visual) or echoic (auditory) sensory registers

Short-term Memory

You will perhaps agree that we do not attend to all the informations that impinge on our senses. Information that is attended to enters the second memory store called the short-term memory (abbreviated as STM), which holds small amount of information for a brief period of time (usually for 30 seconds or less). Atkinson and Shiffrin propose that information in STM is primarily encoded acoustically, i.e. in terms of sound and unless rehearsed continuously, it may get lost from the STM in less than 30 seconds. Note that the STM is fragile but not as fragile as sensory registers where information decays automatically in less than a second.

Long-term Memory

Materials that survive the capacity and duration limitations of the STM finally enter the long-term memory (abbreviated as LTM) which has a vast capacity. It is a permanent storehouse of all informations that may be as recent as what you ate for breakfast yesterday to as distant as how you celebrated your sixth birthday. It has been shown that once any information enters the long-term memory store it is never forgotten because it gets encoded semantically, i.e. in terms of the meaning that any information carries. What you experience as forgetting is in fact retrieval

failure; for various reasons you cannot retrieve the stored information. You will read about retrieval related forgetting later in this chapter.

So far we have only discussed the structural features of the stage model. Questions which still remain to be addressed are how does information travel from one store to another and by what mechanisms it continues to stay in any particular memory store. Let us examine the answers to these questions.

How does information travel from one store to another? As an answer to this question, Atkinson and Shiffrin propose the notion of control processes which function to monitor the flow of information through various memory stores. As suggested earlier, all informations which our senses receive are not registered; if that be the case, imagine the kind of pressure that our memory system will have to cope with. Only that information which is attended to enters the STM from sensory registers and in that sense, selective attention, is the first control process that decides what will travel from sensory registers to STM. Sense impressions, which do not receive attention, fade away quickly. The STM then sets into motion another control process of maintenance rehearsal to retain the information for as much time as required. As the name suggests, these kinds of rehearsals simply maintain information through repetition and when such repetitions discontinue the information is lost. Another control process, which operates in STM to expand its capacity, is Chunking. Through chunking it is possible to expand the capacity of STM which is otherwise 7+2. For example, if you are told to remember a string of digits such as 194719492004 (note that the number exceeds the capacity of STM), you may create the chunks as 1947, 1949, and 2004 and remember them as the year when India became independent, the year when the Indian Constitution was adopted, and the year when the tsunami hit the coastal regions of India and South East Asian countries.

From the STM information enters the longterm memory through elaborative rehearsals. As against maintenance rehearsals, which are carried through silent or vocal repetition, this rehearsal attempts to connect the 'to be retained information' to the already existing information in long-term memory. For example, the task of remembering the meaning of the word 'humanity' will be easier if the meanings of concepts such as 'compassion', 'truth' and 'benevolence' are already in place. The number of associations you can create around the new information will determine its permanence. In elaborative rehearsals one attempts to analyse the information in terms of various associations it arouses. It involves organisation of the incoming information in as many ways as possible. You can expand the information in some kind of logical framework, link it to similar memories or else can create a mental image. Figure 7.1, that presents the stage model of memory, also depicts the arrows to show the manner in which information travels from one stage

to another. Experiments, which were carried out to test the stage model of memory, have produced mixed results. While some experiments unequivocally show that the STM and LTM are indeed two separate memory stores, other evidences have questioned their distinctiveness. For example, earlier it was shown that in the STM information is encoded acoustically, while in LTM it is encoded semantically, but later experimental evidences show that information can also be encoded semantically in STM and acoustically in LTM.

Shallice and Warrington in the year 1970 had cited the case of a man known as KF who met with an accident and damaged a portion of the left side of his cerebral hemisphere. Subsequently, it was found that his long-term memory was intact but the short-term memory was seriously affected. The stage model suggests that informations are committed to the long-term memory via STM and if KF's STM was affected, how can his long-term memory be normal? Several other studies have also shown that memory processes are similar irrespective of whether any information is retained for a few seconds or for many years and that memory can be adequately understood without positing separate memory stores. All these evidences led to the development of another conceptualisation about memory which is discussed below as the second model of memory.

TYPES OF LONG-TERM MEMORY

The short-term memory is now seen as consisting of more than one component (working memory). In the same way it is suggested that long-term memory too is not unitary because it contains a wide variety of information. In view of this, contemporary formulations envisage longterm memory as consisting of various types. For instance, one major classification within the LTM is that of Declarative and Procedural (sometimes called nondeclarative) memories. All information pertaining to facts, names, dates, such as a rickshaw has three wheels or that India became independent on August 15 1947 or a frog is an amphibian or you and your friend share the same name, are part of declarative memory. Procedural memory, on the other hand, refers to memories relating to procedures for accomplishing various tasks and skills such as how to ride a bicycle, how to make tea or play basketball. Facts retained in the declarative memory are amenable to verbal descriptions while contents of procedural memory cannot be described easily. For example, when asked you can describe how the game of cricket is played but if someone asks you how do you ride a bicycle, you may find it difficult to narrate.

Tulving has proposed yet another classification and has suggested that the declarative memory can either be Episodic or Semantic.

Episodic memory contains biographical details of our lives. Memories relating to our personal life experiences constitute the episodic memory and it is for this reason that its contents are generally emotional in nature. How did you feel when you stood first in your class? Or how angry was your friend and what did s/he say when you did not fulfil a promise? If such incidents did actually happen in your life, you perhaps will be able to answer these questions with reasonable accuracy. Although such experiences are hard to forget, yet it is equally true that many events take place continuously in our lives and that we do not remember all of them. Besides, there are painful and unpleasant experiences which are not remembered in as much detail as pleasant life experiences.

Semantic memory, on the other hand, is the memory of general awareness and knowledge. All concepts, ideas and rules of logic are stored in semantic memory. For instance, it is because of semantic memory that we remember the meaning of say 'nonviolence' or remember that 2+6=8 or the STD code of New Delhi is 011 or that the word 'elaphant' is misspelt. Unlike episodic memory this kind of memory is not dated; you perhaps will not be able to tell when you learnt the meaning of non-violence or on which date you came to know that Bangalore is the capital of Karnataka. Since the contents of semantic memory relate to facts and ideas of general awareness and knowledge, it is affect-neutral and not susceptible to forgetting for various other classifications of long-term memory.

Remote memory

Remote memory typically refers **to memory for the distant past, measured on the order of years or even decades**. It encompasses episodic (autobiographical), personal semantic, and general semantic memory involving historical people and events.

Remote memory is the ability to remember things and events from many years earlier. This is a function of long-term memory which the brain stores differently than recent or short-term memories. Short-term memories are stored in different areas of the brain than long-term memories.

Additionally, when a person is young there is a great deal of emotion attached to event memory. The intensity of the emotion involved causes these memory traces to be very intense. This is why, as a person grows older, that they can remember things from childhood and young adulthood yet have difficulty remembering things that have happened recently.

For instance, even when a person is in their 20's or 30's they can look back to their teens and wonder why they can remember every word of a favorite song but have difficulty remembering details about work or recreation.

Picture Memory:

Eidetic memory is the ability to recall an image from memory with high precision for a brief period after seeing it only once, and without using a mnemonic device. Eidetic memory. A person with hyperthymesia can remember nearly every event of their life in a lot of detail. ... People who claim to have a photographic memory state that they can remember a visualization for a long time in the same detail as when they first saw it.

Verbal memory is **the ability to remember what you read or hear**. It includes your ability to hear or read a word, sentence or paragraph and then recall it when needed. If you suspect your child is struggling with verbal memory you should discuss your concerns with a school counselor or your pediatrician. Since difficulties with verbal reasoning, visual memory, or working memory could look like verbal memory difficulties, make sure they use an objective test, not just observation. If you are uncomfortable talking to someone but have concerns, try a verbal memory test at home and then make a decision if you need to speak with a professional.

Why is Verbal Memory so important?

A lot of in-school learning involves reading, listening or studying and then recalling it when needed. Remembering vocabulary definitions, key details in a story, or the name of a species are all examples of verbal memory. While it's true that technology is a good enabler so that students don't need to memorize nearly as much as they did in the past, students still need to efficiently remember. Think about how much longer it would take to read if you had to look up more vocabulary words. Or how difficult it is to write an essay if you are struggling to remember examples. Just as with verbal reasoning, verbal memory can be just as important in math and science class as it is in English and history

Memory Improvement-Role of Memory interventions.

Memory

Human memory can most broadly be defined as a function of the brain that gives us our ability to store and retrieve information.(Wilson, 1991)

Processing Stages of Memory

Analogous to a computer system Input stage: Information must be fed into the system. (attention and encoding)

Storage stage: Maintaining information in memory over time.

Retrieval Stage: Recovering information from memory stores.(Pinel, 2003, Sohlberg & Mateer, 2001).

Memory Interventions

Any intervention strategy or technique which enables patients/clients and their families or caregivers to live with, manage, by-pass, reduce or come to terms with cognitive deficits precipitated by injury in the brain(Wilson, 1991).

Domains of Memory Intervention

Sohlberg and Mateer (2001) classify methods of memory intervention into two main categories

Restorative/Generalized Memory Intervention

Domain-Specific Memory Intervention.

Types of Memory Intervention

Restorative/Generalized

Memory Practice Drills

Mnemonic Strategy Training

Metamemory Training

Prospective Memory Training

Domain-Specific

Mnemonic Strategy Training (for specific information)

Preserved Priming (i.e., Vanishing Cues)

Creating a Personal History

Expanded Rehearsal Time/Spaced Retrieval

Strategies to Improve Memory

Memory consists of recalling information that you have learned or experienced. Many students struggle with remembering what they've read or recalling information for tests. In order to be successful in school, you need to remember the information you have learned. Although there isn't any "magic" memory pill, it *is* possible to improve your memory using the following

strategies.

Short-term and Long-term Memory

- Short-term memory: your mind stores information for a few seconds or a few minutes. This memory is fleeting—typically enough time to dial a phone number or write down an instructor'sthought. Your brain holds an average of seven items in short term memory.
- Long-term memory involves information you make an effort to retain because it is:
    - Personally meaningful to you—such as information about family andfriends.
    - You need it—job procedures or material you're studying for a test.
    - It made an emotional impression on you.
    - Episodic memories—personal memories about experiences.
    - Semantic memories—factual data like the color of your hair.
    - Procedural memory—skills and routines you perform often.

Ways to Improve Memory

The following information from http://www.memory-improvement-tips.com/memory-systems.html focuses on specific memory techniques as well as what you can do to maintain the health of yourbrain. The key to a healthy brain and improved memory is to do these consistently.

- Memorization Techniques: Mnemonics are techniques for remembering information that is otherwise quite difficult to recall: A simple example is the '30 days hath September' rhyme for remembering the number of days in each calendar month. The idea behind using mnemonics isto encode difficult-to-remember information in a way that is much easier to remember.
- Practice study skills for more efficient learning.

  See: http://www.memory-improvement-tips.com/best-study-skills.html
- Weave memory improvement exercises into your daily routine.
- Eat brain foods to improve brain health.
- Take vitamins and supplements to improve brain health: especially B vitamins, antioxidants, and Omega 3 fatty acids. http://www.memory-improvement-tips.com/vitamins-for-memory.html
- Play brain games to strengthen brain skills. Brainteasers and games can be found online as

wellas in print. http://www.memory-improvement-tips.com/brain-games.html

- Perform cardio exercise for healthy brain cells
- Get sufficient restful sleep to consolidate memories
- Engage in mindfulness meditation to improve concentration

Mnemonics ("New-monics")

Our brains are designed to encode and interpret complex stimuli. We use images, colors, structures, sounds, smells, tastes, touch, positions, emotions and language to make sophisticated models of theworld we live in. Our memories store all of these very effectively.

You can do the following things to make your mnemonics more memorable:

- Use positive, pleasant images. Your brain often blocks out unpleasant ones.
- Use vivid, colorful, sensory-laden images to code information– these are easier to remember thanboring ones. Use all your senses; your mnemonic can contain sounds, smells, tastes, touch, movements and feelings as well as pictures.
- Give your image three dimensions, movement and space to make it more vivid. You can use movement either to maintain the flow of association, or to help you to remember actions.
- Exaggerate the size of important parts of the image.
- Use humor! Funny or peculiar things are easier to remember than normal ones.
- Symbols (red traffic lights, pointing fingers, road signs, etc.) can quickly and effectively codecomplex messages.

Three Fundamental Principles Underlying

The Use Of Mnemonics are imagination, associationand location. You can use these principles to generate powerful mnemonic systems.

- Imagination: is what you use to create and strengthen the associations needed to create effective mnemonics. Your imagination is what you use to create mnemonics that are potent for you. The more strongly you imagine and visualize a situation, the more effectively it will stick in your mind for later recall. The imagery you use in your mnemonics can be as violent, vivid, or sensual as youlike, as long as it helps you to remember.
- Association: this is the method by which you link a thing to be remembered to a way of remembering it. You can create associations by:

- Placing things on top of each other.
- Merging images together.
- Wrapping/ Rotating them around each other or having them dancing together.
- Linking using the same color, smell, shape, or feeling.

- Location: this gives you a coherent context into which you can place information so that it hangstogether, and separates one mnemonic from another. You can build the atmosphere of these places into your mnemonics to strengthen the feeling of location.

Exercise and Memory

Regular aerobic exercise can improve your memory. Your brain needs oxygen and other nutrients in order to function well. When your oxygen is low, it affects your ability to concentrate, which in turn makes it more difficult to learn new information or recall information you studied in the past.

It can sometimes be difficult to get enough exercise when you have a lot of classes, or when you're working part-time/full-time jobs or caring for children in addition to school. Exercise helps supply oxygen to the brain, so even walking 15 minutes a day will help you focus better. You might also thinkabout taking a PE class during the day.

Sleep and Memory

Studies have shown that the brain requires 7-8 hours of sleep per night. Sleep strengthens relevant associations and weakens irrelevant associations, thus improving your access to memories. A lack of sleep can disrupt processing speed, long-term memory and emotional stability. Getting too few hourshurts memory almost as much as no sleep at all, because it impacts your concentration, which in turnimpacts your memory. Keep a log of the hours that you sleep, subtracting time that you get up duringthe night; add up the hours at the end of the week to average how much sleep you are getting.

Food and Memory

What you eat can have a huge impact on how your brain functions; certain foods help to improve brain function. Many of these brain foods protect your brain by releasing anti-oxidants, natural chemicals that break down harmful compounds called "oxidants" that your body produces naturally. Most of these foods also contain importantvitamins and nutrients essential for health. Here are the basics of ahealthy diet that will help your memory:

Healthy Fats - Build Your Brain: Your brain cell's structure is made upof "healthy fats". The most important of these are the Omega-3 fatty acids. As your brain repairs itself and grows new neurons, it needs an abundant supply of Omega-3's from your diet: cold-water fish, canola/flax seed oil, soybeans, walnuts, and eggs.

- **Antioxidants -** Protect Your Brain. See the list to the right.
- **High-Tyrosine Proteins -** Spark Your Brain: Neurotransmitters, chemical messengers in your brain, carry signals from one neuron to the next. The best neurotransmitter-building foods for boosting alertness, energy, and concentration include seafood, meat, eggs, soyand dairy.
- **Water -** Hydrate Your Brain: It's important to stay well hydrated. Being even slightly dehydrated decreases your mental energy and can impairyour memory. It can also increase anxiety. You can use the following calculator to help you determine how much water you need per day:
- **Fiber -** Regulate Your Fuel Supply: Fiber helps your brain function at itsbest. Sugar provides fuel for the brain, but it needs to be the right type of sugar—glucose, not refined sugars. Fiber slows the absorption of

  sugar so that it enters your bloodstream gradually, rather than quickly like refined sugars. Foodscontaining healthy amounts of fiber include *dried fruits*, *vegetables*, *legumes*, *nuts and seeds*, *whole fruit*, and *whole grains*.
- **Foods to Avoid:** Because your brain needs a steady stream of glucose to work properly, foods that spike your blood sugar will cause you to crash later and leave you feeling unfocused. Avoidanything made with *white flour*, (i.e. white bread, white pasta, donuts, etc.). Also avoid regular soda, which is a "sugar bomb," delivering too much refined sugar at once.
- **Eat Regularly:** To perform your best mentally, you must keep your blood sugar steady throughoutthe day. The best ways to do this is to eat breakfast, eat smaller meals more frequently throughout the day, and eat a late afternoon protein snack. Studies show that students who eat ahealthy breakfast are more successful than those who do not.

Unit – III: Concept Formation and Problem Solving

Concepts Formation

Concepts are the basic units of thought that underlie human intelligence and communication. From the perspective of cognitive informatics, a layered framework is suggested for concept formation and learning. It combines cognitive science and machine learning approaches. The philosophical issues and various views of concepts are reviewed. Concept learning methods are presented based on the classical view of concepts.

Definitions

- Concept is a process of representing a common property of objects or events – Morgan.
- Concept is a process with represents the similarities in otherwise diverse objects, situations or events – Munn.
- Concepts are those thoughts which mention things, incidents, qualities, etc. – Woodworth.
- Concept is the process of discrimination of the common features and relations in the world of events, things and persons – Hammerton.

Meaning of Concept Formation:

- A concept is the sum total of what we know about the object.
- It refers to a generalized idea about the objects/persons/ events.
- It stands for a general class and not for a particular object/person/event.
- It is a common name given on the basis of similarities or commonness found in different objects/persons/events.
- There are concepts of objects such as cat, tree, chair etc., concepts of persons such as mother, Indian, Negro etc., and concepts of qualities such as honesty, goodness, obedience.
- It is a mental disposition that helps in understanding the meaning of the objects or perceived earlier.
- In one sense, it is general mental image of the objects / persons /events experienced or perceived earlier.

Nature of Concept:

- A concept is not static, it is always changing. For example first a child considers even the walls and doors to be living things. Then it understands they are nonliving, considers cars, buses and running objects to be alive. Later, it learns that only animals and plants are living.

- The concept is not common for all, different persons may have different concept about the same object / events.
- Concept can be formed without the use of language. Concept is a part of thought process.
- Concept formation is the association of certain stimuli and responses.
- Concept is the process of discrimination of the common features.
- Concepts are very useful in recognizing, naming and identifying the objects / persons / events.

Types of Concept Formation:

- **Direct Experience:** It is the first type of concept formation, in which the learner develops concept through direct experience with the particular objects / persons / events. It is developed during from the early childhood onwards. For example, the concept about cow.
- **Indirect Experience:** Here the learner develops concept through pictures, photos and reading descriptions, hearing from other. For example, the concept about Kangaroo.
- Gardening and planting trees are examples of direct experiences with nature; **observing birds and watching nature programs on TV** are examples of indirect experiences.
- **Faulty Concepts:** The concepts or the general ideas we have about the objects, persons or events, are not always adequate and accurate. Small children have so many concepts that are quite erroneous and inadequate. For example, one's anxiety over the crossing of his way by a cat or one's feeling of hatred towards the person belonging to other caste or religion is the result of faulty concepts. Faulty concepts should not be allowed to develop in children.

Problem-solving

Definitions

- Problem-solving is **a mental process that involves discovering, analyzing, and solving problems**. The ultimate goal of problem-solving is to overcome obstacles and find a solution that best resolves the issue.

- The problem solving is a process of overcoming difficulties that appears to interfere with the attainment of goal. It is a procedure of making adjustment in spite of interferences. **SKINNER**
- "Problem solving is a cognitive processing directed at achieving a goal where no solution method is obvious to the problem solver".

Lesson 9 : Concept formation- Process in concept formation – Analytical and non analytical Strategies.

Process of Concept Formation:

The process of concept formation has three important phases.

- **Perception:** Experiences or learning in any form is the starting point of the process of concept formation. Our perceptions or imaginary experiences, formal or informal learning, provide opportunities for getting mental images of the objects, persons or events.
- Formal learning is **education normally delivered by trained teachers in** a systematic intentional way within a school, higher education or university.
- Informal learning refers **to learning that occurs away from a structured, formal classroom environment**. Informal learning comes in many forms, including viewing videos, self-study, reading articles, participating in forums and chat rooms, performance support, coaching sessions and games.
- There are three main types of learning: **classical conditioning, operant conditioning, and observational learning**. Both classical and operant conditioning are forms of associative learning, in which associations are made between events that occur together.
- The four core learning styles include **visual, auditory, reading and writing, and kinesthetic**. Here's an overview of all four leaning style types. Visual - Visual learners are better able to retain information when it's presented to them in a graphic depiction, such as arrows, charts, diagrams, symbols, and more.
- **Abstraction:** The mind analyses the perceived images and synthesizes what is common to all, neglecting what is particular. This process of observing similarities and commonness is named as abstraction.
- **Generalization:** After making such observation in the form of abstraction for a numbers of times the child is able to generalize or form a general idea about the common properties

of some objects or events. On account of this generalization, he will develop a concept about these things or events.

Concept Formation:

- For example, the child perceives a black cow at the first time and is told that it is a cow, he tries to form an idea about it. In the beginning the idea is very particular in nature.
- Later on, when he perceives a white or red cow he does not at once, call it a cow. He again makes an enquiry and comes to that these are cows. He tries to compare the particular mental image the idea of the previously perceived cow with the images he is having, at present, by perceiving white and red cows.
- In this way he compares and contrasts the similarities or dissimilarities of his mental images related to all perceived cows. In spite of the differences in colour, appearance etc., they are found to possess so many common properties or characteristics.

Formation of Concepts: 4 Main Steps

- The process by which we discover the feature or features which are 'common' to a large number of objects and associate these with a symbol which thereafter may be applied to other similar objects is called 'Concept formation'.

Step # 1. Observation:

- The first stage in the formation of concepts is the observation of an event, object or an experience. This can also be called the stage of becoming aware. This can be either direct or indirect. The child can directly see a dog and become aware of it.

- On the other hand, he also hears stories about devils and giants from his parents and grandparents; here the awareness is indirect. Thus, all of us have some knowledge or awareness of primitive people (or at least we believe we have) even though most of us have not seen them. Generally repeated experiences provide the basis for the development of concepts.

Step # 2. Generalisation:

- Repeated experiences or observations of different objects result in a tendency to form a general idea. Thus, a child first sees one dog, then another dog, then a third and so on and begins to form the general idea of a dog. This is called the process of generalisation. The process of generalisation explains how the child acquires many concepts like the concepts of gender, shape, number, etc.

Step # 3. Discrimination or Differentiation:

- Along with generalisation and the observation and organisation of similarities among things and objects, the child also becomes aware of the differences between them. Thus, all dogs are alike and all cows are alike. Dogs run on four legs and cows also do the same.
- At the same time dogs and cows are different from each other and big dogs are different from small dogs, and bulls are different from cows. It is this type of sequential operation of generalisation and differentiation in interaction that leads to the formation of concepts.

Step # 4. Abstraction:

- From the description of the above processes the operation of abstraction becomes evident. The child has seen dogs and he happens to see a cow on a different occasion. He does not observe them at the same time but inwardly he compares his experiences on the two occasions.
- The perceptions and the experiences are now inwardly analysed and re-experienced in the absence of the objects. This results in an appreciation of similarities and differences. This process by which the experience is analysed in the absence of actual situations is known as abstraction. It is abstraction which actually transforms comparable and contrasting experiences into concepts.
- This ability to respond to concrete situations in the absence of the actual situations is known as abstract thinking ability. It can be seen that as the child grows older, the process of abstraction plays an increasingly important role in the development of concepts. It is this process of abstraction which helps us to form ideas of the future and far off objects.

Lesson 10 : Types, functions and Determinants of concept formation.

Concept Formation:

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Lesson 11 : Problem Solving: Functional fixedness, Organization and insight Syllogistic Reasoning; Strategies for problem solving- Role of reasoning

Problem-solving

Definitions

Problem-solving is **a mental process that involves discovering, analyzing, and solving problems**. The ultimate goal of problem-solving is to overcome obstacles and find a solution that best resolves the issue.

The problem solving is a process of overcoming difficulties that appears to interfere with the attainment of goal. It is a procedure of making adjustment in spite of interferences. **SKINNER**

"Problem solving is a cognitive processing directed at achieving a goal where no solution method is obvious to the problem solver".

Functional Fixedness:

Definition of Functional Fixedness

According to Karl Duncker (1945) functional fixedness is the 'mental block against using an object in a new way that is required to solve a problem'.
Functional fixedness is a mental block against using an object in a new way that is required to solve a problem.
Functional fixedness is a physical block against using an object in an old way that is required to solve a problem.

Functional fixedness discovered by Duncker (1935) refers to the tendency to see objects as having only a single, typical use. A hammer is for pounding nails and other things for instance. We categorise objects based on their functional features as well as their features. Generally the prototypical function dominates the way we think.
Functional fixedness is a cognitive bias that limits a person to use an object only in the way it is traditionally used. The concept of functional fixedness originated in Gestalt psychology, a movement in psychology that emphasizes holistic processing.
Functional fixedness is just one type of mental obstacle that can make problem-solving more difficult. Functional fixedness isn't always a bad thing. In many cases, it can act as a mental shortcut allowing you to quickly and efficiently determine a practical use for an object.
For example, imagine that someone has asked you to open a toolbox and find a tool that can be used to loosen a screw. It would take a tremendous amount of time if you had to analyze every item in the box to determine how effective it might be at performing the task. Instead, you are able to quickly grab a screwdriver, the most obvious item for loosening a screw.
Functional fixedness is kind of a mental shortcut that helps you reduce how much you have to think in order to accomplish certain tasks. But functional fixedness can also make you less creative and more fixated on proven solutions rather than thinking about other, possibly more creative or useful solutions.

Reasoning

Definitions of Reasoning:

Reasoning is a stepwise thinking with a purpose or goal in mind" —**Garrett.**
"Reasoning is the term applied to highly purposeful, controlled and selective thinking"—**Gates.**
Sherman defined, **"reasoning is a process of thinking during which the individual is aware of a problem identifies, evaluates, and decides upon a solution"**

Types of Reasoning:

Reasoning may be classified into two types.

Inductive reasoning:

Inductive reasoning, or inductive logic, is a type of reasoning that **involves drawing a general conclusion from a set of specific observations**. Some people think of inductive reasoning as "bottom-up" logic, because it involves widening specific premises out into broader generalizations.

For example, Mohan is mortal, Radha is mortal, Karim is mortal; therefore, all human beings are mortal.

The process of reasoning from specifics to a general conclusion related to those specifics

• Inductive reasoning allows humans to create generalizations about people, events, and things in their environment.

Inductive reasoning works the other way, moving from specific observations to broader generalizations and theories.

For example, the left-handed people I know use left-handed scissors; therefore, all left-handed people use left-handed scissors.

John is an excellent swimmer. His family has a swimming pool. John's sister Mary must also be an excellent swimmer.

Deductive reasoning:

Deductive reasoning works from the more general to the more specific. Sometimes this is informally called a "top-down" approach. We might begin with thinking up a theory about our topic of interest.

Henry has categorized three types of deductive reasoning:

i. Conditioned reasoning:

It is the reasoning tied down by some specific condition such as the following.

For example, if there is a solar eclipse, the street will be dork. There is a solar eclipse

∴ The streets are dark.

ii. Categorical reasoning:

This type of reasoning is based on some categorical statements.

For example, all Robins are birds.

All birds lay eggs.

∴ All Robins lay eggs.

iii. Linear reasoning:

This type of reasoning involves straight forward relationships among elements.

For example, If Ram is taller than Mohan and Mohan is taller than Sohan, Ram is the tallest.

SYLLOGISM

A syllogism is a three-part logical argument, based on deductive reasoning, in which two premises are combined to arrive at a conclusion. So long as the premises of the syllogism are true and the syllogism is correctly structured, the conclusion will be true.

An example of a syllogism is "All mammals are animals. All elephants are mammals. Therefore, all elephants are animals."

In a syllogism, the more general premise is called the **major premise** ("All mammals are animals").

The more specific premise is called the **minor premise** ("All elephants are mammals").

The conclusion joins the logic of the two premises ("Therefore, all elephants are animals").

Some additional key details about syllogisms:

- First described by Aristotle in *Prior Analytics*, syllogisms have been studied throughout history and have become one of the most basic tools of logical reasoning and argumentation.
- Sometimes the word syllogism is used to refer generally to *any* argument that uses deductive reasoning.

- Although syllogisms *can* have more than three parts (and use more than two premises), it's much more common for them to have three parts (two premises and a conclusion). This entry only focuses on syllogisms with three parts.

Structure of Syllogisms

Syllogisms can be represented using the following three-line structure, in which A, B, and C stand for the different terms:

1. All A are B.
2. All C are A.
3. Therefore, all C are B.

Another way of saying the same thing is as follows:

1. If A = B
2. and C = A
3. then C = B

Notice how the "**A" functions as a kind of "middle" for the other term**s. You could, for instance, write the syllogism as: C = A = B, therefore C = B.

Types of Syllogism

Universal Syllogisms

Universal syllogisms are called "universal" because they use words that apply completely and totally, such as "no" and "none" or "all" and "only." The two most common forms of universal syllogisms are:

- "All A are B, and all C are A, so all C are B." (This is the most common type of syllogism.)
    1. All mammals are animals.
    2. All elephants are mammals.
    3. Therefore, all elephants are animals.
- "No A are B, and all C are A, so no C are B."
    1. No mammals are frogs.

2. All elephants are mammals.
3. Therefore, no elephants are frogs.

Particular Syllogisms

Particular syllogisms use words like "some" or "most" instead of "all" or "none." Within this category, there are two main types:

- "All A are B, and some C are A, therefore some C are B."
    1. All elephants have big ears.
    2. Some animals are elephants.
    3. Therefore, some animals have big ears.
- "No A are B, and some C are A, therefore some C are not B."
    1. No doctors are children.
    2. Some immature people are doctors.
    3. Therefore, some immature people are not children.

Enthymemes

Enthymemes are logical arguments in which one or more of the premises is not explicitly stated, but is instead implied. Put another way: an enthymeme is a kind of abbreviated syllogism in which the writer presumes that the audience will accept the implied and unstated premise. For instance, the following statement is an enthymeme:

- "Socrates is mortal because he's human."

This enthymeme is an abbreviation of a famous syllogism:

1. All humans are mortal.
2. Socrates is human.
3. Therefore Socrates is mortal.

The enthymeme leaves out the major premise. It instead assumes that all readers will understand and agree that "Socrates is mortal because he's human" without needing the explicit statement that "all humans are mortal."

Syllogistic Fallacies

A "fallacy" is the name for a mistake in logic. Syllogisms often seem like very simple statements, but you may be surprised how often people make logical mistakes when trying to put together simple syllogisms. For example, it may seem logical to make a statement like "Some A are B, and some C are A, therefore some C are B," such as:

1. Some nice people are teachers.
2. Some people with red hair are nice.
3. Therefore, some teachers have red hair.

Each of these categorical propositions is, after all, true—but in fact the final proposition, while true in itself, is *not* the logical conclusion of the two preceding premises. In other words, the first two propositions, when combined, don't actually *prove* that the conclusion is true. So even though each statement is *independently* true, the "syllogism" above is actually a logical fallacy. Here's an example of a false syllogism whose logical fallacy is a bit easier to see.

1. Some trees are tall things.
2. Some tall things are buildings.
3. Therefore, some trees are buildings.

The error in both of the above examples is called the "fallacy of the undistributed middle," since in each example the A is not "distributed" across the B and C in such a way that the B and C terms actually overlap. Other types of syllogistic fallacies exist, but this is by far the most common logical error people make with syllogisms.

Problem solving

Sometimes, it is not enough to just cope with the problems – they need to be solved.

Most people engage in problem solving every day. It occurs automatically for many of the small decisions that need to be made on a daily basis.

For example, when making a decision about whether to get up now or sleep in for an extra 10 minutes, the possible choices and the relative risks and benefits of obeying the alarm clock or sleeping later come automatically to mind.

Larger problems are addressed in a similar way. For example: “I have tasks that need to be done by the end of the week. How am I going to get them all done on time?”

After considering the possible strategies, 1 is chosen and implemented. If it proves to be ineffective, a different strategy is tried.

People who can define problems, consider options, make choices, and implement a plan have all the basic skills required for effective problem solving.

Sometimes following a step-by-step procedure for defining problems, generating solutions, and implementing solutions can make the process of problem solving seem less overwhelming.

Six step guide to help you solve problems

Step 1: Identify and define the problem

- State the problem as clearly as possible. For example: “I don’t have enough money to pay the bills.”
- Be specific about the behaviour, situation, timing, and circumstances that make it a problem. For example: “I need to pay the phone and gas bills, and I don’t have enough money to cover both this month.”

Step 2: Generate possible solutions

- List all the possible solutions; don’t worry about the quality of the solutions at this stage.
- Try to list at least 15 solutions, be creative and forget about the quality of the solution.
- If you allow yourself to be creative you may come up with some solutions that you would not otherwise have thought about.

Step 3: Evaluate alternatives

- The next step is to go through and eliminate less desirable or unreasonable solutions.
- Order the remaining solutions in order of preference.
- Evaluate the remaining solutions in terms of their advantages and disadvantages.

Step 4: Decide on a solution

- Specify who will take action.
- Specify how the solution will be implemented.
- Specify when the solution will be implemented. For example: tomorrow morning, phone the gas company and negotiate to pay the gas bill next month.

Step 5: Implement the solution

- Implement the solution as planned.

Step 6: Evaluate the outcome

- Evaluate how effective the solution was.
- Decide whether the existing plan needs to be revised, or whether a new plan is needed to better address the problem.
- If you are not pleased with the outcome, return to step 2 to select a new solution or revise the existing solution, and repeat the remaining steps.

Problem solving is something we do every day.

Some problems are small or easily solved - others are more complicated and can seem overwhelming.

One way of tackling problems is to use a specific and systematic problem solving procedure. If you've tried to solve certain problems without much success, try these steps out and see if they help.

Learning to solve problems effectively will help you to minimise the level of stress in your life and improve your overall sense of well-being.

Try it out and see.

Stages of Problem Solving

Gestalt psychologists suggest that problem solving behaviour also follows the stages that are followed in creative thinking: preparation, incubation, illumination or insight and verification. These stages have already been discussed in the earlier section.

According to Polya, there are four stages involved in problem solving and these are as given below.

Stage 1: Define, understand and think about the problem. In this stage, there is identification of the actual problem, attributes of the problem, area of knowledge involved in solving the problem and collecting relevant information.

Stage 2: Devise a plan for solution. This stage includes thinking of alternate ways to solve the problem and preparing a flowchart of solution.

Stage 3: Carry out the plan. This stage of problem solving is to execute the solution of the problem.

Stage 4: Looking back. This involves verifying that the problem solved was the one originally defined and also checking reasonableness, criteria and constraints as well as communicate results.

An information processing translation of the Polya's stages has also been described which is presented in the following model: (See figure below).

Encoding the problem in working memory

↓

Search long-term memory for plan or production system

↓

Execute the production system

↓

Evaluate the results

Strategies of Problem Solving

Even if the operator has all the basic knowledge and background skills, yet there is no guarantee that they will solve the problem successfully.

In order to be successful in solving a problem, an additional element needed is to have a general strategy that can be used for problem solving.

A strategy is a set of sequential steps (or procedure) used by a problem solver in arriving at a solution. The strategy should help the operator by guiding him/her to efficiently extract relevant data from the problem space and by giving a planned approach to solving the problem. Cognitive psychologists have described two major types of strategies generally used by an operator and these are termed as algorithms and heuristics. Let us see what these are

Algorithms

An algorithm is a specific procedure for solving a type of problem. In psychology, one of these problem-solving approaches is known as an algorithm. An algorithm is a defined set of step-by-

step procedures that provides the correct answer to a particular problem. By following the instructions correctly, you are guaranteed to arrive at the right answer.

An algorithm is a strategy that ensures the correct solution of the problem, if the well-defined rule of the solution is properly followed.

In an anagram problem, an algorithm would be attempting all the possible letter sequences until the correct and meaningful word is found.

There are four essential properties of an algorithm:

1) Each step of an algorithm must be exact. An algorithm must be precisely and unambiguously described, so that there remains no uncertainty.

2) An algorithm must terminate. The ultimate purpose of an algorithm is to solve a problem. If the process does not stop when executed, one will not be able to get any result from it. Therefore, an algorithm must contain a finite number of steps in its execution.

3) An algorithm must be effective. It must provide the correct answer to the problem.

4) An algorithm must be general. This means that it must solve every instance of the problem. For example, a program that computes the area of a rectangle should work on all possible dimensions of the rectangle, within the limits of the programming language and the machine.

Although algorithm is a guarantee to reach to the solution, the effort and time involved in using it is so great that a human operator rarely uses this strategy.

Let us take a stack of the cards with the numbers "5 1 4 2 8", and sort it from smallest number to biggest one using this algorithm. In each step, the algorithm compares the elements written in bold. The top of the stack of cards is on the left-hand side.

Firstpass:

(**5** **1** 428) (**1** **5** 428) Here, the algorithm compares the first two elements, and swaps them.

(1 **5** **4** 28) (1 **4** **5** 28)

(14 **5** **2** 8) (14 **2** **5** 8)

(14 2 **5** **8**) (1 4 2 **5** **8**) These elements are already in order, so the algorithm does not swap them.

Secondpass:

(**1** **4** 258) (**1** **4** 258)

(1 **4 2** 58) (1 **2 4** 58)

(12 **4 5** 8) (12 **4 5** 8)

(1 2 4 **5 8**) (1 2 4 **5 8**)

Now, the stack of cards is already sorted, but our algorithm does not know this. The algorithm needs one **whole** pass without **any** swap to know it is sorted.

Thirdpass:

(**1 2** 458) (**1 2** 458)

(1 **2 4** 58) (1 **2 4** 58)

(12 **4 5** 8) (12 **4 5** 8)

(124 **5 8**) (124 **5 8**)

Finally, the array is sorted, and the algorithm can stop.

Heuristics

A heuristic is a mental shortcut that allows people to solve problems and make judgments quickly and efficiently. These rule-of-thumb strategies shorten decision-making time and allow people to function without constantly stopping to think about their next course of action.

What is an example of a heuristic in psychology?

Heuristics can be mental shortcuts that ease the cognitive load of making a decision. Examples that employ heuristics include **using trial and error**, a rule of thumb or an educated guess.



Means-end analysis

Probably the most common and general heuristic is the ‘means-end analysis’. Simply stated, this heuristic says to do something to get a little closer to the goal. This heuristic helps break down a problem into pieces. For example, the ultimate goal is taken in short term goals and each of these short term goals will have to be achieved and doing so helps the person to get closer to the goal and ultimately reach it.

If a large problem is broken down into pieces, it is important to self monitor the sub goals, that is whether the person has achieved it or not. This self-monitoring is known as meta-cognition. Meta-cognition is essential for any extended activity, especially problem solving, because the problem solver needs to be aware of the current activity, of the overall goal, the strategies used to attain the goal and the effectiveness of those strategies.

- Means-Ends Analysis is problem-solving techniques used in Artificial intelligence for limiting search in AI programs.
- It is a mixture of Backward and forward search technique.
- The MEA technique was first introduced in 1961 by Allen Newell, and Herbert A. Simon in their problem-solving computer program, which was named as General Problem Solver (GPS).
- The MEA analysis process centered on the evaluation of the difference between the current state and goal state.

Let's take an example where we know the initial state and goal state as given below. In this problem, we need to get the goal state by finding differences between the initial state and goal state and applying operators.

Analogies

Heuristic of analogy uses experiences of strategies used to solve past problems in solving a current problem. This strategy relies on discovering common attributes among various problems solved at previous occasions and the problem being faced presently.

Analogy is a cognitive process of transferring information or meaning from a particular subject to another, or a linguistic expression corresponding to such a process.

- hammer : nail :: comb : hair.
- white : black :: up : down.
- mansion : shack :: yacht : dinghy.
- short : light :: long : heavy.
- bees : hive :: bears : den.
- speak : sing :: walk : dance.

Types of Analogies	
Part to whole	battery : flashlight :: hard drive : computer
Cause and effect	fatigue : yawning :: itching : scratching
Person to situation	mother : home :: teacher : school
Synonym	obese : fat :: slender : thin
Antonym	poverty : wealth :: sickness : health
Geography	Chicago : Illinois :: Denver : Colorado
Measurement	pound : kilogram :: quart : liter
Time	March : spring :: December : winter

Unit – IV: Creativity and Intelligence

Lesson 12 : Creativity; Creative process, creativity and functional fixity, Investment theory and creativity – Judging creativity'

CREATIVITY

What is Creativity?

Creativity is derived from the word 'creo' meaning – 'to create' or 'to make'.

Spearman (1931)

"Creativity is the power of human mind to create new contents by transforming relations and generating new correlates".

Drevdahl (1956):

"Creativity is the capacity of a person to produce compositions, products or ideas which are essentially new or novel and previously unknown to the producer".

Guilford (1959):

"Creativity is the capacity to produce ideas that are both new and useful through divergent thinking".

Divergent thinking is **a thought process or method used to generate creative ideas by exploring many possible solutions**.

Some other examples of divergent thinking include: **Wondering how many ways you can use a fork**. **Showing a person a photo and asking them to create a caption for the photo**. **Giving a child a stack of blocks** and asking them to see how many shapes they can create with those blocks.

Convergent thinking is the **type of thinking that focuses on coming up with the single, well-established answer to a problem**.

According to Wallach and kogan (1965) creativity lies in producing more associations and in producing more that are unique.

According to david ausubel 1963 creativity is a generalized constellation abilities, personality variables and problem solving traits.

The five-factor model of personality consists of **neuroticism, extraversion, openness to experience, agreeableness, and conscientiousness**, each in turn consisting of six facets. Neuroticism includes anxiety, depression, hostility, impulsiveness, self-consciousness, and vulnerability.

- They have an "attitude"! ...
- They re-define the problem. ...
- They have a system. ...
- They avoid the experience trap. ...
- They consider every position as though it were their own. ...
- They recognize conflict as often a prerequisite to solution. ...
- They listen to their intuition.

Nature of creativity

Creativity is not a single dimension of personality. It is made up of a number of traits which can be determined by the technique of factor analysis.

Creativity indicates the power of an individual to produce something new or unusual. But all individuals may not invent entirely a new idea or object. The ability to reshape or rearrange the already known facts may also be regarded as creativity of an individual. All kinds of planning and construction of new things involve creative thinking.

According to Mackinnon (1963), the most important characteristic of creative persons is that they are intelligent and original in thought and action. But in fact, there is hardly any positive correlation between intelligence and creativity. It has been found that persons having low I.Q. in intelligence test often produce something very original.

According to, Kitano and Kirby (1986), an individual can be extremely bright but uncreative or highly creative but not necessarily intellectually gifted. However a creative person must have a minimum level of intelligence above the average.

Divergent thinking is considered to be an important characteristic of creative persons. Taylor (1963) noted the importance of divergent thinking, especially in production of ideas, fluencies, flexibility and originality.

Creativity is considered as a God-given gift and an innate endowment. Only a very few can think creatively. But we cannot ignore the influence of training and education on nurturing creativity. The old myth that creativity is the function of superior genes has been exploded. Now it has been established that every individual is creative-in a greater or lesser degree. So, the development of creativity cannot be left to chance alone.

Creativity depends upon two factors: (a) Good inborn talent, as well as (b) Good cultural environment. Thus creativity may be considered as the product of both heredity and environment.

Creativity is inborn

According to this theory creativity is innate and hereditary. It cannot be produced by training or education. Great scientists like Einstein or Newton were born with extraordinary innate talent. No training can produce a second Einstein. Dr. Sandra Witelson, the Mc Master University neuro Scientist, studied brain structure of Einstein. He found that although Einstein's brain was no bigger than average men, there was no parietal operculum in it and this led to the inferior parietal lobe becoming 15% bulkier. The inferior pareital lobe is the seat of mathematical and visual reasoning. The genius, therefore, was destined to hit upon big ideas. Does not this recent finding imply that geniuses are only born, not made?

Creativity is acquired

The power of creativity is acquired and nurtured like other qualities of human being. Creativity is the product of culture and environment. Proper environment inspires the creative mind, while lack of opportunities cripples it.

Characteristics of creative person

Capability of divergent thinking.

Flexibility and originality of ideas.

Foresight.

Good memory and intelligence.

A creative imagination.

Ethical sensitiveness.

Enthusiasm for solution of problems.

Ability to work out details of a plan.

Stages in creative process

There are four stages of creative process: 1. Preparation. 2. Incubation.

3. Inspiration. 4. Verification.

Preparation: In this period facts are collected from past experiences.

Incubation: This is a period of no progress. But in the unconscious mind search goes on to find a solution of the problem.

Inspiration: In this stage new pattern towards the solution of the problem seems to emerge all on a sudden, when the individual is occupied with something else.

Verification: In this stage the inspiration is checked to find out whether the solution appeared is consistent with reality.

Creativity and Functional Fixedness

Earlier in this chapter we saw how functional fixedness could impede problem solving. Functional fixedness also may obstruct creativity (which points out the similarity between the concepts of problem solving and creativity). People who do the same old thing over and over again or who think the same thoughts are considered to be rather unimaginative, not to mention being socially boring. On the other hand, creative people see novel relationships or unusual connections among seemingly unrelated things, such as the person who slipped an oversized tire over a small tree so when the tree grew it would have a built-in ring seat.

Functional fixedness **can limit creative thinking and make it difficult to solve problems in a relationship**. It may even limit a person's ability to feel empathy for another person who does not behave in a way that is consistent with their preconceived ideas of them.

Investment Theory of Creativity

The investment theory of creativity, proposed in collaboration with Todd Lubart, holds that creativity is in large part a decision. In particular, it is a decision to buy low and sell high in the world of ideas. Creative people, like good investors, generate ideas that, at the time are viewed as novel and perhaps slightly ridiculous. The creative individuals are metaphorically "buying low." Then, once their ideas have gained some acceptance, the creative individuals "sell high," reaping the profits of their good idea and moving on to the next unpopular idea.

Creative individuals, by their nature, tend to defy the crowd. They resist merely thinking or doing what others are thinking or doing. Rather, they tend to go off in their own direction, seeking to propose ideas that are both novel and useful in some way. The greatest obstacle to creativity, therefore, often is not exactly strictures from others, but rather the limitations one places on one's own thinking. Such limitations, however, may derive from processes of enculturation and socialization, so that it often is not clear whether restrictions on creativity are internal or, down the line, externally imposed.

Creativity is a decision in the same way investing is. People are not born creative or uncreative. Rather, they develop a set of attitudes toward life that characterize those who are willing to go their own way. Examples of such attitudes toward life are willingness to (a) redefine problems in novel ways, (b) take sensible risks, (c) "sell" ideas that others might not initially accept, (d) persevere in the face of obstacles, and (e) examine whether their own preconceptions are interfering with their creative process. Such attitudes are teachable and can be ingrained in students through instruction that encourages students to think for themselves.

Creativity comprises several different aspects: (a) abilities, (b) knowledge, (c) styles of thinking, (d) personality attributes, (e) motivation, and especially intrinsic motivation, and (f) environment. A person can have the creative ability that would allow for creativity, for example, but without a willingness to take sensible risks or an environment that provides at least minimal support for creativity, that individual's potential creativity may be suppressed. It is thus crucially important, especially in schools, to provide an environment that allows creativity to flourish—not just in word, but also in deed. At the same time, an individual can have a creative attitude but without the skills of creativity—such as looking for reconciliation of opposing ideas and dialectical thinking—may not reach his or her full creative potential.

The Investment Theory of Creativity Creative people are willing to "buy low" and "sell high" in the realm of ideas. "Buying low" means pursuing ideas that are unknown or out of favor, but have growth potential. Creative people face resistance and "sell high" then move onto the next idea.

The Investment Theory of Creativity Requires a confluence of six distinct but interrelated resources:

- Intellectual skills
- Knowledge
- Thinking styles
- Personality
- Motivation
- Environment

Judging Creativity

Whether or not we label it, Americans are fond of judging creative acts and individuals. Everything from the latest Italian sports car to the recent Steven Spielberg film, to the performance of ice-skating champions is rated for its originality and creative merit. In most cases judging creative acts is a highly subjective affair. Sometimes the standards are set by an authority in the field, such as a noted professor of design, a film critic, a former Olympic skater, or a very discerning person.

This approach to psychology sounds more like art than science, and, understandably, many scientifically obsessed psychologists would rather put on a white coat, go to their laboratory, and measure blips on an oscilloscope made by a cat looking at a vertical line than try to evaluate a creative act or person. Yet some bold individuals have rushed in where their angelic colleagues fear to tread.

Psychologists thought it might be possible to assess creative ability by measuring how well people see novel connections between seemingly unrelated words. One of these tests, invented by Mednick (1967), was called the Remote Associations Test (RAT); it asked people to generate a single word that would be logically associated with three words. Consider the following two groups of three words: RED, BRIDGE, ANGRY, and HEAD, SICK, PORT. If you said "cross" for the first group, you would be right. What is the common word for the second group?

The RAT measures at least one component of creativity, but it probably measures other things too. Additionally, some very creative people might bomb on the test, which illustrates the slippery concept of creativity. Could it be that we are unconsciously creative, by which we mean

that we have many associates to stimuli, such as a word or a visual scene or a musical piece, but are not consciously aware of them? The remote associates idea was expanded by Bowers and his colleagues (1990) in a task called the dyads of triads. One part of the task is like the RAT in that the words are part of a coherent triad, as are those just presented or those in the threesome GOAT, PASS, GREEN, all of which constellate around the coherent word MOUNTAIN. However, the triad BIRD, PIPE, ROAD is considered incoherent in that no (likely) common element is apparent. In this study subjects were given sets of coherent and incoherent triads and asked to find the common elements, if they could. Also, they were asked to judge which of the triads were coherent. The results showed that they were able to identify the coherent triads even if they were unable to come up with the solution. It was as if the participants knew that there was a common element but could not quite name it. Possibly, people activate part of a solution to a remote associate task, and that may be one phase of a creative solution to a task. Such an idea might be related to the concept of intuition (which is defined by the Oxford English Dictionary as "the immediate apprehension of an object by the mind without the intervention of any reasoning process"), a frequently defamed term in scientific literature. Human intuition may indeed be an important part of the discovery phase of creative acts.

Intelligence: Definition and nature – Factors influencing intelligence: Spearman, Thurston and Guilford's Gardner's Theory of multiple intelligences – Stenberg's Triarchic theory – Measurement of Intelligence.

Lesson 13 : Intelligence: Definition and nature, Differences between Creativity and intelligence.

Intelligence: Definition and nature

WESCHLER: Intelligence is the aggregate or global capacity of an individual to act purposefully, to think rationally and to deal effectively with his environment.

Jean Piaget: intelligence is the ability to adapt to ones surroundings.

Stern's general definition of intelligence was "a general capacity of an individual consciously to adjust his thinking to new requirements

Buckingham: intelligence is the learning ability.

What is Intelligence ?

- This general ability consists of a number of specific abilities.
- Adaptability to a new environment or to change in the current environment.

- Capacity for knowledge and the ability to acquire it.
- Ability to comprehend relationships
- Capacity for reason& abstract thought.
- Ability to evaluate &judge.
- Capacity for original and productive thought.

Some established facts of intelligence

- The relation of intelligence with nature and nurture: Intelligence is the product of heredity and environment. Both are equally important.
- Distribution of intelligence: individual difference is present. A majority are average, few are bright and few are dull.
- Growth of Intelligence: as per the age. It reaches its maximum at 16 to 20 years. There after horizontal growth continues.
- Intelligence and sex differences: No relation.

Misconception of intelligence.

- Intelligence is not memory: Even without memory Intelligence is possible.
- Intelligence can be seen in both normal and abnormal behavior/ in delinquency etc.

Nature of Intelligence:

• It is an innate mental ability which grows and is influenced by the environment.

• It shows the capacity to adapt to new or changed situations quickly and correctly.

• It is the ability to carry on higher mental processes such as reasoning, criticism, rational and judgment.

• It shows the capacity to observe the relationship and absurdities.

• It implies the capacity to learn and solve difficult tasks and problems.

- **Intelligence means all-round mental efficiency:** All round mental efficiency means that it includes all the qualities in mental development of an individual. Through the process of intelligence the whole general and abstract thinking, reasoning powers are revealed

- Mental development includes such abilities as **attending, perceiving, observing, remembering, imagining, thinking, solving problems and growth of intelligence as well as of language**. These abilities change, grow and mature with age and decline in old age.

- **Intelligence is the power or capacity of human being:** Intelligence is the power or capacity of human being which helps to make reason, understanding and well-judgment. Intelligence is the capacity which is innate as well as acquired.

Characteristics of Intelligence

- **Intelligence is the ability of adaptation and adjusting to a new situation**: Intelligence is the property of recombining our behaviour pattern so as to act better in a novel situation. It means that intelligence is the potent factor through which the adjustment and re-adjustment to any situation is easily possible.
- **Intelligence is a power of good responses from the point of truth**: As a global or aggregate capacity of an individual, intelligence is a conglomeration of many powers as a whole in the structure of truth.
- Intelligence is not acquired after sustained labour. It is a gift from nature.
- **Intelligence is the ability for abstract thinking as well as the capacity to learn from the past experience:** Intelligence is the ability to think abstractly. It is also the capacity to learn from the past experience and hence, it makes the profitable use of the past. Intelligence is not knowledge though acquisition of knowledge depends to a great extent on intelligence and vice-versa.
- Intelligence is not a guarantee of a good behaviour of the individual.
- Intelligence is not a skill which a worker acquires after planned practice.
- Intelligence is not memory. An intelligent person may have poor memory.

Types of intelligence: Intelligence is divided into three kinds.

1. Concrete intelligence
2. Social or emotional intelligence
3. Abstract or general intelligence

1.Concrete intelligence: Concrete intelligence is also known as mechanical intelligence. It is the intelligence that has the relation with concrete materials.

It is the ability of an individual to comprehend actual situations and to react to them adequately. The concrete intelligence is evident from various activities in our daily life. This kind of intelligence is measured by performance tests and picture tests in which an individual is asked to manipulate concrete materials.

This type of intelligence is applicable for using machines, tools and instruments.

Ex. Mechanics, engineers are concrete intelligent.

2. Social intelligence:

• It is the ability to understand social situations and act wisely in human relationship.

• Socially intelligent people know the art of winning friends and influence them.

Ex. Leaders, ministers, salesmen, and diplomats are socially intelligent.

Social intelligence: Social intelligence means the ability of an individual to react to social situations of his or her daily life. Social intelligence is possessed by those people who are able to handle people well and also have the ability to make friends easily and understand human relations. Adequate adjustment in social situations is the index of social intelligence.

3. Abstract and general intelligence

• It is the ability to respond to words, numbers and letters etc.

• This type of intelligence is required from study of books literature.

Ex. Teachers, lawyers, doctors and philosophers.

Abstract intelligence: Abstract intelligence indicates the ability to respond to words, numbers, formulas, diagram and letters etc. This formula is conspicuously absent in animals. An abstractly intelligent person is able to discover relations among symbols and to solve the various problems with the help of such aids (symbols, formula, diagram, letter etc.).

Lesson 14 : Factors influencing intelligence: Spearman, Thurston and Guilford's Gardner's Theory of multiple intelligences.

Factors influencing intelligence:

Heredity and Environment:

Heredity provided the physical body to be developed with certain inherent capabilities while environment provides for the maturation and training of the organism. In 1940 Newman concluded that variations in I.Q. were determined about 68% by heredity and 32% by environment.

During an individual's life time variation in I.Q. is due to environment, since the heredity cannot change. Improved nutrition health, an stimulus situations would account for this change. Environment of children ages 2 to 4 years appears to be critical since the child normally learns

language at this time. Better homes, school, medical facilities, less economic discrimination. Provides a mean for achieving a mentally able population.

2) Age:

A person who bright or dull in childhood tends to remain bright or dull throughout his life. Growth I intelligence can continue through the early twenties, person achieves his maximum 10 at about 20 years are and remain stable till 10 years and decreases after that. Some abilities remain constant while others decline rapidly due to decline in physical efficiency. It is observed that the more intelligent person has more rapid of mental growth and continues to develop longer than does the mentally less able.

3) Race and Nationality:

There is no prominent evidence to show that race in a factor for determining intellectual level. Difference exist between families are due to environment. The differences in races are also due to opportunity for training in early years.

All evidence indicates that there is little or no difference in inherited intellectual capacity due to race. There are differences, mostly due to opportunities for training and learning. The idea of inferior races due to heredity is not true.

4) Culture:

Answer in intelligent test question are highly cultural. Cultural will determine the degree of a person's attitude and abilities.

5) Health and Physical Development:

Physical and mental health is related to one's ability to gain desired achievement. A mental activity with delicate health, one may not possess enough energy to engage in mental activity to the extent that he achieves success. Physical defects such as in complete maturation of brain cells sensory and physical handicaps many interfere with observable intelligent behavior. Diseases also affect intelligence. Emotional bodies may interferes one's ability. Unfavorable health affects mental status of the individual.

6) Sex:

It is a popular belief that boys are suppose to be more intelligent than girls. The research studies have shown that there is no significant differences between the sexes, on the average, girls seems to show slight superiority in language, memory and appreciation. Differences in intelligence are caused partly by environmental conditions

7) Social and Economic Conditions:

Home plays a significance role in the early developmental years. Home conditions influence on behaviours attitude. The financial status or the parents, neighborhood and environmental conditions also affect on the intelligence but due to their soci-economic status.

Mentally defective, People (Feeble – minded): Individuals whose I.Q.'s are below 70 they are called as mentally defective or feeble – minded people. They suffer from mamentia or luck of mind.

Thurstone's Group Factor Theory or Anarchic Theory

L L Thurstone, an American psychologist, propounded the group factor theory of intelligence. According to him, intellectual activity is neither an expression of numerous highly specific factors as claimed by Thorndike, nor the expression primarily of a general factor which prevails in all mental tasks as Spearman believed. Instead, as revealed by factor analysis, certain mental operations have in common, a primary factor which gives them psychological and functional unity, and which distinctly separates them from other mental operations. These mental operations are said to constitute a group 'A', similarly, another group of mental operations have their own unifying primary factor and may be said to constitute a group 'B' and so on. Thus, there are a number of groups of mental abilities, each of which has its own primary factor. Thurstone proposed seven factors and called them primary mental abilities. These are:

1. M—Memory: To be able to learn and retain information. Also, to be able to recall the learned material.
2. N—Number: To be able to understand quickly and with accuracy simple arithmetic computations.
3. P—Perceptual: To be able to identify objects quickly and accurately.
4. R—Reasoning: To be able to perceive and utilize abstract relationships. To be able to put together past experiences in the solution of new problems.
5. S—Spatial: To be able to deal with objects in space.
6. V—Verbal: To be able to understand and utilize verbal ideas.

7. W—Word fluency: To be able to think of words rapidly. Spearman's theory is also known as 'electic theory' because it harmonizes elements from all the main types of abilities. Thurstone's theory is also known as the 'anarchic theory' because he conceived that the mind consists of a number of independent facilities.

Spearman two factor theory:

The English psychologist, Charles Spearman (1863-1945), in 1904 proposed his theory of intelligence called two-factor theory. According to him intellectual abilities are comprised of two factors, namely; **the general ability known as G-factor and specific Abilities known as S-factors.**

This theory has greatly influenced the measurement of intelligence. According to Spearman, the frank adoption of two-factor theory would seem to supply an adequate basis for a unified science of all human ability (The Ability of Man).

According to two-factor theory, intelligence consists of two factors- a general ability (g) and a special ability (s).

The general ability or g comes into play in all kinds of activities. A special ability or s comes into play in any particular activity.

There are many special abilities such as verbal ability, ability to deal with numbers, ability to deal with spatial relations, ability in music, ability to deal with mechanical relations and so on. All these specific abilities presuppose a general ability (g) without which they cannot function.

The general factor, g, varies from one individual to another, but is constant for any one individual in all his scores. The specific factor s varies from one individual to another. Any mental test, then, measures g, and s, some measuring 'g' prominently, others **s** predominantly.

Guilford's Model of Structure of Intellect

Guilford (1967) proposed a three dimensional structure of intellect model. According to Guilford every intellectual task can be classified according to its (1) content, (2) the mental operation involved and (3) the product resulting from the operation.

He further classified content into five categories, namely, visual, auditory, symbolic, semantic and behavioural.

He classified operations into five categories, namely, cognition, memory retention, memory recording, Divergent production, Convergent production and evaluation.

He classified products into six categories, namely, units, classes, relations, systems, transformations and implications.

Subsequently, Guilford (1977) modified his model and divided figural factor of the content dimension into visual and auditory factors, thus making five types of the contents dimension, instead of the original four types.

In the modified version the number of cells became 150 (5 contents × 5 Operations × 6 Products= 150 cells). The definition given by Guilford for each of these is too lengthy to present here; they will be described very briefly

The Contents Dimension

This dimension includes the broad areas of information in which operations are applied. It has been divided into four categories.

• Visual - Information arising from stimulation on the retina in the form of an image.

• Auditory – Information arising from stimulation of the cochlea of the ear as image.

• Symbolic - Information perceived as symbols or signs that have no meaning by themselves; for example, Arabic numerals or the letters of an alphabet.

• Semantic - Information perceived in words or sentences, whether oral, written, or silently in one's mind.

• Behavioural – Information perceived as acts of an individual/ individuals.

The Operations Dimension

This consists of five kinds of operation or general intellectual processes:

• Cognition - The ability to understand, comprehend, discover, and become aware.

• Memory - The ability to memorise information.

• Divergent Production - The process of generating multiple solutions to a problem

• Convergent Production - The process of deducing a single solution to a problem.

- Evaluation - The process of judging whether an answer is accurate, consistent, or valid.

The Products Dimension

As the name suggests, this dimension contains results of applying particular operations to specific contents. There are six kinds of products, they are:

• Unit - Represents a single item of information.
• Class - A set of items that share some attributes.
• Relation - Represents a connection between items or variables; may be linked as opposites or in associations, sequences, or analogies.
• System - An organisation of items or networks with interacting parts.
• Transformation - Changes perspectives, conversions, or mutations to knowledge; such as reversing the order of letters in a word.
• Implication - Predictions, inferences, consequences, or anticipations of knowledge.
Some examples may provide a feel for the kinds of distinctions made in this model. Suppose a subject is given a long list of unrelated words to study and is asked to recall them later. The content of this scale is "semantic," since it involves words; the operation is "memory"; and the product is the recall of words as "units."

Gardner's Multiple Intelligences

Following the work of Thurstone, American psychologist Howard Gardner built off the idea that there are multiple forms of intelligence.

He proposed that there is no single intelligence, but rather distinct, independent multiple intelligences exist, each representing unique skills and talents relevant to a certain category.

Gardner (1983, 1987) initially proposed seven multiple intelligences: linguistic, logical-mathematical, spatial, musical, bodily-kinesthetic, interpersonal, and intrapersonal, and he has since added naturalist intelligence.

One of the more recent ideas to emerge is Howard Gardner's theory of multiple intelligences.

Gardner proposed that the traditional idea of intelligence, based on IQ testing, did not fully and accurately depict a person's abilities. His theory proposed eight different intelligences based on skills and abilities that are valued in different cultures:

- **Bodily-kinesthetic intelligence**: The ability to control your body movements and to handle objects skilfully.
- **Interpersonal intelligence**: The capacity to detect and respond appropriately to the moods, motivations, and desires of others.

- **Intrapersonal intelligence**: The capacity to be self-aware and in tune with inner feelings, values, beliefs, and thinking processes.
- **Logical-mathematical intelligence**: The ability to think conceptually and abstractly, and the capacity to discern logically or numerical patterns.
- The ability to use logic and mathematical skills to solve problems.
- **Musical intelligence**: The ability to produce and appreciate rhythm, pitch, and timbre.
- Musical: The ability to perform and enjoy music
- **Naturalistic intelligence**: The ability to recognize and categorize animals, plants, and other objects in nature.
- **Verbal-linguistic intelligence**: Well-developed verbal skills and sensitivity to the sounds, meanings, and rhythms of words.
- **Linguistic: The ability to speak and write well.**
- **Visual-spatial intelligence**: The capacity to think in images and pictures, to visualize accurately and abstractly.
- The ability to think and reason about objects in three dimensions.

Lesson 15 : Stenberg's Triarchic theory –Measurement of Intelligence.

Triarchic Theory of Intelligence

Psychologist Robert Sternberg defined intelligence as "mental activity directed toward purposive adaptation to, selection, and shaping of real-world environments relevant to one's life."

While he agreed with Gardner that intelligence is much broader than a single, general ability, he suggested that some of Gardner's types of intelligence are better viewed as individual talents. Sternberg proposed what he referred to as "successful intelligence," which involves three different factors:

- **Analytical intelligence**: Your ability to evaluate information and solve problems
- **Creative intelligence** is the ability to go beyond what is given to create novel and interesting ideas. This type of intelligence involves imagination, innovation and problem-solving.

- **Practical intelligence** is the ability that individuals use to solve problems faced in daily life, when a person finds the best fit between themselves and the demands of the environment.

Measurement of Intelligence Intelligence tests are used to measure intelligence of an individual. It is important to note that intelligence is inferred from a variety of elements, i.e., behaviour and speed of doing things correctly, etc. An intelligence test is an objective and a standardized measure.

Intelligence is measured through a complicated process. It involves a comparison and establishment of a relationship between CA (Chronological Age) and MA (Mental Age). This relationship is expressed by the term IQ (Intelligence Quotient). When the mental age is divided by the chronological age and the quotient is multiplied by 100, the result is IQ.

$IQ = MA / CA \times 100$

When we want to calculate the mental age of a student, all questions assigned to that age are put to him. (In the individual scale of Binet a certain number of questions are assigned to that age). If he answers all the questions assigned to that age correctly, his mental age is equal to his chronological age and that child is considered to be an average one. Suppose you have to test a child of CA 8 on Binet scale. You will start with questions assigned to the sixth year and then go up. The child may be successful in answering correctly all the questions assigned to years 6, 7, 8 and may stop at 9. His mental age will be 8.

Intelligence is measured on the following factors:

1. Vocabulary—Choosing a synonym or antonym or near-synonym or near antonym.
2. Verbal analogies, e.g., Branch is to a tree as brook is to river.
3. Sentence completion, e.g., India has ……. states.
4. Arithmetic reasoning. Simple arithmetic sums.
5. Number series, e.g., What next? 11, 13, 15 (17, 19, 21).
6. Picture arrangement. Arranging pictures of a story in proper sequence.
7. Comprehension. For testing common sense, certain cards or paragraphs are given in which some absurdity is shown.

8. Similarities, e.g., In what way are cotton and silk alike?

9. General information. From everyday life.

10. Digit span. For testing memory, digits are orally mentioned and the subject is asked to repeat them in the same order.

11. Digit-Symbol substitution. A code is given and substitution is to be carried out.

12. Figure analogies

13. Classification, e.g., Which word on the right belongs to the group on the left? Pen, table, book, stone, pencil, radio

CLASS ROOM ACITIVITY/DEMONSTRATION:

PSY25142

APPLIED SOCIAL PSYCHOLOGY

Objectives:

The concept of social influence and social exchange, the nature of conformity, compliance, obedience and modeling, origin of prejudice and discrimination, meaning of aggression, understating of environmental behaviour.

Unit-1Social Influence and Social Exchange

Lesson -1 Conformity, Asch's conformity studies; explaining conformity
Lesson -2 Compliance - meaning, examples and techniques
Lesson -3 Obedience; Milgram's Studies, obedience and authority
Lesson -4 Modeling; cooperation, competition, bargaining, conflict resolution

Unit-2Prejudices and Discrimination

Lesson -5 Definition and Nature of Prejudice and Discrimination
Lesson -6 Origin of Prejudice (Cognitive and Social) – Challenging Prejudice
Lesson -7 Prejudice based on Gender, prejudice and discrimination
Lesson -8 Forms of Discriminatory behavior

Unit-3Aggression

Lesson -9 Definition and Nature-Theoretical perspectives on Aggression
Lesson -10 The role of biological factors Drive theories and modern theories of aggression
Lesson -11 Determinants of human aggression; social, personal and situational determinants
Lesson -12 Prevention, Control and management of aggression

Unit-4Environment and Behavior

Lesson -13 Urban environment and social behavior
Lesson -14 Effect of noise, Temperature, Weather, Light and Air pollution
Lesson -15 Crowding-Long term -Effects of crowding
Lesson -16 Theories of crowding-Crowd and Behaviour

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Outcomes:

- The nature and origins of prejudice and discrimination
- The nature of Social Identity and Interpersonal Attraction
- The basic aspects of Social Influence and Pro-social behavior
- The meaning and nature of environmental behavour

UNIT. I: SOCIAL INFLUENCE AND SOCIAL EXCHANGE

Lesson -1 Conformity, Asch's conformity studies; explaining conformity

Conformity, Asch's conformity studies; Explaining conformity; Compliance; why people comply; Obedience; Milgram's Studies. modeling; cooperation, competition, bargaining, conflict resolution.

Conformity:

Conformity is a type of social influence involving a change in belief or behavior in order to fit in with a group.

This change is in response to real (involving the physical presence of others) or imagined (involving the pressure of social norms / expectations) group pressure.
Conformity is also known as majority influence (or group pressure).

The term conformity is often used to indicate an agreement to the majority position, brought about either by a desire to '*fit in*' or be liked (normative) or because of a desire to be correct (informational), or simply to conform to a social role (identification).

Conformity can also be simply defined as "*yielding to group pressures*" (Crutchfield, 1955). Group pressure may take different forms, for example bullying, persuasion, teasing, criticism, etc. Conformity is also known as majority influence (or group pressure).

It means changing your behaviour or opinion in order to match with the majority of the group you belong to. People get along with the group behavioursand opinions to avoid being different and abnormal.

Types of Conformity:

1. **Normative** - succumb to group pressure to fit in; afraid of being rejected; involves Compliance.
2. **Informationa**l - when a person is lacking knowledge, and looks for group guidance; socially compares their behaviour with the group; involves Internalization.
3. **Compliance(or group acceptance)** – publicly changes behaviour to fit in while privately disagreeing; conforming to majority.This occurs 'when an individual accepts influence because he hopes to achieve a favourable reaction from another person or group. He adopts the induced behavior because....he expects to gain specific rewards or approval and avoid specific punishment or disapproval by conformity' (Kelman, 1958, p. 53).In other words, conforming to the majority (publicly), in spite of not really agreeing with them (privately). This is seen in Asch's line experiment.Compliance stops when there are no group pressures to conform, and is therefore a temporary behavior change.

4. **Internalization(genuine acceptance of group norms)** – publicly changing behaviour to fit in with the group and agreeing with them privately.This occurs 'when an individual accepts influence because the content of the induced behavior - the ideas and actions of which it is composed - is intrinsically rewarding. He adopts the induced behavior because it is congruent [consistent] with his value system' (Kelman, 1958, p. 53).Internalisation always involves public and private conformity. A person publicly changes their behavior to fit in with the group, while also agreeing with them privately.
   This is the deepest level of conformity were the beliefs of the group become part of the individual's own belief system. This means the change in behavior is permanent. This is seen in Sherif's autokinetic experiment.This is most likely to occur when the majority have greater knowledge, and members of the minority have little knowledge to challenge the majority position.

5. **Ingratiation** – where a person conforms to impress or gain favour/acceptance from the group; similar to Normative influence, but is motivated by the need for social rewards than the threat of rejection; group pressure does not enter decision to conform.This is when a person conforms to impress or gain favor/acceptance from other people.It is similar to normative influence, but is motivated by the need for social rewards rather than the threat of rejection, i.e., group pressure does not enter the decision to conform.

6. **Identification** – conform to the expectations of a social role; similar to compliance, but the does not have to be a change in private opinion.
   This occurs 'when an individual accepts influence because he wants to establish or maintain a satisfying self-defining relationship to another person or group' (Kelman, 1958, p. 53).Individuals conform to the expectations of a social role, e.g. nurses, police officers. It is similar to compliance as there does not have to be a change in private opinion. A good example is Zimbardo's Prison Study.Man (1969) identified an additional type of conformity:

Explanations of Conformity

Deutsch and Gerrard (1955) identified two reasons why people conform.

- Yielding to group pressure because a person wants to fit in with the group. E.g. Asch Line Study.
- Conforming because the person is scared of being rejected by the group.
- This type of conformity usually involves compliance – where a person publicly accepts the views of a group but privately rejects them.
- This usually occurs when a person lacks knowledge and looks to the group for guidance or when a person is in an ambiguous (i.e. unclear) situation and socially compares their behavior with the group. E.g. Sherif's Study.
- This type of conformity usually involves internalization – where a person accepts the views of the groups and adopts them as an individual.

Solomon Asch - Conformity Experiment:

Solomon Asch conducted an experiment to investigate the extent to which social pressure from a majority group could affect a person to conform.
Asch (1951) devised what is now regarded as a classic experiment in social psychology, whereby there was an obvious answer to a line judgment task. If the participant gave an incorrect answer it would be clear that this was due to group pressure.

Aim: Solomon Asch (1951) conducted an experiment to investigate the extent to which social pressure from a majority group could affect a person to conform.

Procedure: Asch used a lab experiment to study conformity, whereby 50 male students from Swarthmore College in the USA participated in a 'vision test'. Using a line judgment task, Asch put a naive participant in a room with seven confederates. The confederates had agreed in advance what their responses would be when presented with the line task. The real participant did not know this and was led to believe that the other seven participants were also real participants like themselves. Each person in the room had to state aloud which comparison line (A, B or C) was most like the target line. The answer was always obvious. The real participant sat at the end of the row and gave his or her answer last. There were 18 trials in total and the confederates gave the wrong answer on 12 trails (called the critical trials). Asch was interested to see if the real participant would conform to the majority view. Asch's experiment also had a control condition where there were no confederates, only a "real participant".

Results: Asch measured the number of times each participant conformed to the majority view. On average, about one third (32%) of the participants who were placed in this situation went along and conformed with the clearly incorrect majority on the critical trials. Over the 12 critical trials about 75% of participants conformed at least once, and 25% of participant never conformed. In the control group, with no pressure to conform to confederates, less than 1% of participants gave the wrong answer.

Conclusion: Why did the participants conform so readily? When they were interviewed after the experiment, most of them said that they did not really believe their conforming answers, but had gone along with the group for fear of being ridiculed or thought "peculiar". A few of them said that they really did believe the group's answers were correct.

Apparently, people conform for two main reasons: because they want to fit in with the group (normative influence) and because they believe the group is better informed than they are (informational influence).

Factors Affecting Conformity:

1. **Group Size:** Asch (1956) found that group size influenced whether subjects conformed. The bigger the majority group (no of confederates) the more people conformed, but only up to a certain point. With one other person (i.e. confederate) in the group conformity was

3%, with two others it increased to 13% and with three or more it was 32% (or 1/3). Optimum conformity effects (32%) were found with a majority of 3. Increasing the size of the majority beyond 3 did not increase the levels of conformity found.

2. **Lack of Group Unanimity / Presence of an Ally**: As conformity drops off with 5 members or more, it may be that it's the unanimity of the group (the confederates all agree with each other) which is more important than the size of the group. In another variation of the original experiment Asch broke up the unanimity (total agreement) of the group by introduced a dissenting confederate. Asch (1956) found that even the presence of just one confederate that goes against the majority choice can reduce conformity as much as 80%.

3. **Difficulty of Task:** When the (comparison) lines (e.g. A, B, C) were made more similar in length it was harder to judge the correct answer and conformity increased. When we are uncertain, it seems we look to others for confirmation. The more difficult the task the greater the conformity.

4. **Answer in Private**: When participants were allowed to answer in private (so the rest of the group does not know their response) conformity decreases. This is because there is less group pressure and normative influence is not as powerful, as there is no fear of rejection from the group.

Suggestions after this experience:

1. Always listen to yourself before listening to others, even if you're wrong, so you can learn from your mistakes.
2. Be confident of your opinion, and stick to your knowledge.
3. Never compare your answers and opinions to others when you know it's right.
4. Never trust people too much, no one is always right.
5. Even if you're the only one is having a different opinion stick to it.

Sherif (1935) Autokinetic Effect Experiment:

Aim: Sherif (1935) conducted an experiment with the aim of demonstrating that people conform to group norms when they are put in an ambiguous (i.e. unclear) situation.

Method: Sherif used a lab experiment to study conformity. He used the autokinetic effect – this is where a small spot of light (projected onto a screen) in a dark room will appear to move, even though it is still (i.e. it is a visual illusion).It was discovered that when participants were individually tested their estimates on how far the light moved varied considerably (e.g. from 20cm to 80cm).

The participants were then tested in groups of three. Sherif manipulated the composition of the group by putting together two people whose estimate of the light movement when alone was very

similar, and one person whose estimate was very different. Each person in the group had to say aloud how far they thought the light had moved.

Results: Sherif found that over numerous estimates (trials) of the movement of light, the group converged to a common estimate. The person whose estimate of movement was greatly different to the other two in the group conformed to the view of the other two.
Sherif said that this showed that people would always tend to conform. Rather than make individual judgments they tend to come to a group agreement.

Conclusion: The results show that when in an ambiguous situation (such as the autokinetic effect), a person will look to others (who know more / better) for guidance (i.e. adopt the group norm). They want to do the right thing, but may lack the appropriate information. Observing others can provide this information. This is known as informational conformity.

Lesson -2 Compliance - meaning, examples and techniques

In psychology, **compliance** refers to changing one's behavior due to the request or direction of another person. It is going along with the group or changing behavior to fit in with the group, while still disagreeing with the group.

- "Compliance refers to a change in behavior that is *requested* by another person or group; the individual acted in some way because others asked him or her to do so (but it was possible to refuse or decline.)" (Breckler, Olson, & Wiggins, 2006)

- "Situations calling for compliance take many forms. These include a friend's plea for help, sheepishly prefaced by the question "Can you do me a favor?" They also include the pop-up ads on the Internet designed to lure you into a commercial site and the salesperson's pitch for business prefaced by the dangerous words "Have I got a deal for you!" Sometimes the request is up front and direct; what you see is what you get. At other times, it is part of subtle and more elaborate manipulation." (Kassin, Fein, & Markus, 2011)

In a situation when you wanted someone to do something for you, you start thinking about the tricks which you can use to get your things done. Social psychologists have put efforts to understand this process, the most frequent form of social influence.

A well-known social psychologist (Robert Cialdini) termed Compliance Professionals- people whose success (financial or otherwise) depends on their ability to get others to say “yes”. These people include salespeople, advertisers, political lobbyists, fundraisers, etc.

Basic Principles underlie many techniques professional use for gaining compliance :

1. Friendship/Liking: We are more willing to comply with requests from friends or from people we like than with requests from strangers or people we do not like.

2. Commitment/Consistency: Once we commit our self to a position or an action, we are more willing to say or do things that fit with committed position in various ways. e.g., if in past, we express our opinion to some specific issue in a official meeting, then we are more inclined to think and act in favour of the opinion expressed in meeting.

3. Scarcity: It is tendency to value, to feel worthy about and to secure, opportunities that are scarce or decreasing.

4. Reciprocity: We are generally more willing to comply with a request from someone who has previously provided a favour or concession to us than to someone who has not.

5. Social Validation: A tendency to comply with a request for some action if this action is consistent with what we believe persons similar to ourselves are doing (or thinking). We want to be correct, and one way to do so is to act and think like others.

6. Authority: We are usually more willing to comply with requests from someone who authority or simply looks like one.

Techniques professionals use for gaining Compliance :

A] Tactics Based on Friendship or Liking :

Ingratiation : There are several techniques for increasing compliance through getting others to like us. This is called as impression management. These impression management techniques are often used for purposes of ingratiation i.e., getting others to like us so that they will be more willing to agree to our requests.

There are many self-enhancing tactics of impression management such as improving one's appearance, emitting positive nonverbal cues (e.g., smiling, a high level of eye contact), and associating oneself with positive events or people the target person already likes. In contrast, other-enhancing tactics include flattery, agreeing with target persons, showing interest in them, and providing them small gifts or favours. Research findings indicate that all of these tactics can be successful, at least to a degree, in increasing others liking for us.

B] Tactics Based on Commitment or Consistency:

The Foot in the Door :A procedure for gaining compliance in which requesters begin with a small request and then, when this is granted, escalate to a larger one (the one they actually desired to be agreed). Once the target person says yes to the small request, it is more difficult for that person to say no to a larger request, because if they don't agree it would be inconsistent with the first response.

The Lowball : It has been seen that auto dealers sometimes use the lowball technique. This involves offering an attractive deal to customers but then, after they accept, changing that offer in some way.

Rationally, customers should refuse: but in fact, they often accept less attractive deal because they feel committed to the decision of buying the car. To explain it futher, in this technique, a very good deal is offered to a customer. After the customer accepts, however, something is manipulated to show that it necessary for the salesperson to change the deal and make it less advantageous for the customer-for example, an "error" in price calculations is found, or the sales manger rejects the deal. The totally rational reaction for customers, of course, is to walk away. Yet often they agree to the changes and accept the less desirable arrangement.

Bait-and-switch tactic :A technique for gaining compliance in which once the customers enters the shop; items offered for sale are showed as unavailable or presented of very low quality. This leads customers to buy a more expensive item that is available. It happens because for customers point of view, changing one's mind and reversing an initial commitment requires hard work, and many people, it appears, would rather pay a higher price than change their minds.

C] Tactics Based on Reciprocity :

The Door in the Face :A procedure for gaining compliance in which requesters begin with a large request and then, when this is refused, retreat to a smaller one (the one they actually desired to be agreed). This is exactly opposite of the foot-in-the-door technique: instead of beginning with a small request and then presenting a larger one,
persons seeking compliance sometimes start with a very large request and then, after this is rejected, shift to a smaller request-the one they wanted all along.

The Foot in the Mouth : When people feel that they are in a relationship with another
person-no matter how trivial or unimportant-they often feel that they are obliged to help or considerate to that person simply because the relationship exists. For example, friends help friends when they need assistance, and persons who perceive themselves as similar in some manner may feel that they should help one another when the need arises.

That's-not-all technique : An initial request is followed, before the target person can
make up or his/her his mind to say yes or no, a small incentive is offered by the person who is using this tactic sweetens the deal. For example, auto dealers sometimes decide to throw in a small additional option to the car for e.g., free full tank fill, offer of seat cover, etc., in the hope that this will help them close the deal; and often, it really helps! Persons on the receiving end of the that's-not all technique view this small extra as a concession on the part of the other person, and so feel obligated to make a concession
themselves.

D] Tactics Based on Scarcity :

Playing Hard to Get : This technique involves the efforts to increase compliance by
suggesting that a person or object is scarce, rare and hard to obtain.

Deadline Technique : This is a technique for increasing compliance in which target persons are told that they have only limited time to take advantage of some offer or to obtain some item. Advertisements using this deadline techniques state a specific time limit during which an item can be purchased for a specific price. After the deadline runs out, the ads suggest, the price will go up.

E] Other Tactics for Gaining Compliance:

Complaining : In the context of compliance, expressing discontent, dissatisfaction, resentment, or regret as a means of exerting social influence on others. Complaining involves expressions of discontent or dissatisfaction with one self or some aspect of the external world, and often such statements are simple expressions of personal states ("I feel lousy!") or comments on the external world ("Wow, is it cold today!"). Sometimes, however, complaining is used as a tactic of social influence:" Why didn't you take out the garbage like you, promised?" "We always see the movie you want; it's not fair," Statements such as these directed toward the goal of getting the recipient to change his or her attitudes or behaviours in some manner.

Putting Others in Good Mood : People's moods often exert a strong effect on their behaviour. And, it seems, this principle also holds with respect to compliance. When individuals are in a good mood, they tend to be more willing to say "Yes" to various request than when they are in a neutral or negative mood.

Factors That Influence Compliance:

- People are more likely to comply when they believe that they share something in common with the person making the request.
- When group affiliation is important to people, they are more likely to comply with social pressure. For example, if a college student places a great deal of importance on belonging to a college fraternity, they are more likely to go along with the group's requests even if it goes against their own beliefs or wishes.
- The likelihood of compliance increases with the number of people present. If only one or two people are present, a person might buck the group opinion and refuse to comply.
- Being in the immediate presence of a group makes compliance more likely.

Lesson -3 Obedience; Milgram's Studies, obedience and authority

Obedience:

Obedience is a form of social influence that involves performing an action under the orders of an authority figure. *Obedience* is a form of social influence where an individual acts in response to a

direct order from another individual, who is usually an authority figure. It is assumed that without such an order the person would not have acted in this way. Obedience occurs when you are told to do something (authority), Obedience involves a hierarchy of power / status. Therefore, the person giving the order has a higher status than the person receiving the order. Although we may be influenced by the people around us more than we recognize, whether we conform to the norm is up to us. But sometimes decisions about how to act are not so easy. Sometimes we are directed by a more powerful person to do things we may not want to do. Researchers who study obedience are interested in how people react when given an order or command from someone in a position of authority. In many situations, obedience is a good thing. We are taught at an early age to obey parents, teachers, and police officers. It's also important to follow instructions from judges, firefighters, and lifeguards.

This is a form of social influence in which a direct order by a person compels others to behave in a particular manner. However, the person giving order can use other means also to influence the behaviours, such as request instead of order (Yukl &Falbe, 1991). Thus, this form of social influence is less frequent than other two forms discussed earlier. Obedience, as a social influence, can sometimes be highly destructive. For example,military troop obeying their command can be brutal towards their target.

Milgram's Experiment:

Obedience to the authority can be seen quite often but experiments by Stanley Milgram (1963, 1974) demonstrated that even people without any legitimate authority can also obtain high levels of submission from others. Participants (all males) were told that the experiment aimed at exploring the effects of punishment on learning. Real participants were then paired up with another participant, who in reality was a fake participant (an assistant of the researcher). In each pair, one participant had to act as a teacher and the other participant had to act as a learner. The role of teacher and learner was decided on the basis of a slip drawn from a hat. However, the slips were drawn in a pre-decided manner so that the real participant always got the role of teacher.

Aim: Milgram (1963) was interested in researching how far people would go in obeying an instruction if it involved harming another person. Stanley Milgram was interested in how easily ordinary people could be influenced into committing atrocities, for example, Germans in WWII.

Apparatus: Apparatus used in the Milgram's experiment was a board containing 30 switches marked from 15 volts to 450 volts. Participants were told that pressing each switch will give an electric shock to the person sitting on the receiving chair. The amount of shock generated by pressing each switch was told to be equivalent to the amount of volt mentioned above that switch. However, in reality, no shock was received by the person sitting on the receiving end. The only real shock ever felt by person sitting on receiving chair was a mild pulse from third button, just to convince participant that the apparatus was real.

Procedure: Volunteers were recruited for a lab experiment investigating "learning" (re: ethics: deception). Participants were 40 males, aged between 20 and 50, whose jobs ranged from unskilled to professional, from the New Haven area. They were paid $4.50 for just turning up. At

the beginning of the experiment, they were introduced to another participant, who was a confederate of the experimenter (Milgram). They drew straws to determine their roles – learner or teacher – although this was fixed and the confederate was always the learner. There was also an "experimenter" dressed in a gray lab coat, played by an actor (not Milgram). Two rooms in the Yale Interaction Laboratory were used - one for the learner (with an electric chair) and another for the teacher and experimenter with an electric shock generator. The "learner" (Mr. Wallace) was strapped to a chair with electrodes. After he has learned a list of word pairs given him to learn, the "teacher" tests him by naming a word and asking the learner to recall its partner/pair from a list of four possible choices. The teacher is told to administer an electric shock every time the learner makes a mistake, increasing the level of shock each time. There were 30 switches on the shock generator marked from 15 volts (slight shock) to 450 (danger – severe shock).The learner gave mainly wrong answers (on purpose), and for each of these, the teacher gave him an electric shock. When the teacher refused to administer a shock, the experimenter was to give a series of orders/prods to ensure they continued. There were four prods and if one was not obeyed, then the experimenter (Mr. Williams) read out the next prod, and so on.

Prod 1: Please continue.
Prod 2: The experiment requires you to continue.
Prod 3: It is absolutely essential that you continue.
Prod 4: You have no other choice but to continue.

Results: Indicated that 65% of the total participants proceeded through the complete series till 450 volt. Few participants protested and asked the experiment to stop but later yielded to the experimenter's demand to proceed. Participants continued even after the assistant acted to become unconscious at the level of 300 volts. In such cases, the participants were asked to consider no response from the learner as an error and hence to continue the punishment to the learner. Other studies have also reported similar results for different cultures and with children and adults also (Kilham& Mann, 1974; Shanab&Yanya, 1977).

Conclusion:Ordinary people are likely to follow orders given by an authority figure, even to the extent of killing an innocent human being. Obedience to authority is ingrained in us all from the way we are brought up. People tend to obey orders from other people if they recognize their authority as morally right and/or legally based. This response to legitimate authority is learned in a variety of situations, for example in the family, school, and workplace**.**

Zimbardo's Prison Experiment

Milgram's controversial experiments generated a great deal of interest in the psychology of obedience. During the early 1970s, social psychologist Philip Zimbardo staged an exploration into the study of prisoners and prison life. He set up a mock prison in the basement of the Stanford University psychology department and assigned his participants to play the roles of either prisoners

or guards, with Zimbardo himself acting as the prison warden.The study had to be discontinued after a mere six days even though it was originally slated to last two weeks. Why did the researchers end the experiment so early? Because the participants had become so involved in their roles, with the guards utilizing authoritarian techniques to gain the obedience of the prisoners. In some cases, the guards even subjected the prisoners to psychological abuse, harassment, and physical torture.The results of the Stanford Prison Experiment are often used to demonstrate how easily people are influenced by characteristics of the roles and situations they are cast in, but Zimbardo also suggested that environmental factors play a role in how prone people are to obey authority.

Lesson -4 Modeling; cooperation, competition, bargaining, conflict resolution

Cooperation

The basic form of human contact and association in society is co-operation. This is the simplest of all social behaviors found in the most elementary forms of life. The importance of the concept is as clear to the student of sociology as of physical and chemical actions to a chemistry student. The customer co-operates with the shopkeeper, though for a short while, for business purposes. The patient with the doctor, the student with the, teacher, the child and her mother, husband and wife, the Imam and his followers, all cooperate together. This vast network of social relationship found in society is resting upon co-operation. Under the pursuit of common Interests, the people cooperate with one another and get themselves associated together. Hence co-operation is a basic necessity of life.

No society can develop its social organization and social system without co-operation. The whole machinery of social life is moving on the wheels of co-operation. We find co-operation in family, education, agriculture, industry, 'business and trade, political activities of a party and a religious sect.

What does the word cooperation means, it means helping one another or sharing in one or more aspects of human social life? In our societies, we find a few important examples of co-operation. In the farms, the rural community people co-operate with their neighbors in watering the crops, ploughing the fields and harvesting the crops. In urban community people, co-operate with one another even in presence of dissimilar social conditions. Such co-operation is found between the customers and shopkeepers, teachers and the students, the owners and the laborers, the doctor and the patients and so many different groups of people.

Cooperation is an integrating activity and is believed to be the opposite of competition. The word “cooperation” is derived from the two Latin words “co” meaning together and “operari” meaning to work. It is thus a joint activity in pursuit of common goals or shared rewards. It is goal oriented and conscious form of social interaction. Actually it involves two elements – (i) common end and (ii) organised effort. It is the process by which individuals or groups combine their effort, in a more or less organised way for the attainment of common objective. Generally co-operation means

working together for a common objective. In many festivals, volunteers work together to collect money from different parts and want to organise the program successfully and everybody want to stretch forward their hands to celebrate the occasion successfully. Among the members of the group, there seems to be indication of good interaction process. All of them behave in co-operative manner. Co-operation is brought about by several factors which
includes the following:

I. desire for individual benefits
II. desire to give and share
III. total decision on common purposes
IV. situational necessity and
V. desire to achieve larger goals.

Definitions of Cooperation

F.E. Merrill: "cooperation is a form of social interaction wherein two or more persons work together to gain a common end."

Fairchild: "cooperation is the process by which the individuals or groups combine their effort, in a more or less organized way for the attainment of common objective."

A.W. Green· "Cooperation is the continuous and common endeavor of two or more persons to perform a task or to reach a goal that is commonly cherished,"

Types of Cooperation

1. **Direct Cooperation:** The task unable to be performed by an individual needs cooperation of other people. Those things which are common to the people and they share in such things is called direct co-operation. It is the co-operation in which the participants have common interest. For example, playing, praying, studying and working together in a farm or road building and house building are the examples of direct so-operation. All the activities of social life in which interest of a group of people is involved falls in direct co-operation.

2. **Indirect Co-operation:** People do unlike things for the achievement of similar goals. Division of labour and specialization in a complex society, create impersonality in human behaviour and the people of one category co-operate with the people of other categories. In this way, labour and the material products are exchanged to the satisfaction of mutual needs and this process involves co-operation among people. People belonging to various' castes and classes have to co-operate, sometimes, in face of common needs and a common situation confronting them.

3. **Primary cooperation:** There is an identity of interest between the individual and the group. It is the cooperation which is found in primary groups such as family. Not only family but also peer groups is also called primary cooperation where people have to choose

face-to-face relations. We can also say that interest of the individual merge with the interest of the primary group.

4. **Secondary cooperation:** This type of cooperation is generally found in the secondary group, viz., government, industry, church and trade union etc.

5. **Tertiary cooperation:** This is the interaction between various big and small group to meet a particular situation. Here the individual or group who wants to compete with one another, come together and cooperate with each other for a specific purpose. In such type of cooperation the attitudes of the cooperating parties are purely opportunistic.

Role of Cooperation

It is a universal phenomenon. Cooperation for human beings is both a psychological and social necessity. Individual as well as collective goals can not be achieved without cooperation. Cooperation is essential for maintenance of social order. We can not do anything without cooperation, if we are to live as members of the society. Family members cooperate with each other in terms of sharing economic, emotional and social requirements of one another. All the activity in each and every family is done in a cooperative way. People learn their first lesion in cooperation as members of the family. The physical, mental and even the spiritual needs of the individual remain unsatisfied if he does not agree to cooperate with his fellow members.

Competition

Competition is the most important form of social struggle. According to Anderson and Parker, "Competition is that form of social action in which we strive against each other for the possession of or use of some limited material and non-material goods." According to Sutherland, Woodward and Maxwell, "Competition is an impersonal, unconscious continuous struggle which, because of their limited supply, all may not have." Competition is a form of interaction. It is the struggle for position to gain economic status. It occurs whenever there is an insufficient supply of anything that human being desire - insufficient in the sense that all cannot have as much of it as they wish. Sometimes competition happens because of limited supply and also difficult for equal distribution.

Characteristics of Competition

The characteristics of competition are:

1. Competition is impersonal struggle. Park and Burgess defined competition as "interaction without social contact." We can say it is inter-individual struggle that is impersonal. It is usually not directed against any individual or group in particular.

2. Competition is universal. There is no society which is devoid of competition. Not only this, degree of competition may vary from society to society. It is very common for society as well as culture.
3. Competition is considered as conducive to progress.
4. Competition provides the individuals better opportunities to satisfy their desires for new experiences and recognitions.
5. Both associative and non-associative dimensions of social processes indicate competition.
6. Competition is mainly an unconscious activity but personal competition or rivalry is a conscious activity.
7. Competition may create emotional disturbances.
8. Competition is an innate tendency.
9. Competition is a social phenomenon.
10. Degree of competition is determined by social values and social structure.

Value of Competition

Competition is indispensable in social life. Competition performs many powerful functions in society. Sometimes competition is extremely dynamic. It performs five positive functions.

1. It helps to determine the status and location of individual members in a system of hierarchy.
2. It tends to stimulate economy, efficiency and inventiveness.
3. It tends to enhance one's ego.
4. It prevents undue concentration of power in an individual or group of individuals.
5. It creates respect for the rules of the same group.

Conflict

Conflict is a conscious action. It is a deliberate intent to oppose. We can say that conflict is universal. Not only this, conflict lacks continuity. Besides this, it is a personal activity. According to Gillin and Gillin, "Conflict is the social process in which individuals or groups seek their ends up directly challenging the antagonist by violence or threat of violence." Green defined, "Conflict is the deliberate attempt to oppose, resist and coerce the will of another or others." Conflict is a competition in its more occasional, personal and hostile forms. It is a process of seeking to obtain rewards by eliminating or weakening the competitors. It is inherent in every society or social system. Conflict is apparently noted when an individual or a group seeks to attain its own end. Conflict is the opposite of cooperation. Conflict may lead to antagonism, violence or threat to peace. Conflict is also regarded as non-associative social process.

Causes of Conflict

According to Freud and some other psychologists, the innate instinct for aggression in man is the main cause of conflicts. Generally it arises from a clash of interest within groups and societies and between groups and societies. The significant causes are:

- **Individual difference** – It is true that, we, the human being, are not alike by nature, attributes, interests, personalities etc. These differences may lead to conflict among the human being.
- **Cultural differences** – The culture of a group differs from the culture of the 55 Cooperation, Competition and Conflicts other group. The cultural differences among the groups sometimes cause tension and lead to conflict.
- **Differences of opinion regarding interest** – In fact, the interests of different people or groups occasionally clash. For example we can say that interests of the employers and employees vary in many respect which may ultimately leads to conflict among them.
- **Social change** – Social changes occur off and on in each and every society. Conflict is an expression of social disequilibrium. Social change is the cultural log which leads to conflict.

Characteristics of Conflict

The characteristics of conflicts are:

1. Although conflict is universal but it's nature vary from society to society. No society is characterized exclusively by conflict.
2. Conflict is a conscious action. To achieve the respective goals the members oppose one another.
3. Conflict leads to social change. It is mainly an expression of disequilibrium.
4. It primarily remains personal activity. Sometimes it acquires greater intensity and involves many people. At first, it is manifested at the level of a particular group but ultimately it involves the entire society.
5. Asymmetrical social, political and economic relations coupled with the sense of relative deprivation as the root cause of social conflict.
6. Sometimes conflict is the life of society and progress emerges from a struggle in which each individual, class or institution seeks to realize its own idea of good. l We the people are organized into groups to seek a common goal, the probability of conflict increases.
7. All conflicts are not the same.
8. We face conflicts on all levels as we have disagreements with family, friends and co-workers.
9. Conflict is an ever-present process in human relations.
10. Unmanaged conflict is a threat to the survival of the group and ultimately tends to make the group less effective.
11. Conflict can occur within groups (intra-group conflict) and among groups (inter-group conflict).

Types of Conflict

Conflict may be of various types,

1. **Overt and covert**: Overt conflict has some manifestation but covert conflict or latent conflict primarily remains invisible.
2. **Personal and corporate**: Personal conflict occurs within the group due to hostility, jealousy etc., on the other hand, corporate conflict occurs among the groups within a society or between two societies.
3. **Temporary and perpetual**: When conflict occurs among the individuals in the bus or in the shop or in the road are called temporary conflict, on the other hand when conflict occurs between two rival groups, then it is called perpetual conflict.

Conflict is not a continuous process. It may stretch for sometime. In each and every conflict there is intermittent period of peace and social harmony.

According to Gillin and Gillin, conflict can be divided into five types, viz.,

1. **Personal conflict** – It is the conflict between two persons within the same group.
2. **Racial conflict** – It is the result of the feeling or racial superiority or inferiority.
3. **Class conflict** – It is the conflict between two classes.
4. **Political conflict** – It is the conflict between parties of political power.
5. **International conflict** – It is the conflict between two nations.

METHODS OF CONFLICT RESOLUTION

Like any other problem Social psychologists have developed number of approachesfor resolution of social conflict. Some of the main methods as follow:

Mutually beneficial goal*:* A common approach to ameliorate social conflict isto establish mutually beneficial goals. Such goals prompt the warring parties towork in close cooperation and help reduce feelings of group struggle. In a classicalexperiment Sheriff and Sherif put two groups in a social situation which involvedcompetition between the two parties. It was observed that very soon theydeveloped feelings of animosity and competition. Things reached to a level whereboth parties raided each other's camps to hurt and damage their prospects ofattaining goal. In the next phase of the experiment both the warring parties wereput in a situation which called for joint efforts to reach the goal. Since neither ofthem had sufficient resources to attain goal on their own, they were left with noalternative but to help each other to surmount problems faced by them. Notlonger than before it was observed that members of both parties began trying tounderstand each other's concerns. They started seeing members of oppositegroup very often and appreciated each other's approach to handle problems.Thus by the end of the experiment both the groups had developed betterunderstanding of each other and there was a marked reduction in feeling ofconflict.

Compromise*:* Reaching a compromise between the two parties also leads toreduction of struggle. Compromise is a situation where no party stands to gain orlose anything. Thus it gradually leads to reduction in struggle. For example, whenGurjars in Rajasthan pressurised the government for

inclusion into scheduledtribes, the Meenas vehemently opposed their demand since they have been thebiggest gainer in Rajasthan. This conflict led to open war between them andresulted in many casualties and bloodshed. Now that location of both communitiesis such that they are found living side by side almost all over Rajasthan, they soonrealised the futility of opposing each other. A compromise was struck between thetwo and an understanding was arrived at under which gurjars dropped theirdemand for inclusion into scheduled tribe but asked for a separate quota forthemselves and Meenas agreed to support Gurjars demand. Thus under the newagreement both parties stood to gain or lose nothing at the cost of each other.

Developing special norms*:* Conflict between two warring factions or groupsmay be reduced through development of special norms. For example, in a game,the question of who will take first turn may be settled by leaving it to the umpire.Thus the bone of contention is removed and thereby the cause of conflict isremoved. Psychologists have delineated social conditions, on the basis of studies,where conflict and struggle can be handled through developing special norms.According to them social conditions where in both the parties have the ability andwill to influence each other lend themselves to such interventions. This techniquehas reportedly been successfully used in a number of situations with differentcommunities.

Prosocial behavior*:* By prosocial behaviour we mean behavior that createssome kind of positive social influence among others. For example, giving charity,working for the welfare of others, helping others in distress are some of the formsof prosocial behaviour. Studies reveal that when members of a community engagein some kind of prosocial behaviour toward members of other community it hasdirect impact on their perception and opinion about them and results significantlyin the reduction of feelings of conflict and struggle.

Psychologists observed that when an individual engages in prosocial behaviour,that is when a person is involved in helping a person in distress, the personpasses through four stages:

1. **First**, the individual takes stock of the seriousness of the condition or situation inwhich help is to be given.
2. **Second,** the individual takes responsibility for the helping another person.
3. **Third,** the individual enters a state where he wishes to help the person in distress.
4. **Finally** the person recognises the fact that the individual is capable of helpinganother person.

Use of scientific approaches*:* Social psychologists suggested some scientificapproaches for reduction of group conflict, which is as follows:

a) Win-Lose approach
b) Lose-Lose approach
c) Lose-Win approach. Let usdewscrfibe these approaches:

Win-Lose approach – In this type of approach one party or group adoptsdifferent methods, means and approaches aimed at ensuring defeat of opponentparty and victory for themselves. In other

words, one group tries to thwartanother group from reaching the target or goal so that they can reach the goal.

This approach has two underlying assumptions: that is (i) conflict among peopleis inevitable and unanimity is impossible. (ii) gain of one party results is loss forthe other party . In other words, only one party can win or reach the goal andanother party is bound to fail in the process. In such situations, conflicts do arise.

Now that conflict is inevitable and there can only be one winner, the besttechnique or approach is leaving no stone unturned to ensure defeat of opponentsand victory for self.Success can be achieved using socially acceptable means like exercising ourdemocratic rights and also to some extent by using subversive methods likethreatening the opponents of dire consequences, making opponent allies crossover the fence by luring them with different types of benefits or even takingrecourse to unlawful activities as advocated. Thus in nutshell this approachadvocates putting in exercise the age old aphorism that every thing is fair in loveand war. It is the end that matters the most and not the means employed toachieve the end.

Lose-Lose approach – This approach is called lose-lose approach becausenone of the parties involved in the struggle stand to gain much and are not ableto cause what they wanted to happen. Underlying assumptions of this approachare:

i) Some thing is better than nothing.
ii) It is better to stay away from struggle than indulging in it and wasting resources.
iii) When groups reconcile with each other after initial bouts of struggle the loseloseapproach is exemplified.
iv) Another characteristic of this approach is that it results in quick solution of theproblem and values and motives of individuals do not find place.

Win-Win approach: This approach is different from the above two approachesmentioned earlier. This is considered ideal for reduction of struggle betweenwarring groups. It involves both parties resorting to different types of cooperativemeasures and techniques to arrive at a conscious solution of the struggle so thatall the concerned parties are benefitted to the maximum.

Underlying assumption in this approach is that it considers struggle as a mutualproblem which can be solved amicably. It lays emphasis on the difficulties andproblems of both parties and not on the means of ensuring victory. Both partiessit together and work on the solution of the problem and whosoever reaches thesolution first acquaints the other party. Thus the solution arrived at is acceptableto all concerned. However successful application of this approach requires skillin human relations otherwise it is difficult to achieve success using this approach.

BLAKE AND MOUTON STRATEGIES

Blake & Mouton on the basis of their research stated that people adopt one ofthe following five strategies for reduction of conflict and struggle which are givenbelow:

Withdrawing*:* In this the person resolves the problem by stepping back from thesituation of struggle (lose-lose approach).

Smoothing: This comprises of putting emphasis on points of common interestsand avoiding discussion on matters of debate or controversy. Thus this approachtries to minimize differences between the two warring factions (lose-lose approach).

Compromising*:* This strategy seeks to decrease differences through discussion(lose-lose approach).

Forcing: This approach is an all or none approach in that it takes a competitivestand. It is this way or that way approach (win-lose approach).

Confrontation and problem solving: Both parties openly discuss all matters andthe best mutually acceptable solution is accepted (win-win approach)

TWO DIMENSIONAL MODEL

This model presumes that all sorts of conflict handling behaviour can be understoodin terms of two dimensions i.e. assertiveness and cooperation. These two basicdimensions of behaviour define five different modes for responding to conflictsituations:

Competing is assertive and uncooperative—an individual pursues his own concernsat the other person's expense. This is a power-oriented mode in which you usewhatever power seems appropriate to win your own position—your ability toargue, your rank, or economic sanctions. Competing means "standing up for yourrights," defending a position which you believe is correct, or simply trying to win.

Accommodating is unassertive and cooperative—the complete opposite ofcompeting. When accommodating, the individual neglects his own concerns tosatisfy the concerns of the other person; there is an element of self-sacrifice in thismode. Accommodating might take the form of selfless generosity or charity, obeyinganother person's order when you would prefer not to, or yielding to another'spoint of view.

Avoiding is unassertive and uncooperative—the person neither pursues his ownconcerns nor those of the other individual. Thus he does not deal with the conflict.Avoiding might take the form of diplomatically sidestepping an issue, postponingan issue until a better time, or simply withdrawing from a threatening situation.

Collaborating is both assertive and cooperative—the complete opposite ofavoiding. Collaborating involves an attempt to work with others to find somesolution that fully satisfies their concerns. It means digging into an issue to pinpointthe underlying needs and wants of the two individuals. Collaborating between twopersons might take the form of exploring a disagreement to learn from eachother's insights or trying to find a creative solution to an interpersonal problem.

Compromising is moderate in both assertiveness and cooperativeness. Theobjective is to find some expedient, mutually acceptable solution that partiallysatisfies both parties. It falls intermediate between competing and accommodating.Compromising gives up more than competing but less than accommodating.Likewise, it addresses an issue more directly than avoiding, but does not exploreit in as much depth as collaborating. In some situations, compromising might meansplitting the difference between the two positions, exchanging concessions, orseeking a quick middle-ground solution.Each of us is capable of using all five conflict-handling modes. None of us can be characterised as having a single style of dealing with conflict. But certainpeople use some modes better than others and, therefore, tend to rely on thosemodes more heavily than others—whether because of temperament or practice.

Third party intervention Another way of resolving group conflict is to seekmediation of an arbitrator or third party. This third party tries to arrive at asolution keeping in view the characteristics of the warring factions. Solution generatedby the third party is binding on all the concerned parties. A good feature of thistype of resolution is that the third party does not enforce a decision but thedecision is arrived at through open discussion and negotiation over points ofproblems. However, this approach requires third party to be very mature andadept at human relations.

Unit. II: Prejudice and Discrimination

Lesson -5 Definition and Nature of Prejudice and Discrimination

Definition and Nature of Prejudice and Discrimination – Origin of Prejudice (Cognitive and Social) – Challenging Prejudice – Prejudice based on Gender – Forms of Discriminatory behavior

Prejudice:

Social Psychologists tried to define prejudice from different viewpoints. Some psychologists define prejudice as a preconceived irrational judgment, while others define it as an expression of dislike against members of some religion, race or group. However, majority of psychologists agree upon the definition given by Secord and Backman "Prejudice is an attitude that predisposes a person to think, perceive, feel and act in favorable and unfavorable ways towards a group or its individual members." According to Baron & Byrne "Prejudice is generally a negative attitude towards the members of some social, ethnic or religious." Prejudice be it negative or positive is

decidedly an attitude and has all the three components of attitude i.e. affective, cognitive and behavioral.

Nature of Prejudice:

1. **Prejudice is acquired**: Like attitude prejudice is acquired through the process of learning and socialization. When born a child is like a blank slate and is free of any kind of prejudice. It is only when the process of socialization begins that he starts imitating his parents and their likes and dislikes. Norms, values, customs, and traditions of the society of which he is member make him prejudiced toward members of other group. Acquisition of prejudice is facilitated by classical conditioning, instrumental and observational learning. A child learns to hate Pakistanis only because he sees significant others in the society hating Pakistanis.

2. **Emotional overtones:** Prejudice is always colored with emotions. It is either for or against some group, community or religion. If favorable, the person would show too much affection, love, care and sympathy for members of another group. But if unfavorable the person would show hatred, dislike and hostility.

3. **Prejudice is irrational**: Prejudice does not lend itself to reason, wisdom, and relevance. The individual does not change his prejudice in the face of information and evidence to the contrary.

4. **Prejudice is functional:** Prejudice helps the individual justify his hostilities, repressed desires and strengthen feelings of self-esteem and prestige. It helps individual justify his exploitation, discrimination of members of other group. For example, in Indian society the upper caste Hindus justified their exploitation of lower castes reasoning that they are like that only and deserve to be exploited and discriminated against.

5. **Prejudice has no connection with reality:** It is primarily based on hearsay, incomplete and wrong information, customs and traditions of the society. It can't stand test of logic and reasoning.

Lesson -6 Origin of Prejudice (Cognitive and Social) – Challenging Prejudice

Origin of Prejudice (Social and Cognitive):

1. **Status and Power structures**: The structure of relations between two groups in terms of relative status and power sometimes gives rise to prejudice. For example, where a dominant group holds another group in a condition of slavery, slaves are likely to be considered lazy, irresponsible and lacking in initiative. These beliefs emerge from the fact that slaves act upon orders from their masters and not given an opportunity to demonstrate initiative or responsibility. Thus the beliefs about them are consonant with their behaviour, which is controlled by structure of relations.

2. **Historical facts**: Prejudice and discrimination develop out of history of economic conflict as well as from political power distribution among groups of people. Historical incidents led to the development of attitude which slowly takes form of prejudice. In our society prejudice against women is one such example. Women have always been considered weak, dependant and tools such prejudice developed out of atrocities perpetrated over women and they kept tolerating them

thinking it to be their duty. Similarly some professions have historically been thought to be fit for men than women. For example, truck driving has never been considered fit for women, Women in this profession and other such professions are looked down upon. Another example of historical reason of prejudice comes from prejudice against Jews. An image of Jews as rich, grasping, and shrewd grew out of their occupational roles as money lenders. The church prohibited Christians from lending money at interest, but did permit them to borrow from Jews. Thus the Jews Prejudice and Discrimination became bankers when this occupation was extremely profitable, and the cognitive image commensurate with the role became firmly established. Besides competitive circumstances produced negative effect against Jews.

3. **Situational Factors**: The number of situational factors in the immediate environment of the individual also lead to development of prejudice:

4. **Social learning**: Every individual during the process of socialization learns and acquires beliefs, values and attitudes through parents, school, religion and church. These agents of socialization invariably transmit prejudices held by them to the child. Besides childrearing practices adopted by parents have been shown to help develop prejudice and discrimination.

5. **Job Competition**: Scarcity of job avenues and abundance of applicants is one important economic factor for development of prejudice. It led to the emergence of sons of soil theory. For example, the Marathi movement against North Indians in Mumbai and other parts of Maharashtra has one of its reasons based in economic factors. North Indians coming to Mumbai are ready to work for longer hours and that too at cheaper wages as against Local people. They have gradually outplace local people in patty and traditional jobs thus rendering many of people jobless and fending for struggle to survive. It has led them to believe that north Indians are responsible for their plight and are replacing them in their own home. Such beliefs lead to development prejudice against North Indians.

6. Conformity to Norms: Once prejudice and discrimination against out-group are well established, the accompanying cognitions and feelings concerning the out group acquire a normative quality. They are shared by members of the in-group and the members expect each other to hold such attitudes. The factors underlying conformity to the norms of prejudice may be explained in terms of the varying reward-cost outcomes ensuing from conformity or nonconformity. If prejudice and discrimination against other group is the norm, then overt expression of prejudice and discrimination will receive approval from other members of the group.

7. **Interaction Patterns:** Prejudice and discrimination create certain interaction patterns that contribute to maintenance of the status quo. Several interaction patterns increase cohesion and thus strengthen the power of the group to enforce conformity to norms of prejudice and discrimination. Any factor that makes members more dependent on the group is likely to increase cohesion. Interaction pattern within the in-group may also increase the economic dependence of members upon each other. Finally, if interaction within each group predominates over interaction across group lines, the development of patterns of thinking, feeling and behaving unique to each group is fostered. Such interaction patterns increase the cultural gulf that separates the two group.

8. **Frustration and Aggression**: Frustration also gives rise to prejudice. The underlying theory in it is displaced aggression. According to this theory when individual finds some obstacle between him and his goal he gets frustrated and becomes aggressive toward the obstacle. Since the interfering agent is stronger and powerful and has the power to punish him this frustration and aggression is displaced toward some weaker object. Thus, the weak person or group becomes scapegoat.

9. **Authoritarian Personality**: Among psychological factors of prejudice authoritarian personality has received much attention from psychologists. People with authoritarian personality exhibit rigid thinking, punitive tendency. These tendencies predispose individual toward prejudice. Besides these people value people on the scale of power, people above them in the power scale are attributed all good characteristics and people below them on power scale are treated as inferior and deserve to be exploited and hated.

10. **Personality needs:** A variety of personality needs may support prejudice and discrimination. One such need is "intolerance for ambiguity". Persons differ in the extent to which they are disturbed by confusing or ambiguous situations. Some persons like to have everything in black and white i.e. they are unable to tolerate least uncertainty or complexity in situation while some persons are least disturbed by confusing or uncertain situations. In general it has been found that individuals who are more intolerant of ambiguity are also likely to be more prejudiced because prejudice for them serves to clarify ambiguity and uncertainty embedded in the situation. Similarly, a need to achieve superior status may be supported by prejudice, which provides a group of persons lower in status than oneself. The need for security may be satisfied through rejection of out-group.

Challenging Prejudice:

Given that prejudice is common in all human societies and exerts damaging effects both on the victims and on those who hold such views, the important question to address is 'Is it possible to remove prejudice? What steps can be taken to do this? Considering the growing prejudice among people over religion, region, language, race, it seems almost like prejudice is inevitable. But stereotyping and prejudice can be definitely reduced to a large extent by becoming aware and educating ourselves. And with concentrated multi-pronged approach we can do many things to counter prejudice.

1. **Teaching children Acceptance Instead of Bigotry (narrow-mindedness)** : Homes and schools are places where children learn to get their generalizations leading to stereotypes. It is the responsibility of parents, teachers to bring their children up with acceptance and positive attitudes towards other groups and be role models. Because often the attitudes and prejudiced notions are not necessarily taught but 'caught' from adults and the environment the children operate in. There is sufficient evidence, today that teachers can be victims of bias and prejudice and can reflect in their classrooms.

2. Teaching children from an early age to respect all groups: We must teach children, including ones very different from their own - prejudice can be nipped in the bud or at least curbed. Valuing Diversity is very important. Ethnocentrism should be discouraged. The education of young

children should promote respect for a multicultural society and tolerance and acceptance virtues that need to be developed quite consciously.

3. Increased Intergroup Contact: This is based on the Contact hypothesis: the theory that direct contact between hostile groups will reduce prejudice. Recent findings indicate that if people merely know that friendly contacts between members of their own group and members of various out groups (groups in which we do not belong) is possible, their prejudice towards these groups can be sharply reduced. E.g., The various Mohalla committees that operate in different areas after the 1992 communal riots in Mumbai are based on this reasoning that increased inter-group contact can remove prejudice and hatred. However, there are some conditions like the groups interacting must be roughly equal in status, the contact situation must involve cooperation and not competition. They should work towards shared goals and the setting should help them understand each other better. In short, direct inter-group contact can be an effective tool to combat cross-group prejudice.

4. Recategorization: redrawing the boundary between 'Us and Them': Once individuals mentally include people, they once excluded from their in-group (groups in which we belong) within it, prejudice toward them may disappear. Reminding people that they are part of large groups - for instance, that they are all Indians, Americans Canadians or even human beings - can help accomplish this kind of recategorization.

5. Undermining-Stereotypes: Stereotypes suggest that all persons belonging to specific social groups alike - they share the same characteristics. Such beliefs can be weakened if people are encouraged to think about others as individuals not simply as members of social groups.
Also, some evidence suggests that affirmative action programs in schools and colleges, offices may actually encourage positive perceptions and the persons who benefit from them will look at people and events more objectively. This will serve to counter prejudice by undermining (deflating) stereotypes. There must be an effort to look down upon people with prejudiced attitudes.

6. Reducing Prejudice through Cognitive interventions: Weakening stereotypes: Stereotypes involve category-driven processing -- thinking about others in terms of their membership in social categories or groups. Stereotypes can be reduced if persons can be made to engage in attribute-driven processing -- thinking about the unique characteristics of individuals and objective judgment.

7. Cooperative activities: Cooperative activities such as team building exercises, workshops involving games that help to remove bias and prejudice should be carried out in schools, colleges, work places, etc. It is a good way to reduce animosity and bitterness that stems from low self esteem and stereotyped attitudes. Thus, non-competitive contact between in and out groups on terms of equal status and the pursuit of common, super ordinate goals which are attainable by cooperation is the way forward.

8. Media responsibility: Instead of perpetuating stereotypes the media should take responsibility to remove prejudices and negative attitudes. It is the responsibility of every individual to be aware of the stereotypes and take care so as to not to discriminate and create a more equitable society.

9. Role of educational institutions: Educational inequality leads to bias and discrimination. High standards of educational experience, should be available to every student. In addition schools and colleges have to deal with so many other situations that cause prejudice and hatred. Efforts to address bias, prejudice and discrimination should be comprehensive in nature.

Lesson -7 Prejudice based on Gender, prejudice and discrimination

Prejudice based on Gender Each society throughout history has distinguished differences in the social roles of its males and females. These gender role differences reflect biases, also known as prejudices (a negative attitude towards others based on a prejudgment about those individuals with no prior knowledge or experience), held by members of a society. Gender prejudice is also referred to as sexism and is based on stereotypes (an oversimplified prejudgment of others, often leading to negative prejudices) held about women and men. Stereotypes of men are usually more positive in societies than stereotypes of women as the males are considered more independent and posing greater physical stamina. Negative stereotypes of women are usually held by both men and women in a society owing to the lack of self-respect and self-confidence imparted to females by societies' prejudices. Changes in gender prejudice through time have been evident in some regions of the world. By the early twenty-first century, the differences in social roles of males and females in some societies, such as those of the United States and Europe, had narrowed. Opportunities for education and employment in addition to simply participation in society as a whole, such political activities and a broad range of social events, became more equal. However, in other societies—such as those in the Middle East—long-held prejudices remained firmly in place. Behavior based on these prejudices normally placed the female role at considerable disadvantage in terms of freedoms and civil rights. These freedoms and rights include the right to vote in elections, associate with others freely, and various legal protections. The long struggle against entrenched gender prejudices to establish social equality for women has led in many directions. Besides the right to vote (suffrage), other concerns include reproductive rights such as access to birth control and right to have abortions (to artificially terminate a pregnancy), freedom from domestic violence, freedom from sexual harassment in the workplace and public, and freedom from overall male dominance, a concept known as patriarchy. The struggle for equality led to the growth of feminism. Feminism is both a belief in the social equality of women including legal, political, and economic equality, and a social movement promoting feminist beliefs, known as a feminist movement. The feminist movement began as an organized movement in 1848. Many leaders of the feminist movement have been women. They view gender prejudice as much like social class prejudice (classism), and racial prejudice (racism). Sexism was just one more form of discrimination (treating some people differently than others or favoring one social group over another based on prejudices) to keep political power in the hands of only a few—primarily white males in the Western world. In many societies, the role of the male is seen as the norm (standard), and females are measured against that norm in terms of abilities. Women who behave as men, such as those who act more direct in

business relations, are often viewed negatively as being overly aggressive. Men who exhibit those same behaviors are often valued as good leaders.

DISCRIMINATION

Discrimination is the differential treatment of individuals belonging to a particular social group or community or religion. It is generally the overt or behavioral expression of prejudice. Generally the person discriminated is denied some privilege or right that is accorded to other members of society who do not belong to the minority group.

According to the exchange theory when the reward-cost outcomes of two separately bounded groups are perceived to be mutually exclusive, so that each group can increase its gains only at the cost of other, members of each group try to protect or increase their outcomes. If the two groups are unequal in power, they will establish different outcomes unless prevented by norms that restrain exploitation of the weaker by the more powerful. These different outcomes create differences in the status of the two groups.

The extent to which the members of the minority group feel discriminated against and dislike or feel hostile toward the majority group is a function of the relation between their comparison level and that of the majority group. If the minority group has the same comparison level as the majority group, it will feel dissatisfied and hostile. But if comparison is sufficiently low relative to majority group no adverse feelings would occur.
However, 'minority groups' use of comparison level relative to majority group depends upon past experiences, the outcomes available in alternative relations, and structural and cultural factors. For example, in Indian society lower caste people were not allowed to take water from wells reserved for people of upper castes. They were not supposed to sit before people of upper castes and there were many more restrictions imposed upon them simply because they were born in shudra castes.

Sometimes discrimination occurs without the accompanying feeling of prejudice. For example, a proprietor may refuse to accept as patrons members of a minority group because he feels it would injure his business. He may not be prejudiced towards those people but he gives priority to his business.

Lesson -8 Forms of Discriminatory behavior

Forms of Discrimination:

People prier to harm the targets of their prejudice without any cost to themselves. This goal is accomplished through the use of several subtle firms of discrimination ones that permit their users to conceal their underlying negative views. The most common of this being (1) withholding aid from persons who need it (2) various forms of tokenism and (3) reverse discrimination.

1. **Reluctance to help n- "Let Someone else do it"**: the first of these tactics withholding aid from the members of disliked group is often used with considerable skill. People are more likely to refrain from aiding members of groups they dislike in situations where other potential helpers are present than in situations where they are absent. But some evidence suggests that the use of this particular subtle form of discrimination may be on the wane. However, it persists in some settings,

and it continues to exact a high cost from persons who could be aided by help but do not receive it simply because they belong to a particular social group.

2. **TOKENISM – Small Benefits, High Costs:** a second form of subtle discrimination involves tokenism. The basic mechanism here is prejudiced individuals engage in trivial, positive actions toward the members of groups they dislike. Then, they use these actions as a rational for refusing other, more important actions or as a justification for later discrimination.

Unfortunately, various forms of tokenism do more than deny groups who are the objects of prejudice important forms of aid. Being the recipient of such grudging help can also play havoc with the self-esteem and career of the persons involved. The impact of tokenism is largely negative, even for those few individuals who seem to profit from its existence.

3. **Reverse Discrimination (Giving with one hand, taking away with the other)**: reverse discrimination occurs in situation where persons holding at least residual prejudice toward the members of some social group actually learn over backwards to treat these persons favorably-more favorably than would be the case if they did not belong to this group. Individuals exposed to reverse discrimination receive raise, promotions and other benefits. At the same time such favorable treatment may actually prove harmful, especially over the long period. For example, while individuals who acquired prejudicial attitudes early in life may successfully hold them in check in many situations. Such reactions may rise to the fore and encourage negative actions toward their target groups when emotions and arousal run high. When reverse discrimination produces negative effects, it can be as harmful as other more obvious forms of discrimination.

Unit -3: Aggression

Lesson -9 Definition and Nature-Theoretical perspectives on Aggression

Aggression is overt or covert, often harmful, social interaction with the intention of inflicting damage or other harm upon another individual. It may occur either reactively or without provocation.

In social sciences and behavioural sciences, aggression is an action or response by an individual that delivers something unpleasant to another person. Some definitions include that the individual must intend to harm another person.

In psychology, the term aggression refers to a range of behaviours that can result in both physical and psychological harm to yourself, others, or objects in the environment. This type of behaviour centers on harming another person either physically or mentally. It can be a sign of an underlying mental health disorder, a substance use disorder, or a medical disorder.

Social psychologists define aggression as *behaviour that is intended to harm another individual who does not wish to be harmed*(**Baron & Richardson, 1994**).

Baron and Byrne: Aggression is Behavior directed towards the goal of harming another living being who is motivated to avoid such treatment.

Atkinsonand Smith et al: Aggression is a Behavior intended to injure someone physically or verbally or destroy someone.

Aggression can take a variety of **forms**, including:

1. Physical 2. Verbal 3. Mental 4. Emotional

Types of Aggression

There are four different types of aggressive behavior: accidental, expressive, instrumental and hostile.

1. Accidental aggression

It is a sporadic display of aggression without any intention of harming someone. Acts of accidental aggression are accidents and not done on purpose.

Examples

- Hitting someone's hand unintentionally while wearing a jacket.
- Stepping on a friend's foot while playing tag.

2. Expressive aggression

The child carries out an aggressive act to feel better or because it makes them feel good. However, the intention is not to hurt someone.

Examples

- The child could enjoy demolishing a sibling's playing blocks without realizing that it could make the sibling feel bad.
- Throwing or spilling a vessel full of liquid could seem delightful for the child but not to the parent.

3. Instrumental aggression

It is aggression shown to gain control of an object, that is, the aggression becomes instrumental in gaining control of an object. A common example is two children fighting for the same object or

purpose, such as a toy, crayon box, or a chance to use the slide at the playground, causing them to hurt one another.

Instrumental aggression is commonly seen between the ages of two and six years. Children of this age group may usually display self-centric personality attributes, making them more likely to quarrel than share.

4. Hostile aggression

It is an extreme form of aggression where the aggressive behavior is directed towards someone with the intention of hurting them. The child's purpose could be to harm someone physically or psychologically.

The child may turn to hostile aggression to gain pleasure or satisfaction — usually seen in bullying. A child may resort to striking, calling names, or ignoring someone to bully them.

Some children may believe that they have been wronged and could resort to hostile aggression to seek revenge. In such cases, the acts of hostile aggression could come across as impulsive or extreme.

Lesson -10 The role of biological factors Drive theories and modern theories of aggression

Factors Influence Aggression

A number of different factors can influence the expression of aggression, including:

1. **Biological Factors:** Men are more likely than women to engage in physical aggression. While researchers have found that women are less likely to engage in physical aggression, they also suggest that women do use non-physical forms, such as verbal aggression, relational aggression, and social rejection.
2. **Environmental Factors:** How you were raised may play a role. People who grow up witnessing more forms of aggression are more likely to believe that such violence and hostility are socially acceptable. Bandura's famous Bobo doll experiment demonstrated that observation can also play a role in how aggression is learned. Children who watched a video clip where an adult model behaved aggressively toward a Bobo doll were more likely to imitate those actions when given the opportunity.

3. **Physical Factors:** Epilepsy, dementia, psychosis, alcohol abuse, drug use, and brain injuries or abnormalities can also influence aggression.

Nature of Aggression

The nature of human aggression is viewed from four explanatory perspectives, derived from the ethological tradition.

The first consists of its a**daptive value,** which can be seen throughout the animal kingdom, involving r**esource competitio**n and pr**otection of the self and offsprin**g, which has been viewed from a cost-benefit perspective. T

The second concerns the **phylogenetic origin of aggression**, which in humans involves **brain mechanisms** that are associated with anger and inhibition, the emotional expression of anger, and how aggressive actions are manifest.

The third concerns the **origin of aggression in development** and its subsequent modification through experience. **An evolutionary approach** to development yields conclusions that are contrary to the influential social learning perspective, notably that physical aggression occurs early in life, and its subsequent development is characterized by learned inhibition.

The fourth explanation concerns the **motivational mechanisms controlling aggression**: approached from an evolutionary background, these mechanisms range from the inflexible reflex-like responses to those incorporating rational decision-making.

Theories of Aggression

- Instinct Theory of Aggression
- Drive theory / Frustration Aggression Hypothesis
- Social Learning Theory

Instinct theory of Aggression

The first theory of aggression goes back to the days of Sigmund Freud. Freud's theories on behavior changed over time. At first, he believed that all behaviors stemmed from Eros, the *life instinct.* These were the instincts that kept us alive and wanting to reproduce.

But in 1920, he wrote that the life instinct just couldn't explain all of the urges or behaviors that humans display. Aggression doesn't always fit into our desire to stay alive and keep others alive. So in *Beyond the Pleasure Principle,* Freud wrote about Thanatos. This was the name he gave to the "death instinct." The death instinct helped to explain why humans become aggressive, why they engage in harmful behaviors, or why they seek to recreate past traumas.

Freud believed that these two instincts exist together in every human mind, and that all behaviors stem from the balance of these two instincts. This may explain why we take anger out on others. Even if this anger or aggression is initially directed inward, Eros overpowers Thanatos. In an effort to keep *ourselves* alive, the aggression is directed outward.

What did Freud propose that we do about these instincts? Not too much. He believed that there wasn't much use in trying to quell aggression when it comes from an innate instinct that is deeply embedded in the human mind. His proposal focused more on structuring society in a way that kept communities small. In order to protect the community, strategies would be devised so aggressive behaviors could be focused outside of the community, protecting all members.

Frustration Aggression Hypothesis

The frustration-aggression hypothesis put forward by **Dollard et al. (1939)** is based on the psychodynamic explanation of catharsis. Freud believed the drive for aggression was innate, like the drive for food. He believed that the only way to reduce aggression is to engage in an activity which released it. We feel better because we have 'got it off of our chest'.

Dollard et al. (1939) proposed that if we experience frustration, this leads to aggression. The aggression is a **cathartic release** of the build-up of frustration. Dollard explains that if the individual is prevented from achieving a goal by some external factor, then this will lead to frustration which will always lead to aggression. The aggression cannot always be directed at the source of aggression, which may be abstract such as lack of money, or too powerful, as the risk of punishment is too high. Psychodynamic theory proposes we have ego defence mechanisms to protect ourselves. Two defence mechanisms that are used in the catharsis of aggression are:

- **Sublimation** – Using aggression in acceptable activities such as sport.
- **Displacement** – Directing our aggression outwards onto something or someone else.

Berkowitz (1969) proposed a revised frustration-aggression hypothesis, where he argued that frustration doesn't always lead to aggression. He stated that aggression would only occur in the presence of certain cues. For example, cues such as the presence of weapons will be more likely to trigger aggression.

Social learning Theory

The last aggression theory comes out of one of the most famous experiments of all time. The Social Learning Theory was proposed by Albert Bandura in the 1960s. Bandura is most known for his Bobo Doll experiment, which attempted to explain how children learn aggression from an early age. He proposed that aggression is a learned behavior. Children exert aggression simply because they are imitating what they have observed.

Here's how the experiment went down. Children were put in a room with an adult, some toys, and a large Bobo Doll. (A Bobo Doll is one of those inflatable clowns that bounces back up when you hit it.) Half of the children sat in the room with the adult as they gently played with the toys, displaying little aggression. The other half sat in the room with the adult as the adult aggressively hit or kicked the Bobo Doll. Some of these adults used explicit language or even hit the Bobo Doll over the head with a mallet.

The children were then separated from the adult and placed in a room alone with more exciting toys. After a few minutes, the toys were taken away in order to upset the children. All of the children were then taken into a third room, where another Bobo Doll stood.

The children who had observed the aggressive adults were more likely to exhibit aggressive behavior themselves. Children were simply imitating and modeling when they also kicked, hit, or were otherwise aggressive toward the Bobo doll. It's from this experiment that Bandura proposed that aggression is learned not only through positive or negative reinforcement, but also through indirect observational learning.

General Aggression model

The GeneralAggressionModel (GAM) is a comprehensive, integrative, framework for understanding aggression. It considers the role of social, cognitive, personality, developmental, and biological factors on aggression. Proximate processes of GAM detail how person and situation factors influence cognitions, feelings, and arousal, which in turn affect appraisal and decision

processes, which in turn influence aggressive or nonaggressive behavioral outcomes. Each cycle of the proximate processes serves as a learning trial that affects the development and accessibility of aggressive knowledge structures. Distal processes of GAM detail how biological and persistent environmental factors can influence personality through changes in knowledge structures. GAM has been applied to understand aggression in many contexts including media violence effects, domestic violence, intergroup violence, temperature effects, pain effects, and the effects of global climate change.

Lesson -11 Determinants of human aggression; social, personal and situational determinants

Determinants of Human Aggression

1. Social determinants:

Frustration: Aggression increases if a person feels blocked from achieving a goal

Direct provocations from others (physical abuse or verbal insults) serve as a powerful determinant of aggression.

Exposure to aggressive models (Observational learning, disinhibition, desensitization)

Lack of social skills (inability to communicate or negotiate)

2. Situational determinants: (Psychological)

Heightened physiological arousal due to stress, anxiety, competition, vigorous exercise or exposure to films that produce arousal.

Sexual arousal: minimal levels of aggression occur in presence of mild sexual stimulation and stronger levels with higher degree of stimulation.

Pain: physical pain may arouse aggressive drive but up to limits. Severe pain may hinder aggression.

3. Personal determinants:

Gender differences

Type A behaviour pattern - Hard driving competitiveness, Time urgency, Hostility

Hostile attribution bias

Lesson -12 Prevention, Control and management of aggression

Prevention of Aggression

The best way to reduce incidents of aggression among young people is to prevent them from occurring in the first place.

1. **Set out clear expectations**

At the start of a new relationship with young people, set out clear ground rules for their behaviour. These might include such simple regulations as "no swearing," "keep your hands and feet to yourselves" and "follow directions." You should also make it clear that those young people who do not follow your ground rules will face consequences; then make sure you follow through when the occasion arises.

2. Build rapport and be understanding

Working well with young people demands more than just maintaining order and discipline. It's also important to establish a bond with them based on trust and mutual respect. To do this, you will need to talk to them to get a sense of their thoughts, feelings and life experiences. You should also show concern and be ready to listen if they bring their own issues or problems to you.

3.Show cultural sensitivity

If you work in a setting in which young people come from different social, cultural and religious backgrounds, be sensitive to that fact. Keep in mind that some behaviours considered unusual or a cause for concern in one culture may be considered normal and acceptable in another. Avoid labelling and stereotyping.

4.Avoid negative talk

Avoid using negative words, facial expressions and tone of voice when talking directly to young people. Also, take care not to overreact emotionally to young people who present challenging behaviours.

5.Don't assume or make judgments

Don't make any assumptions about young people based on what you may have heard from others. Your assumptions could spoil new relationships before they begin. Similarly, be ready to revise earlier judgments you yourself may have made. If you act on assumptions, you may discourage more positive behaviours, or not even notice when a young person's behaviour has begun to change for the better.

6.Be encouraging

Be positive. Offer praise and thanks for positive behaviours, rather than taking them for granted. For example, compliment young people when they do their schoolwork quietly or get along with peers.

7. **Develop strategies to help young people manage their emotions**

Work with young people to help them develop the skills necessary to manage emotions that can lead to aggressive outbursts.

8. **Use positive reinforcement**

Whenever possible, praise and reward positive social behaviours. Praising positive behaviour when it occurs is one of the best ways to encourage children and youth to behave positively in the future.

Controlling Aggressive Behaviour

Excessive Aggressive out of burst is undesirable not only for the person but also for the society. So, it is necessary to prevent and control aggression by the methods/several techniques are effective in reducing aggression.

1. Punishment as a deterrent

The frequency or intensity of aggressive behavior can be reduced by mild forms of punishment like social disapproval and scolding etc. People lacking in basic social skills do not know how to

communicate effectively and hence adopt a type of aggressive self-expression. Aggression in such people can be reduced by giving training on social skill to such persons.

2. Empathy

Humor, drug treatment and clinical monitoring can be helpful for aggressive persons suffering from Psychiatric disorder. Society, Culture and aggression differs according to the cultural variation. Studies of Mead and Benedict serve powerful evidence in support of this view. Kluckhonn views that aggressiveness is a matter of depending upon cultural variation and the nature of free-floating aggression depends upon this.

3. Sublimation of aggression

The reflection of aggression is evident in art and literature. Especially in modern poets we find an angry rejection of the world due to the severe frustration that the individual meets at every stage of life.

4. Utility of aggressiveness

Aggression always carries an overtone of disapproval as violence, hostility and dominancy is condemned in every society. But they are tolerated only when they are justified.

5. By controlling the child's viewing/showing of aggressive and violent models in Television, Video Shows and Films, Pictures a lot of aggressive behavior can be controlled. If the too much aggressive behavior of the child is discouraged through social learning and socialization process, the child will learn to reduce his hostile actions. If aggressive behavior is mildly penalized it will be non- reinforcement action.

6. Loss of Love

When the child is involved in various hostile and violent activities he should be told by his parents, they would not love him or take care of him if he is engaged in hostile activities. Aggression can be reduced effectively in highly aggressive boys by allowing them to observe models who behave in a restrained and non-aggressive manner in the face of provocation. By training the child to be, disciplined and develop positive values and morality a lot of aggressive behaviour can be controlled.

7. Flexibility in Punishment during Childhood and No Rigid Training

The life history of several convicts and criminals in jails has shown that those who have been jailed for their aggressive acts most of them were severely punished during childhood and punished repeatedly.

8. Control of Parental Aggression

It is very often said that angry parents have angry children. So, parents must try to control their frequent anger, irritation and hostility. Parents must try to be an ideal model for their children.

Unit-4 Environment and Behaviour

Lesson -13 Urban environment and social behavior

Urban Environment

The population residing in urban areas in India, according to the 1901 census, was 11.4%, increasing to 28.53% by the 2001 census, and is now currently 34% in 2017 according to The World Bank. ... According to a survey by UN, in 2030 40.76% of country's population is expected to reside in urban areas.

An urban area is the region surrounding a city. Most inhabitants of urban areas have nonagricultural jobs. Urban areas are very developed, meaning there is a density of human structures such as houses, commercial buildings, roads, bridges, and railways. "Urban area" can refer to towns, cities, and suburbs.

As urban populations grow, the quality of the urban environment will play an increasingly important role in public health with respect to issues ranging from solid waste disposal, provision of safe water and sanitation, and injury prevention, to the interface between urban poverty, environment and health.

Social behavior

Social behavior can be defined as all behavior that influences, or is influenced by, other members of the same species.

Social behavior is behavior among two or more organisms within the same species, and encompasses any behavior in which one member affects the other. This means that, in regards to humans, social behavior can be determined by both the individual characteristics of the person, and the situation they are in.

Social behavior is defined as interactions among individuals, normally within the same species, that are usually beneficial to one or more of the individuals. Thus, social behavior is not only displayed by animals possessing well-developed brains and nervous systems.

A study on human behavior has revealed that 90% of the population can be classified into four basic personality types: Optimistic, Pessimistic, Trusting and Envious.

Social behavior is a set of actions performed by individuals of the same species when they interact with each other. Both humans and animals engage in social behavior, and social interaction can be both verbal and nonverbal.

Examples of human social behavior include:

- Shaking hands.
- Flirting.
- Conversation.
- Religious rituals.
- Snubbing or "putting down" another person.
- Exchanging nonverbal signals (like smiles or frowns)
- Offering reassurance or consolation.
- Sharing a meal.

In an urban environment, there does appear to be less social contact between neighbors and less helpfulness and consideration shown toward strangers, while social contact between relatives and between friends is no different from what is found in smaller-sized communities.

Lesson -14 Effect of noise, Temperature, Weather, Light and Air pollution

Effect of Noise

Noise pollution impacts millions of people on a daily basis. The most common health problem it causes is Noise Induced Hearing Loss (NIHL). Exposure to loud noise can also cause high blood

pressure, heart disease, sleep disturbances, and stress. These health problems can affect all age groups, especially children.

Long-term exposure to noise can cause a variety of health effects including annoyance, sleep disturbance, negative effects on the cardiovascular and metabolic system, as well as cognitive impairment in children. Of course, there are many more premature deaths associated with air pollution than for noise.

1. **Hearing Problems**

   Constant exposure to loud levels of noise can easily result in the damage of our eardrums and loss of hearing, causing tinnitus or deafness. It also reduces our sensitivity to sounds that our ears pick up unconsciously to regulate our body's rhythm.

2. **Psychological Issues**

   Excessive noise pollution in working areas such as offices, construction sites, bars and even in our homes can influence psychological health. Studies show that the occurrence of aggressive behavior, disturbance of sleep, constant stress, fatigue, depression, anxiety, hysteria and hypertension in humans as well as animals can be linked to excessive noise levels. The level of irritation increases with increased noise, and people tend to become less and less patient. These, in turn, can cause more severe and chronic health issues later in life.

3. **Physical Problems**

   Noise pollution can cause headaches, high blood pressure, respiratory agitation, racing pulse, and, in exposure to extremely loud, constant noise, gastritis, colitis and even heart attacks may occur.

4. **Cognitive Issues & Behavioral Changes**

   Noise affects brain responses and people's ability to focus, which can lead to low-performance levels over time. Like other sound waves, too much noise when it goes to the brain leads to lower response rates as well as making the mind dull. It is also poor for memory, making it hard to study. The studies have shown that school children living near railway stations or airports have problems in learning.

Research has shown that people, who live near airports or busy roads, usually have a higher incidence of headaches, take more sleeping pills and sedatives, are more prone to minor accidents, and are more likely to seek psychiatric treatment.

5. Sleeping Disorders

While it may not seem like much at this point, excessively high levels of noise are likely to hamper your sleeping pattern, thereby leading to irritation and uncomfortable situations.

Without a good night's sleep, you might experience multiple problems related to fatigue. This will affect your performance in the office as well as at home. It is therefore recommended to take a sound sleep to give your body proper rest.

6. Trouble Communicating

High decibel noise can put trouble and affect free communication between people. This may lead to misunderstanding, and you may get difficult understanding the other person. Constant sharp noise can give you a severe headache and disturb your emotional balance.

Effects of temperature

Higher temperatures are worsening many types of disasters, including storms, heat waves, floods, and droughts. A warmer climate creates an atmosphere that can collect, retain, and drop more water, changing weather patterns in such a way that wet areas become wetter and dry areas drier.

Temperature plays an important part in shaping weather patterns, guiding the life cycle of various organisms and maintaining ocean levels. Shifting the temperature a couple of degrees can throw an entire ecosystem into chaos. Most plants have a range of temperatures in which they will flourish.

Even small differences from seasonal average temperatures are associated with increased illness. Temperature extremes can also worsen chronic conditions, including cardiovascular, respiratory, and cerebrovascular disease and diabetes-related conditions. Heat also has important indirect health effects.

Climate change also alters the life cycles of plants and animals. For example, as temperatures get warmer, many plants are starting to grow and bloom earlier in the spring and survive longer into the fall. Some animals are waking from hibernation sooner or migrating at different times, too.

Effects of Weather

Increased heat, drought and insect outbreaks, all linked to climate change, have increased wildfires. Declining water supplies, reduced agricultural yields, health impacts in cities due to heat, and flooding and erosion in coastal areas are additional concerns.

Weather and weather forecasts affect people's health and lifestyles, the economy, society, and the environment. The degree of impact varies depending upon many factors including weather event type, event timing, event severity, event duration, event location, “unusualness”, etc.

Climate change increases the risk of illness through increasing temperature, more frequent heavy rains and runoff, and the effects of storms. Health impacts may include gastrointestinal illness like diarrhea, effects on the bodies nervous and respiratory systems, or liver and kidney damage.

Effects of Light

Light creates more than just visual effects (image, shape, intensity, perception, contrast, etc.); it also has biological and psychological effects that can impact the health and wellbeing of humans. When light biologically impacts us, it can improve or disrupt our sleep, cognition and overall wellbeing.

Most light pollution comes from outdoor lighting, advertisements and streetlights. Another major source is indoor light from large buildings like office towers.

When there is too much light, too little light, or the wrong kind of light present, it can disrupt the natural harmonies and metabolic cycles of life forms on Earth.

While worrying about whether the ground worm is confused as to whether it is night or day doesn't seem like an issue, consider that human beings have metabolic systems that are also highly light-dependent.

We are as responsive to light as a plant that turns its bloom towards the sun. In fact, Seasonal Affective Disorder (SAD) is thought to be caused by a sharp decrease in exposure to natural light.

Effects of Air pollution

Air pollution can damage crops and trees in a variety of ways. Ground-level ozone can lead to reductions in agricultural crop and commercial forest yields, reduced growth and survivability of tree seedlings, and increased plant susceptibility to disease, pests and other environmental stresses (such as harsh weather).

Causes of air pollution

- Industrial Emission. ...
- Indoor Air Pollution. ...
- Wildfires. ...
- Microbial Decaying Process. ...
- Transportation. ...
- Open Burning of Garbage Waste. ...
- Construction and Demolition.
- Burning of Fossil fuels (Fossil fuels are made from decomposing plants and animals. These fuels are found in the Earth's crust and contain carbon and hydrogen, which can be burned for energy. Coal, oil, and natural gas are examples of fossil fuels.)

Air pollution candirectly contaminate the surface of bodies of water and soil. This can kill crops or reduce their yield. It can kill young trees and other plants. Sulfur dioxide and nitrogen oxide particles in the air can create acid rain when they mix with water and oxygen in the atmosphere.

Air pollution is caused by solid and liquid particles and certain gases that are suspended in the air. These particles and gases can come from car and truck exhaust, factories, dust, pollen, mold spores, volcanoes and wildfires. The solid and liquid particles suspended in our air are called aerosols.

Lesson -15 Crowding-Long term -Effects of crowding

Crowding

A crowd is defined as follows in the various dictionaries and they all have certain common thread of definitions of crowd and crowding running through them which are stated below:

A large number of persons gathered together; a throng.

The common people; the populace.

A group of people united by a common characteristic, as age, interest, or vocation: the over-30 crowd.

A group of people attending a public function; an audience: Theplay drew a small but appreciative crowd.

Floyd Allport(1924) described crowd as "a collection individuals who are all attending and reacting to some common object, their reactions being of a simple prepotent sort and accompanied by strong emotional responses"

Long term effects of crowding

Overcrowding and its associative effects derive from:

- The high 'social density' in homes, schools, clinics, camps;
- The high 'spatial density' of the camp, in particular, congested buildings and roads, and lack of public spaces, including playgrounds and meeting places;
- The burden placed on the already insufficient infrastructure and utilities, including roads, water supply, and electricity, and public services, including health and education.

The social and psychological effects of overcrowding are immediate, long-term, cumulative, multiple, mutually reinforcing, direct as well as indirect, and critical. Overcrowding is likely to worsen.

Health effects:

Overcrowding poses serious direct and indirect health risks to all segments of the population, particularly the elderly, young children, and the disabled:

- overcrowding results in insufficient ventilation in homes, causing or exacerbating respiratory illness;
- susceptibility to disease, the severity of diseases, the spreading of illness, and the mortality due to disease all increase as a result of social and physical overcrowding;
- overcrowding exacerbates health risks related to insufficient and poor water supply and poor sanitation systems in the camps;
- likelihood of accidents in the home and community increases;
- overcrowding physically and emotionally overburdens mothers and other caregivers, increasing health risks of dependents;
- lack of space and overcrowding directly impacts on the physicial development and psychological well being of disabled residents.

Social effects:

Overcrowding contributes to far-reaching social problems:

- it places a strain on social relations within the home and community;
- overcrowding in schools and homes is linked to substandard education and functional illiteracy, and may be related to increased child labour;
- it is a 'push factor' in the decisions leading to girls' early marriage (before the age of 18) which, in turn, leads to serious health and social ramifications for women and children;
- overcrowding affects women's access to social and economic resources; it increases their responsibilities in the home and burdens their time.

Psychological effects:

Direct and indirect psychological effects result from overcrowding:

- lack of privacy is linked to depression and other negative psychological outcomes;

- overcrowding contributes to psychological frustrations which, in turn, have a bearing on behavioural responses and residents' ability to cope with the conditions;
- refugees' perception of options and future prospects is detrimentally affected by overccrowding.

Lesson -16 Theories of crowding-Crowd and Behaviour

Theories of crowding

Some of the most important theories regarding crowd behaviour are as follows: 1. Le Bon's Theory 2. McDougall's Theory 3. Freud's Theory 4. Allport's Theory 5. Turner's Theory. The study of collective behaviour started with the study of crowd behaviour. In the 19th century, crowd behaviour was a significant area of study in the field of psychology.

Many theories have been propounded by various psychologists and sociologists to explain why the crowd behaves in a particular way. We will below examine a few of them. The behaviour of the crowd is always emotionally determined. The things common to all people are such fundamental emotions as fear, anger and rage. In terms of these common elements, crowds form, feel and act.

1. Le Bon's Theory

While explaining the crowd behaviour, Le Bon developed his most important notion of 'group mind'. Group mind makes people feel, think and act in a manner quite different from that in which each individual would feel, think and act where he in a state of isolation.

The group mind is not a mere collection (or the sum) of the minds of all the individual members of a group. It is a mind of its own distinct from minds working on different levels. Its working is based on emotions, appeals, suggestions and slogans.

2. McDougall's Theory

Psychologist William McDougall's theory of the behaviour of unorganized groups or crowd is practically the same as Le Bon's. He explains the two central phenomena of crowd behaviour, namely, the intensification of emotion in a crowd and the lowering of intellectual level, as follows:

The former is due to the principle of direct induction of emotion by way of ‘primitive sympathetic response’.

He says, the greater the number of people in whom the same emotions can be simultaneously observed, the greater the contagion. The individual under the influence of emotion losses the power of criticism and slips into the same emotion.

The collective emotion becomes intensified by mutual interaction. The intensification of emotion and unpreparedness for opposing the authority of the crowd, in their turn, inhibit intellectual processes and the lowering of intellectual level in a crowd.

3. Freud’s Theory

Freud suggests that what holds any group together is a love relationship, i.e., emotional ties. This explains what he considers to be ‘the principal phenomenon of group psychology’. Using psycho-analytic approach of Freud, E. D. Martin interpreted crowd behaviour as the release for repressed drives.

Through a crowd, the restraints of a superego are relaxed and primitive ego-impulses come into play. The ‘censor’ within the individual is set aside in the crowd and the ‘instinct’ or basic ‘id’ impulses, which are normally confined to the inner depths of the personality, come to surface. The crowd thus provides a momentary release of otherwise repressed drives.

How far the Freudian theory is helpful in explaining crowd behaviour, it is argued that it is not substantiated by factual observation. Sometimes, the crowd behaviour may be the expression of repressed drives, but it may not be true of all the crowds. Moreover, it is unable to explain all the features of crowd behaviour.

4. Allport’s Theory

F. H. Allport has also criticized McDougall’s theory of sympathetic induction of emotion and behaviour. He offered the explanation of crowd behaviour by two principles, of which one is the principle of social facilitation.

According to this principle, a common stimulus prepares two individuals for the same response and when they are so prepared, the sight of one making that response releases and heightens that response in the other. The second principle is that of interstimulation.

5. Turner's Theory

Sociologist Ralf Turner has gone beyond inadequate psychological explanation of the crowd behaviour and developed an emergent norm perspective. The central thesis of this perspective is that even in the most violent and dangerous crowds, there is also social interaction, in which a situation is defined, norms for sanctioning behaviour emerge, and lines of action are justified and agreed upon.

Thus, all above explanations throw light on either one or the other factor of the complex phenomenon of crowd behaviour. As such, they are incomplete and insufficient. There are multiple factors, such as anonymity, stimulation, emotionality, suggestibility, initiation, contagion, lack of volition, force of unconscious impulses, etc., which are responsible for the emergence of the typical behaviour of the crowd.

The theory of crowd behaviour has moved away from the older perspective that viewed the individual as coming under the sway of the crowd and losing its capacity for rational judgement before the sweep of an overpowering emotional contagion. Instead, sociologists now explain crowd behaviour by the same sociological concepts that explain social group behaviour.

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: To study the process of reasoning through analysis of subjects behavior while solving similar problems of increasing difficulty.

Activity 2: To determine the relative difficulty of memorizing words and nonsense syllables

PSY 25143

CLINICAL PSYCHOLOGY

Objectives:

To learn about abnormalities in clinical aspects, how to measure psychological problems. Diagnostic the problem and use the counseling techniques and psychological intervention methods.

Unit-1Foundations of Clinical Psychology

Lesson -1 Definition and scope of Clinical Psychology
Lesson -2 History and approaches of Clinical Psychology
Lesson -3 Current status, Careers and Professional issues
Lesson -4 Methods of Clinical Psychology

Unit-2Models of Clinical psychology

Lesson -5 Theories and Models of Clinical psychology
Lesson-6 Psycho-dynamic and cognitive model
Lesson-7 Behaviorist, humanist and gestalt-based and systematic

Unit-3Major clinical disorders

Lesson -8Clinical child Psychology: Clinical issues, Behavioural Disorders
Lesson -9Emotional Disorder, Developmental and Learning Disorder
Lesson-10Clinical Neuro-Psychology:- Symptômes and Syndromes; Delirium, Dementia, Amnesic syndrome
Lesson-11 Neuro-PsychologicalDelusionalsyndrome, Neuro-psychological syndrome, Neuro-Psychologicaldisorderwith HIV – 1 infections

Unit-4Clinical Assessment and Psycho diagnostics

Lesson -12 Signs and symptoms of mental disorders
Lesson -13 Intellectual Assessment; Personality Assessment; Neuropsychological Assessment
Lesson -14 Clinical interviewing and testing- Psycho-diagnostics- intake/ an amnesis – developing hypotheses
Lesson-15 Selecting criteria and materials (tests, questionnaires, observations etc)

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Outcomes:

- To learn about abnormalities in clinical aspects,
- To measure psychological problems.
- Diagnostic the problem and use the counseling techniques
- To use psychological intervention methods

PSY 25143

CLINICAL PSYCHOLOGY

Unit-1Foundations of Clinical Psychology

Lesson -1 Definition and scope of Clinical Psychology

What is Clinical Psychology?

Clinical Psychology is the branch of Psychology. It concerned with the assessment and treatment of mental illness, emotional disorders and behavioural problems.Clinical Psychology promoting human adaptation, adjustment, personal effectiveness and satisfaction. Clinical Psychology focuses on the intellectual, emotional, biological, psychological, social and behavioural aspects of human functioning across the life span, in varying cultures, and at all socioeconomic levels. Clinical Psychology is a science that generates research efforts to discover and validate information about what people are like and why they behave as they do.

Historical Overview of Clinical Psychology

"The official birth of clinical psychology can be traced to 1896 when "Lighter Witmer" opened the first Psychological Clinic in Philadelphia Pennsylvania."

Clinical Psychology has it's roots in 3 sets of historical factors:
1. Use of scientific research methods by Psychologists.
2. Study of human individual differences.
3. The ways in which Behavioural Disorders have been viewed and treated over the years.

Definition of Clinical Psychology

"Clinical psychology involves research, teaching and services relevant to the application of principles, methods and procedures for understanding, predicting and alleviating intellectual, emotional, biological, psychological, social and behavioural maladjustment, disability and discomfort, applied to a wide range of client populations." (APA 1991)

Characteristics of Clinical Psychology:

Emphasis on science
Emphasis on maladjustment
Emphasis on the individual
Emphasis on helping

Scope of Clinical Psychology

Early definitions stressed assessments, evaluation and diagnosis More recent inclusion of intervention in various forms as well as prevention Evidence based practice Active debate on the 'science of clinical psychology

Scope of Clinical Psychology

Early definitions stressed assessments, evaluation and diagnosis
More recent inclusion of intervention in various forms as well as prevention
Evidence based practice
Active debate on the 'science of clinical psychology
Activities of Clinical Psychologist: 1. Intervention And Therapy
Assessment and Diagnosis :

Assessment has long been a critical part of the clinical psychologist's role. Assessment whether through the observation, testing or interview, is a way of gathering information so that an important question can be solved.
Assessment of an individual's development, behavior, intellect, interests, personality, cognitive processes, emotional functioning and social functioning are performed by clinical psychologists as are assessment activities directed towards families, couples and groups.

The process of assessment is very important as it leads to the diagnosis of the clients problems.

• Interpretation of assessment results, and integration of these results with other information available, in a way that is sensitive to the client, is an essential skill of clinical psychologist.

3- Teaching

Clinical Psychologist who have full or part-time academic appointments obviously devote a considerable amount of time to teaching.

Those whose responsibilities are primarily in the area of graduate education, teach course in advance psychopathology, psychological testing, interviewing, intervention, personality theory and so on.

Some also teach abnormal psychology, introduction to clinical psychology.

4- Clinical Supervision

This activity is another form of teaching. However, it typically involves more one-to-one teaching, small group approaches and other less formal, non class room varieties of instruction.

Clinical psychologists often spend significant portions of their time supervising students, interns and others.

In short one learns by doing but under the controlled and secure conditions of a trainee - supervisor relationship.

5- Research

Clinical psychology has grown out of an academic research tradition. As a result, when clinical training programs were first established after world war 2, the scientist - practitioner model was adopted.This meant that is contrast to other mental health workers such as psychiatrist or social worker, all clinicians were to be trained both as scientist and practitioner.Although this research emphasis may not be so prominent in some training programs as it once was, the fact remains that clinical psychologists are in a unique position both to evaluate research conducted by others and to conduct their own research.

6- Consultation

Consultation, regardless of the setting in which it occurs, or the particular purpose it has, is a significant activity of many clinical psychologists. A growing number of clinical psychologists serve as consultants. In consultation, the goal is to increase the effectiveness of those to whom one's efforts are directed by imparting to them some degree of expertise. It takes innumerable forms, in many different settings. Consultation can run the gamut from clinical cases to matter of business, personal and profit. It can deal with individuals or entire organizations. Sometimes it is remedial, other times it is oriented toward prevention.

Lesson -2 History and approaches of Clinical Psychology

Clinical psychology is one of the branches of applied psychology, which combines the psychological theories and scientific discoveries, in order to understand the nature of the human body and the symptoms of mental disorders that affect humans and try to alleviate the symptoms and overcome psychological disorders.
People differ in their understanding and diversity of interests and behaviors. They have their influence on different topics of psychology.

There may be diversity between physical inefficiencies and mental illnesses, and between social psychology and clinical psychology, and within educational, military and industrial psychologies. All this is mixed up with the study of personality traits of intelligence, creativity, and genius.

For a long time, we have seen the attraction of individuals to the subjects of psychology because they relate to the individual's understanding of himself and his social environment.

The more complex the society, the more we need to understand ourselves correctly in a broader and deeper form.

Clinical psychology is needed by individuals in their family relations, and social workers as well as parents in many sectors of society, such as schools, and hospitals as well as mental health institutions.
For the interpretation of diversity and personality traits, the concept of clinical psychology emerged in the world.

Clinical psychology is defined as a science that integrates theories, scientific method, and clinical knowledge.

A Brief History of Clinical Psychology

Clinical psychology refers to an evolutionary stage of rapid psychological development.

Although modern psychology emerged in the late nineteenth century in the continent of Europe, psychotherapy had occupied a place in different cultures.

Despite the use of psychiatric and mental health clinics and hospitals for methods considered primitive compared to the contemporary period, but the academic community was rejecting these methods until Freud introduced a new method of psychotherapy.

Speech therapy was the first practical application of clinical psychology in the world.

The method of brain science was used at the beginning of the 18th century. This science enables the doctor to examine the head through the shape of his external skull.

The first institute specializing in the treatment of mental illness was established in 1879 by the German scientist Wilhelm Griesinger.
The work of the institute focused on psychological research, the pursuit of treatment methods and follow-up of existing cases.

In the United States of America, the first psychotherapy clinic was opened at the University of Pennsylvania that was established in 1896, specializing in the treatment of speech and learning difficulties in children.

The term "clinical psychology" was first used systematically by Witmer-1896 to refer to assessment and diagnostic procedures for mentally retarded and physically disabled children.
This is the first attempt to define the concept of clinical psychology.

Ten years later, the term "clinical psychology" was introduced and defined as a science that deals with studying individuals by observing and experimenting to make a positive change in their lives.

In 1935, the American Psychological Association (APA) defined clinical psychology as an applied branch of psychology that aims to determine the characteristics of an individual's behavior and abilities by using methods of measurement, analysis, and observation.

It also provides tests and recommendations for the purpose of the consensus of the individual after the medical examination, through the use of scientific methods to help people suffering from psychological disorders.

In 1975, the American Psychological Association established the basic rules for the training of specialists in the community psychotherapy.

Medicine and Neurology:
A clinical psychologist is a physician and neurologist who is entrusted with the task of diagnosis and treatment.
No one should consider himself a clinical psychologist unless he has a medical background.

Measurement:
Clinical psychology is concerned with identifying the abnormal conditions of some people, which are the psychological, mental and personal tests and measurements that are the basic tools in the diagnosis.
Without them, the diagnosis will be unsuccessful and its success will not be expected to achieve the desired goal.

Anomalous:
The work of a clinical psychologist is determined by the study of homosexuals and individuals who are less intelligent than ordinary people.

Behavior:
Clinical psychology studies the behavior of the individual and includes diagnosing the psychological state of the individual or his mind through observation and experimentation to achieve an appropriate degree of adaptation to events, attitudes.

Compatibility:
Clinical science links the patient to reality and addresses the problems of psychological compatibility of the individual and his compatibility with his environment in order to achieve his demands.

Clinical Psychologist Job Description
Clinical psychologists are specialized clinicians and licensed professionals who are qualified and specialized in clinical psychology to provide direct services to patients.
Psychologists are experts in or psychological measurement or psychometrics and formal tests.

Psychologists specializing in clinical psychology have a great deal of experience in more than one medical fields as well as clinical psychology.
They are often interested in in-depth scientific research, teaching at universities, counseling in their private clinics or public hospitals.

Clinical psychologists work in a number of settings, including outpatient centers, hospitals, government agencies, private practice offices, and military medical services.

Police also use clinical psychology in investigative and brainwashing requirements, forensic and criminal certificates.

Clinical psychology is the backbone of the mental sciences, and the medical professions concerned with mental health, which are governed by a number of international rules, laws, and standards.

Clinical psychology aims to understand the nervous stresses of man, anxiety, amnesia, Alzheimer's, depression, etc., as a result of the increased difficulty of life.

What Methods Do Clinical Psychologists Use in an Assessment?

Understanding Psychological Testing and Assessment

1 - IQ tests: These tests also interested in measuring the proportion of educational achievement in addition to tests that measure the intelligence of the person.

These evaluation tests measure some of the skills of the child and the proportion of general knowledge and verbal skills and memory and degree of attention and logical thinking and perception of spatial and visual.

2 - Personality tests: The main objective of these tests is to reveal the psychological state of the person and the processes that pass through in addition to describing the patterns of behavior and ideas and feelings of the individual.

These tests often rely on comparisons or on the mysterious stimuli that raise the soul to come out of the ideas and theses.

3 - Neuropsychological tests: This assessment is further examined the neurological condition and if it suffers from problems affecting the health and ability to deal with reality properly, whether this has a psychological impact on the situation or not?

4 - Clinical observation: Clinical observation is carried out in this field by observing the behavior of the individual, observing his general behavior, mood, and his ability to perceive and deal properly with the reality around him.

Problems of children and adolescents

Family problems and family relations counseling

Neuropsychiatric disorders

Treatment of disorders such as (trauma - addiction - eating disorders - sleep - sex - depression - anxiety - phobia)

Clinical psychology is also used in companies, schools, institutions, sports, and forensic backgrounds.

Approaches

Clinical psychologists who work as psychotherapists often utilize different treatment approaches when working with clients. While some clinicians focus on a very specific treatment outlook, many use what is referred to as an "eclectic approach." This involves drawing on different theoretical methods to develop the best treatment plan for each individual client.

Some of the major theoretical perspectives within clinical psychology include:

- **Psychodynamic approach:** This perspective grew out of Freud's work; he believed that the unconscious mind plays an important role in our behavior. Psychologists who utilize psychoanalytic therapy may use techniques such as free association to investigate a client's underlying, unconscious motivations.[3]
- **Cognitive behavioral perspective:** This approach to clinical psychology developed from the behavioral and cognitive schools of thought. Clinical psychologists using this perspective will look at how a client's feelings, behaviors, and thoughts interact. Cognitive-behavioral therapy (CBT) often focuses on changing thoughts and behaviors that contribute to psychological distress.[4]
- **Humanistic perspective:** This approach to clinical psychology grew out of the work of humanist thinkers such as Abraham Maslow and Carl Rogers. This perspective looks at the client more holistically and is focused on such things as self-actualization.[5]

Lesson -3 Current status, Careers and Professional issues

Status of Clinical Psychology in India - The International Journal of Indian Psychology | 190 India regulations. There are very much concern while issuing certificates, in any kind and dealing with medical - legal cases. There psychological report is based upon clinical finding, mental status examination and objective assessment equipments. The clinical psychology has travel a long distance since Dr. GS, Bose a medical practitioner in this field established him and considered as first Calcutta based clinical psychologist in India. He started a course in psychoanalysis for Post Graduated Course in psychology at Calcutta University during twenty centaury. This period was important for professional growth and development in clinical psychology. It is widely recognized in the European country especially in Germany, United State of America and United Kingdom. Government of India established the All India Institute of Mental Health at Bangalore currently famous as National Institute of Mental Health and Allied Neuro – Sciences where a post graduate diploma in medical psychology was started as a full time course with supervised internship training in clinical psychology. Subsequently, various nomenclatures like DMSP, MM&SP and currently M.Phil in clinical psychology are being used for full time training courses in clinical psychology. Central Institute of Psychiatry, Ranchi in (1962) and after that Ranchi Institute of Neuro - Psychiatry and Allied Sciences, Ranchi in (2000) introduced similar courses. Recently more than two dozen centre have started training programmes in clinical psychology in government as well as private set - up recognized by Rehabilitation Council of India, New Delhi and affiliated to Indian universities recognized by University Grant Commission. Some of them are working under centre of excellence by ministry of health and other university department and apex institute under ministry of social justice and empowerment. The services of clinical psychologists are required in psychiatry department of medical colleges, mental hospital and more or less for all disability rehabilitation centers but their services are essentially needed for persons with intellectual disability and mental illness. The areas of responsibility as a clinical psychologist includes child mental health, learning disability, emotional problems, positive health, substance abusers and adult mental health etc. to care and provide services to the needy persons. Clinical psychologists working in rehabilitation centre are an integral part of the medical board member to assess persons with intellectual disability and provide their percentages of disability based on IQ testing and disability assessment scales. Despite of that almost two generations of trained clinical psychologist have spent their life in their professional set – up, moreover still have been lacking behind our professional position at par with other medical colleagues.

Education Requirements

In the United States, clinical psychologists usually have a doctorate in psychology and receive training in clinical settings. The educational requirements to work in clinical psychology are quite rigorous, and most clinical psychologists spend between four to six years in graduate school after earning a bachelor's degree.[6]

Generally speaking, Ph.D. programs are centered on research, while Psy.D. programs are practice-oriented. Students may also find some graduate programs that offer a terminal master's degree in clinical psychology.

Before choosing a clinical psychology program, you should always check to be sure that the program is accredited by the APA. After completing an accredited graduate training program, prospective clinical psychologists must also complete a period of supervised training and an examination.

Specific licensure requirements vary by state, so you should check with your state's licensing board to learn more.

Students in the United Kingdom can pursue a doctorate level degree in clinical psychology (D.Clin.Psychol. or Clin.Psy.D.) through programs sponsored by the National Health Service.[7] These programs are generally very competitive and are focused on both research and practice. Students interested in enrolling in one of these programs must have an undergraduate degree in a psychology program approved by the British Psychological Society in addition to experience requirements.

Opportunities

Clinical psychologists work in a variety of settings (hospitals, clinics, private practice, universities, schools, etc.) and in many capacities. All of them require these professionals to draw on their expertise in special ways and for different purposes.

Some of the job roles performed by those working in clinical psychology can include:

- Assessment and diagnosis of psychological disorders, such as in a medical setting
- Treatment of psychological disorders, including drug and alcohol addiction
- Offering testimony in legal settings
- Teaching, often at the university level
- Conducting research
- Creating and administering programs to treat and prevent social problems

Some clinical psychologists may focus on one of these or provide several of these services. For example, someone may work directly with clients who are admitted to a hospital for psychological disorders, while also running a private therapeutic office that offers short-term and long-term outpatient services to those who need help coping with psychological distress.

Ethical obligations: To minimize harm to participants, steer clear of plagiarism, and avoid fabrication of data, among other things (Fisher & Vacanti-Shova, 2012).

Participants who don't receive treatment: Are placed in one of three conditions (Bjornsson, 2011; Lohr & Adams, 2015)—

1. No treatment (often called a "wait-list control" group).
2. A placebo treatment (some kind of interpersonal interaction with a professional but with presumably therapeutic techniques deliberately omitted).
3. An alternate treatment (the efficacy of which may be unknown).

Challenge: To determine empirically the efficacy of their therapies without unduly exploiting or failing to help some of the participants while doing so (Saks et al., 2002).

Clinical psychologists may find themselves in a tug-of-war between the managed-care companies' profits and their clients' psychological well-being (Alarcón, 2000; Wilcoxon, Remley, Gladding, & Huber, 2007). Moreover, to the extent that clients can perceive that the clinical psychologist has divided loyalties, the therapeutic relationship may suffer (Haas & Cummings, 1991).

Informing the client: The clinical psychologist could tell the client about loyalties to managed-care companies, and much more about managed care, during the informed consent process. One such piece of information is that managed-care plans typically require that a *Diagnostic and Statistical Manual of Mental Disorders* (*DSM*) diagnosis be assigned to a client to qualify for payment (Ackley, 1997; Chambliss, 2000). However, it is difficult to ascertain the extent to which clinical psychologists should discuss this and other facts of managed-care therapy with clients at the outset of therapy.

Ethical dilemma related to *DSM*: Clients must be diagnosed with a *Diagnostic and Statistical Manual of Mental Disorders* (*DSM)* disorder for their managed-care companies to pay for treatment can present another ethical dilemma for clinical psychologists. If a client is struggling with a problem that does not meet the criteria for any *DSM* disorder, or if a family or couple has problems that don't stem directly from an identifiable disorder in one person, the psychologist can face pressure to falsely assign a diagnosis to ensure that the managed-care company will pay the bill (Kielbasa, Pomerantz, Krohn, & Sullivan, 2004; Pomerantz & Segrist, 2006; Wilcoxon et al., 2007). Or the psychologist may be tempted to "upcode," or assign a more serious diagnosis than the client's symptoms actually merit, to increase the number of sessions or amount of money the managed-care company will devote to treatment (C. B. Fisher, 2017; Parry, Furber, & Allison, 2009).

Online assessment raises many more ethical questions (e.g., Knapp & VandeCreek, 2006; Naglieri et al., 2004):

1. As the client is taking the test, are the testing conditions standardized?
2. Will the client keep the test materials secure?
3. Is the client distracted by other tasks?
4. Is the client actually the person completing the test?

5. Are there important behavioral observations the psychologist might be missing because of the remote nature of the assessment?

Online therapy practices: When therapy is done via computer, the clinical psychologist and client may not be able to fully appreciate all aspects of communication (e.g., nonverbals). Moreover, online therapy gives rise to concerns about confidentiality and client identity that don't exist when the clinical psychologist works with the client in person (Fisher & Fried, 2003; Kraus, 2004).

Lesson -4 Methods of Clinical Psychology

Research Methods in Clinical Psychology

Consider the following research methods to create treatment plans more successfully if your line of work involves psychological treatment and therapy

Surveys

To investigate the frequency and root causes of mental health issues, psychologists frequently employ questionnaires. When selecting and surveying sample groups of patients, researchers frequently employ random selection procedures. Survey research might be quantitative or qualitative. To choose a sample that is typical of the population and to generate more trustworthy data for analysis, use random selection. You may adopt treatment solutions by understanding human behavior with the use of survey data.

Case Studies

Case studies are in-depth analyses of a particular individual, group, circumstance, or community. In order to comprehend a patient's behavior and mental health better, psychologists might do a case study research on them. Case study research enables psychologists to interview patients and closely examine their behavior over a predetermined time period, which can help psychologists treat patients more successfully. Case studies may be used by experts in other fields, such as finance, sales, and advertising, to better understand consumer investment and purchasing patterns and to pinpoint the financial services or advertising strategies that will best attract potential clients.

Experimental Studies

Psychologists can use experimental investigations to ascertain if there is a causal link between two variables, such as a patient's behavior and mental health. To determine if a change in one independent variable has an impact on a dependent variable, psychologists may conduct experimental investigations. The psychologist can then gauge how changing one-factor impacts people's performance or behavior. To identify the outside factors influencing a community's patterns of behavior, psychologists could conduct experimental investigations. To better understand collective human behavior as it relates to numerous causes and consequences, these studies can also be helpful in other areas, such as sales psychology.

Content Analysis

To find patterns, meanings, and themes in a patient's written communication styles, content analysis entails examining text-based data. Clinical psychologists are capable of doing content analyses on any type of writing, including handwritten letters, business emails, and whole transcripts of patient interviews or monologues. By carefully examining the patient's word choice

and speech patterns, this study approach can assist you in creating the most effective psychological therapies. Furthermore, content analysis is a useful technique for many forensic psychologists who are trying to solve criminal cases, behavioral difficulties, and serious mental health problems.

Meta-Analysis

A quantitative research technique called meta-analysis examines the outcomes of previously published studies that addressed the same subject and compiles different study findings into a comprehensive set of data. By scanning library databases for all studies that have been published on a certain subject, psychologists may do a meta-analysis and compile the findings into a comprehensive overview. Psychologists can produce an easily available resource that offers a collective appraisal of previous and present research by combining several linked research questions into a single research paper. As a result, meta-analysis can result in suggestions for future studies that will assist the field of psychology to improve.

Correlational Research

A non-experimental research technique called correlational research analyses two variables and evaluates their connection (correlation). In this research strategy, two constant variables—for example, the results of two distinct personality tests—are observed, and their relationship is examined. Given that both tests analyze personality type, the results of the first personality test may correlate with the results of the second.

While correlational research may be used to find connections between two qualities, actions, or events, it does not take into consideration any causal connections between the variables you are studying. As a result, many academics may combine causal research with correlational studies to get more accurate results from their study.

Quasi – Experiment

Similar to a typical experiment, a quasi-experiment does not randomly place individuals in experimental groups. This lends support to studies where psychologists may be more interested in an independent variable that cannot be a random variable, such as a participant's intrinsic quality. For instance, since identified mental health qualities are not random and make it impossible to randomly assign people to research groups, you may utilize a quasi-experiment to examine participants who exhibit these traits.

Naturalistic Observation

Psychologists use the research technique known as naturalistic observation to examine patients' behavior while they are in their natural contexts. Naturalistic observation is a useful technique for seeing how individuals interact and act in familiar situations, such as when they are doing home chores, riding public transit, or shopping in a store. The population-based conclusions that may be drawn from this study technique can help guide future studies that assist innovation in psychology, healthcare, and even technology, sales, and business.



Structured Observation

Structured observation is an observational research technique that assesses behavior in a more controlled context, such as monitoring participants in voluntary psychological tests taking place in a lab. Psychologists may keep an eye on people in controlled environments to better understand particular behaviors. Structured observation techniques are widely used in clinical and medical research investigations to better control experimental variables and guarantee more accurate data analysis. In order to better understand how a new behavioral therapy or medicine impacts patients' behaviors, psychologists may, for example, conduct a structured observation. They may then alter or enhance their treatment strategies in light of the findings.

UNIT-2 MODELS OF CLINICAL PSYCHOLOGY

Lesson -5 Theories and Models of Clinical psychology

Theories of Clinical Psychology

- Psychodynamic
- Behavioral
- Cognitive
- Biological
- Humanistic

One of the most common misconceptions is that clinical psychology is one-size-fits-all. The phrase clinical psychology automatically calls to mind an image of lying stretched out on a couch, pouring out your heart and soul. A stern-faced shrink furiously scribbles down notes until he suddenly stops you and reveals some deeply buried psychic issue or other. This scenario is informed by only one of these five clinical psychology theories and incorrectly informed at that.

1. Psychodynamic

As the oldest clinical psychology theory, the psychodynamic theory bears the strongest similarity to the average preconception of clinical psychology. Founded by Sigmund Freud in the late 1800s, the psychodynamic theory was originally called psychoanalysis. This is because the core tenant of this approach is an analysis of how one's past experiences influence one's current behaviors and emotions. The American Psychoanalytic Association explains that psychoanalysts of the past and

psychodynamic therapists of today agree on this: one cannot fully heal and function with unprocessed underlying traumas. Psychodynamic therapy is typically:

- long-term
- frequent
- grounded mostly in verbal communication.

2. Behavioral

Behavioral psychology is perhaps the most distinct form psychodynamic theory. This branch of behavioral psychology was born in the early 1900s out of Pavlov, Watson, and Skinner's work with animal behaviors. Remember Pavlov's dogs drooling in anticipation of food at the sound of a bell? Behavioral psychologists believe that humans learn to exhibit certain behaviors as a result of mental associations. They are little concerned with patients' pasts and spend little time on talk therapy. Instead, they perform a series of therapeutic tasks. They prescribe tasks for patients to complete at home. These tasks include exposure to stimuli and aversion practices. One task might be initiating punishments every time patients exhibit problematic behaviors. Behavioral therapy tends to be very brief.

3. Cognitive

Cognitive psychology is a field that involves homework. It focuses on the present, as well as the future, rather than the past. Cognitive therapy was founded in the 1950s by Aaron Beck. It focuses on how thoughts guide actions. Cognitive therapists believe that changing one's problematic thought patterns results remedying difficult emotions and troubling behaviors. Some of the most common tactics include thought journals. Another is a reward and punishment system similar to those in behavioral therapy. In fact, many modern psychologists combine cognitive and behavioral theories into one clinical psychology approach.

4. Biological

At many points during the evolution of clinical psychology, various theorists have reminded the field to remember the biological aspect of the human psychological experience. The structure and chemistry of the brain play important roles in human:

- thought
- emotion
- behavior

This biological theory of clinical psychology is known in the field as biopsychology. The Association for Psychological Science offers excellent articles about the most current research in biopsychology.

5. Humanistic

Humanistic theory offers exactly what it sounds like it should. It's a perspective on psychology that focuses on the things that make people distinctly human. Central to this theory is the concept of empowerment through free will. This method uses talk therapy and written and spoken narrative. Humanistic clinicians work with patients to take control of their emotional and behavioral responses to their human experiences. This theory relies heavily on validation and empathy. This is one of the newest theories of clinical psychology.

Related Resource: 10 Best Deals: Master's in Clinical Psychology Online

Each of these clinical psychology theories is like one piece of a whole puzzle. Most therapists specialize in one theory but recognize when and how to pull from other theories or refer patients to a different therapist. The right theory, or combination of theories, is determined by the individual and the issues under treatment. What they all have in common is the goal of helping people live their best lives.

Lesson-6 Psycho-dynamic and cognitive model

Psychodynamic theory (sometimes called *psychoanalytic theory*) explains personality in terms of unconscious psychological processes (for example, wishes and fears of which we're not fully aware), and contends that childhood experiences are crucial in shaping adult personality. Psychodynamic theory is most closely associated with the work of Sigmund Freud, and with psychoanalysis, a type of psychotherapy that attempts to explore the patient's unconscious thoughts and emotions so that the person is better able to understand him- or herself.

Freud's work has been extremely influential, its impact extending far beyond psychology (several years ago *Time* magazine selected Freud as one of the most important thinkers of the 20th century). Freud's work has been not only influential, but quite controversial as well. As you might imagine, when Freud suggested in 1900 that much of our behavior is determined by psychological forces of which we're largely unaware—that we literally don't know what's going on in our own minds—people were (to put it mildly) displeased (Freud, 1900/1953a). When he suggested in 1905 that we humans have strong sexual feelings from a very early age, and that some of these sexual feelings are directed toward our parents, people were more than displeased—they were outraged (Freud, 1905/1953b). Few theories in psychology have evoked such strong reactions from other professionals and members of the public.

Controversy notwithstanding, no competent psychologist, or student of psychology, can ignore psychodynamic theory. It is simply too important for psychological science and practice, and continues to play an important role in a wide variety of disciplines within and outside psychology (for example, developmental psychology, social psychology, sociology, and neuroscience; see Bornstein, 2005, 2006; Solms & Turnbull, 2011). This module reviews the psychodynamic perspective on personality. We begin with a brief discussion of the core assumptions of psychodynamic theory, followed by an overview of the evolution of the theory from Freud's time to today. We then discuss the place of psychodynamic theory within contemporary psychology, and look toward the future as well.

Core Assumptions of the Psychodynamic Perspective

The core assumptions of psychodynamic theory are surprisingly simple. Moreover, these assumptions are unique to the psychodynamic framework: No other theories of personality accept these three ideas in their purest form.

Assumption 1:Primacy of the Unconscious

Psychodynamic theorists contend that the majority of psychological processes take place outside conscious awareness. In psychoanalytic terms, the activities of the mind (or *psyche*) are presumed to be largely unconscious. Research confirms this basic premise of psychoanalysis: Many of our mental activities—memories, motives, feelings, and the like—are largely inaccessible to consciousness (Bargh & Morsella, 2008; Bornstein, 2010; Wilson, 2009).

Assumption 2: Critical Importance of Early Experiences

Psychodynamic theory is not alone in positing that early childhood events play a role in shaping personality, but the theory is unique in the degree to which it emphasizes these events as determinants of personality development and dynamics. According to the psychodynamic model, early experiences—including those occurring during the first weeks or months of life—set in motion personality processes that affect us years, even decades, later (Blatt & Levy, 2003; McWilliams, 2009). This is especially true of experiences that are outside the normal range (for example, losing a parent or sibling at a very early age).

Assumption 3: Psychic Causality

Our every thought and behavior —even something as seemingly random as which seat you choose on the bus —results from biological or psychological influences. [Image: ryuu ji 竜次, https://goo.gl/NrofGI, CC BY 2.0, https://goo.gl/BRvSA7]

The third core assumption of psychodynamic theory is that nothing in mental life happens by chance—that there is no such thing as a random thought, feeling, motive, or behavior. This has come to be known as the principle of **psychic causality**, and though few psychologists accept the principle of psychic causality precisely as psychoanalysts conceive it, most theorists and researchers agree that thoughts, motives, emotional responses, and expressed behaviors do not arise randomly, but always stem from some combination of identifiable biological and psychological processes (Elliott, 2002; Robinson & Gordon, 2011).

The Evolution of Psychodynamic Theory

Given Freud's background in neurology, it is not surprising that the first incarnation of psychoanalytic theory was primarily biological: Freud set out to explain psychological phenomena in terms that could be linked to neurological functioning as it was understood in his day. Because Freud's work in this area evolved over more than 50 years (he began in 1885, and continued until he died in 1939), there were numerous revisions along the way. Thus, it is most accurate to think of psychodynamic theory as a set of interrelated models that complement and build upon each other. Three are particularly important: the topographic model, the psychosexual stage model, and the structural model.

The Topographic Model

In his 1900 book, *The Interpretation of Dreams*, Freud introduced his **topographic model** of the mind, which contended that the mind could be divided into three regions: conscious, preconscious, and unconscious. The conscious part of the mind holds information that you're focusing on at this moment—what you're thinking and feeling right now. The preconscious contains material that is capable of becoming conscious but is not conscious at the moment because your attention is not being directed toward it. You can move material from the preconscious into consciousness simply by focusing your attention on it. Consider, for example, what you had for dinner last night. A moment ago that information was preconscious; now it's conscious, because you "pulled it up" into consciousness. (Not to worry, in a few moments it will be preconscious again, and you can move on to more important things.)

Dreams play an important role in psychodynamic theory, as they are often considered the central route through which the unconscious expresses itself to the conscious mind. [Image: Danmo, CC0 Public Domain, https://goo.gl/m25gce]

The unconscious—the most controversial part of the topographic model—contains anxiety-producing material (for example, sexual impulses, aggressive urges) that are deliberately *repressed* (held outside of conscious awareness as a form of self-protection because they make you uncomfortable). The terms *conscious*, *preconscious*, and *unconscious* continue to be used today in psychology, and research has provided considerable support for Freud's thinking regarding conscious and preconscious processing (Erdelyi, 1985, 2004). The existence of the unconscious remains controversial, with some researchers arguing that evidence for it is compelling and others contending that "unconscious" processing can be accounted for without positing the existence of a Freudian repository of repressed wishes and troubling urges and impulses (Eagle, 2011; Luborsky & Barrett, 2006).

The Psychosexual Stage Model

Freud remained devoted to the topographic model, but by 1905 he had outlined the key elements of his **psychosexual stage model**, which argued that early in life we progress through a sequence of developmental stages, each with its own unique challenge and its own mode of sexual gratification. Freud's psychosexual stages—oral, anal, Oedipal, latency, and genital—are well-known even to non-analytic psychologists. Frustration or overgratification during a particular stage was hypothesized to result in "fixation" at that stage, and to the development of an oral, anal, or Oedipal personality style (Bornstein, 2005, 2006).

Table 1 illustrates the basic organization of Freud's (1905/1953b) psychosexual stage model, and the three personality styles that result. Note that—consistent with the developmental challenges that the child confronts during each stage—oral fixation is hypothesized to result in a dependent personality, whereas anal fixation results in a lifelong preoccupation with control. Oedipal fixation leads to an aggressive, competitive personality orientation.

The Structural Model

Ultimately, Freud recognized that the topographic model was helpful in understanding how people process and store information, but not all that useful in explaining other important psychological phenomena (for example, why certain people develop psychological disorders and others do not). To extend his theory, Freud developed a complementary framework to account for normal and abnormal personality development—the **structural model**—which posits the existence of three interacting mental structures called the id, ego, and superego. The *id* is the seat of drives and instincts, whereas the *ego* represents the logical, reality-oriented part of the mind, and the *superego* is basically your conscience—the moral guidelines, rules, and prohibitions that guide your behavior. (You acquire these through your family and through the culture in which you were raised.)

According to the structural model, our personality reflects the interplay of these three psychic structures, which differ across individuals in relative power and influence. When the id predominates and instincts rule, the result is an impulsive personality style. When the superego is strongest, moral prohibitions reign supreme, and a restrained, overcontrolled personality ensues. When the ego is dominant, a more balanced set of personality traits develop (Eagle, 2011; McWilliams, 2009).

The Ego and Its Defenses

In addition to being the logical, rational, reality-oriented part of the mind, the ego serves another important function: It helps us manage anxiety through the use of **ego defenses**. Ego defenses are basically mental strategies that we use automatically and unconsciously when we feel threatened (Cramer, 2000, 2006). They help us navigate upsetting events, but there's a cost as well: All ego defenses involve some distortion of reality. For example, repression (the most basic ego defense, according to Freud) involves removing from consciousness upsetting thoughts and feelings, and moving those thoughts and feelings to the unconscious. When you read about a person who "blocked out" upsetting memories of child abuse, that's an example of repression.

Another ego defense is denial. In denial (unlike repression), we are aware that a particular event occurred, but we don't allow ourselves to see the implications of that event. When you hear a person with a substance abuse problem say "I'm fine—even though people complain about my drinking I never miss a day of work," that person is using denial. Table 2 lists some common ego defenses in psychodynamic theory, along with a definition and example of each.

Psychodynamic Theories: Where Are We Now?

The topographic model, psychosexual stage model, and structural model continue to influence contemporary psychology, but it is important to keep in mind that psychodynamic theory is never static, ever changing and evolving in response to new ideas and findings. In the following sections we discussion four current trends in the psychodynamic perspective: object relations theory, the empirical testing of psychodynamic concepts, psychoanalysis and culture, and the opportunities and challenges of neuroscience.

Object Relations Theory and the Growth of the Psychodynamic Perspective

Object relations theory holds that the impressions we develop of our parents and how they behave early in our lives serve a scripts that guide our behavior in future relationships. [Image: geralt, CC0 Public Domain, https://goo.gl/m25gce]

In recent years a number of new psychodynamic frameworks have emerged to explain personality development and dynamics. The most important of these is **object relations theory**. (In psychoanalytic language, the term "object" refers to a person, so object relations theory is really something more like "interpersonal relations theory.")

Object relations theory contends that personality can be understood as reflecting the mental images of significant figures (especially the parents) that we form early in life in response to interactions taking place within the family (Kernberg, 2004; Wachtel, 1997). These mental images (sometimes called *introjects*) serve as templates for later interpersonal relationships—almost like relationship blueprints or "scripts." So if you internalized positive introjects early in life (for example, a mental image of mom or dad as warm and accepting), that's what you expect to occur in later relationships as well. If you internalized a mental image of mom or dad as harsh and judgmental, you might instead become a self-critical person, and feel that you can never live up to other people's standards . . . or your own (Luyten & Blatt, 2013).

Object relations theory has increased many psychologists' interest in studying psychodynamic ideas and concepts, in part because it represents a natural bridge between the psychodynamic perspective and research in other areas of psychology. For example, developmental and social psychologists also believe that mental representations of significant people play an important role in shaping our behavior. In developmental psychology you might read about this in the context of *attachment theory* (which argues that attachments—or bonds—to significant people are key to understanding human behavior; Fraley, 2002). In social psychology, mental representations of significant figures play an important role in *social cognition* (thoughts and feelings regarding other people; Bargh & Morsella, 2008; Robinson & Gordon, 2011).

Empirical Research on Psychodynamic Theories

Empirical research assessing psychodynamic concepts has produced mixed results, with some concepts receiving good empirical support, and others not faring as well. For example, the notion that we express strong sexual feelings from a very early age, as the psychosexual stage model suggests, has not held up to empirical scrutiny. On the other hand, the idea that there are dependent, control-oriented, and competitive personality types—an idea also derived from the psychosexual stage model—does seem useful.

Many ideas from the psychodynamic perspective have been studied empirically. Luborsky and Barrett (2006) reviewed much of this research; other useful reviews are provided by Bornstein (2005), Gerber (2007), and Huprich (2009). For now, let's look at three psychodynamic hypotheses that have received strong empirical support.

- *Unconscious processes influence our behavior as the psychodynamic perspective predicts.* We perceive and process much more information than we realize, and much of our behavior is shaped by feelings and motives of which we are, at best, only partially aware (Bornstein, 2009, 2010). Evidence for the importance of unconscious influences is so compelling that it has become a central element of contemporary cognitive and social psychology (Robinson & Gordon, 2011).

- *We all use ego defenses and they help determine our psychological adjustment and physical health.* People really do differ in the degree that they rely on different ego defenses—so much so that researchers now study each person's "defense style" (the unique constellation of defenses that we use). It turns out that certain defenses are more adaptive than others: Rationalization and sublimation are healthier (psychologically speaking) than repression and reaction formation (Cramer, 2006). Denial is, quite literally, bad for your health, because people who use denial tend to ignore symptoms of illness until it's too late (Bond, 2004).

- *Mental representations of self and others do indeed serve as blueprints for later relationships.* Dozens of studies have shown that mental images of our parents, and other significant figures, really do shape our expectations for later friendships and romantic relationships. The idea that you choose a romantic partner who resembles mom or dad is a myth, but it's true that you expect to be treated by others as you were treated by your parents early in life (Silverstein, 2007; Wachtel, 1997).

Psychoanalysis and Culture

The culture in which a person has been raised has a significant influence on self-conceptions. For example, someone raised in North America is likely to describe themselves in very different terms compared to someone raised in India. [Image: Harsha KR, https://goo.gl/cNfV73, CC BY-SA 2.0, https://goo.gl/rxiUsF]

One of Freud's lifelong goals was to use psychoanalytic principles to understand culture and improve intergroup relations (he actually exchanged several letters with Albert Einstein prior to World War II, in which they discussed this issue). During the past several decades, as society has become increasingly multicultural, this effort has taken on new importance; psychoanalysts have been active in incorporating ideas and findings regarding cultural influences into their research and clinical work. For example, studies have shown that individuals raised in individualistic, independence-focused cultures (for example, the United States, Great Britain) tend to define themselves primarily in terms of personal attributes (like attitudes and interests), whereas individuals raised in more sociocentric, interdependent cultures (for example, Japan, India) are more likely to describe themselves in terms of interpersonal relations and connections with others (Oyserman, Coon, & Kemmelmeier, 2002). Our self-representations are, quite literally, a product of our cultural milieu (Markus & Kitayama, 2010).

Cognitive model

The cognitive model of psychology is a methodology that emphasizes a person's perceptions and thoughts as the main influencers of their feelings and behaviors**. The term abnormality refers to a state that is atypical from the norm**

Cognitive Approach in Clinical Psychology

Since the early 1950s psychology has seen many significant developments. On the theoretical side, one of the most dramatic has been the decline of behaviourism and the rise of cognitive psychology, whereas on the practical side there has been an enormous expansion in the clinical applications of psychological knowledge. Much of the subject matter of clinical applications of psychology, such as distressing thoughts and memories, delusions and fixed ideas, is of course cognitive in nature. Cognitive theories are not new but it is only comparatively recently that they have come to play an important role in clinical psychology, backed up by the findings of systematic empirical research.

Cognitive approach or its significance in clinical psychology should first begin with the understanding of what cognitive is and on what basis it was formed.

Cognitive encompasses all acts or processes of knowing, perceiving etc. it relates to the mental processes of perception, memory, judgement and reasoning, as contrasted with emotional and volitional(behavioural) processes. Psychological processes involved in acquisition and understanding of knowledge, formation of beliefs and attitudes and decision making and problem making. Cognition includes the conscious and unconscious, concrete and abstract as well as intuitive thoughts. It involves application of past knowledge and generating new knowledge.

Now that we have understood what cognition is, cognitive approach is the explanation given to any phenomenon through the lens of cognitive processes.

Clinical psychology is a broad branch of psychology that focuses on diagnosing and treating mental, emotional, and behavioural disorders. It also deals with identification and resolving of disorders. Cognitive approach in clinical psychology explains disorders in the perspective of cognition.

The task of the various branches of cognitive psychology is to explain how knowledge is acquired, how it is represented internally, and integrated with the existing information and what kind of knowledge affect people’s feelings and behaviour. The task of the clinical psychologist is to apply this understanding to the enhancement of human potential and to the alleviation of suffering.

Cognitive approach says that people exhibit deviating, dysfunctional behaviour because of faulty thinking. The disorders manifest or continue to prolong because there is distortion in their thinking that prevents people from living life without any difficulties.

We all face cognitive distortion in our life at some moment in our life. But it becomes a disorder when that distortion is never resolved and becomes a central part in which a person thinks or looks at the world around him. For example, we all have this thought at some point in our life that we

are incapable of doing something or when we think that we are not loved. This is cognitive distortion because we have distorted the reality into something which might not be true. It affects us physically, behaviourally, emotionally. This is normal. But It becomes a disorder when we are unable to come out of that thinking. We continue to think that way. Not only does it affect us physically, emotionally etc. it also affects psychologically i.e, it starts to affect other facets of life too.

Cognitive theory in clinical psychology tries to understand and analyse these faulty thinking patterns and find a way to resolve them. The client who is suffering from a psychological disorder undergoes therapy wherein his faulty thinking is changed.

Lesson-7 Behaviorist, humanist and gestalt-based and systematic

Behaviourist.

Behaviorism **focuses on the idea that all behaviors are learned through interaction with the environment**. This learning theory states that behaviors are learned from the environment, and says that innate or inherited factors have very little influence on behavior.

Burrhus Frederic Skinner (March 20, 1904 – August 18, 1990) was an American psychologist, behaviorist, author, inventor, and social philosopher.[2][3][4][5] He was a professor of psychology at Harvard University from 1958 until his retirement in 1974.[6]

Considering free will to be an illusion, Skinner saw human action as dependent on consequences of previous actions, a theory he would articulate as the *principle of reinforcement*: If the consequences to an action are bad, there is a high chance the action will not be repeated; if the consequences are good, the probability of the action being repeated becomes stronger.[7]

Skinner developed behavior analysis, especially the philosophy of *radical behaviorism*,[8] and founded the experimental analysis of behavior, a school of experimental research psychology. He also used operant conditioning to strengthen behavior, considering the rate of response to be the most effective measure of response strength. To study operant conditioning, he invented the operant conditioning chamber (aka the Skinner box),[7] and to measure rate he invented the cumulative recorder. Using these tools, he and Charles Ferster produced Skinner's most influential experimental work, outlined in their 1957 book *Schedules of Reinforcement*.[9][10]

Skinner was a prolific author, publishing 21 books and 180 articles.[11] He imagined the application of his ideas to the design of a human community in his 1948 utopian novel, *Walden Two*,[3] while his analysis of human behavior culminated in his 1958 work, *Verbal Behavior*.[12]

Skinner, John B. Watson and Ivan Pavlov, are considered to be the pioneers of modern behaviorism. Accordingly, a June 2002 survey listed Skinner as the most influential psychologist of the 20th century.[13]

Behaviorism
Main articles: Behaviorism and Radical behaviorism

Skinner referred to his approach to the study of behavior as *radical behaviorism*,[25] which originated in the early 1900s as a reaction to depth psychology and other traditional forms of

psychology, which often had difficulty making predictions that could be tested experimentally. This philosophy of behavioral science assumes that behavior is a consequence of environmental histories of reinforcement (see applied behavior analysis). In his words:

The position can be stated as follows: what is felt or introspectively observed is not some nonphysical world of consciousness, mind, or mental life but the observer's own body. This does not mean, as I shall show later, that introspection is a kind of psychological research, nor does it mean (and this is the heart of the argument) that what are felt or introspectively observed are the causes of the behavior. An organism behaves as it does because of its current structure, but most of this is out of reach of introspection. At the moment we must content ourselves, as the methodological behaviorist insists, with a person's genetic and environment histories. What are introspectively observed are certain collateral products of those histories.… In this way we repair the major damage wrought by mentalism. When what a person does [is] attributed to what is going on inside him, investigation is brought to an end. Why explain the explanation? For twenty-five hundred years people have been preoccupied with feelings and mental life, but only recently has any interest been shown in a more precise analysis of the role of the environment. Ignorance of that role led in the first place to mental fictions, and it has been perpetuated by the explanatory practices to which they gave rise.[25]

Foundations of Skinner's behaviorism

Skinner's ideas about behaviorism were largely set forth in his first book, *The Behavior of Organisms* (1938).[9] Here, he gives a systematic description of the manner in which environmental variables control behavior. He distinguished two sorts of behavior which are controlled in different ways:

- **Respondent behaviors** are elicited by stimuli, and may be modified through respondent conditioning, often called classical (or pavlovian) conditioning, in which a neutral stimulus is paired with an eliciting stimulus. Such behaviors may be measured by their latency or strength.
- ***Operant* behaviors** are 'emitted', meaning that initially they are not induced by any particular stimulus. They are strengthened through operant conditioning (aka *instrumental* conditioning), in which the occurrence of a response yields a reinforcer. Such behaviors may be measured by their rate.

Both of these sorts of behavior had already been studied experimentally, most notably: respondents, by Ivan Pavlov;[26] and operants, by Edward Thorndike.[27] Skinner's account differed in some ways from earlier ones,[28] and was one of the first accounts to bring them under one roof.

The idea that behavior is strengthened or weakened by its consequences raises several questions. Among the most commonly asked are these:

1. Operant responses are strengthened by reinforcement, but where do they come from in the first place?
2. Once it is in the organism's repertoire, how is a response directed or controlled?
3. How can very complex and seemingly novel behaviors be explained?

1. Origin of operant behavior

Skinner's answer to the first question was very much like Darwin's answer to the question of the origin of a 'new' bodily structure, namely, variation and selection. Similarly, the behavior of an

individual varies from moment to moment; a variation that is followed by reinforcement is strengthened and becomes prominent in that individual's behavioral repertoire. *Shaping* was Skinner's term for the gradual modification of behavior by the reinforcement of desired variations. Skinner believed that 'superstitious' behavior can arise when a response happens to be followed by reinforcement to which it is actually unrelated.[*clarification needed*]

2. Control of operant behavior

The second question, "how is operant behavior controlled?" arises because, to begin with, the behavior is "emitted" without reference to any particular stimulus. Skinner answered this question by saying that a stimulus comes to control an operant if it is present when the response is reinforced and absent when it is not. For example, if lever-pressing only brings food when a light is on, a rat, or a child, will learn to press the lever only when the light is on. Skinner summarized this relationship by saying that a discriminative stimulus (e.g. light or sound) sets the occasion for the reinforcement (food) of the operant (lever-press). This three-term contingency (stimulus-response-reinforcer) is one of Skinner's most important concepts, and sets his theory apart from theories that use only pair-wise associations.[28]

3. Explaining complex behavior

Most behavior of humans cannot easily be described in terms of individual responses reinforced one by one, and Skinner devoted a great deal of effort to the problem of behavioral complexity. Some complex behavior can be seen as a sequence of relatively simple responses, and here Skinner invoked the idea of "chaining". Chaining is based on the fact, experimentally demonstrated, that a discriminative stimulus not only sets the occasion for subsequent behavior, but it can also reinforce a behavior that precedes it. That is, a discriminative stimulus is also a "conditioned reinforcer". For example, the light that sets the occasion for lever pressing may also be used to reinforce "turning around" in the presence of a noise. This results in the sequence "noise – turn-around – light – press lever – food." Much longer chains can be built by adding more stimuli and responses.

However, Skinner recognized that a great deal of behavior, especially human behavior, cannot be accounted for by gradual shaping or the construction of response sequences.[29] Complex behavior often appears suddenly in its final form, as when a person first finds his way to the elevator by following instructions given at the front desk. To account for such behavior, Skinner introduced the concept of rule-governed behavior. First, relatively simple behaviors come under the control of verbal stimuli: the child learns to "jump," "open the book," and so on. After a large number of responses come under such verbal control, a sequence of verbal stimuli can evoke an almost unlimited variety of complex responses.[29]

Reinforcement

Main article: Reinforcement

Reinforcement, a key concept of behaviorism, is the primary process that shapes and controls behavior, and occurs in two ways: *positive* and *negative*. In *The Behavior of Organisms* (1938), Skinner defines *negative reinforcement* to be synonymous with *punishment*, i.e. the presentation of an aversive stimulus. This definition would subsequently be re-defined in *Science and Human Behavior* (1953).

In what has now become the standard set of definitions, *positive* reinforcement is the strengthening of behavior by the occurrence of some event (e.g., praise after some behavior is performed), whereas *negative* reinforcement is the strengthening of behavior by the removal or avoidance of

some aversive event (e.g., opening and raising an umbrella over your head on a rainy day is reinforced by the cessation of rain falling on you).

Both types of reinforcement strengthen behavior, or increase the probability of a behavior reoccurring; the difference being in whether the reinforcing event is something applied (positive reinforcement) or something removed or avoided (negative reinforcement). **Punishment** can be the *application* of an aversive stimulus/event (positive punishment or punishment by contingent stimulation) or the *removal* of a desirable stimulus (negative punishment or punishment by contingent withdrawal). Though punishment is often used to suppress behavior, Skinner argued that this suppression is temporary and has a number of other, often unwanted, consequences.[30] **Extinction** is the absence of a rewarding stimulus, which weakens behavior.

Writing in 1981, Skinner pointed out that Darwinian natural selection is, like reinforced behavior, "selection by consequences." Though, as he said, natural selection has now "made its case," he regretted that essentially the same process, "reinforcement", was less widely accepted as underlying human behavior.[31]

Schedules of reinforcement
Main article: Schedules of reinforcement

Skinner recognized that behavior is typically reinforced more than once, and, together with Charles Ferster, he did an extensive analysis of the various ways in which reinforcements could be arranged over time, calling it the *schedules of reinforcement*.[10]

The most notable schedules of reinforcement studied by Skinner were continuous, interval (fixed or variable), and ratio (fixed or variable). All are methods used in operant conditioning.

- **Continuous reinforcement** (**CRF**): each time a specific action is performed the subject receives a reinforcement. This method is effective when teaching a new behavior because it quickly establishes an association between the target behavior and the reinforcer.[32]
- **Interval schedule**: based on the time intervals between reinforcements.[7]
  - **Fixed interval schedule** (**FI**): A procedure in which reinforcements are presented at fixed time periods, provided that the appropriate response is made. This schedule yields a response rate that is low just after reinforcement and becomes rapid just before the next reinforcement is scheduled.
  - **Variable interval schedule** (**VI**): A procedure in which behavior is reinforced after scheduled but unpredictable time durations following the previous reinforcement. This schedule yields the most stable rate of responding, with the average frequency of reinforcement determining the frequency of response.
- **Ratio schedules**: based on the ratio of responses to reinforcements.[7]
  - **Fixed ratio schedule** (**FR**): A procedure in which reinforcement is delivered after a specific number of responses have been made.
  - **Variable ratio schedule** (**VR**):[7] A procedure in which reinforcement comes after a number of responses that is randomized from one reinforcement to the next (e.g. slot machines). The lower the number of responses required, the higher the response rate tends to be. Variable ratio schedules tend to produce very rapid and steady responding rates in contrast with fixed ratio schedules where the frequency of response usually drops after the reinforcement occurs.

Humanistic

Carl Ransom Rogers (January 8, 1902 – February 4, 1987) was an American psychologist and among the founders of the humanistic approach (and client-centered approach) in psychology. Rogers is widely considered one of the founding fathers of psychotherapy research and was honored for his pioneering research with the Award for Distinguished Scientific Contributions by the American Psychological Association (APA) in 1956.

The person-centered approach, Rogers's unique approach to understanding personality and human relationships, found wide application in various domains, such as psychotherapy and counseling (client-centered therapy), education (student-centered learning), organizations, and other group settings.[1] For his professional work he received the Award for Distinguished Professional Contributions to Psychology from the APA in 1972. In a study by Steven J. Haggbloom and colleagues using six criteria such as citations and recognition, Rogers was found to be the sixth most eminent psychologist of the 20th century and second, among clinicians,[2] only to Sigmund Freud.[3] Based on a 1982 survey of 422 respondents of U.S. and Canadian psychologists, he was considered the most influential psychotherapist in history (Freud ranked third).[4]

Gestalt

Rogers's theory of the self is considered humanistic, existential, and phenomenological.[21] It is based directly on the "phenomenal field" personality theory of Combs and Snygg (1949).[22] Rogers's elaboration of his theory is extensive. He wrote 16 books and many more journal articles about it. Prochaska and Norcross (2003) states Rogers "consistently stood for an empirical evaluation of psychotherapy. He and his followers have demonstrated a humanistic approach to conducting therapy and a scientific approach to evaluating therapy need not be incompatible."

Nineteen propositions

Rogers's theory (as of 1951) was based on 19 propositions:[23]

1. All individuals (organisms) exist in a continually changing world of experience (phenomenal field) of which they are the center.
2. The organism reacts to the field as it is experienced and perceived. This perceptual field is "reality" for the individual.
3. The organism reacts as an organized whole to this phenomenal field.
4. A portion of the total perceptual field gradually becomes differentiated as the self.
5. As a result of interaction with the environment, and particularly as a result of evaluative interaction with others, the structure of the self is formed—an organized, fluid but consistent conceptual pattern of perceptions of characteristics and relationships of the "I" or the "me", together with values attached to these concepts.
6. The organism has one basic tendency and striving—to actualize, maintain and enhance the experiencing organism.
7. The best vantage point for understanding behavior is from the internal frame of reference of the individual.
8. Behavior is basically the goal-directed attempt of the organism to satisfy its needs as experienced, in the field as perceived.

9. Emotion accompanies, and in general facilitates, such goal directed behavior, the kind of emotion being related to the perceived significance of the behavior for the maintenance and enhancement of the organism.
10. The values attached to experiences, and the values that are a part of the self-structure, in some instances, are values experienced directly by the organism, and in some instances are values introjected or taken over from others, but perceived in distorted fashion, as if they had been experienced directly.
11. As experiences occur in the life of the individual, they are either (a) symbolized, perceived and organized into some relation to the self, (b) ignored because there is no perceived relationship to the self structure, or (c) denied symbolization or given distorted symbolization because the experience is inconsistent with the structure of the self.
12. Most of the ways of behaving that the organism adopts are those that are consistent with the concept of self.
13. In some instances, behavior may be brought about by organic experiences and needs that have not been symbolized. Such behavior may be inconsistent with the structure of the self, but in such instances the individual does not "own" the behavior.
14. Psychological adjustment exists when the concept of the self is such that all the sensory and visceral experiences of the organism are, or may be, assimilated on a symbolic level into a consistent relationship with the concept of self.
15. Psychological maladjustment exists when the organism denies awareness of significant sensory and visceral experiences, which consequently are not symbolized and organized into the gestalt of the self structure. When this situation exists, there is a basic or potential psychological tension.
16. Any experience that is inconsistent with the organization of the structure of the self may be perceived as a threat, and the more of these perceptions there are, the more rigidly the self structure is organized to maintain itself.
17. Under certain conditions, involving primarily complete absence of threat to the self structure, experiences inconsistent with it may be perceived and examined, and the structure of self revised to assimilate and include such experiences.
18. When the individual perceives and accepts into one consistent and integrated system all his sensory and visceral experiences, he is necessarily more understanding of others and more accepting of others as separate individuals.
19. As the individual perceives and accepts into his self structure more of his organic experiences, he finds that he is replacing his present value system—based extensively on introjections which have been distortedly symbolized—with a continuing organismic valuing process.

In relation to No. 17, Rogers is known for practicing "unconditional positive regard", which is defined as accepting a person "without negative judgment of .... [a person's] basic worth".[24]

Development of the personality

With regard to development, Rogers described principles rather than stages. The main issue is the development of a self-concept and the progress from an undifferentiated self to being fully differentiated.

Self Concept ... the organized consistent conceptual gestalt composed of perceptions of the characteristics of 'I' or 'me' and the perceptions of the relationships of the 'I' or 'me' to others and

to various aspects of life, together with the values attached to these perceptions. It is a gestalt which is available to awareness though not necessarily in awareness. It is a fluid and changing gestalt, a process, but at any given moment it is a specific entity. (Rogers, 1959)[25]

In the development of the self-concept, he saw conditional and unconditional positive regard as key. Those raised in an environment of unconditional positive regard have the opportunity to fully actualize themselves. Those raised in an environment of conditional positive regard feel worthy only if they match conditions (what Rogers describes as *conditions of worth*) that others have laid down for them.

Fully functioning person

Optimal development, as referred to in proposition 14, results in a certain process rather than static state. Rogers calls this *the good life*, where the organism continually aims to fulfill its potential. He listed the characteristics of a fully functioning person (Rogers 1961):

1. A growing openness to experience: they move away from defensiveness and have no need for subception (a perceptual defense that involves unconsciously applying strategies to prevent a troubling stimulus from entering consciousness).
2. An increasingly existential lifestyle: living each moment fully, not distorting the moment to fit personality or self-concept but allowing personality and self-concept to emanate from the experience. This results in excitement, daring, adaptability, tolerance, spontaneity, and lack of rigidity, and suggests a foundation of trust. "To open one's spirit to what is going on now, and discover in that present process whatever structure it appears to have" (Rogers 1961).
3. Increasing organismic trust: they trust their own judgment and ability to choose behavior appropriate for each moment. They do not rely on existing codes and social norms but trust that as they are open to experiences they will be able to trust their own sense of right and wrong.
4. Freedom of choice: not being shackled by the restrictions that influence an incongruent individual, they are able to make a wider range of choices more fluently. They believe they play a role in determining their own behavior and so feel responsible for it.
5. Creativity: it follows that they will feel freer to be creative. They will also be more creative in the way they adapt to their circumstances without feeling a need to conform.
6. Reliability and constructiveness: they can be trusted to act constructively. An individual who is open to all their needs will be able to maintain a balance between them. Even aggressive needs will be matched and balanced by intrinsic goodness in congruent individuals.
7. A rich full life: Rogers describes the life of the fully functioning individual as rich, full and exciting, and suggests that they experience joy and pain, love and heartbreak, fear and courage more intensely. His description of *the good life*:

   This process of the good life is not, I am convinced, a life for the faint-hearted. It involves the stretching and growing of becoming more and more of one's potentialities. It involves the courage to be. It means launching oneself fully into the stream of life. (Rogers 1961

Incongruity

Rogers identified the "real self" as the aspect of a person that is founded in the actualizing tendency, follows organismic values and needs, and receives positive regard from others and self.

On the other hand, to the extent that society is out of sync with the actualizing tendency and people are forced to live with conditions of worth that are out of step with organismic valuing, receiving only conditional positive regard and self-regard, Rogers said that people develop instead an "ideal self". By *ideal*, he was suggesting something not real, something always out of reach, a standard people cannot meet. This gap between the real self and the ideal self, the "I am" and the "I should", Rogers called *incongruity*.

Psychopathology

Rogers described the concepts of *congruence* and *incongruence* as important in his theory. In proposition #6, he refers to the actualizing tendency. At the same time, he recognized the need for *positive regard*. In a fully congruent person, realizing their potential is not at the expense of experiencing positive regard. They are able to lead authentic and genuine lives. Incongruent individuals, in their pursuit of positive regard, lead lives that include falsity and do not realize their potential. Conditions put on them by those around them make it necessary for them to forgo their genuine, authentic lives to meet with others' approval. They live lives that are not true to themselves.

Rogers suggested that the incongruent individual, who is always on the defensive and cannot be open to all experiences, is not functioning ideally and may even be malfunctioning. They work hard at maintaining and protecting their self-concept. Because their lives are not authentic, this is difficult, and they are under constant threat. They deploy *defense mechanisms* to achieve this. He describes two mechanisms: *distortion* and *denial*. Distortion occurs when the individual perceives a threat to their self-concept. They distort the perception until it fits their self-concept. This defensive behavior reduces the consciousness of the threat but not the threat itself. And so, as the threats mount, the work of protecting the self-concept becomes more difficult and the individual becomes more defensive and rigid in their self-structure. If the incongruity is immoderate this process may lead the individual to a state that would typically be described as neurotic. Their functioning becomes precarious and psychologically vulnerable. If the situation worsens it is possible that the defenses cease to function altogether and the individual becomes aware of the incongruity of their situation. Their personality becomes disorganised and bizarre; irrational behavior, associated with earlier denied aspects of self, may erupt uncontrollably.

Applications

Person-centered therapy

Main article: Person-centered therapy

Rogers originally developed his theory as the foundation for a system of therapy. He initially called it "non-directive therapy" but later replaced the term "non-directive" with "client-centered", and still later "person-centered". Even before the publication of *Client-Centered Therapy* in 1951, Rogers believed the principles he was describing could be applied in a variety of contexts, not just in therapy. As a result, he started to use the term *person-centered approach* to describe his overall theory. Person-centered therapy is the application of the person-centered approach to therapy. Other applications include a theory of personality, interpersonal relations, education, nursing, cross-cultural relations and other "helping" professions and situations. In 1946 Rogers co-authored "Counseling with Returned Servicemen" with John L. Wallen (the creator of the behavioral model known as *The Interpersonal Gap*),[27] documenting the application of person-centered approach to counseling military personnel returning from World War II.

The first empirical evidence of the client-centered approach's effectiveness was published in 1941 at the Ohio State University by Elias Porter, using the recordings of therapeutic sessions between Rogers and his clients.[28] Porter used Rogers's transcripts to devise a system to measure the degree of directiveness or non-directiveness a counselor employed.[29] The counselor's attitude and orientation were shown to be instrumental in the decisions the client made.[30][31]

Learner-centered teaching

The application to education has a large robust research tradition similar to that of therapy, with studies having begun in the late 1930s and continuing today (Cornelius-White, 2007). Rogers described the approach to education in *Client-Centered Therapy* and wrote *Freedom to Learn* devoted exclusively to the subject in 1969. *Freedom to Learn* was revised twice. The new Learner-Centered Model is similar in many regards to this classical person-centered approach to education. Before Rogers's death, he and Harold Lyon began a book, *On Becoming an Effective Teacher—Person-centered Teaching, Psychology, Philosophy, and Dialogues with Carl R. Rogers and Harold Lyon*, that Lyon and Reinhard Tausch completed and published in 2013. It contains Rogers's last unpublished writings on person-centered teaching.[32] Rogers had the following five hypotheses regarding learner-centered education:

1. "A person cannot teach another person directly; a person can only facilitate another's learning" (Rogers, 1951). This is a result of his personality theory, which states that everyone exists in a constantly changing world of experience in which they are the center. Each person reacts and responds based on perception and experience. The belief is that what the student does is more important than what the teacher does. The focus is on the student (Rogers, 1951). Therefore, the background and experiences of the learner are essential to how and what is learned. Each student will process what they learn differently depending on what they bring to the classroom.
2. "A person learns significantly only those things that are perceived as being involved in the maintenance of or enhancement of the structure of self" (Rogers, 1951). Therefore, relevancy to the student is essential for learning. The students' experiences become the core of the course.
3. "Experience which, if assimilated, would involve a change in the organization of self, tends to be resisted through denial or distortion of symbolism" (Rogers, 1951). If the content or presentation of a course is inconsistent with preconceived information, the student will learn if they are open to varying concepts. Being open to concepts that vary from one's own is vital to learning. Therefore, gently encouraging open-mindedness is helpful in engaging the student in learning. Also, it is important, for this reason, that new information be relevant and related to existing experience.
4. "The structure and organization of self appears to become more rigid under threats and to relax its boundaries when completely free from threat" (Rogers, 1951). If students believe that concepts are being forced upon them, they might become uncomfortable and fearful. A barrier is created by a tone of threat in the classroom. Therefore, an open, friendly environment in which trust is developed is essential in the classroom. Fear of retribution for not agreeing with a concept should be eliminated. A supportive classroom tone helps to alleviate fears and encourages students to have the courage to explore concepts and beliefs that vary from those they bring to the classroom. Also, new information might

threaten the student's concept of themself; therefore, the less vulnerable the student feels, the more likely they will be able to open up to the learning process.

5. "The educational situation which most effectively promotes significant learning is one in which (a) threat to the self of the learner is reduced to a minimum and (b) differentiated perception of the field is facilitated" (Rogers, 1951). The instructor should be open to learning from the students and working to connect the students to the subject matter. Frequent interaction with the students will help achieve this goal. The instructor's acceptance of being a mentor who guides rather than the expert who tells is instrumental to student-centered, nonthreatening, and unforced learning.

History of Gestalt Psychology

Originating in the work of Max Wertheimer, Gestalt psychology formed in part as a response to the structuralism of Wilhelm Wundt.[2]

While followers of structuralism were interested in breaking down psychological matters into their smallest possible parts, Gestalt psychologists wanted instead to look at the totality of the mind and behavior. Guided by the principle of holism, Wertheimer and his followers identified instances where perception was based on seeing things as a complete whole, not as separate components.

A number of thinkers influenced the development of Gestalt psychology, including Immanuel Kant, Ernst Mach, and Johann Wolfgang von Goethe.[3]

Wertheimer developed Gestalt psychology after observing what he called the phi phenomenon while watching alternating lights on a railway signal.[4] The phi phenomenon is an optical illusion where two stationary objects seem to move if they are shown appearing and disappearing in rapid succession. In other words, we perceive movement where there is none.

Based on his observations of the phi phenomenon, Wertheimer concluded that we perceive things by seeing the whole perception, not by understanding individual parts. In the example of blinking lights at a train station, the whole we perceive is that one light appears to move quickly between two points. The reality is that two separate lights are blinking rapidly without moving at all.

Influential Gestalt Psychologists

Wertheimer's observations of the phi phenomenon are widely credited as the beginning of Gestalt psychology and he went on to publicize the core principles of the field. Other psychologists also had an influence on this school of psychology.[5]

Wolfgang Köhler: Köhler connected Gestalt psychology to the natural sciences, arguing that organic phenomena are examples of holism at work. He also studied hearing and looked at problem-solving abilities in chimpanzees.

Kurt Koffka: Together with Wertheimer and Köhler, Koffka is considered a founder of the field. He applied the concept of Gestalt to child psychology, arguing that infants first understand things

holistically before learning to differentiate them into parts. Koffka played a key role in bringing Gestalt principles to the United States.

Principles of Gestalt Psychology

Gestalt psychology helped introduce the idea that human perception is not just about seeing what is actually present in the world around us. It is also heavily influenced by our motivations and expectations.

Wertheimer created principles to explain how Gestalt perception functions. Some of the most important principles of Gestalt theory are:[26]

- **Prägnanz**: This foundational principle states that we naturally perceive things in their simplest form or organization.
- **Similarity**: This Gestalt principle suggests that we naturally group similar items together based on elements like color, size, and orientation. An example would be grouping dogs based on whether they are small or large, or if they are big or small.
- **Proximity**: The principle of proximity states that objects near each other tend to be viewed as a group.
- **Continuity**: According to this Gestalt principle, we perceive elements arranged on a line or curve as related to each other, while elements that are not on the line or curve are seen as separate.
- **Closure**: This suggests that elements that form a closed object will be perceived as a group. We will even fill in missing information to create closure and make sense of an object. An example of this Gestalt psychology principle is using negative space to give the illusion that a particular shape exists when it doesn't.
- **Common region**: This Gestalt psychology principle states that we tend to group objects together if they're located in the same bounded area. (For example, objects inside a box tend to be considered a group.)

Gestalt Laws of Perception

Uses for Gestalt Psychology

There are several uses for Gestalt psychology today, some of which include those related to therapy, design, product development, and learning.

Gestalt Therapy

Gestalt therapy is based on the idea that overall perception depends on the interaction between many factors. Among these factors are our past experiences, current environment, thoughts, feelings, and needs. Gestalt therapy involves key concepts such as awareness, unfinished business, and personal responsibility.[7]

The main goal of Gestalt therapy is to help us focus on the present. While past context is important for viewing yourself as a whole, a Gestalt therapist will encourage you to keep your focus on your present experience.

Research suggests that Gestalt therapy is effective at treating symptoms of depression and anxiety, and it may help people gain confidence and increase feelings of self-efficacy and self-kindness.[8] It is often a helpful way to structure group therapy.

The therapeutic process is reliant on the relationship between the client and therapist. As a client, you must feel comfortable enough to develop a close partnership with your therapist, and they must be able to create an unbiased environment where you can discuss your thoughts and experiences.

What Is Therapeutic Rapport?

Design

Beginning in the 1920s, designers began incorporating Gestalt principles in their work. Gestalt psychology led these designers to believe that we all share certain characteristics in the way we perceive visual objects and that we all have a natural ability to see "good" design.[9]

Designers embraced Gestalt concepts, using our perception of contrast, color, symmetry, repetition, and proportion to create their work.[10] Gestalt psychology influenced other design concepts, such as:

- **Figure-ground relationship**: This describes the contrast between a focal object (like a word, phrase, or image) and the negative space around it. Designers often use this to create impact.
- **Visual hierarchy**: Designers use the way we perceive and group visual objects to establish a visual hierarchy, ensuring that the most important word or image attracts our attention first.
- **Associativity**: This concept involves the principle of proximity. Designers often use this to determine where to place important objects, including text elements such as headlines, captions, and lists.

Figure-Ground Perception in Psychology

Product Development

Product designers use Gestalt psychology to inform their decisions during the development process. Consumers tend to like products that follow Gestalt principles.[11]

This influence can be seen in the appearance of the products themselves and in their packaging and advertising. We can also see Gestalt principles at work in apps and digital products. Concepts like proximity, similarity, and continuity have become standards of our expected user experience.[12]

Learning and Education

The Gestalt Theory of Learning relies on the law of simplicity. In simple terms, it states that each learning stimulus is perceived in its simplest form.[13]

The psychology behind this learning theory states that we use our senses and previous experiences to gain knowledge about the world around us. It also suggests that we learn from the methods by which we are taught, in addition to being impacted by classroom environments and the academic culture.

Impact of Gestalt Psychology

Gestalt psychology has largely been subsumed by other types of psychology, but it had an enormous influence on the field. Researchers like Kurt Lewin and Kurt Goldstein were influenced by Gestalt concepts before going on to make important contributions to psychology.[2]

Gestalt theory is also important in that the idea of the whole being different than its parts has influenced our understanding of the brain and social behavior. Gestalt theory still impacts how we understand vision and the ways that context, visual illusions, and information processing impact our perception.

UNIT-3 MAJOR CLINICAL DISORDERS

Lesson -8Clinical child Psychology: Clinical issues, Behavioural Disorders

Behavior disorders can greatly impact a person's life. Whether you've been diagnosed since a child, are concerned about warning signs in a loved one, or are recognizing symptoms in your own life, the best way to treat a behavioral disorder is to be fully informed.

Our team of experts led by board-certified psychiatrist Dr. Ronald Winfield at Greater Lowell Psychiatric Associates in North Chelmsford, Massachusetts, specializes in helping you identify, treat, and manage the symptoms of behavioral disorders so you can live a happy and successful life.

Behavioral disorder basics
Behavioral disorders are classified as a pattern of behaviors that are disruptive and harmful to you and those around you. These disorders are typically diagnosed and treated in early childhood, but if left untreated they can affect your adult life making it difficult for you to maintain healthy relationships, regular employment, and other functions of a typical adult.

Some risk factors for developing a behavior disorder include but are not limited to:

- Childhood substance abuse
- Low self-esteem
- Parental substance abuse
- Lack of parental supervision as a childhood

- Traumatic events
- Exposure to toxic substances as a child
- Lack of emotional attachment to parents
- Associating with peers involved in deviant behavior

There are many kinds of behavioral issues. Here are the five most common affecting Americans today:

1. Conduct disorder

Conduct disorder occurs when you struggle with antisocial behaviors, following rules, and showing empathy towards those around you. Signs of this disorder usually appear around the age of 16 and include:

- Starting physical altercations
- Engaging in bullying or threatening your peers
- Stealing
- Destroying or breaking into private property

Other indicators of conduct disorder are lack of remorse, concern, empathy, or any emotional response for your actions.

2. Oppositional defiant disorder (ODD)

Children naturally act out in defiance and throw tantrums. When that behavior translates into adulthood, it's called oppositional defiant disorder. ODD begins in childhood with behaviors like frequent questioning of authority, excessive arguing, refusing to comply, and intentionally engaging in behaviors that upset others.

If not diagnosed and treated at a young age, ODD can present itself in your later life as feeling angry at the world, feeling misunderstood, disliking authority figures, especially at work, becoming defensive when presented with opposition or feedback, and blaming others for your actions.

3. Attention deficit hyperactivity disorder (ADHD)

ADHD is likely one of the most common behavior issues for children, adolescents, and adults. It's signs and symptoms include:

- Impulsivity
- Inability to pay attention
- Lack of time management skills
- Mood swings
- Low frustration threshold and trouble coping with stress

These symptoms can lead to complications like financial instability, unemployment, frequent trouble with law enforcement, substance abuse or misuse, and unstable relationships.

4. Obsessive-compulsive disorder (OCD)

If you have OCD, you likely feel you're drowning in a constant state of worry. OCD is characterized by thoughts, actions, and impulses that are chronic, uncontrollable, and intrusive. You might engage in behaviors like constant washing and cleaning, repeating phrases or prayers, following strict routines, or repetitively counting and checking.

For example, you may have a significant fear of or be worried about germs. This might cause you to wash your hands compulsively to get rid of any potential germs even if your hands become chapped, raw, and blistered.

Other obsessions can include:

- Severe doubt and uncertainty
- The need to have things orderly or symmetrical
- Ideation about losing control or harming yourself or others
- Unwanted aggressive or sexual thoughts

OCD is an overwhelming condition that invades and disrupts your life making it difficult to function normally. OCD can lead to other complications like bipolar disorder, depression, and anxiety.

5. Behavioral addiction

Addiction is not limited to substance abuse. Addiction occurs when your brain associates pleasure and activates its reward mechanism when engaging in an activity. There are many behaviors that you can become addicted to including:

- Gambling
- Sex
- Internet
- Shopping
- Video games
- Plastic surgery
- Binge eating
- Risky behavior

If you're addicted to one or more of these behaviors, you might neglect family and work responsibilities, use the behavior as a way to cope with emotions, deny or hide your problem, and continue the behavior despite wanting to stop or cut back. All of these behavioral addictions can lead to problems with your finances and relationships, and threaten your safety and overall health.

Treating your behavioral issues

No matter what behavioral disorder you're living with, we can help you manage and treat your symptoms. We offer a comprehensive list of treatment options depending on your specific needs including:

- Transcranial magnetic stimulation (TMS)
- Psychotherapy and counseling

- Medication management

Lesson -9 Emotional Disorder, Developmental and Learning Disorder

Are you aware of behavioral and emotional disorders in children? During the developmental years of a child, constant growth and change are signified. This fact should be strictly kept in mind while dealing with behavioral and emotional disorders in children. Brain disorders like **bipolar disorder**, **depression** and anxiety disorders may affect any child during early childhood. The assistance of a mental health professional should be taken while dealing with these problems. On a global scale, almost 15% of children suffer from behavioral disorders. Among the different forms of behavioral disorders, attention deficit **hyperactivity** syndrome accounts for being one of the most common ones.

Here is a list of the different forms of behavioral and emotional disorders, which are observed in children:

1. **ADHD-** This common mental disorder in children is more common in boys than in girls. Commonly, children affected with this disorder are unable to finish familiar tasks, plan or sit still. They may even lose awareness about what goes around them. ADHD-affected children may seem fine on some days, but on other days, they may experience disorganized and frenzied activities. This disorder may continue into adolescence and adulthood as well. There are several treatment options for ADHD, which may include medications, therapy and educational options.
2. **Autism-** Children affected with autism seem to become indifferent, remote and isolated in a world of their own. They are unable to form emotional ties with other people or their peers. Autism is more likely to occur in boys than in girls. The condition may result in language delays, **mental retardation**, and in some cases, a child may suffer high functioning and usual speech and intelligence. This is because the brain of such children functions differently than others. Consistency is very important for managing autism in children.
3. **Bipolar Disorder-** This disorder is most likely to begin in early childhood and it continues into adulthood. It is featured by extreme mood swings in the patient. A child may have extremely high or euphoric feelings, but his mood may turn into depression and sadness all of a sudden. This condition is a genetic illness, and it is often mistaken or misdiagnosed as ADHD.
4. **Anxiety-** Anxiety disorders make children feel uneasy, distressed and unusually scared for no appropriate reason. **Panic** disorders, featured, by intense episodes of **fear**, which occur without any kind of provocation or signal are a common type of anxiety disorder. **OCD** or obsessive compulsive disorder, which is repetitive and compulsive in nature, is another type of anxiety disorder.

I am thinking of how to promote the mental health of our children and help them grow into compassionate, social and competent adults.

One thing I have come to understand as an adult is how powerful and essential it is to give our children words to use to identify feelings. You may have seen the charts that have 20 or so feelings and faces that go with them. That's great for older kids but overwhelming to little ones.

For me, it boils down to these five emotions: mad, glad, sad, lonely and scared. I found that my daughter and I communicated more easily when something difficult was going on if she could identify the feelings.

- **Mad (angry):** When a child is mad, if they can use their words, that is great! We as parents can help them know what is appropriate depending on their age. For a younger child, learning to use words instead of biting, scratching or hitting another person is crucial. One of my favorite things was to ask my daughter to draw about it. We know that coloring is almost meditative. Another tactic is to encourage physically letting out that energy through sports, running, hitting a pillow, etc. One of my personal favorites is to have a screaming minute in the car with windows rolled up to get out that anger in a safe way.

- **Glad (happy):** If a happy moment comes, no matter how small, help your child celebrate that feeling. You can then help them call upon that memory when it is needed. For example, "Nick, do you remember when you had that great soccer game last month and how good you felt? I know you can do this science project and feel as good about yourself!"

- **Sad:** This is a hard one. Children all feel sad once in a while, it comes with being human. If your child is sad for long periods of time, talks about hurting or killing him/herself, or does a self-harming action, it is time to seek professional help. Contact your pediatrician, call a mental health professional and get the needed support. If it's urgent, call 911 or take your child to the nearest emergency room. Depression is bigger than just being sad. The National Suicide Prevention Lifeline number is 1-800-273-TALK (8255).

- **Lonely:** This is another hard experience that children have. Peers, siblings and family are so important to them. If there is a fight with a best friend, they can feel terrible. You know your child best. If there is a time when your child has no friends, does not like going to school, etc., it would be good to investigate further. A real possibility that we know occurs is bullying. If that is happening, help your child by making it easy to be around other kids who have similar interests. Pets can also help with loneliness.

- **Scared:** As an example, I'll share a personal story. My daughter was 7 years old and in a new school when 9/11 happened. The school immediately cancelled classes and went into day care mode. She knew something was wrong as the parade of the other children were picked up. Finally, she and another boy were the only ones left with their teacher. When I could finally leave the hospital and pick her up, she hugged me and cried, telling me how scared she was because she did not know what was happening. We talked a lot about it that evening and for many months. When a child is scared, they need information and reassurance. With all of the upsetting news, it is good to help your child process the things they are hearing about at school, from other people and from the news.

  Understanding your child's feelings is so important. You are their first and best teacher. You can give them the foundation they need as they grow and learn about life. There are so many good resources on the internet for parents. The mental health of our children is one of our best resources for a positive world.

  Children with Severe Emotional Disturbance (SED) are persons who are under the age of 18, who have had a diagnosable mental, behavioral or emotional disorder of sufficient duration to meet diagnostic criteria specified within DSM-V, that resulted in functional impairment which substantially interferes with or limits the child's role or functioning in family, school or community activities.

Now if the clinical definition still has you scratching your head, that's okay. Here are some signs that your child may suffer from SED:

A 'Diagnosed and Pervasive' Developmental Disorder

Pervasive developmental disorders (PDDs) should not be confused with specific developmental disorders (SDDs), such as speech disorders or pediatric epilepsy. For a developmental disorder to be pervasive, it must be characterized by delays in development of multiple basic functions. While not all PDDs are yet identified, the best-known include atypical autism, severe Aspergers, Rett's syndrome and childhood disintegrative disorder.

Childhood Schizophrenia

Childhood schizophrenia is essentially the same as schizophrenia in adults, however, if it occurs early in a child's life, it typically effects serious implications for development. As schizophrenia is a chronic condition that requires lifelong treatment, it is an automatic qualifying condition for your child to receive CCF's assistance.

Conduct Disorder

Conduct disorder is serious behavioral and emotion disorder in a child, teen or youth. While it is normal for children of this age to act out, one may have Conduct Disorder is the behavior is long lasting, violent and disrupts the child's everyday life.

Affective Disorders

Affective disorders are psychiatric diseases that affect mood, such as depression, pediatric bipolar disorder and child anxiety disorders. These illnesses are not qualifiers for CCF assistance by themselves, however, if they are acute enough to where they seriously impede a child's development, they may qualify as SED.

Disruptive Behavior Disorders

Oppositional defiant disorder (ODD) is a serious pattern of negative, defiant, disobedient and hostile behavior to authority figures persisting for a long period of time. Other signs of behavioral disorders include long-term propensities toward ongoing: serious aggression toward people, harming of animals, destruction of property, stealing or self-violence.

Self-destructive behavior with serious medical implications

Any other behavior that could lead to chronic medical implications, such as persistent substance abuse, attempted suicides or eating disorders may mean a child suffers from SED.

Learning Disorders in Children

Many children may struggle in school with some topics or skills from time to time. When children try hard and still struggle with a specific set of skills over time, it could be a sign of a learning disorder. Having a learning disorder means that a child has difficulty in one or more areas of learning, even when overall intelligence or motivation is not affected.

Some of the symptoms of learning disorders are

- Difficulty telling right from left
- Reversing letters, words, or numbers, after first or second grade
- Difficulties recognizing patterns or sorting items by size or shape
- Difficulty understanding and following instructions or staying organized
- Difficulty remembering what was just said or what was just read
- Lacking coordination when moving around
- Difficulty doing tasks with the hands, like writing, cutting, or drawing

- Difficulty understanding the concept of time

Examples of learning disorders include

- Dyslexia – difficulty with reading
- Dyscalculia – difficulty with math
- Dysgraphia – difficulty with writing

Children with learning disorders may feel frustrated that they cannot master a subject despite trying hard, and may act out, act helpless, or withdraw. Learning disorders can also be present with emotional or behavioral disorders, such as attention-deficit/hyperactivity disorder (ADHD), or anxiety. The combination of problems can make it particularly hard for a child to succeed in school. Properly diagnosing each disorder is crucial, so that the child can get the right kind of help for each.

Treatment for learning disorders

Children with learning disorders often need extra help and instruction that are specialized for them. Having a learning disorder can qualify a child for special education services in school. Schools usually do their own testing for learning disorders to see if a child needs intervention. An evaluation by a healthcare professional is needed if there are other concerns about the child's behavior or emotions. Parents, healthcare providers, and the school can work together to find the right referrals and treatment.

Lesson-10 Clinical Neuro-Psychology:- Symptômes and Syndromes; Delirium, Dementia, Amnesic syndrome

Some individuals who seek the services of clinical psychologists aren't exactly sure what is causing their problems or affecting their lives, they just know that things aren't going well, that marriages or relationships are suffering, that, for some reason, depression, anxiety or a number of other conditions have affected either their behaviors, or the behaviors of those they love.

But in some cases, physicians know that certain issues and problems are a direct result of a brain disorder caused either by injury, disease, or developmental dysfunctions. And for these individuals, a specialized form of clinical psychology called clinical neuropsychology focuses on treating those with known structural and functional brain malfunctions.

Clinical neuropsychologists are the professionals who treat neurological patients, often referred to them by neurologists, other physicians, other psychologists, or social agencies. In most cases, the patient exhibits cognitive and behavioral dysfunctions related to a brain injury or disease, and the referring doctor or agency seeks to gather information critical to the patient's rehabilitation.

But in certain cases, such as those of developmental disorders, the clinical neuropsychologist often is consulted to help identify developmental learning disorders, such as dyslexia, attention deficit disorder, and nonverbal disability. Parents, an educational facility or social service agency has already determined that the child's cognition and behaviors suggest a form of brain disorder, and the clinical neuropsychologist helps to accurately assess and diagnose the disorder.

A Clinical Approach

Clinical neuropsychologists consult with patients and their families in healthcare settings, such as hospitals or medical clinics, or in clinical neuropsychology practices. They also see some children in educational settings.

The referring physician gives clinical neuropsychologists the results from neuroimaging tests, such as CT or (computerized tomography) MRIs, (magnetic resonance imaging) and PETs (positron emission tomography).

These tests pinpoint structural brain injuries or tumors within millimeters in the brain, localizing brain dysfunction to specific areas and regions of the brain. Physicians also forward the patient's other medical records and results from laboratory tests to a clinical neuropsychologist to provide a complete case history of the patient.

The referring doctor or agency also develops a question or set of concerns about the patient's cognitive and/or behavioral deficits that must be addressed in order to plan rehabilitative services and interventions. These concerns often involve answering specific "everyday functioning" issues, such as the ability of the patient to drive, independently take care of personal hygiene, finances, and work-related issues.

During the initial examination, the clinical neuropsychologist interviews the patient as well as any family members that are present, asking about the patient's history and the patient's opinion of the nature of the brain injury or deficit. Through this initial interview, the clinical neuropsychologist determines the patient's awareness of problems, and his or her physical capacity and mental ability to undertake a more in-depth neuropsychological assessment consisting of a series of tests.

If the patient is able to sit through and perform a number of cognitive tests, the clinical neuropsychologist assigns a trained technician to administer the tests.

The Clinical Assessment

After the initial interview, the clinical neuropsychologist also develops a hypothesis about the level and depth of cognitive and behavioral impairment. Keeping in mind the rehabilitative concerns and questions of the referring doctor, and synthesizing the results from the other medical tests and the patient's history, the clinical neuropsychologist recommends either a series of neuropsychological tests, or a shortened subset of one or more tests depending on the patient's abilities and needs.

Many valid and reliable neuropsychological tests have been developed over the years designed to assess all areas of cognitive functioning, including:

- Intellectual function
- Memory
- Speech and language

- Visual perception
- Psychomotor/Psychosensory functions
- Planning, judgment, decision-making functions
- Attention

While a technician administers neuropsychological tests, the clinical neuropsychologist assesses the results. Based on the outcomes, the clinical neuropsychologist recommends rehabilitation plans, interventions, a plan to return to work, recommendations for daily living, or any other issues requested by physicians and healthcare personnel.

Monitoring

The neuropsychological tests give psychologists and physicians not only a snapshot of current cognitive and behavioral functioning, but they also provide a baseline – an important element in any rehabilitation program. Periodic re-testing gives both healthcare providers and psychologists a chance to evaluate the success of a particular rehabilitation program or treatment, and to gauge the patient's progress toward stated goals.

Serial testing also gives clinical neuropsychologists valuable information concerning the efficacy of a particular drug used for diseases, such as Parkinson's disease or seizure disorders. And, for patients initially diagnosed with mild dementia, a baseline test followed by periodic re-testing provides important information regarding the disease's progression.

Legal Situations

Attorneys and the courts also hire clinical neuropsychologists. Brain injuries resulting from car accidents, exposure to toxic chemicals, carbon monoxide poisoning, electrical or other forms of construction injuries, often end up in the courts to determine monetary settlements to the plaintiff. And in many of cases of accidental injury, brain dysfunction isn't apparent through neuroimaging techniques or on an EEG.

Data gathered from neuropsychological tests in addition to the interview with the client that the clinical neuropsychologist conducts often provide the essential data for the courts to decide these complex cases.

Careers in Clinical Neuropsychology

If you are interested in becoming a clinical neuropsychologist, in treating individuals suffering from brain dysfunctions, contact psychology schools. Most positions in this field require a Ph.D.

Additionally, most clinical neuropsychologists must have certification by the American board of Clinical Neuropsychology (ABCN) or the American Academy of Clinical Neuropsychology (AACN). Many clinical positions also require psychologist licensure.

Cognitive Disorders: Delirium, Dementia, and Amnestic Disorders

Dementia, delirium, and amnestic and other cognitive disorders are biologic illnesses with two defining characteristics in common. The first characteristic is a clinically significant *decrease in cognitive function* that is a marked change from the person's previous level of function. *Cognition* refers to multiple, interrelated mental processes that include memory, reasoning, judgment, orientation, problem solving, abstraction, and acquisition and use of knowledge and language. When these processes fail to be synchronized, dysfunction is inevitable.

The second characteristic common to delirium, dementia, and amnestic categories is that each disorder is caused by (1) a general medical condition or (2) use of a substance or (3) a combination of these. Specific medical conditions may be diagnosed at the time of symptom eruption, and may or may not be identified during the course of the disorder (e.g., Alzheimer's).

Substances may be in the following forms, with examples:

1. **Toxins**: Industrial chemicals used in manufacturing are inhaled or absorbed through skin; ingested gardening or cleaning products in the home cause inadvertent poisoning of a very young child.
2. **Medications**: Elderly person who is seeing more than one physician uses multiple incompatible prescription medications; a person with chronic depression, disease, or pain intentionally overdoses on a prescribed narcotic in a suicide attempt.
3. **Drugs of abuse**: Young people experiment with and mix illegal designer drugs at a rave party; heroin abuser unintentionally overdoses on substances that are more potent than expected; housewife drinks excessive amounts of alcohol every day.

Each decade, people are living longer. Over the next decade the mental disorders in the categories of delirium, dementia, and amnestic and other cognitive disorders will result in extraordinary losses in terms of human despair, interrupted relationships, functional disturbances, and economic costs that will affect individuals, businesses, and governments. The most notable monetary loss will be attributed to *cost of treatment* for involved individuals and their families and to the *loss of productivity* associated with these disorders.

Productivity is also lost by caregivers, who often must interrupt or completely stop their occupations to care for a client. Cognitive disruption will occur especially in individuals diagnosed with dementia, a disorder that primarily targets the population with the greatest projected growth over the next several decades—the elderly. Currently more than 4 million Americans are affected by Alzheimer's disorder. That number will swell in direct relation to that population increase unless preventive measures or a cure are found.

ETIOLOGY

As noted, delirium, dementia, and amnestic and other cognitive disorders have an organic etiology, and the cause is always either a medical condition or a substance, or a combination of these (Box 8-1). Primary and secondary causes of dementia appear in

Etiology of Delirium, Dementia, and Amnestic and Other Cognitive Disorders*

DELIRIUM

- Central nervous system: head trauma, epilepsy, brain tumor
- Vascular: stroke, hypertensive encephalopathy
- Degenerative: Pick's disease, infections
- Metabolic: renal/hepatic disease, fluid/electrolyte imbalance, hypoxia

▪ Cardiopulmonary: congestive heart failure, shock, arrhythmias
▪ Systemic: septicemia, urinary tract infection, pneumonia, neoplasms
▪ Sensory deprivation: postoperative state, visual/hearing impairment
▪ Substance induced: intoxication or withdrawal
▪ Medications
▪ Anticholinergics: traditional antidepressants, antiparkinsonian agents, neuroleptics
▪ Antihistamines: over-the-counter remedies for cold, flu, sleep
▪ Antiarrhythmics: quinidine
▪ Sedative-hypnotics: benzodiazepines
▪ Narcotic analgesics
▪ Histamine-2 receptor blockers: cimetidine
▪ Anticonvulsants
▪ Beta-blockers
▪ Antihypertensives
▪ Corticosteroids
▪ Antibiotics
▪ Toxins/street drugs

DEMENTIA

▪ Genetic factors: chromosomal mutations
▪ Acetylcholine loss
▪ Alzheimer's disease: neuritic plaques, neurofibrillary tangles
▪ Vascular impairment: multiinfarct disorder
▪ Human immunodeficiency virus (HIV)
▪ Head trauma
▪ Parkinson's disease
▪ Huntington's disease (chorea)
▪ Pick's disease
▪ Creutzfeldt-Jakob disease
▪ Substance induced, persisting: drug abuse, medications, toxins

AMNESTIC AND OTHER COGNITIVE DISORDERS

▪ Primary systemic: hypoxia
▪ Cerebrovascular disease
▪ Closed head trauma
▪ Brain surgery
▪ Herpes simplex encephalopathy
▪ Seizures
▪ Substance induced: chronic alcoholism

*Etiology is attributed to biologic factors.

Causes of Dementia

Delirium

Delirium may occur in any age group. The onset for delirium is typically rapid, and the course fluctuates. One of the most preventable, reversible, undiagnosed, underdiagnosed, and untreated mental conditions, delirium is often mistaken for other disorders, such as depression or dementia (Table 8-1). Delirium may be superimposed on other disorders, making it even more difficult to

identify, and then may be missed. Although difficult to identify because of its fluctuating nature, delirium is one of the more treatable mental conditions when recognized. When unrecognized, clients can suffer irreversible brain damage or die.

Early recognition of delirium by nurses and other medical professionals is key in the prevention and treatment of delirium. Risk factors for delirium include the following:

- Accidental or intentional poisoning
- Elderly clients
- Recent major surgery
- Preexisting cognitive dysfunction
- Multiple drug therapy, especially polypharmacy of high-dose psychotropic drugs
- High doses of hypnotics

Delirium is a prevalent disorder. Study results vary because of the different populations studied who develop delirium, including *children* who swallow poisons or have high fevers, *adolescents* or *adults* who abuse drugs, and *elderly persons* who are hospitalized for surgeries or who have dementia. Depending on the study, estimates for elderly clients who are hospitalized for acute physical illnesses and then develop delirium range from 14% to 80%. Clients who have delirium while hospitalized have longer hospital stays than those without delirium, experience greater decline in function, and have higher rates of nursing home admissions. Hospital mortality rates for clients who have delirium are high. The belief that delirium is rapid, transient, and temporary sometimes is a myth. Approximately two thirds of elderly clients who experience delirium while hospitalized demonstrate the symptoms after discharge. More than half of these clients live alone and are expected to carry out postdischarge medical and nursing orders in addition to their daily activities while in this compromised state. In addition, the cost of caring for clients who are delirious is increased because of the need for more demanding and continuous intervention from staff or other caregivers.

Dementia

Dementia of the *Alzheimer's type* is the most prevalent dementia. The onset and course for this disorder is slow, insidious, progressive, and terminal. It usually takes 8 to 10 years from onset of symptoms to death; however, some individuals live for 15 years or more with this stealthy, relentless thief of life.

The trend of lengthening life span will continue throughout the twenty-first century. By 2010, approximately 300 million people will live in the United States, 15% of whom will be older than age 65. Alzheimer's-type dementia, one of the diseases most dreaded by elderly persons, their families, and the government, will strike 8% to 15% of those older than age 65. That number doubles every 5 years after age 65. Costs of caring for clients with dementia are currently estimated at more than $100 billion annually, and costs will continue to increase.

The recent Surgeon General's Report on Mental Health states that education, measured by number of years of schooling, is a reliable predictor of late-life cognitive functioning. Highest percentages of Alzheimer's-type dementia occur in those with the least number of years of education. Also, those with higher education who remain cognitively active and engaged in their environment and life interests have less incidence or have symptoms later in life

Amnestic and Other Cognitive Disorders

Onset and course for amnestic and other cognitive disorders vary widely depending on the primary cause. For example, traumatic brain injury, sudden stroke, or neurotoxic poisoning from any source

may erupt into an *acute amnestic syndrome*. On the other hand, chronic alcoholism, other prolonged drug abuse, malnutrition, or continuous exposure to toxic chemicals over an extended period usually results in a more insidious onset and persistent course. Amnestic disorder may be *temporary*, as in a brain injury from which the individual may fully recover, or *permanent*, as in individuals who chronically abuse alcohol and fail to eat a nutritional diet.

ASSESSMENT AND DIAGNOSTIC CRITERIA

A shared symptom in delirium, dementia, and amnestic disorders is **memory impairment.** As noted, making a definitive diagnosis within this category of disorders may be difficult, and a thorough assessment helps to distinguish the cause of delirium (Box 8-3). The disorders differ in regard to additional symptoms that accom-pany memory disturbance, defining characteristics, and in some cases etiology

Delirium Assessment

Physiologic assessment is based on the following workup:

- Vital signs/electrocardiogram (ECG)/oxygen saturation
- Pain (as fifth vital sign)
- Medication/alcohol/substance use history (see Chapter 9)
- Neurologic examination
- Mental status examination
- Step-by-step physical examination by organ system
- Electroencephalogram (EEG)

Vital signs/ECG offer a baseline of the client's physical status (e.g., blood pressure, cardiac problems).

Medication/alcohol/substances can contribute to the client's delirium and confusion.

Neurologic examination can rule out a stroke or transient ischemic attack (TIA). Examination of the cranial nerves and extraocular (eye) movements can reveal neurologic impairment. Older adults with cataracts are at risk for delirium, which is why patching one eye at a time prevents "black patch" delirium.

Mental status examination can rule out depression and other treatable mental disorders. Persons with delirium or acute confusion are often distracted during this examination, whereas persons with dementia struggle to come up with answers that they often nearly miss.

Step-by-step physical examination by organ system searches for metabolic disturbances: cardiovascular and pulmonary (arterial blood gases, chest radiograph, oxygen saturation/hypoxia), renal (electrolyte imbalance, uremia, dehydration), endocrine (thyroid panel, diabetes, glucose, calcium), and liver (hepatic function tests). A search for infection (sepsis, occult abscess, meningitis, urinary tract infection, pneumonia) is critical.

EEG may be normal in dementia but is often abnormal in delirium (shows diffuse slowing); therefore it is a very sensitive test for identifying delirium, although not necessarily pinpointing a cause. EEG abnormalities may show different patterns (low amplitude, fast activity) in alcohol withdrawal and sedative-hypnotic withdrawal. EEG abnormalities may continue after the clinical symptoms of the brain syndrome are gone.

N OTE: Finding the underlying cause of delirium or confusion is important for treatment to be effective. Although there are many causes, the assessment process is consistent.

Delirium

The defining characteristics of delirium are disturbances of consciousness and cognition. Delirium typically occurs over a short period, usually hours or days, and levels of consciousness fluctuate

without following a set pattern, as evidenced by increased or decreased awareness of the environment. Ability to focus, sustain focus, or voluntarily shift attention is greatly impaired; therefore the client may be easily distracted by any stimulus or may be unable to refocus attention away from a task or thought if asked. The client may become comatose.

In regard to the common symptom of impaired memory, recent memory is affected in delirium. For example, when asked to remember a specific item, the client may be unable to recall it after only a few minutes. The client may repeat within minutes what he just told the nurse. The client may ask, "When will the doctor be here?" after the physician has just left.

Disorientation usually occurs in regard to time. For example, the client may arise in the middle of the night and start to dress for breakfast. The client may tell the nurse he is going to sleep for the night when it is noon.

Language and speech disturbances vary widely, as evidenced by impaired ability to articulate *(dysarthria)*, to name objects *(dysnomia)*, or to write *(dysgraphia)*. The client may ramble or may be completely unable to communicate through speech, writing, or signs *(aphasia)*. Aphasia may be sensory (receptive) or motor (expressive).

DSM-IV-TR

Association **Delirium**

Delirium due to a general medical condition (name condition)
Substance-induced delirium
▪ Substance intoxication delirium
▪ Substance withdrawal delirium
Delirium due to multiple etiologies (e.g., cardiac failure, sedative intoxication)
Delirium not otherwise specified

DSM-IV-TR

Association **Dementia**

Dementia of the Alzheimer's type (early onset by age 65)
▪ Uncomplicated
▪ With delirium
▪ With delusions
▪ With depressed mood
Dementia of the Alzheimer's type (late onset after age 65)
▪ Uncomplicated
▪ With delirium
▪ With delusions
▪ With depressed mood
Vascular dementia
▪ Uncomplicated
▪ With delirium
▪ With delusions
▪ With depressed mood
Dementia due to HIV disease
Dementia due to head trauma
Dementia due to Parkinson's disease
Dementia due to Huntington's disease
Dementia due to Pick's disease
Dementia due to Creutzfeldt-Jakob disease

Dementia due to other general medical conditions
Substance-induced persisting dementia
Dementia due to multiple etiologies
Dementia not otherwise specified

Sensory-perceptual disturbances are common in delirium and may manifest as *illusions* (e.g., thinks slippers are rodents), *hallucinations* (e.g., hears voice of dead mother, sees people who are not actually there walk into the room), or *misinterpretations* (e.g., thinks loud noise in hallway was automobile accident). Delusional thinking may develop around the sensory disturbances as the individual attempts to make sense of the changing symptoms.

Associated **behavioral disturbances** frequently follow cognitive disturbances and range from psychomotor *hyperactivity* (e.g., restlessness, pacing, screaming, cursing) to psychomotor *hypoactivity* (e.g., lethargy, inability to move from bed or chair, moaning).

Emotional disturbances usually occur in delirium. Labile moods with rapid, unpredictable changes from one mood state to another, often without provocation, are common. Shifts among anxiety, depression, irritability, elation, agitation, and fear may occur in no set pattern.

Sleep pattern disturbances frequently occur in delirium. The sleep/wake cycle may be completely reversed, with the client unable to fall asleep at night, lying awake and agitated, then sleeping all day long. In the phenomenon called *sundown syndrome*, the client is relatively alert and has marginally intact cognition, then suddenly becomes confused, agitated, and restless as night approaches. Symptoms may be triggered by changes in light and other environmental stimuli coupled with fatigue at the end of the day and compromised age-related senses (decreased vision/hearing). Sundown syndrome occurs frequently in clients who have dementia. Delirium may also be superimposed on dementia.

Dementia

The defining characteristics of dementia are multiple cognitive deficits, especially memory impairment, and one or more of the following: aphasia, apraxia, agnosia, and impaired executive function.

As with delirium, symptoms for dementia are directly related to either a medical condition, ongoing effects of a substance, or a combination of the two. All symptoms represent a decline from previous function and interfere with social function (e.g., family, friends, organizations) and occupational function (e.g., job, homemaker).

The symptoms of dementia are progressive and develop slowly through four stages (Box 8-4). As a result, clients frequently cover up the symptoms as they deny the inevitable and attempt continued functioning. Close family members are usually aware of changes long before casual acquaintances, who may argue that the client "seems normal."

Stages of Dementia

STAGE 1

Client:
- Demonstrates minor short-term memory problems.
- Displays subtle mental and physical decline.
- Covers up errors.
- Has difficulty focusing attention.
- Starts to lose interest in environment, events, and situations.
- Demonstrates deterioration of social courtesies.
- Becomes uncertain in actions.

▪ Probes for right words.
▪ Shows difficulty making decisions.

STAGE 2

Client:
▪ Demonstrates obvious memory deficits.
▪ Hesitates with verbal initiation or responses.
▪ Is disoriented to time.
▪ Complains of neglect.
▪ Forgets appointments, routines, and events.
▪ Accuses other clients of stealing.
▪ Hides possessions, then forgets location.

STAGE 3

Client:
▪ Demonstrates severe memory loss.
▪ Is disoriented to time, place, person.
▪ Displays motor deterioration.
▪ Wanders, then cannot find way back.
▪ Licks lips; uses chewing motions.
▪ Becomes hyperoral.
▪ Demonstrates immodest behavior.
▪ Has unsteady gait; shows physical incompetence.
▪ Displays poor communication; repeats words.
▪ Cannot read or write.
▪ Does not recognize self in mirror.
▪ Needs help with all daily activities.
▪ Has catastrophic reactions.

STAGE 4 (TERMINAL STAGE)

Client:
▪ Demonstrates global memory loss.
▪ Displays ataxic movements.
▪ Demonstrates psychotic behavior.
▪ Displays extreme psychomotor retardation.
▪ Recognizes no one.
▪ Needs complete assistance.
▪ Cannot eat; tube-fed.
▪ Becomes susceptible to infection.
▪ Loses weight.
▪ Is incontinent.

As in delirium and amnestic disorders, the hallmark symptom for dementia that must be present to make the diagnosis is *impaired memory*, which is described as (1) inability to learn new material or (2) inability to recall previously learned material. Clients with dementia usually have both deficits. At first, long-term memory often is retained as clients begin to forget immediate events, situations, and everyday routines. In early stages they forget appointments, where they put the car keys, or where the car is parked after shopping. This progresses to inability to remember their

birthday or how to return home from a local familiar place, which causes fright and anxiety. Progressively, the client fails to identify relatives or own name, eventually experiences global memory loss, and is unable to recall either remote or recent events or situations.

Mild cognitive impairment(MCI) describes early neurocognitive dysfunction in which the person can still compensate for deficits in memory, executive function, (making plans, balancing checkbook), attention, language, and perceptual motor ability (integration of information with motor activity). Several studies are in progress for MCI because 15% to 20% of these clients develop dementia of the Alzheimer's type (Surgeon General's Report on Mental Health, 1999).

Speech and language deteriorate noticeably through-out the course of dementia, and *aphasia* develops. An extensive vocabulary acquired over a lifetime diminishes as the disorder progresses. When describing events, the client cannot think of appropriate words and begins to call many items "that thing" or may use extraneous words while searching for the right ones. Poverty of speech occurs, and sometimes unusual patterns of speech occur. For example, the client may repeat words of the person talking to them *(echolalia).* Eventually all speech may disappear and the client becomes mute.

Another symptom that occurs in dementia is *apraxia*, referring to the inability to use objects correctly or perform intentional motor function and purposive movements, even though comprehension and sensorimotor systems are intact. The client may name objects correctly (fork, shoes, comb) but may be unable to use them as intended.

The loss of ability to recognize, comprehend, and identify objects is known as *agnosia.* The client's vision and sensorimotor systems may be intact, but the client may be unable to name specific indicated objects (e.g., light, table, flowers). As the disorder progresses, the client is unable to recognize own spouse or even own image in a mirror.

Disturbance of executive function refers to the inability to carry out higher-level thinking such as planning, organizing, initiating functions, performing tasks in a new way, or sequencing objects or tasks. Clients also have difficulty thinking abstractly. For example, clients find they are unable to take care of finances, organize a trip, or comprehend and participate in discussions unless the topics are concrete.

Sensory-perceptual alterations frequently occur in the later stages of dementia. Psychotic symptoms may manifest as hallucinations, delusions, or illusions.

All symptoms of dementia represent a decline from the client's usual or normal functioning and culminate in an inability to participate in life that eventually leads to death. In addition to cognitive impairments, multiple physical symptoms arise as part of the progressive decline (see Box 8-4).

N OTE: Delirium may occur in dementia.

Amnestic and Other Cognitive Disorders

The defining characteristic of amnestic disorder is *impaired memory* directly related to effects of a general medical condition or to ongoing effects of one or more substances (drugs of abuse, medications, toxins), or a combination of the two. The ability to learn and recall new information is always impaired in amnestic and other cognitive disorders. Depending on the areas of the brain affected, clients are unable to recall verbal or visual information that was communicated a few minutes before. Recent memory is impaired. Remote memory is frequently intact; the client is able to remember stories from the distant past, but may be unable to tell the nurse how he got to the health care facility, who brought him, or the name of the agency. The client may *confabulate* (create information to fill in memory gaps) in early stages of amnestic disorder.

Disorientation to person and place is common in amnestic and other cognitive disorders; disorientation to self is usually associated with dementia. Because of lack of insight in amnestic

disorder, the client with severe memory impairment may become argumentative when confronted, even though situational facts and events confirm the memory deficit.

Lesson-11 Neuro-Psychological Delusional syndrome, Neuro-psychological syndrome,

Neuro-Psychological disorder with HIV – 1 infections

Amnesia refers to the loss of memories, including facts, information and experiences. Movies and television tend to depict amnesia as forgetting your identity, but that's not generally the case in real life.

Instead, people with amnesia — also called amnestic syndrome — usually know who they are. But they may have trouble learning new information and forming new memories.

Amnesia can be caused by damage to areas of the brain that are vital for memory processing. Unlike a temporary episode of memory loss, called transient global amnesia, amnesia can be permanent.

There's no specific treatment for amnesia, but treatment can be directed at the underlying cause. Tips to help enhance memory and get support can help people with amnesia and their families cope.

Overview. Amnesia refers to the loss of memories, including facts, information and experiences. Movies and television tend to depict amnesia as forgetting your identity, but that's not generally the case in real life. Instead, people with amnesia — also called amnestic syndrome — usually know who they are.

Symptoms

The two main features of amnesia are:

- Trouble learning new information.
- Trouble remembering past events and previously familiar information.

Most people with amnesia have problems with short-term memory, so they can't retain new information. Recent memories are most likely to be lost. More-remote or deeply ingrained memories may be spared.

For example, people may recall experiences from childhood or know the names of past presidents. But they may not be able to name the current president, know the month or remember what they ate for breakfast.

Isolated memory loss doesn't affect a person's intelligence, general knowledge, awareness or attention span. It also doesn't affect judgment, personality or identity. People with amnesia usually

can understand written and spoken words and can learn skills such as bike riding or piano playing. They may understand they have a memory disorder.

Amnesia isn't the same as dementia. Dementia often includes memory loss but also involves other problems with thinking that lead to a decline in daily functioning. These problems include having trouble with language, judgment and visual-spatial skills.

Memory loss also is a common symptom of mild cognitive impairment. This disorder involves memory and other cognitive problems that aren't as severe as those experienced in dementia.

Additional symptoms

Depending on the cause of the amnesia, other symptoms may include:

- False memories that are either completely invented or are real memories misplaced in time.
- Confusion or disorientation.

Causes

Typical memory function involves many parts of the brain. Any disease or injury that affects the brain can affect memory.

Amnesia can result from damage to brain structures that form the limbic system, which controls emotions and memories. They include the thalamus found deep within the center of the brain. They also include the hippocampal formations found within the temporal lobes of the brain.

Amnesia caused by brain injury or damage is known as neurological amnesia. Possible causes of neurological amnesia include:

- Stroke.
- Brain inflammation, which may be due to an infection with a virus such as herpes simplex virus. Or inflammation may be a result of an autoimmune reaction to cancer somewhere in the body. It also may be due to an autoimmune reaction in the absence of cancer.
- Not enough oxygen in the brain. This may happen as a result of a heart attack, respiratory distress or carbon monoxide poisoning.
- Long-term alcohol misuse that leads to too little vitamin B-1, known as thiamin, in the body. When this happens, it's called Wernicke-Korsakoff syndrome.
- Tumors in areas of the brain that control memory.
- Alzheimer's disease and other diseases that involve the degeneration of nerve tissue.
- Seizures.

- Certain medicines such as benzodiazepines or others that act as sedatives.

Head injuries that cause a concussion, whether from a car accident or sports, can lead to confusion and problems remembering new information. This is especially common in the early stages of recovery. Mild head injuries typically don't cause lasting amnesia, but more-severe head injuries may cause permanent amnesia.

Another rare type of amnesia, called dissociative amnesia, stems from emotional shock or trauma. It can result from being the victim of a violent crime or experiencing other trauma. In this disorder, people may lose personal memories and information about their lives. The memory loss is usually brief.

Risk factors

The chance of developing amnesia might increase if you've experienced:

- Brain surgery, head injury or trauma.
- Stroke.
- Alcohol abuse.
- Seizures.

Complications

Amnesia varies in severity and scope. But even mild amnesia takes a toll on daily activities and quality of life. The syndrome can cause problems at work, at school and in social settings.

It may not be possible to recover lost memories. Some people with severe memory problems need to be supervised or need to live in a care facility.

Prevention

Damage to the brain can be a root cause of amnesia. It's important to take steps to minimize your chance of a brain injury. For example:

- Don't drink large amounts of alcohol.
- Wear a helmet when bicycling and a seat belt when driving.
- Treat infections quickly so that they don't have a chance to spread to the brain.
- Get immediate medical treatment if you have symptoms that suggest a stroke or brain aneurysm. Those symptoms include a severe headache, feeling numb on one side of the body or not being able to move one side of the body.

Amnesia is a general term that describes memory loss. The loss can be temporary or permanent, but 'amnesia' usually refers to the temporary variety. Causes include head and brain injuries, certain drugs, alcohol, traumatic events, or conditions such as Alzheimer's disease. Anterograde amnesia means that the person can't learn anything new, while retrograde amnesia means the person forgets events from their past. In many mild cases, such as those caused by concussion, the person can't recall the blow to the head or their recovery time, but the rest of their memory is intact. Infantile amnesia, or the inability to remember anything at all from the first few months or years of life, is universal. It is thought that the memory processes of the brain take time to develop.

Symptoms of amnesia

The symptoms of amnesia depend on the cause, but generally include:

- Memory loss
- Confusion
- Inability to recognise familiar faces or places
- Once the person recovers, they typically have no memory of their amnesia episode.

Causes of amnesia

Memory loss can be caused by a wide range of conditions, including:

- Head injury
- Severe illness
- High fever
- Seizures
- Emotional shock or hysteria
- Alcohol-related brain damage
- Certain drugs, such as barbiturates or heroin
- General anaesthetics
- Electroconvulsive therapy
- Stroke
- Transient ischaemic attack (a 'mini stroke')
- Alzheimer's disease
- Brain surgery.

Memory is mysterious

The way the brain codes and stores information remains a mystery. Current and often conflicting theories include:

- Widespread distribution of memories across the outermost layer of the brain (cortex)
- Widespread distribution of memories, with certain structures (such as the hippocampus) playing important roles
- Localised memory areas in certain structures of the brain only
- Memory storage in brain molecules

- Memory storage in clusters of nerve cells (neurones).

Different types of memory

It seems that the brain has a number of different memory forms, including:

- **Short term** - new information is stored for a brief time. If the information isn't further processed, it will soon be forgotten.
- **Long term** - information from the short term memory is shifted to the long term memory.
- **Declarative** - conscious memories of information and events.
- **Non-declarative** - once learned, habits such as driving a car are ingrained and automatic.

The severity varies

The severity of amnesia depends on the cause. Concussion, seizures and electroconvulsive therapy seem to temporarily disrupt the electrical activity of the brain and prevent the short term memory from working properly. The stress of a traumatic event may also interfere with the processing of short term memories. These episodes of amnesia are usually temporary. In more severe cases, memory loss may be permanent. Brain trauma or some types of brain surgery, may damage vital memory structures. The type of memory loss experienced depends on which parts of the brain are affected. In Alzheimer's disease, the person typically remembers past events, but experiences memory losses dating from the onset of their condition.

Diagnosis of amnesia

Diagnosing the cause of amnesia involves a range of tests, including:

- Medical history
- General examination
- Tests for short and long term memory recall
- Other tests related to thought processing
- Head x-ray
- Blood tests
- Computed tomography (CT) scan
- Cerebral angiography (scans taken after a special dye is injected).

Treatment for amnesia

Treatment depends on the cause. For example, a person who has suffered a traumatic event may benefit from sedation, plenty of love and care, and (perhaps) psychiatric treatment. Concussion needs rest, once complications have been ruled out. If alcoholism is the cause, then abstinence, emotional support and addressing dietary deficiencies are recommended. In the case of Alzheimer's disease, a range of new medications that enhance the cholinergic function of the brain are now available; however, nursing homes or other extended care options may eventually be needed as the person becomes less able to look after themselves.

Neuropsychiatric disorder is a blanket medical term that encompasses a broad range of medical conditions that involve both neurology and psychiatry.

Common neuropsychiatric disorders include:

- seizures
- attention deficit disorders
- cognitive deficit disorders
- palsies
- uncontrolled anger
- migraine headaches
- addictions
- eating disorders
- depression
- anxiety

Neuropsychiatric disorders in children can range widely in cause, course, and severity.

What causes neuropsychiatric disorders in children?

Some of the most common causes include:

- traumatic head injury (TBI)
- infections
- medication side effects
- attention deficit hyperactivity syndrome (ADHD)
- genetic predisposition
- environmental factors

In some cases, cause may be unknown, may be related to other factors.

What are the symptoms of neuropsychiatric disorders?

The key component of neuropsychiatric disorders is that the symptoms tend to impact brain function, emotion and mood.

These can range from problems with focus and learning in the case of ADHD to sadness, irritability, memory problems, mood problems, depression to a variety of other psychiatric and/or neurological symptoms.

What are neuropsychiatric disorder care options?

Medications, psychological and/or psychiatric therapy are common treatments for neuropsychiatric disorders. Specific treatments will vary based on the precise cause, clinical presentation and severity.

UNIT-4 CLINICAL ASSESSMENT AND PSYCHO DIAGNOSTICS

Lesson -12 Signs and symptoms of mental disorders

erious mental illness (SMI) is a term used by health professionals to describe the most severe mental health conditions. These illnesses significantly interfere with or limit one or more major life activities. Two of the most common SMIs are bipolar disorder and schizophrenia.

Types

There are hundreds of mental illnesses listed in the fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), the American Psychiatric Association's diagnostic manual. The DSM-5 puts illnesses into categories based on their diagnostic criteria.[3]

Anxiety Disorders

This group of mental illnesses is characterized by significant feelings of anxiety or fear, accompanied by physical symptoms such as shortness of breath, rapid heartbeat, and dizziness.

Three major anxiety disorders are:

- Generalized anxiety disorder (GAD)
- Panic disorder
- Social anxiety disorder (SAD)

Bipolar and Related Disorders

Formerly known as manic depression, bipolar disorders are characterized by alternating episodes of mania, hypomania, and major depression.

There are three broad types of bipolar disorder:

- Bipolar I
- Bipolar II
- Cyclothymia

How Mania Varies Between the Bipolar Types

Depressive Disorders

The common feature of all depressive disorders is the presence of sad, empty, or irritable mood, accompanied by physical symptoms and cognitive changes that significantly affect a person's capacity to function.

Examples include major depressive disorder and premenstrual dysphoric disorder (PMDD).

Disruptive, Impulse-Control, and Conduct Disorders

A group of psychiatric conditions that involve problems with the self-control of emotions and behaviors.

Disorders in this group include:

- Intermittent explosive disorder
- Kleptomania
- Oppositional defiant disorder (ODD)
- Pyromania

Dissociative Disorders

This group of psychiatric syndromes is characterized by an involuntary disconnection between consciousness, memories, emotions, perceptions, and behaviors—even one's own identity or sense of self.

Elimination Disorders

Children with elimination disorders repeatedly void urine or feces at inappropriate times and in inappropriate places, whether the action is involuntary or not.

Feeding and Eating Disorders

Eating disturbances are characterized by a persistent disturbance of eating patterns that leads to poor physical and psychological health.

Three major eating disorders include:

- Anorexia nervosa
- Binge-eating disorder
- Bulimia nervosa

Gender Dysphoria

Formerly known as gender identity disorder, gender dysphoria occurs when a person feels extreme discomfort or distress because their gender identity is at odds with the gender they were assigned at birth.

Neurocognitive Disorders

These disorders are characterized by a decrease in a person's previous level of cognitive function. In addition to Alzheimer's disease, other conditions in this category include:

- Huntington's disease
- Neurocognitive issues due to HIV infection
- Traumatic brain injury (TBI)

Medication or Substance-Induced Neurocognitive Disorder
Neurodevelopmental Disorders

These disorders typically manifest early in development, often before a child enters grade school. They are characterized by impairments of personal, social, academic, or occupational functioning.

Examples of neurodevelopmental disorders include:

- Attention-deficit/hyperactivity disorder (ADHD)
- Autism
- Learning and intellectual disabilities

Obsessive-Compulsive and Related Disorders

As the name suggests, these disorders are characterized by the presence of obsessions and/or compulsions.

Examples obsessive-compulsive and related disorders include:

- Body dysmorphic disorder
- Hoarding disorder
- Obsessive-compulsive disorder (OCD)

Paraphilic Disorders

Describes intense or persistent sexual interests that cause distress or impairment. These may involve recurrent fantasies, urges, or behaviors involving atypical sexual interests.

Personality Disorders

These disorders are characterized by an enduring inflexible pattern of experience and behavior that causes distress or impairment. There are currently 10 recognized personality disorders.

Schizophrenia Spectrum and Other Psychotic Disorders

These disorders are defined by abnormalities in one or more of the following areas:

- Delusions
- Disorganized thinking
- Disorganized or abnormal motor behavior
- Hallucinations
- Negative symptoms

Sexual Dysfunctions

This heterogeneous group of disorders is characterized by a person's inability to fully engage in or experience sexual pleasure.

Some of the most common sexual dysfunctions include:

- Delayed ejaculation
- Erectile disorder
- Female orgasmic disorder
- Female sexual interest/arousal disorder

Sleep-Wake Disorders

There are several different types of sleep-wake disorders, and all involve problems falling asleep or staying awake at desired or socially appropriate times.

These disorders are characterized by misalignment of circadian rhythms with the surrounding environment or abnormalities of the circadian system itself. Common sleep-wake disorders include insomnia and narcolepsy.

Somatic Symptom and Related Disorders

People with these disorders feel extreme, exaggerated anxiety about physical symptoms—such as pain, weakness, or shortness of breath. This preoccupation is so intense that it disrupts the person's daily life.

Substance-Related and Addictive Disorders

All substance-related disorders are characterized by a cluster of behavioral and physical symptoms, which can include withdrawal, tolerance, and craving. Substance-related disorders can result from the use of 10 separate classes of drugs.

Trauma and Stressor-Related Disorders

This group includes disorders that were related to exposure to a traumatic or stressful event. The most common is post-traumatic stress disorder (PTSD).

Signs and Symptoms

Everyone experiences peaks and valleys in their mental health. A stressful experience, such as the loss of a loved one, might temporarily diminish your psychological well-being. In general, in order to meet the criteria for mental illness, your symptoms must cause significant distress or interfere with your social, occupational, or educational functioning and last for a defined period of time.

Each disorder has its own set of symptoms that can vary greatly in severity, but common signs of mental illness in adults and adolescents can include:[4]

- **Excessive fear or uneasiness**: Feeling afraid, anxious, nervous, or panicked
- **Mood changes**: Deep sadness, inability to express joy, indifference to situations, feelings of hopelessness, laughter at inappropriate times for no apparent reason, or thoughts of suicide
- **Problems thinking**: Inability to concentrate or problems with memory, thoughts, or speech that are hard to explain
- **Sleep or appetite changes**: Sleeping and eating dramatically more or less than usual; noticeable and rapid weight gain or loss
- **Withdrawal**: Sitting and doing nothing for long periods of time or dropping out of previously enjoyed activities

It's important to note that the presence of one or two of these signs alone doesn't mean that you have a mental illness. But it does indicate that you may need further evaluation.

If you're experiencing several of these symptoms at one time and they're preventing you from going about your daily life, you should contact a physician or mental health professional.

Causes

There is no single cause of mental illness. Instead, it’s thought that they stem from a wide range of factors (sometimes in combination). The following are some factors that may influence whether someone develops a mental illness:[5]

- **Biology**: Brain chemistry plays a major role in mental illnesses. Changes and imbalance in neurotransmitters, the chemical messengers within the brain, are often associated with mental disorders.
- **Environmental exposures**: Children exposed to certain substances in utero may be at higher risk of developing mental illness. For example, if your mother drank alcohol, used drugs, or was exposed to harmful chemicals or toxins when she was pregnant with you, you may be at increased risk.
- **Genetics**: Experts have long recognized that many mental illnesses tend to run in families, suggesting a genetic component. People who have a relative with a mental illness—such as autism, bipolar disorder, major depression, and schizophrenia—may be at a higher risk of developing it, for example.
- **Life experiences**: The stressful life events you've experienced may contribute to the development of mental illness. For example, enduring traumatic events might cause a condition like PTSD, while repeated changes in primary caregivers in childhood may influence the development of an attachment disorder.

Diagnosis

Diagnosis of a mental illness is a multi-step process that may include more than one healthcare provider, often starting with your primary care physician.

Physical Exam

Before a diagnosis is made, you may need to undergo a physical exam to rule out a physical condition. Some mental illnesses, such as depression and anxiety, can have physical causes. Thyroid problems and other physical diseases can also sometimes be misdiagnosed as mental health disorders due to overlapping or similar symptoms; this is why a thorough physical exam is essential.

Your doctor will take a lengthy history and may order lab tests to rule out physical issues that could be causing your symptoms. If your doctor doesn't find a physical cause for your symptoms, you'll likely be referred to a mental health professional so you can be evaluated for mental illness.

Is There a Blood Test for Depression?
Psychological Evaluation

A mental health professional, such as a psychiatrist or psychologist, will ask you a series of questions related to your symptoms and family history. They may even ask one of your family members to participate in the interview so they can describe the symptoms they see.

Sometimes, the mental health professional will administer tests and other psychological evaluation tools to pinpoint your exact diagnosis or help determine the severity of your illness. Most psychiatrists and psychologists use the APA's Diagnostic and Statistical Manual of Mental Disorders (DSM) to diagnose mental health illnesses.

This manual contains descriptions and symptoms for all of the different mental illnesses. It also lists criteria like what symptoms must be present, how many, and for how long (along with conditions that should not be present) in order to qualify for a particular diagnosis. This is known as the diagnostic criteria.

It's not uncommon to be diagnosed with more than one mental illness. Some conditions increase the risk of other disorders. For instance, sometimes an anxiety disorder can develop into a depressive disorder.

Lesson -13 Intellectual Assessment; Personality Assessment; Neuropsychological Assessment

Intellectual assessment and intelligence testing refer to the evaluation of an individual's general intellectual functioning and cognitive abilities. Intellectual Functioning Intelligence tests provide at least one measure of ''general intellectual functioning'' and are usually administered by clinical psychologists in community settings and by school psychologists in schools. General intellectual functioning typically refers to one's global or overall level of intelligence, often referred to as IQ (intelligence quotient). Higher IQ scores are assumed to mean that the individual has higher intellectual functioning. Unfortunately, this single score indicates general functioning. It does not necessarily explain, for example, why a given student does not know how to read even though he or she is in the fifth grade, or if the student has some special skills in an area such as art, music, or learning a foreign language. More importantly, because a global score cannot tell us in what specific area a child has difficulty or talent, it is not useful for drawing conclusions about how this child learns or should be taught. Performance on a single measure of intellectual ability might be useful as a starting point in efforts to understand a student's skills and needs, when used in combination with other sources of information including measures of cognitive abilities.

Cognitive Abilities One way psychologists have tried to gather information about the specific abilities that explain children's learning and learning problems is by focusing on cognitive abilities rather than general intelligence. Cognitive abilities are those skills that make up an individual's general intelligence. Although past theories focused on intelligence as a single global ability, modern theories view intelligence as composed of many different abilities. According to recent research, this concept of multiple abilities seems to better explain Helping Children at Home and School III | S2H12–1 why individuals may perform very well on some types of tasks and poorly on others. See the Recommended Resources below for more information about this view of cognitive ability. M

ETHODS OF ASSESSING INTELLIGENCE Intellectual or cognitive abilities are typically assessed using a combination of standardized tests and ecological measures. Standardized and Norm-Referenced Tests The most common measures of intellectual ability are standardized, norm-referenced intelligence tests, many of which yield a score where average performance is set at 100. The term IQ (or Full Scale IQ–FSIQ) is the traditional designation for these scores, which represent global intelligence and are associated most closely with the Wechsler Scales of Intelligence. Other tests use different terms for scores that mean essentially the same thing, such as General Intellectual Ability (GIA) or Mental Processing Index (MPI).

Standardized tests. Intelligence tests or batteries are made up of a series of tasks or subtests that are usually administered on an individual basis, although some group-administered tests are available. These tasks, intended to provide samples of a person's intelligence or cognitive abilities, with each yielding a score, are referred to as standardized because each task is presented to each examinee in the same or standardized way. When standardized tests are used, performance is thought to show a person's unique cognitive abilities, taking into account any error resulting from factors such as the individual having a bad day or imperfections in the test itself. Norm-referenced tests. To understand how the individual compares to others, the test scores are then compared to the test's norms. Norms are established when tests are first developed, using large groups of individuals (often reflecting characteristics of the general population in terms of age, gender, race/ethnicity, geographic region, and socioeconomic status) to determine the range of typical performance. For example, if a child gets 3 questions correct on a test of vocabulary and the norms tell us that most children of the same age correctly respond to 8 to 10 questions, we could say that this child's vocabulary skills are poor in comparison to most children of the same age. Most intelligence tests are both norm-referenced and standardized. Typically, norm referenced intelligence tests are designed so that the mean or average score falls at 100, and about two-thirds of the population taking the test obtains scores between 85 and 115, which is considered the normal range.

Common intelligence tests. Commonly used normreferenced tests include the Wechsler Preschool and Primary Scale of Intelligence (WPPSI), Wechsler Intelligence for Children (WISC), Woodcock–Johnson Tests of Cognitive Abilities (WJ), the Stanford–Binet Intelligence Scale (SB), the Differential Abilities Scale (DAS), and the Kaufman Assessment Battery for Children (KABC). There also exist brief measures of intelligence such as the Wechsler Abbreviated Scale of Intelligence (WASI) and the Kaufman Brief Intelligence Test (K–BIT). All of these measures have been updated and renormed over time. Assessments should be conducted using the most recent edition of the test to ensure up-to-date test norms. Ecological Assessment Procedures In addition to formal intelligence tests, intellectual assessment often includes ecological information—that is, information about the student's environment and history, or the context of the student's learning and development. These procedures typically include a review of educational records; observations of the child in different environments (such as classroom, playground, gym); interviews with parents, teachers, and other professionals involved with the student; developmental history; classroom work samples (reading assignments, class tests, workbooks); and measures of adaptive behavior (the student's degree of independence in various situations). Intelligence test scores can easily be misinterpreted if ecological information is overlooked. For example, an extremely low overall IQ score on an intelligence test could lead one to conclude that a student might have mental retardation (very limited intellectual ability). However, there are many factors that could prevent an individual from demonstrating optimal performance on a given test. These factors include illness, fatigue, poor motivation, missed instruction due to frequent absences from school, frequent moves or school changes, poor vision or hearing, inattention, or difficulties with the English language (a common issue for immigrant students). Without evaluating the possible presence of these kinds of influences, it is imp

Intellectual assessment has changed a great deal in the twentieth century. It has moved from assessments based on language and speech patterns to sensory discrimination, with most early assessments being for the mentally deficient. Gradually, the assessment instruments developed into the precursors of the standardized instruments used in the 1990s, which measure more complex cognitive tasks for all levels of cognitive ability. The most commonly used tests, the Wechsler scales, were not developed out of theory, but were guided rather by clinical experience. In the progression of test development, the relative alternates to the Wechsler have been more theory-based. The direction in test development in the 1990s seems to be continuing to lead to theory being at the base of intelligence tests, instead of being just purely clinically driven. Psychometric theories and neurological theories are growing as the basis for new instruments. However, despite this proliferation of new theory-driven tests, there is a definite conservatism that holds on to the past, reluctant to let Wechsler tests be truly rivaled. Part of this hold on the past is research-based and part is clinically-based. Because the Wechsler tests have been reigning supreme for so long, there has been a mountain of research studies using the Wechsler scales. Thus, clinicians have a good empirical basis to form their understanding of what a specific Wechsler profile may be indicating. Clinically, psychologists are also quite comfortable and familiar with the Wechsler scales. A good clinician who has done many assessments may be familiar enough with every nook and cranny of the WAIS-R to barely need the manual to administer it. Thus, the field so far has changed relatively slowly. Computer based technology is likely going to ultimately shape the field of assessment by 2020. Computer scoring programs and computer assisted reports are already in use, and the future is likely to include a much greater progression of technologically advanced instruments for assessing intelligence.

Neuropsychological Testing
Glenn T. Stebbins, in Textbook of Clinical Neurology (Third Edition), 2007
Clinical Situations for Tests of Intellectual Abilities
Intellectual assessment is very important for patients presenting with impairments of higher cognitive functioning such as memory, language, and reasoning. Because these higher cognitive functions are highly correlated with intelligence, intelligence assessment allows the examiner to interpret scores of memory, language, and reasoning performance in light of basic intellectual abilities. Intellectual assessments are not particularly helpful in patients with disturbances in level of consciousness (see Chapter 1), disrupted sleep/wake cycles (see Chapter 2), or neglect and dyspraxia (see Chapter 4). Additionally, the comparison of intellectual abilities and academic achievement is helpful in identifying patients who may have learning disabilities. In children with developmental or neurodegenerative disorders affecting the cortex or subcortex (see Chapters 28 and 34Chapter 28Chapter 34) and those with chromosome alterations, enzyme defects, or storage abnormalities (see Chapters 30–32Chapter 30Chapter 31Chapter 32), these assessments help in tailoring educational efforts and expectations. When intelligence testing is requested, the referring physician should ask for a reporting of subscale performance as well as summary IQ scores.

Neuropsychological assessment is a performance-based method to assess cognitive functioning. This method is used to examine the cognitive consequences of brain damage, brain disease, and severe mental illness.

What is neuropsychology?

Neuropsychology is a specialty field that joins the medical fields of neurology, psychology and psychiatry. Neuropsychology involves determining how well the brain is working when it is disrupted by a brain injury or psychological disorder. A neuropsychological assessment is a comprehensive test of a wide range of mental functions including behavior.

What mental functions are assessed in a neuropsychological exam?

If you are undergoing a neuropsychological assessment, the mental functions tested include:

- General intellect.
- Reading/reading comprehension.
- Language usage and understanding of what others say.
- Attention/concentration.
- Processing speed.
- Learning and memory.
- Reasoning.
- Executive functions, which are higher-level skills you use to organize and plan, manage your time, problem solve, multi-task, make judgments and maintain self-control.
- Visuospatial skills.
- Motor speed and dexterity.
- Mood and personality.

Why has a neuropsychological assessment been requested?

A neuropsychological assessment can be requested for a number of reasons including:

- **To help with diagnosis**: Test results are sometimes used to help understand the cause of problems with your thinking and understanding. For example, test results might be used to determine if your cognitive (mental) changes are due to normal aging, a neurological illness, depression, anxiety or other causes. Your healthcare provider can then use the results of your neuropsychological examination along with the results of other tests, such as brain scans, EEGs, and blood tests, to arrive at a diagnosis that will help to guide your healthcare.
- **To determine cognitive strengths and weaknesses**: In some cases, a healthcare provider may order tests if you've had a known neurological event or injury, such as a stroke or traumatic brain injury, to find which cognitive functions have changed and how much they have changed.
- **To establish a baseline**: In some instances, an exam is performed before and after a medical or surgical treatment to determine if cognitive abilities were affected by the intervention. Re-examination at some point after the baseline can also demonstrate how well you're recovering from a stroke or traumatic brain injury and whether or not you're ready to return to work, resume driving or take on other duties.
- **To help plan a treatment or other intervention**: Test results can be used to identify which cognitive abilities should be the focus of rehabilitation if you've had a brain injury. Results

also help therapists determine which strengths might be able to compensate for weaknesses. The evaluation can provide the basis for making decisions and/or adjustments to school or work schedules and determine the skills to work on that are most important to you.

How will I know if my cognitive abilities have changed if I have not had an exam in the past?

Some cognitive abilities tend to be very stable despite neurologic illnesses or injuries. Those abilities often provide an estimate of the level of your other cognitive abilities if no injury or illness had occurred. Your results will be compared to the pattern of results associated with various illnesses or injuries to help determine if changes have occurred.

What common complaints might signal the need for a neuropsychological test?

See your healthcare professional for a referral for a neuropsychological assessment if you or a loved one show any of the following signs:

- Changes in short-term memory, asks the same question repeatedly.
- Frequently loses items, gets lost easily.
- Confusion.
- Poor attention and concentration. Doesn't appear to listen. Gets confused in conversation.
- Language difficulty. Has difficulty speaking or finding words or cannot understand what others say.
- Can't recognize things.
- Visuospatial difficulties including difficulty drawing or using a map.
- Poor judgment/decision-making.
- An unexplained change in personality, increase in anxiety or depression, development of delusions or hallucinations.
- New difficulty with understanding or managing bills or finances.
- Can't recognize familiar persons.

What does a neuropsychological exam involve?

The neuropsychologist will talk with you to understand any concerns you and your family members might have about your cognitive (mental) functioning. He or she will also review your medical and psychological history and educational background. If a family member comes to the evaluation with you, the neuropsychologist may ask for your permission to interview him or her as well.

Your neuropsychologist will choose the tests that you are given. The tests are given and scored by a trained technician called a psychometrist who works under the supervision of the neuropsychologist. The tests typically involve writing or drawing, solving puzzles or answering questions, and responding to things presented on a computer. Most people find some of the tests to be quite easy and others to be difficult. It is important to work as hard as possible on all of the tests in order for the results to be most informative.

You will also complete questionnaires about mood and psychological symptoms. Parents of children referred for neuropsychological examinations often complete questionnaires about their child's behavior.

Finally, the neuropsychologist writes a report that summarizes the results and includes recommendations for improving cognition (e.g., attention, memory) and possibly referrals to other professionals.

How long does a neuropsychological exam take?

The length of time for testing varies considerably based on the nature of the reason for the examination. Depending on the situation, testing can take anywhere between one and eight hours, although two to four hours is typical. The testing time depends on which tests need to be administered and how quickly you are able to work comfortably. You will be allowed to take some breaks depending on how you are feeling and the length of the test.

What should I do to prepare for the neuropsychological examination?

- Get a good night's sleep.
- Try to eat a good breakfast.
- Take all of your medications as usual unless you are directly instructed to do otherwise.
- If you use glasses, contact lenses, or hearing aids, make sure you have them with you.
- If you have had any neuropsychological, psychological or academic testing done in the past, bring those records with you.
- If your child is undergoing the testing, and he or she has completed an intellectual evaluation, psychoeducational evaluation, multifactored evaluation (MFE), or individual education program (IEP), bring copies of the results of those evaluations.

How are test results interpreted?

Most of the tests used in neuropsychology are standardized, which means they are given the same way to everybody. The tests are also norm-referenced, which means that a patient's performance on those tests will be compared to the performance of other people who are about the same age and, sometimes, people who have the same educational background. The test results are used to answer many types of questions.

Lesson -14 Clinical interviewing and testing- Psycho-diagnostics- intake/ an amnesis – developing hypotheses

All clinical interviews follow a common process or outline. Shea (1998) offered a generic or atheoretical model, including five stages: (1) introduction, (2) opening, (3) body, (4) closing, and (5) termination. Each stage includes specific relational and technical tasks.

Introduction

The introduction stage begins at first contact. An introduction can occur via telephone, online, or when prospective clients read information about their therapist (e.g., online descriptions, informed consents, etc.). Client expectations, role induction, first impressions, and initial rapport-building are central issues and activities.

First impressions, whether developed through informed consent paperwork or initial greetings, can exert powerful influences on interview process and clinical outcomes. Mental health professionals who engage clients in ways that are respectful and culturally sensitive are likely to facilitate trust and collaboration, consequently resulting in more reliable and valid assessment data (Ganzini et al., 2013). Technical strategies include authentic opening statements that invite collaboration. For example, the clinician might say something like, "I'm looking forward to getting to know you better" and "I hope you'll feel comfortable asking me whatever questions you like as we talk together today." Using friendliness and small talk can be especially important to connecting with diverse clients (Hays, 2016; Sue & Sue, 2016). The introduction stage also includes discussions of (1) confidentiality, (2) therapist theoretical orientation, and (3) role induction (e.g., "Today I'll be doing a diagnostic interview with you. That means I'll be asking lots of questions. My goal is to better understand what's been troubling you."). The introduction ends when clinicians shift from paperwork and small talk to a focused inquiry into the client's problems or goals.

Opening

The opening provides an initial focus. Most mental health practitioners begin clinical assessments by asking something like, "What concerns bring you to counseling today?" This question guides clients toward describing their presenting problem (i.e., psychiatrists refer to this as the "chief complaint"). Clinicians should be aware that opening with questions that are more social (e.g., "How are you today?" or "How was your week?") prompt clients in ways that can unintentionally facilitate a less focused and more rambling opening stage. Similarly, beginning with direct questioning before establishing rapport and trust can elicit defensiveness and dissembling (Shea, 1998).

Many contemporary therapists prefer opening statements or questions with positive wording. For example, rather than asking about problems, therapists might ask, "What are your goals for our meeting today?" For clients with a diverse or minority identity, cultural adaptations may be needed to increase client comfort and make certain that opening questions are culturally appropriate and relevant. When focusing on diagnostic assessment and using a structured or semi-structured interview protocol, the formal opening statement may be scripted or geared toward obtaining an overview of potential psychiatric symptoms (e.g., "Does anyone in your family have a history of mental health problems?"; Tolin et al., 2018, p. 3).

Body

The interview purpose governs what happens during the body stage. If the purpose is to collect information pertaining to psychiatric diagnosis, the body includes diagnostic-focused questions. In contrast, if the purpose is to initiate psychotherapy, the focus could quickly turn toward the history of the problem and what specific behaviors, people, and experiences (including previous therapy) clients have found more or less helpful.

When the interview purpose is assessment, the body stage focuses on information gathering. Clinicians actively question clients about distressing symptoms, including their frequency, duration, intensity, and quality. During structured interviews, specific question protocols are

followed. These protocols are designed to help clinicians stay focused and systematically collect reliable and valid assessment data.

Closing

As the interview progresses, it is the clinician's responsibility to organize and close the session in ways that assure there is adequate time to accomplish the primary interview goals. Tasks and activities linked to the closing include (1) providing support and reassurance for clients, (2) returning to role induction and client expectations, (3) summarizing crucial themes and issues, (4) providing an early case formulation or mental disorder diagnosis, (5) instilling hope, and, as needed, (6) focusing on future homework, future sessions, and scheduling (Sommers-Flanagan & Sommers-Flanagan, 2017).

Termination

Termination involves ending the session and parting ways. The termination stage requires excellent time management skills; it also requires intentional sensitivity and responsiveness to how clients might react to endings in general or leaving the therapy office in particular. Dealing with termination can be challenging. Often, at the end of an initial session, clinicians will not have enough information to establish a diagnosis. When diagnostic uncertainty exists, clinicians may need to continue gathering information about client symptoms during a second or third session. Including collateral informants to triangulate diagnostic information may be useful or necessary.

in psychodiagnostic the data or information gathering can be divided into three main phases: 1) Collection of data. 2) Processing and interpretation of data. 3) Decision making that is psychodiagnosis and prognosis.

Korchin and Schuldberg (1981) define psychodiagnosis as a process that :/1 a) uses a number of procedures, b) intended to tap various areas of psychological functions, c) both at a conscious and unconscious level, d) using projective techniques as well as more objective and standardized tests, e) in both cases, interpretation may rest on syinbolic signs as well as scoreable responses, 6 f) with the goal of describing individuals in personological rather than normative terms. (p. 1147) 1) Individual tests of general mental ability, ,7 The term psychodiagnosis might be applied more aptly to the neutral term clinical assessment. The central difference between clinical assessment and other testing applications is that the clinician, rather than the test, is at the center of the assessment process. The clinician has two distinct functions, both of which are essential to the assessment process. First, the clinician must gather data. Although standardized tests are used in clinical assessment, projective tests, interviews, and behavioural observations represent the clinician's most important measurement tools. Second, the clinician must integrate data from various tests, interviews, and observations to form an overall assessment of the individual. Introduction to Psychodiagnostics, Deflnition, Concept and Description The data gathering function has clear implications for the quality of psychological measurement. A clinician who makes inaccurate observations, conducts poorly structured interviews, or misinterprets or misrecords responses to open ended questions or ambiguous stimuli (e.g., responses to Rorschach cards) is not likely to produce valid assessments. The clinician often functions as a measurement instrument, and it is important to assess the reliability and validity of the clinical data he or she gathers. Although it may not be immediately obvious, the clinician's second function the integrationof clinicaldata-alsoaffectsthe qualityof psychological measurement in clinical settings. Assessment represents an attempt to arrive at a valid classification of each individual patient or client. In some cases,

clinicians may be called on to diagnose or assist in the diagnosis of mental or behavioural disorders. In others, the clinician must make recommendations regarding the placement of children or adults in remedial education or in therapeutic programs. In anycase, the classification of individuals represents a fundamental type of measurement, and the clinician's skill in integrating diverse sources of data may be a critical factor in determining the validity of his or her classifications and assessments of individuals.

A hypothesis is a tentative statement about the relationship between two or more variables. It is a specific, testable prediction about what you expect to happen in a study.

One hypothesis example would be a study designed to look at the relationship between sleep deprivation and test performance might have a hypothesis that states: "This study is designed to assess the hypothesis that sleep-deprived people will perform worse on a test than individuals who are not sleep-deprived."

This article explores how a hypothesis is used in psychology research, how to write a good hypothesis, and the different types of hypotheses you might use.

The Hypothesis in the Scientific Method

In the scientific method, whether it involves research in psychology, biology, or some other area, a hypothesis represents what the researchers think will happen in an experiment.[1] The scientific method involves the following steps:

1. Forming a question
2. Performing background research
3. Creating a hypothesis
4. Designing an experiment
5. Collecting data
6. Analyzing the results
7. Drawing conclusions
8. Communicating the results

The hypothesis is a prediction, but it involves more than a guess. Most of the time, the hypothesis begins with a question which is then explored through background research. It is only at this point that researchers begin to develop a testable hypothesis. Unless you are creating an exploratory study, your hypothesis should always explain what you *expect* to happen.

In a study exploring the effects of a particular drug, the hypothesis might be that researchers expect the drug to have some type of effect on the symptoms of a specific illness. In psychology, the hypothesis might focus on how a certain aspect of the environment might influence a particular behavior.

Remember, a hypothesis does not have to be correct.[1] While the hypothesis predicts what the researchers expect to see, the goal of the research is to determine whether this guess is right or wrong. When conducting an experiment, researchers might explore a number of factors to determine which ones might contribute to the ultimate outcome.

In many cases, researchers may find that the results of an experiment *do not* support the original hypothesis. When writing up these results, the researchers might suggest other options that should be explored in future studies.

Hypothesis Format

In many cases, researchers might draw a hypothesis from a specific theory or build on previous research. For example, prior research has shown that stress can impact the immune system. So a researcher might hypothesize: "People with high-stress levels will be more likely to contract a common cold after being exposed to the virus than people who have low-stress levels."

In other instances, researchers might look at commonly held beliefs or folk wisdom. "Birds of a feather flock together" is one example of folk wisdom that a psychologist might try to investigate. The researcher might pose a specific hypothesis that "People tend to select romantic partners who are similar to them in interests and educational level."

Elements of a Good Hypothesis

So how do you write a good hypothesis? When trying to come up with a hypothesis for your research or experiments, ask yourself the following questions:

- Is your hypothesis based on your research on a topic?
- Can your hypothesis be tested?
- Does your hypothesis include independent and dependent variables?

Before you come up with a specific hypothesis, spend some time doing background research. Once you have completed a literature review, start thinking about potential questions you still have. Pay attention to the discussion section in the journal articles you read. Many authors will suggest questions that still need to be explored.

To form a hypothesis, you should take these steps:

- Collect as many observations about a topic or problem as you can.
- Evaluate these observations and look for possible causes of the problem.
- Create a list of possible explanations that you might want to explore.
- After you have developed some possible hypotheses, think of ways that you could confirm or disprove each hypothesis through experimentation. This is known as falsifiability.

Falsifiability of a Hypothesis

In the scientific method**,** falsifiability is an important part of any valid hypothesis.[1] In order to test a claim scientifically, it must be possible that the claim could be proven false.

Students sometimes confuse the idea of falsifiability with the idea that it means that something is false, which is not the case. What falsifiability means is that *if* something was false, then it is possible to *demonstrate* that it is false.

One of the hallmarks of pseudoscience is that it makes claims that cannot be refuted or proven false.

Operational Definitions

A variable is a factor or element that can be changed and manipulated in ways that are observable and measurable. However, the researcher must also define how the variable will be manipulated and measured in the study.

For example, a researcher might operationally define the variable "test anxiety" as the results of a self-report measure of anxiety experienced during an exam. A "study habits" variable might be defined by the amount of studying that actually occurs as measured by time.

These precise descriptions are important because many things can be measured in a number of different ways. One of the basic principles of any type of scientific research is that the results must be replicable.[1] By clearly detailing the specifics of how the variables were measured and manipulated, other researchers can better understand the results and repeat the study if needed.

Some variables are more difficult than others to define. How would you operationally define a variable such as aggression? For obvious ethical reasons, researchers cannot create a situation in which a person behaves aggressively toward others.

In order to measure this variable, the researcher must devise a measurement that assesses aggressive behavior without harming other people. In this situation, the researcher might utilize a simulated task to measure aggressiveness.

Hypothesis Checklist

- Does your hypothesis focus on something that you can actually test?
- Does your hypothesis include both an independent and dependent variable?
- Can you manipulate the variables?
- Can your hypothesis be tested without violating ethical standards?

Types of Hypotheses

The hypothesis you use will depend on what you are investigating and hoping to find. Some of the main types of hypotheses that you might use include:

- **Simple hypothesis**: This type of hypothesis suggests that there is a relationship between one independent variable and one dependent variable.
- **Complex hypothesis**: This type of hypothesis suggests a relationship between three or more variables, such as two independent variables and a dependent variable.
- **Null hypothesis**: This hypothesis suggests no relationship exists between two or more variables.
- **Alternative hypothesis**: This hypothesis states the opposite of the null hypothesis.
- **Statistical hypothesis**: This hypothesis uses statistical analysis to evaluate a representative sample of the population and then generalizes the findings to the larger group.
- **Logical hypothesis**: This hypothesis assumes a relationship between variables without collecting data or evidence.

Hypotheses Examples

A hypothesis often follows a basic format of "If {this happens} then {this will happen}." One way to structure your hypothesis is to describe what will happen to the dependent variable if you change the independent variable.

The basic format might be: "If {these changes are made to a certain independent variable}, then we will observe {a change in a specific dependent variable}."

A few examples of simple hypotheses:

- "Students who eat breakfast will perform better on a math exam than students who do not eat breakfast."
- Complex hypothesis: "Students who experience test anxiety before an English exam will get lower scores than students who do not experience test anxiety."
- "Motorists who talk on the phone while driving will be more likely to make errors on a driving course than those who do not talk on the phone."

Examples of a complex hypothesis include:

- "People with high-sugar diets and sedentary activity levels are more likely to develop depression."
- "Younger people who are regularly exposed to green, outdoor areas have better subjective well-being than older adults who have limited exposure to green spaces."

Examples of a null hypothesis include:

- "Children who receive a new reading intervention will have scores different than students who do not receive the intervention."
- "There will be no difference in scores on a memory recall task between children and adults."

Examples of an alternative hypothesis:

- "Children who receive a new reading intervention will perform better than students who did not receive the intervention."
- "Adults will perform better on a memory task than children."

Collecting Data on Your Hypothesis

Once a researcher has formed a testable hypothesis, the next step is to select a research design and start collecting data. The research method depends largely on exactly what they are studying. There are two basic types of research methods: descriptive research and experimental research.

Descriptive Research Methods

Descriptive research such as case studies, naturalistic observations, and surveys are often used when it would be impossible or difficult to conduct an experiment.[1]

These methods are best used to describe different aspects of a behavior or psychological phenomenon.

Once a researcher has collected data using descriptive methods, a correlational study can then be used to look at how the variables are related. This type of research method might be used to investigate a hypothesis that is difficult to test experimentally.

Experimental Research Methods

Experimental methods are used to demonstrate causal relationships between variables. In an experiment, the researcher systematically manipulates a variable of interest (known as the independent variable) and measures the effect on another variable (known as the dependent variable).

Unlike correlational studies, which can only be used to determine if there is a relationship between two variables, experimental methods can be used to determine the actual nature of the relationship—whether changes in one variable actually *cause* another to change.

Lesson-15 Selecting criteria and materials (tests, questionnaires, observations etc)

Collecting data is when you gather information in mathematics which you can then organise and analyse to answer questions.
To do this effectively you need to know about different data collection methods.

First let us consider the different types of data you can collect because these may be asked about in GCSE Mathematics.

Primary data	Primary data collection can be done by direct observation using data collection sheets or two-way tables. This data is collected directly by the person who is going to analyse it.
Secondary data	Data that is taken from a secondary source rather than collected directly. Secondary data sources could be books or the internet.
Qualitative data	Qualitative data is data in words, for example the colour of student's hair.
Quantitative data	Quantitative data is in numbers, for example the age of students.
Discrete data	Quantitative data which can only take certain values, for example shoe size.
Continuous data	Quantitative data which can take any value, for example time.

You can collect data from the whole group. The whole group is known as the **population**. An example of this is a **census** which takes place every 1010 years in the UK to collect data about the people who live in the UK. The process of gathering this data is time-consuming but is very useful for planning public services such as health care.
A more practical way to gather information for data analysis is to carry out a survey on a smaller section of the population. This smaller selection is known as a **sample**.
There are some important points to consider when taking a sample,

- **Representation** – ideally you want a sample to accurately represent the whole population. Certain sampling methods, such as stratified sampling, are better for this than others.
- **Sample size** – the bigger the sample size, the more accurately it will represent the population. However, a bigger sample size means more time and money will need to be spent on the sample.
- **Bias** – bias in data collecting refers to data which may distort the results of research as it is not truly representative of the situation you are studying. It is very important to avoid bias in data collecting. This might happen when you collect data from a small subset of the population.

Methodology

A **methodology** is an explanation of a **data collection process**. When carrying out research studies, the data collection process can have a significant impact on the **data quality**. Different methods of data collection have their own advantages and disadvantages.
Each research method must be considered to ensure that the person that is collecting the data (the **data processor**) can make an informed decision about the method of data collection, to provide high-quality data for their research.
Data collection techniques include,

- Questionnaires (including online surveys)
- Interviews
- Focus groups
- Observations

- Case studies

  Questionnaires and online surveys

  A **questionnaire** or **online survey** consists of a list of standardised questions about a specific topic. They can be distributed to a large sample of the population and returned to the data processor quickly using data collection tools such as the internet.
  Questionnaires are one of the **most common** types of data collection as they can be **cost-effective** and, when the questionnaire is well structured, can provide enough detail to satisfy the requirements of the researcher.
  Marketing campaigns regularly use the internet, specifically **social media** to broadcast new ideas and products to the market, often followed up by a questionnaire to provide **feedback** about the product or service that has been sold.
  A questionnaire can contain a mixture of closed and open questions. **Closed questions** have a **fixed response** that can be answered by ticking a box or selecting a value on a scale. Closed questions are used to collect **quantitative data** as you can quickly determine the number of participants that select each option.
  For example, the question below is a closed question because the response can only be one of the options provided.

  Open-ended questions are used for **qualitative research**, where the answer to the question is more descriptive and therefore it cannot be answered using a simple tick box.
  For example, the question below is an open-ended question because the response is more descriptive.

  There are **limitations** with questionnaires compared to other data gathering methods. One of the main issues is the lack of **respondents**. Participants in research studies usually need an **incentive** to complete a questionnaire.
  This may take the form of an external motivator such as a discount or free entry into a prize draw, or there could be an intrinsic motivator such as a strong emotional response or feeling of charity. Without incentives, research studies can struggle to collect enough data for their research to be of any significance.

  Another limitation of questionnaires and online surveys is the **fixed scheme** of questions which does not allow participants to **elaborate** (explain further) on their choice. It is also important that the participant **understands** the question that is being asked which the data processor cannot check if they are not in attendance when the questionnaire is being answered.
  This can lead to **unanswered questions** or answers that are based on a misinterpretation of the question.
  interviews and focus groups

  Interviews are usually conducted between the researcher and an **interviewee**. Interviews provide a chance for the interviewee to elaborate on their answers in more detail, which may provide **more informative** data however interviews are **time consuming** and are **more costly** than other qualitative methods of data collection.
  Focus groups allow a group of participants to answer questions whilst being able to listen and share ideas with other members within the group. Focus groups are useful as they can be more **time efficient** and **informative** compared to other data collection techniques however the dynamics of the individual members of the group can **skew** the results.

To minimise the risk of a **biased data set**, the participants should be selected using an appropriate **sampling technique**. This is important for any research study being carried out.
Observations and case studies

Participant observation is a useful data collection technique to collect **real time** qualitative research.
For example,

A social experiment is carried out on a group of 33 year old children into their communication techniques with one another.
By observing their behaviours around each other, researchers can recognise different methods of communication between the children, rather than through difficult processes such as an interview where the children may not understand research questions or may not be able to explain communication techniques with their peers to the research providers.

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: To assess Mental health status the subject

Activity 2: To determine the intelligence of an individual using Alexander Pass Along test.

SEMESTER - II

PSY25144 POSITIVE PSYCHOLOGY

Objectives:

The meaning and approaches Positive Psychology and the classifications, The nature of Positive Emotional States and its Experiences, The Positive cognitive states and Universal Virtues, The nature of Optimal Experiences and Pro-social behaviour, The meaning of Attachments and Changing Human behavior

Unit I: Introduction

Lesson-1 Positive Psychology: Concept, History, Nature, Dimension and Scope

Lesson-2 Western and Eastern views of Positive Psychology-Athenian Views Judeo Christianity – Confucianism – Taoism – Buddhism – Hinduism

Lesson-3 Positive Psychology in India-Value Systems – Orientation to Time – Thought Processors – Different Ways to Positive Outcomes: Hope – Compassion – Harmony

Lesson-4 Theories: PERMA model, Self-determination theory, Hope theory, Broaden and Build theory.

Unit II: Happiness and Wellbeing

Lesson-5 Introduction to Psychology of happiness, well-being and scope

Lesson-6Age old definition of Happiness – Subjective Wellbeing as a Synonym for Happiness – Types of happiness – Eudemonic and Hedonic, History of Happiness, Traits associated with Happiness

Lesson-7 Happiness across the life span; gender and happiness; marriage and happiness; other facts of life

Lesson-8 Measurement of Happiness (The Steen Happiness Index, Subjective Happiness Scale, Happiness Scale).

Unit III: Wellbeing and Emotions

Lesson-9 Positive emotions, positive relationships and wellbeing, Hope and optimism, love and belongingness, love and kindness.

Lesson-10 Gratitude, forgiveness, empathy and altruism

Lesson-11Positive emotions and health resources - Expanding the Repertoire of Pleasure - Happiness and Subjective Well being- Influence of positive Emotions.

Lesson-12Determinants of Subjective Well being – 21st Century Definition of Happiness – Complete Mental Health: Emotional – Social and Psychological Well being – Increasing Happiness in Life

Lesson-13 Positive traits, personality, emotions and biology, positive beliefs.

Unit IV: Wellbeing and Spirituality

Lesson-14 Religion, spirituality and wellbeing: role of religion and spirituality to maintain subjective wellbeing – A special focus to Indian spirituality and well being

Lesson-15 Understanding a Changing Human Behavior Primary Prevention – Components of Effective Primary Prevention – Head Start – Primary Preventions for Racial & Ethnic Minorities – Primary Preventions for Children and Elders – Secondary Preventions – Common Components of Secondary Preventions – Secondary Preventions for Racial & Ethnic Minorities – For Children and Elders .

Lesson-16 Virtues in Indian culture - Primary Enhancement – Psychological Health – Physical Health – Secondary Enhancement – Psychological and Physiological Health – Balancing the Preventions and Enhancement systems

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Outcome:

The meaning and approaches Positive Psychology and the classifications

The nature of Positive Emotional States and its Experiences

The Positive cognitive states and Universal Virtues

The nature of Optimal Experiences and Pro-social behaviour

The meaning of Attachments and Changing Human behaviour

UNIT I: INTRODUCTION

Lesson-1 Positive Psychology: Concept, History, Nature, Dimension and Scope

Positive psychology has been described in many ways and with many words, but the commonly accepted definition of the field is this: —Positive psychology is the scientific study of what makes life most worth living (Peterson, 2008).

To push this brief description a bit further, positive psychology is a scientific approach to studying human thoughts, feelings, and behavior, with a focus on strengths instead of weaknesses, building the good in life instead of repairing the bad, and taking the lives of average people up to —great instead of focusing solely on moving those who are struggling up to —normal

In today‘s world, society is facing extremely tough challenges in the form of global warming, natural disasters, economic recession, unprecedented homelessness, terrorism and the draining continuation of war. With all this sadness and horror, where in the world does a science based on testing happiness, wellbeing, personal growth and the good life‘ fit into the modern-day agenda? Psychology after World War II became a science largely devoted to healing. It concentrated on repairing damage using a disease model of human functioning. This almost exclusive attention to pathology neglected the idea of a fulfilled individual and a thriving community, and it neglected the possibility that building strength is the most potent weapon in the arsenal of therapy.

Positive psychology focuses on wellbeing, happiness, flow, personal strengths, wisdom, creativity, imagination and characteristics of positive groups and institutions. Furthermore, the focus is not just on how to make individuals happy, thereby perpetuating a self-centered, narcissistic approach, but on happiness and flourishing at a group level as well. Positive Psychology looks at how individuals and groups thrive and how increasing the wellbeing of one will have a positive effect on the other, leading to a win-win situation.

Positive psychology is not simply the focus on positive thinking and positive emotions. It‘s much more than that. Indeed, the area of positive psychology is focused on what makes individuals and communities flourish, rather than languish. Flourishing is defined as _a state of positive mental health; to thrive, to prosper and to fare well in endeavors free of mental illness, filled with emotional vitality and function positively in private and social realms.‘

According to positive psychologists, for most of its life mainstream psychology (sometimes also referred to as _psychology as usual‘) has been concerned with the negative aspects of human life. There have been pockets of interest in topics such as creativity, optimism and wisdom, but these have not been united by any grand theory or a broad, overarching framework. This rather negative state of affairs was not the original intention of the first psychologists, but came about through a historical accident. Prior to the Second World War, psychology had three tasks, which were to: cure mental illness, improve normal lives and identify and nurture high talent. However, after the war the last two tasks somehow got lost, leaving the field to concentrate predominantly on the first one.

How did that happen? Given that psychology as a science depends heavily on the funding of governmental bodies, it is not hard to guess what happened to the resources after World War II. Understandably, facing a human crisis on such an enormous scale, all available resources were poured into learning about and the treatment of psychological illness and psychopathology.

This is how psychology as a field learnt to operate within a disease model. This model has proven very useful. Seligman highlights the victories of the disease model, which are, for example, that 14 previously incurable mental illnesses (such as depression, personality disorder, or anxiety

attacks) can now be successfully treated. However, the costs of adopting this disease model included the negative view of psychologists as _victimologists‘ and _pathologisers‘, the failure to address the improvement of normal lives and the identification and nurturance of high talent. Just to illustrate, if you were to say to your friends that you were going to see a psychologist, what is the most likely response that you would get? _What ‘s wrong with you? ‘How likely are you to hear something along the lines of: _Great! Are you planning to concentrate on self-improvement? ‘

Many psychologists admit that we have little knowledge of what makes life worth living or of how normal people flourish under usual, rather than extreme, conditions. In fact, we often have little more to say about the good life than self-help gurus. But shouldn‘t we know better? The Western world has long overgrown the rationale for an exclusively disease model of psychology. Perhaps now is the time to readdress the balance by using psychology resources to learn about normal and flourishing lives, rather than lives that are in need of help. Perhaps now is the time to gather knowledge about strengths and talents, high achievement (in every sense of this word), the best ways and means of self-improvement, fulfilling work and relationships, and a great art of ordinary living carried out in every corner of the planet. This is the rationale behind the creation of positive psychology.
However, positive psychology is still nothing else but psychology, adopting the same scientific method. It simply studies different (and often far more interesting) topics and asks slightly different questions, such as _what works?‘ rather than _what doesn‘t?‘ or _what is right with this person?‘ rather than _what is wrong?

Assumptions and Goals

The aim of positive psychology is to catalyze a change in psychology from a preoccupation only with repairing the worst things in life to also building the best qualities in life. To redress the previous imbalance, we must bring the building of strength to the forefront in the treatment and prevention of mental illness. The field of positive psychology at the subjective level is about positive subjective experience: well-being and satisfaction (past); flow, joy, the sensual pleasures, and happiness (present); and constructive cognitions about the future—optimism, hope, and faith.
At the individual level it is about positive personal traits—the capacity for love and vocation, courage, interpersonal skill, aesthetic sensibility, perseverance, forgiveness, originality, future-mindedness, high talent, and wisdom. At the group level it is about the civic virtues and the institutions that move individuals toward better citizenship: responsibility, nurturance, altruism, civility, moderation, tolerance, and work ethic.
Positive psychology concentrates on positive experiences at three time points: (1) the past, centering on wellbeing, contentment and satisfaction; (2) the present, which focuses on concepts such as happiness and flow experiences; (3) the future, with concepts including optimism and hope. Not only does positive psychology distinguish between wellbeing across time points but it also separates the subject area into three nodes:

- the subjective node, which encompasses things like positive experiences and states across past, present and future (for example, happiness, optimism, wellbeing);
- the individual node, which focuses on characteristics of the _good person‘ (for example, talent, wisdom, love, courage, creativity); and
- the group node, which studies positive institutions, citizenship and communities (for example, altruism, tolerance, work ethic)

A Brief History of Positive Psychology

While crucial to the introduction and development of positive psychology, Seligman wasn’t alone in introducing the approach and theories.

Indeed, 2,000 years ago, the Ancient Greek philosophers Aristotle and Epicurus were sharing their thoughts on what makes up the "good life." For the psychology perspective, we must travel forward to the late 1800s, when psychologist William James argued that free will played a vital role in happiness and satisfaction in life (Hart, 2021).

For many, humanistic psychology is the forerunner of positive psychology. Led by well-known psychologists at the time, including Abraham Maslow and Carl Rogers, humanistic theory recognized that "people are motivated by a desire to grow and fulfill their potential—a need to self-actualize" (Hart, 2021, p. 10).

"Unfortunately, humanistic psychology did not attract much of a cumulative empirical base, and it spawned myriad therapeutic self-help movements" (Seligman & Csikszentmihalyi, 2000, p. 7). As a result, when Seligman, Csikszentmihalyi, Ed Diener (researching subjective wellbeing), Barbara Fredrickson (exploring positive emotions), Christopher Peterson (studying character strengths and virtues), and others established positive psychology as a discipline, it was on the back of a continually growing body of research (Hart, 2021; Fredrickson, 2010; Seligman, 2011). And it has continued to expand ever since, resulting in a wealth of data, knowledge sharing, and an improved understanding of what makes life worthwhile and which factors underpin and contribute to our wellbeing (Hart, 2021; Fredrickson, 2010; Seligman, 2011).

Before World War II, psychology had three distinct missions: curing mental illness, making the lives of all people more productive and fulfilling, and identifying and nurturing high talent. Right after the war, two events— both economic—changed the face of psychology. In 1946, the Veterans Administration was founded, and thousands of psychologists found out that they could make a living treating mental illness. At that time the profession of clinical psychologist came into its own. In 1947, the National Institute of Mental Health (which was based on the American Psychiatric Association's disease model and is better described as the National Institute of Mental Illness) was founded, and academics found out that they could get grants if their research was described as being about pathology.

This arrangement brought many substantial benefits. There have been huge strides in the understanding of and therapy for mental illness: At least 14 disorders, previously intractable, have yielded their secrets to science and can now be either cured or considerably relieved. But the downside was that the other two fundamental missions of psychology— making the lives of all people better and nurturing genius—were all but forgotten. It was not only the subject matter that altered with funding but also the currency of the theories underpinning how we viewed ourselves. Psychology came to see itself as a mere subfield of the health professions, and it became a Victimology. We saw human beings as passive foci: stimuli came on and elicited responses (what an extraordinarily passive word). External reinforcements weakened or strengthened responses, or drives, tissue needs, or instincts. Conflicts from childhood pushed each of us around.

Psychology's empirical focus then shifted to assessing and curing individual suffering. There has been an explosion in research on psychological disorders and the negative effects of environmental stressors such as parental divorce, death, and physical and sexual abuse. Practitioners went about treating mental illness within the disease-patient framework of repairing damage: damaged habits, damaged drives, damaged childhood, and damaged brains. The message of the positive psychology movement is to remind our field that it has been deformed. Psychology is not just the study of disease, weakness, and damage; it also is the study of strength and virtue. Treatment is not just fixing what is wrong; it also is building what is right. Psychology is not just about illness or health; it also is about work, education, insight, love, growth, and play. And in this quest for what is best, positive psychology does not rely on wishful thinking, self-deception, or hand waving; instead, it tries to adapt what is best in the scientific method to the unique problems that human behavior presents in all its complexity.

Thus, positive psychology is not a selfish psychology. At its best, positive psychology has been able to give the scientific community, society and individuals a new perspective on existing ideas as well as providing empirical evidence to support the phenomenon of human flourishing. Above all, though, positive psychology has challenged and rebalanced the deficit approach to living while connecting its findings to many different disciplines

Goals of Positive Psychology:

According to Martin Seligman's goal of positive psychology was

- To refocusing the entire field of psychology.
- To find elements of positive psychology represented in so many different areas of psychology, from physiological to clinical psychology.
- To restore balance within the discipline of Psychology which was too much focused on negative aspects.
- To catalyze a change in psychology from a preoccupation only with repairing the worst things in life to also building the best qualities in life.
- To improve understanding of positive human behaviors to balance the negative focus of much mainstream research & theory (Sheldon & King, 2001).
- To develop an empirically-based conceptual understanding and language for describing healthy human functioning that parallels our classification and understanding of mental illness (Keyes, 2003)

Lesson-2 Western and Eastern views of Positive Psychology-Athenian Views Judeo Christianity – Confucianism – Taoism – Buddhism – Hinduism

Positive psychology scholars aim to define specific strengths and highlight the many paths that lead to better lives (Aspinwall & Staudinger, 2002; Keyes & Haidt, 2003; Lopez & Snyder, 2003; Lopez & Snyder, 2009; Peterson & Seligman, 2004). As Western civilization and European events and values shaped the field of psychology as we know it today in the United States, it is not surprising that the origins of positive psychology have focused more on the values and experiences of Westerners. Constructs such as hope, optimism, and personal self-efficacy, among others, are particularly valued in these cultures and have been prominent throughout Western history. Increasingly, however, scholars are taking the broader historical and cultural contexts into account to understand strengths and the practices associated with living well (see, e.g., Leong & Wong, 2003; Schimmel, 2000; Sue & Constantine, 2003). Today, the previously neglected wisdoms of the Eastern traditions are being consulted in addition to those originating in the West, with the goal of adding different viewpoints about human strengths. "A good fortune may forebode a bad luck, which may in turn disguise a good fortune." This Chinese proverb exemplifies the Eastern perspective that the world and its inhabitants are in a perpetual state of flux. Thus, just as surely as good times occur, so, too, will bad times visit us. This expectation of and desire for balance distinguishes Easterners' views of optimal functioning from the more linear path taken by Westerners to resolve problems and monitor progress. Ever adaptive and mindful, Easterners move with the cycle of life until the change process becomes natural and enlightenment (i.e., being able to see things clearly for what they are) is achieved. While Westerners might search for rewards in the physical plane, Easterners seek to transcend the human plane and rise to the spiritual one.

In this chapter, we discuss and contrast both Western and Eastern historical and philosophical traditions that demonstrate how these different groups characterize important strengths and life outcomes. Next, we discuss some of the inherent and fundamental differences between Eastern and Western value systems, thought processes, and life outcomes sought. We also articulate the idea of the "good life" from both perspectives and discuss the associated strengths that assist each group in attaining positive life outcomes. We then delve into a discussion of some specific concepts that are deemed to be necessary qualities for achieving the "good life" in each group. It is important to note that what is viewed as the "good life" may be different in each cultural group. Though we will not always enclose this term in quotation marks as we do here, please note that

it is always culturally bound. In closing we talk about the ME perspective and the WE perspective and give our thoughts on trying to see things from more than one perspective.

Historical and Philosophical Traditions

To summarize thousands of years of Western and Eastern ideology and traditions is obviously beyond the scope of this chapter. Therefore, we highlight the basic tenets of three influential Western traditions: (1) Athenian, (2) JudeoChristian, and (3) Islam, as well as four influential Eastern disciplines: (1) Confucianism, (2) Taoism (these two traditions are generally associated with China), (3) Buddhism (associated with Japan), and (4) Hinduism (rooted in traditions of Southeast Asia). Within both Western and Eastern historical contexts, the concept of the "good life" has existed for many centuries. While Western cultures emphasize optimal functioning as it occurs intrapsychically, Eastern cultures hold that an optimal life experience is a spiritual journey involving others and resulting in transcendence and enlightenment. The Eastern search for spiritual transcendence parallels the Westerner's hopeful pursuits for a better life on Earth.

Western Influences: Athenian, Judeo-Christian, and Islamic Traditions

Athenian Views

Discussion of virtue and human strength is something on which both Plato and Aristotle focused heavily in their teachings in Ancient Greece. Aristotle, after expanding on Plato's ideas regarding virtue, detailed 11 moral virtues: courage, moderation, generosity, munificence (this relates to money spending at an appropriate level), magnificence (described as "greatness of soul"), even temper, friendliness, truthfulness, wit (describing an ability to laugh and have fun at an appropriate level), justice, and friendship (Solomon, 2006). In addition to these moral virtues, Aristotle described intellectual virtues (mainly associated with ideas regarding wisdom) and believed that "strength of character, as inculcated by the political community, would lead to enduring human excellence" (Solomon, 2006, p. 9).

Aristotle and Plato also emphasized the influence the political community, termed polis, has on the development and maintenance of these virtues (Euben, Wallach, & Ober, 1994; Solomon, 2006). Aristotle discussed this community as being a necessity in helping the average individual to self-actualize with regard to virtue; he stated it was only within a life of order and sanction that one could rise above hedonistic desire and become truly virtuous (Peterson & Seligman, 2004; Solomon, 2006). In this view, people with good human virtue create such a community and then can provide a good model for others so that the masses also develop such human excellence. In addition, Aristotle believed that government should be charged with the development of virtue in a particular society via early education (i.e., in childhood) and training (Solomon, 2006).

Judeo-Christianity

In thinking about virtue in general, the religious teachings of Judaism and Christianity often come directly to mind. The Bible contains discussions of virtues in many chapters and verses. In the Old Testament, the virtues of faith, hope, and charity are highlighted and encouraged and were later discussed as part of the "Seven Heavenly Virtues" by Thomas Aquinas (Williams & Houck, 1982). According to historians, Aquinas lists these virtues as fortitude (courage), justice, temperance, wisdom (these four are often called the cardinal virtues; Peterson & Seligman, 2004), faith, hope, and charity (Williams & Houck, 1982). Other scholars cite the Ten Commandments given by Moses in the Old Testament as directives toward cultivating certain strengths within the Jewish tradition. Peterson and Seligman (2004) interpret the acts that the commandments prohibit as falling under the category of particular cardinal virtues: "Justice is implied in prohibitions against murder, theft, and lying; temperance in those against adultery and covetousness; and transcendence generally within the divine origin of the commands" (p. 48). Other mentions are made of various gifts and strengths throughout the New Testament. For example, the Book of Romans describes the "gifts" that are valued by God and includes strengths such as leadership, faith, mercy, love, joy, hope, patience, hospitality, and others (12:3–21). In addition, the Book of Proverbs has many affirmations of specific virtuous behaviors (Peterson &

Seligman, 2004). In the prologue of this book of the Bible, the following words are given as the purpose and theme of Proverbs: 1. The proverbs of Solomon, son of David, king of Israel: 2. for attaining wisdom and discipline; for understanding words of insight; 3. for acquiring a disciplined and prudent life, doing what is right and just and fair; 4. for giving prudence to the simple, knowledge and discretion to the young5. let the wise listen and add to their learning, and let the discerning get guidance6. for understanding proverbs and parables, the sayings and riddles of the wise. 7. The fear of the LORD is the beginning of knowledge, but fools despise wisdom and discipline. (Proverbs 1: 1–7) These words caution followers to live virtuous lives, giving particular weight to the virtue of wisdom. Finally, the Beatitudes discussed in the Book of Matthew give a series of virtuous traits (e.g., meekness, being a "peacemaker," mercy, righteousness, etc.) that are said to be pleasing to God (Matthew 5: 1–11). The Talmud also provides instructions about living a virtuous life. In the Pirke Avot, or Ethics of the Fathers, directives are given on how to live life as an ethical follower of Judaism (N. Mendel, personal communication, February 3, 2010). The lessons here include being a hospitable host, particularly to the poor, being fair in decision-making and judgments, and seeking peace in everyday life (Bokser, 1989). In addition, the Talmud states, "You shall administer truth, justice and peace within your gates" (Zech 8:16), showing similar value to other religious traditions for these specific virtues (Bokser, 1989).

Islam

Though we have added Islam to the "Western" heading in this section as is commonly done in texts that discuss both Western and Eastern religions, it is important to note that scholars disagree as to whether Islam should be considered a Western or an Eastern religion (S. Lloyd-Moffet, personal communication, November 21, 2013). Islam is practiced by both Western and Eastern individuals and groups, and thus its virtues and practices may be influenced by more than one context.

Islam incorporates many virtues recognizable in other philosophical traditions and categorizes them as moral obligations. Among others, gratitude (e.g., to Allah for His benevolence), love (of Allah because of His forgiveness), kindness (especially toward parents), justice (emphasizing fraternity and equality of all), and courage (acts of bravery) are valued (Farah, 1968). In addition, there is a strong component of looking out for one's brother, particularly if one has more than one needs. This emphasis is especially directed toward the wealthy in terms of their support of the poor as "[the wealthy] are obligated . . . to aid the poor as a duty, not a privilege" (Farah, 1968, p. 127). Giving to the poor is a requirement in the Islamic faith reflected in the third pillar, zakat (alms), and it is something that is to be done secretly as opposed to directly if possible so that the giver maintains his or her humility and the recipient is not embarrassed by having to accept the gift (Ahmed, 1999). Abiding by these moral obligations and pillars assists the faithful in pleasing Allah in this tradition.

Eastern Influences: Confucianism, Taoism, Buddhism, and Hinduism

Confucianism

Confucius, or the Sage, as he is sometimes called, held that leadership and education are central to morality. Born during a time when his Chinese homeland was fraught with strife, Confucius emphasized morality as a potential cure for the evils of that time (Soothill, 1968), and the tenets of Confucianism are laden with quotations that encourage looking out for others. In fact, one of Confucius's most famous sayings is a precursor of the Golden Rule and can be translated as, "You would like others to do for you what you would indeed like for yourself" (Ross, 2003; Analects 6:28). In some ways, these teachings are parallel to thoughts put forth by Aristotle and Plato regarding the responsibility of leaders to take charge of the group, though there is less emphasis in Western writings on the collectivist ideal of taking care of others in the group.

The attainment of virtue is at the core of Confucian teachings. The five virtues deemed central to living a moral existence are jen (humanity, the virtue most exalted by Confucius and said to encompass the other four virtues); yi (duty to treat others well); li (etiquette and sensitivity for

others' feelings); zhi (wisdom), and xin (truthfulness). Confucian followers must strive to make wise decisions based on these five virtues; this continual striving leads the Confucian follower to enlightenment, or the good life.

Taoism

The Way Ancient Taoist beliefs are difficult to discuss with Western audiences partly because of the untranslatable nature of some key concepts in the tradition of Taoism. Lao-Tzu (the creator of the Taoist tradition) states in his works that his followers must live according to the Tao (pronounced "dow" and roughly translated as "the Way"). The Chinese character portraying the concept of the Way is a moving head and "refers simultaneously to direction, movement, method, and thought" (Peterson & Seligman, 2004, p. 42; Ross, 2003); Tao is the energy that surrounds everyone and is a power that "envelops, surrounds, and flows through all things" (Western Reform Taoism, 2003, p. 1).

In this regard, Lao-Tzu (1994) described the Way in the following lines:
The Way can be spoken of,
But it will not be the constant way;
The name can be named,
But it will not be the constant name.
The nameless was the beginning of the myriad creatures;
The named was the mother of the myriad creatures.
Hence constantly rid yourself of desires in order to observe its subtlety;
But constantly allow yourself to have desires in order to observe what it is after.
These two have the same origin but differ in name.
They are both called dark,
Darkness upon darkness
The gateway to all is subtle. (p. 47)

According to Taoist traditions, the difficulty in understanding the Way stems from the fact that one cannot teach another about it. Instead, understanding flows from experiencing the Way for oneself by fully participating in life. In this process, both good and bad experiences can contribute to a greater understanding of the Way. Achieving naturalness and spontaneity in life is the most important goal in Taoist philosophy. Thus, the virtues of humanity, justice, temperance, and propriety must be practiced by the virtuous individual without effort (Cheng, 2000). One who has achieved transcendence within this philosophy does not have to think about optimal functioning but behaves virtuously naturally.

Buddhism

Seeking the good of others is woven throughout the teachings of "the Master" or "the Enlightened One" (i.e., the Buddha). In one passage, the Buddha is quoted as saying, "Wander for the gain of the many, for the happiness of the many, out of compassion for the world" (Sangharakshita, 1991, p. 17). At the same time, the Buddha teaches that suffering is a part of being and that this suffering is brought on by the human emotion of desire. In the Buddhist philosophy, Nirvana is a state in which the self is freed from desire for anything (Schumann, 1974). It should be noted that both premortal and postmortal nirvana states are proposed as possible for the individual. More specifically, the premortal nirvana may be likened to the idea of the ultimate good life in this philosophy. Postmortal nirvana may be similar to the Christian idea of heaven.

Like the other Eastern philosophies, Buddhism gives an important place to virtue, which is described in several catalogs of personal qualities. Buddhists speak of the Brahma Viharas, those virtues that are above all others in importance, described by Peterson and Seligman as "universal virtues" (2004, p. 44). These virtues include love (maitri), compassion (karuna), joy (mudita), and equanimity (upeksa) (Sangharakshita, 1991). The paths to achieving these virtues within

Buddhism require humans to divorce themselves from the human emotion of desire to put an end to suffering.

Hinduism

The Hindu tradition differs somewhat from the other three philosophies discussed previously in that it does not appear to have a specific founder, and it is not clear when this tradition began in history (Stevenson & Haberman, 1998). The main teachings of the Hindu tradition emphasize the interconnectedness of all things. The idea of a harmonious union among all individuals is woven throughout the teachings of Hinduism, which refer to a "single, unifying principle underlying all of Earth" (Stevenson & Haberman, 1998, p. 46).

One's goal within this tradition would be to live life so fully and so correctly that one would go directly to the afterlife without having to repeat life's lessons in a reincarnated form (Stevenson & Haberman, 1998). Hindu teachings are very clear about the qualities one must embody to avoid reincarnation: "To return to this world is an indication of one's failure to achieve ultimate knowledge of one's self" (Stevenson & Haberman, 1998, p. 53). Thus, the quest of one's life is to attain ultimate self-knowledge and to strive for ultimate self-betterment (notably also a Western concept). Individuals are encouraged to be good to others as well as to improve themselves; the Upanishads state, "A man turns into something good by good action and something bad by bad action" (Stevenson & Haberman, 1998, p. 54). "Good action" is also encouraged in the sense that, if one does not reach ultimate self-knowledge in one's life and thus does have to return to Earth via reincarnation after death, the previous life's good actions correlate directly with better placement in the world in the subsequent life (Stevenson & Haberman, 1998). This process is known as karma. The good life in the Hindu tradition, therefore, encompasses individuals who are continually achieving knowledge and continually working toward good actions (Dahlsgaard, Peterson, & Seligman, 2005; Peterson & Seligman, 2004; Stevenson & Haberman, 1998).

Summary of Eastern and Western Philosophies

Each of the philosophies discussed here incorporates ideas about the importance of virtue, along with human strengths, as people move toward the good life. Similarities also can be drawn among the different ideologies, especially in the types of human qualities and experiences that are valued, though there are also differences in terms of which traits are particularly valued. Thus, it is important to contrast these Eastern beliefs with Western ideology to understand the differences in positive psychology viewed from each perspective.

Lesson-3 Positive Psychology in India-Value Systems – Orientation to Time – Thought Processors – Different Ways to Positive Outcomes: Hope – Compassion – Harmony

Lesson-4 Theories: PERMA model, Self-determination theory, Hope theory, Broaden and Build theory.

PERMA Theory of Well-Being

What is human flourishing and what enables it? Dr. Seligman's PERMA™ theory of well-being is an attempt to answer these fundamental questions. There are five building blocks that enable flourishing – **P**ositive Emotion, **E**ngagement, **R**elationships, **M**eaning, and **A**ccomplishment (hence **PERMA™**) – and there are techniques to increase each.

Different people will derive well-being from each of these five building blocks to varying degrees. A good life for one person is not necessarily a good life for another. There are many different routes to a flourishing life. Positive Psychology is descriptive, not prescriptive. In other words, we are not telling people what choices to make or what to value, but research on the factors that enable flourishing can help people make more informed choices to live a more fulfilling life that is aligned with their values and interests.

Here is a brief definition of each of the five building blocks:

Positive Emotion: This route to well-being is hedonic – increasing positive emotion. Within limits, we can increase our positive emotion about the past (e.g., by cultivating gratitude and forgiveness), our positive emotion about the present (e.g., by savoring physical pleasures and mindfulness) and our positive emotion about the future (e.g., by building hope and optimism).

Unlike the other routes to well-being described below, this route is limited by how much an individual can experience positive emotions. In other words, positive affectivity is partly heritable and our emotions tend to fluctuate within a range. Many people are, by disposition, low in experiencing positive emotion. Traditional conceptions of happiness tend to focus on positive emotion, so it can be liberating to know that there are other routes to well-being, described below.

Engagement: Engagement is an experience in which someone fully deploys their skills, strengths, and attention for a challenging task. According to Mihaly Csikszentmihalyi, this produces an experience called "flow" that is so gratifying that people are willing to do it for its own sake, rather than for what they will get out of it. The activity is its own reward. Flow is experienced when one's skills are just sufficient for a challenging activity, in the pursuit of a clear goal, with immediate feedback on progress toward the goal. In such an activity, concentration is fully absorbed in the moment, self-awareness disappears, and the perception of time is distorted in retrospect, e.g., time stops. Flow can be experienced in a wide variety of activities, e.g., a good conversation, a work task, playing a musical instrument, reading a book, writing, building furniture, fixing a bike, gardening, sports training or performance, to name just a few.

Relationships: Relationships are fundamental to well-being. The experiences that contribute to well-being are often amplified through our relationships, for example, great joy, meaning, laughter, a feeling of belonging, and pride in accomplishment. Connections to others can give life purpose and meaning. Support from and connection with others is one of the best antidotes to "the downs" of life and a reliable way to feel up. Research shows that doing acts of kindness for others produces an increase in well-being.

From an evolutionary perspective, we are social beings because the drive to connect with and serve others promotes our survival. Developing strong relationships is central to adaptation and is enabled by our capacity for love, compassion, kindness, empathy, teamwork, cooperation, self-sacrifice, etc.

Meaning: A sense of meaning and purpose can be derived from belonging to and serving something bigger than the self. There are various societal institutions that enable a sense of meaning, such as religion, family, science, politics, work organizations, justice, the community, social causes (e.g., being green), among others.

Accomplishment: People pursue achievement, competence, success, and mastery for its own sake, in a variety of domains, including the workplace, sports, games, hobbies, etc. People pursue accomplishment even when it does not necessarily lead to positive emotion, meaning, or relationships.

Each of these five building blocks contributes to well-being and:

- Is pursued for its own sake, not as a means to an end
- Is defined and measured independently of the other elements

Self-determination theory

Self-determination theory argues, like Maslow's hierarchy of needs, that there is an evolutionary adaptive function of three basic psychological needs. Autonomy is the tendency to self-regulate one's behaviour inaccordance with personal volition (rather than external control). It is also the tendency to resist coercion, pressure, and control; to regulate one's behaviour in accordance with one's own needs (and situational affordances), which promotes better survival than organizing behaviour tomeet external demands.

Competence is the tendency to be interested and open, to seek learning/mastery opportunities (promote acquisition of new skills). The need for competence manifests in early motor play, manipulation ofobjects, and exploration of surroundings. The tendency to experience satisfaction from learning for its own sake – and the tendency to explore and seek challenges. This need is shared to some degree with othermammals. Thus, competence is the ability to affect the environment and attain desired outcomes.

Relatedness is the tendency to feel connection and caring with group members (it promotes group cohesion and mutual protection). It is similar to Baumeister and Leary's 'need to belong' and overlaps with Bowlby's attachment need. This need can at times conflict with need for autonomy but normally it is complementary

Hope theory

According to Snyder et al. (1991) hope is a human strength manifested in our perceived capacities to clearly conceptualize goals (goals thinking), develop strategies to reach those goals (pathways thinking), andinitiate and sustain the motivation for using those strategies (agencythinking).
Hope is conceptualized as a *dynamic* cognitive *motivational system*. According to this conceptualization, emotions follow cognitions and not the other way round. Hope leads to pursuit of *learning goals* and learninggoals are positively related to success.

Hope theory has the following major components: goals, pathway thoughts, agency thoughts and barriers.

Goals: Goals are the targets of mental action sequences and are thecognitive component in hope theory. Goals provide direction for hopeful thinking.
Pathways Thinking: Pathway thoughts refer to the routes we take to achieve our goals and the perceived ability to produce these routes.

Agency Thinking: This is the motivational component in hope theory. Agentic thinking reflects the self-referential thoughts regarding moving along a pathway as well as continuing to progress along that pathway.

Barriers: Barriers block the attainment of our goals. The individual then has to make a choice to either give up or use pathway thoughts to create new routes. Snyder (2000) suggests that hope develops over the course ofinfancy, childhood and adolescence. Snyder proposes that there are no hereditary contributions to hope and it is a learned cognitive set. The basiccause and effect thinking contained in pathways thinking is acquired fromparents and others. Snyder also proposes that strong attachment to caregivers is crucial for the development of hope. Traumas during the lifecourse contribute to the lessening of hope.

Broaden and Build theory.

Barbara Fredrickson (2000) has developed a new theoreticalframework, the broaden-and-build model, that may provide someexplanations for the robust social and cognitive effects of positiveemotional experiences. In Fredrickson's review of models of emotions (Smith, 1991), she found that responses to positive emotions have not beenextensively studied and that, when researched, they were examined in avague and underspecified manner.

Furthermore, action tendencies generally have been associated withphysical reactions to negative emotions (again, imagine "fight or flight"), whereas human reactions to positive emotions often are more cognitive than physical. For these reasons, she proposes discarding the specific action tendency concept (which suggests a restricted range of possible behavioral options) in favor of newer, more inclusive term, momentary thought-action repertoires (which suggest a broad range of behavioral options; imagine "taking off blinders" and seeing available opportunities).

To illustrate the difference in that which follows positive and negative emotions, consider the childhood experience of one of the authors(SJL). Notice how positive emotions (e.g., excitement and glee) lead to cognitive flexibility and creativity, whereas negative emotions (e.g., fear and anxiety) are linked to a fleeing response and termination of activities.In testing her model of positive emotions, Fredrickson (2000) demonstrated that the experience of joy expands the realm of what a personfeels like doing at the time; this is referred to as the *broadening* of an individual's momentary thought-action repertoire. Following an emotion-eliciting film clip (the clips induced one of five emotions: joy, contentment, anger, fear, or a neutral condition), research participants wereasked to list everything they would like to do at that moment. Those participants who experienced joy or contentment listed significantly moredesired possibilities than did the people in the neutral or negative conditions. In turn, those expanded possibilities for future activities shouldlead the joyful individuals to initiate subsequent actions.
Those who expressed more negative emotions, on the other hand, tended to shut down their thinking about subsequent possible activities. Simply put, joy appears to open us up to many new thoughts and behaviors, whereas negative emotions dampen our ideas and actions.

Joy also increases our likelihood of behaving positively toward other people, along with developing more positive relationships. Furthermore, joy ~duces playfulness (Frijda, 1994), which is quite important because such behaviors are evolutionarily adaptive in acquisition of necessary resources. Juvenile play builds (1) enduring social and intellectual resources by encouraging attachment, (2) higher levels of creativity, and (3) brain development (Fredrickson, 2002). It appears that, through the effects of broadening processes, positive emotions also can help *build* resources. In 2002, Fredrickson and her colleague, Thomas Joiner, demonstrated this building phenomenon by assessing people's positive andnegative emotions and broad-minded coping (solving problems with creative means) on two occasions 5 weeks apart. The researchers found that initial levels of positive emotions predicted overall increases in creative problem solving.

These changes in coping also predicted further increases in positive emotions. Similarly, controlling for initial levels of positive emotion, initial levels of coping predicted increases in positive emotions, which in turn predicted increases in coping. These results held true only for positive emotions, *not* for negative emotions. Therefore, positive emotions such as joy may help generate resources, maintain a sense of vital energy (i.e., more positive emotions), and create even more resources. Fredrickson (2002) referred to this positive sequence as the "upward spiral" of positive emotions.

Extending her model of positive emotions, Fredrickson and colleagues examined the "undoing" potential of positive emotions (Fredrickson, Mancuso, Branigan, & Tugade, 2000) and the ratio of positive to negative emotional experiences that is associated with human flourishing (Fredrickson & Losada, 2005). Fredrickson et al. (2000) hypothesized that, given the broadening and building effects of positive emotions, joy and contentment might function as antidotes to negative emotions.

UNIT II: HAPPINESS AND WELLBEING

Lesson-5 Introduction to Psychology of happiness, well-being and scope

Happiness seems to be one of those terms that people understand until they try to define it. Indeed, defining happiness has turned out to be much more contentious than defining positive psychology. Even in philosophy, defining happiness is a troublesome issue. Positive psychologists often refer to two types of happiness: hedonic and eudaimonic. Briefly, *hedonic happiness* refers to how happy you feel whereas *eudaimonic happiness* involves being true to your authentic self. Researchers interested in eudaimonic happiness are most interested in whether individuals are using their capacities—both their general human capacities and their own peculiar individual strengths—to flourish in life. Most positive psychologists follow Aristotle in their understanding of eudemonia; nevertheless, it is important to acknowledge that Aristotle was one of several different ancient Greek philosophers who took the eudaimonic approach to happiness. All of these approaches, however, emphasized the importance of fulfilling one's *true nature.*

In this sense, eudaimonic happiness is achieved when we fulfill our true human potential. We flourish when our life fulfills our human capacities. Aristotle tended to emphasize the importance of becoming a virtuous person. Importantly for Aristotle, it was the virtues that should be pursued, not happiness in and of itself. There are several advantages to the eudaimonic tradition of happiness. Perhaps most importantly, this approach does not divorce morality and virtue from the study of happiness. Most of us would agree with Aristotle in that a life simply filled with consumptive pleasure is not a happy life. Cows may have all the pleasure of eating grass and grain to their heart's content, but are they really happy? No. Aristotle would claim that the authentically happy life is also the virtuous life. The eudaimonic view of happiness is an important tradition that positive psychologists would be foolish to neglect. But there are a number of problems with this approach to defining happiness. First, it can come across as being overly prescriptive or even paternalistic. The eudaimonic definition of happiness basically tells people that they have to be doing certain things or they have to be developing certain qualities.

The rich Greek tradition of happiness involves much more than Aristotle and the eudaimonic approach described earlier. For example, many Greek philosophers followed a hedonistic approach to happiness. But even the most famous hedonists— the Epicureans— had a much more nuanced view of happiness than popular hedonism would suggest. Indeed, those of the Epicurean camp were much more likely to represent happiness as the —pleasures of tranquility, ¦ rather than the accumulation of raw unadulterated pleasure. One thing that

seemed to distinguish the Epicureans from the disciples of Aristotle, however, was that this tranquil,happy life could be an explicit goal, whereas for Aristotle any kind ofpsychic harmony was simply the result of the virtuous life. Even theStoics—who strove to push pleasure and pain toward the margins of life—felt that inner psychological harmony was an important aspect of thevirtuous life. Thus, for virtually all philosophers, the good life was apleasant life, although they would emphasize different paths to this end.

McMahon has emphasized (2006, 2008), there was alwayssomething of a tragic theme in the Greek approach to happiness for it seemed that happiness for the Greeks was never completely under one‘s control; for one never knew when luck, fate, or the capricious act of some god might completely derail one‘s journey to happiness. McMahon emphasizes that this theme seems to come through in early Judeo– Christian thought as well, albeit in a notably different form.

Happiness underlying factors are considerable from two dimensions: endogenic factors (biological, cognitive, personality and ethical sub-factors) and exogenic factors (behavioral, social cultural, economical, geographical, life events and aesthetics sub-factors). Among all endogenic factors, biological sub-factors are the significant predictors of happiness.

Genetic factors

Among all related genes with mood and emotional characteristics, effects of two genes investigated directly on happiness: *5-HTTLPRand MAO-A* studies suggested an association between *5-HTTLPR* and life satisfaction as a cognitive dimension of happiness. This gene is coding serotonin distribution in brain cells and therefore leads to mood regulation.

Brain and neurotransmitters

The emotion circuitry of the brain is complex, involving primarilystructures in the prefrontal cortex, amygdale, hippocampus, anterior cingulated cortex, and insular cortex. These structures normally work together to process and generate emotional information and emotional behavior. Research has particularly focused on the prefrontal cortex which,unlike most other brain regions involved in emotion processing, shows asymmetric activation in relation to positive and negative emotions.

At the other hand, studies showed that positive and negative moodsaffected by brain chemical in several ways. Two of the most important neuro-transmitters that involved in mood are dopamine and serotonin. Positive mood and negative mood mediated by dopamine and serotonin levels.

Adrenaline

Adrenaline (also known as Epinephrine) is a hormone and a neurotransmitter that releases from Adrenal glands. Adrenaline has many functions in the body, regulating heart rate, blood vessel and air passage diameters, and metabolic shifts; Adrenaline release is a crucial componentof the fight-or-flight response of the sympathetic nervous system.
Therefore, adrenaline produces similar effects to Cortisol, such as increased heart rate and immune system suppression. Researches indicatethat urinary adrenaline is a good predictor of happiness. Individuals with higher levels of “personal growth” and “purpose in life” registered lower and more stable levels of salivary Cortisol and urinary Adrenaline.

Physical health

Medicine and psychological findings showed that positive emotion reactions against life events can influence in various ways on physiologic characteristics. Therefore, several studies investigate the association between physical health and happiness. A group of researchers studied thegeneral association between physical health and happiness and others investigate the relation between physical illness (hypertension, bulimia…) and happiness.

Recently, several studies concluded that positive mood in individuals is a strong predictor of physical health an there is a significant correlation between positive mood and physical health. Researchers stated that peoplewith happiness experience a long life.

Subjective well-being

Subjective well-being emphasizes peoples' reports of their lifeexperiences. Accordingly, the subjective report is taken at face value. Thissubjective approach to happiness assumes that people from many culturesare comfortable in focusing on individualistic assessments of their affects and satisfaction and that people will be forthright in such personal analyses (Diener et al., 2002). These assumptions guide the researchers' attempts tounderstand a person's subjective experiences in light of his or her objectivecircumstances

DETERMINANTS OF SUBJECTIVE WELL-BEING

When examining college students' (from 31 nations) satisfaction invarious life domains, financial status was more highly correlated with satisfaction for students in poor nations than for those in wealthy nations (Diener & Diener, 1995). Moreover, the people in wealthy nations generally were happier than those in impoverished nations. Within-nation examination of this link between income and well-being reveals that, oncehousehold income

Psychological well-being

Psychological well-being refers to the achievement of one‘s full psychological potential. The construct is central to the humanistic tradition. Professor **Carol Ryff** is the leading researcher in this area. She hasdeveloped a scale called The ***Ryff Scales of Psychological Well-Being***. Itmeasures multiple facets of psychological wellbeing as the following:

- autonomy
- environmental mastery
- personal growth
- positive relations with others
- purpose in life
- self-acceptance

Keyes et al. (2002) in a factor analytic study involving over 3,000 Americans aged 25-74 found that psychological well-being and subjectivewell-being were related, but distinct, constructs which correlateddifferentially with socio-demographic variables and personality.

Social well-being

Social well-being refers to positive states associated with optimal functioning within one‘s social network and community. It refers to an endstate in which basic human needs are met and people coexist peacefully. There is an equal access to basic facilities such as water, food, shelter, health services etc.

Complete Mental Health: Emotional, Social, and Psychological Well-Being

Ryff and Keyes (1995; Keyes & Lopez, 2002; Keyes & Magyar- Moe, 2003) combine many principles of pleasure to define completemental health.

Specifically, they view optimal functioning as the combination of emotional well-being (as they refer to subjective well-being; defined as thepresence of positive affect and satisfaction with life and the absence of negative affect), social well-being (incorporating acceptance, actualization, contribution, coherence, and integration), and psychological well-being (combining self-acceptance, personal growth, purpose in life, environmental mastery, autonomy, positive relations with others). Taking the symptoms of mental illness into consideration, they define "complete mental health" as the combination of "high levels of symptoms of emotional well-being, psychological wellbeing, and social well-being, as well as the absence of recent mental illness" (Keyes & Lopez, 2002, p. 49).This view of mental health combines all facets of well-being into a model that is both dimensional (because extremes of mental health and illness symptomatology are reflected) and categorical (because assignment to distinct diagnostic categories is possible). This complete state model (Keyes & Lopez, p. 49; see Figure 7.5) suggests that combined mental health and mental illness symptoms may be ever-changing, resulting in fluctuations in states of overall well-being ranging from complete mental illness to complete mental health.

Lesson-6 Age old definition of Happiness – Subjective Wellbeing as a Synonym for Happiness – Types of happiness – Eudemonic and Hedonic, History of Happiness, Traits associated with Happiness

The term happiness comes from the Old Norse term happ meaning “luck” or “chance.” It's also related to the Old English word hæpic meaning “equal.” While early senses of happiness dating from the 1500s are still very much in use, such as “good luck,” “success,” and “contentment,” Happiness is often known by another name in positive psychology research: subjective wellbeing, or SWB.

Some believe happiness is one of the core components of SWB, while others believe happiness is SWB. Regardless, you’ll frequently find SWB used as a shorthand for happiness in the literature. And speaking of the literature, you will find references to SWB everywhere.

Hedonic and Eudemonic Happiness

Positive psychologists often refer to two types of happiness: hedonic and eudemonic. Briefly, *hedonic happiness* refers to how happy you feel whereas *eudemonic happiness* involves being true to your authentic self. Researchers interested in eudemonic happiness are most interested in whether individuals are using their capacities—both their general human capacities and their own peculiar individual strengths—to flourish in life.

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Most of us would agree with Aristotle in that a life simply filled with consumptive pleasure is not a happy life. Aristotle would claim that the authentically happy life is also the virtuous life. The eudemonic view of happiness is an important tradition that positive psychologists would be foolish to neglect. But there are a number of problems with this approach to defining happiness. First, it can come across as being overly prescriptive or even paternalistic. The eudemonic definition of happiness basically tells people that they have to be doing certain things or they have to be developing certainqualities or they aren‘t happy, regardless of whether they think they are happy.

But the real rub involves the issue of human nature. What really areour capacities as humans? What is my personal potential? What are the human needs that need to be satisfied in order to flourish? On this issue, even among eudemonic theorists, there is little agreement. A related issuehas to do with the measurement of this kind of happiness. Although measures have been developed for this construct (often referred to as measures of *psychological wellbeing*), are they really measuring eudemonic happiness? A good measure of Aristotle 's eudemonia would accurately assess whether we really are fulfilling our human potential.

Lesson-7 Happiness across the life span; gender and happiness; marriage and happiness; other facts of life

Happiness is more than a good feeling or a yellow smiley face. It's the feeling of truly enjoying your life, and the desire to make the very best of it. Happiness is the "secret sauce" that can help us be and do our best. Here's what researchers found when they studied happy people: Happy people are more successful. In general, happiness can be thought of as an emotional state that reflects a high level of mental and or emotional well-being. Current scientific perspectives typically frame happiness as a complex binary construct that encompasses subjective elements of both affect and cognition that contribute to wellbeing. Subjective well-being (SWB).

The positive psychology movement, which has its own historical roots in the human potential movement of Rogers and Maslow, has unabashedly adopted the Aristotelian concepts of hedonia and eudaimonia in an attempt to frame the exploration of happiness within the field of psychology. Indeed, positive psychology defines itself as the scientific study of human flourishing (Seligman & Csikszentmihalyi, 2000), and Martin Seligman's (the oft-noted founder of the positive psychology movement) theoretical model of happiness is based on the good life which contains three elements: The Pleasant Life, the Meaningful Life, and the Engaged Life (Seligman, Rashid & Parks, 2006)

Gender and Happiness

Who's happier, men or women? Research shows it's a complicated question and that asking whether males or females are happier isn't really that helpful, because essentially, happiness is different for women and men.

Women's happiness has been declining for the past 30 years, according to recent statistics. And research shows that women are twice as likely to experience depression compared with men. Gender differences in depression are well established and studies have found that biological, psychological and social factors contribute to the disparity.

But research also shows that women are more likely to experience intense positive emotions – such as joy and happiness – compared to men. So, it seems that women's more intense positive emotions balance out their higher risk of depression. Research also shows women are more likely to try and get help and access treatment – allowing them to also recover sooner.

Early studies on gender and happiness found men and women were socialized to express different emotions. Women are more likely to express happiness, warmth and fear, which helps with social bonding and appears more consistent with the traditional role as primary caregiver, whereas men display more anger, pride and contempt, which are more consistent with a protector and provider role.

Brain research

Recent research suggests that these differences are not just social, but also in the brain. In numerous studies females score higher than males in standard tests of emotion recognition, social sensitivity and empathy.

Neuroimaging studies have investigated these findings further and discovered that females utilize more areas of the brain containing mirror neurons than males when they process emotions. Mirror neurons allow us to experience the world from other people's perspective, to understand their actions and intentions. This may explain why women can experience deeper sadness.

Psychologically it seems men and women differ in the way they process and express emotions. With the exception of anger, women experience emotions more intensely and share their emotions more openly with others. Studies have found in particular that women express more pro-social emotions – such as gratitude – which has been linked to greater happiness. This supports the theory that women's happiness is more dependent on relationships than men's.

Marriage and Happiness

Marriage: Married people have been found to be happier than unmarried people, be they divorced, separated or never married (Myers, 2000). However, the least happy of all are people in unhappy marriages. Marriages in which people communicate clearly and respectfully and forgive each other's faults are usually associated with higher levels of satisfaction.

A good marriage is one of the life factors most strongly associated and consistently associated with happiness. Good relationships make people happy because dependable companionship is a basic human need. Improving social relationships will bring our happiness score up. There is a strong consensus in the field of positive psychology that the number and depth of personal relationships have the greatest effect of all on happiness. And the relationship where vast numbers of people derive that greatest boost to their well-being is in their marriage. All marriages require the couples to make adjustments to accommodate each other. The marriages arranged through parents and elders attempt to reduce the number of future points of friction, starting with religion, food habits, compatibility in income, social status, education, age and so on; often leaving the final decision to the singles themselves. In 2002, two pioneers of Positive Psychology, Ed Diener and Martin Seligman, conducted a study at the University of Illinois on the 10% of students with the highest scores recorded on a survey of personal happiness. They found that the most salient characteristics shared by students who were very happy and showed the fewest signs of depression were "their strong ties to friends and family and commitment to spending time with them." The New Wallis, 2005).

Lesson-8 Measurement of Happiness (The Steen Happiness Index, Subjective Happiness Scale, Happiness Scale).

Measuring Happiness Research conducted by the Gallup organization is representative of how happiness is often measured. Individuals are surveyed about their levels of well-being and the information is compiled to develop national indices. The World Happiness Report states that "among various measures of subjective well-being, the primary distinction to be made is between cognitive life evaluations (represented by questions asking how happy or satisfied people are with their lives as a whole), and emotional reports" (The World Happiness Report, p. 11). Emotional reports are further distinguished from those that assess current emotional status (how happy are you now) and those that are retrospective (how happy were you yesterday). These questions can be seen as an attempt to get at the dichotomous nature of happiness as discussed earlier. Counseling and Wellness: A Professional Counseling Journal 96 2013 Volume 4 Psychometric measures of personal happiness and well-being most popular with psychological researchers also reflect this duality. Early measures such as the PANAS (Positive and Negative Affect Scale) attempted to gauge happiness as relative levels of positive and negative emotion (Kercher, 1992). Currently popular psychological measures of happiness tend to be more evaluative in nature, and include the Satisfaction with Life Scale (Diener, Emmons, Larsen & Griffen, 1985) and the Subjective Happiness Scale (Lyubomirsky & Lepper, 1997). In recent past, some researchers have attempted to tease apart the differences underlying the various concepts related to well-being, e.g., life satisfaction, positive and negative affect, self-esteem, and optimism.

The Steen Happiness Index (SHI; Seligman et al, 2005) was modeled after the Beck Depression Inventory (BDI; Beck et al, 1961). Just as the BDI is sensitive to changes in depressive symptoms, the SHI was created to be sensitive to changes, particularly upward changes, in

happiness levels. The SHI contains 20 items and requires participants to read a series of statements and pick the one from each group that describes them at the present time. The items on the SHI reflect the three kinds of happy lives (the pleasant life, the engaged life, and the meaningful life): experiencing and savoring pleasures, losing the self in engaging activities, and participating in meaningful activities. Response choices range from a negative (1) to an extreme positive (5), as in the following example: Most of the time I am bored (1); Most of the time I am neither bored nor interested in what I am doing (2); Most of the time I am interested in what I am doing (3); Most of the time I am quite interested in what I am doing (4); and Most of the time I am fascinated by what I am doing (5). Pilot work with several hundred adult respondents showed that scores on this measure converged substantially with scores on other measures of happiness (r .79 with Lyubomirsky & Lepper's (1999) General Happiness Scale, and r .74 with Fordyce's (1977) Happiness Scale). (APA PsycTests Database Record, 2019 APA,). Using Campbell and Fiske's (1959) multi trait-multimethod matrix analyses, Lucas, Diener and Suh (1996) found high discriminant validity for life satisfaction from positive and negative affect and also from optimism and self-esteem. Self-report questionnaire assessments have long been the mainstay of research in subjective well-being and happiness, as in many areas of psychological research. For some time now, researchers have been developing hybrid methods to measure happiness in the hope that such measures might increase validity. Although technically still a measure of self-report, Csikszentmihalyi (1997) introduced the Experience Sampling Method (ESM) in the 1970s. The ESM was novel in that it used electronic prompts to signal participants to record their immediate levels of happiness (or other variables such as motivation) rather than relying on retrospective reports. Csikszentmihalyi's ESM utilizes random prompting to generate large quantities of data, that when compiled is assumed to be highly reliable. A recent attempt to improve upon self-report data in the measurement of happiness is described by Zhou (2011) as an approach that combines a happiness evaluation method developed by Ng with the day reconstruction method by Kahneman and associates. According to Zhou, the combined method moves happiness assessment toward more accuracy and standardization.

UNIT III: WELLBEING AND EMOTIONS

Lesson-9: Positive emotions, positive relationships and wellbeing- Hope and optimism, love and belongingness, love and kindness.

Every humans wish to lead a happy life. The happiness is influences by the inner qualities what we have ourselves. Optimism is the main characteristics of happiness. Those who have optimistic belief can lead a peaceful life. Optimistic thought brings hope in everything and develops self confidence.

Hope

Hope reflects the will as well as the ways. The confidence aspect in hope is similar to optimism however there is more emphasis on personal agency. Pessimism has been found to resemble the construct of neuroticism. Neuroticism involves a tendency to worry, and experience unpleasant

emotions. Therefore, it can be said that a sense of pessimism is a part of neuroticism. Optimism and pessimism relate to people‘s expectations for the future.

Due to this they are linked to the expectancy-value theories of motivation. Expectancy-value theories assume that behavior is directed on the pursuit of goals. When a goal is important to someone, the greater is its value. Expectancy is a sense of confidence or doubt about whether goal can be attained or not.

Optimism

Optimism and pessimism refer to the positive and negative predictions that people make regarding their future. Some people look usually on the bright side of events, while others look on the dark side. Therefore, it can be said that optimists expect good things to happen to them while pessimists expect bad things to happen to them. Many studies have provided evidence for the optimistic bias. Optimism has been linked to psychological and physical well- being. The study of optimism has contributed a lot to what we understand about optimism today.

There are several other constructs that are similar or related to optimism and they pose a methodological issue. Two of these concepts are hope and self-efficacy. People with high self-efficacy expectancies believe that their personal efforts determine the outcome of events. Whereas self- efficacy looks at self as a causal agent, optimism takes a broader view of the potential causes at work.

Love and Belongingness:

Theories of love are difficult to classify. Nevertheless, they may be grouped under two broad headings; naturalistic/biological and psychological/social.

Naturalistic/Biological Approaches: Recent studies have begun to explore some of the biological underpinnings of our need for belongingness. For example, oxytocin is a pituitary hormone which sometimes is referred to as the —cuddle hormone¦ because close physical contacts such as touching, hugging and kissing stimulate its release (Hazan, Campa & Gur-Yaish, 2006).

Oxytocin, along with vasopressin helps humans and about 3 percent of other mammal species to experience lasting, monogamous love. These two chemicals are associated with our ability to form memories of others and help us recognize other people. They're also released, along with dopamine, during sex. Oxytocin levels however are at their highest during sexual orgasm, in both men and women (Uvnas-Moberg, 1997).

Love and Kindness:

Virtually all people act, and often talk, as if they have some inkling about love. We speak about loving food, falling in love, loving God, feeling loved, and loving a type of music. We say that love hurts, love waits, love stinks, and love means never having to say you’re sorry. We use the word and its derivatives in a wide variety of ways (Oord, 2003).

Trying to define the verb love by Oord (2003), “love is to act intentionally, in sympathetic response to others (including God), to promote well-being”. The word intentionally referring to three aspects of love: deliberateness, motive, and self-determination.

Webster’s Dictionary defines it as “the state, quality, or habit of being kind; kind act or treatment; kind feeling; affection; good will.”5 Synonyms of kindness are compassion, gentleness, benevolence, thoughtfulness, mercy, consideration, and helpfulness. Kindness manifests itself in words, actions, and nonverbal behavior. A cheerful mother serving supper to her family and asking them about their day is a picture of kindness (Beeke, 2007).

Trying to relate these two definitions, there is a high possibility to confuse these terms. Love and kindness are complementary words when it comes to actions. Where there is love, there must be kindness.

The components of this essay are:

1. The Act of Kindness

2. The journey of kindness

3. Loving Kindness According to Buddha and Developing Loving Kindness and Compassion

Love and kindness include:

1. Relating love and kindness

2. Explaining the act (activities) of love and kindness applicable to human society.

3. Relating love and God will

2. Situation analysis

2.1. The act of kindness

Human beings are created by God, the Creator of the universe. This relation with God has different dimensions which affect them as persons and all of their relationships. Rashmi Raj in his book says that people were given a very precious gift known as unique emotion side. But unfortunately human beings are using it for personal interest, the reason behind wars all over the world.

Naturally, referring to humanity definition, for every human being, kindness might be a must. So many examples can be given to express kindness. Some are:

(1)Bring treats to police officers or fire fighters. (2) Give up your seat on a bus, bench, or other public place to a pregnant woman or someone older than you. (3) Contribute to or pay the bill of the car behind you in the drive thru. (3) The next time you go out for breakfast, buy breakfast for a senior citizen or another family with young kids. Pick up trash in your neighborhood. (3) Leave $5 in an envelope with a note to the effect of “Do something good with this” in a public place. (6) Help a neighbor who could need a hand. Etc.

These acts of kindness contribute to the improvement of humanity good actions of the society. When somebody commits an act of kindness, he/she not only touches the life of the recipient, but he/she inspires others to be kind, too. When people participate in acts of kindness, it encourages individuals to thoughtfully commit acts of kindness in their workplaces, schools, communities, and homes. The movement is intended to fill the world with kindness and goodwill.

2.2 Why to help people in need

In one small step by EJ Peterson, a child asked to her mom: “What happened to helping people in need? What happened to putting others before yourself?” It can be good if everyone can answer positively to this question. After the mother’s positive response to help a woman and her little girl in need, the child said: “My heart felt as if it was being filled with water, it was just overflowing with lover for this one little girl who represented over a million of children all over the world , all reaching out to love, compassion, and kindness. After this kind of action, the child tried to thing the way we people we are not help each other, we are acting as some of us are treated in stranger in our shared world. After a good action of kindness and compassion, the child felt very empty soul, because he does not understand why people do not help each other.

Anik et all. (2009) in Working Paper on Feeling Good about Giving: The Benefits (and Costs) of Self-Interested Charitable Behavior, the author says that lay intuitions and pop psychology suggest that helping others leads to higher levels of happiness. This explains the feelings of the little child after they helped the young girl.

Helping others takes countless forms, from giving money to charity to helping a stranger dig his car out of the snow, and springs from countless motivations, from deep-rooted empathy to a more calculated desire for public recognition. Indeed, social scientists have identified a host of ways in which charitable behavior can lead to benefits for the giver, whether economically via tax breaks (Clotfelter, 1985, 1997; Reece & Zieschang 1985), socially via signaling one's wealth or status (Becker 1974; Glazer & Konrad 1996; Griskevicius et al., 2007) or psychologically via experiencing well-being from helping (Andreoni, 1989, 1990; Dunn, Aknin, & Norton, 2008).

Batson (1987; 1991) cited by Anik et all. (2009) said that the existing evidence suggests that happier people do indeed help more in a variety of contexts. Studies using random assignment to experimentally induce positive mood have provided important evidence that happiness causes increased helping behavior. Supporting the external validity of these findings, naturally occurring positive moods have also been shown to facilitate prosocial behavior. While we have focused on the impact of positive mood on giving, however, another well-documented area of inquiry has documented the impact of negative mood on helping as well, a seeming contradiction.

Based on above literatures, we can conclude saying that there is a relationship between helping others and happiness. People that help others are likely to be happier than others. If so why some people are not happy to see neighbors enjoying healthier life. There is a socio-economic inequality between people in world. Some are the richest (financially) in the world; others are the poorest of the world. Why we people we do not share what we have! Can we say that there is no enough resources or can we say that the available resources are among few hands not ready to share with others?

Sometimes I see people lying in middle ways suffering from hungry where I pass going to job, and even in my area where I live. Are these people most sinners compared to others? This was the side of financial.

The term kindness is not only applied to financial support. A kind person even the one who do not higher financial means can act positively to support people in need. Let me use an example based on my experience. After I finish secondary school in Groupe Scolaire Saint Joseph de Nyamasheke, in Western Province of Rwanda, July 2006, few months before I join University, I went to Southern Province in Nyamagabe. It was my first time to go there to pay a visit to my Sister. Around 7:00PM, I went out by foot to a nearby center to discover the area. On my way coming back the sister's house, I met an old woman with at least 85 years old; she was very weak with a heavy package on head. After I greet her, I asked gently if I can help, she replied yes kindly. I took the package up to her house. I did not go inside but she blessed me with very sweet words, wishing me all kinds of greetings from Heaven.

I gave personal experience to express how much I was happy after doing this. Today almost 10 years were passed, but one thing I know is that One among blessings which make me strong today in different corner of my life is the one from that old woman which I did not see again. Based on this I c an conclude that be kind, do good things, helping others bring blessings in your life, and this makes you happy ever.

2.2 Love and Kindness Vs and Religion

By Wilson, 2015, in Kindness in World Religions, a man once asked the Prophet what was the best thing in Islam, and the latter replied, "It is to feed the hungry and to give the greeting of peace both to those one knows and to those one does not know." This is on side of Islam. Nothing is different to the theory of loving-kindness.

When referring to God love, it inspires everybody on earth who believes in God to practice kindness to his/her friends, family, neighbors, etc. All God believers know that whoever does not love does not know God, because God is love. But anyone who does not love does not

know God, for God is love. Anyone who does not love does not know God, because God is love. The one who does not love does not know God, for God is love. From this, a question rise: Does we God believers can practice this love? If yes the world will be safe from war, hunger, etc. If no, do we God believers really know in what we believe? Maybe people think they know, but maybe they don't.

To be kind, the love has to be unconditional. Usual meaning of the word unconditional is `without conditions or reservations; absolute. To be a child of God, people have to love people. The unconditional love is a way to help people to have kindness to support each other when in need.

Lesson-10 Gratitude, forgiveness, empathy and altruism

Gratitude:

The word 'Gratitude' is derived from the Latin term *gratia*, which means grace, gratefulness, and graciousness. It's a concept that is close to kindness. Gratitude has been a very popular topic amongst the self-help book authors and also it attracts a lot of attention as it is a virtue that is extremely likeable and desired. However, gratitude like a few other positive psychology constructs e.g. courage hasn't received much attention. One reason that could be attributed to the lack of empirical research in this area could be the fact that this concept is difficult to empirically measure and put to test. Therefore, relationship of gratitude with health and well being variables stand speculative and are under scrutiny. The systematic study of gratitude had its auspicious beginning in 2000 under the purview of positive psychology.

Gratitude is different from the feeling of indebtness, which is an obligation of repayment. Gratitude can be seen as a motivator of behaviourwherein people work towards enhancing and improving relationships withthe benefactor. Robert Emmons, a leader in the field of gratitude research,defines gratitude as the feeling that occurs when a person attributes a benefit received by him to another. According to McCullough, Kilpatrick, Emmons and Larson (2001) gratitude is seen as a moral affect as both the origins and consequences of gratitude are oriented toward the well-being ofanother person. It is a prosocial behaviour that has positive feelings both for the beneficiary and the benefactor.

Functions of Gratitude

McCullough and colleagues have put forth 3 functions of gratitude. First, as a **moral barometer** wherein gratitude signals a change in one's social relationships, for both the people and instills both these people withpositive feelings. Second, as a **moral motive**, due to this act of kindness, the beneficiary is also reminded of other acts of gratitude that may have happened in past and may take a chance to reciprocate these benefactors for their kindness, Third, as a **moral reinforcer**, wherein the beneficiary isinstilled with an obligation to continue the tradition of helping other and thus reciprocate grateful acts to other people, which starts a chain of positive acts toward mankind.

Benefits of Gratitude

Gratitude is an emotional expression of thankfulness. It also has a very strong spiritual component wherein gratitude is seen as a source of connecting to each other. Gratitude allows an individual to appreciate everyday mundane activities with a sense of wonder and appreciation for life. Life becomes worth living if it has moments of gratitude. It has great impacts on an individual's psychological and emotional well-being as it provides continuous intrinsic rewards.

Gratitude has been linked with positive states such as happiness, pride and hope (Overwalle, Mervielde, & De Schuyter, 1995), experiences of extreme happiness (Gallup, 1998) and experiences of a meaningful and productive life (Emmons & Crumpler, 2000). It has empathy at its foundation in an interpersonal emotion that exists in an interpersonal context. Gratitude imbibes feelings of love and care. In this interpersonal context it is highly imperative to acknowledge the cost of the gift, the recipient must identify with the psychological state of the one who has provided it. The benefactor's giving is interpreted by

the recipient as freelyoffered, and with that comes the acknowledgment that such offering might prove costly to or incur hardship for the benefactor. Such an understanding blends fittingly with some object relations formulations of gratitude, where it is seen as a major derivative of the capacity for love (Klein, 1957; Emmons & Shelton, 2002).
Gratitude is related to a host of positive outcomes, including subjective well-being, relational support, and prosocial behavior.

Feelings of gratitude are associated with less frequent negative emotions and more frequent positive emotions such as feeling energized, alert, and enthusiastic (McCullough, Emmons, & Tsang, 2002).

It appears to broaden people‘s modes of thinking as they creativelyconsider a wide array of actions that might benefit others. This positive emotion also builds psychological and social resources (Emmons & Shelton, 2002). Gratitude also builds people‘s skill for loving and showingappreciation. Their creativity enhances and they then find new ways to express their gratitude.

Cultivating Gratitude

It is important to understand that whether the virtue of gratitude can be cultivated? Does developing gratitude provide a depressed individual witha hope to live for? Or can it be used to alleviate distress and provide avenues for better positive living? There have been researches that have put forth steps through which an individual can develop the value of gratitude. Miller (1995) offers a 4 step guide: (1) Identify non-grateful thoughts; (2) formulate gratitude-supporting thoughts; (3) substitute the gratitude-supporting thoughts for non-grateful thoughts and (4) translate the inner feelings into outward action.

Shelton (2000) believed that developing a healthy moral life involves (1) a self awareness that one is moral, (2) developing gratitude (3) self examination of one‘s day and (4) encouraging one‘s self to increase moral responsibility. All these ways place a high degree of initiative that the individual will have to take to develop. Also engaging in and keeping a daily moral inventory also helps in developing gratitude.

Remembering one‘s acts of gratitude instills positive feelings that further motivate an individual to indulge in more subsequent acts ofgratitude. This therefore, starts a cycle of gratitude. The acts that have beenbuilt over a period of time also act as buffers in protecting an individual from negative emotions such as guilt, depression, shame etc.
Making a personal commitment that one wants to invest psychic energy indeveloping a personal schema, outlook or worldview of one‘s life as a giftor one‘s very self as gifted holds considerable sway from the standpoint ofpositive psychology.

Indeed numerous groups (like ĴAlcoholics Anonymous‘) haveabsorbed this insight wherein people get together and view themselves as agift. Thus, setting aside time on daily basis to recall moments of gratitude associated with even mundane or ordinary events, personal attributes one has, or valued people one encounters has the potential to interweave and thread together a sustainable life theme of highly cherished personalmeaning just as it nourishes a fundamental life stance whose thrust is decidedly positive (Emmons & Shelton, 2002).

Thus, this highly required virtue of gratitude needs to be inculcatedin infants since their childhoods. They should be taught the importance ofbeing thankful and gratitude. This virtue develops over a period that requires constant motivation and an environment that is provided byteachers and parents.

FORGIVENESS

Forgiveness is one of the supreme human virtues. The idea of forgiveness is integral to many religious texts, scriptures and philosophical discourses. William Shakespeare in his classic work
The Merchant of Venice has considered mercy as the finest of all human qualities. In the Hindu religious mythology, a defining quality of all gods isthe ability to forgive and punish. The idea and ability of forgiveness has been more often connected with the idea of divinity as compared to the ability to punish. In Buddhism a form of forgiveness called compassion has been considered a way of achieving Nirvana or salvation.

There is an increasing need in the society for an absolute understanding of forgiveness. Traditionally, forgiveness was conceptualized as acting on a mere moral or religious intention.

Although several empirical articles have been published in research journals about forgiveness, those with a philosophical or psychological perspective appeared only a few decades ag(McCollough, Pargament & Thoresen, 2000). Adequate financial resources and research teams working on scientific research on forgiveness could create a global awareness for the significance of forgiveness research.

Forgiveness is a comprehensive word and it comes in many formsand varieties. Words like mercy, compassion, excusing, forgetting, etc., can be associated with the idea of forgiveness. Forgiveness is a positive pro-social behavior and the chances of its being confounded with the otherkind of pro-social constructs like compassion or mercy are significantly high. As previously stated, forgiveness is a positive pro-social behavior which involves the presence of the victim, a perpetrator and a criminal actor wrongdoing from the perpetrator to the victim. If in response to the perpetrated wrongdoing the victim forgives the wrong doer then we call itan instance of forgiveness. Thompson et al (2005) descries forgiveness‘ asrelieving oneself of the negativity fastened to the transgressor. The target of the consequent forgiveness may be either oneself or another person or the situation.

Further, a look at the existing research literature suggests that forgiveness has been defined either as a response, or as a personality characteristic, or as a characteristic of the social unit. These three aspects of definitions have been discussed below:

□ *As response*, forgiveness can be defined as a pro-social change in thoughts, emotions and/or behaviors of a victim toward the wrongdoer. Conceptualizations of forgiveness as a response are based on one primaryfeature: when people forgive, their thoughts and behaviors towards the offenders become less negative and more pro-social eventually. (McCollough et al, 2000).

□ *As a personality characteristic*, forgiveness may be explained as a tendency to forgive others in diversified circumstances. Although most people fall around the mean of the population in a forgiveness unforgiveness continuum, the disposition to forgive itself may have certainaspects (Mullet et al, 1998).

□ *As a characteristic of social units*, forgiveness may be described as a virtue similar to intimacy, trust or commitment. Some social institutions like families or marriages are characterized by a higher intensity offorgiveness, whereas other institutions, which cast out members whotransgress, are characterized by less forgiveness.

Types of Forgiveness

Forgiveness can be distinguished between positive forgiveness andnegative forgiveness. Positive (or emotional) forgiveness is a therapeutic process of absolute forgiveness, which also involves reinstituting positive feelings and thoughts toward the offender. Negative forgiveness, on the other hand, is a situation in which forgiveness is extended while brooding over the act of transgression. Negative forgiveness, also known as decisional forgiveness, involves mere overt inhibition of a retributive response, mostly in the interests of social acceptance. Yet, grudges are formed and sustained.

The two types of forgiveness indicate two distinct psychological processes. In pure positive forgiveness, cognitive restructuring is used to create the belief that retribution, of any kind is unnecessary. Retribution is instead a coping mechanism in situations where the hurt avoidance failed.Emotional forgiveness is not expected to improve psychological well-being but to restore well-being following the negative feelings or grudges.Negative or decisional forgiveness, on the contrary, involves a dissonancein the thought process of the victim; his positive well-being maybe enhanced by the thought that he has forgiven the transgressor, but the negative effect of grudge holding and anger would sustain. However, the process of adapting to consonant thoughts about the transgression will occur in varying degrees, depending on variables such as the degree of hurt, or victim‘s capacity for forgiveness. The meta-analysis of

Wade et al. (2005) confirms a positive relationship between the time spent in group counseling and the intensity of forgiveness. It was concluded thatchanges in forgiveness patterns reflect the passage of time and strength oftraining to reduce negative forgiveness.

Applications

The concept of forgiveness has been applied in many situations anddisciplines. Below are listed some of the major areas of human life whereforgiveness has its applications.

□ ***Forgiveness & Health:*** Worthington et al (2006) pointed out that unforgiving responses lead to poor health, especially cardiovascular activity and blood pressure. People with high blood pressures, generally are less forgiving than those who are forgiving.
Researches also explain that forgiving people have better social support, are less likely to depend on alcohol or drugs, and are less prone to anxietyand depression. Acquiring the capacity to forgive can prove to be beneficial to families dealing with health issues (Friedman, 2009). Hence,forgiveness enhances your health.

□ ***Forgiveness in Marriage:*** Researchers studying forgiveness in interpersonal relationships are of the opinion that forgiveness is essential part of marriage and relationship dynamics. People who are capable to forgive experience greater levels of marital satisfaction and longevity, better communication, and enhanced intimacy. It also reduces the negative affect in the relationship. It is necessary to learn to forgive your spouse. Itsaves one from repression of emotions of sadness or anger toward the spouse, which could instead result in more negativity and frustration.

□ ***Forgiveness as a Healing Process:*** Forgiveness can help alleviate psychological problems like anxiety, stress, depression, etc. Also, forgiveness has an impact on psychological well-being of an individual. It can be used to recover from psychological issues and to boost one‘s self- esteem and hopefulness (Enright, 2001). It is important to confront the issue and let go of the self-inflicted emotional pain and hurt.

Empathy & Altruism:

Human behavior is holistically guided by an interdependent natureand we spend a lot of time in our lives helping others in many different ways. From a small act of helping someone locate an address to volunteering to help victims of natural disasters, we engage in a lot activity, consciously or unconsciously that basically benefit someone else- a friend or stranger for that matter. But in terms of Psychology, the question lies in exploring the reasons behind this human nature to even put themselves in danger to help someone else. However, when social psychologists started unfolding the pages of helping behavior in the late 1960s, they didn‘t ponder over the decision of why people helped, but rather on why people fail to help when we think they should.

Over the past 50 years however, the focus of helping behavior and its related ideas have changed and we find under its wraps the processes of empathy and altruism. Empathy is a kind of emotional response to the plight of another person. It is the ability to put oneself in the place of another person and feel what she/he does. Rather than mimicking someoneelse‘s emotions, it is about having an accurate sense of what she/he mightbe feeling like.
The term *altruism* is used to describe two distinct phenomena. The first is *behavioral altruism*, which refers to helping behavior that is eithervery costly to the helper or conveys no self-benefit for the helper. The second is *psychological altruism*, which refers to a motivation to increasethe welfare of another as an end in itself. Although the latter definition is commonly used among psychologists and philosophers, the former is popular among researchers who study non-human animals.
Altruism is when we promote someone else‘s welfare even at the cost or risk of our own interests. It stands in opposition to being selfish and refersto a specific form of motivation that leads people to engage in behaviors that are aimed at benefitting others. It aims at increasing another‘s welfare. The reason for altruistic motivation is empathic concern. Empathic concern is another oriented emotional response driven by the perceived welfare of someone in need. Universal egoism states that everything we do are ultimately aimedat self-benefit. Egoism is the motive to pursue some sort of personal benefit through targeted behavior. This personal benefit can take the formof relieving the negative emotions that an undesirable situation produces inus. Proponents of pure altruism agree that the motivation for much of

what we do is egoistic. But some of us to some extent are capable of beingmotivated with an ultimate goal of benefiting else. So they propose that altruism, if not completely, is a part of basic human nature, one that arises out of empathic motions like compassion, sympathy, tenderheartedness and the like.

Empathic Emotion:

An important source of altruistic motivation is other-oriented emotional reaction elicited in congruence with the perceived welfare of them. This reaction has variously been called empathy; sympathy; sympathetic distress; and compassion.
Eight uses of the term Empathy (Batson, 2009): In relation to the concept of empathy, there seem to emerge eight different standpoints, as given by Daniel Batson. These are being discussed below-

Knowing the thoughts and feelings of another person: Various researchersstate knowing another person's internal state as empathy, cognitive empathy, or understanding.

Assuming the posture of an observed other or mimicking the neuralresponse: Preston & de Waal (2003) put forth the idea that while trying totake the posture of an observed other, or trying to feel what the other person goes through, one goes through a state of mimicking the neural response. This not necessarily means imitating the motor behavior, but having similar kind of neural responses. The neural responses become similar because perception and action work on the same neural circuits. Soas the perceptions become similar, so do the neural response and thereby one starts to get a better glimpse of the other person's internalstates.

Feeling as another person would feel: Feeling the same emotion that another person would feel is a common understanding of empathy. Philosophers call coming to feel as the other feels as sympathy.

Intuiting or projecting oneself into another's situations: This state also hasbeen called —projective empathy¦ (Becker, 1931). We try to projectourselves into another's situation and with the help of intuitive feelings, tryto feel as they do.

*Imagining how another is feeling:*Wispe (1968) called imagining how another is feeling psychological empathy. Stotland (1969) spoke of an—imagine other¦ perspective.

Imagining how one would feel in the other's place: Adam Smith (1759/1853) referred to this act of imagination as —changing places in fancy.¦ Mead (1934) called it —role taking¦. It is also known as—perspective taking.

Being upset by another's suffering: The state of personal distress evoked by seeing another in distress has been given a variety of names such as sympathetic pain; promotive tension; unpleasant arousal and empathy.

Feeling for another person who is suffering: The term empathy refers to anemotion that is *other-oriented* and is *congruent* with the perceived welfareof someone else. Congruence refers to the valence of the emotion- positiveor negative. However, congruence of emotion doesn't 't imply that thecontent of what you feel for the other person has to be the same.

Lesson-11 Positive emotions and health resources - Expanding the Repertoire of Pleasure - Happiness and Subjective Well-being- Influence of positive Emotions.

Psychologists often employ the notion of affect as an umbrella term for various positive and negative emotions, feelings and moods we frequently experience and easily recognize. Seligman (2002) in his book Authentic Happiness classifies positive emotions into three categories: those associated with the past such as satisfaction, contentment etc, the present and the future such as optimism, and hope. Momentary pleasures and more enduring gratifications are related to the present. Gratifications involve states of absorption that result from engagement in activities in which the unique signature strengths are utilized. Positive emotions have received little empirical attention since traditionally the focus was more on psychological problems and their remedies. Another reason could be that there is a habit among emotion theorists of creating general models of emotions which typically focus on negative emotions with positive emotions added later. Positive emotions facilitate approach behaviors. They motivate individuals to engage with their environments and undertake activities which facilitate adaptation and survival.

The value of positive emotions

For years, psychology turned its attention to the study of negative emotions or negative effect, including: depression, sadness, anger, stress and anxiety. Not surprisingly, psychologists found them interesting because they may often lead to, or signal the presence of, psychological disorders. However, positive emotions are no less fascinating, if only because of many common sense misconceptions that exist about positive affect. We tend to think, for example, that positive affect typically, by its very nature, distorts or disrupts orderly, effective thinking, that positive emotions are somehow simple‘ or that, because these emotions are short lived, they cannot have a long term impact. Research has shown the above not to be the case but it took it a while to get there. It is only relatively recently that psychologists realized that positive emotions can be seen as valuable in their own right, and started studying them.

Negative emotions, like anxiety or anger, are associated with tendencies to act in specific ways, which are adaptive in evolutionary terms, i.e. the fight and flight response. Thus, fear contributes to a tendency to escape and anger to a tendency to attack. If our ancestors were not equipped with such effective emotional tools, our own existence could have been doubtful. Moreover, negative emotions seem to narrow our action repertoires (or actual behaviors) when running from danger we are unlikely to appreciate a beautiful sunset. This function of negative emotions can help minimize distractions in an acute situation. Positive emotions, on the other hand, are not associated with specific actions. So what good are they, apart from the fact that they merely feel good? What is the point in feeling happy or joyful, affectionate or ecstatic?

The broaden and build theory of positive emotions, developed by Barbara Fredrickson, shows that positive affective experiences contribute and have a long lasting effect on our personal growth and development .

(a) Positive emotions broaden our thought action repertoires: First of all, positive emotions broaden our attention and thinking, which means that we have more positive and a greater variety of thoughts. When we are experiencing positive emotions, like joy or interest, we are more likely to be creative, to see more opportunities, to be open to relationships with others, to play, to be more flexible and open minded.

(b) Positive emotions undo negative emotions: It‘s hard to experience both positive and negative emotions simultaneously, thus a deliberate experience of positive emotions at times when negative emotions are dominant can serve to undo their lingering effects. Mild joy and contentment can eliminate the stress experienced at a physiological level.

(c) Positive emotions enhance resilience: Enjoyment, happy playfulness, contentment, satisfaction, warm friendship, love, and affection all enhance resilience and the ability to cope, while negative emotions, in contrast, decrease them. Positive emotions can enhance problem focused coping, positive reappraisal, or infusing negative events with positive meaning, all of which facilitate fast bouncing back after an unpleasant event.

(d) Positive emotions build psychological repertoire: Far from having only a momentary effect, positive emotions help to build important physical, intellectual, social and psychological resources that are enduring, even though the emotions themselves are temporary. For example, the positive emotions associated with play can build physical abilities and self mastery, enjoyable times with friends – increase social skills.

(e) Positive emotions can trigger an upward developmental spiral: More than that, just as negative emotions can lead one into downward spirals of depression, positive emotions can

trigger upward developmental spirals towards improved emotional wellbeing and transform people into better versions of themselves.

Lesson-12 Determinants of Subjective Wellbeing – 21st Century Definition of Happiness – Complete Mental Health: Emotional – Social and Psychological Wellbeing – Increasing Happiness in Life

DETERMINANTS OF SUBJECTIVE WELL-BEING

When examining college students' (from 31 nations) satisfaction in various life domains, financial status was more highly correlated with satisfaction for students in poor nations than for those in wealthy nations (Diener & Diener, 1995). Moreover, the people in wealthy nations generally were happier than those in impoverished nations. Within-nation examination of this link between income and well-being reveals that, once household income

Psychological well-being

Psychological well-being refers to the achievement of one‘s full psychological potential. The construct is central to the humanistic tradition. Professor **Carol Ryff** is the leading researcher in this area. She has developed a scale called The ***Ryff Scales of Psychological Well-Being***. It measures multiple facets of psychological wellbeing as the following:

□ autonomy

□ environmental mastery

□ personal growth

□ positive relations with others

□ purpose in life

□ self-acceptance

Keyes et al. (2002) in a factor analytic study involving over 3,000 Americans aged 25-74 found that psychological well-being and subjective well-being were related, but distinct, constructs which correlated differentially with socio-demographic variables and personality.

Social well-being

Social well-being refers to positive states associated with optimal functioning within one‘s social network and community. It refers to an end state in which basic human needs are met and people coexist peacefully. There is an equal access to basic facilities such as water, food, shelter, health services etc.

Complete Mental Health: Emotional, Social, and Psychological Well-Being

Ryff and Keyes (1995; Keyes & Lopez, 2002; Keyes & Magyar- Moe, 2003) combine many principles of pleasure to define complete mental health.

Specifically, they view optimal functioning as the combination of emotional well-being (as they refer to subjective well-being; defined as the presence of positive affect and satisfaction with life and the absence of negative affect), social well-being (incorporating acceptance, actualization, contribution, coherence, and integration), and psychological well-being (combining self acceptance, personal growth, purpose in life, environmental mastery, autonomy, positive relations with others). Taking the symptoms of mental illness into consideration, they define "complete mental health" as the combination of "high levels of symptoms of emotional well-being, psychological wellbeing, and social well-being, as well as the absence of recent mental illness" (Keyes & Lopez, 2002, p. 49). This view of mental health combines all facets of well-being into a model that is both dimensional (because extremes of mental health and illness symptomatology are reflected) and categorical (because assignment to distinct diagnostic categories is possible). This complete state model (Keyes & Lopez, p. 49; see Figure 7.5) suggests that combined mental health and mental illness symptoms may be ever-changing, resulting in fluctuations in states of overall well-being

Lesson-13 Positive traits, personality, emotions and biology, positive beliefs.

Well-being is a dynamic concept that includes subjective, social, and psychological dimensions. Psychological well-being, social well-being and health-related quality of life are constructs

related to, but distinct from, subjective well-being. A number of factors that contribute to happiness are as follows:

Personality traits and happiness: Personality studies of happiness show that happy and unhappy people have distinctive personality profiles (Diener et al., 1999). Happy people in western cultures are found to be extraverted, optimistic, having high self-esteem and an internal locus of control. In contrast, unhappy people are found to be high on neuroticism. However, a significant relationship has not been found between intelligence and happiness.

Cultural factors partially determine the types of personality factors associated with happiness. In western individualistic cultures such as the USA, self-esteem and acting in congruence with one's personal beliefs and personality factors is associated with high levels of subjective well-being. However, subjective wellbeing is not correlated with these factors in eastern collectivist societies. Therefore, it can be said that cultural values.

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Positive beliefs:

Positive psychology is a relatively new form of psychology. It emphasizes the positive influences in a person's life. These might include character strengths, optimistic emotions, and constructive institutions. This theory is based on the belief that happiness is derived from both emotional and mental factors. Positive psychology aims to help people identify happiness from moment to moment. It emphasizes this over only appreciating happy moments when looking back on them. People seeking therapy who desire to experience a greater sense of joy and liberation from their current circumstances may find this approach helpful. Many find it easier to focus on positive emotions they experience in the present after treatment has ended.

Role of Positive beliefs:

- Negative beliefs will lead to negative outcomes.
- Neutral beliefs won't do much of anything.
- Positive beliefs create positive outcomes.

Negative beliefs can create action. They can create both antipathy or apathy, evoke anger, tear down buildings, get people fired, or bring relationships to an end. But they will not create calming, collaborative and creative results going forward. Which means that if we want to create long-term positive outcomes, we must lead from a positive place. It's a short sentence and a simple concept but it's a big statement. If we want to build meaningful, healthy, sustainable organizations, organizations that generate positive energy and leave their communities better than we found them, we need to begin our work with positive beliefs.

Like many things in life, it's easier to understand this concept intellectually than it is to put it into daily practice. Negative beliefs are all around us. Most of us grew up, unknowingly, with plenty of them. They're on the news, they're in social media, they're in notes from meetings we go to. It could be thinking ill of coworkers or customers. Or about our neighbors, our in-laws, our partners, our spouses. It could be about entire groups of people—racism, anti-Semitism, the belief that women can't lead, or young people won't read. Negative beliefs are so pervasive that we're often not even aware that they've entered our minds—criticism and complaining, gossip and negative thinking can be so pervasive as to pass, pretty much unnoticed, as "normal." The more we hear them, the deeper the roots go, and the harder it is to get them out of our heads. It can be done though. We have the freedom and power to choose our own beliefs. And small shifts that we make from negative beliefs to positive beliefs actually make positive differences. In the long run, a *big* difference.

To be clear, sticking to positive beliefs does not mean ignoring problems. We face big issues in our organizations and society at large every day. So, no, I'm not suggesting we nominate Pollyanna for President. But, it turns out, we can have negative beliefs about a problem ("There's nothing we can do. We're at the mercy of others."), or we can have positive beliefs about problems ("This is a serious issue—let's start working on how we can make things better"). Starting with positive beliefs won't guarantee good results, but they sure will increase the odds that we can make good things happen.

Unit IV: Wellbeing and Spirituality

Lesson-14 Religion, spirituality and wellbeing: role of religion and spirituality to maintain subjective wellbeing – A special focus to Indian spirituality and well being

Religion and spirituality: Moderate correlations have been found between happiness and involvement in religious activity in North American studies (Myers, 2000). This could be due to the following reasons: *First*, religion provides a belief system through which people to find meaning and hope (Seligman, 2002). *Second*, the involvement religious activities like visiting the place of worship, doing charity etc provides people with social support. *Third*, involvement in religion leads to a healthier lifestyle.

The term *spirituality* comes from the word *spirit* (to breathe) and there is general agreement that spirituality is a living, dynamic process that is oriented around whatever the individual may hold sacred. The *sacred* refers to concepts of God and transcendent reality as well as other aspects of life that take on divine character and significance by virtue of their association with the holy. Thus, the sacred can encompass material objects

(e.g., crucifix, American flag), special times (e.g., the Sabbath, birth and death), special places (e.g., cathedral, the outdoors), relationships (e.g., marriage, parenting), and psychological attributes (e.g., soul, virtues).

Spirituality can be defined as a search for the sacred, that is, an attempt to discover and hold onto the sacred and, when necessary, transform the sacred. In their search for the sacred, people take a variety of spiritual pathways. These paths include traditional or nontraditional organized religious beliefs (e.g., God, afterlife, karma), practices (e.g., prayer, meditation, rituals), experiences (e.g., mysticism, conversion), and institutions (e.g., church attendance, Bible study). Pathways to the sacred may also take nonreligious forms, such as walking in the outdoors, journaling, listening to music, scientific study, intimate relations with others, or participating in sociopolitical action.

Over the course of the lifespan, the search for the sacred can unfold in many directions. For some, spirituality is a relatively smooth, stable process. For others, spirituality involves sharp shifts in spiritual pathways and understandings of the sacred itself. Some embed their spirituality in a traditional religious milieu. Others leave traditional religious settings and pursue more individualized spiritual pathways and destinations. Still others join and leave a variety of religious contexts, traditional and nontraditional, over their lives. Spirituality is, in short, a rich, complex, and multiform process.

How Spirituality Works

Having shown that many people in the United States are spiritually involved and that spirituality is linked to health and well-being, researchers are currently taking a closer look at how spirituality works in people's lives. Although this research is still in process, several preliminary conclusions appear to be warranted.

The Sacred Elicits Spiritual Emotions: The experience of sacredness is accompanied by a sense of transcendence (i.e., being connected to something that goes beyond oneself), boundlessness (i.e., infinite time and space), and ultimacy (i.e., being in touch with what is —really real¦). Theologian Rudolf Otto noted that the idea of the divine is often accompanied by a *mysterium*, a complex of feelings of attraction

love, adoration, gratitude) and repulsion (e.g., repulsion, fear, dread). More recently, researchers have linked perceptions of the sacred to a variety of emotion-based responses, including peak experiences, mystical experiences, and feelings of responsibility, duty, humility, awe, elevation, and uplift.

The Sacred becomes an Organizing Force: As the source of powerful emotions, the sacred becomes a passion and a priority. People feel drawn to, or even grasped by, the sacred and, as a result, they begin to invest more and more of themselves in sacred pursuits. For example, studies have shown that people who sanctify the environment (i.e., view the environment as sacred) are more likely to invest financially in environmental causes. Similarly, people direct more of their time and energy to sacred strivings than non-sacred strivings. In the process of building their lives around the sacred, people look to the sacred to lend greater coherence to disparate thoughts, feelings, actions, and goals by integrating their competing aspirations into a unified life plan.

The Sacred becomes a Resource: Sacred beliefs, practices, experiences, values, and relationships often serve as resources that people can draw on for strength, support, and satisfaction in their lives. A large body of research points to the positive implications of several spiritual pathways for individual health and well-being. Prayer and meditation, beliefs in a loving God, attendance at religious services, positive spiritual coping, and perceptions of various aspects of life as sacred have all been associated with indices of greater psychological, social, physical, and spiritual well-being. For example, in one study of patients undergoing kidney transplant surgery and their loved ones, those who engaged in positive religious coping reported greater life satisfaction 3 months and 12 months after transplantation. In a meta-analysis of 42 studies on the effects of transcendental meditation, this practice was tied to a greater number of transcendental experiences and these experiences were, in turn, associated with significant psychological and physical change.

Lesson-15 Understanding a Changing Human Behavior Primary Prevention – Components of Effective Primary Prevention – Head Start – Primary Preventions for Racial & Ethnic Minorities – Primary Preventions for Children and Elders – Secondary Preventions – Common Components of Secondary Preventions – Secondary Preventions for Racial & Ethnic Minorities – For Children and Elders.

Understanding a changing human behaviour takes place through the determinants like social diagnosis, epidemiological diagnosis, educational diagnosis, and environmental and behavioural diagnosis.

The focus of **social diagnosis** is to identify and evaluate the social problems which impact on the quality of life within your population. Doing a 'social diagnosis' will help you to gain an understanding of the social problems which affect the quality of life of people in your community, and how local people see those problems. This understanding is followed by the establishment of a link between the social problems and specific health problems that will become the focus of your health education activities.

Epidemiological diagnosis will help you to determine the specific health issues that affect the people in your community. The focus of this phase is to identify both the health problems and the non-health factors which are associated with a poor quality of life. Describing these health problems can help establish a relationship between health problems and the quality of life. It can also lead to the setting of priorities which will guide your health education programmes and show you how to best use your resources.

Educational diagnosis

This phase of assessment pinpoints the factors that must be changed to initiate and maintain behavioural change. **Educational diagnosis** looks at the specific features that hinder or promote behaviour related to the health issues that are important in your community. An

example would be to identify why people are behaving in a way that is dangerous to their life? As you will see later in the study session there are a number of ways of examining these sorts of questions as they relate to predisposing, enabling and reinforcing factors.

Environmental and behavioural diagnosis

Environmental diagnosis is a parallel analysis of factors in the social and physical environment that could be linked to health problems.

Behavioural diagnosis is the analysis of behavioural links to the health problems that are identified in the epidemiological or social diagnosis.

This phase focuses on the systematic identification of health practices or behaviours which cause health problems in your community. If you take the example of HIV/AIDS, once you reach this stage your focus will be to examine why people in your community are highly affected by HIV/AIDS. Is it because of their behaviours — or is it due to other environmental factors such as lack of HIV prevention and counselling services? At this stage you need to be able to identify the factors — and as in many societies the most important factor responsible for higher level of HIV/AIDS in your community may be high-risk sexual behaviours.

Types of Prevention:

Primary prevention:

Primary prevention aims to prevent disease or injury before it ever occurs. This is done by preventing exposures to hazards that cause disease or injury, altering unhealthy or unsafe behaviours that can lead to disease or injury, and increasing resistance to disease or injury should exposure occur. Examples include:

- legislation and enforcement to ban or control the use of hazardous products (e.g. asbestos) or to mandate safe and healthy practices (e.g. use of seatbelts and bike helmets)
- education about healthy and safe habits (e.g. eating well, exercising regularly, not smoking)
- immunization against infectious diseases.

Secondary prevention

Secondary prevention aims to reduce the impact of a disease or injury that has already occurred. This is done by detecting and treating disease or injury as soon as possible to halt or slow its progress, encouraging personal strategies to prevent reinjury or recurrence, and implementing programs to return people to their original health and function to prevent long-term problems. Examples include:

- regular exams and screening tests to detect disease in its earliest stages (e.g. mammograms to detect breast cancer)
- daily, low-dose aspirins and/or diet and exercise programs to prevent further heart attacks or strokes
- suitably modified work so injured or ill workers can return safely to their jobs.

Tertiary prevention

Tertiary prevention aims to soften the impact of an ongoing illness or injury that has lasting effects. This is done by helping people manage long-term, often-complex health problems and injuries (e.g. chronic diseases, permanent impairments) in order to improve as much as possible their ability to function, their quality of life and their life expectancy. Examples include:

- cardiac or stroke rehabilitation programs, chronic disease management programs (e.g. for diabetes, arthritis, depression, etc.)
- support groups that allow members to share strategies for living well

- vocational rehabilitation programs to retrain workers for new jobs when they have recovered as much as possible.

For many health problems, a combination of primary, secondary and tertiary interventions are needed to achieve a meaningful degree of prevention and protection. However, as this example shows, prevention experts say that the further "upstream" one is from a negative health outcome, the likelier it is that any intervention will be effective.

Lesson-16 Virtues in Indian culture - Primary Enhancement – Psychological Health – Physical Health – Secondary Enhancement – Psychological and Physiological Health – Balancing the Preventions and Enhancement systems

Indian thinkers, like their counterparts elsewhere, recognized morality's pervasiveness throughout human life and culture, and did not shy away from enquiry into the nature of morality, right and wrong, and good and bad. On the side of "good" they placed such values as happiness, health, survival, progeny, pleasure, calmness, friendship, knowledge and truth. h e "bad" were, more or less, opposites or disvalues: misery, suffering, sickness, injury, death, barrenness, pain, anger, enmity, ignorance, error and untruth. h ese positive and negative qualities are universalized, in principle at least, for all sentient beings, for it was felt that the highest good (summum bonum) is possible when the whole world can enjoy the good things that the cosmos has to of er. Reflecting upon the meaning or applications of these judgements has been the way of bringing theory to bear upon the time- worn practices intended to yield the highest good. h ese may extend from ritual performances and collective desires to affective responses to challenging situations from their surroundings. In much of Indian philosophy, the beginning point is not discursive theorizing on i rst principles, axiomatic propositions, rational intuitions and so on. Rather, the starting point is the practices that are embedded or grounded in all human cognitive, affective, social- political and even aesthetic efforts. Over time these practices come to be embodied in a tradition and only then in comprehensive doctrines, or articulated first in oral and later in written texts foreshadowed by, and preiguring, other scriptural texts. Questions about their own beginnings – historical or conceptual – seem prosaic and unwarranted (Mohanty 1995: 8). The subsequent hermeneutic and critical exegesis of these practices via the texts encapsulating their intentionalities (bhāvanas) lend themselves to interpretations, protocols or laws in the sense of practical wisdom (akin to Aristotle's phronēsis), but not rigidly in the sense of theoria or purely discursive judgement as, for example, in Kant.

One term in classical Indian ethical discourse is perhaps the crucial one, and the closest Indian equivalent to the notion of ethics: "dharma". h is notion embodies the traditional pursuit of moral values and constitutes a distinctively essential aspect of Indian ethical culture, nearer to (if not a direct influence on) Hegel's concept of Sittlichkeit (the actual ethical order that regulates the conduct of the individual, family, civil life and state) than it is to Kant's ideal conception of the moral law (Tripathi 2004: 122). It has both a theoretical and a practical aspect. "Dharma" is semantically connected with the idea of r. ta, the underlying natural order in the Hindu belief system that connects individuals, society and the universe as a whole. h is constitutes an overarching social, moral and natural order that must be maintained, and that relies on human compliance with respect to renouncing selfish desires and accepting individual responsibility. In this regard the self that is anchored in dharma is inexorably a "relational social self " (N. Gier, cited in Puri 2013: n. 4). h is highest good is the telos, the creative purpose and motivation that underpins human behaviour. The summum bonum, however, expresses itself in the total harmony or homology of the cosmic and natural order thus characterized. R. ta, then, conveys the struggle for balance in the world and for the welfare of all beings, including gods, humans and animals. The interconnectedness of humanity and nature is a recurring idea and the foundation of the traditional ethical system. The belief is that the all- encompassing cosmic order must be upheld by dharma, the embodiment of rights, duties, laws, justice, virtue and truth. Dharma is the fixed position of duty and of right, in the sense of what is proper and normative. It is by no means

restricted to the realm of personal ethics, but also designates religious observance and secular law, prescribing the individual's social and legal standing within the wider domains of community, caste and station. Expanding its range even further, dharma connotes a general principle or law of nature to which the individual is bound in a two- fold sense, both in terms of supporting the cosmic, social and personal orders, and deriving from them a corresponding obligation. Thus, as universal order, it assigns to each entity, personal or impersonal, its specific place within the wider community

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: To assess the self esteem of the subject.

Activity 2: To assess the Depression level of the subject by using suitable test.

PSY 25145 **Psychology of Stress**

Objectives:

- To discuss the techniques of stress management like relaxation technique, meditation, Yoga, mindfulness and biofeedback.
- To discuss the nature of coping and explain its goals; and explain coping styles.
- To explain cognitive restructuring and discuss time management.

Unit-I: Introduction on stress

Lesson 1: Concept of Stress: Definition of Stress, History and present status of the stress concept. Indian view of stress and coping- Physiology of stress response: Fight or Flight- General adoption syndrome.

Lesson2: Generic diversity, Endocrinology, Role of nervous system, Somatic Disorders, Stress and environment.

Lesson 3: Stress and environment

Lesson 4: Models of Stress: Person-Environment fit theory, Cybernetic theory of stress, Coping and well-being, A multidimensional theory of Burnout.

Unit-II: Sources and Effects of Stress

Lesson 5: Sources of Stress: Physiological, Physiological and Sociological-

Lesson 6: Effects of Stress: Physiological effects, Physiological effects- Learned helplessness-

Lesson 7: Stress- Coping and Burnout.

Lesson 8: Moderators of stress: Age, Personality, Intelligence, and Social Support.

UNIT – III: STRESS MANAGEMENT INTERVENTION

Lesson – 9: Effects of Occupational Stress Effect on Job Behavior, Job Satisfaction, Performance, Absenteeism. Effects on Employee's Physical and Psychological Health.

Lesson – 10: Individual - Centered Stress Management Techniques Cognitive, Hypno-suggestive, Behavioural and Physical (Yoga and Aerobic Exercises) Interventions; Coping Strategies.

Lesson – 11: Organization Centered Stress Management Interventions: Preventing, Mitigating and Moderating Stress at Organizational Level.

UNIT – IV: STRESS MANAGEMENT- WORK AND AREAS

Lesson – 12: Work Related Stress and Stress Management What is Stress? - Stressors - Individual Differences in Stress - Consequences of Distress - Managing Work - Related Stress - Any Two Stress Reduction Techniques - Case Studies

Lesson – 13: Areas of Counselling Educational, Career, Marital and Gerontological; Stress Management Oriented Counselling; Counselling for Terminal Disease Patients: Cancer and HIV/AIDS.

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Unit-I: Introduction on stress

Lesson 1: Concept of Stress: Definition of Stress, History and present status of the stress concept. Indian view of stress and coping- Physiology of stress response: Fight or Flight- General adoption syndrome.

Stress is a state of mind that reflects certain biochemical reactions in the human body and is projected by a sense of anxiety, tension and depression and is caused by such demands by the environmental forces or internal forces that cannot be met by the resources available to the person. The intensity of such demands that require a readjustment of resources or operational styles would determine the extent of stress. Such environmental events or conditions that have the potential to induce stress are known as 'stressors'. In this unit, we will discuss the concept of stress, along with various models and theories on it. We will also focus on stress reactions as well as pain and its management.

Medical researcher Hans Selye first used the term 'stress' to describe the body's biological response mechanisms. He defined stress as 'the nonspecific response of the body to any demand'. It must be understood that for the stress to occur, the response should be non-specific. All responses require utilization of energy. Any demand made on the body that is for some specific activity that is natural, expected and a part of daily routine, does not necessarily create stress. Even walking, thinking, writing and doing physical activities that are a part of personal and organizational existence, require energy consumption of the body but are not necessarily stress producing forces.

Stress does not necessarily occur due to undesirable developments. All situations that produce increased demand on a vital activity requiring adaptation to a new situation, produce stress in the form of a stereotyped pattern of biochemical, functional and structural changes in the human organism. These situations could be fear, pain, fatigue, emotional arousal, humiliation, frustration, need for concentration, loss of blood, drugs, loss of a loved one, non-occurrence of an expected event and even unexpected successes that require a change in the operational style.

The stress created by desirable and successful events is called ‘eustress’ and the stress created by undesirable outcomes is known as ‘distress’. It is primarily the distress form of stress that requires examination and steps to cope with it. Eustress is a positive, healthy and developmental stress response. Thus, just as tension on muscles causes them to strengthen, some level of stress may lead to better performance and a more adjusted personality. Since we learn how to deal with our problems better, it improves our capacity to confront distress better. However, even though some levels of stress are necessary for psychological growth, creative activities and the acquisitions of new skills such as learning to drive a car or learning the use of a computer, it is the highly stressful situations that weaken a person’s physical and psychological capacity to cope with the stressors that have dysfunctional consequences. Just as high-level stress is damaging to the physical and psychological well-being of the person, extremely low levels of stress are equally undesirable for they cause boredom, and result in lack of stimulation, innovation and challenges. Thus, moderate level of stress is necessary for higher level of performance. The following diagram shows the relationship between the level of performance and degree of stress.

Theories of Stress

Stress has been explained with three ‘Traditions of Thought.

- The Epidemiological Tradition
- The Psychological Tradition
- The Biological Tradition

The Epidemiological Tradition The definition of what may constitute a stressful life event has varied over a generation of time. According to the epidemiological perspective, stress has been explained primarily by how others, judge the negative impact of particular events. These explanations are suggestively appropriate in prediction of morbidity due to depression, respiratory infections, and coronary heart disease, disease progression, wound healing, and autoimmune diseases and mortality. The focus of the epidemiological tradition of stress is in determination of circumstances, along with the experiences that may be stressful to a person on and constitute as threats to social or physical well-being of the individual or society.

The Psychological Tradition The psychological perspective concentrates upon particular observations, that is, the perspective that incidents of the same event can be stressful for some individuals but may not necessarily have any known repercussions on others. This theory dictates that hazard judgment of events is influenced by the immediacy of danger. Moreover, intensity, duration, potential accountability of the event, along with the self-esteem and the individual’s immediate surroundings, ethics, vices, values and commitments, and related personality dispositions all are considered in the psychological tradition of stress. The psychological perspective has defined stress as an experience that may occur when individuals simultaneously appraise life events as hazardous or life threatening and their coping resources or tools to manage the situation maybe imperfect or inadequate.

The Biological Tradition In the biological approach, the impact of identified stressors is indexed through perturbations of physiological systems that are otherwise important for homeostatic regulation and metabolic control. An assumption in the biological tradition is that these physiological perturbations or responses provide assistance over the short term for adaptive behavioural action or coping. Over the long term, however, such physiological reactions can prove maladaptive and contribute to risk for disease. The biological tradition of stress believes that stress is equivalent to chronic activation of one of the body’s principal neuroendocrine axes, the hypothalamic–pituitary– adrenal (HPA) system. This school of thought is rooted in the experimental psychophysiology and psychosomatic medicine. According to this, responses of the autonomic nervous system have long figured prominently as markers of stress. These days the biological context of human stress is gaining popularity with concepts characterizing the brain and appraising the psychological and social stressors,

which generate physiological reactions that might relate to disease risk. The focal point of all these three traditions of stress have been formed as the elements of a stage model of disease, wherein events appraised as stressful are viewed as triggering affective states that in turn engender behavioural and biological responses having possible implications for disease.

Types of stress:

1. Eustress: It is healthy, positive and developmental stress response. This is a motivational stress that leads a person to work hard enough to get ample amount of success in life.

2. Distress: It is a negative stress. It is caused due to tension and unhealthy relationships. Errors increase and this result in bad decision making that causes uneasiness.

3. Hyper stress: This is also a part of negative stress. It occurs when the individual is unable to cope with the workload. This results in headaches, person gets irritate on small-small issues.

4. Hypo stress: Hypo stress is an extreme low side of Eustress. This stress occurs when a person has nothing to do with his/her time and feels constantly bored and unmotivated. This reduces productivity of a person and causes idleness

STRESS REACTIONS

The process of stress elicits three types of responses. These are: physiological, psychological and behavioural responses.

Physiological responses As soon as stress appears, the brain reacts, and immediate biochemical changes take place in heart beat and heightening of practically all the senses. The long terms physiological effects are more disturbing. Serious health problems occur as body confronts stress over a long period of time. The stress could lead to breakdowns in the body's immune systems and may result in serious health problems such as high blood pressure, ulcers and heart attack. In general, according to Baron, 'taking all evidence into account though, it seems reasonable to conclude that high level of stress can result in physical changes that threaten our health and well-being'.

Psychological responses There are some people who can handle stress better than others. People who tend to be highly affected by stress tend to be depressed and lack self-confidence and self-esteem. They tend to believe that they are helpless and elicit sympathy from others. They have greater fear of the unknown and an increased sense of futility, tension and neurotic tendencies. They become irritated quickly, are impatient and tend to blame everybody else for their own problems. They are more worried about their job security and their job commitment is very low.

Behavioural responses According to Cohen, people under constant stress behave differently as compared to people who are emotionally well-balanced. Stress is usually associated with increased use of alcohol, smoking, eating and sometimes drugs. People under stress may gain weight and thus behave differently. Their behaviour becomes highly defensive or highly aggressive towards others and inter-personal relationships are highly affected. Stress induces irritation and lack of patience and these elements are exhibited in behavioural patterns. The person may become an introvert, may withdraw from social situations and may avoid communication with others resulting in social isolation.

History and present status of the stress concept

In addition to providing the first clear definition of stress, Hans Selye was also the first to recognize that homeostasis could not by itself ensure stability of body systems under stress.

He coined the term heterostasis (from the Greek heteros or other) as the process by which a new steady state was achieved by treatment with agents that stimulate the physiological adaptive mechanisms. Heterostasis, could be regarded as the precursor for the concept of allostasis, first advanced by Peter Sterling and Joseph Eyer in the 1980s (Sterling and Eyer, 1988; McEwen, 1998; Schulkin, 2004). That is, homeostasis, which has dominated physiological and medical thinking since the 19th century, is thought to provide "stability through constancy." Allostasis, on the other hand, provides "stability through change" brought about by regulation of the set-points that adjust physiological parameters to meet the stress/challenge.

A different tack, focused on cognition, was taken by the eminent and influential Berkeley University Psychologist, Richard Lazarus. At a time when psychology tried to understand human behavior by first understanding simple organisms engaging in simple behaviors learned by associations, rewards or punishments, Lazarus instead emphasized the importance of studying cognition which he extended into stress and coping (Lazarus, 2000, 2006).

Epidemiology Epidemiology has been and remains central to stress studies. Numerous epidemiological studies have tried to define the effects of social, workplace and lifestyle on stress, health and well being. Indeed, job stress is by far the major source of stress for American adults and has escalated progressively over the past few decades. Increased levels of job stress as assessed by the perception of having little control but many demands have been demonstrated to be associated with increased rates of heart attack, hypertension, obesity, addiction, anxiety, depression and other disorders. In New York, Los Angeles and other municipalities, the relationship between job stress and heart attacks is recognized, so that any police officer that suffers a coronary event on or off the job is assumed to have a work related injury and is compensated accordingly.

Thrifty Phenotype Hypothesis However, of the recent epidemiological studies that have generated new stress concepts, the most important perhaps remains that of David Barker (Barker, 1995), which led to the hypothesis that fetal undernutrition in middle to late gestation programs later coronary heart disease. This concept was soon extended by Hales and Barker (2001) in their "thrifty phenotype hypothesis." The latter proposes an association between poor fetal and infant growth and the subsequent development of type 2 diabetes and the metabolic syndrome, which afflict communities in epidemic proportions. Poor nutrition in early life, it is postulated, produces permanent changes in glucose-insulin metabolism. These changes include insulin resistance (and possibly defective insulin secretion) which, combined with effects of obesity, aging and physical inactivity, are the most important factors in determining type 2 diabetes. Many studies worldwide have confirmed Barker's initial epidemiological evidence, although the strength of the relationships has varied between studies. As well as intrauterine growth restriction, maternal obesity and diabetes can be associated with disease of the individual later in life (Rando and Simmons, 2015). Furthermore, paternal exposures can result in the later development of metabolic disorders in the offspring, and human and animal studies suggest intergenerational transmission of the maternal or paternal phenotype (Rando and Simmons, 2015). Several changes in epigenetic marks identified in offspring provide a likely mechanism by which early molecular changes result in persistent phenotypic changes (Keating and El-Osta, 2015), although additional mechanisms yet to be uncovered cannot be excluded.

Brain Plasticity Early life exposures, fetal and perinatal, could result in biological embedding by way of at least three plausible mechanisms: brain plasticity, epigenetics (especially DNA methylation; Demetriou et al., 2015), and hormonal action. There are numerous examples of brain plasticity, perhaps one of the most dramatic being the recruitment of new neurons and the increase in hippocampal gray matter in scatter-hoarding

animals which need to remember the location of food caches (Barnea and Nottebohm, 1994; Barnea et al., 2006) and London Cab drivers (Maguire et al., 2006; Woollett et al., 2009) for whom spatial memory is of paramount importance. Hormonal effects are exemplified by the possible effect of excess androgen in causing the polycystic ovary syndrome phenotype (Gur et al., 2015).

Lesson2: Generic diversity, Endocrinology, Role of nervous system, Somatic Disorders, Stress and environment.

Genetic Susceptibility to Stress: Gene x Environment Interaction With recent advances in genomics, the concept of susceptibility genes that increase the vulnerability of individuals to stressful life events has attracted considerable research interest. Thus, for example, the work of Avshalom Caspi and associates (Caspi et al., 2002) suggested that a polymorphism in the MAOA (monoamine oxidase A) gene promoter, which reduces MAOA expression, influences vulnerability to environmental stress, and that this biological process can be initiated by childhood abuse. Furthermore, in what seemed to be a landmark paper Caspi et al. (2003) reported that a polymorphism in the promoter of the serotonin transporter (SERT or 5-HTT) gene can render individuals more susceptible to stressful life events. Because SERT is also the therapeutic target for popular antidepressant drugs, the selective serotonin reuptake inhibitors, the publication by Caspi et al. (2003) led to a flurry of research publications and subsequent meta-analyses to assess the replicability of the findings. However, the overall conclusions from primary and meta-analyses on genetic x environmental interactions suggest that it is unlikely that there is a stable gene x environment interaction involving the SERT, life stress and mental disorders (Fergusson et al., 2011). Whether the uncertainty regarding the suggested link beween the SERT gene promoter polymorphism and susceptibility to stress reflects an artifact or problems in posthoc analysis is discussed by Rutter et al. (2009) who conclude that “the totality of the evidence on GXE is supportive of its reality but more work is needed to understand properly how 5HTT allelic variations affect response to stressors and to maltreatment.” In fact, recent data suggest that there is a stable gene x environment interaction involving MAOA, abuse exposure and antisocial behavior across the life course (Fergusson et al., 2012) as proposed in the earlier paper by Caspi et al. (2002).

Single nucleotide polymorphisms in the gene for the CRF receptor-1 have been tentatively linked to suicidal thoughts and behavior, depression, panic disorder and fear (Ben-Efraim et al., 2011; Ishitobi et al., 2012; Wasserman et al., 2008, 2009, 2010; Weber et al., 2015). Recent genetic and functional magnetic resonance imaging (fMRI) studies have shown that a functional deletion variant of ADRA2B, the gene encoding the alpha 2b-adrenergic receptor, is related to enhanced emotional memory in healthy humans and enhanced traumatic memory in war victims (Rasch et al., 2009). The ADRA2B deletion variant, which acts primarily as a loss-of-function polymorphism, is related to increased responsiveness and connectivity of brain regions implicated in emotional memory (Rasch et al., 2009; de Quervain et al., 2007). In normal subjects, only carriers of ADRA2B deletion variant showed increased phasic amygdala responses to acute psychological stress, illustrating that genetic affects on brain function can be context (state) dependent (Cousijn et al., 2010). In twin studies, Tambs et al. (2009) have investigated the genetic interactions and the 5 types of anxiety disorder. A recent report by Zannas et al. (2015) suggests that cumulative lifetime stress may accelerate epigenetic aging, possibly driven by glucocorticoid-induced DNA methylation. Taken

together, the above examples illustrate that our understanding of gene x environment interactions in stress and stressrelated disorders is still at an early phase. Nonetheless, these findings have heuristic importance for further thinking and research on genetic-environmental interactions that determine the response to stress and the development of mental disorders.

Neuroendocrine and Autonomic Nervous Control of Stress Response In parallel with these stress concepts, major neuroendocrine advances revealed the physiological substrate for homeostasis, allostasis and the stress response mechanisms. The autonomic nervous and the hypothalamic-pituitary-adrenocortical (HPA) systems subserve the afferent and efferent limbs of the stress response in vertebrates, and are also central to maintaining homeostasis and effecting allostasis. The term autonomic nervous system was coined in 1898 by the Cambridge Physiologist, John Newport Langley, who was also renowned for his development (in parallel with Paul Ehrlich) of receptor theory. Controlled by the brain, and utilizing as neurotransmitters epinephrine and norepinephrine (sympathetic nervous component) or acetylcholine (parasympathetic component), the role of the autonomic nervous system in fight-or-flight and homeostasis (especially cardiovascular) was clearly explained by Walter Cannon.

The story of our understanding of the HPA, and the concept of the neurohumoral hypothesis of anterior pituitary control, is tortuous (Fink, 2012, 2015). The pituitary gland had long been regarded, by luminaries such as the great Harvard neurosurgeon Harvey Cushing, as the autonomous controller of the adrenal cortex, the thyroid and the gonads. That is, the anterior pituitary gland was considered to be the "conductor of the endocrine orchestra." This view was reinforced by the dramatic effects of experimental hypophysectomy in rodents made feasible by the parapharyngeal surgical approach to the pituitary developed by PE Smith around 1930. However, at about the same time experiments were in progress that triggered a cascade of studies which would eventually prove that the anterior pituitary gland is not autonomous; rather, it is controlled by the brain. These experiments, carried out by William Rowan working alone in Edmonton Alberta in the late 1920s, showed that migration in birds was controlled by the gonads, and that gonadal size in birds was increased many-fold by increases in day-length. Day-length and the effects of other exteroreceptive factors, such as stress, on endocrine function together with the effects of brain tumors and trauma in the human, lead to an acceptance of the concept that the pituitary gland is under central nervous system (CNS) control.

The neural lobe of the pituitary gland is comprised of nerve projections from the paraventricular and supraoptic nuclei of the hypothalamus: these projections terminate on systemic blood vessels into which they release the nanopeptide neurohormones, vasopressin and oxytocin. In contrast, the anterior pituitary gland receives no direct innervation from the brain. Rather, the CNS control of the anterior pituitary gland is mediated by neurohormones synthesized and released from hypothalamic neurons and transported to the anterior pituitary gland by the hypophysial portal vessels. Proof of the neurohumoral hypothesis of anterior pituitary control came, first, from the elegant pituitary stalk section and pituitary grafting experiments of Geoffrey Harris and Dora Jacobsohn, secondly, the characterization of some of the neurohormones by Andrew Schally and Roger Guillemin, for which they were awarded the 1977 Nobel Prize for Physiology and Medicine, and, thirdly, the demonstration, first by my group, that these neurohormones were indeed released into hypophysial portal blood (Fink, 2012, 2015; Fink and Sheward, 1989; Fink et al., 1971). Corticotropin releasing hormone (CRH), a 41 amino acid peptide that mediates neural control of adrenocorticotropic hormone (ACTH) release from pituitary corticotropes, was isolated and sequenced by Wylie Vale et al. (1981). A series of physiological studies, including measurements of neurohormone release into hypophysial portal blood in vivo, have confirmed earlier views

that arginine vasopressin acts synergistically with CRH to control ACTH release (Fink, 2012; Fink et al., 1988).

Glucocorticoids – Discovery and Clinical Importance Finally, no outline of the history of stress concepts would be complete without mention of the characterization of the adrenocortical glucocorticoids and their function. The glucocorticoids are steroid hormones whose secretion by the adrenal cortex is controlled by ACTH. The hormones of the adrenal cortex were isolated, identified, and synthesized independently by Edward Kendall (at the Mayo Foundation) and Tadeus Reichstein (at Zurich) and their associates. The availibility of large amounts of synthetic steroids enabled their physiological effects to be studied. Ultimately, Philip Hench was able to test the glucocorticoid, cortisone, in the human and demonstrated that it was a powerful anti-inflammatory agent. Also from the Mayo Foundation, Hench had previously observed that rheumatoid arthritis was sometimes relieved during pregnancy and in some patients with jaundice, leading him to conclude that the pain-alleviating substance was a steroid. Kendall, Hench, and Reichstein jointly were awarded the Nobel Prize for Physiology and Medicine for 1950, and synthetic glucocorticoids continue to be used to treat arthritis, asthma, autoimmune conditions and other inflammatory disorders in the human (Raju, 1999).

Indian view of stress and coping

The origin of the concept of stress predates antiquity. Even prehistoric man must have recognised a common element in the loss of vigour and a sense of exhaustion that overcame him after various kinds of strenuous exertion. However, empirical research could not begin until Hans Selye wrote his first article on Stress in 1956.

The popularity of the stress concept has dwindled in the physiological field, where it first started and the use of stress terminology continues to flourish in the psychological and social fields, and during the past 15 years the term “stress” has come into wide use in relation to work organisations. Stress management refers to the wide spectrum of techniques and psychotherapies aimed at controlling a person's levels of stress, especially chronic stress, usually for the purpose of improving everyday functioning.

In this context, the term 'stress' refers only to a stress with significant negative consequences, or distress in the terminology advocated by Hans Selye, rather than what he calls eustress, a stress whose consequences are helpful or otherwise positive.

Stress produces numerous symptoms which vary according to persons, situations, and severity. These can include physical health decline as well as depression. The process of stress management is named as one of the keys to a happy and successful life in modern society. Although life provides numerous demands that can prove difficult to handle, stress management provides a number of ways to manage anxiety and maintain overall well-being. Despite stress often being thought of as a subjective experience, levels of stress are readily measurable using various physiological tests, similar to those used in polygraphs.

Many practical stress management techniques are available, some for use by health practitioners and others for self help, which may help an individual to reduce stress, provide positive feelings of being in control of one's life and promote general wellbeing.

The effectiveness of the different stress management techniques can be difficult to assess, as few of them have received significant attention from researchers. Consequently, the amount and quality of evidence for the various techniques varies widely. Some are accepted as effective treatments for use in psychotherapy, whilst others with less evidence favouring them are considered alternative therapies. Many professional organisations exist to promote and provide training in conventional or alternative therapies.

There are several models of stress management, each with distinctive explanations of mechanisms for controlling stress. Much more research is necessary to provide a better understanding of which mechanisms actually operate and are effective in practice

The concept of stress in modern science is not easily found in the traditional texts of Indian culture and tradition such as “Chark Samhita”, “Patanjali’s Yoga sutras” and “Bhagwad Gita”. However, there are a number of concepts, developed ancient Indian scholars which relate to or appear similar to the phenomenon of stress. Some of these, for example are “Dukha” (pain, misery or suffering), “Klesa” (afflictions), “Kama” or “Trisna” (desires), “atman” and “ahamkara” (self and ego), “adhi” (mental aberrations) and “parjnaparadha” (failure or lapse of consciousness). It is interesting to note that body-mind relationship, characteristic of modern stress studies, is emphasised in the ayurvedic (Indian) system of medicine.

Palsane, Bhavsar, Goswami and Evans (1986) noted that the Indian tradition is characterised by a holistic approach to human phenomena. Behaviour is interpreted in terms of a totality of individual's life styles and total body-mind relationship. Mindbody level of analysis in the Indian tradition is treated as less significant than analysis at a still higher transpersonal level of self, “atman” or soul, which in turn is equaled with “Brahman” the ultimate reality

Rao (1983) very succinctly spoke about the origin of stress in the Indian thought. Tracing it back to “sankhya” and “yoga” systems, he points out that there are two Sanskrit words “klesa” and “dukha” which approximate stress. The Word “klesa” has its origin in the root “khis” which means to torment, cause pain or “to afflict”. The “klesas” are not mental processes but are a set of “hindering load” on our mental process; they produce agitations which act as restrictions or hindrances. The “samkhya yoga” explains that the fundamental non-cognition which leads to phenomenological stress is “avidya”. This “avidya” leads to “asmita” (self appraisal), “raga” (object appraisal), “dvesha” (threat appraisal) and “athinivesa” (coping orientation), Faulty evaluation in either or all of the three appraisals, namely, those concerning the self, object and the threat, can cause stress and torment.

The samkhya system postulates that the feeling of “dukh” or stress is experienced by the individual in the course of his interaction with the world around him. There are three types of stresses which samkhya speaks of: personal (adhyatmik), situational (adhibhotik) and environmental (adhidevik).

The personal stresses can again be of two types, namely, physiological and psychological (or mental). The physiological stresses are born out of imbalances between the three fundamental physiologic constituents, namely, “vata”, “pitta”, and “kaph”. The psychological stresses are caused by emotional states of lust, hatred, greed, fear, jealousy and depression. The situational stresses are usually caused by unwholesome interpersonal transactions which might include conflicts, competitiveness, aggression etc. The third type of stresses namely the environmental stress are occasioned by natural calamities, extremes of temperatures, storms etc.

The “abhinivesa” indicates the commencement of coping behaviours by arranging the behaviours in a proper response sequence. Looking back at the concepts of “asmita”, “raga” and “dvesha”, we find that they clearly indicate the ‘increasing relevance of transactional cognitive processes to life situations’ and also the increasing role of ‘energy-dynamics’.

The “klesa” as stress has been defined operates at four different modes. The first is “prosupta” or dormant. Given the right type of conditions, any mental process can become a stressor. The analogy is given of a seed which can flower into a tree provided the facilitating conditions are present. The second is “tonu” or tenuous denoting comparatively weak stressors which are held in check by more powerful stressors. They are present but without sufficient intensity and urgency. The third stressors are “vichchinna” or intercepted; these

lack continuities due to conflict with competing responses. Their demand character is high but they alternate between the levels of 'high operation' to 'dormant'.
Naturally, they surrender their stressor value when in dormant stage. The fourth mode is "udara" or operative stressors. These are potent stress responses which have found full expression in clearly observable behavioural modes. They have overcome the weaknesses of the first three modes
The above given model proposed in the "Yoga-Sutra" is a comprehensive one incorporating cognitive structuring affective or emotional stages and adaptive reactions. It also presents the concept of "kriya-yoga" which is aimed at reducing the 'number and intensity of the stressors' and to facilitate related conservation of mental energy devoid of tension which is designated as "Samadhibhavana".
It is evident from the above description that like western researchers, Indian scholars also differently approached the problem of stress. They viewed this phenomenon from various angles ranging from stimulus oriented to response and psychodynamic point of view. The system of yoga is analytical and not only helps the individuals in understanding their own stresses but also leads them to the roots of these stresses. There are subtle variations in the intensity with which different stressors operate and it is not unlikely that the powerful stressors become less or even dormant after a period of time whereas the less powerful ones may vanish or return with greater vigour.

Fight Or Flight Response

The fight or flight response is an automatic physiological reaction to an event that is perceived as stressful or frightening. The perception of threat activates the sympathetic nervous system and triggers an acute stress response that prepares the body to fight or flee. These responses are evolutionary adaptations to increase chances of survival in threatening situations. Overly frequent, intense, or inappropriate activation of the fight or flight response is implicated in a range of clinical conditions including most anxiety disorders. A helpful part of treatment for anxiety is an improved understanding of the purpose and function of the fight or flight response. This client information sheet describes the bodily consequences of the fight or flight response.

The three stages of fight-or-flight are:

- **The alarm stage**: During this stage, the central nervous system is ramped up, preparing your body to fight or flee.
- **The resistance stage**: This is the stage in which the body attempts to normalize and recover from the initial elevated fight-or-flight response.
- **The exhaustion stage**: If the first two stages occur repeatedly over time, such as when under chronic stress, this can cause the body to feel exhausted and begin to break down.

The fight or flight reaction is associated with activation of the sympathetic nervous system. The chain reaction brought about by the fight or flight response can result in the following physical effects:

- **Heart:** Increased heart rate; Dilation of coronary blood vessels; Increase in blood flow Increased availability of oxygen and energy to the heart.

- **Circulation:** Dilation of blood vessels serving muscles; Constriction of blood vessels serving digestion; Increased availability of oxygen to skeletal muscles; Blood shunted to skeletal muscles and brain.
- **Lungs:** Dilation of bronchi; Increased respiration rate; Increased availability of oxygen in blood.
- **Liver:** Increased conversion of glycogen to glucose; Increased availability of glucose in skeletal muscle and brain cells.
- **Skin:** Skin becomes pale or flushed as blood flow is reduced; Increased blood flow to muscles and away from non-essential parts of the body such as the periphery.
- **Eyes:** Dilation of the pupils; Allows in more light so that visual acuity is improved to scan nearby surroundings.

In addition to physiological reactions there is also a psychological component to fight or flight response. Automatic reactions include a quickening of thought and an attentional focus on salient targets such as the source of the threat and potential avenues for escape. Secondary psychological responses can include appraisals about the meaning of the body reactions. For example, patients with panic disorder often misinterpret fight or flight responses as signs of impending catastrophe (*"I'm having a heart attack"*, *"If this carries on I'll go mad"*).

History of the fight or flight response

The fight or flight response was originally described by American physiologist Walter Bradford Cannon in the book *Bodily changes in pain, hunger, fear and rage* (1915). He noted that when animals were threatened, by exposure to a predator for example, their bodies released the hormone adrenaline / epinephrine which would lead to a series of bodily changes including increased heart rate and respiration. The consequences of these changes are increases in the flow of oxygen and energy to the muscles. Canon's interpretation of this data was that there were *emergency functions* of these changes. He noted that they happened automatically and they served the function of helping the animal to survive threatening situations by readying the body for fighting or running.

A more modern understanding of the fight or flight response is reflected in the work of Schauer & Elbert (2010). Their more elaborated model of physiological / psychological / behavioral responses to threat is termed the 'defense cascade'. They describe a series of stages which individuals exposed to threat or trauma may go through, including: freeze, flight, fight, fright, flag, and faint.

IMPORTANCE

The physiological responses associated with fight or flight can play a critical role in surviving truly threatening situations. However, many patients suffering from anxiety disorders or other conditions may have threat systems which have become over-active, or which are insufficiently counterbalanced by activity in the parasympathetic nervous system.

Practically, many patients who suffer from anxiety will benefit from a deeper understanding of the fight or flight response. For example, patients with panic attacks or panic disorder often misinterpret the bodily signs associated with fight or flight as signs of impending catastrophe and understanding the fight or flight response is therefore a helpful 'decatastrophizing' technique. Similarly, patients with post-traumatic stress disorder (PTSD) may mistake the increased physiological arousal as an indicator that there is a genuine threat

present: understanding more about the fight or flight response can help them to feel safer, and to implement relaxation and grounding strategies.

General Adaptation Syndrome

What happens to the body when stress is prolonged? Selye studied this issue by subjecting animals to a variety of stressors such as high temperature, X-rays and insulin injections, in the laboratory over a long period of time. He also observed patients with various injuries and illnesses in hospitals. Selye noticed a similar pattern of bodily response in all of them. He called this pattern the General Adaptation Syndrome (GAS). According to him, GAS involves three stages: alarm reaction, resistance, and exhaustion

1. Alarm reaction stage : The presence of a noxious stimulus or stressor leads to activation of the adrenalpituitary-cortex system. This triggers the release of hormones producing the stress response. Now the individual is ready for fight or flight.
2. Resistance stage : If stress is prolonged, the resistance stage begins. The parasympathetic nervous system calls for more cautious use of the body's resources. The organism makes efforts to cope with the threat, as through confrontation.
3. Exhaustion stage : Continued exposure to the same stressor or additional stressors drains the body of its resources and leads to the third stage of exhaustion. The physiological systems involved in alarm reaction and resistance become ineffective and susceptibility to stress-related diseases such as high blood pressure becomes more likely.

Selye's model has been criticised for assigning a very limited role to psychological factors in stress. Researchers have reported that the psychological appraisal of events is important for the determination of stress. How people respond to stress is substantially influenced by their perceptions, personalities and biological constitutions.

Endocrinology

When someone perceives a situation to be challenging, threatening, or uncontrollable, the brain initiates a cascade of events involving the hypothalamic-pituitary-adrenal (HPA) axis, which is the primary driver of the endocrine stress response. This ultimately results in an increase in the production of steroid hormones called glucocorticoids, which include cortisol, often referred to as the "stress hormone".

The HPA axis

During times of stress, the hypothalamus, a collection of nuclei that connects the brain and the endocrine system, signals the pituitary gland to produce a hormone, which in turn signals the adrenal glands, located above the kidneys, to increase the production of cortisol.

Cortisol increases the level of energy fuel available by mobilizing glucose and fatty acids from the liver. Cortisol is normally produced in varying levels throughout the day, typically increasing in concentration upon awakening and slowly declining throughout the day, providing a daily cycle of energy.

During a stressful event, an increase in cortisol can provide the energy required to deal with prolonged or extreme challenge.

Stress and health

Glucocorticoids, including cortisol, are important for regulating the immune system and reducing inflammation. While this is valuable during stressful or threatening situations where injury might result in increased immune system activation, chronic stress can result in impaired communication between the immune system and the HPA axis.

This impaired communication has been linked to the future development of numerous physical and mental health conditions, including chronic fatigue, metabolic disorders (e.g., diabetes, obesity), depression, and immune disorders.

Role of nervous system

The nervous system has several divisions: the central division involving the brain and spinal cord and the peripheral division consisting of the autonomic and somatic nervous systems. The autonomic nervous system has a direct role in physical response to stress and is divided into the sympathetic nervous system (SNS), and the parasympathetic nervous system (PNS). When the body is stressed, the SNS contributes to what is known as the "fight or flight" response. The body shifts its energy resources toward fighting off a life threat, or fleeing from an enemy.

The SNS signals the adrenal glands to release hormones called adrenalin (epinephrine) and cortisol. These hormones, together with direct actions of autonomic nerves, cause the heart to beat faster, respiration rate to increase, blood vessels in the arms and legs to dilate, digestive process to change and glucose levels (sugar energy) in the bloodstream to increase to deal with the emergency.

The SNS response is fairly sudden in order to prepare the body to respond to an emergency situation or acute stress—short term stressors. Once the crisis is over, the body usually returns to the pre-emergency, unstressed state. This recovery is facilitated by the PNS, which generally has opposing effects to the SNS. But PNS over-activity can also contribute to stress reactions, for example, by promoting bronchoconstriction (e.g., in asthma) or exaggerated vasodilation and compromised blood circulation.

Both the SNS and the PNS have powerful interactions with the immune system, which can also modulate stress reactions. The central nervous system is particularly important in triggering stress responses, as it regulates the autonomic nervous system and plays a central role in interpreting contexts as potentially threatening.

Chronic stress, experiencing stressors over a prolonged period of time, can result in a long-term drain on the body. As the autonomic nervous system continues to trigger physical reactions, it causes a wear-and-tear on the body. It's not so much what chronic stress does to the nervous system, but what continuous activation of the nervous system does to other bodily systems that become problematic.

Somatic Disorders

When mental factors such as stress cause physical symptoms the condition is known as somatisation. **Somatoform disorders** are a severe form of **somatisation** where physical symptoms can cause great distress, often long-term. However, people with somatoform disorders are usually convinced that their symptoms have a physical cause.

When physical symptoms are caused by mental (psychological) or emotional factors it is called somatisation. For example, many people have occasional headaches caused by mental stress. But, stress and other mental health problems can cause many other physical and bodily symptoms such as:

- Chest pains.
- Tiredness.
- Dizziness.
- Back pain.
- Feeling sick (nauseated).

It is common. Sometimes we can relate the physical symptoms to a recent stress or mental health problem. For example, you may realise that a bout of neck pain or headache is due to stress. Anxiety and depression are also common reasons to develop physical symptoms such as a 'thumping heart' (palpitations), aches and pains, etc. Often the physical symptoms go when emotional and mental factors ease. However, often we do not realise the physical symptom is due to a mental factor. We may think we have a physical disease and see a doctor about it.

The somatoform disorders are the extreme end of the scale of somatisation. So, the physical symptoms persist long-term, or are severe but no physical disease can fully explain the symptoms. Somatoform disorders include:

- Somatisation disorder.
- Hypochondriasis.
- Conversion disorder.
- Body dysmorphic disorder.
- Pain disorder.

They are classed as mental health disorders, as the cause of the symptoms is thought to be mental factors. However, they cannot be fully explained by depression, substance abuse, or other recognised mental health disorders. There has recently been a renaming of these mental health disorders and they have all been put under the main heading of 'somatic symptom disorder'. Doctors used to make the diagnosis based on strict patterns of symptoms. However, they now rely much more on how much the symptoms affect the person's quality of life and well-being. It is now also recognised that somatic symptom disorder can occur in people who also have physical diseases such as arthritis or cancer.

Somatisation disorder

People with this disorder have many physical symptoms from different parts of the body - for example:

- Headaches.
- Feeling sick (nauseated).
- Tummy (abdominal) pain.
- Bowel problems.
- Period problems.
- Tiredness.
- Sexual problems.

The main symptoms may vary at different times. Affected people tend to be emotional about their symptoms. So they may describe their symptoms as 'terrible', 'unbearable', etc and

symptoms can greatly affect day-to-day life. The disorder persists long-term although the symptoms may wax and wane in severity.

The cause is not known. It may have something to do with an unconscious desire for help, attention or care. It runs in some families. The disorder usually first develops between the ages of 18 years and 30 years. More women than men are affected.

It is difficult for a doctor to diagnose somatisation disorder. This is because it is difficult to be sure that there is no physical cause for the symptoms. So, people with this disorder tend to be referred to various specialists, and have many tests and investigations. However, no physical disease is found to account for the symptoms.

Hypochondriasis

This is a disorder where people fear that minor symptoms may be due to a serious disease. For example, that a minor headache may be caused by a brain tumour, or a mild rash is the start of skin cancer. Even normal bodily sensations such as 'tummy rumbling' may be thought of as a symptom of serious illness. People with this disorder have many such fears and spend a lot of time thinking about their symptoms.

This disorder is similar to somatisation disorder. The difference is that people with hypochondriasis may accept the symptoms are minor but believe or fear they are caused by some serious disease. Reassurance by a doctor does not usually help, as people with hypochondriasis fear that the doctor has just not found the serious disease.

Conversion disorder

Conversion disorder is a condition where a person has symptoms which suggest a serious disease of the brain or nerves (a neurological disease) - for example:

- Total loss of vision (severe sight impairment).
- Deafness.
- Weakness, paralysis or numbness of the arms or legs.

The symptoms usually develop quickly in response to a stressful situation. You unconsciously convert your mental stress into a physical symptom.

Conversion disorder tends to occur between the ages of 18 years and 30 years. Symptoms often last no longer than a few weeks but persist long-term in some people. In many cases

there is only ever one episode and no treatment is needed once symptoms have gone. Some people have repeated episodes of conversion disorder from time to time.

Body dysmorphic disorder

Body dysmorphic disorder is a condition where a person spends a lot of time worried and concerned about their appearance. See the separate leaflet called Body Dysmorphic Disorder for more details.

Pain disorder

Pain disorder is a condition where a person has a persistent pain that cannot be attributed to a physical disorder.

Somatoform disorders can affect anyone of any age. The exact number of people affected is difficult to determine, as many cases are probably not diagnosed. Somatisation disorder is thought to be quite rare, perhaps affecting about 1 in 1,000 people. Hypochondriasis and body dysmorphic disorder are perhaps more common.

It is not clear why some people develop somatoform disorders. Genetic 'makeup' and environmental factors both probably play a part. Genetic makeup is the material inherited from your parents which controls various aspects of your body. This genetic makeup combined with factors such as how you were brought up, your parental and peer influences, etc, may all contribute.

Somatoform disorders are more common in people who misuse alcohol and drugs. However, drugs and alcohol may be factors in both cause and effect. For example, some people may turn to alcohol or other drugs to ease the distress of their somatic symptoms. On the other hand, excess alcohol or illicit drugs may make the symptoms worse.

Lesson 3: Stress and environment

Environmental stress can be defined as an undesirable transaction between the person and the environment. The demands made by environmental factors on us constitute the sources of environmental stress. To understand environmental stress better, let us learn about some issues linked to the human-environment relationship.
As indicated earlier considerable part of human behaviour is influenced by our environment. Environment has been used to include the physical world and sociocultural surrounding in which we live and interact with others. The field of environmental psychology deals with the diverse effects of environment on behaviour. We all wonder at times that in the relationship between human and environment who is superior? We all have read in the newspapers about the remarkable achievement of human beings over nature like going to the moon, cloning etc.

But does this mean that human beings are free to play with nature? What answer can one give to the natural calamities which befall human lives?
To answer these questions, let us try to understand the different ways in which the human-environment relationship can be understood. Let us understand these views in some detail.

a) **Humans Subjugated to Nature:** During the early part of history human beings looked at the environment from a different perspective. They believed that the nature rules human beings. The various components of nature were considered supernatural and objects of worship. However, this view is still held in some cultures. The sun, moon, stars, animals, birds, trees etc. all are worshipped and also considered to be objects to be afraid of. This view did not find place in the modern scientific view in which reality in the physical form became important and human capabilities were highly valued. This led to another model which favoured human beings i.e. power or control over nature.
b) **Humans Control over Nature:** With the advent of science and technology, nature no longer remained a thing to be afraid of. Experiments were conducted by scientists in the various facets of life. We all have heard about kidney transplant, open heart surgery etc. In the same way we all see that nowadays the vegetables grown at a specific time of the year are now available throughout the year. A field called genetic engineering is bringing changes in the pattern of crop production and medicines. Human life has been elevated scientifically but again it was felt that some questions remained unanswered. Devastating earthquakes and global warming (rise in the temperature) which have brought climatic changes and depletion of the ozone layerall are now matters of great concern. Nuclear power has become a constant threat to the world. These problems lead us to think about environment differently in which a balance between man and environment is warranted.
c) **Symbiotic Relationship between Human and Nature:** We all read slogans like 'conserve oil', 'plant trees' and 'save water' but have you ever thought why it is said so? Do you know that many of the changes in our environment are happening due to various acts of human beings? Human beings and nature have always shared an interdependent relationship which means that no one is sufficient in itself. Human beings derive a lot from nature in various forms and it also becomes a necessity that a sense of duty towards nature is generated. People blindly exploit various natural resources without thinking about their impact on others in present time and future. Many natural resources (e.g. oil, coal) are unreplenishable and therefore should not be wasted.
   The imbalance created due to blind exploitation of nature is also resulting in various forms of pollution and threats to life. You will read more about the various types of pollution and their impact in a later section of this lesson. The quality of air, water and eatables all are influenced by human intervention. The symbiotic relationship between human beings and nature is a balanced view which tells us that we must recognize our limitations and do not blindly harm environment by our activities. It recognizes the natural interdependence of the two.

Lesson 4: Models of Stress: Person-Environment fit theory, Cybernetic theory of stress, Coping and well-being, A multidimensional theory of Burnout.

Person-Environment fit theory

Person-environment (PE) fit refers to the degree of match between individuals and some aspect of their work environment. The concept of PE fit is firmly rooted in the tradition of Kurt Lewin's maxim that B = /(PE); behavior is a function of both person and environment.

The early interactional psychologists emphasized Lewin's perspective and developed a perspective that individuals' behaviors and attitudes are determined jointly by personal and environmental conditions. On the person side, characteristics may include interests; preferences; knowledge, skills, and abilities (KSAs); personality traits; values; or goals. On the environment side, characteristics may include vocational norms, job demands, job characteristics, organizational cultures and climates, and company or group goals. Various synonyms have been used to describe fit, including congruence, match, similarity, interaction, correspondence, and need fulfillment.

The basic premise of PE fit research is that for each individual there are particular environments that are most compatible with that person's personal characteristics. If a person works in those environments, positive consequences including improved work attitudes and performance, as well as reduced stress and withdrawal behaviors, will result. Although the premise is straightforward, research on PE fit is one of the most eclectic domains in organizational psychology. In part this is because of the wide variety of conceptualizations, content dimensions, and measurement strategies used to assess fit.
What do we mean by fit?

Although terms such as congruence or match seem to imply similarity, multiple conceptualizations of PE fit have been discussed in the literature. Supplementary fit exists when the individual and the environment are similar on a particular characteristic. The underlying mechanism is one of similarity-attraction, such that people tend to like interacting with other people and with environments that are similar to themselves in some way. Alternatively, complementary fit occurs when individuals' characteristics fill a gap in the current environment or the environment meets a need in the person. Complementary fit is based on the underlying process of need fulfillment, resulting in positive attitudinal and behavioral outcomes.

Research on stress and coping, which describes fit as adjustment, has elaborated on two distinct forms of complementary fit. The first is needs-supplies fit, which exists when a person's needs are met by the resources in the environment. The second is demands-abilities fit, which generally focuses on individuals' KSAs meeting environmental demands.

Fit on what?

PE fit research has generally concentrated on matching the individual to one of four levels of the environment:

1. Vocation
2. Job
3. Organization
4. Group

Each of these subtypes of PE fit emphasizes different person and environment characteristics as relevant to fit. Each type is briefly reviewed in the following text.

Person-Vocation Fit

The broadest form of PE fit is the fit between individuals and their vocations or occupations, generally labeled person-vocation (PV) fit. Vocational choice theories, such as those by John L. Holland, Rene Dawis, and L. H. Lofquist fall into this category. Holland proposed in 1985 that the RIASEC typology (people and vocations are characterized as realistic, investigative, artistic, social, enterprising, or conventional) suggests that people will be most satisfied if they pursue careers that are compatible with their interests. Fit is defined by the degree of match between an individual's interests and those of others who generally make up the person's chosen vocation. Dawis and Lofquist's theory of work adjustment posits that individuals and careers are compatible to the extent that personal traits (including skills, abilities, needs, and values) correspond with the requirements imposed by the environment, and personal needs are simultaneously met by the environment.

Reviews of the PV literature generally report moderate positive correlations between PV congruence and individual measures of well-being such as job and career satisfaction, stability, and personal achievement. Correlations are higher when focusing on the congruence with specialty areas within vocations. Consistent negative relationships have been found between fit and mental distress, somatic symptoms, changing vocations, and seeking satisfaction through leisure activities unrelated to work.

Person-Job Fit

A second type of fit concerns the relationship between an individual and a specific job. Labeled person-job (PJ) fit, this includes the match between a person's KSAs and the demands of a job (demands-abilities fit), or the person's needs and interests and the resources provided by the job (needs-supplies fit). Traditional notions of personnel selection, which began during World War II with the selection of soldiers into specific positions in the army, emphasized the importance of hiring people who possessed the requisite KSAs for particular jobs. Thus, PJ fit was defined from the organization's perspective, such that the most appropriately qualified people would be hired.

PJ fit from the needs-supplies perspective was the emphasis of work done by J. R. P. French, Jr., and colleagues on stress and adjustment. Their research presented a model that described psychosocial stress as the outcome of a discrepancy between the subjective environment and the subjective person (i.e., subjective PE fit), which in turn was the result of the fit between the objective environment and objective self. Thus, fit was equated to adjustment. PJ fit is also assessed in early definitions of job satisfaction, which emphasized satisfaction as the result of personal needs being met by a job. Over time scholars have separated job satisfaction (the affective outcome) from fit (the objective or perceived match that leads to the outcome). Yet because of the close relationship, PJ fit and job satisfaction are generally found to have a moderate to strong positive relationship. Other outcomes associated with PJ fit include organizational commitment, intent to quit, task performance, and strain.

Person-Organization Fit

Person-organization (PO) fit, defined broadly as the compatibility between people and organizational characteristics, is a third type of PE fit. Benjamin Schneider popularized this approach to fit with the attraction-selection-attrition (ASA) model, used to explain how

homogeneity naturally results from organizational recruitment and selection processes. Although the ASA model emphasizes the antecedents and consequences of homogeneity at the organizational level, Jennifer Chatman proposed a model that emphasized PO fit from the individual's perspective. This interactive model of PO fit emphasizes the objective fit between individuals' values and those that senior management believe best represent the organization. Many researchers have used the notion of value congruence to assess PO fit but have followed French and colleagues' approach of emphasizing subjective or perceived fit, rather than objective fit. PO fit has also been assessed using personality traits, goals, and needs. Meta-analytic estimates demonstrate that PO fit is most strongly associated with feelings of attachment to the organization, such as organizational commitment and intent to quit. Additional influences on contextual performance, or extrarole behavior, and turnover have been found.

Person-Group Fit

Finally, a fourth type of PE fit is the match between individuals and members of their immediate work groups. Most of the emphasis on person-group (PG) or person-team fit has been on demographic variables. The concept of relational demography suggests that individuals' attitudes and behaviors are influenced by the demographic similarity among teammates or coworkers. However, more recent studies have moved beyond demographic similarity to examine fit on deeper, less directly observable characteristics, including personality traits, goals, and KSAs. Outcomes most strongly associated with PG fit are group-level attitudes, including cohesion and satisfaction with coworkers, as well as contextual performance.

Cybernetic theory of stress

What Is Cybernetics?

Cybernetics is a complex science that deals with how self-regulating systems function. Systems include electronic, mechanical and biological systems, which makes the topic very complicated. When dealing with stress responses, the focus is solely on the the theory's application to biological systems, most notably, the endocrine and the nervous systems.

Norbert Wiener was a mathematician who coined the word cybernetics in 1948. The word stems from the Greek word, *kybernetes*, meaning *steersman*. Metaphorically, the term also relates to guiding or governing.

The Cybernetic Theory of Stress

The body has self-regulating systems that maintain an internal equilibrium, or state of constancy. The state of constancy helps maintain physiological health, but stress can disrupt the balance. This causes the body to produce neurotransmitters and hormones to restore balance.

Compare a person's self-regulating system to a reservoir that maintains a certain level of water (healthy stress). If too much water flows into the reservoir, the levels increase, possibly to an unmanageable level (anxiety). The system has to respond to the increase by finding a way to lessen it (coping).

Sometimes, the system has to constantly adapt to stress, which leads to allostasis. Over time, the endocrine and nervous systems may become less effective in maintaining balance, which leads to allostatic load. How does the body (machine) maintain allostasis? The answer is negative feedback, which is a fundamental element in organizational stress theories, and the cybernetic theory of stress.

Organizational Stress and Negative Feedback

Work-related stress appears to be a major concern for many people, including employers. The negative feedback loop is a basic principal that can help relate cybernetics to different theories of organizational stress (OS).

Normally, people associated the word negative with something that is detrimental on some level. However, when it comes to regulating stress, negative feedback yields positive results. A negative feedback loop negates the effects of stress-causing input.

J.R. Edwards' Theory

J.R. Edwards has developed a cybernetic theory that relates to stress, coping and wellbeing in organizations. We can look at objectives in the workplace, or demands, as stressors. The stress cycle in the workplace is complex, and it involves many feedback loops that are working at the same time.

Edwards has developed a theory that adopts perception as a key element in workplace stress. He postulates that stress is "a discrepancy between the employee's perceived state and desired state." The discrepancy should be important to the employee, as it begins a loop.

- Perceived state differs from desired state.

    - The discrepancy is important, and leads to stress.
- Stress can damage wellbeing, which leads to coping mechanisms.
- Coping may involve changing perceptions and desires.
- Coping may involve lessening the importance of the discrepancy.

Perceptions, Desires and Importance

In this theory, the individual's cognitive construction of reality (perception) and social information (behaviors, beliefs and opinions) are important factors. When the employee has little information, he or she constructs reality based on social information and personal perceptions.

The theory presents desires as conditions the employee consciously wants. Stress occurs when a discrepancy between the employee's desires and perception exists. The Importance of the discrepancy is a factor as well. If the discrepancy is detrimental to the employee's wellbeing, it is a greater cause of stress.

UNIT-II: SOURCES AND EFFECTS OF STRESS

Lesson 5: Sources of Stress: Physiological, Physiological and Sociological-

SOURCES OF STRESS

As we have developed a clear idea about the concept and nature of stress, we will now move on to the sources of stress. As such, the sources of stress can be categories in to three main sources, namely, Frustration, Conflict of motives and Pressure (Coleman, 1970).

Frustration: Frustration can be described as occurring when a goal oriented behaviour of an individual is thwarted. As stated by Mangal (1984, pg. 46) “a wide range of environmental obstacles, both physical and social and the internal factors in the form of personal limitations, biological conditions and psychological barriers may lead to frustration of our needs, motives

and efforts". For example, Ravi expected his promotion, however, when he did not receive the same, he experienced frustration.

Conflict of Motives: The next source is conflict of motives that can cause stress as an individual has to choose between alternatives and decision making in this regard can lead to stress. Conflict of motives can be of four different types, approach - approach conflict, avoidance - avoidance conflict, approach - avoidance conflict and double approach - avoidance conflict.

1) **Approach-approach conflict:** In this type of conflict the individual has to choose between two goals that are positive and are similar. For example, an individual may have to choose between two similar job opportunities.
2) **Avoidance- avoidance conflict:** The next type of conflict of motives is avoidance-avoidance conflict. Here again there are two goals that are similar but as opposed to approach- approach conflict, the goals here are negative. For example, an individual may have to take decision whether he/ she wants to remain unemployed or take up a mediocre job that he/ she is not interested in.
3) **Approach- avoidance conflict:** Here there is a single goal which is both positive and negative. For example, an individual taking up a job abroad may be excited about the opportunity but also has to face the fact that he/ she will have to stay away from the family. Thus, this goal has both negative and positive consequences and thus the individual may experience stress.
4) **Multiple approach-avoidance conflict:** It may so happen that the conflicts faced by us are quite complex and they are combinations of approach and avoidance conflicts. In this conflict the individual has to choose between the options that have both positive and negative consequences. For example, a student may have to choose between two educational opportunities, one of pursuing music that he/ she is interested in which will make his/her parents unhappy and the other of pursuing engineering, which will make his/ her parents happy but he/ she is really not interested in it.

Pressure: Pressure is yet another source of stress that can be external or internal. External pressures are result of demands from the environment, responsibilities and obligations that are mainly social in nature as well as demands and expectations of the significant persons in our lives. With regard to internal pressures, Mangal (1984, pg. 46) states that "Internal pressures are caused by our own self for maintaining the picture of ourselves - as we think we could and should be". An example of external pressure is when a child is pressurised by parents to do well in examination and an example of internal pressure is when a child himself/ herself feels that he/ she needs to study and do well in examination.

Thus, any stress can be categorised into the above three sources, though a person may experience stress due to more than one or all the three sources as well.

Besides the above sources, stress can also be as a result of personality factors that may make an individual prone to stress. Past experiences, basic temperamental factors, perception of control over the situation causing stress play an important role in determining the reaction of individual to stress (Parmeswaran and Beena, 2004). In the context of personality, the type A and type B personalities can be discussed. Individuals with Type A personality display hurriedness, restlessness and are often involved in carrying out various activities at the same time. They are also competitive, anxious and are often achievement oriented. The individuals with type B personality, on the other hand, display behaviour that is relaxed. It can be said that individuals with type A personality are comparatively more prone to stress when compared with individuals with type B personality (Parmeswaran and Beena, 2004).

Further, certain life events, like death of a near or dear one, divorce, pregnancy, child leaving home, change in conditions of living, retirement, marriage, losing one's job and so on can also lead to stress (Nolen- Hoeksema, 2009).
Frustration, conflict of motives and pressure can be termed as broad sources of stress. Though stress can also result due to ineffective interpersonal relationships or interpersonal conflicts, family related issues, lack of work life balance, work pressures and work overload and even environmental pressures that could be due to noise pollution, crowding and so on.
Let us now discuss some of the specific sources of stress.
Life events: Stress can be caused due to varied life events. These could be death of a loved one, divorce or separation, losing one's job, marriage and so on. These could be termed as significant life events in one's life that may put pressure on the adjustment and coping resources of the person as they expose the individual to certain new and unique challenges. Thus, life events can lead to stress which in turn can have detrimental effects on the day to day functioning as well as physical and psychological health of the individual.
Daily hassles: Similar to life events, daily hassles can also create stress. Daily hassles like lack of time, work overload, daily roles, responsibilities and duties and so on can lead to stress.
Interpersonal relationships: Stress can also be caused due to issues related to interpersonal relationships. Interpersonal relationships could be in the context of family, work or it could be with peer groups. Conflicts could occur between individuals that can have an impact on the interpersonal relationship, which in turn may lead to stress. Issues in interpersonal relationship could range from misunderstandings to violence and abuse and can have negative impact on the physical and psychological wellbeing of the person.
Stress as a result of social conditions: Social conditions in which the individual exists can also lead to development of stress. These social conditions can be related to crowding, discrimination, technological developments and changes, pollution and so on. The individual today exists in a world that is much more complex. There is globalisation, increased competition, and high amount of social pressure. These have a negative impact on the individual and makes him/ her prone to stress.

Lesson 6: Effects of Stress: Physiological effects, Physiological effects- Learned helplessness-

Physiological effects of stress
Though stress originates in the brain, it affects all body systems including nervous, cardiovascular, respiratory, musculoskeletal, immune, gastrointestinal, endocrine and reproductive systems
Effect of stress on nervous system The stress system not only triggers the fight or flight response through sympathetic nervous system, but also interacts with mesocortical and mesolimbic components of dopaminergic system, the amygdala/ hippocampus complex and the arcuate nucleus proopiomelanocortin neuronal system.9 Mesocortical and mesolimbic system forms the reward system and mediates euphoria and dysphoria.2 Stress activates amygdala/hippocampus system, which activates central stress system and mesocorticolimbic dopaminergic system. Corticotropin-releasing hormone neurons of amygdala are connected to paraventricular nuclei and parabrachial nucleus and elicit autonomic and behavioral effects. Activation of hippocampus causes shutting off of stress response by inhibiting PVN, CRH and LC/NE system. Stress induces release of opioid peptides especially α-melanocyte-

stimulating hormone and β-endorphin from hypothalamus. Both the opioid peptides inhibit components of central stress system

Effect of stress on cardiovascular system Stress induced cardiovascular diseases include hypertension, myocardial ischaemia, myocardial infarction, ventricular arrhythmia, and sudden cardiac death. Stress causes hazardous effects on cardiovascular system through activation of sympathetic nervous system and vagal withdrawal. Stress induces certain mechanisms like rapid increase in arterial pressure and heart rate, increased serum lipid levels, altered blood coagulation, atherosclerosis and vascular changes that may lead to cardiovascular diseases.

Effect of stress on respiratory system Stress also induces respiratory changes like increase in respiratory rate and decrease in end-tidal Carbon dioxide that is hyperventilation. Changes in respiratory quotient and energy expenditure were observed in stressful conditions. However, the change depends on type of the stress. Stress influences respiration through modulating the activity of respiratory neurons. Excessive stress alters the membrane properties, strength of neuro-modulation in respiratory neurons.

Effect of stress on musculoskeletal system Increase in muscle tension is a reflex phenomenon to protect an individual from injury and pain. However chronic exposure to stress causes prolonged increase in muscle tension and promotes musculoskeletal disorders. The muscles of shoulders, muscles of the neck and jaw, accessory muscles of respiration and diaphragm are commonly affected due to stress.

Effect of stress on blood Stress induced changes in blood cell parameters are well documented. It was studied that, stress induced release of cortisol decreases the eosinophils by increasing their sequestration in spleen and also decreases the number of basophils whereas stress increases number of neutrophils, platelets and red blood cells.

Effect of stress on immune system Activation of HPA axis during stress modulates both innate and adaptive immunity through increase in secretion of glucocorticoids. Cortisol modulates number and function of leucocytes, decreases production of cytokines, inhibits pro-inflammatory pathways in target tissues. Hence, chronic activation of HPA axis increases susceptibility to infections and tumors. Activation of autonomic system during stress has direct suppressive action on immune cells/organs. Glucocorticoids and catecholamines modulate balance between T helper-1 versus T helper-2 responses.

Effect of stress on gastro intestinal system Stress suppresses gastric emptying by inhibiting dorsal vagal complex and increases colonic motor activity through stimulation of sacral parasympathetic system. Further stress weakens gastro intestinal barrier and increases gastrointestinal permeability and contributes to inflammatory bowel disease. Chronic stress decreases pain threshold for visceral sensations in patients with gastro intestinal disorders. Stress has negative impact on oral health as there is a decrease in the salivary flow rates, total protein concentration was observed in stressful conditions.28 Further, stress contributes to increase in the salivary cortisol and PH.

Effect of stress on metabolism Stress increases catabolism of glucose, lipids and proteins and simultaneously inhibits anabolic actions of growth hormone, insulin and sex steroids. This action of stress may be beneficial for short term basis to cope up with stressor. However, chronic activation of stress axis leads to metabolic syndrome through activation of HPA, LC/NE sympathetic adrenal system and changes in health behavior. HPA axis activation increases gluconeogenesis in liver, increases plasma glucose levels, increases lipolysis, increases protein breakdown at skeletal muscles, bone and skin, which leads to visceral obesity, insulin resistance, atherosclerosis, decreases osteoblastic activity, muscle and bone mass. Further, chronic stress causes changes in physical activity like following sedentary life style, increased sleeping hours, changes in diet habits, which further worsen regulation of metabolic functions and increases the incidence of metabolic syndrome.

Effect of stress on endocrine system Stimulation of pituitary-adrenal axis in stress increases secretion of corticotrophin releasing hormone (CRH) which increases secretion of adrenocorticotropin (ACTH), 8-lipotropin and 3-endorphin. Increase in vasopressin secretion during stress potentiates the effect of CRH. Stress decreases levels of thyroid hormones through inhibition of thyroid-stimulating hormone secretion. Increase in the growth hormone secretion was reported followed by physical stress. GH responses for psychological stress was not well reported. Prolactin levels may increase or decrease during stress depending on local regulatory mechanisms. However, the role of altered prolactin levels is still unclear. Decrease in insulin secretion was observed in stress which may contribute for stress induced increase in blood glucose levels.
Effect of stress on reproductive system Though stress was reported to have negative impact on reproductive system, but the underlying mechanism was not clear. CRH and glucocorticoids exerts inhibitory effect on gonadotropin releasing hormone (GnRH) neurons, pituitary gonadotrophs and gonads directly and indirectly. As a result, the production of steroid hormones will be decreased at gonads that lead to amenorrhea of stress in females and decreased libido and hypo-fertility in males.

Physiological effects of stress stress have a psychological impact in every individual. Some of them are mentioned below:
Bored
Urge to cry
Urge to hide
Suspiciousness
Negative attitude/thoughts
Loneliness
Nightmares
Excessive daydreaming about "getting away from it all"
Use of sleeping pills
Impairment of intellectual functioning
Difficulty in making decisions
Poor concentration
Decrease in creativity
Formation of unhealthy defense patterns

Learned helplessness
Learned helplessness is a state that occurs after a person has experienced a stressful situation repeatedly. They come to believe that they are unable to control or change the situation, so they do not try — even when opportunities for change become available.
According to the American Psychological Association, learned helplessness occurs when someone repeatedly faces uncontrollable, stressful situations, then does not exercise control when it becomes available.

They have "learned" that they are helpless in that situation and no longer try to change it, even when change is possible.

Once a person having this experience discovers that they cannot control events around them, they lose motivation. Even if an opportunity arises that allows the person to alter their circumstances, they do not take action.

Individuals experiencing learned helplessness are often less able to make decisions.

Learned helplessness can increase a person's risk of depression.

Prof. Martin Seligman, one of the psychologists credited with defining learned helplessness, has detailed three key features:

1. becoming passive in the face of trauma
2. difficulty learning that responses can control trauma
3. an increase in stress levels

Origins Of The Theory

The concept of learned helplessness was discovered accidentally by psychologists Martin Seligman and Steven F. Maier. They had initially observed helpless behavior in dogs that were classically conditioned to expect an electrical shock after hearing a tone.
Later, the dogs were placed in a shuttlebox that contained two chambers separated by a low barrier. The floor was electrified on one side, and not on the other. The dogs previously subjected to the classical conditioning made no attempts to escape, even though avoiding the shock simply involved jumping over a small barrier.
To investigate this phenomenon, the researchers then devised another experiment.

- **In group one**, the dogs were strapped into harnesses for a period of time and then released.
- **In group two**, the dogs were placed in the same harnesses but were subjected to electrical shocks that could be avoided by pressing a panel with their noses.
- **In group three**, the dogs received the same shocks as those in group two, except that those in this group were not able to control the shock. For those dogs in the third group, the shocks seemed to be completely random and outside of their control.

The dogs were then placed in a shuttlebox. Dogs from the first and second group quickly learned that jumping the barrier eliminated the shock. However, those from the third group made no attempts to get away from the shocks.

Causes of Learned Helplessness

Learned helplessness is frequently the result of experiencing stress or trauma. People may feel that they have little to no control over the situation. Because of the lack of control, people may feel helpless and unmotivated to take action.

Common causes that can lead to learned helplessness include:

- Abuse
- Childhood neglect
- Difficult
- Domestic violence
- Natural disasters
- Trauma

Overparenting can also contribute to the development of learned helplessness in children. When children are not allowed to try things independently, they may develop a poor sense of personal agency. Instead of trying, they believe that they are unable to do things and do not put forth any effort.

Symptoms of Learned Helplessness

Everyone can struggle at times, especially when coping with adversity and setbacks. Learned helplessness is characterized by more lasting symptoms such as:

- Avoiding decisions
- Bad attitude
- Giving up quickly
- Inability to tolerate frustration
- Lack of effort
- Low motivation
- Passive behavior
- Poor self-esteem
- Procrastination
- Refusing to try

Learned helplessness is not a mental health condition, but it can sometimes be a sign of a mental disorder such as depression or anxiety.

Impact of Learned Helplessness

The impact of learned helplessness has been demonstrated in different animal species, but its effects can also be seen in people.

Consider one often-used example: A child who performs poorly on math tests and assignments will quickly begin to feel that nothing they do will have any effect on their math performance. When later faced with any type of math-related task, they might feel hopeless and unable to do the work.

For example, feeling shy in social situations can cause people to feel that there is nothing they can do to overcome their symptoms. Because symptoms feel out of control, people may stop trying to engage themselves in social situations, thus making their shyness even more pronounced
Researchers have found, however, that learned helplessness does not always generalize across all settings and situations.

A student who experiences learned helplessness with regards to math class will not necessarily experience that same helplessness when faced with performing calculations in the real world. In other cases, people may experience learned helplessness that generalizes across a wide variety of situations.

Learned Helplessness in Children

Learned helplessness often originates in childhood, and unreliable or unresponsive caregivers can contribute to these feelings. This learned helplessness can begin very early in life. Children raised in institutionalized settings, for example, often exhibit symptoms of helplessness even during infancy.

When children need help but no one comes to their aid, they may be left feeling that nothing they do will change their situation. Repeated experiences that bolster these feelings of helplessness and hopelessness can result in growing into adulthood ultimately feeling that there is nothing one can do to change his or her problems.

Some common symptoms of learned helplessness in children include:

- Failure to ask for help
- Frustration
- Giving up
- Lack of effort
- Low self-esteem
- Passivity
- Poor motivation
- Procrastination

Learned helplessness can also result in anxiety, depression, or both.[11] When kids feel that they've had no control over the past events of their lives, they gain the expectation that future events will be just as uncontrollable. Because they believe that nothing they do will ever change the outcome of an event, kids are often left thinking that they should not even bother trying.

Since nothing they do seems to make any difference, they will stop trying and their grades may suffer even more. Such problems can also affect other areas of the child's life. Their poor performance in school can make them feel that nothing they do is right or useful, so they may lose the motivation to try in other areas of their life as well.

Learned Helplessness and Mental Health

Learned helplessness may also contribute to feelings of anxiety and may influence the onset, severity, and persistence of conditions such as generalized anxiety disorder (GAD).

When you experience chronic anxiety, you may eventually give up on finding relief because your anxious feelings seem unavoidable and untreatable. Because of this, people who are experiencing mental health issues such as anxiety or depression may refuse medications or therapy that may help relieve their symptoms.

As people age learned helplessness can become something of a vicious cycle. When encountering problems such as anxiety or depression, people may feel that nothing can be done to ease these feelings.

Lesson 7: Stress- Coping and Burnout-

DEFINITION AND NATURE OF COPING

Coping in simple terms can be described as ways in which an individual tries to deal with stress experienced by him/ her. And in this context, the individual may be able to deal effectively with stress if he/ she adopts effective coping skills and he/ she may not be able to deal effectively with stress if the coping skills adopted by him/ her are ineffective.

Coping refers to "the individual's response to a psychological stressor which is often related to a negative event" (Roncaglia, 2014, pg. 137). Coping can also be described as deliberate efforts that are directed towards decreasing the negative effects, that could be psychological, physical or even social, of the stressful situation.

Coping has been defined by Lazarus and Folkman in 1980 as efforts, both cognitive and behavioural, that are directed towards overcoming, decreasing or enduring the internal and external demands. Thus, there is an effort to deal with the demands that are created on the resources of an individual as a result of stress. Coping in this context could be instrumental or palliative. Instrumental coping is denoted by results of cognitive appraisal or conflicts related to emotions. Palliative coping, on the other hand, is denoted by regulation of emotions as a result of cognitive reappraisal of the stressful event or situation. Thus, either the problem is altered or the emotional regulation is carried out during the coping process (Ghosh, 2015).

Matheny et al. (1986) defined coping as conscious or unconscious, healthy or unhealthy effort that is directed towards either prevention or elimination of stressor or to weaken it or be able to endure its effects in a manner that is least harmful.

Haan (1993) explained coping as an effort directed towards using the resources to overcome the difficulties as the obstacles are encountered.

As described by Bartram and Gardner (2008, pg. 228) "Coping is the process of thoughts and behaviours that people use to manage the internal and external demands of situations they appraise as being stressful or exceeding their own resources. Coping efforts seek to manage, master, tolerate, reduce or minimise the demands of a stressful environment".

From the above definitions it is clear that coping is an effort to deal with a stressor. The coping may aim at dealing with the problem that is the cause of any distress in the individual or to deal with the negative emotions experienced by him/ her. Though coping could either be adaptive or maladaptive.

An adaptive coping not only reduces stress but has long term positive effects. Whereas, maladaptive coping may provide respite from stress for a short period of time but it could have negative impact on physical and psychological health of the individual. Some of the maladaptive coping strategies are consumption of alcohol or drugs, smoking, risky behaviours, isolating oneself, being overcritical of oneself and so on.

Individual differences exist in the way individuals cope with stressful situations. Variations could exist based on personality of an individual, his/ her tolerance level for stimulation, psychological hardiness, the style of attribution, learned helplessness and his/ her sense of coherence. Difference may also exist due to gender (Ghosh, 2015).

Personality not only determines how stress is perceived by an individual but also how he/ she reacts and copes with stress. Individuals having varied personalities will display different coping styles and strategies. Similarly people with high or low need for stimulation will also

vary in their coping strategies. Persons with low need for stimulation are not able to cope with stress as do people with high need for stimulation (Ghosh, 2015).
Hardiness and resilience also play a role in determining the coping ability of the individuals. Psychological hardiness can be described as an individual's ability to deal with stressful situations with resilience (Bartone, 1999; Kobasa, 1979). Hardiness can be related with three main aspects, namely, commitment, control and challenge and these can determine the response to stress or coping strategy employed by the individual (Ghosh, 2015).
With reference to the style of attribution, individuals could be optimistic or pessimistic and this can also determine the coping strategy employed by the individuals. Optimistic individuals are likely to cope better with stressful situations when compared with individuals who are pessimistic.
The term learned helplessness was introduced by Seligman. The term is used to "describe the interference with adaptive responding produced by inescapable shock and also as a shorthand to describe the process which we believe underlies the behavior" (Seligman, 1972, pg. 408). The coping process that an individual goes through may differ based on learned helplessness. Anttonovosky, 1987 (as cited in Ghosh, 2015, pg. 169) described sense of coherence as "enduring through dynamic, feeling of confidence that:
(1) The stimuli deriving from ones internal and external environments in the course of living are structured, predictable and explicable,
(2) The resources are available to one to meet the demands posed by these stimuli, and
3. These demands are challenges worthy of investment and engagement". Men and women also differ in the way they cope with stress. Women are more likely to employ coping strategies like positive self-talk, seeking support from others, continuously worrying about the stressful event and so on, though they are also more prone to develop learned helplessness. Men on the other hand could engage in maladaptive coping strategies like consumption of alcohol and drugs (Ghosh, 2015). Further, women are more prone to using the coping strategies that target at bringing about change in their emotional reactions to a situation that is stressful and men are more likely to focus on the problem (Endler and Parker, 1990; Matud, 2004; Ptacek et al., 1994). Further, women also experience more psychological distress as well as display symptoms of depression and anxiety when compared with men and this can also be attributed to the coping strategies that are emotion focused that are more frequently used by women when compared with men (Kelly et al, 2008). Coping can also differ or may depend on various external aspects including social support received by the individual.

Goals of Coping

As we now have a clear idea about the meaning and definition of coping, let us now discuss the goals of coping. As was explained earlier, the main goal of coping is dealing with the stressor so that its impact on the individual is minimal. Further, individual differences exist in the coping styles adopted by the persons. Also, different coping strategies may be effective in different situations. Thus, during the coping process either internal resources or external resources are utilized by the individuals (Ghosh, 2015). Some of the major goals of coping are as follows (Ghosh, 2015):

- To enhance the possibility of recovery by decreasing the negative environmental conditions.
- To be able to adjust to the negative situation.
- To maintain a self-image that is positive and maintain emotional balance.
- To ensure positive interpersonal relationship.

Coping is termed as effective when it leads to recovery and when the individual is able to adapt to the stressful situation by means of maintaining a positive image about one's self, emotional balance and has effective interpersonal relationships. Whether a coping has been effective or not can be assessed based on the results of the coping. For instance, physiological

and biochemical changes and functioning can be assessed in order ascertain whether the coping has been adaptive or not. An adaptive coping will lead to regulation of various physiological indicators. Besides the physiological measures, if there is decrease in psychological distress displayed by the individual, then the coping strategy can be termed as adaptive. The duration of time taken by individuals to achieve the same state of functioning before they experienced a negative event (for instance, divorce, job loss and so on) or display of an improved state of functioning than before the negative event was experienced can also be an indicator of successful coping.

COPING STYLES

As the nature of coping is now clear let us focus on the coping styles and strategies. Coping styles can be categoried as proactive coping, avoidant coping, emotion focused coping and problem focused coping. Proactive coping and avoidant coping are based on the method of coping and the emotion focused coping and problem focused coping are based on the focus of the coping. Let us discuss each of these styles in detail.

Proactive coping and avoidant coping In proactive coping the individual will directly confront the stressful situation or event. An individual adopting this type of coping will take direct action by developing a better idea about the stress creating situation.

There are various stages of proactive coping (Ghosh, 2015):

Stage 1 Accumulation of resources: In order to be able to deal with the stressful situation, the individual will make attempts to accumulate resources, this could also be in terms of gathering information so as to understand the stressful situation in better manner.

Stage 2 Identifying or anticipating the potential stressor: A potential stressor is anticipated or identified by the individual. For instance, if it is expected that one's superior may ask for a certain report, the employee will anticipate the stressor and start working on the report before hand.

Stage 3 Initial appraisal: Initial appraisal of the stress creating situation is carried out.

Stage 4 Preliminary efforts to cope with the stressful situation: Based on the initial appraisal, preliminary efforts to cope with the stressful situation are carried out. In case these efforts do not show any positive results then, some other action is taken in order to deal with the situation.

Stage 5 Seeking feedback and using the same: Based on the stages three and four, feedback is sought and used so that modifications and changes can be carried out.

In avoidant coping, as the name suggests, the individual will try to avoid the stress creating situation or may give less importance to the stressful event. Thus, the behaviour that is displayed by the individual is directed towards avoiding certain thoughts or feelings that could arise due to the stressful situation. Avoidance coping can result in individuals experiencing not only stress and anxiety but it can have an impact on self-confidence of the individual as well (Boyes, 2013). As stated by Holahan et al. (2005) "Avoidance coping involves cognitive and behavioral efforts oriented toward denying, minimising, or otherwise avoiding, dealing directly with stressful demands and is closely linked to distress and depression. In avoidant coping certain behaviours that may stir up negative memories are avoided". Situations that may stir up negative feelings are also avoided. Individuals who use avoidant coping may not continue pursuing their goal if they experience thoughts that create anxiety (Boyes, 2013).

WELL-BEING

People have always aspired for a long, dignified and meaningful life. However, the stresses in contemporary life are increasing in such a way that it is becoming difficult to cope with them. The hectic pace of life, environmental factors, and increasing greed are making people's lives

more and more stressful. These factors influence our life style and combine to create serious health hazards. Keeping this in view efforts are being made to achieve health and well-being for all. The study of this area involves many subjects such as psychology, medicine, sociology, anthropology etc. Much concern is shown nowadays for the study of the factors influencing health, well-being, quality of life, healthy life style, and happiness. In this lesson you will learn about the concepts of health and well-being and how important they are for good life and for happiness. Awareness, adoption and practice of healthy lifestyle are the keys to our well being.

The term 'well-being' includes many aspects such as, happiness, energy, leisure, peace with oneself, contentment and sensitivity to environment. Well-being involves not only happiness and pleasure but also experiencing/feeling satisfaction in life, presence of positive feeling or affect (e g. interest, love, surprise, pleasure) and the absence of negative feelings (e.g. anxiety, depression, stress). A very rich person may be unhappy and a poor person may be happy. Well-being is a matter of how one experiences and interprets one's life conditions. Money, property, power and prestige provide us with limited sense of well-being. It has been found that material prosperity is positively related to life satisfaction to some extent. But indefinite increase in material prosperity does not lead to significant life satisfaction.

Factors Influencing Well-Being We can achieve a sense of well-being through a healthy lifestyle and by adopting good habits. The basic principles of healthy life style include good diet, adequate exercise, sleep, proper recreation and a set routine. Balance and avoidance of extremes is very important for leading a good life.

Nutrition: Do you remember when you were unable to eat for a long period of time? You must have felt weak, irritable and hungry. Sufficient calorie intake is necessary for good health. Since we need energy through out the day we should not skip meals or go on diets or under eat. We should eat a balanced diet of carbohydrates, proteins, fats and vitamins. The reverse is also true as some people consume too many calories and became overweight. Such people are also likely to have a lower self-esteem particularly if the culture approves a slim body.

Exercise: Talk to your grandparents, they will tell you that they have led a physically active life. They used their body to work and to go from place to place, whereas now we prefer to use cars, buses and other vehicles. Decreased physical activity affects our health and well-being. Lack of exercise can lead to headache, high blood pressure, diabetes, backache, irritability, anxiety and depression. Today obesity is another serious problem faced by the youth and adults alike. Exercise has a positive effect on mood. People who exercise regularly are able to control stress and enjoy high quality of life.

Sleep: Adequate sleep is required to rejuvenate and refresh both the mind and body. Sleep deficiency leads to irritability, anxiety, depression and disturbed thinking. How many hours of sleep do you get? We need about 8-9 hours of sleep to function effectively. We should try to establish a regular sleep routine and not sleep at irregular hours. Remember that people who exercise regularly also sleep well.

Recreation: Just as we need proper nutrition, exercise and sleep, we also need to relax and enjoy life. Indulging in certain activities, personally important to us, give us joy, fulfilment, satisfaction and act as a diversion from the hectic life we lead in current times.

Yoga and meditation: These activities bring discipline and peace in life. In addition there are many health related benefits which make a person fit and active.

Having friends and maintaining relationships: Friends and other people provide support and give opportunities to develop network, cooperation and enriching social life.

Optimism, resilience and hope: People with optimism, resilience and hope are found to be healthier and happier than others who lack these characteristics. Having these positive

attitudes makes a person effective and productive. Optimism refers to being positive while resilience is the ability to be strong in difficult times.
Routine: Routines help establish healthy habits that bring structure to our lives. A daily routine provides us with a framework and orders our life world. We know what is expected from us; otherwise we get alarmed when we are unclear about the expectations from others. The positive effect of routine is that it provides a sense of organization, control and responsibility to the person. Bad habits can be turned into good habits. You may love to sleep late but you know that you need to keep your body in good shape. Do you have a routine? Pick a routine that will keep you happy

BURNOUT is proposed as a concept by Freudenberger for the first time (1974). The concept of burnout is conceptualized as a composition of long-term emotional burnout, physical fatigue, not participating in business, removing those receiving service from humanity and low business success. Freudenberger created burnout syndrome as a result of the studies he gained from his clinical experience. In this context, it has been concluded that burnout is widely seen in the service sector and in occupational groups where face-to-face service is provided. However, it was later observed that the burnout was also observed in the blue-collar workers, called the working class (Pines and Aronson 1988). Later on, the concept of burnout was defined as a three-dimensional structure in the form of emotional burnout, desensitization and personal failure and a measurement tool (Maslach Burnout Inventory) was developed in this direction (Maslach and Jackson 1986, Hobfoll and Shirom 2001).

The multidimensional theory of burnout proposes that the greater the mismatch between the person and the job, the greater the chances are that burnout will occur.

Six mismatches which have a distinct relationship with burnout

1. Work overload
2. Lack of control
3. Insufficient reward
4. Breakdown of community
5. Absence of fairness
6. Value conflict

1. Work overload
   A mismatch in workload is generally found as excessive overload, through the simple formula that too many demands exhaust an individual's energy to the extent that recovery becomes impossible. A workload mismatch may also result from the wrong kind of work, as when people lack the skills or inclination for a certain type of work, even when it is required in reasonable quantities. Emotional work is especially draining when the job requires people to display emotions inconsistent with their feelings. Generally, workload is most directly related to the exhaustion aspect of burnout.
2. Lack of control
   A mismatch in control is generally related to the inefficacy or reduced personal accomplishment aspect of burnout. Mismatches in control most often indicate that individuals have insufficient control over the resources needed to do their work or have insufficient authority to pursue the work in what they believe is the most effective manner. Individuals who are overwhelmed by their level of responsibility may experience a crisis in control as well as in workload. This mismatch is reflected as one of responsibility exceeding one's authority. It is distressing for people to feel responsible for

producing results to which they are deeply committed while lacking the capacity to deliver on that mandate.

3. Insufficient reward
   A third type of mismatch involves a lack of appropriate rewards for the work people do. Sometimes these may be insufficient financial rewards, as when people are not receiving the salary or benefits commensurate with their achievements. Even more important at times is the lack of social rewards, as when one's hard work is ignored and not appreciated by others. This lack of recognition devalues both the work and the workers. In addition, the lack of intrinsic rewards (such as pride in doing something of importance and doing it well) can also be a critical part of this mismatch. Lack of reward is closely associated with feelings of inefficacy.
4. Breakdown of community
   The fourth mismatch occurs when people lose a sense of positive connection with others in the workplace. People thrive in community and function best when they share praise, comfort, happiness, and humor with people they like and respect. In addition to emotional exchange and instrumental assistance, this kind of social support reaffirms a person's membership in a group with a shared sense of values. Unfortunately, some jobs isolate people from each other, or make social contact impersonal. However, what is most destructive of community is chronic and unresolved conflict with others on the job. Such conflict produces constant negative feelings of frustration and hostility, and reduces the likelihood of social support.
5. Absence of fairness
   A serious mismatch between the person and the job occurs when there is not perceived fairness in the workplace. Fairness communicates respect and confirms people's self-worth. Mutual respect between people is central to a shared sense of community. Unfairness can occur when there is inequity of workload or pay, when there is cheating, or when evaluations and promotions are handled inappropriately. If procedures for grievance or dispute resolution do not allow for both parties to have a voice, then those will be judged as unfair. A lack of fairness exacerbates burnout in at least two ways. First, the experience of unfair treatment is emotionally upsetting and exhausting. Second, unfairness fuels a deep sense of cynicism about the workplace.
6. Value conflict
   The sixth area of mismatch occurs when there is a conflict between values. In some cases, people might feel constrained by the job to do things that are unethical and not in accord with their own values. For example, in order to make a sale or to obtain a necessary authorization, they might have to tell a lie or be otherwise deceptive or not forthcoming with the truth. In other instances, there may be a mismatch between their personal aspirations for their career and the values of the organization. People can also be caught between conflicting values of the organization, as when there is a discrepancy between the lofty mission statement and actual practice, or when the values are in conflict (e.g. high quality service and cost containment do not always co-exist).

   Interaction of the Six Areas Research on this model is beginning to elucidate the relationship between these six areas, as well as their relation to the three dimensions of burnout. Preliminary evidence suggests that the area of values may play a central mediating role for the other areas. Another possibility is that the weighting of the importance of the six areas may reflect an important individual difference. For example, some people might place a higher weight on rewards than on values, and thus might be more distressed by insufficient rewards than by value conflicts. It is not clear how much of a mismatch people are willing to tolerate, and this may depend on both the particular

area of mismatch and the pattern of the other five areas. For example, people may be willing to tolerate a mismatch in workload if they receive praise and good pay, work well with their colleagues, feel their work is valuable and important, and so on. Thus, the mismatches in these six critical areas of organizational life are not simply a list summarizing research findings from burnout studies. Rather, they provide a conceptual framework for the crises that disrupt the relationships people develop with their work. This approach emphasizes the importance of looking at the person in context, in terms of his or her fit with the key domains of work life.

Lesson 8: Moderators of stress: Age, Personality, Intelligence, and Social Support.

Stress moderators are the internal elements skills, behavioral patterns, qualities, and traits that reduce the stress level. Stress moderators also include such external elements like, yoga, exercises, social support, and more that perform the same function.

Age: Stress and aging are two things in life that you're going to have to deal with. While we are young, handling stress can be a little easier though. The effects of stress on our bodies actually change as we add on the years. The good news is there are some easy ways to identify and relieve stress so you can improve your overall health. Before learning how stress affects the older population differently, you first have to ask how does stress affect the body? For decades it was a truism that stress increased in later life. After all, older people are more likely to develop chronic illnesses, become widowed, lose friends and loved ones, experience middle-aged adults. Middle-aged adults were more likely to list family problems, parents' health and other health problems, reflecting their position as the "sandwich" generation, while young adults were coping with work problems and "angst," a general category that included problems with one's identity or mental health. The number of major problems does not seem to decrease much, if at all, with age, but the types of problems may change, as stressors are often linked to our social roles. Young adults struggle to establish their own careers, while in middle adulthood and late life, individuals often have more generative concerns about taking care of their families and older relatives.

Self-efficacy expectations are beliefs to the effect that one can handle a task or manage stress • Research show that high self efficacy expectations contribute to lower adrenaline levels in the blood stream • How does this contribute to fight or flight response? • Psychological Hardiness is a set of traits that buffer stress and are characterized by commitment, challenge, and control • Researcher Kobasa (1990) fond that those who are psychologically hardy have these characteristics • Commitment to their goals and feel involved in their lives • Challenge is viewed as normal and see it as a way to grow • Control over their lives where they feel that they can prepare for the challenges

Personality: Those who are psychologically hardy also see themselves as having an internal locus of control • Control allows us to feel that we are not at the mercy of randomness and that we can influence our lives • Internals are those who believe that they can exercise the control necessary to obtain reinforcement • Externals are those who believe that other people or the situation exercises control • Social support is also vital

Social support: Social support is also vital in moderating stress and acting as a buffer against the effects of stress • Sources of social support include the following: • Emotional Concern—listening to the problems of others and expressing feelings of empathy • Instrumental Aid—having material support and services available to help (after hurricane, the government provides resources) • Information—Guidance and advice that enhances people's ability to cope • Appraisal—feedback from others about how one is doing to make sense of what has happened • Socializing—simple conversation or recreation with another

Cognitive Reframing • "It's not the problem that's the problem, but how we see the problem that is the problem" • Aaron Beck—Cognitive Restructuring in response to Cognitive distortions such as catastrophizing.

UNIT – III: STRESS MANAGEMENT INTERVENTION

Lesson – 9: Effects of Occupational Stress Effect on Job Behavior, Job Satisfaction, Performance, Absenteeism. Effects on Employee's Physical and Psychological Health.

Stressors in workplaces are categorised as physical and psychosocial. Noise, inadequate lighting, poor office or job layout, and ergonomic factors, such as bad working postures, represent physical stressors. The most predominant stress factors are arguably psychosocial stressors. These include high demands for jobs, inflexible working hours, poor management of jobs, poor design and layout of work, bullying, abuse, and insecurity of employment. Stress in the workplace not only affects the worker, it also has detrimental effects on the output of the business. The consequences of work-related stress are visible in the physical health, mental health and behaviour of workers. We have already touched upon the effects of job stress in the Unit 4. In this unit, we will further our understanding regarding the effects of occupational stress. This includes effect on job behaviour, job satisfaction, performance, absenteeism, as well as effects on employee's physical and psychological health.

OVERVIEW OF EFFECTS OF OCCUPATIONAL STRESS

Unhealthy chronic stress can result in physiological, psychological, and behavioural changes in individuals. Some of these symptoms remain unnoticed for a very long period of time. Individuals who rely on fast-track living tend to assume these symptoms as a part of their life style and, therefore, keep avoiding it for a very long period. Since the symptoms of stress are so varied, they may be overlooked or mistaken. Special attention should be paid to prolonged headaches, elevated blood pressure, fatigue, and depression (Schwimmer, 1991). The physiological, psychological, and behavioural consequences of stress are depicted.

Effects on Employee's Physical Health

The physiological consequences of stress include heightened blood pressure, increased pulse rate, sweating, hot and cold flashes, headaches, and gastrointestinal disorders. Medical experts have attributed more that 50% of illnesses such as ulcers, arthritis, and allergies to stress. A very high correlation has been observed between job dissatisfaction and heart diseases among workers in more than 40 different occupations (Fletcher 1988). Besides this, stress-related illnesses are a heavy load on individuals and organizations. The cost is borne not only by individuals but by organizations as well. The cost to the employer can be visualized from three

different perspectives. The first cost could be in terms of health insurance that the employer has to pay for very serious illnesses such as heart disease. Secondly, the majority of the accidents caused by employees in the workplace setting could be because of the emotional turmoil the employee is going through. Thirdly, there could be legal hassles between the employer and the employee because of stressrelated worker compensation claims.

Effect on Employee's Psychological Health

Excessive amount of stress in an organization can cause boredom, dissatisfaction, anxiety, tension, irritability, and depression. All of these take the feeling of wellbeing away from the individual and contribute to poor concentration, indecision, and decreased attention span at work (Watson and Tellegen, 1985). If individuals are not able to change or move away from the stressors, they may resort to psychological substitutes such as anger, feeling of persecution, displacement, criticism, negativism, denial, apathy, fantasy, hopelessness, withdrawal, forgetfulness, and procrastination (Fletcher, 1988).

Effect on Job Satisfaction

Excessive and unmanageable amount of stress depletes the individual's effort to combat stress, resulting in job burnout. Job burnout refers to the adverse effects of working conditions in which the ability to confront and manage stress reduces significantly and job satisfaction and relief from stress seem impossible. According to Lee and Ashforth (1996), the burnout phenomenon typically contains three components:

I A state of emotional exhaustion
II Depersonalization of individuals
III Feeling of low personal accomplishment

A state of emotional exhaustion occurs most commonly among individuals who have both a high frequency and a high intensity of interpersonal contact. This kind of interpersonal contact may lead to emotional exhaustion, a key component of job burnout (Moore 2005). Most vulnerable to this problem are teachers, police officers, and social workers. Depersonalization refers to the treatment of people as objects. When a teacher refers to her student by roll number and not by name, she has actually dissociated herself with the student as a person. The student is, then, treated not as a human being but as a numbered object. These factors combined together lead to a feeling of low personal accomplishment. According to a study conducted by Advani et al. (2005), it was found that organizational politics and coordination expertise was significant in influencing burnout level among software professionals. While organizational politics had a greater influence on emotional exhaustion and depersonalization aspects, coordination expertise had a greater influence on the personal accomplishment aspect of burnout. According to Etzion et al. (1998), the characteristics which are found to be associated with high degree of burnout are as follows:

- Burnout candidates experience a great deal of stress as a result of jobrelated stressors.
- They tend to be idealistic and self-motivated achievers.
- They often seek unattainable goals.

Absenteeism

Absenteeism refers to a habitual pattern of absence from a duty or obligation without good reason. Generally, absenteeism is unplanned absences. Traditionally, workplaces viewed absenteeism as an indicator of poor individual performance, as well as a breach of an implicit contract between employee and employer. It was seen as a management problem, and framed in economic or quasi-economic terms. However, more recent analysis seeks to examine absenteeism as an indicator of psychological, medical, or social adjustment to work. Heavy workloads, stressful meetings/presentations and feelings of being unappreciated can cause employees to avoid going into work. Personal stress outside of work can lead to absenteeism.

Effect on Job Behaviour

When stress becomes distress, job performance gets affected and workplace accidents become very common. High stress levels impair our ability to remember information, make effective decisions, and take appropriate actions (Jamal 1984). Some of the initial behavioural consequences of distress are sleep disorders, change in eating habits, increase in smoking and alcohol consumption, and nervous behaviour such as rapid speech, stuttering and rude behaviour towards others. Negative stress causes even very polite and sober people to shout at their colleagues or subordinates. The most extreme form of distress results in workplace violence directed towards self or others resulting in suicide attempts or physical attacks on others.

Effect on Job Performance

Performance decline is another major outcome of negative stress. It is estimated that nearly 75% of all work loss is due to stress (Bell 1990). The relationship between stress and performance can be explained with the help of a violin string. For a violin, an optimum degree of tension is essential to play on it; similarly for an effective performance, optimum level of stress is mandatory. In the absence or low levels of stress, there will be boredom, apathy, and the lack of desire to do any work. This can be easily understood with an example of a student who does not study consistently all through the year because of lack of tension. Just before the commencement of the examination, the tension or stress starts building up and this forces the student to pick up the books and study so that he is able to perform effectively in the examination. But if the stress continues to increase beyond that and persists for a long period of time, the ability to perform effectively will decrease because of depleted energy, overload, and anxiety. In that state, the student who otherwise has prepared well will suffer from examination anxiety and will not remember the material that he knows so well for his answer and might perform miserably. This process can be easily explained with the help of the Yerkes–Dodson Law (1908), which states that stress leads to improved performance up to an optimum point. After that, stress has a detrimental impact on performance. McGrath (1976) suggests that performance declines beyond the midpoint in the Yerkes– Dodson curve because of the increased difficulty level of the task to be performed.

Lesson – 10: Individual - Centered Stress Management Techniques Cognitive, Hypnosuggestive, Behavioural and Physical (Yoga and Aerobic Exercises) Interventions; Coping Strategies.

Stress management is necessary to enhance and improve one's response to different stressors. Stressors can be both external and internal. Optimal level of stress is healthy for success and productivity. However, the inability to manage stress can have a detrimental effect on individuals. Various stress management techniques are in practice today to increase stress tolerance and avoid the vicious cycle of maladaptive behaviour. This unit will discuss in detail these stress management techniques and also delve into the types of coping strategies individuals can adopt.

STRESS MANAGEMENT TECHNIQUES

Stress seems to be as old as the mankind. Stress in multiple ways touches each of us in various ways. It is an integral part of our life. It can be seen both as a stimulus (as a property of the event or situation we face, for example, natural disaster, noise, crowding etc) and as a response (to a stimulus event known as a stressor that threatens to disrupt one's physical and social functioning).

Stressors can be external or internal. Examples of external stressors are earthquake, pollution, etc. and internal stressor can be one's thinking pattern. Stressors threaten one's well-being and demand some kind of an adaptive response. Stress is usually experienced in terms of three components, that is, emotion (such as anxiety or fear), thought (such as pessimistic self-talk)

and behaviour (such as smoking). All situations, positive and negative, that require adjustment can be stressful. Thus, according to Hans Selye (1956), there are two kinds of stress: Eustress (refers to stress caused by positive situations such as marriage, promotion, etc.) and Distress (refers to stress caused by negative situations such as death, divorce, loss of a job, etc.). Though both eustress and distress tax an individual's coping skills and resources but distress has more potential to cause damage. While stress is considered a major cause of various mental and physical health problems, its effects is not always undesirable. In fact, stress is a basic ingredient of life and has several advantages as well. They are as follows:

- Our biological system is equipped with some stress alarms that are essential for survival and allow one to function effectively in many situations.
- Stress seems to underlie all creative and constructive activities. A certain level of stress is necessary to perform better. Stress quite often increases our efficiency and makes us search for new coping strategies.
- It improves our adaptive system, thereby making us better equipped to deal with such situations in future. It is seen that individuals who have not experienced any stress in their lives have a poor adaptive mechanism and may succumb to even mild forms of stress. On the other hand, there are people who thrive on stress and show greater efficiency in handling crisis. In essence, intermittent stress (i.e., occasional exposure but with recovery periods) leads to enhanced stress tolerance

Thus, optimal level of stress is natural and healthy for success, productivity and effectiveness. It is advisable that each individual finds out his or her optimal level of stress as too little stress results in a boring and apathetic existence whereas too much of stress leads to undue wear and tear, making us more vulnerable to illness.

Stress Management Techniques

Generally speaking, increased levels of stress threaten a person's well-being. It usually results in the individuals taking some steps to do away with stress, which can have a detrimental effect on their well-being. What action an individual takes often depends on a complex interplay between internal factors like a person's frame of reference, motive, competency, or stress tolerance and external factors like one's social demand and expectation.

Ironically, some people create stress for themselves by engaging in maladaptive behaviours and cognitions. Some individuals get caught in the vicious cycle of generating life events that in turn produce adjustment problems. There are various ways to manage stress. Some of the techniques are discussed below.

Cognitive Techniques

Stress is subjective in nature and is more about how we perceive or look at a situation. Our thoughts and cognitions are key determinants in the degree and the amount of stress we experience. Therefore, one of the key ways to overcome stress is to replace negative maladaptive thoughts, behaviours and actions with more adaptive and functional ones.

Altering one's negative thoughts: Research has shown that our mind listens to what we say to ourselves. The more negatively we talk to ourselves, the more negatively we think, thereby becoming more anxious. This will lead to poor performance. Therefore, it is important to rectify the statements we tell ourselves. One should try to talk positively to oneself by focusing more on the solutions to the problem rather than procrastinating or avoiding the problem. For example, instead of saying that lot of syllabus is still left, I will surely fail; it is better to say that I can choose at this point to cover those concepts properly that carry more marks in exams. Similarly, instead of repeatedly telling oneself that one would forget everything, it's better to say that I have studied hard and I know the concept; hence I will be able to write it in my own language.

Problem solving and Goal setting: To deal with the current situation at hand, we must set adequate achievable, clear, and specific goals. One should work towards solving the issue at

hand. Once the key concerns are resolved, we feel less stressed. Prioritizing one's goals is essential. One can make use of following steps to set effective goals as a way of resolving key problems and stressors we face every day:

- Set clear concrete goals by breaking larger tasks into simpler ones and try to achieve them bit by bit.
- Reward yourself. After achievement of each goal, one should reward oneself. The rewards can be small, like treating oneself with ice cream, or buying a new outfit or playing a game, etc.
- One needs to regularly review the extent to which one has achieved their goals.

Hypno-Suggestion

Hypnosis is used as a stress management technique by making people get into a deep state of relaxation. Hypnosis can also be used to prevent health problems that may result from chronic stress. Hypnotic techniques are also seen quite useful in enabling people to bring about various healthy changes in their life style. Hypnosis comes very handy in reducing anxiety levels and can help one to overcome negative habits like smoking or compulsive eating. It can also be used to induce a state of relaxation and reduce pain in situations like child birth which can at times be very painful and stressful for few.

Hypnosis has many benefits. It potentially has no major side effects and its positive impact can be quite lasting. But, some people find it difficult to get into the state of trance.

Behavioural and Physical Interventions

Stress management refers to the use of various methods that have been designed to reduce the impact of potential stressful experiences. The various stress management techniques can be broadly divided into eastern and western techniques. The eastern techniques include meditation, Yoga, pranayam and self-awareness (Anashakti). The western techniques include relaxation techniques (Jacobson's Progressive Muscle Relaxation Technique, Autogenic Training), biofeedback, assertiveness, time management, and effective communication

Meditation: The various relaxation techniques are closely associated with meditation practice in eastern religion. Earliest evidence of formal relaxation was seen in Hinduism in India between 3000 and 4000 BC. Religious rituals stressed on meditation. This meditation aspect represented the beginning of Yoga. Meditation means directing one's attention to an object. It may provide direct access to inner conscious energies. For meditation, one requires a quiet room free from distraction, a comfortable place to sit, a mental desire to focus one's attention, and a passive attitude. It is effective as it slows metabolism, lowers blood pressure and reduces arousal.

Yoga: Yoga literally means to unite. It is the unity of physical and mental energy. It may mean different things to different people. For some, it is a way to keep the body free from ailments by reducing stress but for others it is a way of life. It helps one to control one's intellect, emotion, and behaviour.

Yoga prescribes some asana (bodily postures) to restore balance. 'Asana' is a Sanskrit word meaning at ease and relaxed. Shavasana is the simplest asana and a quite effective relaxation technique. It requires a quiet room with subdued lighting free from distraction. In it, the individual is asked to lie down straight on his back by keeping his head in a comfortable position. He asked to focus on his breathing, on the air that passes in and out of the nostril. It focuses on passive observation of one thoughts, which is quite difficult but essential for total relaxation to take place. Individual is instructed to release tension from the body from time to time.

Those who cannot relax in Shavasana because of its resemblance to a dead body can benefit from Makarasana (crocodile position). Depending upon needs and capabilities of each individual, different technique can be prescribed. Research has shown that it is effective in regulating blood pressure, anxiety, stress, insomnia, etc.

Pranayam: It refers to breathing exercise. It is based on the principle that if one could harmonize breathing then one can alter their thought, behaviour and emotion. The lungs take in air which is rich in oxygen and hence energies both the body and the mind.

Self-awareness or Anasakti: It is an Indian concept based on selfrealization/self-actualization. But it is a health promoting attitude that gradually develops. It is a relative concept. It is emotional detachment from something/ someone. In spiritual love, it means detachment of the spiritual principal (basis of consciousness) from body and ego and body movement that create tension and interfere in lifetime function of internal organs and may result in psychosomatic illness.

Anasakti person tends to perceive his work as duty and shows lack of concern for the consequences of action. He appraises his own success and failure in objective rather than egoistic terms and is less governed by external standards like social approval and concerns. He does not insist on seeking pleasure or avoiding pain. He shows absence of egoism and maintains an emotional equipoise both in the face of positive and negative experience. He is a man of stable wisdom with little mood swings. He shows total absorption in the work/task at hand and shows heightened concentration, trying to make efforts towards achieving task excellence. The statement that 'this too shall pass' very well expresses the attitude of an anasakti person.

Relaxation technique: Relaxation is often misunderstood as rest or lying down. It is rest after effort, or more accurately, conscious rest after conscious effort. Stebbins (1960) has defined relaxation as the complete resignation of body to law of gravity and of the mind to the law of nature. It involves the total energy transformation into deep dynamic breathing. It is both a means and an end to self realization.

The main goal of relaxation is not to eliminate stressors or to prevent its recurrence but to increase one's tolerance. Though it is suitable for all people, when and how it is introduced varies. It possesses both curative and preventive characteristics. For the relaxation techniques to be effective, they should be administered live by the therapist. The client should be told about its nature and how it works. The clients should be motivated and encouraged to use it on regular basis. The benefits of the technique should be transferred to real life situations and the client should be made to see its several beneficial applications. Some of the relaxation techniques commonly used are Jacobson's muscle relaxation technique and Autogenic Training.

Biofeedback: It converts physiological response into electrical activity and provides a visual/auditory feedback about them. It is based on the principle that when we make a response, then feedback about the consequence of the response enable us to make appropriate adjustment. It requires making an individual aware of a metabolic response, drawing his attention to a signal that indicates desirable changes in internal responses, trying to control biofeedback signal and thereby controlling physiological response. It involves transferring control from laboratory to everyday setting. Two majorly used biofeedback technique are EMG (electromyography), which measures muscle tension and Thermal biofeedback, which measure skin temperature. It is based on the principle that under stress, blood vessels constrict and temperature becomes cold.

The method of biofeedback has been criticized for being expensive; also it is difficult to transfer control from laboratory to real life settings. Till date it is not certain whether the change is due to biofeedback only or anything else. One does not know how it exactly works and its effects are short lived. Biofeedback has been found to be effective in treating chronic tension, headaches, blood pressure, muscle tension and lower back pain.

Assertiveness training: Individuals who are unable to say 'NO', that is, those who are non-assertive, are likely to experience stress as they are unable to express their feelings and tend to suppress their emotions like anger, hostility, disgust which are associated with chronic stress.

Hence, assertiveness training is likely to benefit such individuals. Assertiveness training involves teaching individuals using modeling, instruction, rehearsal and feedback. Assertiveness training involves three components namely, refusal (the ability to say no for things that you do not wish to do), commendatory (i.e., the ability to express positive emotions), and request (i.e., asking someone to do things that help you to accomplish your goal).

Time management: It helps individuals to learn to prioritize their tasks and do them efficiently. One should first make a list of the task that one needs to, rate them in terms of their importance, assign the amount of time you wish to spend at it and review how much you were able to achieve. Stress usually occurs when various tasks pile up and one is not able to finish them on time.

Effective communication or calming self-talk: It involves engaging in silent, relaxing, reasoning that aims at telling oneself that stress is temporary and it shall soon go away. It also helps individuals to preserve a sense of personal control, such as telling oneself that one handle the stress.

Diet: Eating a well balanced healthy diet is seen to reduce stress. One should avoid fast foods and drinks like tea and coffee which contain caffeine as it lessens the process of grasping. Drinking plenty of water keeps the brain hydrated.

Exercise and rejuvenation: Exercise such as walking, jogging, swimming or cycling 15 minutes a day helps by keeping one physically active. Taking short breaks helps one refocus and gives time for relaxation, rejuvenation and assimilation. Physical exercise and sports help improve general circulation, facilitate increased blood flow to the brain, and is instrumental in raising the levels of nor-epinephrine and endorphins – all of which may reduce stress, improve mood, and perhaps as a result improve achievement.

Sleep: Adequate sleep routine helps you feel fresh for longer duration and reduces stress. Regular seven or eight hours of sleep is mandatory for the body to function well and keeps one alert and responsive. Staying up late a few nights in a row and then trying to catch up on the weekend does not help.

COPING STRATEGIES

Individuals tend to cope with stress at three levels: at the biological or the physiological level (through the use of immunological defenses and damage-repair mechanisms), the psychological or the interpersonal level (through the use of learned coping patterns, self-defenses, and support from family and friends) and at the socio-cultural level (though group resources, such as labor unions, religious organizations, and law-enforcement agencies). The failure of coping efforts at any of these levels may seriously increase a person's vulnerability on other levels and also to other stressors. In order to effectively cope with stress, individuals are seen to engage in various coping strategies. Coping strategies refer to various cognitive, behavioural and emotional ways people engage in to manage stress. They are dynamic processes which neither eliminate a stressor nor prevent its re-occurrence but increase one's tolerance of the situations. All the coping strategies an individual engages in are not equally effective. Lazarus has given two kinds of coping strategies: emotion focused coping strategy and problem focused coping strategy.

Emotion focused coping It involves the use of cognitive and behaviour strategy to manage one's emotional reaction to stress. Emotion focused coping primarily aims at diverting attention from unpleasant situations, stress evoking events and problems. An individual tends to make use of them when nothing significant can be done to alter the stressor or stress evoking situation and events and when he or she lacks the skills or resource to meet demands posed by the stressors. Three types of emotion focused coping strategies that are frequently used are:

1. **Escape avoidance:** In it, the individual physically/psychologically separates himself or herself from the stressors. For instance, to avoid failing one may either not give the exam or may engage in excessive sleeping.

2. **Distancing:** It refers to psychological detachment of oneself from the stressor. For instance, over-weight people may stop thinking about their weight.
3. **Position reappraisal:** It refers to reinterpreting the situation to turn the negative aspects of the situation or the stressor into its positive aspects. For example, one may look at the loss of job as an opportunity to get something better.

Problem focused coping It involves directly dealing with the stressful situation by either reducing its demands, or by increasing one's capacity to deal with it. The types of problem focused coping strategies that are frequently used are:

Proactive coping (preventive coping): It anticipates potential stressors and acts in advance to either prevent their occurrence or to reduce its impact. To achieve this goal, it may make use of several mechanism like, improving problem solving skills, developing stronger social support network, etc. For example, the fear of failing can be prevented by studying in advance for one's exams. This coping strategy usually has a long term effect and may bring about a change in one's attitudes, cognitive styles and behaviours.

Combating coping: It involves escaping from stressors that cannot be avoided. It involves the active use of Relaxation techniques, meditation and eating nutritious diet.

Research has shown that women in general and individuals from a low socioeconomic status are more likely to use emotion focused strategy than men and people from a high socioeconomic status. The latter make use of problem focused strategies. One reason behind this could be that women react more emotionally to stress than men and past experiences may create feelings of helplessness and hopelessness in individuals from a low socio-economic status. In fact, in dealing with various stressors, both the above mentioned coping strategies are often used together. In addition to the various coping strategies, certain factors that can affect one's ability to cope with stress are as follows:

Hardiness: It is associated with better coping. It is seen that individuals high on Hardiness are more likely to make use of problem focused coping strategy.

Resilience: Some individuals are more resilient than others. For instance, it is seen that some children have the ability to become competent and well-adjusted despite being raised in an extremely disadvantaged environment. Resilience may be a result of child's personality trait and the available social support. Resilient children tend to have well-developed social, academic and creative skill; an easy temperament; high self esteem, high self description; better personal control; healthy social relationship with others; and at least one consistently supportive person in their life.

Explanatory style: It can be of two types, namely, positive explanatory style and a negative explanatory style. Individuals who adopt a positive explanatory style tend to attribute outcomes always to a positive event such as one's personality, hard work, etc. whereas, individuals who adopt a negative explanatory style tend to attribute outcomes always to a negative event. These individuals tend to feel that they are always surrounded by failures. Pessimists have a negative explanatory style and are vulnerable to experience negative emotions like depression, anger, anxiety and hostility. They are also likely to have suppressed immune systems. However, in contrast, optimists have a positive explanatory style and are more likely to experience positive emotions and are likely to increase one's social, physical and cognitive resources. They are also likely to have healthy attitudes and healthy habits. The functioning of their immune system is also seen to be better.

Self-regulation: It refers to the ability to modulate thoughts, behaviours, and emotions in each and every situation. However, too much of self control is not healthy as it leads to suppression of anger and may make an individual vulnerable to develop ulcers later in life.

Repression: It is a defense mechanism adopted by some individuals who tend to repress or block the awareness of negative stress emerging out of consciousness. It is unhealthy and may result in complications.

Learned helplessness: It is a phenomenon, in which after experiencing a series of negative uncontrolled events, the individual comes to an understanding that he is helpless in the face of adverse circumstances and hence does not make effort to overcome his difficulties.
Social support: The presence of adequate social support is likely to reduce stress. There are two hypotheses, namely buffering hypothesis and the direct effect hypothesis, which have been proposed to explain how social support reduces the negative effects of stress. According to the buffering hypotheses, social support reduces stress by providing resources on the spot to cope with the stress effectively. People with good social support tend to ruminate less and this further minimizes the negative impact of the stressor on the individual. According to the direct effect hypotheses, social support enhances the physical response to challenging situations. For example, pressure of others may reduce Sympathetic Nervous System arousal and may decrease the release of CRH. Individuals with better social skills tend to create stronger social network and are likely to receive more social support.

Lesson – 11: Organization Centered Stress Management Interventions: Preventing, Mitigating and Moderating Stress at Organizational Level.

It is important to minimize stress at the organizational level as doing so benefits both the individual and the organization. Reduced stress enhance performance of employees, leads to reduction in anxiety and burnout, and improves well-being. Stress management interventions can be grouped into individual-level intervention and organization-level intervention. These intervention change the relationship between the individuals and the organization. This unit will discuss in detail the stress preventing, stress mitigating and stress moderating practices at the organizational level.

STRESS MANAGEMENT INTERVENTIONS AT ORGANIZATIONAL LEVEL

Working in an organization can be a very stressful experience for many. Research has shown that there are several benefits of minimizing stress at the organizational level both for the organization and for its employees. For instance, reduced stress level is associated with enhanced performance, improved relationships, lesser instances of burnout, anxiety and depression among employees, less number of sick leaves and reduced absenteeism. It is also associated with positive psychological well-being of the employees. To achieve these benefits, every organization adopts several stress management interventions. These interventions adopt certain set of activities that are aimed at improving employee well-being and reducing stress at the organizational level.
Stress management interventions can be largely grouped into individuallevel intervention and organizational-level intervention. Interventions that help employees develop the skills to manage, cope and reduce stress are called individual-level interventions. Whereas, organizational-level interventions refer to those activities that make systematic changes in organizational practices and are aimed at reducing stress of all employees or a specific group of workers in an organization. Both these interventions focus mainly on changing the relationship between the individual and the organization. Both these interventions can be primary, secondary and tertiary in nature.
Stress Preventing Activities The basic aim of primary interventions is to remove the sources of stress and enhance well-being. It includes several activities. They are as follows:
Selection and Assessment activities: It aims at selecting those applicants who have skills and abilities so that the demands of the job can be managed. It also aims at screening out those applicants who might be susceptible to experiencing stress in the target role, particularly in highly-stressful occupations (Bartone, Roland, Picano & Williams, 2008).

Pre-employment medical examination of employees: This is done to screen those employees who may be undergoing several medical concerns that can interfere with their ability to work and thereby increase their stress level.

Job Redesign: It refers to reviewing job responsibilities and tasks so as to re-allocate them among staff to improve output. This can lead to improvements in productivity and lead to job satisfaction through job enrichment, job enlargement and job rotation. Job enrichment provides each employee with more tasks to do as a part of their job and also gives them the responsibility and authority to complete these additional tasks on time. These tasks can provide variety and new challenges to employees, thus leading to employee satisfaction. Job design also involves changing organizational practices and policies, especially those which are concerned with leadership, working time, and occupational health and safety, as a way of reducing stress.

Working time and schedules: This guarantees that tasks are done adequately on time. Only achievable goals are set and they are prioritized to ensure that schedule is maintained.

Management training: Training, for example, mentoring given to employees at different times in an organization is also seen to be an effective way of reducing stress at the organizational level.

Stress Mitigating Activities The aim of these secondary interventions is to reduce the severity or duration of stress once it occurs and also to prevent the level of stress from becoming problematic. They include the following activities:

Mindfulness training: It aims at promoting a state of mindfulness by focusing on the current experiences, thoughts and emotions in a nonjudgmental, compassionate, accepting and non-reactive way. It helps to promote psychological well-being by primarily helping the person to dissociate negative thoughts and emotions from maladaptive behavioral and emotional responses.

Health promotion: Encouraging employees to actively engage in physical exercise every day is quite effective in reducing stress.

Cognitive behavioural therapy: It aims at identifying and altering maladaptive cognitions, feelings and behaviors with more adaptive and functional thoughts, feelings and behaviors. It also aims to identify misconceptions about the nature and causes of stress in order to test the validity of existing thoughts and understandings. This helps in developing new conceptions about stress. Stress can also be reduced by encouraging employees to adopt a healthier lifestyle involving exercise, better diet, effective support system, etc.

Relaxation: Various relaxation techniques aim to reduce negative states of anxiety, tension and anger, etc. (Russell, 1979). One of the most popular relaxation technique is progressive muscle relaxation technique. This technique consists of tensing and then relaxing different muscles in our entire body in a prescribed order. For example, one might start by clenching and unclenching one hand then the other, then focus on the muscles in one forearm and then the other, and so on. The length, duration and range of muscles covered can vary, but once the employees practice it, they can induce a deep state of muscle relaxation in minutes (Murphy, 2003).

Meditation: It focuses on relaxing the mind and involves mental processes rather than physical processes. Most of the meditative techniques typically involve sitting in a quiet place and then repeating a word or sound while maintaining a passive mental state that excludes intrusive thoughts. It also involves focusing on the air that we inhale and exhale.

Personal and interpersonal skill training: It aims at enhancing communication, goal setting and time management skills, which are seen to reduce stress. It helps employees by preventing stressful situations such as conflict, high work load, etc. from occurring.

Acceptance and commitment therapy: Also known as ACT, it is a kind of behavioral therapy that works on inculcating the practice of selfacceptance with mindful actions in an individual,

so that one could live a value based life. It focuses on enhancing psychological flexibility in an individual. It makes use of 6 core processes. They are as follows:

- Attention to the present moment
- Values o Committed Action
- Self-as-Context
- Defusion
- Acceptance

These processes are together known as ACT hexaflex. Through these processes, an individual is guided by their therapist to become consciously aware of the present moment and take steps to make things happen with full purpose and engagement. It encourages people to embrace their thoughts and feelings instead of fighting or feeling guilty for them. ACT when paired with mindfulness-based therapy is quite effective in reducing stress.

Psychosocial intervention training: It focuses on enhancing an individual's psychological, behavioral or social aspects of functioning to reduce the amount of stress they experience and to minimize the negative impact of the same on the individual. It is a broad heading under which a wide range of interventions are included. Some of these interventions are psychotherapies (e.g., psychodynamic therapy, CBT, interpersonal therapy, problem solving therapy, etc.); community based treatments (e.g., assertive community treatment, first episode psychosis interventions); vocational rehabilitation, peer support services, and integrated care interventions.

Coping skills training: To deal with stress, conflicts, and issues at hand, we all make use of certain strategies which may be adaptive (seeking social support, resolving concerns, etc.) or maladaptive (drinking, escapeavoidance, etc.) in nature. Various mechanisms which one adopts to deal with stress are broadly called as coping skills. The effectiveness of the coping skills depends on the type of stress, the individual, and the circumstances. How one copes with issues at hand partially depends on the circumstance and the personality of the individual. The most commonly used six adaptive coping strategies are as follows:

Appraisal-focused coping strategies: It modifies the way we look at a thing or a circumstance or event in our life. One may modify their thinking by altering their goals and values. o **Adaptive behavioral coping strategies**: It focuses on dealing with the cause of the problem. Learning new skills is one of the ways the problem can be taken care of. The three problem-focused coping strategies identified by Folkman and Lazarus are: taking control, information seeking, and evaluating the pros and cons.

Emotion-focused coping strategies: It focuses on releasing pent-up emotions, distracting oneself, managing hostile feelings, meditating, mindfulness practices, using systematic relaxation procedures, etc. These strategies primarily focus on changing the meaning of the stressor (by trying to drive a positive meaning out the stressful event that has happened) or transferring attention away from it. These strategies are very effective in dealing with stressors that are uncontrollable.

Reactive and proactive coping: It focuses on anticipating a future stressor and becoming prepared for the stressor in advance.

Social coping: It refers to actively seeking social support from others as a way of dealing with current stressors at hand.

Humor: This minimizes the stress that one experiences. With humor, people can develop a sense of control over a more powerless situation and this helps them to temporarily escape from feelings of helplessness.

Resilience training: Resilience training focuses on four areas, namely, emotional, cognitive and mental, physical, and spiritual resilience. The key aim of resilience training is to enable people to deal with a crisis, enhance their quality of life, and decrease their stress and anxiety. Resilience training helps one view the inevitable challenges of life as opportunities where they

can grow and learn. According to Dr Ginsburg, there are seven integral and interrelated components – called as 7 C's – that make one resilient. The details of the same are, as follows:
Competence: It refers to one's ability to handle stressful situations effectively by facing the challenges at hand by developing the skills that are needed.
Confidence: It refers to one's belief in their own abilities.
Connection: It refers to having a stronger sense of security and sense of belonging with our fellow beings.
Character: It refers to a strong sense of self-worth and confidence that some of us may feel within.
Contribution: It implies the valuable contributions one makes to one's world.
Coping: It means using a wide variety of skills and strategies that allow one to cope and overcome life's challenges more effectively.
Control: It refers to a sense of realization that one has control over one's decisions and actions. This allows them to bounce back from life's challenges more effectively.

- **Improving communication and decision making:** Effective Communication is significant for managers and employees in the organization to perform basic functions of management, such as Planning, Organizing, Leading and Controlling. It serves many functions such as promoting motivation, offering adequate information to make effective decisions, altering an individual's attitude, enhancing the socialization experience, and assisting in controlling process, thus minimizing conflict and stress as far as possible.
- **Conflict management:** Conflict management is the process by which disputes are resolved, wherein negative results are minimized and positive results are prioritized. This key management skill involves using different tactics depending on the situation, negotiation, and creative thinking. The most widely used five effective conflict management styles are as follows:

(a) Accommodating: It refers to simply putting the needs of others before one's own. It primarily aims at ensuring peace and not putting in more effort than the issue is worth. It is an effective way to resolve small conflicts.
(b) Avoiding: It refers to reducing conflict by ignoring it, removing the conflicted parties, or evading it in some manner, such as by removing team members who are in conflict, pushing the deadlines or reassigning people to other departments.
(c) Compromising: It aims to find the middle ground by asking both parties to concede some aspects of their desires so that a solution can be agreed upon.
(d) Competing: It means rejecting a compromise and not giving in to viewpoints or wants of others.
(e) Collaboration: It involves coming up with solutions that cater to needs and wants of each person to arrive at a win-win solution so that everyone is satisfied in the end. This technique produces the best longterm results and is often the most difficult and time-consuming approach of resolving conflict.

- **Peer support groups**: It refers to regular meetings of people or employees in an organization, where people talk about their experiences, struggles and challenges so that they can learn from one another. These groups are usually moderated by a trained peer support specialist.
- **Coaching and career planning**: It is quite beneficial for everyone including leaders, managers, and individual employees, as it increases work engagement, boosts confidence, and improves overall productivity, leading to an increase in retention and a higher profit margins for the organization.

Stress Moderating Activities Also known as Tertiary interventions, it aims at rehabilitating and maximizing the functioning for those employees in an organization who are already experiencing or suffering from psychological ill-health. They include the following activities:

- **Employee Assistance Programs (EAP):** It provides counselling and advice to those employees that are experiencing high-levels of stress or mental health problems, irrespective of whether these issues are related to work or not.
- **Counseling:** Some organization hire counselors who provide various counseling services to employees to deal effectively with the problems or concerns they may be experiencing at the work or the personal front. Counselors help employees to deal with issues related to family, relationships, drug abuse, etc.
- **Post-traumatic stress assistance**: Employees who have experienced a traumatic experience in their life and are suffering from post-traumatic stress disorder (PTSD) can be referred to a qualified clinical psychologist who can deal with their concerns in a far better manner.
- **Disability management**: It offers help to people with disability. It involves combined and coordinated use of medical, social, educational, and vocational measures for training or retraining the individual to the highest possible level of functional ability especially if they have become disabled in an organizational hazard or have met with an accident while at work. This enables people to restart their life in a far more meaningful manner.
- **Vocational rehabilitation**: It is a process which enables persons with functional, psychological, developmental, cognitive, and emotional disabilities, impairments or health disabilities to overcome barriers to accessing, maintaining, or returning to employment or other useful occupation. It primarily focuses on job retention.
- **Outplacement**: It refers to finding new employment for some employees who have become redundant for an organization. Outplacement services are also called as career transition services as they help employees to find better career paths and employment opportunities that can bring a greater sense of satisfaction in their life. By providing such services, the organization is able to save the organizational or legal costs that may arise while terminating an employee from work. The key benefit this service offers to employees is that it helps them survive financially during the transition, and do it swiftly and successfully.

UNIT – IV: STRESS MANAGEMENT

Lesson – 12: Work Related Stress and Stress Management What is Stress? - Stressors - Individual Differences in Stress - Consequences of Distress - Managing Work - Related Stress - Any Two Stress Reduction Techniques - Case Studies

Work-related stress can be caused due to poor work organization, bad work design, lack of resources, poor management, dissatisfying work conditions, and lack of support. The level of stress varies from person to person, depending upon the personality, attitude, emotional intelligence, etc. Not only does work related stress affect the performance of the organization, it also influences the emotional and physical well-being of employees. This unit will discuss in detail the stressors at work and its impact on employees. The various stress reduction techniques will also be discussed in detail.

WHAT IS STRESS?

Stress is a psychological and physical response of the body towards the demands of the situation. No individual can claim that he/she has never experienced stress in their lifetime. The feeling of stress occurs in case of unfulfilled or unfinished tasks. In a stressful situation, the body releases hormones which prepares the body to fight or flight. The body prepares itself to either confront the situation or get to the safe zone as soon as possible. Stress can arise due to a situation that makes you feel uncomfortable and as soon as you deal with the situation the feeling of stress starts reducing. Stress can occur due to many reasons and these causes vary with each individual. Stress can also be understood as a state of mind and depends on the interpretation of situation. Stress cannot always be termed negative, as it also forces the

individual to finish the task in time. If the needs are not met within the given period of time, stress starts making negative impressions. To understand it in more precise manner, stress can be categorized into two categories as per experience:

Positive form of stress or Eustress: It is the force that motivates the individual to meet the target.

Negative form of stress or Distress: It is the force that reduces the energy and efficiency of the individual. There could be many causes of stress, namely, financial losses, personal life, relationships and work-related stress. Many individuals all over the world report about workplace stress. With the change in work style and increasing expectations, the stress level among employees is also increasing day by day. Work related stress can be due to poor work organization, poor work design, lack of resources, poor management, dissatisfying work conditions, over expectations and lack of support.

Stressors

Stress associated to work, is a rising problem around the globe that people are experiencing on a continued basis. Work related stress not only affects the mental and physical health of the individual but also affects the productivity or output of the organization. There is no single reason for stress. Work stress might arise due to number of different reasons which basically depends upon the psychological sensitivity of the individual and on personal and professional experiences as well. The following factors could be considered as potential stressors at the workplace:

- **Work overload**: Repeated assigning of heavy workload can lead to tiredness and eventually stress. Workload also reduces the work capacity of employees. Overwork leads to a kind of stagnation in the brain which further influences the performance of employees.
- **Deadlines:** Meeting the deadlines engages the individual physically and psychologically. Completion of task on time leads to anxiety which in return can cause stress and restlessness. The efforts to meet the deadline increases the blood pressure of the individual and tires the body easily which further causes increased pressure to meet the deadline.
- **Lack of support**: If an individual does not get needed support at the workplace, it results in helplessness, which in turn causes stress. Several kinds of supports are required at work. Psychological and physical support can be considered as the example of resources needed. Inability to get support increases mental and physical stress.
- **Lack of resources**: Resources are a must for any employee to function properly. A task cannot be completed without the required resources; lack of resources can affect the speed or quality of the task, leading to a stressful outcome. Lack of resources causes the delay in meeting target.
- **Unclear/undefined targets**: When the target is unclear, the chances of taking a wrong path of work is more. This can lead to a stressful situation. Proper direction of work makes it easy to finish it in time.
- **Organizational change:** Changes in the organizational structure and job description causes uncertainty regarding job security. The style of work and the processes also change with the change in the organizational structure. The alteration in the structure should be properly communicated to employees.
- **Role conflict**: When the roles, responsibilities and power is not defined, it can lead to a lot of confusion. The capacity of work and power should always be clear to all the employees. When the hierarchy is defined, then the process of work become easier, and the number of obstacles reduces. Role conflicts also becomes an obstacle in meeting organizational goals.
- **Lack of learning outcomes**: When an individual is doing the same work for a long time and does not see any further learning from the same, it can become a stressor. Employees usually want to grow with the organization, but when the learning becomes stagnant then the trouble starts. The repetition of same kind of work reduces the interest in the job and then becomes a

reason of dissatisfaction. Ambiguity about the future also leads to stress. Employees need to know where they stand and where the current job is going to take them; uncertainty about it causes high attrition rate.

- **Lack of recognition**: Recognition is one of the most important factors that increases motivation among employees. If the work is not recognized, the employees would not be able to know whether they are going in the right direction. Recognition also increases enthusiasm among employees. With less recognition, employees do not feel like working with passion, and so the performance decreases.
- **Relationship at work**: Healthy relations at workplace leads to a positive work environment and decreases employee absenteeism.
- **Organizational environment:** Health work environment impacts psychological and physical wellbeing which further leads to high efficiency of the employee.
- **Harassment:** Psychological or physical (sexual) harassment has become a looming realty of workplaces. This unwanted behavior of the senior authority has become a major concern for the employees.
- **Lack of interest in work**: When an individual does not get the profile of his/her choice, it leads to dissatisfaction at work.
- **Work-life balance:** Personal and professional sides of life should be maintained well for a healthier experience. Work overload or overtime creates an imbalance which in turns leads to stressful life.

INDIVIDUAL DIFFERENCES IN STRESS

Experience of stress varies according to the sensitivity level of the individual. Many factors play role in experience of stress.

- **Personality**: The type of personality determines how a person responds to stress. Type A personality individuals generally exhibit a high sense of competitiveness and a tendency of working against the clock. On the other hand, type B personality people are comparatively less competitive and are more relaxed in their approach towards work. Both the personality types have some pros and cons. Highly competitive individual gets stressed very quickly due to running with the clock whereas the individual who is easy going also faces stress issues as he delays the task and completes it at the last moment.
- **Attitude:** The attitude of observing our surrounding impacts our experience of stress. Optimistic point of view can lead to better understanding of environment, which in turn makes us focus on good in people and not in bad. An optimistic point of view leads to a positive approach in life and enhances motivation.
- **Emotional Intelligence**: If a person has high emotional intelligence then the ability to cope with stress automatically increases. An emotionally intelligent person majorly focuses on positive feelings and is more capable of restraining negative feelings like stress and anxiety. Emotional intelligence helps an individual to cope with distress.
- **Previous Experiences**: Unpleasant experience with the previous job or boss can also cause a pessimistic approach towards new work or employer. Bad experience with the earlier job shakes the motivation or confidence of the employee and makes an individual doubtful about his/her abilities to do well in future.

Consequences of Distress

Experiencing a high level of stress for a long period can reduce the body's immunity and may generate many biological symptoms. The common side effects of stress could be the following:

- **Biological Consequences**: Prolonged and persistent stress might cause many biological reactions. Elevated levels of stress hormones can cause increased heart rate and high blood pressure. Frequent occurrence of stress might increase the risk for hypertension, stroke or heart

attack. Recurring episodes of stress might cause inflammation in the respiratory system. Cholesterol level of the individual also gets impacted by the way they respond to stress.

- **Headache**: Headache is a very common trigger of stress. Complexity of life and irregular daily routine disturbs the diet and sleep of the individual which in turns might cause headache.
- Muscle tension and body ache: Stress can cause pain in neck and back. Stress can also cause sensitivity on scalp and shoulder muscles. Chronic fatigue syndrome may arise due to prolonged muscle tension.
- Weakened immune system: Stress and immunity go hand in hand. As stress increases, it reduces the ability to fight against germs or antigens. Immunity deteriorates every time we experience stress. Exceeding stress conditions may lead an individual to illness due to compromised immunity.
- Insomnia: Stress is one of the most common causes of insomnia. Work stress, performance stress, financial stress, etc. increases the activity of the brain, which in turn disturbs our sleep pattern. Lack of sleep cause various malfunctions in the brain.
- Change in appetite: Appetite has a surprising connection with stress. Stress triggers emotional and psychological changes in the body to deal with the pressure. These changes might affect a person's digestive system. Sometimes high metabolism causes excessive hunger, or it might also lead to lack of appetite. In some cases, stress causes people to overeat and on the other hand, it can also cause loss of appetite.
- **Psychological consequences:** Stress can create many psychological concerns like emotional strain, anxiety, and depression, etc. Emotional instability can trouble an individual's personal and professional relationship. Emotional connections, motivation, drive for achievements, etc. gets impacted by stress.
- **Reduced performance:** Motivation to perform better reduces due to the unpleasant state the individual is in. Lack of interest in work and doubt about one's own capabilities might occur due to stress. Lack of self-esteem might cause difficulty in performance efficiency. A stressed individual cannot perform their best at work which in turns aggravates the feelings of stress.
- **Increased absenteeism**: Stress reduces the interest in work which causes increased absenteeism. Efficiency of the individual gets negatively impacted and it leads to reduced motivation. Stress increases the chances of experiencing burnout and increases unpredictability, which can cause employees to avoid going to work. Stress has also become the reason of increasing attrition rate among employees.

MANAGING WORK RELATED STRESS

The following measures can be taken to manage work related stress:

- **Take a break**: Take a small break after every hour of work. Distract your mind from work for a few minutes. Stretch your back, shoulder and all body muscles to reduce physical stress, which will further reduce mental stress. Never skip your meal due to work.
- **Pen down your stressors**: Always try to write about the issues which are causing stress. Writing about them can help you develop a better understanding of the problem.
- **Understand the side effects of stressors**: Our body gives signals in response to the stressors. Psychological and physical responses can be noticed if observed properly, e.g. body ache, change in appetite, sleep disturbance, reduced work efficiency, mood fluctuations.
- **Balance your work and personal life**: Keeping balance between professional and personal life can be very challenging but it is vital for a balanced life. A properly differentiated work and personal responsibilities enhances one's work efficiency. Employers who are committed to provide an environment which provides work-life balance for their employees experience more loyal work force, reduced absenteeism, and reduced expenses.
- **Prioritize and organize**: Finishing the work in time increases confidence and satisfaction towards the job. Try making a priority list ranking them according to the importance at the

beginning of the work, enlist the estimated time required for the same and try to abide by the timing allotted for each task.

- **Reduce the gap between real and ideal self:** We have so many expectations from our life and in order to meet those expectations we often set unrealistic targets which creates an added pressure on our life. This pressure keeps on increasing when we are not able to meet the target. It is better for you that do not burden yourself with unrealistic targets; understand your efficiency and set your targets and keep on setting higher target with your achievements.
- **Talk to your supervisor**: Sometimes, discussion of the current situation helps in resolving many concerns. An employee's mental and physical health is related to the productivity and in order to perform best at work, an effective plan of work can be developed.
- **Build positive relations at work:** Relationships at work are equally important to develop interest at workplace. When you build positive relationships, you feel more comfortable with your interactions and less intimidated by others. You feel a closer bond to the people you spend the majority of your time working with.
- **Maintain a healthy diet and exercise routine**: A healthy diet has a significant role in living a stress-free life. Eating healthy, small and frequent meals helps in maintaining blood sugar level in stressful situation. A proper diet prevents mood swings and maintains your energy and focus. Minimize the craving for food intake that adversely effects the mood and body. Along with healthy diet, a proper routine of exercise is also important. Simple stretching, yoga, brisk walk, running, swimming, etc. can also help. Deep breathing for 5-10 mins can help very well. Do not smoke when you feel stressed.

Stress Reduction Techniques:

Case Studies Some of the commonly used stress reduction techniques are discussed below: Deep Breathing Deep breathing is a relaxation technique that is used in other relaxation practices such as yoga, meditation, and visualization. It involves using the lungs and the abdomen or diaphragm. Most of us take shallow breaths from our upper chest instead of using our diaphragms. Shallow breathing limits the amount of oxygen we take in and when we are stressed our breathing becomes shallower. This makes us feel more tense and anxious. Deep breathing encourages full oxygen intake throughout the chest and lungs which calms the mind and the body. The easy way of doing deep breathing exercise is by doing the following:

- Breathe into a slow count of four.
- Hold your breath to a slow count of eight.
- Exhale to a slow count of twelve. If you have trouble breathing from your diaphragm while sitting up, lie on the floor, put a small book on your stomach, and try to breathe so that the book rises when you inhale and drops when you exhale.

Progressive Muscle Relaxation Progressive muscle relaxation involves a two-step process in which you systematically tense and relax different muscle groups in the body. We are often unaware that we are holding our bodies in tension until it becomes very painful. This technique teaches you to become very familiar and aware of how tension and complete relaxation feels in different parts of the body. It is designed to teach you to recognize the first signs of muscular tension that accompanies stress so that you can quickly counteract the effects of tension by intentionally relaxing. The following is a procedure for achieving deep muscle relaxation quickly. Whole muscle groups are simultaneously tensed and then relaxed. Repeat each procedure at least once, tensing each muscle group from five to seven seconds and then relaxing from 20-30 seconds. Remember to notice the contrast between the sensations of tension and relaxation. 1. Curl both fists, tightening biceps and forearms. Relax. 2. Wrinkle up your forehead. At the same time, press your head as far back as possible, roll it clockwise in a complete circle, reverse. Now wrinkle up the muscles of your face like a walnut: frowning, eyes squinted, lips pursed, tongue pressed to the roof of your mouth, and shoulders hunched. Relax. 3. Arch your back as you take a deep breath into the chest. Hold. Relax. Take a deep

breath, pressing out the stomach. Hold. Relax. 4. Pull feet and toes back toward face, tightening shins. Hold. Relax. Curl toes, simultaneously tightening calves, thighs, and buttocks.

Lesson – 13: Areas of Counselling Educational, Career, Marital and Gerontological; Stress Management Oriented Counselling; Counselling for Terminal Disease Patients: Cancer and HIV/AIDS.

It is the people in the organization who are the most important and valuable commodity. They have emotions, thoughts and a tender spirit. Often emotions and feelings cause difficulties and people fail to manage them, losing their motivation they struggle to put their mind in the job. These issues do not inherently apply to the job. They could be family issues, financial problems, marital problems, etc. All have a negative effect on their careers. The need of the hour is to eliminate the issue and help the person to recover his or her self-confidence and esteem. Counselling can do the magic and help a person to resolve their lost confidence. In this unit, we will discuss the different areas of counselling, along with the stress management oriented counselling. We will also focus on the counselling for terminal disease patients.

Education Counselling

Educational counselling is concerned with providing advice and support to students in the development of their educational plans, choice of fitting courses, and choice of college or technical school. Counselling can also help in improving study skills or provide support in case of school-related problems that interfere with performance (e.g., learning disabilities). Due to the relationship between educational preparedness and occupational choices, vocational counselling is also associated with educational counselling. An educational counsellor usually works with academic institutions like school and college. Nowadays, most of the schools have their own educational counsellor for an easy access to consultation regarding their higher education and career-related queries, and to stay informed regarding study choices especially for special needs children. To resolve student's issues, sometimes an educational counsellor also interacts with the student's primary caregivers (parents, guardian, or teachers) to help them find the best solutions for the student's issues or problems. An educational counsellor also follows up with each student that is in his/her charge until they successfully complete their program or school year. In case of a primary and secondary educational counsellor, one will also be responsible in part for assisting parents and teachers determine a student's readiness to move to the next grade. Education counsellors work with students of all ages on issues specific to their academic, vocational, and professional success. Education counsellors understand the barriers to success that many students experience such as learning disabilities, lack of motivation, language barriers, lack of accountability, test anxiety, and myriad of other things that hold students back from achieving their academic potential. They also typically have detailed knowledge of specific teachers, subjects, and resources available to help students achieve their goal. Educational counsellors also provide detailed assistance about the professional opportunities that students might not be aware of, and also the pros and cons of opting any particular program. Educational counselling might get associated with other education-based occupations, which include:

- Teacher
- Guidance counsellor
- Rehabilitation counsellor.

Career Counselling

Career counselling can help to choose suitable career track. Since people spend most of their time as a professional, the choice of career becomes significant in ensuring happiness. The

purpose of the career counselling is that people should enjoy the work they do and the profession should suit their goals and personality. A career should reflect the goals and the personal values of the individual and should not be something which creates discomfort.

Functioning of a career counsellor In this form of counselling, interests, preferences and possibilities of the career path of the people are discussed. Many standardized tests and tools are generally used such as personality and aptitude tests to know more about the person's inclinations. The values, beliefs, temperament, and natural tendencies of the client are studied by the career counsellor. These topics are discussed based on oneon-one interactions with the client. Career counsellor might have very diverse client base depending on his/her expertise. They can work with varied organizations like educational institutions, corporates, etc. Many career counsellors work in private practice, employment agencies and social service offices.

Need of a career counsellor In society many important roles are played by career counsellors. For school and college going students, career counsellors help to explore the areas of interest and availability of opportunities and options. As a result of counsellor's interaction, students can realize the possible course for their career and set their goals accordingly. The people who have left their job, are planning to change the job, or are fired from the job can also receive help from a career counsellor. Career counsellors provide guidance in finding the resources people need to retrain for a new job. In order to regenerate the interest and enthusiasm in the existing job, workers can take help of a career counsellor. A career counsellor can guide a person in maintaining a better work-life balance, improved positive attitude towards current job and to enjoy their careers.

Marriage Counselling

Marriage counsellors provides guidance to the couples to help manage their marital relationship in case they are facing any conflict. With the help of conversation with a couple, a counsellor assists by helping to uncover their conflicts and finding the roots of their problems. However, counsellors offer little to no suggestions to the couple and helps only via making them understand their own situation. With the discussion and better clarity about the situation, the couple can begin to think of better strategies for coping with these areas of conflict, resolving them together. By managing and solving the continuing issues, marriage counsellors can develop long term connections with the couples. Marriage counsellors are usually referred in case of serious marital disputes such as divorce or separation. Though marriage counsellors work with only a specific population, they do offer a multiple type of services to the society. They also provide individual counselling for both the partners, couple counselling and if required they can call for family counselling with children. The onus of the counselling is on the relationship and its effects on the family. Some programs like martial enrichment courses to improve communication skills are being conducted by the counsellor to build trust between the couple and determine the cause of the problems between marriage partners.

Relationships impacted by alcohol, drugs, or domestic violence are also being addressed by the marriage counsellors. Marriage counsellors can also work with spouses that have been abused by their loved one to help them overcome the trauma of the abuse. Yet another duty of marriage counsellor is to work with spouses that have committed abusive acts in order to help them work through the anger or other underlying issues that led to their behaviour. Wherever, there is a communication gap, which leaves couples at odds with each other, a marriage counsellor can provide better options for bridging that gap. Marriage counsellors can diagnose and treat the mental and emotional disturbances experienced by the couple due to the ongoing marital conflicts.

Gerontological Counselling

Gerontological counselling provides help related to mental or emotional health issues to people during the difficult stages of advanced aging and loss of independence. Common problems of aged individuals are addressed to make their lives manageable and comfortable. The unique

needs of the client are understood in depth by the gerontological counsellors. They review the client's case history, medical conditions, and other related conditions to support them properly. They will also seek to understand their client's social, personal, and financial situation. To improve the client's quality of life, they will create the concise plan according to their needs. The services might involve health care, financial management, daily life activities and client's hospital social workers. The most common jobs in gerontology include nursing home administrator, case management and hospital executive. Probable workplaces for gerontological counsellors include old age home or adult care facilities community, retirements and human services.

Stress Management Oriented Counselling

Any cause of stress which is left unaddressed may become overwhelming and trouble causing which can further lead to other mental, physical, social, and emotional problems. A counsellor can provide numerous ways to manage and support employees experiencing stress or other associated concerns at work. Every type of work-related stress can be addressed by the counsellor.

While stress sometime is a normal part of life and in some cases even manageable without the need for professional support, this might not be true in every situation. The process of talking to the professional counsellor about the issues and concerns that an individual is experiencing is called as stress management counselling. Many workplaces organize employee assistance programs to provide them support related to stress creating issues. These services are kept confidential and according to the need of the employees the trained counsellors can be hired. Counselling can help employee to understand the causes of their stress and find the solution to reduce the experience of the same. The stress can be worked out by understanding and identifying the source. The counsellors create judgment free environment to make individual more comfortable. Experience of stress at workplace is very common for people of different ages and genders, but it can disturb one's daily routine, mood, and social interactions with family, friends, and colleagues. This impact can in turn increase their stress. If the stress stays for a very long period of time, then it might cause various risks for individual such as anxiety and depression. This is the situation when the counsellor is required to be consulted. The counsellor can help in identifying the cause of the stress and then look for the solutions of the problems. In the process of counselling, employee can talk openly with the counsellor and develop better understanding to manage their concerns. The counsellor can use a specific technique of interaction depending on the need of the client.

Benefits of stress management

counselling Discussion with the counsellor is an approved step towards finding solution of stress, anxiety, and depression. With the help of Cognitive Behavioural Therapy (CBT), a counsellor suggests the client to open up and express their feeling and negative thoughts to reduce stress and manage their mental health. Other benefits include:

- Better self-understanding: A counsellor can help in developing better selfunderstanding, which leads people to find solutions to their problems. When the issues are resolved then self-confidence of the individual also increases.
- More engagement at workplace: It is proven that expression of the feelings might help in reducing stress, if not eliminating it fully. Employees might feel more comfortable and relaxed once they have dealt with their concerns.
- Reduced absences and employee turnover: When an employee is aware of one's concerns and issues, it becomes easier to deal with them. When individual is less stressed about their organization, then they are more motivated to work with them, thus reducing absenteeism.

Suggestions for managing stress

There are multiple ways to release stress. Numerous methods can be suggested by counsellors which can help people to get free from stress that affects them physically, mentally, and emotionally. Some tips for reducing stress in all of these areas are mentioned below:

- Biological:
  - o Deep breathing exercise
  - o Hot water bath
  - o Exercise such as running, dancing, swimming, or yoga
  - o Acupuncture
  - o Playing with a pet
  - o Going for a nature walk
  - o Listening to music
  - o Body massage
- Mental:
  - o Checking on to the negative thoughts and disastrous thinking
  - o Organizing and planning
  - o Concentrating on positive life events
  - o Meditation
  - o Positive self-talk or affirmations
- Emotions:
  - o Laughing or crying
  - o Articulating emotions with the help of art or writing
  - o Sharing thought with someone you connect well
  - o Engaging yourself in hobby or interesting activities such as dancing, cooking, singing, etc.

Counselling for Terminal Disease: Cancer and HIV/AIDS

A terminal illness is a disease that cannot be treated or cured and thus can cause death within a few years. The individual diagnosed with such illness is most likely to experience a varied range of emotions such as regrets, grief, depression or sadness. The similar kind of feeling can also be experienced by family members, spouse/romantic partners, friends and relatives. In such a situation, seeking help from a health professional can be beneficial for both individual and the people related with the patient. Terminal illness is a broad term that can include many different medical diagnoses. Some diseases, such as Amyotrophic Lateral Sclerosis (ALS), also called Lou Gehrig's disease, are known to be terminal from the time of diagnosis, while others may not necessarily be terminal at first onset. Cancer, for example, can be highly treatable and people can experience complete remission, while in others it is terminal. Late-stage cancer may progress quickly, however, and the period of time between diagnosis and death may be relatively short (from a few weeks to a few months). Some illnesses for which there is no cure, such as HIV/AIDS, have a long period of progression, and in many cases, those who receive early treatment may not see a significant decrease in quality of life or life expectancy. It is often recommended that those diagnosed with a terminal illness join a support group and/ or see a counsellor who can help them understand and come to terms with their diagnosis. In counselling or support groups, people may be able to express the complex emotions and feelings that are often likely to accompany a terminal diagnosis.

Coping with a terminal diagnosis

The individuals who are experiencing any terminal illness may go through some conflicting and diverse emotions. After knowing about the diagnosis for the first time, an individual might feel numb, calm, or accept it as fate as the reality of life. Many people might experience doubts as to how long they are going to live and how their body will react in the coming days. As the symptoms of the disease increases, the feeling of sadness, stress, depression, frustration, anger or regret might aggravate. The uncertainty about life and death might cause grief in a person and in such a situation an individual needs experts' support to manage all these emotions and

to stay calm and positive. People who are at the end of their lives typically face several decisions: What type of care they will need; where they should seek care; how they should manage their finances; what type of funerary services they want; how they would like to spend the remaining duration of their life (is in general the most important decision to make by individual who have a terminal illness). Some individuals might seek to fill the rest of their lives with joy and may pursue activities that provide them with satisfaction and happiness. Others might feel the need to play a meaningful and quality role before saying goodbye to the world.

Impact of diagnosis on family and friends

The knowledge about terminal illness not only affects the individual who is suffering, but also their family and friends. Loved ones may find it difficult to deal with the feeling of loss and grief. The primary caretaker of the patient might experience high level of anxiety and depressing feeling and may be guilt, if they are not able to devote adequate time for the caring of a terminally ill loved one. This might lead to stress, resentment, or caregiver burnout. The family members and the dependent individuals also need emotional support to be ready to face the situation. Other challenges that loved ones may face include feelings of isolation, insomnia or other sleep issues, and general irritability. Therapy for a person with a terminal illness typically differs from other types of therapies. It may be more time-limited and focused, and the goals of therapy may be more modest, especially if the individual is restricted by their medical condition. Therapy may be a collaborative approach that involves medical providers, spiritual advisers, and family members. Counselling may use a variety of therapeutic approaches for end-of-life situation, but it frequently involves the work of Elizabeth Kubler-Ross, who identified the stages of grief that a person is likely to go through while they are slowly dying that is: Denial, Anger, Bargaining, Depression, and Acceptance (DABDA). The individuals with terminal illness might be suggested to attend support groups for managing their remaining life. These support groups can be very helpful for some people and the patient can get opportunity to share their concerns and seek companionship from others who are going through the same situation.

CLASS ROOM ACITIVITY/DEMONSTRATION:

Activity 1: To assess the Coping styles of the subject.

Activity 2: To assess the stress level of the subject by using suitable test.

PSY 25145

STATISTICS IN PSYCHOLOGY

Objectives:

To understand the concepts of frequency distribution, NPC; to comprehend tests of hypothesis, Non-Parametric tests and to understand the concepts of Correlation and MRA

Unit I: Introduction to statistics
Lesson 1 Parametric and Nonparametric Statistics
Lesson 2 Descriptive and Inferential Statistics
Lesson 3 Types I and Type II Errors
Lesson 4 Setting Up The Level of Significance

Unit II: Correlation and Regression
Lesson 5 Product Moment Coefficient of Correlation
Lesson 6 Other Types of Correlation (including Phi-coefficient)
Lesson 7 Partial and Multiple Correlations
Lesson 8 Bivariate and Multiple Regression.

Unit III: Normal Distribution
Lesson 9 Characteristics of Normal Distribution
Lesson 10 Significance of Mean Differences, Standard Error of the Mean
Lesson 11 Anova (One-Way)
Lesson 12 Two-Way Anova

Unit IV: Nonparametric Statistics
Lesson 13 Rationale
Lesson 14Mann Whitney 'U' Test For Two Sample Test
Lesson 15 Kruskal Wallis Analysis of Variance
Lesson 16 Chi Square and Kendall Rank Correlation.

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Outcomes:

To understand the concepts of frequency distribution, NPC
To comprehend tests of hypothesis, Non-Parametric tests
To understand the concepts of Correlation and MRA

Unit I: Introduction to statistics

Lesson 1 Parametric and Nonparametric Statistics

The word statistics has been derived from Latin word ‘status’ or Italian ‘Statista’ meaning statesman. Professor Gott Fried Achenwall used it in the 18th century. During early period, these words were used for political state of the region. The word ‘Statista’ was used to keep the records of census or data related to wealth of a state. Gradually, its meaning and usage extended and thereonwards its nature also changed.

The word statistics is used to convey different meanings in singular and plural sense. Therefore it can be defined in two different ways

Statistics in Singular Sense

In singular sense, ‘Statistics’ refers to what is called statistical methods. It deals with the collection of data, their classification, analysis and interpretations of statistical data. Therefore, it is described as a branch of science which deals with classification, tabulation and analysis of numerical facts and make decision as well. Every statistical inquiry should pass through these stages.

Statistics in Plural Sense

‘Statistics’ used in plural sense means that quantitative information is available called ‘data’. For example, information on population or demographic features, enrolment of students in Psychology programmes, and the like. According to Websters “Statistics are the classified facts representing the conditions of the people in a State specifically those facts which can be stated in number or in tables of number or classified arrangement.

Definition of Statistics In this unit emphasis is on the term statistics as a branch of science. It deals with classification, tabulation and analysis of numerical facts. Different statistician defined this aspect of statistics in different ways. For example.

A. L. Bowley gave several definitions of Statistics:

i) “Statistics may be called the science of counting” . This definition emphasises enumeration aspect only.

ii) In another definition he describes it as “ Statistics may rightly be called the science of average”.

iii) At another place Statistics is defined as, “Statistics is the science of measurement of social organism regarded as a whole in all its manifestations”.

All three definitions given by Bowely seem to be inadequate because these do not include all aspects of statistics.

According to Selligman “Statistics is the science which deals with the methods of collecting, classifying, presenting , comparing and interpreting numerical data collected to throw some light on any sphere of enquiry”.

Croxton and Cowden defined “statistics as the collection , presentation, analysis,and interpretation of numerical data”.

Among all the definitions , the one given by Croxton and Cowden is considered to be most appropriate as it covers all aspects and field of statistics.

These aspects are given below:

Collection of Data : Once the nature of study is decided , it becomes essential to collect information in form of data about the issues of the study. Therefore, the collection of data is the first basic step. Data may be collected either from primary source or secondary or from both the sources depending upon the objective/s of the investigation

Classification and Presentation : Once data are collected , researcher has to arrange them in a format from which they would be able to draw some conclusions. The arrangement of data in groups according to some similarities is known as classification.

Tabulation is the process of presenting the classified data in the form of table. A tabular presentation of data becomes more intelligible and fit for further statistical analysis. Classified and Tabulated data can be presented in diagrams and graphs to facilitate the understanding of various trends as well as the process of comparison of various situations.

Analysis of Data : It is the most important step in any statistical enquiry . Statistical analysis is carried out to process the observed data and transform it in such a manner as to make it suitable for decision making.

Lesson 2 Descriptive and Inferential Statistics

Interpretation of Data : After analysing the data, researcher gets information partly or wholly about the population. Explanation of such information is more useful in real life. The quality of interpretation depends more and more on the experience and insight of the researcher.

Types of Statistics:

After knowing the concept and definition of statistics, let us know the various types of statistics. Though various bases have been adopted to classify statistics, following are the two major ways of classifying statistics: (i) on the basis of function and (ii) on the basis of distribution.

On the Basis of Functions

As statistics has some particular procedures to deal with its subject matter or data, three types of statistics have been described.

A) Descriptive statistics: The branch which deals with descriptions of obtained data is known as descriptive statistics. On the basis of these descriptions a particular group of population is defined for corresponding characteristics. The descriptive statistics include classification, tabulation measures of central tendency and variability. These measures enable the researchers to know about the tendency of data or the scores, which further enhance the ease in description of the phenomena.

B) Correlational statistics: The obtained data are disclosed for their inter Introduction to Statistics correlations in this type of statistics. It includes various types of techniques to compute the correlations among data. Correlational statistics also provide description about sample or population for their further analyses to explore the significance of their differences.

C) Inferential statistics: Inferential statistics deals with the drawing of conclusions about large group of individuals (population) on the basis of observations of few participants from them or about the events which are yet to occur on the basis of past events. It provide tools to compute the probabilities of future behaviour of the subjects.

On the Basis of Distribution of Data

Parametric and nonparametric statistics are the two classifications on the basis of distribution of data. Both are also concerned to population or sample. By population we mean the total number of items in a sphere. In general it has infinite number therein but in statistics there is a finite number of a population, like the number of students in a college. According to Kerlinger (1968) “the term population and universe mean all the members of any well-defined class of people, events or objects.” In a broad sense, statistical population may have three kinds of properties – (a) containing finite number of items and knowable, (b) having finite number of articles but unknowable, and (c) keeping infinite number of articles.

Sample is known as a part from population which represents that particular population’s properties. As much as the sample selection will be unbiased and random, it will be more representing its population. “Sample is a part of a population selected (usually according to some procedure and with some purpose in mind) such that it is considered to be representative of the population as a whole”.

Parametric statistics is defined to have an assumption of normal distribution for its population under study. “Parametric statistics refers to those statistical techniques that have been developed on the assumption that the data are of a certain type. In particular the measure should be an interval scale and the scores should be drawn from a normal distribution”.

There are certain basic assumptions of parametric statistics. The very first characteristic of parametric statistics is that it moves after confirming its population’s property of normal distribution. The normal distribution of a population shows its symmetrical spread over the continuum of –3 SD to +3 SD and keeping unimodal shape as its mean, median, and mode coincide. If the samples are from various populations then it is assumed to have same variance ratio among them. The samples are independent in their selection. The chances of occurrence of any event or item out of the total population are equal and any item can be selected in the sample. This reflects the randomized nature of sample which also happens to be a good tool to avoid any experimenter bias.

In view of the above assumptions, parametric statistics seem to be more reliable and authentic as compared to the nonparametric statistics. These statistics are more powerful to establish the statistical significance of effects and differences among variables. It is more appropriate and

reliable to use parametric statisticsin case of large samples as it consist of more accuracy of results. The data to be analysed under parametric statistics are usually from interval scale.
However, along with many advantages, some disadvantages have also been noted for the parametric statistics. It is bound to follow the rigid assumption of normal distribution and further it narrows the scope of its usage. In case of small sample, normal distribution cannot be attained and thus parametric statistics cannot be used. Further, computation in parametric statistics is lengthy and complex because of large samples and numerical calculations. T-test, F-test, r-test, are some of the major parametric statistics used for data analysis.
Nonparametric statistics are those statistics which are not based on the assumption of normal distribution of population. Therefore, these are also known as distribution free statistics. They are not bound to be used with interval scale data or normally distributed data. The data with non-continuity are to be tackled with these statistics. In the samples where it is difficult to maintain the assumption of normal distribution, nonparametric statistics are used for analysis. The samples with small number of items are treated with nonparametric statistics because of the absence of normal distribution. It can be used even for nominal data along with the ordinal data. Some of the usual nonparametric statistics include chisquare, Spearman's rank difference method of correlation, Kendall's rank difference method, Mann-Whitney U test, etc.

Having an understanding of the concepts discussed in this article will allow healthcare providers to accurately and thoroughly assess the results and validity of medical research. Without an understanding of type I and II errors and power analysis, clinicians could make poor clinical decisions without evidence to support them.

Lesson 3 Types I and Type II Errors

Type I and Type II Errors

Type I and Type II errors can lead to confusion as providers assess medical literature. A vignette that illustrates the errors is the Boy Who Cried Wolf. First, the citizens commit a type I error by believing there is a wolf when there is not. Second, the citizens commit a type II error by believing there is no wolf when there is one.

A type I error occurs when in research when we reject the null hypothesis and erroneously state that the study found significant differences when there indeed was no difference. In other words, it is equivalent to saying that the groups or variables differ when, in fact, they do not or having false positives. An example of a research hypothesis is below:

Drug 23 will *significantly reduce symptoms associated with Disease A* compared to Drug 22.

For our example, if we were to state that Drug 23 significantly reduced symptoms of Disease A compared to Drug 22 when it did not, this would be a type I error. Committing a type I error can be very grave in specific scenarios. For example, if we did, move ahead with Drug 23 based on our research findings even though there was actually was no difference between groups, and the drug costs significantly more money for patients or has more side effects, then we would raise healthcare costs, cause iatrogenic harm, and not improve clinical outcomes. If a p-value is used to examine type I error, the lower the p-value, the lower the likelihood of the type I error to occur.

A type II error occurs when we declare no differences or associations between study groups when, in fact, there was. As with type I errors, type II errors in certain cause problems. Picture an example with a new, less invasive surgical technique that was developed and tested in comparison to the more invasive standard care. Researchers would seek to show no differences between patients receiving the two treatment methods in health outcomes (noninferiority study). If, however, the less invasive procedure resulted in less favorable health outcomes, it would be a severe error. Table 1 provides a depiction of type I and type II errors.

As we saw in the examples on the previous page, the consequences of Type I and Type II errors vary depending on the situation. Researchers take into account the consequences of each when they are setting their level before data are even collected.

In many disciplines an level of 0.05 is standard, for example in the social sciences. There are some situations when a higher or lower level may be desirable. Pilot studies (smaller studies performed

before a larger study) often use a higher level because their purpose is to gain information about the data that may be collected in a larger study; pilot studies are not typically used to make important decisions.

Studies in which making a Type I error would be more dangerous than making a Type II error may use smaller levels. For example, in medical research studies where making a Type I error could mean giving patients ineffective treatments, a smaller level may be set in order to reduce the likelihood of such a negative consequence. Lower levels mean that smaller p-values are needed to reject the null hypothesis; this makes it more difficult to reject the null hypothesis, but this also reduces the probability of committing a Type I error.

Unit II: Correlation and Regression

Lesson 5 Product Moment Coefficient of Correlation

Correlation is a measure of association between to variables. Typically, one variable is denoted as X and the other variable is denoted as Y. The relationship between these variables is assessed by correlation coefficient. Look at the earlier example of optimism and happiness. It states the relationship between one variable, optimism (X) and other variable, happiness (Y). Similarly, following statements are example of correlations: As the intelligence (IQ) increases the marks obtained increases. As the introversion increases number of friends decreases. More the anxiety a person experiences, weaker the adjustment with the stress. As the score on openness to experience increases, scores on creativity test also increase. More the income, more the expenditure. On a reasoning task, as the accuracy increases, the speed decreases. As the cost increases the sales decrease. Those who are good at mathematics are likely to be good at science. As the age of the child increases, the problems solving capacity increase. More the practice, better the performance. All the above statements exemplify the correlation between two variables. The variables are shown in italics. In this first section, we shall introduce ourselves to the concept of correlation.

Correlation: Linear and Non-Linear Relationship The relationship between two variables can be of various types. Broadly, they can be classified as linear and nonlinear relationships. In this section we shall try to understand the linear and nonlinear relationships. Linear Relationship One of the basic forms of relationship is linear relationship. Linear relationship can be expressed as a relationship between two variables that can be plotted as a straight line. The linear relationship can be expressed in the following equation.

$Y = á + â X$ (eq. 1.1)

In the equation 1.1, z

Y is a dependent variable (variable on y-axis),

á (alpha) is a constant or Y intercept of straight line, z

â (beta) is slope of the line and z

X is independent variable (variable on x-axis)

Non-linear Relationship There are other forms of relationships as well. They are called as curvilinear or nonlinear relationships. The Yorkes-Dodson Law, Steven's Power Law in Psychophysics, etc. are good examples of non-linear relationships. The relationship between stress and performance is popularly known as Yorkes-Dodson Law. It suggests that the performance is poor when the stress is too little or too much. It improves when the stress is moderate. Figure 3 shows this relationship. The non-linear relationships, cannot be plotted as a straight line.

The performance is poor at extremes and improves with moderate stress. This is one type of curvilinear relationship.

The characteristics of a relationship

A correlation measures three characteristics of the relationship between X and Y. The three characteristics are as follows

The direction of the relationship: Correlations can be classified into two basic categories positive and negative. In positive correlation, the two variables tend to move in the same direction. When the X variable increases, the Y variable also increases, if the X variable decreases, the Y variable also decreases. In negative correlation, the two variables tend to go in opposite direction. As the X variable increases, the Y variable decreases that is, it is an inverse relationship. The direction of

a relationship is identified by the sign of the correlation. A positive value (+) indicates a positive relationship; a negative value (-) indicates a negative relationship.

The Form of Relationship The most common use of correlation is to measure straight - line relationships. However, others forms of relationships do exist and that there are special correlations used to measure them. For example the relationship between reaction time and age. Reaction time improves with age until the late teens, when it reaches a peak, after that reaction time starts to get worse. This shows a curved relationship.

The Degree or the strength of Relationship Finally, a correlation measures how well the data fit the specific form being considered. For example, a linear correlation measures how well the data points fit on a straight line. A perfect correlation always is identified by a correlation of 1.00 and indicates a perfect fit. At the other extreme, a correlation of 0 indicates no fit at all. Intermediate values represent the degree to which the data points approximate the perfect fit. The numerical value of the correlation also reflects the degree to which there is a consistent, predictable relationship between the two variables. Again, a correlation of 1.00 (or - 1.00) indicates a perfect consistent relationship.

computing Co-efficient of correlation

For expressing the degree of relationship quantitatively between two sets of measures or variables, we usually take the help of an index that is known as co-efficient of correlation. It is a ratio which expresses the extent to which changes in one variable are accompanied by chance in the other variable. It involves no units and varies from - 1 to +1.In case the coefficient of correlation is zero. it indicates zero correlation between two sets of measures.

Check your Progress -1

1. For each of the following, indicate whether you would expect a positive or a negative correlation. a. Model year and price for a used Honda

b. IQ and grade point average for high school students

c. Daily high temperature and daily energy consumption for 30 winter days in New York City

2. The data points would be clustered more closely around a straight line for a correlation of –0.80 than for a correlation of _0.05. (True or false?)

Pearson Product moment correlation

The Pearson's correlation measures the degree and the direction of linear relationship between two variables.

The Pearson's correlation is identified by the letter r and computed by the formula.

$$r = \frac{\text{deg ree to which x and y var y together}}{\text{deg ree to which x and y var y separately}}$$

When there is a perfect linear relationship, every change in the X variable is accompanied by a corresponding change in the Y variable. To calculate the Pearson's correlation, it is necessary to compute the sum of products of deviations. The sum of products provides a parallel procedure for measuring the amount of co - variability between two variables. The Pearson's correlation consists of a ratio comparing the covariability of X and Y (the numerator) with the variability of X and Y separately (the denominator).

Computation correlation of Pearson Product moment

Individuals	**X**	**Y**	**X**	**Y**	**Xy**	**x^2**	y^2
A	15	60	-10	10	-100	100	100
B	25	70	0	20	0	0	400
C	20	40	-5	-10	50	25	100
D	30	50	5	0	0	25	0
E	35	30	10	-20	-200	100	400
					∑xy = 250	∑x 2 =250	∑y 2 =1000

Mean of series x, Px = 25

Mean of series y, Py = 50

xy- sum of product of deviations from the mean

Understanding and interpreting the Pearson correlation.

There are three additional consideration that we need to keep in mind while dealing with Pearson correlation.

Correlation simply describes a relationship between two variables. It does not explain why the two variables are related. The correlation cannot be interpreted as proof of a cause and effect relationship between two variables.

The value of a correlation can be affected greatly by the range of scores represented in the data.

When judging "how good" a relationship is, it is tempting to focus on the numerical value of the correlation for example, a correlation of +.5 is halfway between 0 and 1.00 and therefore appears to represent a moderate degree of relationship. However correlation should not be interpreted as a proportion. Although a correlation of 1.00 does mean that there is a 100% perfectly predictable relationship between X and Y, a correlation of 0.5 does not mean that you can make predictions with 50% accuracy. To describe how accurately one variable predicts the other, you must square the correlation. Thus, a correlation 0.5 provides only r2 = 0.52 = 0.25 or 25% accuracy. The value of r2 is called as the co-efficient of determination because it measures the proportion of variability in one variable that can be determined from the relationship with the other variable. Computed correlation co-efficient may be understood by summarizing it as follows.

The range of computed correlation co-efficient	Interpretation
1. 0 (zero value)	Zero correlation, absolutely no relationship
2. From 0.00 to + 0.20	Slight, almost negligible relationship
3. From + 0.21 to + 0.40	Low correlation, definite but small relationship
4. From + 0.41 to + 0.70	Moderate correlation, substantial but small relationship
5. From + 0.71 to + 0.90	High correlation, marked relationship
6. From + 0.91 to 0.99	Very high correlations, quite dependable relationship
7. +1	Perfect correlation, almost identical and opposite relationship

The use of the product moment method for the computation of correlation co-efficient between two variable is based on the following assumptions.

1. Linearity of relationship: The relationship should be linear (described by a straight line).
2. Homoscedasticity: The standard deviation of the scores in the different columns and rows should be equal and fairly homogenous.
3. Continuity of the variables: The two variables for which correlation is to be computed should be continuous variables in terms of measurement.
4. Normality of the distribution: Distributions in two variables should be fairly symmetrical and unimodal. In other words it should not be badly skewed.

Regression

The statistical technique for finding the best-fitting straight line for a set of data is called regression, and the resulting straight line is called the regression line. Regression is used to denote the estimation and prediction of the average value of one variable for a specified value of the other variable. This estimation is done by deriving a suitable equation on the basis of available bivariate data. This equation is called Regression equation and its geometrical representation is called Regression curve. The regression equation requires the Regression coefficient.

Regression Line: When the bivariate data are plotted on graph paper, the concentration of points shows certain pattern showing the relationship. When the trend points are found to be linear then by least square method we can obtain the regression line. If two variables are linearly related then the relation can be expressed as y = bx + a, where 'b' is the slope of the line and 'a' is the intercept of that line. This line serves several purposes. 1.The line makes the relationship between the two variables easier to see. 2. The line identifies the center, or central tendency, of the relationship, just as the mean describes central tendency for a set of scores. 3. Finally, the line can be used for prediction.

Regression Equation

In general, a linear relationship between two variables X and Y can be expressed by the equation Y = bX+ a (where a and b are fixed constants) For example, a local video store charges a

membership fee of $5 per month, which allows you to rent videos and games for $2 each. With this information, the total cost for 1 month can be computed using a linear equation that describes the relationship between the total cost (Y) and the number of videos and games rented (X).

Y =2X + 5

In the general linear equation, the value ofb is called the slope. The slope determines how much the Y variable changes when X is increased by 1 point. For the video store example, the slope is b = 2 and indicates that your total cost increases by $2 for each video you rent. The value of a in the general equation is called the Y-intercept because it determines the value of Y when X = 0. (On a graph, the avalue identifies the point where the line intercepts the Y-axis.).In other words, Constant a is distance between the point of origin and the point where the regression line touches Y axis. Constant b is the steepness or slope of line and is also called coefficient of regression. In the video store example, a = 5; there is a $5 membership charge even if you never rent a video.

Regression Equation through Least Square Method: Regression equation can be derived through least square method. Linear regression equations are the equations of those lines of best fit which are made on the basis of least square method. To determine how well a line fits the data points, the first step is to define mathematically the distance between the line and each data point. For every X value in the data, the linear equation determines a Y value on the line. This value is the predicted Y and is called ̑a(“Y hat"). The distance between this predicted value and the actual Y value in the data is determined by distance = Y- Y

Note that we simply are measuring the vertical distance between the actual data point. This distance measures the error between the line and the actual data. Because some of these distances are positive and some are negative, the next step is to square each distance to obtain a uniformly positive measure of error. Finally, to determine the total error between the line and the data, we add the squared editors for all of the data points. The result is a measure of overall squared error between the line and the data: total squared error = 6(Y - ̑aฑ 2 Now we can define the best-fitting line as the one that has the smallest total squared - error. For obvious reasons, the resulting line is commonly called the least-squared-error solution. In symbols, we are looking for a linear equation of the for ̑a = bX + a

USING THE REGRESSION EQUATION FOR PREDICTION

As we noted at the beginning of this section, one common use of regression equations is for prediction. For any given value of X, we can use the equation to compute a predicted value for Y. For the equation from Example, an individual with a scope ofX = 1 would be predicted to have a Y score of ̑a=X+4= 1+4=5 Although regression equations can be used for prediction, a few cautions should be considered whenever you are interpreting the predicted values:

USING THE REGRESSION EQUATION FOR PREDICTION

1. The predicted value is not perfect (unless r = +1.00 or -1.00). Although the amount of error varies from point to point, on average the errors are directly related to the magnitude of the correlation. With a correlation near 1.00 (or - 1.00), the data points generally are clustered close to the line and the error is small. As the correlation gets nearer to zero, the points move away from the line and the magnitude of the error increases.
2. The regression equation should not be used to make predictions for X values that fall outside of the range of values covered by the original data.

Lesson 6 Other Types of Correlation (including Phi-coefficient)

PHI COEFFICIENT (φ)

The Pearson’s correlation between one dichotomous variable and another continuous variable is called as point-biserial correlation. When both the variables are dichotomous, then the Pearson’s correlation calculated is called as Phi Coefficient (φ). For example, let us say that you have to compute correlation between gender and ownership of the property. The gender takes two levels, male and female. The ownership of property can be measured as either the person owns a property and the person do not own property. Now you have both the variables measured as dichotomous variables. Now if you compute the Pearson’s correlation between these two variables is called as Phi Coefficient (φ). Both the variables take value of either of 0 or one. Look at the data given in the table below.

The relationship is negative, is function of the way we have assigned the number 0 and 1 to each of the variable. If we assign 0 to females and 1 to males, then we will get the same value of correlation with positive sign. Nevertheless, this does not mean that sign of the relationship cannot be interpreted. Once these numbers have been assigned, then we can interpret the sign. Male is 0 and female is 1; whereas 0 = no ownership and 1 is ownership. The negative relation can be interpreted as follows: as we move from male to female we move negatively from no ownership to ownership. Meaning that male have more ownership than females. We can also calculate the proportion of variance shared by these two variables. That is r2 = φ2 = – .4652 = 0.216 percent.

Significance Testing of Phi (φ)

The significance can be tested by using the Chi-Square (÷2) distribution.

The φ can be converted into the ÷2 by obtaining a product of n and φ2 .

The Chi-Square of nφ2 will have df = 1.

The null and alternative hypothesis are as follows

HO: ñ = 0

HA: ñ ‘“ 0

÷2 = nφ2 = 12 × .216 = 2.5

The value of the chi-square at 1 df is 3.84. the obtained value is less than the tabled value. So we accept the null hypothesis which states that the population correlation is zero.

One need to know that this is primarily because of the small sample size. If we take a larger sample, then the values would be significant. Quickly note the relationship between ÷2 and φ.

2 n χ φ

So one can compute the chi-square and then calculate the phi coefficient

Lesson 7 Partial and Multiple Correlations

PARTIAL CORRELATION (rP)

Two variables, A and B, are closely related. The correlation between them is partialled out, or controlled for the influence of one or more variables is called as partial correlation. So when it is assumed that some other variable is influencing the correlation between A and B, then the influence of this variable(s) is partialled out for both A and B. Hence it can be considered as a correlation between two sets of residuals. Here we discuss a simple case of correlation between A and B is partialled out for C. This can be represented as AB C. r which is read as correlation between A and B partialled out for C. the correlation between A and B can be partialled out for more variables as well.

Formula and Example For example, the researcher is interested in computing the correlation between anxiety and academic achievement controlled from intelligence. Then correlation between academic achievement (A) and anxiety (B) will be controlled for Intelligence (C). This can be represented as: Academic Achievement(A) Anxiety (B) . Intelligence (C) r . To calculate the partial correlation (rP) we will need a data on all three variables. The computational formula is as follows: . 2 2 (1)(1) AB AC BC p AB C AC BC r rr r r r r − = = − − (eq. 3.1) Look at the data of academic achievement, anxiety and intelligence. Here, the academic achievement test, the anxiety scale and intelligence test is administered on ten students. The data for ten students is provided for the three variables in the table below.

In order to compute the partial correlation between the academic achievement and anxiety partialled out for Intelligence, we first need to compute the Pearson’s Product moment correlation coefficient between all three variables. We have already learned to compute it in the first Unit of this Block. So I do not again explain it here. The correlation between anxiety (B) and academic achievement (A) is – 0.369. The correlation between intelligence (C) and academic achievement (A) is 0.918. The correlation between anxiety (B) and intelligence (C) is – 0.245. Give the correlations, we can now calculate the partial correlation . . 22 2 2 0.369 (0.918 0.245) .1441 0.375 (1)(1) (1 0.918)(1 (0.245)) 0.499 AB AC BC AB C AC BC r rr r r r − − − ×− − = = = =− − − − —— (eq.3.2) The partial correlation between the two variables, academic achievement and anxiety controlled for intelligence, is -0.375. You will realise that the correlation between academic achievement and anxiety is -0.369. Whereas, after partialling out for the effect of intelligence, the correlation between them has almost remained unchanged. While computing this correlation, the effect of intelligence on both the variables, academic achievement and anxiety, was removed.

We can test the significance of the partial correlation for the null hypothesis H0 : ñP = 0 and the alternative hypothesis H0 : ñP = 0 Where, the ñP denote the population partial correlation coefficient. The t-distribution is used for this purpose. Following formula is used to calculate the t-value. 2 1 P P r nv t r − = −

Where, r p = partial correlation computed on sample, rAB.C n = sample size, v = total number of variables employed in the analysis. The significance of the rP is tested at the df = n – v. In the present example, we can employ significance testing as follows: 2 2 .375 10 3 0.992 1.69 1 1 (.375) 0.927 P P r nv t r −− −− = = == − −− We test the significance of this value at the df = 7 in the table for t-distribution in the appendix. You will realise that at the df = 7, the table provides the critical value of 2.36 at 0.05 level of significance. The obtained value of 1.69 is smaller than this value. So we accept the null hypothesis stating that H0 : ñP = 0

MULTIPLE CORRELATION COEFFICIENT (R)

The multiple correlation coefficient denoting a correlation of one variable with multiple other variables. The multiple correlation coefficient is denoted as RA.BCD…k which denotes that A is correlated with B, C, D, up to k variables. For example, we want to compute multiple correlation between A with B and C then it is expressed as RA.BC. In this case we create a linear combination of the B and C which is correlated with A. We continue with the same example which we have discussed for partial and semipartial correlations. This example has academic achievement, anxiety and intelligence as three variables. The correlation between academic achievement with the linear combination of anxiety and intelligence is multiple correlation. This denotes the proportion of variance in academic achievement explained by intelligence and anxiety. We denote this as R (Academic Achievement. Intelligence, Anxiety), which is a multiple correlation. Often, it is used in the context of regression, where academic achievement is a criterion variable and intelligence and anxiety are called as predictors.

We are not using regression equation since you have not learned it. The Multiple R can be calculated for two predictor variable as follows. 2 2 A . BC 2 2 1 AB AC AB AC BC BC r r rrr R r + − = − (eq. 3.7) Where, R A . BC = is multiple correlation between A and linear combination of B and C. rAB = is correlation between A and B rAC = is correlation between A and C rBC = is correlation between B and C

Lesson 8 Bivariate and Multiple Regression

BIVARIATE AND MULTIPLE REGRESSION.

We always see that the meteorology department predicts the rain, the economists predict the outcome of a particular policy, financial experts predict the share market, the election experts predict the outcome of voting and so on. Similarly using statistical method, psychologists too can predict certain human behaviours. For example, one can predict the examination marks after writing the examination by checking what questions we have attempted and how we have faired in it and what marks we can expect for each question and so on. Most of us are interested in this exercise of prediction. On many occasions, we also predict the behaviour of our friends, colleagues and family. Predicting and trying to speculate about what might happen in future is integral part of human curiosity. While there are many theories of psychology and personality that would help us predict behaviours, one can also predict a certain phenomenon in terms of statistics. This method is called regression in statistics. The simplest form of the regression is simple linear regression (at times also called as bivariate regression). Carl Frederick Gauss discovered a method of least squares (1809) and later on developed Gauss-Markov theorem (1821). Sir Fransis Galton contributed to the method of regression and also gave the name. Let us see what this is all about. Let us say we have data on two variables (Y and X), and we create an equation, called regression equation, which later on helps us in predicting the score of one variable (Y) by simply using the scores on another variable (X). Let us learn about the utility of the regression analysis, how to do it, how to test the significance and issues surrounding it. Regression analysis tries to predict Y variable from X variable. In the general form, it tries to predict Y from a X1 , X2 , …, Xk , where k is number of predictor variables. Initially we will learn about two variable prediction, one of which is a predictor and the other one will be predicted. Then we will look at the general form of Regression. Just think of the variables that can be used in prediction in psychology. Look at the following statements.

MULTIPLE REGRESSION

When we have multiple predictors than a single predictor variable, the regression carried out is called as multiple regression. So we have a dependent variable and a set of independent variables. Suppose we have X1 , X2 , X3 , … up to Xk as k independent variables, and Y as a dependent variable, then the regression equation for sample can be written as: 11 2 2 ... Y a bX bX bK =+ + ++ k k (eq. 4.32) The same equation for the population can be written as 11 2 2 ... Y XX K =+ + ++ αβ β β k k (eq. 4.33) Look at the following data. The data is about three variables, number of appointments missed, stigma scores, and the distance between the hospital and home. Generally, one would expect that if the stigma is high, then the appointments would be missed. Similarly if the hospital is far away, then the appointments may be missed.

The equation for which we carry out the regression analysis is as follows: Appointmnets Missed ( ) Stigma Score Distance from Home Y ab b e =+ + + 1 2 We will solve the numerical for this problem. I shall directly provide you with the answer

The Multiple R2 for this problem is 0.81. which means that 81 percent information in appointments missed is explained by these two variables. The adjusted value for the same is .76. The value of intercept is –7.88. The slope for stigma is 0.22 and The slope for distance is 0.40.

The obtained F-value tells us that the overall model we have tested for is turning out to be significant. We can actually test the significance of each of the b separately. When that is done, the b of stigma turned out to be significant (t = 4.61, p .05). Here too the size of the sample appears to be the problem leading to non significant results. The multiple regression equation can be solved hierarchically or directly. When the equation is solved directly, all the predictors are entered into the equation simultaneously. When the equation is solved hierarchically, then the predictors are entered one after another depending on the theory or simply depending on their statistical ability to predict the Y. The multiple regression is very useful technique in psychological research.

Unit III: Normal Distribution

Lesson 9 Characteristics of Normal Distribution

Concept of Normal Distribution Carefully look at the following hypothetical frequency distribution, which a teacher has obtained after examining 150 students of class IX on a Mathematics achievement test.

Characteristics or Properties of Normal Probability Curve (NPC):

The characteristics of the normal probability curve are:

The Normal Curve is Symmetrical: The normal probability curve is symmetrical around it's vertical axis called ordinate. The symmetry about the ordinate at the central point of the curve implies that the size, shape and slope of the curve on one side of the curve is identical to that of the other. In other words the left and right halves to the middle central point are mirror images, as shown in the figure given here.

The Normal Curve is Unimodel: Since there is only one maximum point in the curve, thus the normal probability curve is unimodel, i.e. it has only one mode.

The Maximum Ordinate occurs at the Center: The maximum height of the ordinate always occur at the central point of the curve, that is the mid-point. In the unit normal curve it is equal to 0.3989.

The Normal Curve is Asymptotic to the X Axis: The normal probability curve approaches the horizontal axis asymptotically; i.e. the curve continues to decrease in height on both ends away from the middle point (the maximum ordinate point); but it never touches the horizontal axis. Therefore its ends extend from minus infinity (- ∞) to plus infinity (+ ∞).

The Height of the Curve declines Symmetrically: In the normal probability curve the height declines symmetrically in either direction from the maximum point.

The Points of Influx occur at point ±1 Standard Deviation (± 1 σ): The normal curve changes its direction from convex to concave at a point recognised as point of influx. If we draw the perpendiculars from these two points of influx of the curve to the horizontal X axis; touch at a distance one standard deviation unit from above and below the mean (the central point).

The Total Percentage of Area of the Normal Curve within Two Points of Influxation is fixed: Approximately 68.26% area of the curve lies within the limits of ± 1 standard deviation (± 1 σ) unit from the mean.

The Total Area under Normal Curve may be also considered 100 Percent Probability: The total area under the normal curve may be considered to approach 100 percent probability; interpreted in terms of standard deviations. The specified area under each unit of standard deviation are shown in this figure.

The Normal Curve is Bilateral: The 50% area of the curve lies to the left side of the maximum central ordinate and 50% of the area lies to the right side. Hence the curve is bilateral.

The Normal Curve is a mathematical model in behavioural Sciences Specially in Mental Measurement: This curve is used as a measurement scale. The measurement unit of this scale is ± 1σ (the unit standard deviation).

Suppose you have administered a verbal test of intelligence on a group of 50 students studying in class VIII of a school of your city. Further, suppose you find the mean I.Q. of this specified group is "105". Can you from this data or information obtained on the relatively small group, say any thing about the I.Q. of all the VIII class students studying in your city. The answer is "Yes" but under certain conditions. The specified condition is "the degree to which sample mean (M) which is also known as "Estimate" represents its parent population mean which is known as "True Mean" or "Parameter". Therefore the two terms Estimates and Parameters are defined as given below.

Lesson 10 Significance of Mean Differences, Standard Error of the Mean

SIGNIFICANCE AND LEVEL OF SIGNIFICANCE OF THE STATISTICS

Ordinarily, we draw only a single sample from its parent population. However, our problem becomes one of determining how we can infer or estimate the mean of the population (Mpop) on the basis of the sample mean (M). Thus the degree to which a sample mean (M) represents its parameter is an index of the "Significance" or Trustworthiness of the computed sample mean.

When we draw a sample from the population, the observed statistics or estimate that is the mean of the sample obtained, may be some time large or small to the mean of the population (Mpop). The difference may have arisen "by chance" due to the differences in the composition of our sample, or due to its selection method or the procedure followed in the sample selection. The gap between the two measures sample mean (M) and population mean (Mpop), if is low and negligible the sample mean is considered to be trustworthy and we can forecast or estimate the population mean (Mpop) successfully. Therefore, a sample mean (M) is statistically trustworthy or significant to forecast the mean of the population (Mpop), depending upon the probability that the difference between the two measures i.e. Mpop and M could have been arisen "by chance". And the confidence level to which this forecast has been made is known as level of confidence or level of significance. In simpler terms, the level of significance or level of confidence is a degree to which we accept or reject or predict a happening or incidence with confidence.

There are a number of levels of confidence or levels of significance e.g. 100%, 99%, 95%, 90% ..... 50% etc. In psychology and other behavioural sciences, generally, we consider only two levels of significance viz. the 99% level of significance or level of confidence and 95% level of significance a level of confidence. The amount 99% and 95% confidence is also termed as 0.01 and 0.05 level of confidence. The 0.01 level means, if we repeatedly draw a sample or conduct an experiment 100 times, only on one occasion, the obtained sample mean or results will fallout side the limits Mpop ± 2.58 S.E. Here the term S.E. means the standard error exists in the estimate or sample statistics. Similarly 0.05 level means, if repeatedly draw a sample or conduct an experiment 100 times, only on five occasions the obtained sample mean will fall out side the limits Mpop ± 1.96 S.E. The value 1.96 and 2.58 have been taken from the 't' distribution or 't' table (P...). Keeping large size sample in view. The 0.01 level is more rigorous and higher in terms of standard, as compared to the 0.05 level and would require a high level of accuracy and precision. Hence, if an obtained value (on the basis of a sample or an experiment) is significant at 0.01 level, it is automatically significant at 0.05 level but the reverse is not always true.

SAMPLING ERROR AND STANDARD ERROR

The score of an individual of a group obtained on a certain test consists of two types of errors (i) measurement error and (ii) statistics error. In other words, True Score XT = observed score or obtained score (X0) ± error (E) and error E is = Measurement Error + Statistics Error The measurement error is caused by a measuring instrument used to measure a trait or variable and personal observation made by the individual on the instrument.

The measurement error is due to the reliability of a test, as no test is perfectly reliable specially in behavioural sciences. In other words no test or a measuring instrument gives us 100% accurate measurement. The personal error is dependent upon the accurate perception and attention of the individual to take observations or measurement on the measuring instrument. The statistics error refers to the errors of sample statistical measurements or estimates obtained on the basis of a sample drawn from a population. As it is not possible to have perfectly true representative sample of a population in behaviourable sciences. The statistics error is of two types: (i) Sampling Error (ii) Standard Error of statistics i.e. statistical measurements. Now let us see what are these two errors in detail.

Sampling Errors Sampling error refers to the difference between the mean of the entire population and the mean obtained of the sample taken from the population. Thus sampling Error = Mpop ~ M or M - M As the difference is low the mean obtained on the basis of sample is near to the population mean and sample mean is considered to be representing the population mean (or Mpop)

Standard Error The standard error is nothing but the intra differences in the sample measurements of number of samples taken from a single population. As the intra differences in the number of sampling observation i.e. the statistics of the same parent population is less and tending to zero, we may say the obtained sample statistics is quite reliable and can be considered as representative of the Mpop or M . For more clarification, suppose you wish to determine the I.Q. level of the high school going students studying in the various schools of your district. Is it possible for you to administer the intelligence test on all the high school going students of your district and get the Average I.Q.? Your answer may be certainly not.

The easiest method is to select a sample of 10 schools each from urban and rural areas of your district by using random method and administer the intelligence test to the students studying in high school class of these selected 20 schools in total. In such condition, you may have approximately 20 samples of high school going students and have 20 means of intelligence scores obtained on the intelligence test which you have administered. It is possible that all the mean values you have obtained are not equal. Some may be small and some may be large in their values. Theoretically, there should be no difference in the mean value obtained and all the mean values should be equal as all the samples are taken from the same parent population by using random method of sample selection. Thus the inter variation lies within the values of 20 mean values indicating that the error lies within the various observations taken. Further, to have the mean I.Q. of all the high school going students you may calculate combined mean or average mean of all 20 means obtained on the basis of samples taken. This obtained combined or average mean value is the I.Q. of the parent population i.e. the high school going students of your district. If you compare all the 20 sample means to the obtained combined mean value i.e. the population mean, you will find that some of sample means are lesser than this Mpop value, and some are higher. Further one step more, you calculate the difference of these sample means from the population mean obtained, i.e. find (Mpop – M1), (Mpop – M2) ..... (Mpop – M20). You will find that some of these difference values are negative and some are positive. The mean of all these differences should be zero and the standard deviation of all these differences should be 1. As the errors are normally distributed in the universe, in simple terms we can say that we have a normal distribution of specific statistics or sample statistical measurement, which is also known as sampling distribution. Therefore, theoretically the standard error of the statistics (sample statistical measurements) is the standard deviation of the sampling distribution of the statistics and is represented by the symbol S.E.M. Standard Error of sampling measurements on statistics is calculated by using the formula given below :

S.E.M or $\sigma M = N \sigma$ Where, S.E.M = Standard Error of sampling measurement σ = Standard deviation of the scores obtained from the population. N = Size of the sample or total number of

units in a sample Look carefully and study the formula given above, you will find, the standard error of any statistics depends mathematically upon two characteristics. i) the variability or spread of scores around the mean of the population and ii) the number of units or cases in the sample taken from the population. As there is low variability in the scores of population, i.e. the population is homogeneous on the trait being measured, and also the number of cases in the sample are too large, the standard error of the statistics is tending to zero. In the formula, standard error of statistics is directly proportionate to the standard deviation (σ) of the scores of population and inversely proportionate to the size of sample or number of cases in the sample (N). Thus in brief, it can be said that if the population is homogeneous to the variable or trait being measured and a large size of sample (say more than the 500 units), taken from the population; in such condition the sample drawn will be representative to its parent population and is highly reliable.

STANDARD ERROR OF SAMPLE STATISTICS – THE SAMPLE MEAN

The standard error of sample statistics or the statistical measurements of a sample has great importance in inferential statistics. With the help of the standard error statistics we can determine the reliability or trustworthiness of the descriptive statistics e.g. proportion percentage, measures of central tendency (mean, median and mode) measures of variability (standard deviation & quartile deviation), Measures of correlation (r, p and R) etc. For convenience here we discuss only the significance of means which are detained as under :

Meaning of Standard Error of Mean

The Standard error of mean measures the degree to which the mean is affected by the errors of measurement as well as by the errors of Sampling or Sampling fluctuations from one random sample to the other. In other words how dependable is the mean obtained from a sample to its parameter i.e. population Mean (M pop). Keeping in mind the Sample Size; there are two situations: i) Large Sample ii) Small Sample.

The Standard Error of Mean of Large Sample When we say large sample, the number of items in the sample will be more than 30. That is N>30. In such condition the Standard error of the Mean is determined by using the formula given below:

This formula is used when the population parameter of standard deviation (σ) is known. But in practice it is not possible to have the value of Ïó. Having the situation that the sample is selected from the population by using random method of sample selection, σ can be replaced by the ó. The value of the standard deviation of the scores of the sample taken. Therefore, in the above formula the σ can be replaced by ó.

Degree of Freedom Before to proceed for the standard error of small sample mean, it is imperative here to understand the concept of Degrees of Freedom. The expression degrees of freedom is abbreviated from the full expression “Degrees of Freedom to Vary”. A point in the space has unlimited freedom to move in any direction, but a point on a straight line has only one freedom to move on the line i.e. it is free to move in one dimension only. A point on the plane has two degrees of freedom. A point on a three dimension space having three degree of freedom to move. This shows that the degree of freedom is associated to the number of restrictions imposed upon the observations. The degree of freedom is a mathematical concept and is a of key importance in inferential statistics. Almost all test of significance require the calculation of degree of freedom. When a sample statistics is used to estimate a parameter, the number of degrees of freedom depends upon the number of restrictions placed upon the scores, each restriction reducing one degree of freedom (df).

For example, we have four numbers 4, 5, 8 and 3, the sum of these numbers is 20 which is fixed. e.g. here we have restricted freedom to get sum 20, as we have only to change the values of first three figures and not the last one, as it depends upon the fixed sum 20, i.e. 6 + 6 + 9 +———-20 Or 2 + 7 + 10 + ———20 Or 6 + 7 + 5 +———20 In the above sum expressions we have last one freedom to determine the value of last 4th numeral figure as to get fixed sum 20. Therefore in the above expressions, we are bound to take forcefully the numeral figures 4,1 and 2 respectively. Like wise, in Statistics, when we calculate Mean or S.D. of a given distribution of scores we lose one degree of freedom to get fixed sums. Therefore we have N-1 df to compute statistical measures , specifically the standard deviation of the scores given and N-2 in case to compute the co-efficient

of correlation. It means the degree of freedom is not always (N-1) however, but will vary with the problem and the restrictions imposed. In the case of very large sample used in behavioural sciences or social sciences no appreciable difference takes place in the value of óM by N-1 instead of N. The use of N or N-1 thus remains a matter of arbitrary decision. But in the case of small samples having number of units or cases below 30 óM no correction (N-1) has been applied in computation of S.D and thus a considerable variation occurs on the value of óM … Therefore, it is imperative to use N-1 in place of N in computation of óM of the small sample. Further you have to study the t" distribution table (table No-2.5.1) very carefully. In the process, you will find that as the size of sample or degree of freedom approaches to 500 or above, the "t" value approaches to the value of 95% and 99% are 1.96 and 2.58 respectively and remain constant. It means the "t" distribution becomes normal distribution or Z-distribution. When the size of sample decreases especially below 30 you will find the "t" values are gradually increasing at 95% level and 99% level considerably. In such condition the óM values gives us wrong information and we may interpret the results inappropriately.

Lesson 11 Anova (One-Way)

ANALYSIS OF VARIANCE

The analysis of variance is an important method for testing the variation observed in experimental situation into different part each part assignable to a known source, cause or factor. In its simplest form, the analysis of variance is used to test the significance of the differences between the means of a number of different populations. The problem of testing the significance of the differences between the number of means results from experiments designed to study the variation in a dependent variable with variation in independent variable. Thus the analysis of variance, as the name indicates, deals with variance rather than with standard deviations and standard errors. It is a method of dividing the variation observed in experimental data into different parts, each part assignable to a known source, cause or factor therefore.

The technique of analysis of variance was first devised by Sir Ronald Fisher, an English statistician who is also known as the father of modern statistics as applied to social and behavioural sciences. It was first reported in 1923 and its early applications were in the field of agriculture. Since then it has found wide application in many areas of experimentation.

Meaning of the Variance

Before to go further the procedure and use of analysis of variance to test the significance difference between the means of various populations or groups at a time, it is very essential, first to have the clear concept of the term variance. In the terminology of statistics the distance of scores from a central point i.e. Mean is called deviation and the index of variability is known as the mean deviations or standard deviation ( ). In the study of sampling theory, some of the results may be some what more simply interpreted if the variance of a sample is defined as the sum of the squares of the deviation divided by its degree of freedom (N-1) rather than as the mean of the squares deviations.

The variance is the most important measure of variability of a group. It is simply the square of S.D. of the group, but its nature is quite different from standard deviation, though formula for computing variance is same as standard deviation (S.D.).

∴ Variance = S.D.2 or 2 σ = ( )2 X M N Σ –

Where X : are the raw scores of a group, and M : Mean of the raw scores. Thus we can define variance as "the average of sum of squares of deviation from the mean of the scores of a distribution."

Characteristics of Variance

The following are the main features of variance:

The variance is the measure of variability, which indicates the among groups or between groups difference as well as within group difference.

The variance is always in plus sign.

The variance is like an area. While S.D. has direction like length and breadth has the direction.

The scores on normal curve are shown in terms of units, but variance is a area, therefore either it should be in left side or right side of the normal curve.

The variance remain the some by adding or subtracting a constant in a set of data.

The Procedure of Analysis of Variance (ANOVA)

In its simplest form the analysis of variance can be used when two or more than two groups are compared on the basis of certain traits or characteristics or different treatments of simple independent variable is studied on a dependent variable and having two or more than two groups. Before we discuss the procedure of analysis of variance, it is to be noted here that when we have taken a large group or a finite population, to represent its total units the symbol 'N' will be used. When the large group is divided into two or more than two sub groups having equal number of units, the symbol 'n' will be used and for number of groups the symbol 'k' will be used. Now, suppose in an experimental study, three randomly selected groups having equal number of units say 'N' have been assigned randomly, three kinds of reinforcement viz. verbal, kind and written were used. After a certain period, the achievement test was given to three groups and mean values of achievement scores were compared. The mean scores of three groups can then be compared by using ANOVA. Since there is only one factor i.e. type of reinforcement is involved, the situation warrants a single classification or one way ANOVA, and can be arranged as below:

Step 1 Correction tem Cx = ( )2 x N ∑ = ( )2 123 abx x x x nnn ∑ +∑ +∑ + +

Step 2 Sum of Squares of Total SST = 2 ∑ − x Cx = ( )2 2 x x N ∑ ∑ − = ( ) 222 (3) abc x xxx N ∑ ∑ +∑ +∑ −

Step 3 Sum of Squares Among the Groups SSA = ( )2 x Cx N ∑ − = ( ) ( ) ( ) ( ) 2 2 22 123 abc x xxx nn nN

Step 4 Sum of Squares Within the Groups SSW = SST – SSA

Step 5 Mean Scores of Squares Among the Groups MSSA = 1 A SS k − Where k = number of groups.

Step 6 Mean Sum of Squares Within the Groups MSSW = W SS n k − Where N = Total number of units.

Step 7 F Ratio i.e. F = A W MSS MSS

Step 8 Summary of ANOVA

The obtained F ratio in the summary table, furnishes a comprehensive or overall test of the significance of the difference among means of the groups. A significant F does not tell us which mean differ significantly from others. If F-Ratio is not significant, the difference among means is insignificant. The existing or observed differences in the means is due to chance factors or some sampling fluctuations.

To decide whether obtain F-Ratio is significant or not we are taking the help of F table from a statistics book. The obtained F-Ratio is compared with the F value given in the table keeping in mind two degrees of freedom k-1 which is also known as greater degree of freedom or df1 and N-k, which is known as smaller degree of freedom or df2 .Thus, while testing the significance of the F ratio, two situations may arise.

The obtained F Ratio is Insignificant: When the obtained F ratio is found less than the value of F ratio given in F table for corresponding lower degrees of freedom df1 that is, k-1 and higher degree of freedom df that is, (df=N-K) (See F table in a Statistics Book) at .05 and .01 level of significance it is found to be significant or not significant. Thus the null hypothesis is rejected retained. There is no reason for further testing, as none of the mean difference will be significant. When the obtained 'F Ratio' is found higher than the value of F ratio given in F table for its corresponding df1 and df2 at .05 level of .01 level, it is said to be significant. In such condition, we have to proceed further to test the separate differences among the two means, by applying 't' test of significance. This further procedure of testing significant difference among the two means is known as post-hoc test or post ANOVA test of difference.

Steps of One Way Analysis of Variance

From the foregoing two illustration, it is clear that following steps are to be followed when we use analysis of variance.

Step 1 : Set up null hypothesis.

Step 2 : Set the raw scores in table form as shown in the two illustrations.

Step 3 : Square the individual scores of all the sets and write the same in front of the corresponding raw score.

Step 4 : Obtain all the sum of raw scores and the squares of raw scores. Write them at the end of each column.
Step 5 : Obtain grand sums of raw scores as and square of raw square as 2 ∑ x
Step 6 : Calculate correction term by using the formula Cx = 2 x N ∑ Or Cx = (
Step 7 : Calculate sum of squares i.e. SST by using the formulaSST = 2 ∑ x Cx –
Step 8 : Calculate sum of squares among the groups i.e. SSA by using the formulaSSA = 2 – x Cx n ∑ Or SSA = ( ) ( ) ( ) ( ) 1 2 3 2 2 2 2 22 2 2 123 ..... k k xx x x Cx nnn n ∑∑ ∑ ∑ ++++– Step 9 : Calculate sum of squares within the groups i.e. SSw by using the formula SSW = SST – SSA
Step 10 : Calculate the degrees of freedom as greater degree of freedom df1 = k – 1 (where k is number of groups) Smaller degree of freedom df2 = N-k (where N is the total number in the group)
Step 11 : Find the value of Mean sum of squares of two variances asMean sum of squares between the group MSSA = 1 A SS k − Mean sum of squares within the groups MSSW = W SS N K–
Step 12 : Prepare summary table of analysis of variance as shown in 3.2.5 or 3.2.8.
Step 13 : Evaluate obtained F Ratio with the F ratio value given in F table (Table no. 3.3.1) keeping in mind df1 and df2 .
Step 14 : Retain or Reject the Null Hypothesis framed as in step no-I.
Step 15 : If F ratio is found insignificant and null hypothesis is retained, stop further calculation, and interpret the results accordingly. If F ratio is found significant and null hypothesis is rejected, go for further calculations and use post-hoc comparison, find the t values and interpret the results accordingly.

Merits or Advantages of Analysis of Variance The analysis of variance technique has the following advantages:

It is the improved technique over the ‘t’ test or ‘z’ test, it evaluates both types of variance ‘between’ and ‘within’.

This technique is used for ascertaining the difference among several groups or treatments at a time.

It is an economical device.

It can involve more than one variable in studying their main effects and interaction effects.

In some of the experimental design e.g. simple random design and levels X treatment designs are based on one way analysis of variance.

If ‘t’ is not significant, F test must be followed, to analyse the difference between two means.

Demerits or Limitations of Analysis of Variance The analysis of variance technique has following limitations also:

We have seen that analysis of variance techniques is based on certain assumptions e.g. normality and homogeneity of the variances among the groups. The departure of the data from these assumptions may effect adversely on the inferences.

The F value provides global findings of difference among groups, but it can not specify the inference. Therefore, for complete analysis of variance, the ‘t’ test is followed for specifying the statistical inference.

It is time consuming process and requires the knowledge and skills of arithmetical operations as well the high vision for interpretations of the results.

For the use of ‘F’ test, the statistical table of ‘F’ value is essential without it results can not we interpreted.

Lesson 12 Two-Way Anova

TWO WAY ANALYSIS OF VARIANCE

In two way analysis of variance, usually the two independent variables are taken simultaneously. It has two main effects and one interactional or joint effect on dependent variable. In such condition we have to use analysis of variance in two way i.e. vertically as well as horizontally or we have to use ANOVA, column and row wise. To give an example , suppose you are interested to study the intelligence i.e. I.Q. level of boys and girls studying in VIII class in relation to their level of socio economic status (S.E.S.). in such condition you have the following 3 x 2 design.

Table 4.2.1: SES, Intelligence and Gender factors

Levels of S.E.S.

Groups	High	Average	Low	Total
Boys	MHB	MAB	MLB	MB
Girls	MHG	MAG	MLG	MG
Total	MH	MA	ML	M

In the table above,
M : Mean of intelligence scores.
MHB, MAB, & MLB : Mean of intelligence scores of boys belonging to different levels of S.E.S. i.e. High, Average & Low respectively.
MHG, MAG, & MLG : Mean of intelligence scores of girls belonging to different levels of S.E.S. respectively.
MH, MA, & ML : Mean of the intelligence scores of students belonging to different levels of S.E.S. respectively.
MB, MG : Mean of the intelligence scores of boys and girls respectively.
From the above 3 x 2 contingency table, it is clear, first you have to study the significant difference in the means column wise or vertically, i.e. to compare the intelligence level of the students belonging to different categories of socio-economic status (High, Average and Low). Second you have to study the significant difference in the means row wise or horizontally, i.e. to compose the intelligence level of the boys and girls. Then you have to study the interactional or joint effect of sex and socio-economic status on intelligence level i.e. we have to compare the significant difference in the cell means of columns and rows. Obviously, you have more than two groups, and to study the independent as well as interaction effect of the two variables viz. socio-economic status and sex on dependent variable viz. intelligence in terms of I.Q., you have to use two way analysis of variance i.e. to apply analysis of variance column and row wise. Therefore, in two way analysis of variance technique, the following type of effects are to be tested:
Significance of the effect of A variable on D.V.
Significance of the effect of B variable on A.V.
Sgnificance of the interaction effect of A x B variables on D.V.
In two way analysis of variance, the format of summary table after applying the analysis of variance is as under.
For interpretation of the obtained F ratios, we have to evaluate each F ratio value with the F ratio given in F table (refer to statistics book) keeping in view the corresponding grater and smaller df and the level of confidence. There may be two possibilities. All the obtained F ratios may be found insignificant even at .05 level. This shows that there is no independent (i.e. individual) as well as interaction (i.e. joint) effect of the two independent variables on dependent variable. Hence null hypothesis will retain. There is no need to do further calculations.
All the three obtained F ratio's may be found significant either at .05 level of significance or at .01 level of significance. This shows that there is a significant independent (i.e. individual) as well as interactional (i.e. joint) effect of the independent variables on the dependent variable. Therefore the null hypothesis is rejected. In such condition if the two independent variables have more than two levels i.e. three or four, we have to go for further calculations and use post-hoc comparisons by finding out various 't' values by pairing the groups. Similarly the significant interactional effect will also be studied further by applying 't' test of significance or by applying graphical method.
At least one or two obtained F ratio will be found significant either at .05 level of significance or at .01 level of significance. Thus the null hypothesis may partially be retained. In such condition too we have to do further calculations, by making posthoc comparisons and use 't' test of significance, if the independent variables have more than two levels.
For more clarification, go through the following illustration carefully.

Merits of Two Way Analysis of Variance

The following are the advantages of two way analysis of variance.
This technique is used to analyse two types of effects viz. main effects and Interaction Effects.
More than two factors effects are analysed by this technique.
For analysing the data obtained on the basis of factorial designs, this technique is used.
This technique is used to analyse the data for complex experimental studies.

Demerits or Limitations of Two Way ANOVA

The following limitations are found in this technique:
When there are more than two classification of a factor or factors of study. F ratio value provides global picture of difference among the main treatment effects. The inference can be specified by using 't' test in case when F ratio is found significant for a treatment.
This technique also follows the assumptions on which one way analysis of variance is based. If these assumptions are not fulfilled, the use of this technique may give us spurious results.
This technique is difficult and time consuming.

As the number of factors are increased in a study, the complexity of analysis in increased and interpretation of results become difficult.
This technique requires high level arithmetical and calculative ability. Similarly it also requires high level of imaginative and logical ability to interpret the obtained results.

Unit IV: Nonparametric Statistics

Lesson 13 Rationale

DEFINITION OF NON-PARAMETRIC STATISTICS

Non-parametric statistics covers techniques that do not rely on data belonging to any particular distribution. These include (i) distribution free methods (ii) non structural models. Distribution free means its interpretation does not depend on any parametrized distributions. It deals with statistics based on the ranks of observations and not necessarily on scores obtained by interval or ratio scales.
Non-parametric statistics is defined to be a function on a sample that has no dependency on a parameter. The interpretation does not depend on the population fitting any parametrized distributions. Statistics based on the ranks of observations are one example of such statistics and these play a central role in many non-parametric approaches. Non-parametric techniques do not assume that the structure of a model is fixed. Typically, the model grows in size to accommodate the complexity of the data. In these techniques, individual variables are typically assumed to belong to parametric distributions, and assumptions about the types of connections among variables are also made. Non-parametric methods are widely used for studying populations that are based on rank order (such as movie reviews receiving one to four stars). The use of non-parametric methods may be necessary when data have a ranking but no clear numerical interpretation. For instance when we try to assess preferences of the individuals, (e.g. I prefer Red more than White colour etc.), we use non parametic methods. Also when our data is based on measurement by ordinal scale, we use non-parametric statistics.
As non-parametric methods make fewer assumptions, their applicability is much wider than those of parametric methods. Another justification for the use of non-parametric methods is its simplicity. In certain cases, even when the use of parametric methods is justified, non-parametric methods may be easier to use. Due to the simplicity and greater robustness, non-parametric methods are seen by some statisticians as leaving less room for improper use and misunderstanding. A statistic refers to the characteristics of a sample, such as the average score known as the mean. A parameter, on the other hand, refers to the characteristic of a population such as the average of a whole population. A statistic can be employed for either descriptive or inferential purposes and one can use either of the two types of statistical tests, viz., parametric Tests and non-parametric Tests (assumption free test).
The distinction employed between parametric and non-parametric test is primarily based on the level of measurement represented by the data that are being analysed. As a general rule, inferential statistical tests that evaluate categorical / nominal data and ordinal rank order data are categorised as non-parametric tests, while those tests that evaluate interval data or ratio data are categorised as parametric tests.

ASSUMPTIONS OF PARAMETRIC AND NONPARAMETRIC STATISTICS

Assumptions to be met for the use of parametric tests are given below:
Normal distribution of the dependent variable
A certain level of measurement: Interval data
Adequate sample size (>30 recommended per group)
An independence of observations, except with paired data
Observations for the dependent variable have been randomly drawn
Equal variance among sample populations
Hypotheses usually made about numerical values, especially the mean Assumptions of Non-parametric Statistics test are fewer than that of the parametric tests and these are given below.:
An independence of observations, except with paired data
Continuity of variable under study Characteristics of non-parametric techniques:
Fewer assumptions regarding the population distribution
Sample sizes are often less stringent
Measurement level may be nominal or ordinal
Independence of randomly selected observations, except when paired
Primary focus is on the rank ordering or frequencies of data
Hypotheses are posed regarding ranks, medians, or frequencies of data
There are three major parametric assumptions, which are, and will continue to be routinely violated by research in psychology: level of measurement, sample size, and normal distribution of the

dependent variable. The following sections will discuss these assumptions, and elucidate why much of the data procured in health science research violate these assumptions, thus implicating the use of non-parametric techniques.

Sample Size

Adequate sample size is another of the assumptions underlying parametric tests. In a large number of research studies, we do use small sample size and in certain cases we just use one case study and observe that case over a period of time. Some times, we take small sample sizes from a certain place and such samples are called as convenience samples, and limited funding. Thus, the assumption of large sample size is often violated by such studies using parametric statistical techniques. The sample size required for a study has implications for both choices of statistical techniques and resulting power. It has been shown that sample size is directly related to researchers' ability to correctly reject a null hypothesis (power). As such, small sample sizes often reduce power and increase the chance of a type II error. It has been found that by using non-parametric techniques with small sample sizes, it is possible to gain adequate power. However, there does not seem to be a consensus among statisticians regarding what constitutes a small sample size. Many statisticians argue that if the sample size is very small, there may be no alternative to using a non-parametric statistical test, but the value of 'very small' is not delineated. It has been suggested by Wampold et al. (1990) , that the issue of sample size is closely related to the distribution of the dependent variable, given that as sample size increases, the sampling distribution approaches normal($n>100$)

At the same time, one can state that if the distribution of the dependant variable resembles closely the normal distribution, then it will amount to the sampling distribution of the mean being approximately normal. For other distributions, 30 observations might be required. Furthermore, in regard to decision about the statistical technique to be used, there is no clear cut choice but one can choose a technique depending on the nature of the data and sample size. Thus even the choice of parametric or non-parametric tests 'depends' on the nature of the data, the sample size, level of measurement, the researcher's knowledge of the variables' distribution in the population, and the shape of the distribution of the variable of interest. If in doubt, the researcher should try using both parametric and non-parametric techniques.

Normality of the Data According to Pett (1997), in choosing a test we must consider the shape of the distribution of the variable of interest. In order to use a parametric test, we must assume a normal distribution of the dependent variable. However, in real research situations things do not come packaged with labels detailing the characteristics of the population of origin. Sometimes it is feasible to base assumptions of population distributions on empirical evidence, or past experience. However, often sample sizes are too small, or experience too limited to make any reasonable assumptions about the population parameters. Generally in practice, one is only able to say that a sample appears to come from say, a skewed, very peaked, or very flat population. Even when one has precise measurement (ratio scale), it may be irrational to assume a normal distribution, because this implies a certain degree of symmetry and spread. Non-parametric statistics are designed to be used when we know nothing about the distribution of the variable of interest. Thus, we can apply non-parametric techniques to data from which the variable of interest does not belong to any specified distribution (i.e. normal distribution). Although there are many variables in existence that are normally distributed, such as weight, height and strength, this is not true of all variables in social or health sciences.

The incidence of rare disease and low prevalence conditions are both non-normally distributed populations. However, it seems that most researchers using parametric statistics often just 'assume' normality. Micceri et al. (1989) states that the naïve assumption of normality appears to characterise research in many fields. However, empirical studies have documented non normal distributions in literature from a variety of fields. Micceri et al. (1989) investigated the distribution in 440 large sample achievement and psychometric measures. It was found that all of the samples were significantly non-normal ($p<0.01$).

It was concluded that the underlying tenets of normality assuming statistics appeared to be fallacious for the commonly used data in these samples. It is likely that if a similar study, investigating the nature of the distributions of data were to be conducted with some of the measures commonly used in health science research, a similar result would ensue, given that not all variables are normally distributed.

THE USE OF NON-PARAMETRIC TESTS

It is apparent that there are a number of factors involved in choosing whether or not to use a non-parametric test, including level of measurement, sample size and sample distribution. When the choice of statistical technique for a set of data is not clear, there is no harm in analysing the data with both these methods, viz., parametric and on parametric methods. It must be remembered that for each of the main parametric techniques there is a nonparametric test available. Also,

experiments with the data would also determine which test provides the best power, and the greatest level of significance. In general, these tests fall into the following categories:

- Tests of differences between groups (independent samples);
- Tests of differences between variables (dependent samples);
- Tests of relationships between variables

Advantages and Disadvantages of Non-parametric Statistics

Advantages

1) Non-parametric test make less stringent demands of the data. For standard parametric procedures to be valid, certain underlying conditions or assumptions must be met, particularly for smaller sample sizes. The one-sample t test, for example, requires that the observations be drawn from a normally distributed population. For two independent samples, the t test has the additional requirement that the population standard deviations be equal. If these assumptions/conditions are violated, the resulting P-values and confidence intervals may not be trustworthy. However, normality is not required for the Wilcoxon signed rank or rank sum tests to produce valid inferences about whether the median of a symmetric population is 0 or whether two samples are drawn from the same population.
2) Non-parametric procedures can sometimes be used to get a quick answer with little calculation. Two of the simplest non-parametric procedures are the sign test and median test. The sign test can be used with paired data to test the hypothesis that differences are equally likely to be positive or negative, (or, equivalently, that the median difference is 0). For small samples, an exact test of whether the proportion of positives is 0.5 can be obtained by using a binomial distribution. For large samples, the test statistic is: $(\text{plus} - \text{minus})^2 / (\text{plus} + \text{minus})$, where plus is the number of positive values and minus is the number of negative values. Under the null hypothesis that the positive and negative values are equally likely, the test statistic follows the chi-square distribution with 1 degree of freedom. Whether the sample size is small or large, the sign test provides a quick test of whether two paired treatments are equally effective simply by counting the number of times each treatment is better than the other. Example: 15 patients given both treatments A and B to test the hypothesis that they perform equally well. If 13 patients prefer A to B and 2 patients prefer B to A, the test statistic is $(13 - 2)^2 / (13 + 2)$ [= 8.07] with a corresponding P-value of 0.0045. The null hypothesis is therefore rejected. The median test is used to test whether two samples are drawn from populations with the same median. The median of the combined data set is calculated and each original observation is classified according to its original sample (A or B) and whether it is less than or greater than the overall median. The chi-square test for homogeneity of proportions in the resulting 2-by-2 table tests whether the population medians are equal
3) Non-parametric methods provide an air of objectivity when there is no reliable (universally recognized) underlying scale for the original data and there is some concern that the results of standard parametric techniques would be criticized for their dependence on an artificial metric. For example, patients might be asked whether they feel extremely uncomfortable / uncomfortable / neutral / comfortable / very comfortable. What scores should be assigned to the comfort categories and how do we know whether the outcome would change dramatically with a slight change in scoring? Some of these concerns are blunted when the data are converted to ranks.
4) A historical appeal of rank tests is that it was easy to construct tables of exact critical values, provided there were no ties in the data. The same critical value could be used for all data sets with the same number of observations because every data set is reduced to the ranks 1,...,n. However, this advantage has been eliminated by the ready availability of personal computers.
5) Sometimes the data do not constitute a random sample from a larger population. The data in hand are all there are. Standard parametric techniques based on sampling from larger populations are no longer appropriate. Because there are no larger populations, there are no population parameters to estimate. Nevertheless, certain kinds of non-parametric procedures can be applied to such data by using randomization models.

Disadvantages

Such a strong case has been made for the benefits of non-parametric procedures that some might ask why parametric procedures are not abandoned entirely in favour of non-parametric methods! The major disadvantage of non-parametric techniques is contained in its name. Because the procedures are non-parametric, there are no parameters to describe and it becomes more difficult to make quantitative statements about the actual difference between populations. (For example,

when the sign test says two treatments are different, there is no confidence interval and the test does not say by how much the treatments differ.
However, it is sometimes possible with the right software to compute estimates (and even confidence intervals!) for medians, differences between medians. However, the calculations are often too tedious for pencil-and-paper. A computer is required. As statistical software goes though its various iterations, such confidence intervals may become readily available, but are not there.
The second disadvantage is that non-parametric procedures throw away information. The sign test, for example, uses only the signs of the observations. Ranks preserve information about the order of the data but discard the actual values. Because information is discarded, non-parametric procedures can never be as powerful (able to detect existing differences) as their parametric counterparts when parametric tests can be used.

Lesson 14Mann Whitney 'U' Test For Two Sample Test

The Mann-Whitney (Wilcoxon) rank-sum test is a non-parametric analog of the twosample t test for independent samples. The Mann-Whitney U test is a non-parametric test that can be used to analyse data from a two-group independent groups design when measurement is at least ordinal. It analyses the degree of separation (or the amount of overlap) between the Experimental (E) and Control (C) groups. The null hypothesis assumes that the two sets of scores (E and C) are samples from the same population; and therefore, because sampling was random, the two sets of scores do not differ systematically from each other. The alternative hypothesis, on the other hand, states that the two sets of scores do differ systematically. If the alternative is directional, or one-tailed, it further specifies the direction of the difference (i.e., Group E scores are systematically higher or lower than Group C scores).
The statistic that is calculated is either U or U'.
U1 = the number of Es less than Cs
U2 = the number of Cs less than Es
U = the smaller of the two values calculated above
U' = the larger of the two values calculated above
When you perform these tests, your data should consist of a random sample of observations from two different populations. Your goal is to compare either the location parameters (medians) or the scale parameters of the two populations. For example, suppose your data consist of the number of days in the hospital for two groups of patients: those who received a standard surgical procedure and those who received a new, experimental surgical procedure. These patients are a random sample from the population of patients who have received the two types of surgery. Your goal is to decide whether the median hospital stays differ for the two populations.

STEP BY STEP PROCEDURE FOR 'U' TEST FOR SMALL SAMPLE

Step-by-step procedure

Mann Whitney U Test for Small Sample case (not more than 20 items in each set), use U if the data is
a) in the form of ranks or
b) not normally distributed
c) there is an obvious difference in the variance of the two groups
STEP 1: Rank the data (taking both groups together) giving rank 1 to the lowest score, and the highest rank to then highest score.
STEP 2: Find the sum of the ranks for the smaller sample
STEP 3: Find the sum of the ranks for the larger sample
STEP 4: Find U applying the formula given below:
$U = N1\ N2 + [N1\ (N1 + 1) / 2] - \Sigma\ R1$ and
$U' = N1\ N2 + [\ N2\ (N2 + 1) / 2\ ] - \Sigma\ R$
STEP 5: Look up the smaller of U and U' in Table H. There is a significant difference if the observed value is equal to or more than the table value.
STEP 6: Translate the results of the test back in the terms of experiment.

Lesson 15 Kruskal Wallis Analysis of Variance

When there are more than two groups or k number of groups to be compared, ANOVA is utilised but, again since ANOVA is a parametric statistics and requires assumption of normality as a key assumption, we need to also be aware of its non-parametric counterpart. The Kruskal-Wallis test compares the medians of several (more than two) populations to see whether they are all the same or not. The Kruskal Wallis test is a non-parametric analogue to ANOVA. It can be viewed as ANOVA based on rank transformed data. That is, the initial data are transformed to their associated

ranks before being subjected to ANOVA. In other words, it's like ANOVA, except that it is computed with medians and not means. It can also be viewed as a test of medians.
The null and alternative hypotheses may be stated as:
H0: the population medians are equal
H1: the population medians differ

STEP BY STEP PROCEDURE FOR KRUSKAL WALLIS ANOVA

1) Rank all the numbers in the entire data set from smallest to largest (using all samples combined); in the case of ties, use the average of the ranks that the values would have normally been given.
2) Total the ranks for each of the samples; call those totals T1, T2, . . ., Tk, where k is the number of groups or populations.
3) Calculate the Kruskal-Wallis test statistic
H = [12 / N (N+1)] [Σ((ΣR)2 / n)] – 3(N + 1)
N = the total number of cases
n = the number of cases in a given group
(ΣR)2 = the sum of the ranks squared for a given group of subjects
4) Find the p-value.
5) Make your conclusion about whether you can reject Ho by examining the p-value.

Lesson 16 Kendall Rank Correlation and Chi Square.

KENDALL'S RANK ORDER CORRELATION (KENDALL'S TAU): (ð)

Kendall's tau (ð) is one of a number of measures of correlation or association. Measures of correlation are not inferential statistical tests, but are, instead, descriptive statistical measures which represent the degree of relationship between two or more variables. Upon computing a measure of correlation, it is a common practice to employ one or more inferential statistical tests in order to evaluate one or more hypotheses concerning the correlation coefficient. The hypothesis stated below is the most commonly evaluated hypothesis for Kendall's tau. Null Hypothesis Ho : ð = 0 (In the underlying population the sample represents, the correlation between the ranks of subjects on Variable X and Variable Y equals 0.)

STEP BY STEP PROCEDURE FOR KENDALL RANK-ORDER CORRELATION

These are the steps in use of the Kendall rank order Correlation coefficient ð(tau): Rank the observations on the X variable from 1 to N. Rank the observations on the Y variable from 1 to N. Arrange the list of N subjects so that the rank of the subjects on variable X are in their natural order, that is, 1, 2, 3,….N. Observe the Y ranks in the order in which they occur when X ranks are in natural order. Determine the value of S, the number of agreements in order minus the number of disagreements in order, for the observed order of the Y ranks.
If there are no ties among either the X or the Y observations then we use the formula:
T = 2S / (N (N -1))
Where:
S = (score of agreement – score of disagreement on X and Y)
N = Number of objects or individuals ranked on both X and Y
If there are ties then the formula would be:
T= 2S / [N (N-1) – Tx T= 2S / [N (N-1) – Ty
Where:
S and N are as above
Tx = Σ t (t – 1), t being the number of tied observations in each group of the ties on the X variable
T y = Σ t (t – 1), t being the number of tied observation in each group of the ties on the Y variable
If the N subjects constitute a random sample from some population, one may test the hypothesis that the variable X and Y are independent in that population. The method for doing so depends on the size of N:
For N ≤ 10, Table — Upper tail probabilities for T, the Kendall rank order correlation coefficient
For N > 10, but less than 30, Table – Critical value for T, the Kendall rank order correlation coefficient
For N < 30 (or for intermediate significance levels for 10 < N ≤ 30) compute the value of z associated with T by using formula given below and use the z table
z = 3T N (N – 1) / 2 (2N+5)
If the probability yielded by the appropriate method is equal to or less than the critical value, null hypothesis may be rejected in the favour of alternative hypothesis.
Without Ties:
Suppose we ask X and Y to rate their preference for four objects and give points out of 10. Now to see whether their preferences are related to each other we may use the following steps.
Data:

	A	B	C	D
X	6	8	5	2
Y	8	4	9	6

Step 1: Ranking the data of X and Y

	A	B	C	D
X	3	4	2	1
Y	3	1	4	2

Step 2: Rearrange the data of X in order of 1 to N (4 in this case)

	D	C	A	B
X	1	2	3	4

Step 3: Put the corresponding score of Y in order of X and Determine number of agreements and disagreements.

	D	C	A	B
X	1	2	3	4
Y	2	4	3	1

To calculate S we need number of agreements and disagreements. This can be calculated by Using the Y scores, starting from left and counting the number of ranks to its right that are larger, these are agreements in order. We subtract from this the number of ranks to its right that are smaller- these are the disagreements in order. If we do this for all the ranks and then sum the results we obtain S:

Y	2	4	3	1	Total
	2	+	+	-	+1
		4	-	-	-2
			3	-	-1
				1	0
				Grand Total= S – 2	

Step 4: Calculate T

T = 2S / (N (N -1))

T = 2 (– 2) / (4 (4 – 1))

T = – 4 / 12 T= – 0.33 Thus,

T = – 0.33 is a measure of the agreement between the preferences of X and Y

CHI-SQUARE TEST

The chi-square (X2) test measures the alignment between two sets of frequency measures. These must be categorical counts and not percentages or ratios measures (for these, use another correlation test). Note that the frequency numbers should be significant and be at least above 5 (although an occasional lower figure may be possible, as long as they are not a part of a pattern of low figures). Chi Square performs two types of functions.

1) Goodness of fit A common use is to assess whether a measured/observed set of measures follows an expected pattern. The expected frequency may be determined from prior knowledge (such as a previous year's exam results) or by calculation of an average from the given data. The null hypothesis, H0 is that the two sets of measures are not significantly different.

2) Measure of Independence The chi-square test can be used in the reverse manner to goodness of fit. If the two sets of measures are compared, then just as you can show they align, you can also determine if they do not align.

The null hypothesis here is that the two sets of measures are similar. The main difference in goodness-of-fit vs. independence assessments is in the use of the Chi Square table. For goodness of fit, attention is on 0.05, 0.01 or 0.001 figures. For independence, it is on 0.95 or 0.99 figures (this is why the table has two ends to it).

STEP BY STEP PROCEDURE FOR CHISQUARE TEST

1) Write the observed frequencies in column O

2) Figure the expected frequencies and write them in column E.

Expected Frequencies:

When you find the value for chi square, you determine whether the observed frequencies differ significantly from the expected frequencies. You find the expected frequencies for chi square in three ways:

1) You hypothesize that all the frequencies are equal in each category. For example, you might expect that half of the entering freshmen class of 200 at Tech College will be identified as women and half as men. You figure the expected frequency by dividing the number in the sample by the number of categories. In this exam pie, where there are 200 entering freshmen and two categories, male and female, you divide your sample of 200 by 2, the number of categories, to get 100 (expected frequencies) in each category.

2) You determine the expected frequencies on the basis of some prior knowledge. Let us use the Tech College example again, but this time pretend we have prior knowledge of the frequencies of men and women in each category from last year's entering class, when 60% of the freshmen were men and 40% were women. This year you might expect that 60% of the total would be men and 40% would be women. You find the expected frequencies by multiplying the sample size by each of the hypothesized population proportions. If the freshmen total were 200, you would expect 120 to be men (60% x 200) and 80 to be women (40% x 200).
3) Use the formula to find the chi-square value: Chi Square = Σ [(O - E)2 / E] Where: O is the Observed Frequency in each category E is the Expected Frequency in the corresponding category
4) Find the df. (N-1)
5) Find the table value (consult the Chi Square Table.)
6) If your chi-square value is equal to or greater than the table value, reject the null hypothesis: differences in your data are not due to chance alone